



US 20250282877A1

(19) **United States**

(12) **Patent Application Publication**  
**SEEHRA et al.**

(10) **Pub. No.: US 2025/0282877 A1**

(43) **Pub. Date: Sep. 11, 2025**

(54) **METHODS OF USING ALK2 AND ALK3  
ANTIBODIES**

**Publication Classification**

(51) **Int. Cl.**  
**C07K 16/28** (2006.01)

(52) **U.S. Cl.**  
**CPC .... C07K 16/2863** (2013.01); **C07K 2317/565**  
(2013.01); **C07K 2317/76** (2013.01); **C07K**  
**2317/94** (2013.01)

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(21) Appl. No.: **18/289,039**

(22) PCT Filed: **May 11, 2022**

(86) PCT No.: **PCT/US22/28712**

§ 371 (c)(1),

(2) Date: **Oct. 31, 2023**

**Related U.S. Application Data**

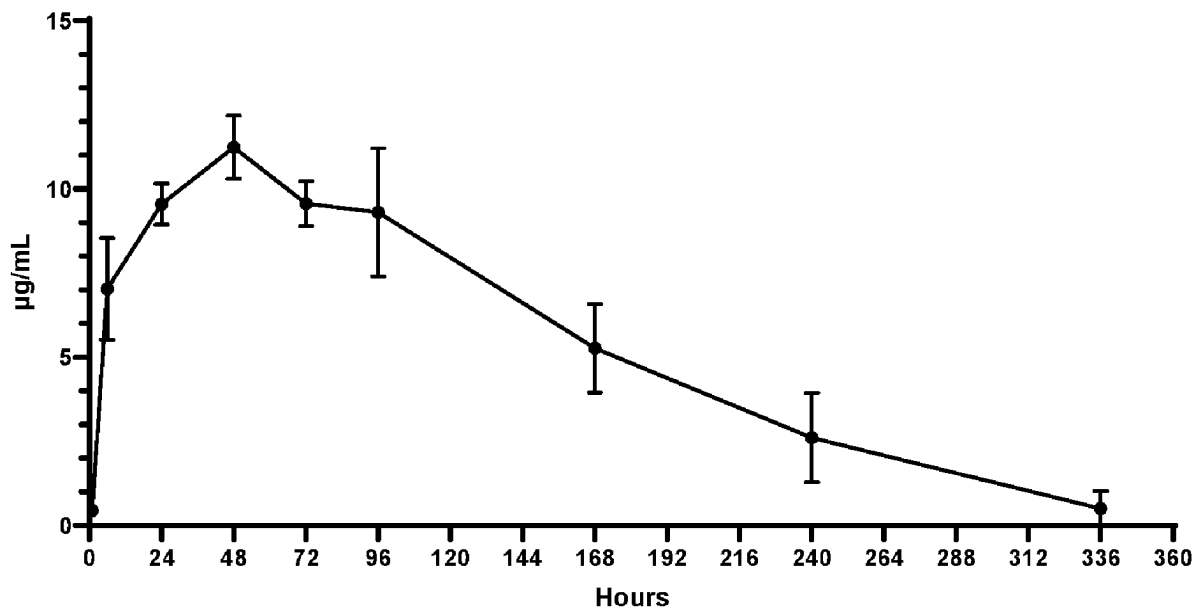
(60) Provisional application No. 63/187,350, filed on May  
11, 2021.

(57) **ABSTRACT**

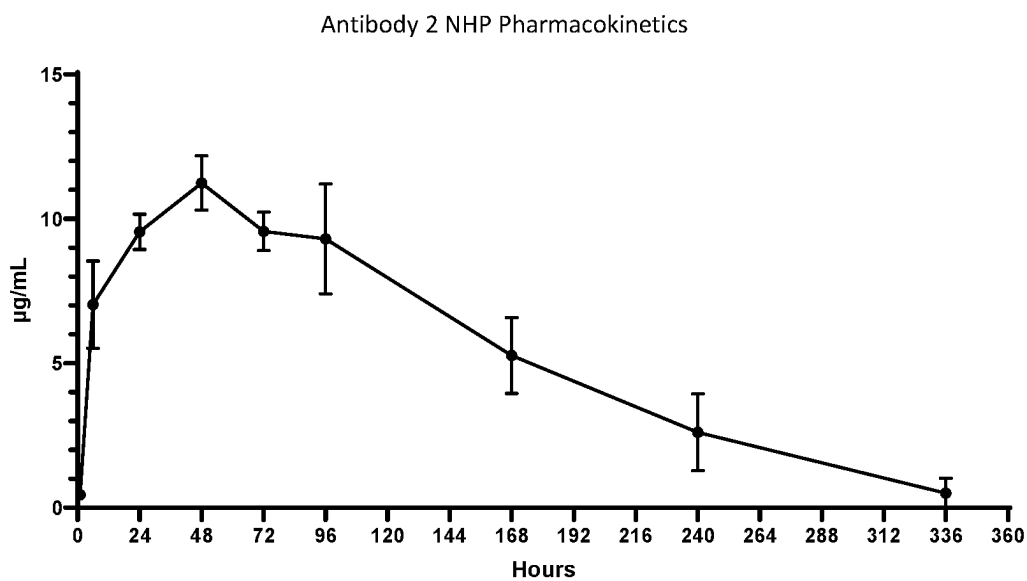
The invention features methods of treating a disease or condition that can be treated with an ALK2 or ALK3 inhibitor including a bone disease or bone damage, low red blood cell levels (e.g., anemia or blood loss), heterotopic ossification (e.g., heterotopic ossification resulting from fibrodysplasia ossificans progressiva), Sjogren's syndrome (e.g., dry eye associated with Sjogren's syndrome), multiple osteochondroma, diffuse intrinsic pontine glioma, posterior capsule opacification, or cardiac hypertrophy and/or cardiac fibrosis by administering to a subject an isolated ALK2 or ALK3 antibody, or antigen binding fragment thereof, in an amount of 0.002 mg/kg to 3 mg/kg.

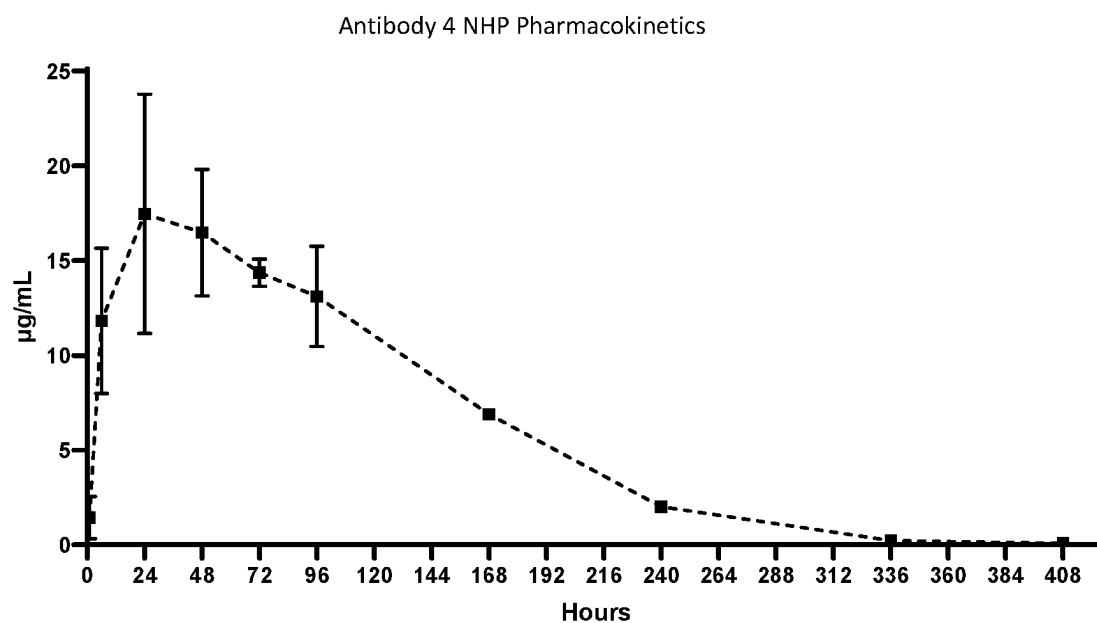
**Specification includes a Sequence Listing.**

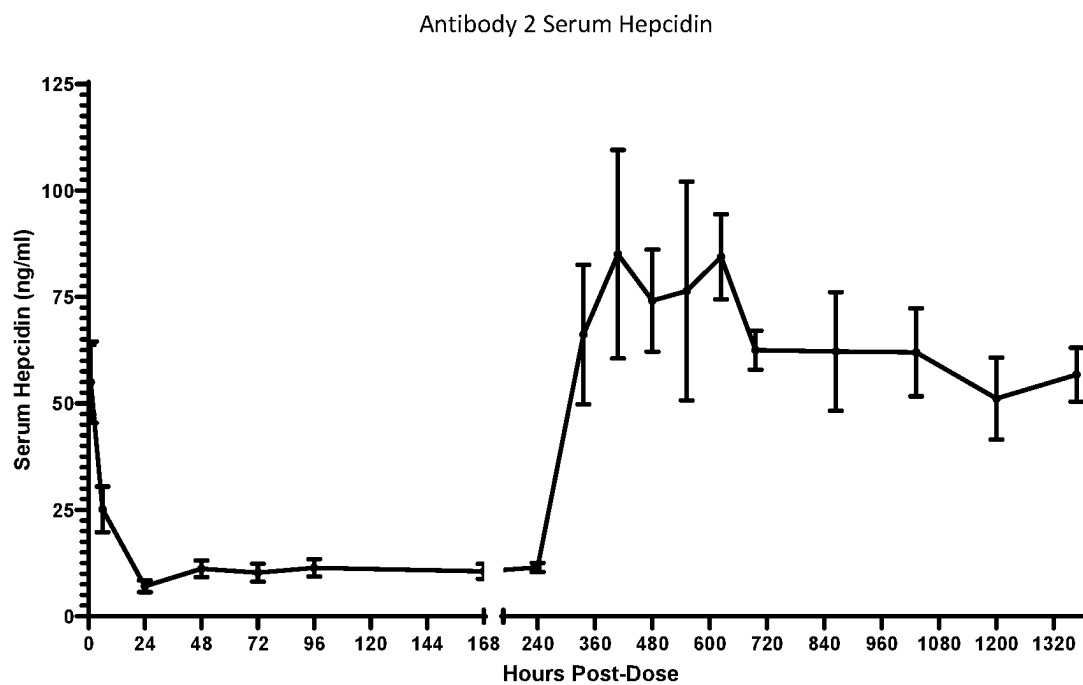
Antibody 2 NHP Pharmacokinetics



**FIG. 1A**



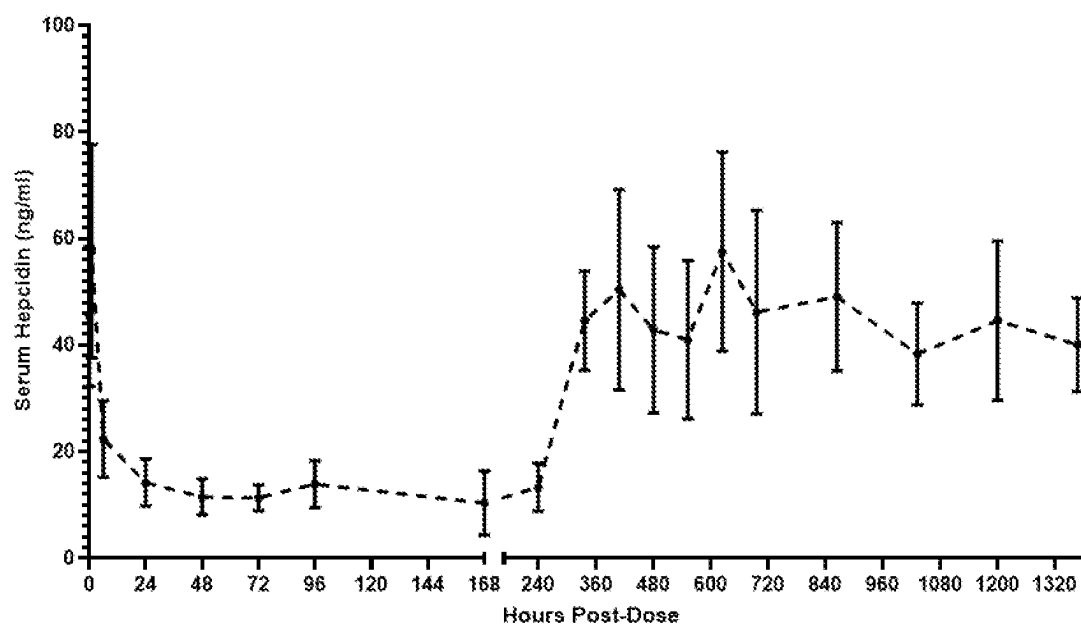
**FIG. 1B**

**FIG. 2A**



**FIG. 2B**

Antibody 4 Serum Hepcidin



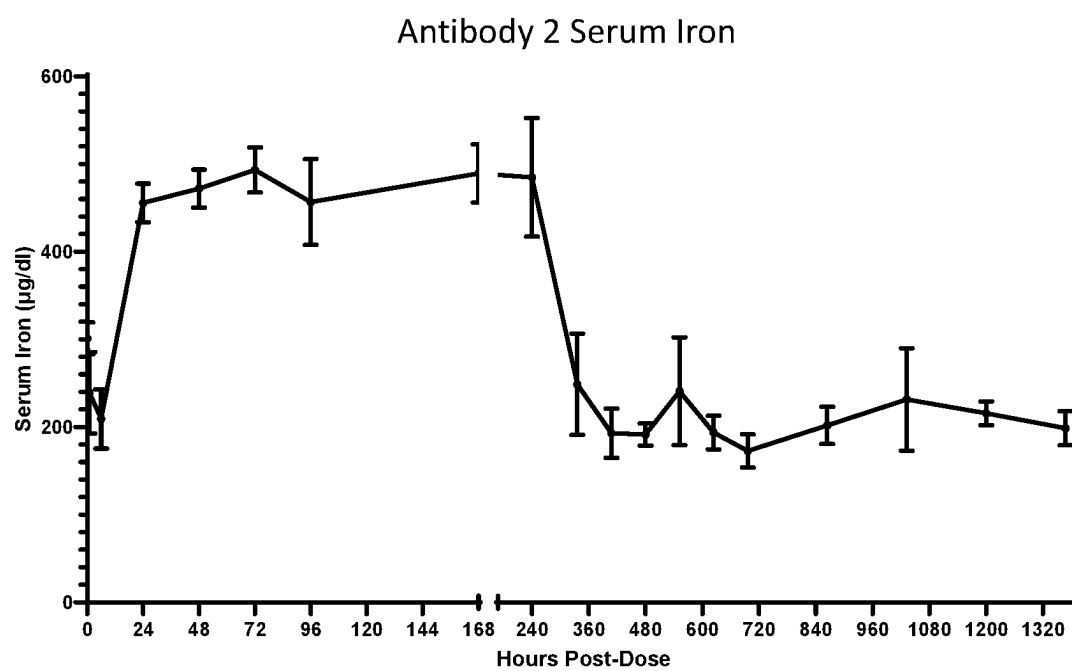
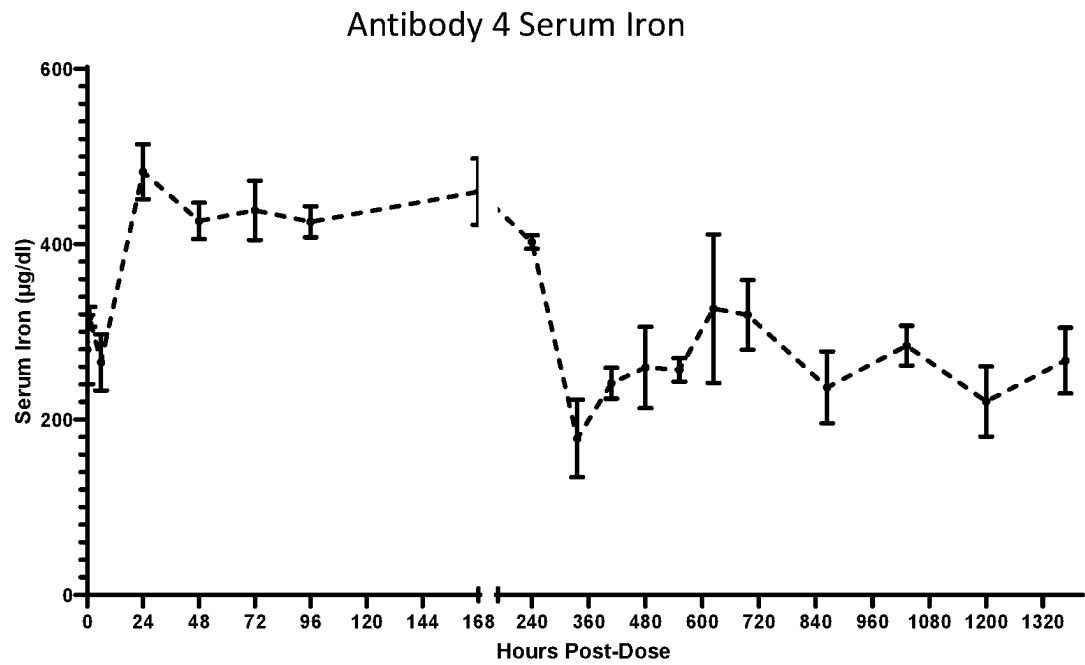
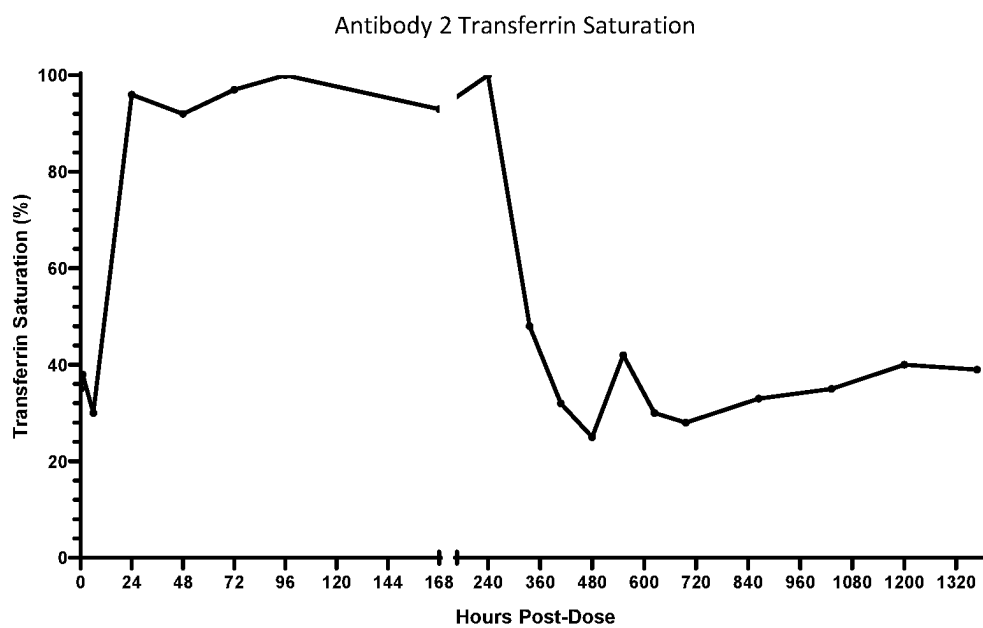
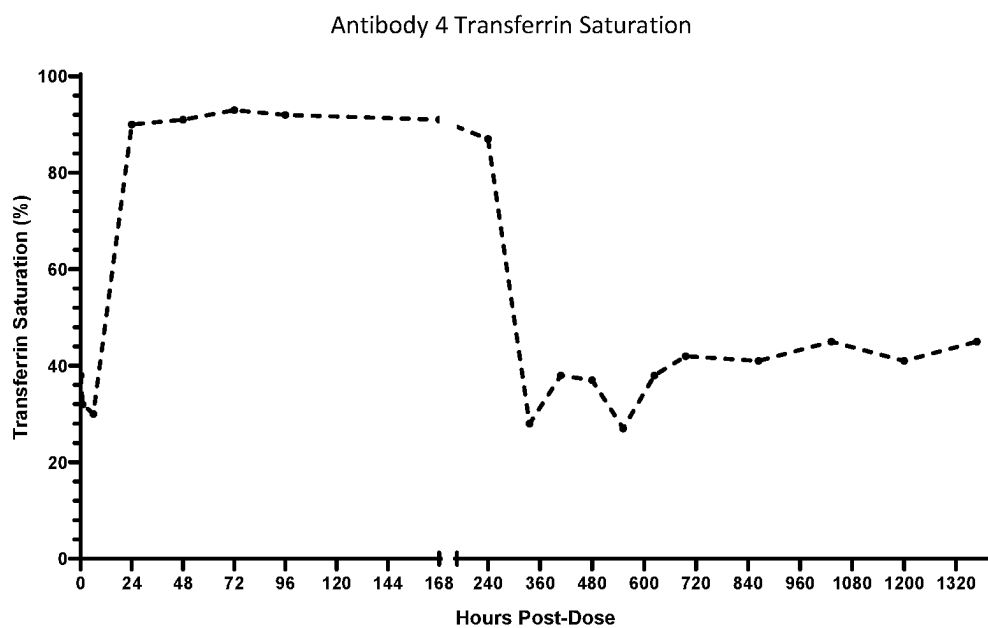
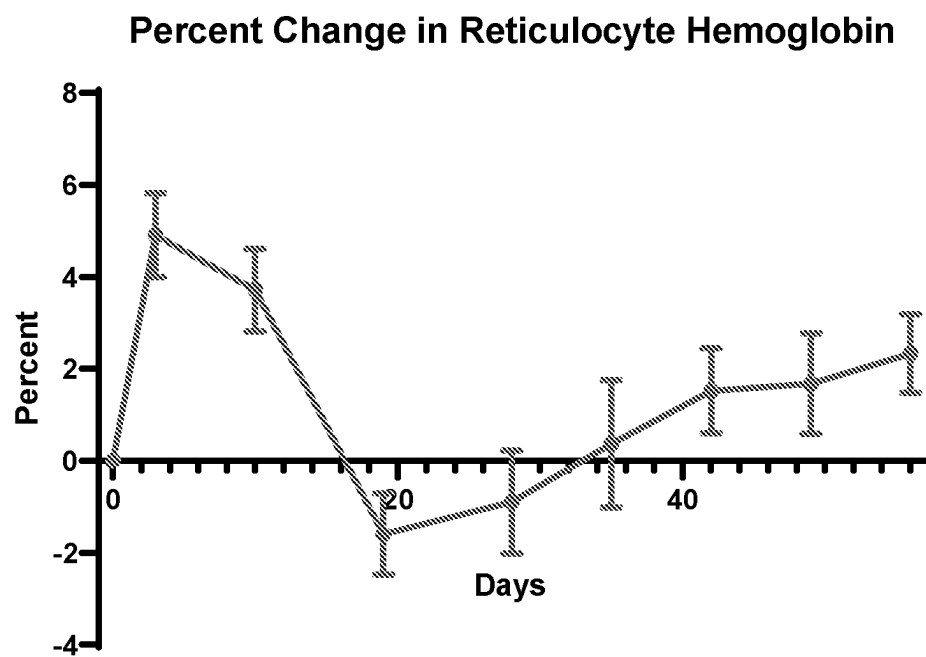
**FIG. 3A**

FIG. 3B

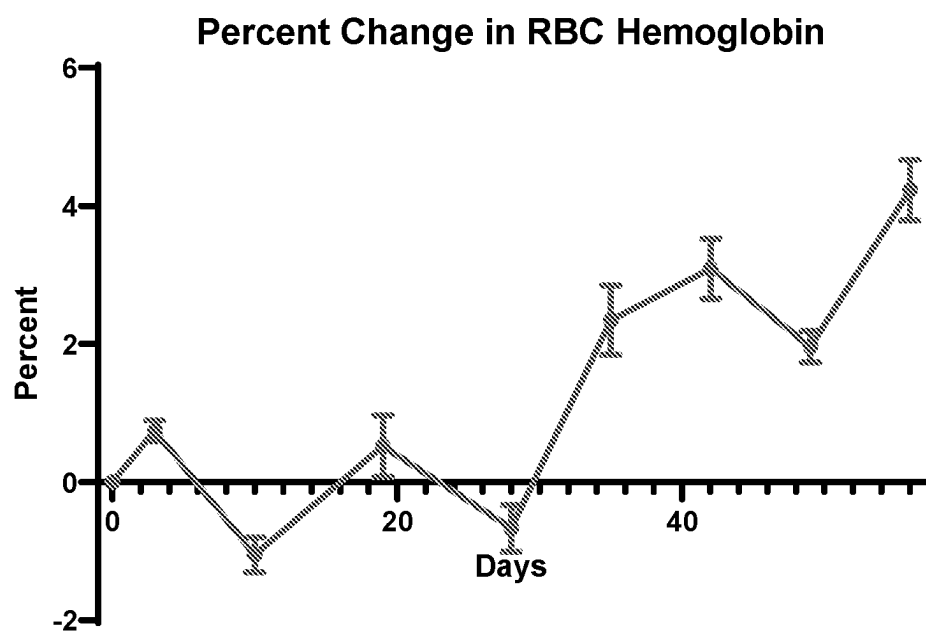


**FIG. 4A**

**FIG. 4B**

**FIG. 5A**

**FIG. 5B**



**FIG. 5C**

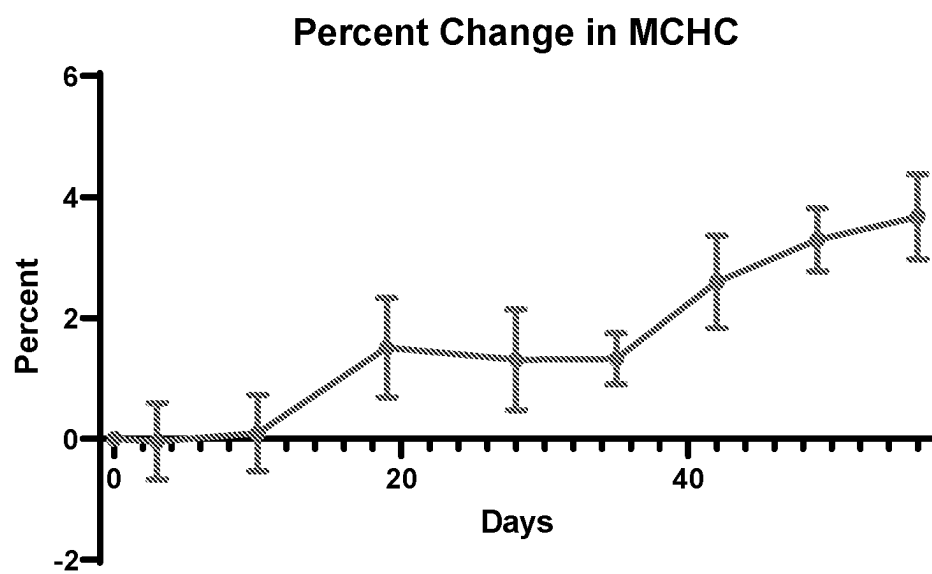




FIG. 6A

Exposure

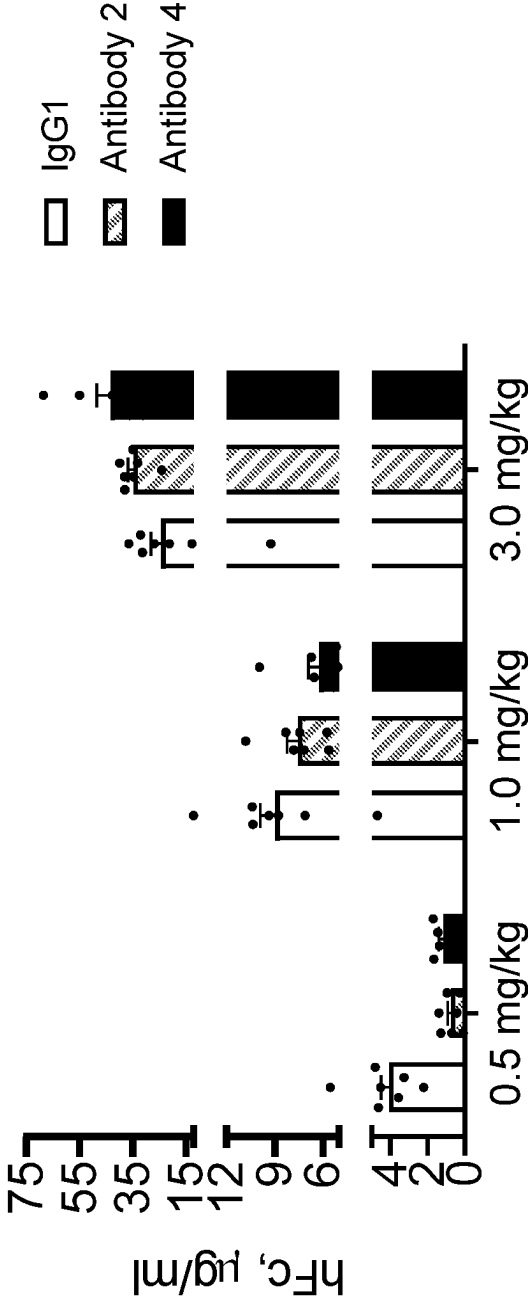
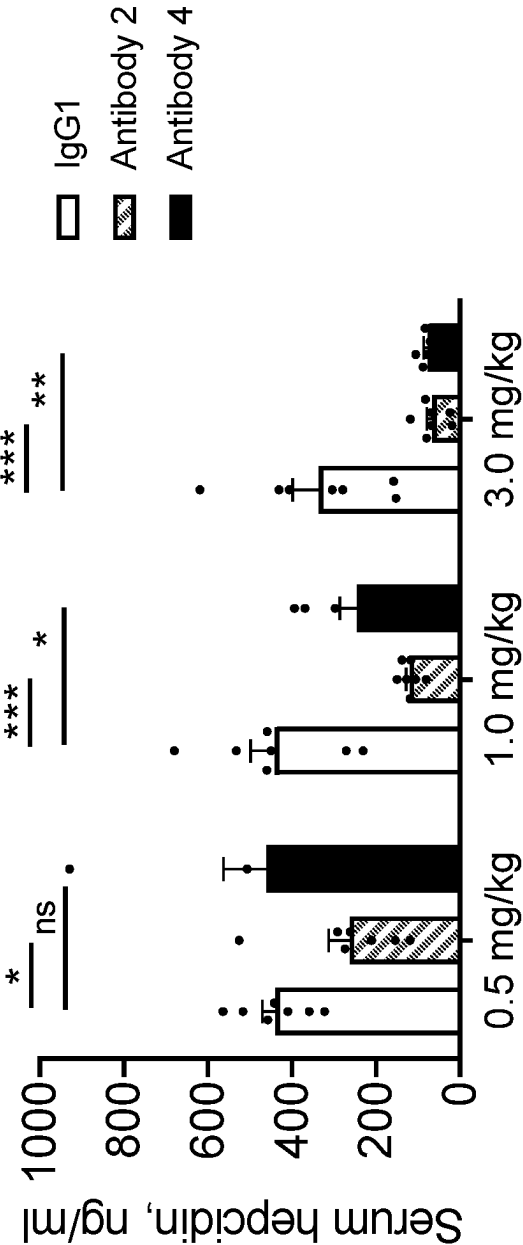
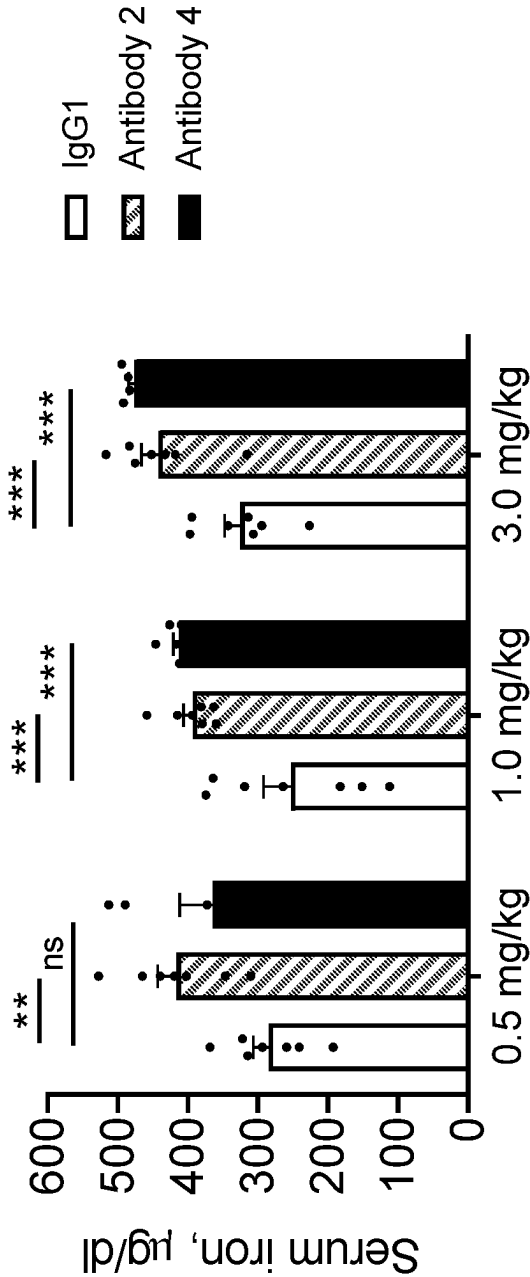


FIG. 6B



ns - not significant; \*p<0.05; \*\*p<0.01; \*\*\*p<0.001

FIG. 6C



ns - not significant; \*p<0.05; \*\*p<0.01, \*\*\*p<0.001

## METHODS OF USING ALK2 AND ALK3 ANTIBODIES

### BACKGROUND OF THE INVENTION

**[0001]** Healthy bone undergoes a constant remodeling that involves both bone breakdown and bone growth. Bone growth is mediated by the osteoblast cell type whereas the osteoclasts resorb the bone. Pathology occurs when these systems fall out of balance either through downregulation of the anabolic program, upregulation of the catabolic system or a combination of both, resulting in a net bone loss. Therefore, controlling the balance in bone remodeling can be useful for promoting the healing of damage to bone as well as the treatment of disorders, such as osteoporosis, associated with loss of bone mass and bone demineralization.

**[0002]** Bone damage can result from a range of root causes, including age- or cancer-related bone loss, genetic conditions, or adverse side effects of drug treatment. The World Health Organization estimates that osteoporosis alone affects 75 million people in the U.S., Europe, and Japan, and is a significant risk factor in bone damage. In general, the whole of bone loss represents pathological states for which there are few effective treatments. Treatment instead focuses on immobilization, exercise, and dietary modifications rather than agents that directly promote bone growth and increase bone density. With respect to osteoporosis, estrogen, calcitonin, osteocalcin with vitamin K, or high doses of dietary calcium are all used as therapeutic interventions. Other therapeutic approaches to osteoporosis include bisphosphonates, parathyroid hormone, parathyroid hormone related protein, calcimimetics, statins, anabolic steroids, lanthanum and strontium salts, and sodium fluoride. Such therapeutics, however, are often associated with undesirable side effects.

**[0003]** Anemia is a global health problem with health implications that affect both morbidity and mortality. In the United States alone, the prevalence of anemia nearly doubled from 2003 to 2012. Symptoms of anemia include fatigue, weakness, shortness of breath, heart palpitations, and reduced cognitive performance, and children, pregnant women, women of reproductive age, and the elderly have been found to have the highest risk of developing anemia. The most common form of anemia is iron deficiency anemia, but anemia can also be caused by chronic diseases, blood loss, and red blood cell destruction. While iron deficiency anemia can be treated with iron supplements, many other forms of anemia, such as aplastic anemia, anemia of chronic disease, and hemolytic anemia may require blood transfusions.

**[0004]** Heterotopic ossification (HO) is the pathologic formation of extra-skeletal bone in soft tissues. This process occurs in patient populations with severe trauma, including large-surface area burns, musculoskeletal injury, orthopedic operations, and spinal cord injury, and in patient populations with a genetic disease known as fibrodysplasia ossificans progressiva (FOP). FOP is caused by a hyper-activating mutation in the type I bone morphogenetic protein (BMP) receptor activin A receptor type 1 (ACVR1), and patients with FOP can develop ectopic bone lesions in muscles, tendons, ligaments, and other connective tissues in the absence of any substantial trauma. The clinical sequela of these pathologic ectopic bone formations, whether in the setting of trauma or genetic mutations, include non-healing

wounds, chronic pain, and joint immobility. In the case of FOP, the formation of ectopic bone can lead to bridges of bone that develop across joints and progressively restrict movement and progressive ossification may lead to death due to loss of thoracic cage compliance.

**[0005]** Treatment options for HO are limited as bone often recurs following surgical resection and attempts to surgically remove extra bone can result in further bone growth. Moreover, some patients may have non-resectable HO due to its sensitive location. The risk of an operation may outweigh the benefits of excision, especially in the face of recurrence. There is a need for therapeutic options that can prevent HO before its initial occurrence in at-risk patients and/or reduce the amount of HO, reduce the recurrence of HO, or prevent additional ectopic bone formation in patients with FOP. There is currently no approved treatment for FOP.

**[0006]** Sjogren's syndrome is a systemic autoimmune disorder identified by its two most common symptoms—dry eye and dry mouth. It may also feature joint pain, swelling, and stiffness, swollen salivary glands, skin rashes or dry skin, vaginal dryness, persistent dry cough, and prolonged fatigue. Sjogren's can also cause dysfunction of organs such as the kidneys, gastrointestinal system, blood vessels, lungs, liver, pancreas, and the central nervous system. It is one of the most prevalent autoimmune disorders with over four million Americans suffering from the disorder, nine out of ten of which are women. Dry eye associated with Sjogren's syndrome may be treated with immunosuppressants and steroid eye drops; however, due to side effects such as increased intraocular pressure and susceptibility to infection, steroid eye drops should not be used for an extended period of time.

**[0007]** Diffuse intrinsic pontine glioma (DIPG) is a pediatric brain tumor that originates in the pons and accounts for approximately 20% of all pediatric brain tumors. DIPG remains incurable with a median survival of 10-11 months. Failure to identify a successful therapy for DIPG likely stems from the lack of biological understanding of the disease, as biopsies were not commonly performed until recently due to the sensitive location of the tumor. As a result, many trials have been based upon genetic alterations found in adult glioblastomas, which have recently been found to be molecularly distinct from DIPG.

**[0008]** Multiple osteochondroma (MO), also called hereditary multiple exostoses, is a rare genetic disorder characterized by the development of multiple benign (non-cancerous) bone tumors called osteochondromas, often on the growing end (metaphysis) of the long bones of the legs, arms, and digits and on flat bones such as the hip and shoulder blade. The number of osteochondromas and the bones on which they are located vary greatly among affected individuals. The osteochondromas are not present at birth, but approximately 96% of affected individuals develop multiple osteochondromas by the time they are 12 years old. These osteochondromas usually continue to grow until shortly after puberty and may lead to bone deformities, skeletal abnormalities, differences in limb length, short stature, pain, decreased range of motion, and pressure on nerves, blood vessels, the spinal cord, and tissues surrounding the osteochondromas. Hereditary multiple osteochondromas is inherited as an autosomal dominant genetic condition and is associated with mutations in the exostosin-1 (EXT1) or exostosin-2 (EXT2) gene. Current treatment options include a "watch and wait" approach (when no

symptoms are present), surgical removal of the osteochondroma, corrective osteotomy, and growth plate arrest or limb-lengthening procedures.

**[0009]** Posterior capsule opacification (PCO) is the most frequent complication associated with decreased vision after cataract surgery and intraocular lens implantation. During cataract surgery, the natural lens of the eye is replaced with an artificial intraocular lens, which is placed inside the lens capsule. PCO occurs when lens epithelial cells migrate toward the posterior capsule, proliferate, and differentiate, growing over the back of the capsule and causing it to thicken and become slightly opaque. This reduces the amount of light that can reach the retina and may lead to blurred or cloudy vision or problems with bright lights and glare. Although PCO can be treated using Nd:YAG laser capsulotomy, the potential complications and significant cost of treatment underscore the importance of identifying new PCO therapies.

**[0010]** Cardiac hypertrophy is the abnormal enlargement or thickening of the heart muscle resulting from increases in cardiomyocyte size and changes in other heart muscle components, such as extracellular matrix. Cardiac hypertrophy may be physiological, occurring in response to exercise or pregnancy, or pathological, occurring in response to stress (e.g., hemodynamic stress), such as hypertension or valvular disease. Cardiac fibrosis may also feature an abnormal thickening of the heart valves due to inappropriate proliferation of cardiac fibroblasts or excess deposition of extracellular matrix in the cardiac muscle, which may result in stiffer, less compliant muscle and occur during the progression to heart failure. There is a lack of efficacious therapies for reducing or reversing cardiac fibrosis, and existing therapies for cardiac hypertrophy, such as adrenergic antagonists, renin-angiotensin system modulators such as angiotensin-converting enzyme (ACE) inhibitors, and angiotensin receptor blockers may have a ceiling effect, characterized by a lack of efficacy, and even regression, in some patients.

**[0011]** There exists a need for novel and effective treatments for bone diseases, anemia, HO (e.g., HO resulting from FOP), Sjogren's syndrome (e.g., dry eye associated with Sjogren's syndrome), DIPG, MO, PCO, and cardiac hypertrophy and/or cardiac fibrosis.

#### SUMMARY OF THE INVENTION

**[0012]** Described herein are methods of treating a subject having a disease or condition that can be treated with an ALK2 or ALK3 inhibitor by administering to the subject an activin receptor-like kinase-2 (ALK2, also known as activin A receptor type I (ACVR1)) antibody, such as an ALK2 neutralizing antibody or an ALK2 binding fragment thereof, or an activin receptor-like kinase-3 (ALK3) antibody, such as an ALK3 neutralizing antibody or an ALK3 binding fragment thereof, in an amount of 0.002 mg/kg to 3 mg/kg. The methods described herein can be used to treat a subject, such as a human subject, that has a bone disease (e.g., a disease or condition involving bone damage, such as primary osteoporosis, secondary osteoporosis, osteopenia, osteopetrosis, osteogenesis imperfecta, bone fracture, bone cancer or cancer metastasis-related bone loss, Paget's disease, renal osteodystrophy, treatment-related bone loss, neuromuscular disease-related bone loss, burn-induced bone loss, anorexia-related bone loss, diet-related bone loss, bone loss associated with the treatment of obesity, low gravity-related bone loss, or immobility-related bone loss.), low red

blood cell levels (e.g., low hemoglobin levels or low red blood cell count, e.g., anemia), heterotopic ossification (HO), such as heterotopic ossification resulting from Fibrodysplasia ossificans progressiva (FOP), Sjogren's syndrome (e.g., dry eye associated with Sjogren's syndrome), diffuse intrinsic pontine glioma (DIPG), multiple osteochondroma (MO), posterior capsule opacification (PCO), or cardiac hypertrophy and/or cardiac fibrosis.

**[0013]** In a first aspect, the disclosure provides a method of treating a subject having a disease or condition that can be treated with an ALK2 or ALK3 inhibitor, the method comprising administering to the subject an isolated ALK2 or ALK3 antibody, or an antigen-binding fragment thereof, in an amount of 0.002 mg/kg to 3 mg/kg (e.g., 0.005 mg/kg to 3 mg/kg, 0.01 mg/kg to 3 mg/kg, 0.05 mg/kg to 3 mg/kg, 0.1 mg/kg to 3 mg/kg, 0.5 mg/kg to 3 mg/kg, 1 mg/kg to 3 mg/kg, 1.5 mg/kg to 3 mg/kg, 2 mg/kg to 3 mg/kg, 2.5 mg/kg to 3 mg/kg, 0.002 mg/kg to 2.5 mg/kg, 0.002 mg/kg to 2 mg/kg, 0.002 mg/kg to 1.5 mg/kg, 0.002 mg/kg to 1 mg/kg, 0.002 mg/kg to 0.5 mg/kg, 0.002 mg/kg to 0.25 mg/kg, 0.002 mg/kg to 0.1 mg/kg, 0.002 mg/kg to 0.05 mg/kg, 0.002 mg/kg to 0.01 mg/kg, 0.002 mg/kg to 0.005 mg/kg, 0.1 mg/kg to 0.3 mg/kg, or 0.1 mg/kg to 1 mg/kg). In some embodiments, the subject is administered an ALK2 antibody, or an antigen-binding fragment thereof. In some embodiments, the ALK2 antibody, or antigen-binding fragment thereof, is a composition of Table 2. In some embodiments, the ALK2 antibody, or antigen-binding fragment thereof, has the CDR sequences of an antibody described in Table 3 (e.g., any of the light chain CDR1, CDR2, and CDR3 sequences and heavy chain CDR1, CDR2, and CDR3 sequences described in Table 3). In some embodiments, the ALK2 antibody or antigen binding fragment thereof includes a light chain CDR1, CDR2, and CDR3 sequence and a heavy chain CDR1, CDR2, and CDR3 sequence from the same row of Table 3 (e.g., a light chain CDR1, CDR2, and CDR3 sequence and a heavy chain CDR1, CDR2, and CDR3 sequence of any one of Antibodies 1-38). In some embodiments, the ALK2 antibody, or antigen-binding fragment thereof, has a variable light chain (VL) sequence and a variable heavy chain (VH) sequence of an antibody described in Table 4. In some embodiments, the ALK2 antibody, or antigen-binding fragment thereof, has a VL sequence and a VH sequence having at least 90% sequence identity (e.g., 90%, 91%, 92%, 93%, 94%, 95%, 95%, 97%, 98%, 99% or 100% sequence identity) to a VL sequence and a VH sequence of Table 5 (e.g., any VL sequence and any VH sequence of Table 5). In some embodiments, the ALK2 antibody, or an antigen binding fragment thereof, has a VL sequence and a VH sequence from the same row of Table 5. In some embodiments, the VH and VL have the sequence of SEQ ID NOs: 107 and 108; 109 and 110; 109 and 115; 111 and 112; 113 and 114; 116 and 118; 116 and 119; 117 and 118; 131 and 118; 117 and 119; 131 and 119; 117 and 115; 131 and 115; 117 and 120; 131 and 120; 121 and 115; 122 and 126; 132 and 126; 122 and 127; 132 and 127; 122 and 128; 132 and 128; 123 and 126; 133 and 126; 123 and 127; 133 and 127; 123 and 128; 133 and 128; 124 and 126; 124 and 127; 124 and 128; 124 and 129; 125 and 128; 125 and 129; 125 and 130; 134 and 129; 135 and 127; 139 and 129; 140 and 136; 140 and 137; 140 and 138; 201 and 202; 203 and 204; 205 and 206; 207 and 208; 304 and 305; 306 and 307; 308 and 309; 310 and 311; 306 and 312; 313 and 314; 315 and 316; 317 and 318; 319 and 320; 321 and 322; 323

and 324; 325 and 326; 327 and 328; 329 and 330; 331 and 332; 333 and 334; 335 and 336; 337 and 338; 339 and 340; or 341 and 342. In some embodiments, the ALK2 antibody, or an antigen-binding fragment thereof, has a heavy chain sequence and light chain sequence having at least 90% sequence identity (e.g., 90%, 91%, 92%, 93%, 94%, 95%, 95%, 97%, 98%, 99% or 100% sequence identity) to a heavy chain and light chain sequence of Table 6 (e.g., any heavy chain sequence and any light chain sequence of Table 6). In some embodiments, the ALK2 antibody or an antigen binding fragment thereof, has a heavy chain sequence and light chain sequence from the same row of Table 6. In some embodiments, the subject is administered an ALK3 antibody, or an antigen-binding fragment thereof. In some embodiments, the ALK3 antibody, or antigen-binding fragment thereof, contains an antigen binding fragment (Fab) described in Harth et al., PLoS ONE 5: e13049, 2010, such as AbD1556 or AbD1564. In some embodiments, the ALK3 antibody, or antigen binding fragment thereof, has a heavy chain CDR1 including TGYMK (SEQ ID NO: 79); a heavy chain CDR2 including RINPDNGGRTYNQIFKDK (SEQ ID NO: 80); and a heavy chain CDR3 including RERGQYGNYGGSFSD (SEQ ID NO: 81). In some embodiments, the ALK3 antibody, or antigen binding fragment thereof, contains a heavy chain variable region having at least 90% (e.g., at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or more) sequence identity to SEQ ID NO: 77 or SEQ ID NO: 78.

**[0014]** In some embodiments, the antibody is administered in an amount of 0.01 mg/kg to 2 mg/kg (e.g., 0.05 mg/kg to 2 mg/kg, 0.1 mg/kg to 2 mg/kg, 0.5 mg/kg to 2 mg/kg, 1 mg/kg to 2 mg/kg, 1.5 mg/kg to 2 mg/kg, 0.01 mg/kg to 1.5 mg/kg, 0.01 mg/kg to 1 mg/kg, 0.01 mg/kg to 0.5 mg/kg, 0.01 mg/kg to 0.25 mg/kg, 0.01 mg/kg to 0.1 mg/kg, or 0.01 mg/kg to 0.05 mg/kg). In some embodiments, the antibody is administered in an amount of 0.05 mg/kg to 1 mg/kg (e.g., 0.1 mg/kg to 1 mg/kg, 0.2 mg/kg to 1 mg/kg, 0.3 mg/kg to 1 mg/kg, 0.4 mg/kg to 1 mg/kg, 0.5 mg/kg to 1 mg/kg, 0.6 mg/kg to 1 mg/kg, 0.7 mg/kg to 1 mg/kg, 0.8 mg/kg to 1 mg/kg, 0.9 mg/kg to 1 mg/kg, 0.05 mg/kg to 0.9 mg/kg, 0.05 mg/kg to 0.08 mg/kg, 0.05 mg/kg to 0.8 mg/kg, 0.05 mg/kg to 0.7 mg/kg, 0.05 mg/kg to 0.6 mg/kg, 0.05 mg/kg to 0.5 mg/kg, 0.05 mg/kg to 0.4 mg/kg, 0.05 mg/kg to 0.3 mg/kg, 0.05 mg/kg to 0.2 mg/kg, or 0.05 mg/kg to 0.1 mg/kg). In some embodiments, the antibody is administered in an amount of 0.01 mg/kg to 1 mg/kg (e.g., 0.05 mg/kg to 1 mg/kg, 0.1 mg/kg to 1 mg/kg, 0.2 mg/kg to 1 mg/kg, 0.3 mg/kg to 1 mg/kg, 0.4 mg/kg to 1 mg/kg, 0.5 mg/kg to 1 mg/kg, 0.6 mg/kg to 1 mg/kg, 0.7 mg/kg to 1 mg/kg, 0.8 mg/kg to 1 mg/kg, 0.9 mg/kg to 1 mg/kg, 0.01 mg/kg to 0.9 mg/kg, 0.01 mg/kg to 0.8 mg/kg, 0.01 mg/kg to 0.7 mg/kg, 0.01 mg/kg to 0.6 mg/kg, 0.01 mg/kg to 0.5 mg/kg, 0.01 mg/kg to 0.4 mg/kg, 0.01 mg/kg to 0.3 mg/kg, 0.01 mg/kg to 0.2 mg/kg, 0.01 mg/kg to 0.1 mg/kg, or 0.01 mg/kg to 0.05 mg/kg). In some embodiments, the antibody is administered in an amount of 0.01 mg/kg to 0.3 mg/kg (e.g., 0.01 mg/kg to 0.3 mg/kg, 0.025 mg/kg to 0.3 mg/kg, 0.05 mg/kg to 0.3 mg/kg, 0.075 mg/kg to 0.3 mg/kg, 0.1 mg/kg to 0.3 mg/kg, 0.125 mg/kg to 0.3 mg/kg, 0.15 mg/kg to 0.3 mg/kg, 0.175 mg/kg to 0.3 mg/kg, 0.2 mg/kg to 0.3 mg/kg, 0.225 mg/kg to 0.3 mg/kg, 0.25 mg/kg to 0.3 mg/kg, 0.275 mg/kg to 0.3 mg/kg, 0.01 mg/kg to 0.275 mg/kg, 0.01 mg/kg to 0.25 mg/kg, 0.01 mg/kg to 0.225 mg/kg, 0.01 mg/kg to 0.2 mg/kg, 0.01 mg/kg to 0.225 mg/kg, 0.01 mg/kg to 0.2

mg/kg, 0.01 mg/kg to 0.175 mg/kg, 0.01 mg/kg to 0.15 mg/kg, 0.01 mg/kg to 0.125 mg/kg, 0.01 mg/kg to 0.1 mg/kg, 0.01 mg/kg to 0.075 mg/kg, 0.01 mg/kg to 0.05 mg/kg, or 0.01 mg/kg to 0.025 mg/kg). In some embodiments, the antibody is administered in amount of 0.1 mg/kg to 0.3 mg/kg (e.g., 0.1 mg/kg to 0.25 mg/kg, 0.1 mg/kg to 0.2 mg/kg, 0.1 mg/kg to 0.15 mg/kg, 0.15 mg/kg to 0.3 mg/kg, 0.2 mg/kg to 0.3 mg/kg, or 0.25 mg/kg to 0.3 mg/kg).

**[0015]** In some embodiments, the antibody is administered to the subject at a frequency that is weekly to semi-annually. For example, the antibody may be administered once a week, once every 14 days, once a month, once every two months, once every three months, once every four months, once every five months, or once every six months. In some embodiments, the antibody may be administered to the subject at a frequency of once every 6 months. In some embodiments, the antibody may be administered to the subject at a frequency of once every 4 months. In some embodiments, the antibody may be administered to the subject at a frequency of once every 2 months. In some embodiments, the antibody may be administered to the subject at a frequency of once every month. In some embodiments, the antibody may be administered to the subject at a frequency of once biweekly. In some embodiments, the antibody is administered to the subject at a frequency of once every 10 to 14 days (e.g., once every 10 days, once every 11 days, once every 12 days, once every 13 days, or once every 14 days). In some embodiments, the antibody is administered to the subject at a frequency of once a week.

**[0016]** In some embodiments, the antibody, or antigen-binding fragment thereof, is administered subcutaneously. In some embodiments, the antibody, or antigen-binding fragment thereof, is administered intravenously.

**[0017]** In some embodiments, the disease or condition that can be treated with an ALK2 or ALK3 inhibitor is anemia. In some embodiments, the subject being treated for anemia is administered an isolated ALK2 or ALK3 antibody, or an antigen-binding fragment thereof, in an amount of 0.002 mg/kg to 3 mg/kg (e.g., 0.005 mg/kg to 3 mg/kg, 0.01 mg/kg to 3 mg/kg, 0.05 mg/kg to 3 mg/kg, 0.1 mg/kg to 3 mg/kg, 0.5 mg/kg to 3 mg/kg, 1 mg/kg to 3 mg/kg, 1.5 mg/kg to 3 mg/kg, 2 mg/kg to 3 mg/kg, 2.5 mg/kg to 3 mg/kg, 0.002 mg/kg to 2.5 mg/kg, 0.002 mg/kg to 2 mg/kg, 0.002 mg/kg to 1.5 mg/kg, 0.002 mg/kg to 1 mg/kg, 0.002 mg/kg to 0.5 mg/kg, 0.002 mg/kg to 0.25 mg/kg, 0.002 mg/kg to 0.1 mg/kg, 0.002 mg/kg to 0.05 mg/kg, 0.002 mg/kg to 0.01 mg/kg, 0.002 mg/kg to 0.005 mg/kg, 0.1 mg/kg to 0.3 mg/kg, or 0.1 mg/kg to 1 mg/kg) once every 6 months. In some embodiments, the subject is administered an isolated ALK2 or ALK3 antibody, or an antigen-binding fragment thereof, in an amount of 0.002 mg/kg to 3 mg/kg (e.g., 0.005 mg/kg to 3 mg/kg, 0.01 mg/kg to 3 mg/kg, 0.05 mg/kg to 3 mg/kg, 0.1 mg/kg to 3 mg/kg, 0.5 mg/kg to 3 mg/kg, 1 mg/kg to 3 mg/kg, 1.5 mg/kg to 3 mg/kg, 2 mg/kg to 3 mg/kg, 2.5 mg/kg to 3 mg/kg, 0.002 mg/kg to 2.5 mg/kg, 0.002 mg/kg to 2 mg/kg, 0.002 mg/kg to 1.5 mg/kg, 0.002 mg/kg to 1 mg/kg, 0.002 mg/kg to 0.5 mg/kg, 0.002 mg/kg to 0.25 mg/kg, 0.002 mg/kg to 0.1 mg/kg, 0.002 mg/kg to 0.05 mg/kg, 0.002 mg/kg to 0.01 mg/kg, 0.002 mg/kg to 0.005 mg/kg, 0.1 mg/kg to 0.3 mg/kg, or 0.1 mg/kg to 1 mg/kg) once every 4 months. In some embodiments, the subject is administered an isolated ALK2 or ALK3 antibody, or an antigen-binding fragment thereof, in an amount of

0.002 mg/kg to 3 mg/kg (e.g., 0.005 mg/kg to 3 mg/kg, 0.01 mg/kg to 3 mg/kg, 0.05 mg/kg to 3 mg/kg, 0.1 mg/kg to 3 mg/kg, 0.5 mg/kg to 3 mg/kg, 1 mg/kg to 3 mg/kg, 1.5 mg/kg to 3 mg/kg, 2 mg/kg to 3 mg/kg, 2.5 mg/kg to 3 mg/kg, 0.002 mg/kg to 2.5 mg/kg, 0.002 mg/kg to 2 mg/kg, 0.002 mg/kg to 1.5 mg/kg, 0.002 mg/kg to 1 mg/kg, 0.002 mg/kg to 0.5 mg/kg, 0.002 mg/kg to 0.25 mg/kg, 0.002 mg/kg to 0.1 mg/kg, 0.002 mg/kg to 0.05 mg/kg, 0.002 mg/kg to 0.01 mg/kg, 0.002 mg/kg to 0.005 mg/kg, 0.1 mg/kg to 0.3 mg/kg, or 0.1 mg/kg to 1 mg/kg) once every 3 months. In some embodiments, the subject is administered an isolated ALK2 or ALK3 antibody, or an antigen-binding fragment thereof, in an amount of 0.002 mg/kg to 3 mg/kg (e.g., 0.005 mg/kg to 3 mg/kg, 0.01 mg/kg to 3 mg/kg, 0.05 mg/kg to 3 mg/kg, 0.1 mg/kg to 3 mg/kg, 0.5 mg/kg to 3 mg/kg, 1 mg/kg to 3 mg/kg, 1.5 mg/kg to 3 mg/kg, 2 mg/kg to 3 mg/kg, 2.5 mg/kg to 3 mg/kg, 0.002 mg/kg to 2.5 mg/kg, 0.002 mg/kg to 2 mg/kg, 0.002 mg/kg to 1.5 mg/kg, 0.002 mg/kg to 1 mg/kg, 0.002 mg/kg to 0.5 mg/kg, 0.002 mg/kg to 0.25 mg/kg, 0.002 mg/kg to 0.1 mg/kg, 0.002 mg/kg to 0.05 mg/kg, 0.002 mg/kg to 0.01 mg/kg, 0.002 mg/kg to 0.005 mg/kg, 0.1 mg/kg to 0.3 mg/kg, or 0.1 mg/kg to 1 mg/kg) once every 2 months. In some embodiments, the subject is administered an isolated ALK2 or ALK3 antibody, or an antigen-binding fragment thereof, in an amount of 0.002 mg/kg to 3 mg/kg (e.g., 0.005 mg/kg to 3 mg/kg, 0.01 mg/kg to 3 mg/kg, 0.05 mg/kg to 3 mg/kg, 0.1 mg/kg to 3 mg/kg, 0.5 mg/kg to 3 mg/kg, 1 mg/kg to 3 mg/kg, 1.5 mg/kg to 3 mg/kg, 2 mg/kg to 3 mg/kg, 2.5 mg/kg to 3 mg/kg, 0.002 mg/kg to 2.5 mg/kg, 0.002 mg/kg to 2 mg/kg, 0.002 mg/kg to 1.5 mg/kg, 0.002 mg/kg to 1 mg/kg, 0.002 mg/kg to 0.5 mg/kg, 0.002 mg/kg to 0.25 mg/kg, 0.002 mg/kg to 0.1 mg/kg, 0.002 mg/kg to 0.05 mg/kg, 0.002 mg/kg to 0.01 mg/kg, 0.002 mg/kg to 0.005 mg/kg, 0.1 mg/kg to 0.3 mg/kg, or 0.1 mg/kg to 1 mg/kg) once every month. In some embodiments, the subject being treated for anemia is administered an isolated ALK2 or ALK3 antibody, or an antigen-binding fragment thereof, in an amount of 0.01 mg/kg to 2 mg/kg (e.g., 0.05 mg/kg to 2 mg/kg, 0.1 mg/kg to 2 mg/kg, 0.5 mg/kg to 2 mg/kg, 1 mg/kg to 2 mg/kg, 1.5 mg/kg to 2 mg/kg, 0.01 mg/kg to 1.5 mg/kg, 0.01 mg/kg to 1 mg/kg, 0.01 mg/kg to 0.5 mg/kg, 0.01 mg/kg to 0.25 mg/kg, 0.01 mg/kg to 0.1 mg/kg, or 0.01 mg/kg to 0.05 mg/kg). In some embodiments, the subject being treated for anemia is administered an isolated ALK2 or ALK3 antibody, or an antigen-binding fragment thereof, in an amount of 0.05 mg/kg to 1 mg/kg (e.g., 0.1 mg/kg to 1 mg/kg, 0.2 mg/kg to 1 mg/kg, 0.3 mg/kg to 1 mg/kg, 0.4 mg/kg to 1 mg/kg, 0.5 mg/kg to 1 mg/kg, 0.6 mg/kg to 1 mg/kg, 0.7 mg/kg to 1 mg/kg, 0.8 mg/kg to 1 mg/kg, 0.9 mg/kg to 1 mg/kg, 0.05 mg/kg to 0.9 mg/kg, 0.05 mg/kg to 0.8 mg/kg, 0.05 mg/kg to 0.7 mg/kg, 0.05 mg/kg to 0.6 mg/kg, 0.05 mg/kg to 0.5 mg/kg, 0.05 mg/kg to 0.4 mg/kg, 0.05 mg/kg to 0.3 mg/kg, 0.05 mg/kg to 0.2 mg/kg, or 0.05 mg/kg to 0.1 mg/kg). In some embodiments, the subject being treated for anemia is administered an isolated ALK2 or ALK3 antibody, or an antigen-binding fragment thereof, in an amount of 0.01 mg/kg to 1 mg/kg (e.g., 0.05 mg/kg to 1 mg/kg, 0.1 mg/kg to 1 mg/kg, 0.2 mg/kg to 1 mg/kg, 0.3 mg/kg to 1 mg/kg, 0.4 mg/kg to 1 mg/kg, 0.5 mg/kg to 1 mg/kg, 0.6 mg/kg to 1 mg/kg, 0.7 mg/kg to 1 mg/kg, 0.8 mg/kg to 1 mg/kg, 0.9 mg/kg to 1 mg/kg, 0.01 mg/kg to 0.9 mg/kg, 0.01 mg/kg to 0.8 mg/kg, 0.01 mg/kg to 0.7 mg/kg, 0.01 mg/kg to 0.6 mg/kg, 0.01 mg/kg to 0.5 mg/kg, 0.01

mg/kg to 0.4 mg/kg, 0.01 mg/kg to 0.3 mg/kg, 0.01 mg/kg to 0.2 mg/kg, 0.01 mg/kg to 0.1 mg/kg, or 0.01 mg/kg to 0.05 mg/kg). In some embodiments, the subject being treated for anemia is administered an isolated ALK2 or ALK3 antibody, or an antigen-binding fragment thereof, in an amount of 0.01 mg/kg to 0.3 mg/kg (e.g., 0.01 mg/kg to 0.3 mg/kg, 0.025 mg/kg to 0.3 mg/kg, 0.05 mg/kg to 0.3 mg/kg, 0.075 mg/kg to 0.3 mg/kg, 0.1 mg/kg to 0.3 mg/kg, 0.125 mg/kg to 0.3 mg/kg, 0.15 mg/kg to 0.3 mg/kg, 0.175 mg/kg to 0.3 mg/kg, 0.2 mg/kg to 0.3 mg/kg, 0.225 mg/kg to 0.3 mg/kg, 0.25 mg/kg to 0.3 mg/kg, 0.275 mg/kg to 0.3 mg/kg, 0.01 mg/kg to 0.275 mg/kg, 0.01 mg/kg to 0.25 mg/kg, 0.01 mg/kg to 0.225 mg/kg, 0.01 mg/kg to 0.2 mg/kg, 0.01 mg/kg to 0.175 mg/kg, 0.01 mg/kg to 0.15 mg/kg, 0.01 mg/kg to 0.125 mg/kg, 0.01 mg/kg to 0.1 mg/kg, 0.01 mg/kg to 0.075 mg/kg, 0.01 mg/kg to 0.05 mg/kg, or 0.01 mg/kg to 0.025 mg/kg). In some embodiments, the subject being treated for anemia is administered an isolated ALK2 or ALK3 antibody, or an antigen-binding fragment thereof, in an amount of 0.1 mg/kg to 0.3 mg/kg (e.g., 0.1 mg/kg to 0.25 mg/kg, 0.1 mg/kg to 0.2 mg/kg, 0.1 mg/kg to 0.15 mg/kg, 0.15 mg/kg to 0.3 mg/kg, 0.2 mg/kg to 0.3 mg/kg, or 0.25 mg/kg to 0.3 mg/kg). In some embodiments, the anemia is associated with cancer, cancer treatment, chronic kidney disease, acute renal disease or failure, chronic renal disease or failure, myelofibrosis treatment, a myelodysplastic syndrome, thalassemia, a nutritional deficit, adverse reaction to medication, ineffective hematopoiesis, an inflammatory or autoimmune disease, splenomegaly, porphyria, vasculitis, hemolysis, a bone marrow defect, bone marrow transplantation, myelofibrosis, diabetes, acute liver disease, chronic liver disease, acute bleeding, chronic bleeding, infection, hemoglobinopathy, drug use, alcohol abuse, Churg-Strauss syndrome, Felty syndrome, Pearson syndrome, dyskeratosis congenita, graft versus host disease, hematopoietic stem cell transplantation, osteomyelofibrosis, pancytopenia, pure red-cell aplasia, purpura Schoenlein-Henoch, Shwachman syndrome, advanced age, contraindication to transfusion, surgery, trauma, a wound, an ulcer, urinary tract bleeding, digestive tract bleeding, frequent blood donation, or heavy menstrual bleeding. In some embodiments, the anemia is aplastic anemia, iron deficiency anemia, vitamin deficiency anemia, anemia of inflammation, IRIDA, anemia associated with bone marrow disease, hemolytic anemia, sickle cell anemia, microcytic anemia, hypochromic anemia, sideroblastic anemia, congenital dyserythropoietic anemia, Diamond Blackfan anemia, Fanconi anemia, or refractory anemia with excess of blasts. In some embodiments, the anemia is aplastic anemia, iron deficiency anemia, vitamin deficiency anemia, anemia of inflammation, IRIDA, anemia associated with bone marrow disease, hemolytic anemia, sickle cell anemia, microcytic anemia, hypochromic anemia, sideroblastic anemia, Diamond Blackfan anemia, Fanconi's anemia, or refractory anemia with excess of blasts. In some embodiments, the anemia is IRIDA. In some embodiments, the anemia is anemia of inflammation. In some embodiments, the anemia is associated with elevated hepcidin levels.

**[0018]** In some embodiments, the disease or condition that can be treated with an ALK2 or ALK3 inhibitor is fibrodysplasia ossificans progressiva (FOP). In some embodiments, the subject being treated for FOP is administered an





mg/kg, 0.002 mg/kg to 0.25 mg/kg, 0.002 mg/kg to 0.1 mg/kg, 0.002 mg/kg to 0.05 mg/kg, 0.002 mg/kg to 0.01 mg/kg, 0.002 mg/kg to 0.005 mg/kg, 0.1 mg/kg to 0.3 mg/kg, or 0.1 mg/kg to 1 mg/kg) once every 4 months. In some embodiments, the subject is administered an isolated ALK2 or ALK3 antibody, or an antigen-binding fragment thereof, in an amount of 0.002 mg/kg to 3 mg/kg (e.g., 0.005 mg/kg to 3 mg/kg, 0.01 mg/kg to 3 mg/kg, 0.05 mg/kg to 3 mg/kg, 0.1 mg/kg to 3 mg/kg, 0.5 mg/kg to 3 mg/kg, 1 mg/kg to 3 mg/kg, 1.5 mg/kg to 3 mg/kg, 2 mg/kg to 3 mg/kg, 2.5 mg/kg to 3 mg/kg, 0.002 mg/kg to 2.5 mg/kg, 0.002 mg/kg to 2 mg/kg, 0.002 mg/kg to 1.5 mg/kg, 0.002 mg/kg to 1 mg/kg, 0.002 mg/kg to 0.5 mg/kg, 0.002 mg/kg to 0.25 mg/kg, 0.002 mg/kg to 0.1 mg/kg, 0.002 mg/kg to 0.05 mg/kg, 0.002 mg/kg to 0.01 mg/kg, 0.002 mg/kg to 0.005 mg/kg, 0.1 mg/kg to 0.3 mg/kg, or 0.1 mg/kg to 1 mg/kg) once every 3 months. In some embodiments, the subject is administered an isolated ALK2 or ALK3 antibody, or an antigen-binding fragment thereof, in an amount of 0.002 mg/kg to 3 mg/kg (e.g., 0.005 mg/kg to 3 mg/kg, 0.01 mg/kg to 3 mg/kg, 0.05 mg/kg to 3 mg/kg, 0.1 mg/kg to 3 mg/kg, 0.5 mg/kg to 3 mg/kg, 1 mg/kg to 3 mg/kg, 1.5 mg/kg to 3 mg/kg, 2 mg/kg to 3 mg/kg, 2.5 mg/kg to 3 mg/kg, 0.002 mg/kg to 2.5 mg/kg, 0.002 mg/kg to 2 mg/kg, 0.002 mg/kg to 1.5 mg/kg, 0.002 mg/kg to 1 mg/kg, 0.002 mg/kg to 0.5 mg/kg, 0.002 mg/kg to 0.25 mg/kg, 0.002 mg/kg to 0.1 mg/kg, 0.002 mg/kg to 0.05 mg/kg, 0.002 mg/kg to 0.01 mg/kg, 0.002 mg/kg to 0.005 mg/kg, 0.1 mg/kg to 0.3 mg/kg, or 0.1 mg/kg to 1 mg/kg) once every 2 months. In some embodiments, the subject is administered an isolated ALK2 or ALK3 antibody, or an antigen-binding fragment thereof, in an amount of 0.002 mg/kg to 3 mg/kg (e.g., 0.005 mg/kg to 3 mg/kg, 0.01 mg/kg to 3 mg/kg, 0.05 mg/kg to 3 mg/kg, 0.1 mg/kg to 3 mg/kg, 0.5 mg/kg to 3 mg/kg, 1 mg/kg to 3 mg/kg, 1.5 mg/kg to 3 mg/kg, 2 mg/kg to 3 mg/kg, 2.5 mg/kg to 3 mg/kg, 0.002 mg/kg to 2.5 mg/kg, 0.002 mg/kg to 2 mg/kg, 0.002 mg/kg to 1.5 mg/kg, 0.002 mg/kg to 1 mg/kg, 0.002 mg/kg to 0.5 mg/kg, 0.002 mg/kg to 0.25 mg/kg, 0.002 mg/kg to 0.1 mg/kg, 0.002 mg/kg to 0.05 mg/kg, 0.002 mg/kg to 0.01 mg/kg, 0.002 mg/kg to 0.005 mg/kg, 0.1 mg/kg to 0.3 mg/kg, or 0.1 mg/kg to 1 mg/kg) once every month. In some embodiments, the subject being treated for a bone disease is administered an isolated ALK2 or ALK3 antibody, or an antigen-binding fragment thereof, in an amount of 0.01 mg/kg to 2 mg/kg (e.g., 0.05 mg/kg to 2 mg/kg, 0.1 mg/kg to 2 mg/kg, 0.5 mg/kg to 2 mg/kg, 1 mg/kg to 2 mg/kg, 1.5 mg/kg to 2 mg/kg, 0.01 mg/kg to 1.5 mg/kg, 0.01 mg/kg to 1 mg/kg, 0.01 mg/kg to 0.5 mg/kg, 0.01 mg/kg to 0.25 mg/kg, 0.01 mg/kg to 0.1 mg/kg, or 0.01 mg/kg to 0.05 mg/kg). In some embodiments, the subject being treated for a bone disease is administered an isolated ALK2 or ALK3 antibody, or an antigen-binding fragment thereof, in an amount of 0.05 mg/kg to 1 mg/kg (e.g., 0.1 mg/kg to 1 mg/kg, 0.2 mg/kg to 1 mg/kg, 0.3 mg/kg to 1 mg/kg, 0.4 mg/kg to 1 mg/kg, 0.5 mg/kg to 1 mg/kg, 0.6 mg/kg to 1 mg/kg, 0.7 mg/kg to 1 mg/kg, 0.8 mg/kg to 1 mg/kg, 0.9 mg/kg to 1 mg/kg, 0.05 mg/kg to 0.9 mg/kg, 0.05 mg/kg to 0.8 mg/kg, 0.05 mg/kg to 0.7 mg/kg, 0.05 mg/kg to 0.6 mg/kg, 0.05 mg/kg to 0.5 mg/kg, 0.05 mg/kg to 0.4 mg/kg, 0.05 mg/kg to 0.3 mg/kg, 0.05 mg/kg to 0.2 mg/kg, or 0.05 mg/kg to 0.1 mg/kg). In some embodiments, the subject being treated for a bone disease is administered an isolated ALK2 or ALK3 antibody, or an antigen-binding fragment

thereof, in an amount of 0.01 mg/kg to 1 mg/kg (e.g., 0.05 mg/kg to 1 mg/kg, 0.1 mg/kg to 1 mg/kg, 0.2 mg/kg to 1 mg/kg, 0.3 mg/kg to 1 mg/kg, 0.4 mg/kg to 1 mg/kg, 0.5 mg/kg to 1 mg/kg, 0.6 mg/kg to 1 mg/kg, 0.7 mg/kg to 1 mg/kg, 0.8 mg/kg to 1 mg/kg, 0.9 mg/kg to 1 mg/kg, 0.01 mg/kg to 0.9 mg/kg, 0.01 mg/kg to 0.8 mg/kg, 0.01 mg/kg to 0.7 mg/kg, 0.01 mg/kg to 0.6 mg/kg, 0.01 mg/kg to 0.5 mg/kg, 0.01 mg/kg to 0.4 mg/kg, 0.01 mg/kg to 0.3 mg/kg, 0.01 mg/kg to 0.2 mg/kg, 0.01 mg/kg to 0.1 mg/kg, or 0.01 mg/kg to 0.05 mg/kg). In some embodiments, the subject being treated for a bone disease is administered an isolated ALK2 or ALK3 antibody, or an antigen-binding fragment thereof, in an amount of 0.01 mg/kg to 0.3 mg/kg (e.g., 0.01 mg/kg to 3 mg/kg, 0.025 mg/kg to 0.3 mg/kg, 0.05 mg/kg to 0.3 mg/kg, 0.075 mg/kg to 0.3 mg/kg, 0.1 mg/kg to 0.3 mg/kg, 0.125 mg/kg to 0.3 mg/kg, 0.15 mg/kg to 0.3 mg/kg, 0.175 mg/kg to 0.3 mg/kg, 0.2 mg/kg to 0.3 mg/kg, 0.225 mg/kg to 0.3 mg/kg, 0.25 mg/kg to 0.3 mg/kg, 0.275 mg/kg to 0.3 mg/kg, 0.01 mg/kg to 0.275 mg/kg, 0.01 mg/kg to 0.25 mg/kg, 0.01 mg/kg to 0.225 mg/kg, 0.01 mg/kg to 0.2 mg/kg, 0.01 mg/kg to 0.225 mg/kg, 0.01 mg/kg to 0.2 mg/kg, 0.01 mg/kg to 0.175 mg/kg, 0.01 mg/kg to 0.15 mg/kg, 0.01 mg/kg to 0.125 mg/kg, 0.01 mg/kg to 0.1 mg/kg, 0.01 mg/kg to 0.075 mg/kg, 0.01 mg/kg to 0.05 mg/kg, or 0.01 mg/kg to 0.025 mg/kg). In some embodiments, the subject being treated for a bone disease is administered an isolated ALK2 or ALK3 antibody, or an antigen-binding fragment thereof, in an amount of 0.1 mg/kg to 0.3 mg/kg (e.g., 0.1 mg/kg to 0.25 mg/kg, 0.1 mg/kg to 0.2 mg/kg, 0.1 mg/kg to 0.15 mg/kg, 0.15 mg/kg to 0.3 mg/kg, 0.2 mg/kg to 0.3 mg/kg, or 0.25 mg/kg to 0.3 mg/kg). In some embodiments, the bone disease is osteoporosis (e.g., primary or secondary osteoporosis), osteopenia, osteopetrosis, osteogenesis imperfecta, bone fracture, bone cancer or cancer metastasis-related bone loss, Paget's disease, renal osteodystrophy, treatment-related bone loss, neuromuscular disease-related bone loss, burn-induced bone loss, anorexia-related bone loss, diet-related bone loss, bone loss associated with the treatment of obesity, low gravity-related bone loss, or immobility-related bone loss.

**[0020]** In some embodiments, the disease or condition that can be treated with an ALK2 or ALK3 inhibitor is Sjogren's syndrome, multiple osteochondroma, diffuse intrinsic pontine glioma, posterior capsule opacification, cardiac hypertrophy, or cardiac fibrosis. In some embodiments, the subject being treated for Sjogren's syndrome, multiple osteochondroma, diffuse intrinsic pontine glioma, posterior capsule opacification, cardiac hypertrophy, or cardiac fibrosis is administered an isolated ALK2 or ALK3 antibody, or an antigen-binding fragment thereof, in an amount of 0.002 mg/kg to 3 mg/kg (e.g., 0.005 mg/kg to 3 mg/kg, 0.01 mg/kg to 3 mg/kg, 0.05 mg/kg to 3 mg/kg, 0.1 mg/kg to 3 mg/kg, 0.5 mg/kg to 3 mg/kg, 1 mg/kg to 3 mg/kg, 1.5 mg/kg to 3 mg/kg, 2 mg/kg to 3 mg/kg, 2.5 mg/kg to 3 mg/kg, 0.002 mg/kg to 2.5 mg/kg, 0.002 mg/kg to 2 mg/kg, 0.002 mg/kg to 1.5 mg/kg, 0.002 mg/kg to 1 mg/kg, 0.002 mg/kg to 0.5 mg/kg, 0.002 mg/kg to 0.25 mg/kg, 0.002 mg/kg to 0.1 mg/kg, 0.002 mg/kg to 0.05 mg/kg, 0.002 mg/kg to 0.01 mg/kg, 0.002 mg/kg to 0.005 mg/kg, 0.1 mg/kg to 0.3 mg/kg, or 0.1 mg/kg to 1 mg/kg) once every month. In some embodiments, the subject is administered an isolated ALK2 or ALK3 antibody, or an antigen-binding fragment thereof, in an amount of 0.002 mg/kg to 3 mg/kg (e.g., 0.005 mg/kg to 3 mg/kg, 0.01 mg/kg to 3 mg/kg, 0.05 mg/kg to 3 mg/kg,

0.1 mg/kg to 3 mg/kg, 0.5 mg/kg to 3 mg/kg, 1 mg/kg to 3 mg/kg, 1.5 mg/kg to 3 mg/kg, 2 mg/kg to 3 mg/kg, 2.5 mg/kg to 3 mg/kg, 0.002 mg/kg to 2.5 mg/kg, 0.002 mg/kg to 2 mg/kg, 0.002 mg/kg to 1.5 mg/kg, 0.002 mg/kg to 1 mg/kg, 0.002 mg/kg to 0.5 mg/kg, 0.002 mg/kg to 0.25 mg/kg, 0.002 mg/kg to 0.1 mg/kg, 0.002 mg/kg to 0.05 mg/kg, 0.002 mg/kg to 0.01 mg/kg, 0.002 mg/kg to 0.005 mg/kg, 0.1 mg/kg to 0.3 mg/kg, or 0.1 mg/kg to 1 mg/kg) once biweekly. In some embodiments, the subject is administered an isolated ALK2 or ALK3 antibody, or an antigen-binding fragment thereof, in an amount of 0.002 mg/kg to 3 mg/kg (e.g., 0.005 mg/kg to 3 mg/kg, 0.01 mg/kg to 3 mg/kg, 0.05 mg/kg to 3 mg/kg, 0.1 mg/kg to 3 mg/kg, 0.5 mg/kg to 3 mg/kg, 1 mg/kg to 3 mg/kg, 1.5 mg/kg to 3 mg/kg, 2 mg/kg to 3 mg/kg, 2.5 mg/kg to 3 mg/kg, 0.002 mg/kg to 2.5 mg/kg, 0.002 mg/kg to 2 mg/kg, 0.002 mg/kg to 1.5 mg/kg, 0.002 mg/kg to 1 mg/kg, 0.002 mg/kg to 0.5 mg/kg, 0.002 mg/kg to 0.25 mg/kg, 0.002 mg/kg to 0.1 mg/kg, 0.002 mg/kg to 0.05 mg/kg, 0.002 mg/kg to 0.01 mg/kg, 0.002 mg/kg to 0.005 mg/kg, 0.1 mg/kg to 0.3 mg/kg, or 0.1 mg/kg to 1 mg/kg) once every 10 to 14 days (e.g., once every 10 days, once every 11 days, once every 12 days, once every 13 days, or once every 14 days). In some embodiments, the subject is administered an isolated ALK2 or ALK3 antibody, or an antigen-binding fragment thereof, in an amount of 0.002 mg/kg to 3 mg/kg (e.g., 0.005 mg/kg to 3 mg/kg, 0.01 mg/kg to 3 mg/kg, 0.05 mg/kg to 3 mg/kg, 0.1 mg/kg to 3 mg/kg, 0.5 mg/kg to 3 mg/kg, 1 mg/kg to 3 mg/kg, 1.5 mg/kg to 3 mg/kg, 2 mg/kg to 3 mg/kg, 2.5 mg/kg to 3 mg/kg, 0.002 mg/kg to 2 mg/kg, 0.002 mg/kg to 1.5 mg/kg, 0.002 mg/kg to 1 mg/kg, 0.002 mg/kg to 0.5 mg/kg, 0.002 mg/kg to 0.25 mg/kg, 0.002 mg/kg to 0.1 mg/kg, 0.002 mg/kg to 0.05 mg/kg, 0.002 mg/kg to 0.01 mg/kg, 0.002 mg/kg to 0.005 mg/kg, 0.1 mg/kg to 0.3 mg/kg, or 0.1 mg/kg to 1 mg/kg) once a week. In some embodiments, the subject being treated for Sjogren's syndrome, multiple osteochondroma, diffuse intrinsic pontine glioma, posterior capsule opacification, cardiac hypertrophy, or cardiac fibrosis is administered an isolated ALK2 or ALK3 antibody, or an antigen-binding fragment thereof, in an amount of 0.01 mg/kg to 2 mg/kg (e.g., 0.05 mg/kg to 2 mg/kg, 0.1 mg/kg to 2 mg/kg, 0.5 mg/kg to 2 mg/kg, 1 mg/kg to 2 mg/kg, 1.5 mg/kg to 2 mg/kg, 0.01 mg/kg to 1.5 mg/kg, 0.01 mg/kg to 1 mg/kg, 0.01 mg/kg to 0.5 mg/kg, 0.01 mg/kg to 0.25 mg/kg, 0.01 mg/kg to 0.1 mg/kg, or 0.01 mg/kg to 0.05 mg/kg). In some embodiments, the subject being treated for Sjogren's syndrome, multiple osteochondroma, diffuse intrinsic pontine glioma, posterior capsule opacification, cardiac hypertrophy, or cardiac fibrosis is administered an isolated ALK2 or ALK3 antibody, or an antigen-binding fragment thereof, in an amount of 0.05 mg/kg to 1 mg/kg (e.g., 0.1 mg/kg to 1 mg/kg, 0.2 mg/kg to 1 mg/kg, 0.3 mg/kg to 1 mg/kg, 0.4 mg/kg to 1 mg/kg, 0.5 mg/kg to 1 mg/kg, 0.6 mg/kg to 1 mg/kg, 0.7 mg/kg to 1 mg/kg, 0.8 mg/kg to 1 mg/kg, 0.9 mg/kg to 1 mg/kg, 0.05 mg/kg to 0.9 mg/kg, 0.05 mg/kg to 0.8 mg/kg, 0.05 mg/kg to 0.7 mg/kg, 0.05 mg/kg to 0.6 mg/kg, 0.05 mg/kg to 0.5 mg/kg, 0.05 mg/kg to 0.4 mg/kg, 0.05 mg/kg to 0.3 mg/kg, 0.05 mg/kg to 0.2 mg/kg, or 0.05 mg/kg to 0.1 mg/kg). In some embodiments, the subject being treated for Sjogren's syndrome, multiple osteochondroma, diffuse intrinsic pontine glioma, posterior capsule opacification, cardiac hypertrophy, or cardiac fibrosis is administered an isolated ALK2 or

ALK3 antibody, or an antigen-binding fragment thereof, in an amount of 0.01 mg/kg to 1 mg/kg (e.g., 0.05 mg/kg to 1 mg/kg, 0.1 mg/kg to 1 mg/kg, 0.2 mg/kg to 1 mg/kg, 0.3 mg/kg to 1 mg/kg, 0.4 mg/kg to 1 mg/kg, 0.5 mg/kg to 1 mg/kg, 0.6 mg/kg to 1 mg/kg, 0.7 mg/kg to 1 mg/kg, 0.8 mg/kg to 1 mg/kg, 0.9 mg/kg to 1 mg/kg, 0.01 mg/kg to 0.9 mg/kg, 0.01 mg/kg to 0.8 mg/kg, 0.01 mg/kg to 0.7 mg/kg, 0.01 mg/kg to 0.6 mg/kg, 0.01 mg/kg to 0.5 mg/kg, 0.01 mg/kg to 0.4 mg/kg, 0.01 mg/kg to 0.3 mg/kg, 0.01 mg/kg to 0.2 mg/kg, 0.01 mg/kg to 0.1 mg/kg, or 0.01 mg/kg to 0.05 mg/kg). In some embodiments, the subject being treated for Sjogren's syndrome, multiple osteochondroma, diffuse intrinsic pontine glioma, posterior capsule opacification, cardiac hypertrophy, or cardiac fibrosis is administered an isolated ALK2 or ALK3 antibody, or an antigen-binding fragment thereof, in an amount of 0.01 mg/kg to 0.3 mg/kg (e.g., 0.01 mg/kg to 0.3 mg/kg, 0.025 mg/kg to 0.3 mg/kg, 0.05 mg/kg to 0.3 mg/kg, 0.075 mg/kg to 0.3 mg/kg, 0.1 mg/kg to 0.3 mg/kg, 0.125 mg/kg to 0.3 mg/kg, 0.15 mg/kg to 0.3 mg/kg, 0.175 mg/kg to 0.3 mg/kg, 0.2 mg/kg to 0.3 mg/kg, 0.225 mg/kg to 0.3 mg/kg, 0.25 mg/kg to 0.3 mg/kg, 0.275 mg/kg to 0.3 mg/kg, 0.01 mg/kg to 0.275 mg/kg, 0.01 mg/kg to 0.25 mg/kg, 0.01 mg/kg to 0.225 mg/kg, 0.01 mg/kg to 0.2 mg/kg, 0.01 mg/kg to 0.225 mg/kg, 0.01 mg/kg to 0.2 mg/kg, 0.01 mg/kg to 0.175 mg/kg, 0.01 mg/kg to 0.15 mg/kg, 0.01 mg/kg to 0.125 mg/kg, 0.01 mg/kg to 0.1 mg/kg, 0.01 mg/kg to 0.075 mg/kg, 0.01 mg/kg to 0.05 mg/kg, or 0.01 mg/kg to 0.025 mg/kg). In some embodiments, the subject being treated for Sjogren's syndrome, multiple osteochondroma, diffuse intrinsic pontine glioma, posterior capsule opacification, cardiac hypertrophy, or cardiac fibrosis is administered an isolated ALK2 or ALK3 antibody, or an antigen-binding fragment thereof, in an amount of 0.1 mg/kg to 0.3 mg/kg (e.g., 0.1 mg/kg to 0.25 mg/kg, 0.1 mg/kg to 0.2 mg/kg, 0.1 mg/kg to 0.15 mg/kg, 0.15 mg/kg to 0.3 mg/kg, 0.2 mg/kg to 0.3 mg/kg, or 0.25 mg/kg to 0.3 mg/kg).

**[0021]** In some embodiments of any of the foregoing aspects, the method includes administering an ALK2 antibody or an antigen binding fragment thereof. In some embodiments, the ALK2 antibody or antigen binding fragment thereof, is an antibody of Table 2. In some embodiments, the ALK2 antibody, or antigen-binding fragment thereof, has the CDR sequences of an antibody described in Table 3 (e.g., any of the light chain CDR1, CDR2, and CDR3 sequences and heavy chain CDR1, CDR2, and CDR3 sequences described in Table 3). In some embodiments, the ALK2 antibody or antigen binding fragment thereof includes a light chain CDR1, CDR2, and CDR3 sequence and a heavy chain CDR1, CDR2, and CDR3 sequence from the same row of Table 3 (e.g., a light chain CDR1, CDR2, and CDR3 sequence and a heavy chain CDR1, CDR2, and CDR3 sequence of any one of Antibodies 1-38). In some embodiments, the ALK2 antibody, or antigen-binding fragment thereof, has a variable light chain (VL) sequence and a variable heavy chain (VH) sequence of an antibody described in Table 4. In some embodiments, the ALK2 antibody, or antigen-binding fragment thereof, has a VL sequence and a VH sequence having at least 90% sequence identity (e.g., 90%, 91%, 92%, 93%, 94%, 95%, 97%, 98%, 99% or 100% sequence identity) to a VL sequence and a VH sequence of Table 5 (e.g., any VL sequence and any VH sequence of Table 5). In some embodiments, the ALK2 antibody, or an antigen binding fragment thereof, has a VL

sequence and a VH sequence from the same row of Table 5. In some embodiments, the VH and VL have the sequence of SEQ ID NOs: 107 and 108; 109 and 110; 109 and 115; 111 and 112; 113 and 114; 116 and 118; 116 and 119; 117 and 118; 131 and 118; 117 and 119; 131 and 119; 117 and 115; 131 and 115; 117 and 120; 131 and 120; 121 and 115; 122 and 126; 132 and 126; 122 and 127; 132 and 127; 122 and 128; 132 and 128; 123 and 126; 133 and 126; 123 and 127; 133 and 127; 123 and 128; 133 and 128; 124 and 126; 124 and 127; 124 and 128; 124 and 129; 125 and 128; 125 and 129; 125 and 130; 134 and 129; 135 and 127; 139 and 129; 140 and 136; 140 and 137; 140 and 138; 201 and 202; 203 and 204; 205 and 206; 207 and 208; 304 and 305; 306 and 307; 308 and 309; 310 and 311; 306 and 312; 313 and 314; 315 and 316; 317 and 318; 319 and 320; 321 and 322; 323 and 324; 325 and 326; 327 and 328; 329 and 330; 331 and 332; 333 and 334; 335 and 336; 337 and 338; 339 and 340; or 341 and 342. In some embodiments, the ALK2 antibody, or an antigen-binding fragment thereof, has a heavy chain sequence and light chain sequence having at least 90% sequence identity (e.g., 90%, 91%, 92%, 93%, 94%, 95%, 95%, 97%, 98%, 99% or 100% sequence identity) to a heavy chain and light chain sequence of Table 6 (e.g., any heavy chain sequence and any light chain sequence of Table 6). In some embodiments, the ALK2 antibody or an antigen binding fragment thereof, has a heavy chain sequence and light chain sequence from the same row of Table 6. In some embodiments of any of the foregoing aspects, the method includes administering an ALK3 antibody or an antigen binding fragment thereof. In some embodiments, the ALK3 antibody, or antigen-binding fragment thereof, contains an antigen binding fragment (Fab) described in Harth et al., PLoS ONE 5: e13049, 2010, such as AbD1556 or AbD1564. In some embodiments, the ALK3 antibody, or antigen binding fragment thereof, has a heavy chain CDR1 including TGYMK (SEQ ID NO: 79); a heavy chain CDR2 including RINPDNGGRTYNQIFKDK (SEQ ID NO: 80); and a heavy chain CDR3 including RERGQYGNYGFS (SEQ ID NO: 81). In some embodiments, the ALK3 antibody, or antigen binding fragment thereof, contains a heavy chain variable region having at least 90% (e.g., at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or more) sequence identity to SEQ ID NO: 77 or SEQ ID NO: 78.

**[0022]** In some embodiments, the subject is a human.

#### Definitions

**[0023]** As used herein, the term “about” refers to a value that is within 10% above or below the value being described.

**[0024]** As used herein, “administration” refers to providing or giving a subject a therapeutic agent (e.g., an ALK2 or ALK3 antibody or antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment) described herein), by any effective route. Exemplary routes of administration are described herein below.

**[0025]** The term “antibody” is used in the broadest sense and specifically covers intact monoclonal antibodies, polyclonal antibodies, multispecific antibodies (e.g., bispecific antibodies) formed from at least two intact antibodies, and antibody fragments so long as they exhibit the desired biological activity.

**[0026]** “Antibody fragments” include a portion of an intact antibody, preferably the antigen binding or variable region of the intact antibody. Examples of antibody fragments include Fab, Fab', F(ab')<sub>2</sub>, and Fv fragments; diabodies;

linear antibodies (Zapata et al. Protein Eng. 8(10):1057-1062 (1995)); single-chain antibody molecules; and multispecific antibodies formed from antibody fragments.

**[0027]** The term “monoclonal antibody” as used herein refers to an antibody obtained from a population of substantially homogeneous antibodies, i.e., the individual antibodies included in the population are identical except for possible naturally occurring mutations that may be present in minor amounts. Monoclonal antibodies are highly specific, being directed against a single antigenic site.

**[0028]** The term “monoclonal antibody” as used herein specifically includes “chimeric” antibodies in which a portion of the heavy and/or light chain is identical with or homologous to corresponding sequences in antibodies derived from a particular species or belonging to a particular antibody class or subclass, while the remainder of the chain(s) is identical with or homologous to corresponding sequences in antibodies derived from another species or belonging to another antibody class or subclass, as well as fragments of such antibodies, so long as they exhibit the desired biological activity (U.S. Pat. No. 4,816,567; Morrison et al., Proc. Natl. Acad. Sci. USA, 81:6851-6855 (1984)).

**[0029]** “Humanized” forms of non-human (e.g., murine) antibodies are chimeric antibodies, antibody chains or fragments thereof, (such as Fv, Fab, Fab', F(ab')<sub>2</sub> or other antigen-binding subsequences of antibodies) which contain minimal sequence derived from non-human antibody. For the most part, humanized antibodies are human antibodies (recipient antibody) in which residues from a complementarity-determining region (CDR) of the recipient are replaced by residues from a CDR of a non-human species (donor antibody) such as mouse, rat or rabbit having the desired specificity, affinity, and capacity. In some instances, Fv framework region (FR) residues of the human antibody are replaced by corresponding non-human residues. Further, humanized antibodies may include residues which are found neither in the recipient antibody nor in the imported CDR or framework sequences.

**[0030]** As used herein, the terms “bone mineral density (BMD),” “bone density,” and “bone mass” refer to a measure of the amount of bone mineral (e.g., calcium) in bone tissue. BMD may be measured by well-established clinical techniques known to one of skill in the art (e.g., by single- or dual-energy photon or X-ray absorptiometry (DEXA)). The concept of BMD relates to the mass of mineral per volume of bone, although clinically it is measured by proxy according to optical density per square centimeter of bone surface upon imaging. BMD measurement is used in clinical medicine as an indirect indicator of osteoporosis and fracture risk. In some embodiments, BMD test results are provided as a T-score, where the T-score represents the BMD of a subject compared to the ideal or peak bone mineral density of a healthy 30-year-old adult. A score of 0 indicates that the BMD is equal to the normal reference value for a healthy young adult. Differences between the measured BMD of subject and that of the reference value for a healthy young adult are measured in standard deviations units (SDs). Accordingly, a T-score of between +1 SD and -1 SD may indicate a normal BMD, a T-score of between -1 SD and -2.5 SD may indicate low bone mass (e.g., osteopenia), and a T-score lower than -2.5 SD may indicate osteoporosis or severe osteoporosis. In some embodiments, an ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g.,

an ALK2 or ALK3 binding fragment), of the invention is administered to a subject in need thereof, wherein the patient has low bone mass (e.g., a T-Score of between  $-1$  SD and  $-2.5$  SD). In some embodiments, an ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment), of the invention is administered to a subject in need thereof, wherein the patient has osteoporosis (e.g., a T-Score of less than  $-2.5$  SD). In some embodiments, administration of an ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment), of the invention treats the subject by increasing their BMD (e.g., increasing the T-Score of the subject).

**[0031]** As used herein, the term “bone strength” refers to a measurement of bone that is determined by bone quality in addition to bone mineral density. Bone quality is influenced by bone geometry, microarchitecture, and the properties of constituent tissues. Bone strength can be used to assess the bone’s risk of fracture.

**[0032]** As used herein, the term “bone disease” refers to a condition characterized by bone damage (e.g., decreased bone mineral density, decreased bone strength, and/or bone loss). Such diseases or conditions may be caused by an imbalance in osteoblast and/or osteoclast activity (e.g., increased bone resorption or reduced bone formation). Bone diseases include primary osteoporosis, secondary osteoporosis, osteopenia, osteopetrosis, osteogenesis imperfecta, bone fracture, bone cancer or cancer metastasis-related bone loss (e.g., bone loss associated with multiple myeloma), Paget’s disease, renal osteodystrophy, treatment-related bone loss, neuromuscular disease-related bone loss, burn-induced bone loss, anorexia-related bone loss, diet-related bone loss, bone loss associated with the treatment of obesity, low gravity-related bone loss, and immobility-related bone loss.

**[0033]** As used herein, the term “neuromuscular disease-related bone loss” refers to bone loss that occurs in a subject having a neuromuscular disease. Poor bone health is often a significant problem for patients with neuromuscular disease. Deficiency of bone mineral density and increased incidence of bone fractures, for example, are a well-recognized clinical consequence of diseases such as DMD, ALS, and SMA.

**[0034]** As used herein, the terms “bone remodeling” or “bone metabolism” refer to the process for maintaining bone strength and ion homeostasis by replacing discrete parts of old bone with newly synthesized packets of proteinaceous matrix. Bone is resorbed by osteoclasts and is deposited by osteoblasts in a process called ossification. Osteocyte activity plays a key role in this process. Conditions that result in a decrease in bone mass, can either be caused by an increase in resorption, or a decrease in ossification. In a healthy

individual, during childhood, bone formation exceeds resorption. As the aging process occurs, resorption exceeds formation. Bone resorption rates are also typically much higher in post-menopausal older women due to estrogen deficiency related to menopause.

**[0035]** As used herein, the terms “bone resorption” or “bone catabolic activity” refer to a process by which osteoclasts break down the tissue in bones and release the minerals, resulting in a transfer of the mineral (e.g., calcium) from bone tissue to the blood. Increased rates of bone resorption are associated with aging, including in post-menopausal women. High rates of bone resorption, or rates of bone resorption that exceed the rate of ossification, are associated with bone disorders, such as decreased bone mineral density, including osteopenia and osteoporosis, and result in bone loss. In some embodiments, an ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment), of the invention is administered to a subject in need thereof, to decrease bone resorption in the subject (e.g., the rate of bone resorption in the subject).

**[0036]** As used herein, the terms “bone formation,” “ossification,” “osteogenesis,” or “bone anabolic activity” refer to the process of forming new bone tissue by osteoblasts. In some embodiments, an ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment), of the invention is administered to a subject in need thereof, to increase bone formation (e.g., increase the rate of bone formation or osteogenesis in the subject).

**[0037]** As used herein, the term “cell type” refers to a group of cells sharing a phenotype that is statistically separable based on gene expression data. For instance, cells of a common cell type may share similar structural and/or functional characteristics, such as similar gene activation patterns and antigen presentation profiles. Cells of a common cell type may include those that are isolated from a common tissue (e.g., epithelial tissue, neural tissue, connective tissue, bone tissue, or muscle tissue) and/or those that are isolated from a common organ, tissue system, blood vessel, or other structure and/or region in an organism.

**[0038]** As used herein, the terms “conservative mutation,” “conservative substitution,” and “conservative amino acid substitution” refer to a substitution of one or more amino acids for one or more different amino acids that exhibit similar physicochemical properties, such as polarity, electrostatic charge, and steric volume. These properties are summarized for each of the twenty naturally-occurring amino acids in Table 1 below.

TABLE 1

| Representative physicochemical properties of naturally occurring amino acids |               |               |                     |                                                   |                            |
|------------------------------------------------------------------------------|---------------|---------------|---------------------|---------------------------------------------------|----------------------------|
| Amino Acid                                                                   | 3 Letter Code | 1 Letter Code | Side-chain Polarity | Electrostatic character at physiological pH (7.4) | Steric Volume <sup>†</sup> |
| Alanine                                                                      | Ala           | A             | nonpolar            | neutral                                           | small                      |
| Arginine                                                                     | Arg           | R             | polar               | cationic                                          | large                      |
| Asparagine                                                                   | Asn           | N             | polar               | neutral                                           | intermediate               |
| Aspartic acid                                                                | Asp           | D             | polar               | anionic                                           | intermediate               |
| Cysteine                                                                     | Cys           | C             | nonpolar            | neutral                                           | intermediate               |

TABLE 1-continued

| Representative physicochemical properties of naturally occurring amino acids |               |               |                     |                                                          |                            |
|------------------------------------------------------------------------------|---------------|---------------|---------------------|----------------------------------------------------------|----------------------------|
| Amino Acid                                                                   | 3 Letter Code | 1 Letter Code | Side-chain Polarity | Electrostatic character at physiological pH (7.4)        | Steric Volume <sup>†</sup> |
| Glutamic acid                                                                | Glu           | E             | polar               | anionic                                                  | intermediate               |
| Glutamine                                                                    | Gln           | Q             | polar               | neutral                                                  | intermediate               |
| Glycine                                                                      | Gly           | G             | nonpolar            | neutral                                                  | small                      |
| Histidine                                                                    | His           | H             | polar               | Both neutral and cationic forms in equilibrium at pH 7.4 | large                      |
| Isoleucine                                                                   | Ile           | I             | nonpolar            | neutral                                                  | large                      |
| Leucine                                                                      | Leu           | L             | nonpolar            | neutral                                                  | large                      |
| Lysine                                                                       | Lys           | K             | polar               | cationic                                                 | large                      |
| Methionine                                                                   | Met           | M             | nonpolar            | neutral                                                  | large                      |
| Phenylalanine                                                                | Phe           | F             | nonpolar            | neutral                                                  | large                      |
| Proline                                                                      | Pro           | P             | nonpolar            | neutral                                                  | intermediate               |
| Serine                                                                       | Ser           | S             | polar               | neutral                                                  | small                      |
| Threonine                                                                    | Thr           | T             | polar               | neutral                                                  | intermediate               |
| Tryptophan                                                                   | Trp           | W             | nonpolar            | neutral                                                  | bulky                      |
| Tyrosine                                                                     | Tyr           | Y             | polar               | neutral                                                  | large                      |
| Valine                                                                       | Val           | V             | nonpolar            | neutral                                                  | intermediate               |

<sup>†</sup>based on volume in Å<sup>3</sup>: 50-100 is small, 100-150 is intermediate, 150-200 is large, and >200 is bulky

**[0039]** From this table it is appreciated that the conservative amino acid families include (i) G, A, V, L and I; (ii) D and E; (iii) C, S and T; (iv) H, K and R; (v) N and Q; and (vi) F, Y and W. A conservative mutation or substitution is therefore one that substitutes one amino acid for a member of the same amino acid family (e.g., a substitution of Ser for Thr or Lys for Arg).

**[0040]** As used herein, the term a “disease or condition that can be treated with an ALK2 or ALK3 inhibitor” refers to any disease or condition that could be expected to benefit from ALK2 or ALK3 inhibition based on studies performed in cell culture conditions, animal models, or human trials. Such diseases and conditions include anemia, bone diseases, fibrodysplasia ossificans progressiva; Sjogren’s syndrome, multiple osteochondroma, diffuse intrinsic pontine glioma, posterior capsule opacification, cardiac hypertrophy, and cardiac fibrosis.

**[0041]** As used herein, the term an “isolated antibody” refers to an antibody that is substantially free of other antibodies having different antigenic specificities (e.g., an isolated antibody that binds to ALK2 or ALK3 is substantially free of contaminants, e.g., antibodies that do not bind to ALK2 or ALK3). In addition, an “isolated” antibody is one that has been identified and separated and/or recovered from a component of its natural environment. Contaminant components of its natural environment are materials that could interfere with diagnostic or therapeutic uses for the antibody, and may include enzymes, hormones, and other proteinaceous or non-proteinaceous solutes.

**[0042]** As used herein, the terms “increasing” and “decreasing” refer to modulating resulting in, respectively, greater or lesser amounts, of function, expression, or activity of a metric relative to a reference. For example, subsequent to administration of an ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment), of the invention in a method described herein, the amount of a marker of a metric (e.g., red blood cell levels, bone mineral density, tear production, survival time) as described herein may be increased in a subject relative to the amount of the marker prior to administration

or relative to an untreated subject, or the amount of a marker of a metric (e.g., heterotopic ossification, or osteochondroma size, number, or formation) as described herein may be decreased in a subject relative to the amount of the marker prior to administration or relative to an untreated subject. Generally, the metric is measured subsequent to administration at a time that the administration has had the recited effect, e.g., at least one week, one month, 3 months, or 6 months, after a treatment regimen has begun.

**[0043]** As used herein, the terms “increase red blood cell levels” and “promote red blood cell formation” refer to clinically observable metrics, such as increases in hematocrit, red blood cell counts, and hemoglobin measurements, and are intended to be neutral as to the mechanism by which such changes occur. The term “low red blood cell levels” as used herein refers to red blood cell counts, hematocrit, and/or hemoglobin measurements that are below the range of values that is considered normal for the subject’s age and gender. In some embodiments, an ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment), of the invention is administered to a subject (e.g., a subject having anemia) to increase red blood cell levels (e.g., increase hemoglobin levels or red blood cell counts).

**[0044]** As used herein, the terms “red blood cell formation” and “red blood cell production” refer to the generation of red blood cells, such as the process of erythropoiesis in which red blood cells are produced in the bone marrow.

**[0045]** As used herein, the term “anemia” refers to any abnormality in hemoglobin or red blood cells that leads to reduced oxygen levels in the blood. Anemia can be associated with abnormal production, processing, or performance of erythrocytes and/or hemoglobin. The term anemia refers to any reduction in the number of red blood cells and/or level of hemoglobin in blood relative to normal blood levels.

**[0046]** As used herein the term “anemia of inflammation” refers to a type of anemia driven by inflammatory cytokines and characterized by hypoferrremia despite adequate iron stores, which is largely due to elevated hepcidin. Bone morphogenetic proteins (BMP) upregulate hepcidin by acti-

vating the SMAD signaling pathway through ALK2. Anemia of inflammation is also called anemia of chronic disease. Chronic conditions that cause anemia of inflammation include infection (e.g., chronic infection, such as HIV/AIDS or tuberculosis), autoimmune disease (e.g., rheumatoid arthritis or lupus), cancer (e.g., cancer or cancer treatment), inflammatory bowel disease (e.g., Crohn's disease or ulcerative colitis), and chronic kidney disease.

**[0047]** As used herein the terms “iron refractory iron deficiency anemia” and “IRIDA” refer to an inherited form of iron deficiency anemia. Key features of IRIDA include lifelong anemia (hemoglobin 6-9 g/dL); very low red blood cell size (microcytic), with a mean corpuscular volume (MCV) of 45-65 fL; very low iron levels in the blood (transferrin saturation <5%); abnormal oral iron absorption—no response to oral iron supplements or failure of an “oral iron challenge;” abnormal iron utilization—a slow, incomplete, and transient response to parenteral iron (iron injected intravenously); and other affected family members with an autosomal recessive inheritance pattern. A diagnosis of IRIDA can be confirmed by measuring the level of hepcidin in the blood. Mutations in TMPRSS6 have been found to be associated with IRIDA.

**[0048]** As used herein, the term “low transfusion burden” refers to a condition of a subject that has received less than four units of red blood cells (RBCs) within eight weeks (e.g., 3, 2, 1, or 0 units of RBCs within eight weeks) prior to treatment with an antibody described herein. A subject with a low transfusion burden can be identified as having anemia based on measurements of mean hemoglobin concentration. A subject with a low transfusion burden and a mean hemoglobin concentration of less than 10.0 g/dL of two measurements performed at least one week apart prior to treatment with an antibody described herein (e.g., one measurement performed within one day prior to treatment and the other performed 7-28 days prior, not influenced by RBC transfusion within seven days of measurement) is defined as having anemia. In some embodiments, a subject with a low transfusion burden receives 1-3 units of RBCs (1-3 RBC transfusions) within eight weeks prior to treatment with an antibody described herein. In some embodiments, a subject with a low transfusion burden does not receive any units of RBCs (0 RBC transfusions) within eight weeks prior to treatment with an antibody described herein.

**[0049]** As used herein, the term “high transfusion burden” refers to a condition of a subject requiring greater than or equal to four units of RBCs (e.g., 4, 5, 6, 7, 8, or more units) within eight weeks prior to treatment with an antibody described herein. A subject with a high transfusion burden can be identified as having anemia based on measurements of mean hemoglobin concentration. A subject with a high transfusion burden and a mean hemoglobin concentration of less than or equal to 9.0 g/dL is defined as having anemia.

**[0050]** As used herein, the term “ineffective hematopoiesis” refers to the failure to produce fully mature hematopoietic cells (e.g., the failure to produce red blood cells, platelets, and neutrophils). Ineffective hematopoiesis may be due to single or multiple defects, such as abnormal proliferation and/or differentiation of progenitor cells (e.g., an excessive production of progenitors that are unable to complete differentiation), that can lead to a hyperproliferation or a shortage of progenitor cells.

**[0051]** As used herein, the terms “erythropoiesis stimulating agent” and “ESA” refer to a class of drugs that act on the

proliferation stage of red blood cell development by expanding the pool of early-stage progenitor cells. Examples of erythropoiesis-stimulating agents are epoetin alfa and darbepoetin alfa.

**[0052]** As used herein, the terms “heterotopic ossification,” “heterotopic bone formation,” and “ectopic bone formation” refer to the abnormal growth of bone in non-skeletal tissues, such as muscles, tendons, and other soft tissue. “Ectopic bone” refers to bone that has formed in non-skeletal tissues. Heterotopic ossification can occur in subjects with FOP.

**[0053]** As used herein, the terms “fibrodysplasia ossificans progressiva” and “FOP” refer to a disorder in which skeletal muscle and connective tissue, such as tendons and ligaments, are gradually replaced by bone (ossified). This condition leads to bone formation outside the skeleton (extra-skeletal or heterotopic bone) that restricts movement. This process generally becomes noticeable in early childhood, starting with the neck and shoulders and moving down the body and into the limbs. People with FOP are born with abnormal big toes (hallux valgus) which can be helpful in making the diagnosis. Trauma, such as a fall or invasive medical procedure, or a viral illness may trigger episodes of muscle swelling and inflammation (myositis). These flare-ups last for several days to months and often result in permanent bone growth in the injured area. FOP is caused by mutation of the ACVR1 gene and is inherited in an autosomal dominant manner.

**[0054]** As used herein, the term “fibrosis” refers to the pathological process of excess formation of fibrous connective tissue. Fibrosis is characterized by fibroblast accumulation and collagen deposition in excess of normal deposition in any particular tissue. In response to inflammation or an injury to a tissue, nearby fibroblasts can migrate into the wound, proliferate, and produce large amounts of collagenous extracellular matrix. When fibrosis occurs in response to injury, the term “scarring” can be used as synonym. Fibrosis may occur in many tissues of the body, including, e.g., lungs, skin, liver, kidney, heart, eye, lacrimal gland, tendon, cartilage, pancreatic tissue, uterine tissue, neural tissue, testis, ovary, adrenal gland, artery, vein, colon, small and large intestine, biliary tract, and gut.

**[0055]** As used herein, the term “cardiac hypertrophy” refers to the abnormal enlargement, or thickening, of the heart muscle resulting from a process in which adult cardiac myocytes respond to stress through hypertrophic growth. Such growth is characterized by cell size increases without cell division, assembling of additional sarcomeres within the cell to maximize force generation, and an activation of a fetal cardiac gene program. Cardiac hypertrophy is often associated with increased risk of morbidity and mortality and has been found to be associated with fibrous tissue deposition in the cardiac interstitium accompanied by alterations in the extracellular matrix scaffold (e.g., cardiac fibrosis).

**[0056]** As used herein, the term “cardiac fibrosis,” also known as myocardial fibrosis, refers to fibrosis of the myocardium and is characterized by wall stiffening, reduced contractility, and impaired overall heart performance. Cardiac myofibroblasts mediate fibrosis by the excessive deposition of connective tissue (e.g., extracellular matrix) in the interstitial space. Cardiac fibrosis is often found in association with cardiac hypertrophy.

**[0057]** As used herein, the term “dry eye” refers to a chronic condition that occurs when the eyes do not produce enough tears (e.g., reduced tear production or tear volume) or when the tears evaporate too quickly. Dry eye may be accompanied by eye discomfort and abnormal visual function.

**[0058]** The term “Sjogren’s syndrome” as used herein refers to a systemic inflammatory disorder characterized by dry mouth, decreased tearing, and other dry mucous membranes. Dryness of the eyes and mouth are the most common symptoms of this syndrome.

**[0059]** As used herein, “dry eye associated with Sjogren’s syndrome” refers to dry eye associated with primary Sjogren’s syndrome and dry eye associated with secondary Sjogren’s syndrome. Dry eye associated with Sjogren’s syndrome is generally classified as dry eye with reduced tear secretion.

**[0060]** As used herein, the term “multiple osteochondroma” or “MO” refers to a condition or disease associated with formation of osteochondromas on bones, e.g., at the ends of long bones or on flat bones. Subjects with MO often carry a loss-of-function mutation in an exostosin gene, e.g., EXT1 or EXT2. MO is also known as multiple hereditary exostoses, Bessel-Hagen disease, diaphyseal aclasis, multiple cartilaginous exostoses, multiple congenital exostosis, and hereditary multiple osteochondroma.

**[0061]** As used herein, the term “osteochondroma” refers to a benign (noncancerous) tumor that develops during childhood or adolescence. It is an abnormal overgrowth of cartilage and bone that typically forms on the surface of a bone near the growth plate. Osteochondromas most often form on the long bones of the leg and arm and on flat bones such as the pelvis and shoulder blade (scapula).

**[0062]** As used herein, the term “diffuse intrinsic pontine glioma” refers to a highly aggressive and difficult to treat brain tumor that originates in the pons and accounts for approximately 20% of all pediatric brain tumors. The median overall survival for children with diffuse intrinsic pontine glioma (DIPG) is 9-11 months, with one- and two-year survival rates of approximately 30% and less than 10%, respectively. DIPG has been found to be molecularly distinct from adult gliomas, and frequently harbors mutations in genes encoding histone variants. Mutations in ALK2 (ACVR1) have also been observed in DIPG.

**[0063]** As used herein, the term “posterior capsule opacification” refers to the most common postoperative complication of cataract surgery that occurs in 20% to 40% of patients. In posterior capsule opacification (PCO), the posterior capsule undergoes secondary opacification due to the migration, proliferation, and differentiation of lens epithelial cells. PCO can cause significant visual symptoms, particularly when it involves the central visual axis, and can be identified by the formation of fibrosis (e.g., scar tissue) on the posterior capsule (e.g., behind a lens implant). Symptoms of PCO include a gradual decrease of vision, blurred vision, sensitivity to sunlight, an observation of a glare or halo around lights.

**[0064]** “Percent (%) amino acid sequence identity” with respect to a reference polypeptide sequence is defined as the percentage of amino acid residues in a candidate sequence that are identical with the amino acid residues in the reference polypeptide sequence, after aligning the sequences and introducing gaps, if necessary, to achieve the maximum percent sequence identity over the length of comparison, and

not considering any conservative substitutions as part of the sequence identity. Alignment for purposes of determining percent amino acid sequence identity can be achieved in various ways that are within the skill in the art, for instance, using publicly available computer software such as BLAST, BLAST-2, ALIGN or Megalign (DNASTAR) software. In certain embodiments, the present amino acid sequence identity is at least 80%, 90%, 95%, 98%, or 99%.

**[0065]** As used herein, the term “affinity” or “binding affinity” refers to the strength of the binding interaction between two molecules. Generally, binding affinity refers to the strength of the sum total of non-covalent interactions between a molecule and its binding partner, such as an ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment), and ALK2 or ALK3 protein. Unless indicated otherwise, binding affinity refers to intrinsic binding affinity, which reflects a 1:1 interaction between members of a binding pair. The binding affinity between two molecules is commonly described by the dissociation constant ( $K_D$ ) or the affinity constant ( $K_A$ ). Two molecules that have low binding affinity for each other generally bind slowly, tend to dissociate easily, and exhibit a large  $K_D$ . Two molecules that have high affinity for each other generally bind readily, tend to remain bound longer, and exhibit a small  $K_D$ . The  $K_D$  of two interacting molecules may be determined using methods and techniques well known in the art, e.g., surface plasmon resonance.  $K_D$  is calculated as the ratio of  $k_{off}/k_{on}$ .

**[0066]** As used herein, the term “specific binding” of an antibody or ALK2 binding fragment thereof (e.g., a Fab, Fab', F(ab')<sub>2</sub>, or Fv fragment), is binding to a target molecule that is measurably different from binding to molecules that are not target molecules. As used herein, specific binding refers to a greater than 95% preference for binding a particular antigen versus background (“non-specific”) binding. “Substantially specific” binding refers to a greater than about 80% preference for binding a particular antigen versus background. Binding can be measured using a variety of methods including, but not limited to, Western blot, immunoblot, enzyme-linked immunosorbent assay (“ELISA”), radioimmunoassay (“RIA”), immunoprecipitation, surface plasmon resonance, bio-layer interferometry, chemiluminescence, fluorescent polarization, phosphorescence, immunohistochemical analysis, matrix-assisted laser desorption/ionization time-of-flight (“MALDI-TOF”) mass spectrometry, microcytometry, microarray, microscopy, fluorescence activated cell sorting (“FACS”) and flow cytometry. In certain embodiments, an antibody that specifically binds to a target (e.g., ALK2 or ALK3) has a dissociation constant ( $K_D$ ) no more than 0.1 nM, 0.15 nM, 0.2 nM, 0.3 nM, 0.5 nM, 0.7 nM, 1.0 nM, 2.0 nM, 4.0 nM, 5.0 nM, 7.0 nM, 10.0 nM, 14.0 nM, or 15.0 nM.

**[0067]** As used herein, the term “polypeptide” describes a single polymer in which the monomers are amino acid residues which are covalently conjugated together through amide bonds. A polypeptide is intended to encompass any amino acid sequence, either naturally occurring, recombinant, or synthetically produced.

**[0068]** As used herein, the terms “effective amount,” “therapeutically effective amount,” and “sufficient amount” of a composition or ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment), described herein refer to a quantity sufficient to, when administered to the subject effect beneficial or desired

results, including clinical results, and, as such, an “effective amount” or synonym thereto depends upon the context in which it is being applied. For example, in the context of treating patient having a or at risk of developing a disease, such as bone disease (e.g., osteoporosis, or a condition involving bone damage, e.g., primary osteoporosis, secondary osteoporosis, osteopenia, osteopetrosis, osteogenesis imperfecta, bone fracture, bone cancer or cancer metastasis-related bone loss (e.g., bone loss associated with multiple myeloma), Paget’s disease, renal osteodystrophy, treatment-related bone loss, neuromuscular disease-related bone loss, burn-induced bone loss, anorexia-related bone loss, diet-related bone loss, bone loss associated with the treatment of obesity, low gravity-related bone loss, and immobility-related bone loss), a disease or condition involving low red blood cell levels (e.g., anemia or blood loss), heterotopic ossification (e.g., heterotopic ossification resulting from FOP), Sjogren’s syndrome (e.g., dry eye associated with Sjogren’s syndrome), DIPG, MO, PCO, or cardiac hypertrophy and/or fibrosis, it is an amount of the composition or ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment), sufficient to achieve a treatment response as compared to the response obtained without administration of the composition or ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment). The amount of a given composition described herein that will correspond to such an amount will vary depending upon various factors, such as the given agent, the pharmaceutical formulation, the route of administration, the type of disease or disorder, the identity of the subject (e.g. age, sex, weight) or host being treated, and the like, but can nevertheless be routinely determined by one skilled in the art by routine methods known in the art. Dosage regimen may be adjusted to provide the optimum therapeutic response.

**[0069]** As used herein, “locally” or “local administration” means administration at a particular site of the body intended for a local effect and not a systemic effect. Examples of local administration are epicutaneous, inhalational, intra-articular, intrathecal, intravaginal, intravitreal, intrauterine, intra-lesional administration, lymph node administration, intratumoral administration, and administration to a mucous membrane of the subject, wherein the administration is intended to have a local and not a systemic effect.

**[0070]** As used herein, the term “pharmaceutical composition” refers to a mixture containing a therapeutic agent, optionally in combination with one or more pharmaceutically acceptable excipients, diluents, and/or carriers, to be administered to a subject in order to prevent, treat or control a particular disease or condition affecting or that may affect the subject. The pharmaceutical composition may be in tablet or capsule form for oral administration or in aqueous form for intravenous or subcutaneous administration.

**[0071]** As used herein, the term “pharmaceutically acceptable carrier or excipient” refers to an excipient or diluent in a pharmaceutical composition. The pharmaceutically acceptable carrier must be compatible with the other ingredients of the formulation and suitable for contact with the tissues of a subject without excessive toxicity, irritation, allergic response, and other problem complications commensurate with a reasonable benefit/risk ratio. In the present invention, the pharmaceutically acceptable carrier or excipient must provide adequate pharmaceutical stability to the

ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment) of the invention. The nature of the carrier or excipient differs with the mode of administration. For example, for intravenous administration, an aqueous solution carrier is generally used; for oral administration, a solid carrier is preferred.

**[0072]** As used herein, the term “sample” refers to a specimen (e.g., blood, blood component (e.g., serum or plasma), urine, saliva, amniotic fluid, cerebrospinal fluid, tissue (e.g., neural tissue, placental tissue, or dermal tissue), pancreatic fluid, chorionic villus sample, and cells (e.g., blood cells or bone cells)) isolated from a subject.

**[0073]** As used herein, “treatment” and “treating” in reference to a disease or condition, refer to an approach for obtaining beneficial or desired results, e.g., clinical results. Beneficial or desired results can include, but are not limited to, alleviation or amelioration of one or more symptoms or conditions; diminishment of extent of disease or condition; stabilized (i.e., not worsening) state of disease, disorder, or condition; preventing spread of disease or condition; delay or slowing the progress of the disease or condition; amelioration or palliation of the disease or condition; and remission (whether partial or total), whether detectable or undetectable. “Ameliorating” or “palliating” a disease or condition means that the extent and/or undesirable clinical manifestations of the disease, disorder, or condition are lessened and/or time course of the progression is slowed or lengthened, as compared to the extent or time course in the absence of treatment. “Treatment” can also mean prolonging survival as compared to expected survival if not receiving treatment. Those in need of treatment include those already with the condition or disorder, as well as those prone to have the condition or disorder or those in which the condition or disorder is to be prevented.

**[0074]** As used herein, the terms “subject” and “patient” refer to a mammal, e.g., a human. Mammals include, but are not limited to, humans and domestic and farm animals, such as monkeys (e.g., a cynomolgus monkey), mice, dogs, cats, horses, and cows, etc. A subject to be treated according to the methods described herein may be one who has been diagnosed with bone disease (e.g., a disease or condition involving bone damage, e.g., osteoporosis, osteopenia, osteopetrosis, osteogenesis imperfecta, bone fracture, bone cancer or cancer metastasis-related bone loss, Paget’s disease, renal osteodystrophy, treatment-related bone loss, neuromuscular disease-related bone loss, burn-induced bone loss, anorexia-related bone loss, diet-related bone loss, bone loss associated with the treatment of obesity, low gravity-related bone loss, and immobility-related bone loss), a disease involving low blood cell levels (e.g., anemia or blood loss), heterotopic ossification (e.g., heterotopic ossification resulting from FOP), Sjogren’s syndrome (e.g., dry eye associated with Sjogren’s syndrome), DIPG, MO, PCO, or cardiac hypertrophy and/or fibrosis or one at risk of developing these conditions. Diagnosis may be performed by any method or technique known in the art. One skilled in the art will understand that a subject to be treated according to the present disclosure may have been subjected to standard tests or may have been identified, without examination, as one at risk due to the presence of one or more risk factors associated with the disease or condition.



## DESCRIPTION OF THE DRAWINGS

**[0075]** FIGS. 1A-1B are a series of graphs showing the pharmacokinetic properties of ALK2 antibodies after administration to cynomolgus monkeys. FIG. 1A is a graph of the pharmacokinetic properties of Antibody 2 over a 14-day period. FIG. 1B is a graph of the pharmacokinetic properties of Antibody 4 over a 17-day period. Error bars represent SEM, and the pharmacokinetic values were calculated using a standard non-compartmentalized model.

**[0076]** FIGS. 2A-2B are a series of graphs showing the effect of treatment with an ALK2 antibody on serum hepcidin after administration to cynomolgus monkeys. FIG. 2A is a graph of hepcidin serum concentration after administration of Antibody 2. FIG. 2B is a graph of hepcidin serum concentration after administration of Antibody 4. Error bars represent SEM.

**[0077]** FIGS. 3A-3B are a series of graphs showing the effect of treatment with an ALK2 antibody on serum iron levels after administration to cynomolgus monkeys. FIG. 3A is a graph of serum iron levels after administration of Antibody 2. FIG. 3B is a graph of serum iron levels after administration of Antibody 4. Error bars represent SEM.

**[0078]** FIGS. 4A-4B are a series of graphs showing the effect of treatment with an ALK2 antibody on transferrin saturation after administration to cynomolgus monkeys. FIG. 4A is a graph of percent transferrin levels after administration of Antibody 2. FIG. 4B is a graph of percent transferrin saturation after administration of Antibody 4. The serum from three monkeys was combined for each time point before the assay.

**[0079]** FIGS. 5A-5C are a series of graphs showing the effect of treatment with an ALK2 antibody on reticulocyte hemoglobin, red blood cell hemoglobin, and mean corpuscular hemoglobin content in cynomolgus monkeys. The graphs show the percent change in reticulocyte hemoglobin (FIG. 5A), percent change in red blood cell (RBC) hemoglobin (FIG. 5B), and percent change in mean corpuscular hemoglobin concentration (MCHC) (FIG. 5C) over time after administration of Antibody 2 or Antibody 4. Data from Antibody 2 and Antibody 4 were combined for each metric. Error bars represent SEM.

**[0080]** FIGS. 6A-6C are a series of graphs showing the effect of single dose administration of an ALK2 antibody on CD1 mice. The graphs show drug exposure (FIG. 6A), hepcidin levels (FIG. 6B), and iron levels (FIG. 6C) from mice treated with 0.5 mg/kg, 1.0 mg/kg, or 3.0 mg/kg of an IgG1 isotype control, Antibody 2, or Antibody 4. ns—not significant; \* $p < 0.05$ ; \*\* $p < 0.01$ ; \*\*\* $p < 0.001$ . Error bars represent SEM.

## DETAILED DESCRIPTION OF THE INVENTION

**[0081]** Described herein are methods for treating for a disease or condition that can be treated with an ALK2 or ALK3 inhibitor, including bone disease, anemia, heterotopic ossification (e.g., heterotopic ossification resulting from FOP), Sjogren's syndrome (e.g., dry eye associated with Sjogren's syndrome), DIPG, MO, PCO, or cardiac hypertrophy and/or cardiac fibrosis. The invention features methods of treating these diseases and conditions by administering an ALK2 or ALK3 antibody (e.g., an ALK2 or ALK3 neutralizing antibody) or an antigen binding fragment

thereof (e.g., an ALK2 or ALK3 binding fragment), in an amount of 0.002 mg/kg to 3 mg/kg.

## ALK2 and ALK3 Antibodies

**[0082]** ALK2 (also known as ACVR1) and ALK3 (also known as BMPR1A) are bone morphogenetic protein (BMP) receptors in the transforming growth factor  $\beta$  (TGF- $\beta$ ) superfamily. ALK2 and ALK3 are widely expressed in many tissues during embryonic development and highly expressed in bones during postnatal development and are thought to have essential roles in both osteogenesis and chondrogenesis. Gain of function mutations in ALK2, including c.617G>A (p.R206H), c.619C>G (p.Q207E), c.1067G>A (p.G356D), c.982G>T (p.G328W), c.983G>A (p.G328E), c.982G>A (p.G328R), c.774G>C/c.774G>T (p.R258S), c.1124G>C (p.R375P), c.587T>C (p.L196P), c.590-592delICTT (p.P197\_F198delinsL), and c.605G>T (p.R2021), have been found in subjects with FOP, and studies using ALK2<sup>R206H</sup> mutant mice suggest that ALK2 regulates the osteogenic differentiation of mesenchymal stem cells. Activating mutations in ALK2 have also been observed in approximately 25% of DIPG patients and small molecule ALK2 inhibitors have been found to extend survival and reduce cellularity in orthotopic DIPG xenograft models. ALK2 may also play a role in anemia, as BMP activation of the SMAD signaling pathway through ALK2 induces the upregulation of hepcidin, a master regulator of iron homeostasis that is implicated in anemia of inflammation (also known as anemia of chronic disease) and iron refractory iron deficiency anemia (IRIDA). ALK2 also promotes the proliferation of lens epithelial cells during development; therefore, inhibition of ALK2 may prevent or reduce the aberrant proliferation of lens epithelial cells that leads to PCO. In addition, small molecule ALK2 inhibitors have been found to decrease inflammation and treat symptoms of Sjogren's syndrome in an established mouse model, and to reduce osteochondroma formation, growth, and size in multiple studies using mouse models of MO. Furthermore, both treatment with a small molecule ALK2 inhibitor and cardiomyocyte-specific deletion of ALK2 mitigated cardiac hypertrophy and left ventricular fibrosis in mice. Moreover, the effect of ALK2 ligands in promoting fibrosis suggests that inhibiting ALK2 can be used to treat diseases or conditions in which fibrosis contributes to the pathology (e.g., fibrosis of the posterior capsule in PCO, fibrosis of the lacrimal gland in Sjogren's syndrome, and fibrosis in cardiac fibrosis and/or cardiac hypertrophy). Downregulation of ALK3 signaling in young and old mice has been found to increase bone mass and/or prevent bone loss, suggesting that inhibiting ALK3 could be used to treat bone diseases. In addition, knockout of ALK3 has been observed to partly protect from anemia of inflammation, suggesting that ALK3 may be a possible therapeutic target for treatment. In addition, inhibition of ALK3 may also disrupt ALK2 signaling.

**[0083]** The present invention is based, in part, on the discovery that administration of an ALK2 antibody described herein to non-human primates in an amount of 3 mg/kg resulted in reduced serum hepcidin levels within six hours of administration that was sustained for at least 10 days and increased serum iron and increased transferrin saturation for at least 10 days after administration. Increased reticulocyte hemoglobin and red blood cell hemoglobin were also observed.

[0084] The ALK2 antibody administered to the subject may be an ALK2 antibody or an antigen-binding fragment thereof as described in Table 2. In some embodiments, the ALK2 antibody administered to the subject is an ALK2 antibody or antigen-binding fragment thereof, as described in International Patent Application Publication Nos. WO2020086730, WO2020118011, and WO2021163170,

US Patent Application Publication Nos. US20210253720, US20220098310, and US20210253716, and U.S. Pat. No. 10,428,148, which are incorporated herein by reference. In some embodiments, the ALK2 antibody administered to the subject is a Mab or Fab described in Aykul et al., J Clin Invest., e-publication doi: 10.1172/JCI153792, 2022.

TABLE 2

| Row | Composition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | An isolated antibody, or ALK2 binding fragment thereof, comprising (1) a light chain variable domain comprising a light chain complementarity determining region (CDR)1 comprising an amino acid sequence selected from the group consisting of SGSSSNIGSNYVS (SEQ ID NO: 1) and SGDX <sub>1</sub> X <sub>2</sub> X <sub>3</sub> X <sub>4</sub> X <sub>5</sub> X <sub>6</sub> X <sub>7</sub> X <sub>8</sub> (SEQ ID NO: 2, wherein X <sub>1</sub> is S or N, X <sub>2</sub> is I or L, X <sub>3</sub> is P, G, or R, X <sub>4</sub> is S, T, or K, X <sub>5</sub> is F, K, or Y, X <sub>6</sub> is F, Y, or S, X <sub>7</sub> is A or V, and X <sub>8</sub> is S, Y, or H); a light chain CDR2 comprising the amino acid sequence X <sub>1</sub> X <sub>2</sub> YX <sub>3</sub> X <sub>4</sub> X <sub>5</sub> X <sub>6</sub> RPS (SEQ ID NO: 3, wherein X <sub>1</sub> is V or L, X <sub>2</sub> is V or L, X <sub>3</sub> is K, R, G or Y, X <sub>4</sub> is N or D, X <sub>5</sub> is N or S, and X <sub>6</sub> is H, N, D, or K); and a light chain CDR3 comprising an amino acid sequence selected from the group consisting of ASWDHSDRFYV (SEQ ID NO: 4), YVTAPWKSIV (SEQ ID NO: 5), YSADAQQMKA (SEQ ID NO: 6), QVYASVHRM (SEQ ID NO: 7), and QTYDWSHFGW (SEQ ID NO: 8); and (2) a heavy chain variable domain comprising a heavy chain CDR1 comprising the amino acid sequence GX <sub>1</sub> TFX <sub>2</sub> SX <sub>3</sub> X <sub>4</sub> X <sub>5</sub> X <sub>6</sub> (SEQ ID NO: 9, wherein X <sub>1</sub> is G or F, X <sub>2</sub> is S or N, X <sub>3</sub> is Y, H, S, or A, X <sub>4</sub> is G or A, X <sub>5</sub> is V, M, or I, and X <sub>6</sub> is S or H); a heavy chain CDR2 comprising an amino acid sequence selected from the group consisting of WMGX <sub>1</sub> IIPX <sub>2</sub> FGX <sub>3</sub> ANYAQKFQG (SEQ ID NO: 10, wherein X <sub>1</sub> is G or R, X <sub>2</sub> is H or D, and X <sub>3</sub> is I or T), WVGRIKSKX <sub>1</sub> DX <sub>2</sub> X <sub>3</sub> TTDYAAPVKG (SEQ ID NO: 11, wherein X <sub>1</sub> is A or R, X <sub>2</sub> is S or G, and X <sub>3</sub> is G or Y), and WVSVSSDGGSTYYADSVKG (SEQ ID NO: 12); and a heavy chain CDR3 comprising an amino acid sequence selected from the group consisting of EIGSLDI (SEQ ID NO: 13), DYGVAFAY (SEQ ID NO: 14), DYGGGLKFDY (SEQ ID NO: 15), GPTQAIHYFAY (SEQ ID NO: 16), and AGFILGSLGVAWMDV (SEQ ID NO: 17). |
| 2   | The isolated antibody, or ALK2 binding fragment thereof, of row 1, wherein the light chain CDR1 comprises the sequence SGSSSNIGSNYVS (SEQ ID NO: 1).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 3   | The isolated antibody, or ALK2 binding fragment thereof, of row 1, wherein the light chain CDR1 comprises the sequence SGDSIPSFAS (SEQ ID NO: 18).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 4   | The isolated antibody, or ALK2 binding fragment thereof, of row 1, wherein the light chain CDR1 comprises the sequence SGDNLTKYAY (SEQ ID NO: 19).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 5   | The isolated antibody, or ALK2 binding fragment thereof, of row 1, wherein the light chain CDR1 comprises the sequence SGDNLTKYSAH (SEQ ID NO: 20).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 6   | The isolated antibody, or ALK2 binding fragment thereof, of row 1, wherein the light chain CDR1 comprises the sequence SGDSLGSKSVH (SEQ ID NO: 21).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 7   | The isolated antibody, or ALK2 binding fragment thereof, of any one of rows 1 to 6, wherein the light chain CDR2 comprises the sequence VLIYKNNHRPS (SEQ ID NO: 24).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 8   | The isolated antibody, or ALK2 binding fragment thereof, of any one of rows 1 to 6, wherein the light chain CDR2 comprises the sequence LVTYRDSNRPS (SEQ ID NO: 25).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 9   | The isolated antibody, or ALK2 binding fragment thereof, of any one of rows 1 to 6, wherein the light chain CDR2 comprises the sequence LVTYGDSDRPS (SEQ ID NO: 26).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 10  | The isolated antibody, or ALK2 binding fragment thereof, of any one of rows 1 to 6, wherein the light chain CDR2 comprises the sequence LVIYYDNKRPS (SEQ ID NO: 27).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 11  | The isolated antibody, or ALK2 binding fragment thereof, of any one of rows 1 to 6, wherein the light chain CDR2 comprises the sequence LVTYRDSKRPS (SEQ ID NO: 28).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 12  | The isolated antibody, or ALK2 binding fragment thereof, of any one of rows 1 to 11, wherein the light chain CDR3 comprises the sequence ASWDHSDRFYV (SEQ ID NO: 4).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 13  | The isolated antibody, or ALK2 binding fragment thereof, of any one of rows 1 to 11, wherein the light chain CDR3 comprises the sequence YVTAPWKSIV (SEQ ID NO: 5).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 14  | The isolated antibody, or ALK2 binding fragment thereof, of any one of rows 1 to 11, wherein the light chain CDR3 comprises the sequence YSADAQQMKA (SEQ ID NO: 6).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 15  | The isolated antibody, or ALK2 binding fragment thereof, of any one of rows 1 to 11, wherein the light chain CDR3 comprises the sequence QVYASVHRM (SEQ ID NO: 7).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 16  | The isolated antibody, or ALK2 binding fragment thereof, of any one of rows 1 to 11, wherein the light chain CDR3 comprises the QTYDWSHFGW (SEQ ID NO: 8).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 17  | The isolated antibody, or ALK2 binding fragment thereof, of any one of rows 1 to 16, wherein the heavy chain CDR1 comprises the sequence GGTFSSYGV (SEQ ID NO: 31).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 18  | The isolated antibody, or ALK2 binding fragment thereof, of any one of rows 1 to 16, wherein the heavy chain CDR1 comprises the sequence GFTFSSHAMS (SEQ ID NO: 32).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 19  | The isolated antibody, or ALK2 binding fragment thereof, of any one of rows 1 to 16, wherein the heavy chain CDR1 comprises the sequence GFTFNSSAMS (SEQ ID NO: 33).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 20  | The isolated antibody, or ALK2 binding fragment thereof, of any one of rows 1 to 16, wherein the heavy chain CDR1 comprises the sequence GGTFSSYAIH (SEQ ID NO: 34).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 21  | The isolated antibody, or ALK2 binding fragment thereof, of any one of rows 1 to 16, wherein the heavy chain CDR1 comprises the sequence GFTFSSAAMH (SEQ ID NO: 35).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 22  | The isolated antibody, or ALK2 binding fragment thereof, of any one of rows 1 to 21, wherein the heavy chain CDR2 comprises the sequence WMGX <sub>1</sub> IIPX <sub>2</sub> FGX <sub>3</sub> ANYAQKFQG (SEQ ID NO: 36).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 23  | The isolated antibody, or ALK2 binding fragment thereof, of any one of rows 1 to 21, wherein the heavy chain CDR2 comprises the sequence WVGRIKSKADSGTTDYAAPVKG (SEQ ID NO: 37).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

TABLE 2-continued

| Row | Composition                                                                                                                                                                                                                                                                                                                                                  |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 24  | The isolated antibody, or ALK2 binding fragment thereof, of any one of row s1 to 21, wherein the heavy chain CDR2 comprises the sequence WVGRIKSKRDGYTTDYAAPVKG (SEQ ID NO: 38).                                                                                                                                                                             |
| 25  | The isolated antibody, or ALK2 binding fragment thereof, of any one of rows 1 to 21, wherein the heavy chain CDR2 comprises the sequence WMGRIIPDFGTANYAQKFQG (SEQ ID NO: 39).                                                                                                                                                                               |
| 26  | The isolated antibody, or ALK2 binding fragment thereof, of any one of rows 1 to 21, wherein the heavy chain CDR2 comprises the sequence WVSIVSSDGGSTYYADSVKG (SEQ ID NO: 12).                                                                                                                                                                               |
| 27  | The isolated antibody, or ALK2 binding fragment thereof, of any one of rows 1 to 26, wherein the heavy chain CDR3 comprises the sequence EIGSLDI (SEQ ID NO: 13).                                                                                                                                                                                            |
| 28  | The isolated antibody, or ALK2 binding fragment thereof, of any one of row 1 to 26, wherein the heavy chain CDR3 comprises the sequence DYGVAFAFAY (SEQ ID NO: 14).                                                                                                                                                                                          |
| 29  | The isolated antibody, or ALK2 binding fragment thereof, of any one of row 1 to 26, wherein the heavy chain CDR3 comprises the sequence DYGLKFDY (SEQ ID NO: 15).                                                                                                                                                                                            |
| 30  | The isolated antibody, or ALK2 binding fragment thereof, of any one of row 1 to 26, wherein the heavy chain CDR3 comprises the sequence GPTQAIHYFAY (SEQ ID NO: 16).                                                                                                                                                                                         |
| 31  | The isolated antibody, or ALK2 binding fragment thereof, of any one of row 1 to 26, wherein the heavy chain CDR3 comprises the sequence AGFILGSLGVAWMDV (SEQ ID NO: 17).                                                                                                                                                                                     |
| 32  | The isolated antibody, or ALK2 binding fragment thereof, of row 1, wherein the light chain CDR2 comprises the sequence LVIYX <sub>1</sub> DX <sub>2</sub> X <sub>3</sub> RPS (SEQ ID NO: 22, wherein X <sub>1</sub> is R, G, or Y, X <sub>2</sub> is S or N, and X <sub>3</sub> is N, D, or K).                                                              |
| 33  | The isolated antibody, or ALK2 binding fragment thereof, of row 1, wherein the light chain CDR2 comprises the sequence LVIYRDSX <sub>1</sub> RPS (SEQ ID NO: 23, wherein X <sub>1</sub> is N or K).                                                                                                                                                          |
| 34  | The isolated antibody, or ALK2 binding fragment thereof, of row 1, wherein the heavy chain CDR1 comprises the sequence GFTFSSX <sub>1</sub> AMX <sub>2</sub> (SEQ ID NO: 29, wherein X <sub>1</sub> is H or A, and X <sub>2</sub> is S or H).                                                                                                                |
| 35  | The isolated antibody, or ALK2 binding fragment thereof, of row 1, wherein the heavy chain CDR1 comprises the sequence GFTFX <sub>1</sub> SX <sub>2</sub> AMS (SEQ ID NO: 30, wherein X <sub>1</sub> is S or N, and X <sub>2</sub> is H or S).                                                                                                               |
| 36  | The isolated antibody, or ALK2 binding fragment thereof, of row 1, wherein the light chain CDR1 comprises the amino acid sequence SGSSNIGSNYYVS (SEQ ID NO: 1); the light chain CDR2 comprises the amino acid sequence VLIYKNNHRPS (SEQ ID NO: 24); and the light chain CDR3 comprises the amino acid sequence ASWDHSDRFYV (SEQ ID NO: 4).                   |
| 37  | The isolated antibody, or ALK2 binding fragment thereof, of row 1 or 36, wherein the heavy chain CDR1 comprises the amino acid sequence GGTFFSSYGVS (SEQ ID NO: 31); the heavy chain CDR2 comprises the amino acid sequence WMGGIIPHFGLIANYAQKFQG (SEQ ID NO: 36); and the heavy chain CDR3 comprises the amino acid sequence EIGSLDI (SEQ ID NO: 13).       |
| 38  | The isolated antibody, or ALK2 binding fragment thereof, of row 1, wherein the light chain CDR1 comprises the amino acid sequence SGDSIPSFAS (SEQ ID NO: 18); the light chain CDR2 comprises the amino acid sequence LVIYRDSNRPS (SEQ ID NO: 25); and the light chain CDR3 comprises the amino acid sequence YVTAPWKSIW (SEQ ID NO: 5).                      |
| 39  | The isolated antibody, or ALK2 binding fragment thereof, of row 1 or 38, wherein the heavy chain CDR1 comprises the amino acid sequence GFTFSSHAMS (SEQ ID NO: 32); the heavy chain CDR2 comprises the amino acid sequence WVGRIKSKADSGTTDYAAPVKG (SEQ ID NO: 37); and the heavy chain CDR3 comprises the amino acid sequence DYGVAFAFAY (SEQ ID NO: 14).    |
| 40  | The isolated antibody, or ALK2 binding fragment thereof, of row 1, wherein the light chain CDR1 comprises the amino acid sequence SGDNLGTYAY (SEQ ID NO: 19); the light chain CDR2 comprises the amino acid sequence LVIYGDSRPS (SEQ ID NO: 26); and the light chain CDR3 comprises the amino acid sequence YSADAQMQKA (SEQ ID NO: 6).                       |
| 41  | The isolated antibody, or ALK2 binding fragment thereof, of row 1 or 40, wherein the heavy chain CDR1 comprises the amino acid sequence GFTFNSSAMS (SEQ ID NO: 33); the heavy chain CDR2 comprises the amino acid sequence WVGRIKSKRDGYTTDYAAPVKG (SEQ ID NO: 38); and the heavy chain CDR3 comprises the amino acid sequence DYGLKFDY (SEQ ID NO: 15).      |
| 42  | The isolated antibody, or ALK2 binding fragment thereof, of row 1, wherein the light chain CDR1 comprises the amino acid sequence SGDNLRKYSVAH (SEQ ID NO: 20); the light chain CDR2 comprises the amino acid sequence LVIYYDNKRPS (SEQ ID NO: 27); and the light chain CDR3 comprises the amino acid sequence QVYASVHRM (SEQ ID NO: 7).                     |
| 43  | The isolated antibody, or ALK2 binding fragment thereof, of row 1 or 42 wherein the heavy chain CDR1 comprises the amino acid sequence GGTFFSSYAIH (SEQ ID NO: 34); the heavy chain CDR2 comprises the amino acid sequence WMGRIIPDFGTANYAQKFQG (SEQ ID NO: 39); and the heavy chain CDR3 comprises the amino acid sequence GPTQAIHYFAY (SEQ ID NO: 16).     |
| 44  | The isolated antibody, or ALK2 binding fragment thereof, of row 1, wherein the light chain CDR1 comprises the amino acid sequence SGDSLGSKSVH (SEQ ID NO: 21); the light chain CDR2 comprises the amino acid sequence LVIYRDSKRPS (SEQ ID NO: 28); and the light chain CDR3 comprises the amino acid sequence QTYDWSHFGW (SEQ ID NO: 8).                     |
| 45  | The isolated antibody, or ALK2 binding fragment thereof, of row 1 or 44, wherein the heavy chain CDR1 comprises the amino acid sequence GFTFSSAAMH (SEQ ID NO: 35); the heavy chain CDR2 comprises the amino acid sequence WVSIVSSDGGSTYYADSVKG (SEQ ID NO: 12); and the heavy chain CDR3 comprises the amino acid sequence AGFILGSLGVAWMDV (SEQ ID NO: 17). |

TABLE 2-continued

| Row | Composition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 46  | The isolated antibody of row 1, wherein the antibody, apart from the light chain CDR1, CDR2, and CDR3 and the heavy chain CDR1, CDR2, and CDR3, has at least 90% sequence identity (e.g., 90%, 91%, 92%, 93%, 94%, 95%, 95%, 97%, 98%, 99% or more sequence identity) to amino acids 1 to 331 of the sequence of SEQ ID NO: 67, amino acids 1 to 332 of the sequence of SEQ ID NO: 68, amino acids 1 to 333 of the sequence of SEQ ID NO: 69, amino acids 1 to 332 of the sequence of SEQ ID NO: 70, or amino acids 1 to 337 of the sequence of SEQ ID NO: 71.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 47  | The isolated antibody of row 1, wherein the antibody comprises amino acids 1 to 433 of the sequence of SEQ ID NO: 67, amino acids 1 to 434 of the sequence of SEQ ID NO: 68, amino acids 1 to 435 of the sequence of SEQ ID NO: 69, amino acids 1 to 434 of the sequence of SEQ ID NO: 70, or amino acids 1 to 439 of the sequence of SEQ ID NO: 71.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 48  | An isolated antibody, or ALK2 binding fragment thereof, comprising (1) a light chain variable domain comprising a light chain complementarity determining region (CDR)1 comprising an amino acid sequence selected from the group consisting of RASQGISGNWLT (SEQ ID NO: 40), SGDX <sub>1</sub> X <sub>2</sub> RX <sub>3</sub> X <sub>4</sub> X <sub>5</sub> X <sub>6</sub> H (SEQ ID NO: 64, wherein X <sub>1</sub> is N or A, X <sub>2</sub> is I or L, X <sub>3</sub> is K or Y, X <sub>4</sub> is K or Y, X <sub>5</sub> is Y or I, and X <sub>6</sub> is V or A), and SGSSSNIGQNYVS (SEQ ID NO: 58); a light chain CDR2 comprising the amino acid sequence LX <sub>1</sub> IYX <sub>2</sub> X <sub>3</sub> X <sub>4</sub> X <sub>5</sub> X <sub>6</sub> X <sub>7</sub> S (SEQ ID NO: 65, where X <sub>1</sub> is V or L, X <sub>2</sub> is D, R, or Y, X <sub>3</sub> is A, D, or N, X <sub>4</sub> is S or N, X <sub>5</sub> is K or N, X <sub>6</sub> is L or R, and X <sub>7</sub> is Q or P); and a light chain CDR3 comprising an amino acid sequence selected from the group consisting of HQSYRGPM (SEQ ID NO: 42), SSAGRDNY (SEQ ID NO: 48), QSYGPGSV (SEQ ID NO: 54), and SSWDLLSKSR (SEQ ID NO: 60); and (2) a heavy chain variable domain comprising a heavy chain CDR1 comprising the amino acid sequence GX <sub>1</sub> TFX <sub>2</sub> X <sub>3</sub> X <sub>4</sub> X <sub>5</sub> X <sub>6</sub> X <sub>7</sub> (SEQ ID NO: 66, wherein X <sub>1</sub> is F or G, X <sub>2</sub> is G or S, X <sub>3</sub> is R, S, D, or T, X <sub>4</sub> is F, S, Y, or H, X <sub>5</sub> is V or A, and X <sub>6</sub> is M or I, and X <sub>7</sub> is H or S); a heavy chain CDR2 comprising an amino acid sequence selected from the group consisting of WVSX <sub>1</sub> IX <sub>2</sub> YX <sub>3</sub> X <sub>4</sub> SYX <sub>5</sub> TYADSVKG (SEQ ID NO: 76, wherein X <sub>1</sub> is V or S, X <sub>2</sub> is G, H, or F, X <sub>3</sub> is S or D, X <sub>4</sub> is G or S, and X <sub>5</sub> is S, E, or N), and WMGLIQPRFGTANYAQKFQR (SEQ ID NO: 62); and a heavy chain CDR3 comprising an amino acid sequence selected from the group consisting of EPGYYPSGYRGPYWMVDV (SEQ ID NO: 45), DRYFFDV (SEQ ID NO: 51), PKSYASGPFAY (SEQ ID NO: 57), and DYYGGMAY (SEQ ID NO: 63). |
| 49  | The isolated antibody, or ALK2 binding fragment thereof, of row 48, wherein the light chain CDR1 comprises the sequence RASQGISGNWLT (SEQ ID NO: 40).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 50  | The isolated antibody, or ALK2 binding fragment thereof, of row 48, wherein the light chain CDR1 comprises the sequence SGDNIKKYVH (SEQ ID NO: 46).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 51  | The isolated antibody, or ALK2 binding fragment thereof, of row 48, wherein the light chain CDR1 comprises the sequence SGDALRYIAH (SEQ ID NO: 52).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 52  | The isolated antibody, or ALK2 binding fragment thereof, of row 48, wherein the light chain CDR1 comprises the sequence SGSSSNIGQNYVS (SEQ ID NO: 58).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 53  | The isolated antibody, or ALK2 binding fragment thereof, of any one of rows 48 to 52, wherein the light chain CDR2 comprises the sequence LLIYDASNLS (SEQ ID NO: 41).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 54  | The isolated antibody, or ALK2 binding fragment thereof, of any one of rows 48 to 52, wherein the light chain CDR2 comprises the sequence LVIYRDSNRPS (SEQ ID NO: 47).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 55  | The isolated antibody, or ALK2 binding fragment thereof, of any one of rows 48 to 52, wherein the light chain CDR2 comprises the sequence LVITYNNNRPS (SEQ ID NO: 53).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 56  | The isolated antibody, or ALK2 binding fragment thereof, of any one of rows 48 to 52, wherein the light chain CDR2 comprises the sequence LLIYDNSKRPS (SEQ ID NO: 59).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 57  | The isolated antibody, or ALK2 binding fragment thereof, of any one of rows 48 to 56, wherein the light chain CDR3 comprises the sequence HQSYRGPM (SEQ ID NO: 42).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 58  | The isolated antibody, or ALK2 binding fragment thereof, of any one of rows 48 to 56, wherein the light chain CDR3 comprises the sequence SSAGRDNY (SEQ ID NO: 48).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 59  | The isolated antibody, or ALK2 binding fragment thereof, of any one of rows 48 to 56, wherein the light chain CDR3 comprises the sequence QSYGPGSV (SEQ ID NO: 54).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 60  | The isolated antibody, or ALK2 binding fragment thereof, of any one of rows 48 to 56, wherein the light chain CDR3 comprises the sequence SSWDLLSKSR (SEQ ID NO: 60).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 61  | The isolated antibody, or ALK2 binding fragment thereof, of any one of rows 48 to 60, wherein the heavy chain CDR1 comprises the sequence GFTFGRFVMH (SEQ ID NO: 43).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 62  | The isolated antibody, or ALK2 binding fragment thereof, of any one of rows 48 to 60, wherein the heavy chain CDR1 comprises the sequence GFTFSSSAMH (SEQ ID NO: 49).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 63  | The isolated antibody, or ALK2 binding fragment thereof, of any one of rows 48 to 60, wherein the heavy chain CDR1 comprises the sequence GFTFSDYAMH (SEQ ID NO: 55).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 64  | The isolated antibody, or ALK2 binding fragment thereof, of any one of rows 48 to 60, wherein the heavy chain CDR1 comprises the sequence GGTFSTHAIS (SEQ ID NO: 61).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 65  | The isolated antibody, or ALK2 binding fragment thereof, of any one of rows 48 to 64, wherein the heavy chain CDR2 comprises the sequence WVSIVIGYSSTYYADSVKG (SEQ ID NO: 44).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 66  | The isolated antibody, or ALK2 binding fragment thereof, of any one of rows 48 to 64, wherein the heavy chain CDR2 comprises the sequence WVSIVHYDSSTYYADSVKG (SEQ ID NO: 50).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 67  | The isolated antibody, or ALK2 binding fragment thereof, of any one of rows 48 to 64, wherein the heavy chain CDR2 comprises the sequence WVSSIFYSGSNTYYADSVKG (SEQ ID NO: 56).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 68  | The isolated antibody, or ALK2 binding fragment thereof, of any one of rows 48 to 64, wherein the heavy chain CDR2 comprises the sequence WMGLIQPRFGTANYAQKFQR (SEQ ID NO: 62).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

TABLE 2-continued

| Row | Composition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 69  | The isolated antibody, or ALK2 binding fragment thereof, of any one of rows 48 to 68, wherein the heavy chain CDR3 comprises the sequence EPGYYYPGYYRGPYWMVDV (SEQ ID NO: 45).                                                                                                                                                                                                                                                                                                                           |
| 70  | The isolated antibody, or ALK2 binding fragment thereof, of any one of rows 48 to 68, wherein the heavy chain CDR3 comprises the sequence DRYFFDV (SEQ ID NO: 51).                                                                                                                                                                                                                                                                                                                                       |
| 71  | The isolated antibody, or ALK2 binding fragment thereof, of any one of rows 48 to 68, wherein the heavy chain CDR3 comprises the sequence PKSYASGPFAY (SEQ ID NO: 57).                                                                                                                                                                                                                                                                                                                                   |
| 72  | The isolated antibody, or ALK2 binding fragment thereof, of any one of rows 48 to 68, wherein the heavy chain CDR3 comprises the sequence DYYGGMAY (SEQ ID NO: 63).                                                                                                                                                                                                                                                                                                                                      |
| 73  | The isolated antibody, or ALK2 binding fragment thereof, of row 48, wherein the light chain CDR1 comprises the amino acid sequence RASQGISGNWLT (SEQ ID NO: 40); the light chain CDR2 comprises the amino acid sequence LLIYDASNLQS (SEQ ID NO: 41); and the light chain CDR3 comprises the amino acid sequence HQSYRGPM (SEQ ID NO: 42).                                                                                                                                                                |
| 74  | The isolated antibody, or ALK2 binding fragment thereof, of row 48 or 73, wherein the heavy chain CDR1 comprises the amino acid sequence GFTFGRFVMH (SEQ ID NO: 43); the heavy chain CDR2 comprises the amino acid sequence WWSVIGYSGSSTYYADSVKG (SEQ ID NO: 44); and the heavy chain CDR3 comprises the amino acid sequence EPGYYYPGYYRGPYWMVDV (SEQ ID NO: 45).                                                                                                                                        |
| 75  | The isolated antibody, or ALK2 binding fragment thereof, of row 48, wherein the light chain CDR1 comprises the amino acid sequence SGDNRKKYVH (SEQ ID NO: 46); the light chain CDR2 comprises the amino acid sequence LVTYRDSNRPS (SEQ ID NO: 47); and the light chain CDR3 comprises the amino acid sequence SSAGRDNY (SEQ ID NO: 48).                                                                                                                                                                  |
| 76  | The isolated antibody, or ALK2 binding fragment thereof, of row 48 or 75, wherein the heavy chain CDR1 comprises the amino acid sequence GFTFSSSAMH (SEQ ID NO: 49); the heavy chain CDR2 comprises the amino acid sequence WWSVIHYDSSETYYADSVKG (SEQ ID NO: 50); and the heavy chain CDR3 comprises the amino acid sequence DRYFFDV (SEQ ID NO: 51).                                                                                                                                                    |
| 77  | The isolated antibody, or ALK2 binding fragment thereof, of row 48, wherein the light chain CDR1 comprises the amino acid sequence SGDALRYIAH (SEQ ID NO: 52); the light chain CDR2 comprises the amino acid sequence LVTYNNNRPS (SEQ ID NO: 53); and the light chain CDR3 comprises the amino acid sequence QSYGPGSV (SEQ ID NO: 54).                                                                                                                                                                   |
| 78  | The isolated antibody, or ALK2 binding fragment thereof, of row 48 or 77, wherein the heavy chain CDR1 comprises the amino acid sequence GFTFSDYAMH (SEQ ID NO: 55); the heavy chain CDR2 comprises the amino acid sequence WWSIFYSGSNTYYADSVKG (SEQ ID NO: 56); and the heavy chain CDR3 comprises the amino acid sequence PKSYASGPFAY (SEQ ID NO: 57).                                                                                                                                                 |
| 79  | The isolated antibody, or ALK2 binding fragment thereof, of row 48, wherein the light chain CDR1 comprises the amino acid sequence SGSSNIGQNYVS (SEQ ID NO: 58); the light chain CDR2 comprises the amino acid sequence LLIYDNSKRPS (SEQ ID NO: 59); and the light chain CDR3 comprises the amino acid sequence SSWDLLSKSR (SEQ ID NO: 60).                                                                                                                                                              |
| 80  | The isolated antibody, or ALK2 binding fragment thereof, of row 48 or 79, wherein the heavy chain CDR1 comprises the amino acid sequence GGTFSTHAIS (SEQ ID NO: 61); the heavy chain CDR2 comprises the amino acid sequence WMGLIQRFGTANYAQKFQR (SEQ ID NO: 62); and the heavy chain CDR3 comprises the amino acid sequence DYYGGMAY (SEQ ID NO: 63).                                                                                                                                                    |
| 81  | The isolated antibody of row 48, wherein the antibody, apart from the light chain CDR1, CDR2, and CDR3 and the heavy chain CDR1, CDR2, and CDR3, has at least 90% sequence identity (e.g., 90%, 91%, 92%, 93%, 94%, 95%, 95%, 97%, 98%, 99% or more sequence identity) to amino acids 1 to 344 of the sequence of SEQ ID NO: 72, amino acids 1 to 327 of the sequence of SEQ ID NO: 73, amino acids 1 to 331 of the sequence of SEQ ID NO: 74, or amino acids 1 to 332 of the sequence of SEQ ID NO: 75. |
| 82  | The isolated antibody of row 48, wherein the antibody comprises amino acids 1 to 446 of the sequence of SEQ ID NO: 72, amino acids 1 to 429 of the sequence of SEQ ID NO: 73, amino acids 1 to 433 of the sequence of SEQ ID NO: 74, or amino acids 1 to 434 of the sequence of SEQ ID NO: 75.                                                                                                                                                                                                           |

**[0085]** In some embodiments, the ALK2 antibody administered to the subject is an ALK2 antibody or an antigen-binding fragment thereof, having CDR sequences described in Table 3 (i.e., a light chain CDR1, CDR2, and CDR3 and a heavy chain CDR1, CDR2, and CDR3). In some embodiments, the ALK2 antibody or antigen binding fragment thereof includes a light chain variable CDR1 sequence having at least 90% (e.g., at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100%) sequence identity to a light chain variable CDR1 sequence in Table 3, such as any one of SEQ ID NOs: 1, 18-21, 40, 46, 52, 58, 98, 85, 104, 91, 180, 186, 192, 198, 220, 226, 231, 239, 245, 252, 257, 262, 267, 278, 286, 289, 295, and 301; a light chain variable CDR2 sequence having at least 90% (e.g., at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100%)

sequence identity to a light chain variable CDR2 sequence in Table 3, such as any one of SEQ ID NOs: 24-28, 41, 47, 53, 59, 99, 86, 105, 92, 94, 181, 187, 193, 199, 221, 227, 232, 234, 240, 246, 248, 253, 263, 273, 281, 290, 296, and 302; a light chain variable CDR3 sequence having at least 90% (e.g., at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100%) sequence identity to a light chain variable CDR3 sequence in Table 3, such as any one of SEQ ID NOs: 4-8, 42, 48, 54, 60, 100, 87, 106, 93, 87, 182, 188, 194, 200, 222, 228, 235, 241, 247, 254, 258, 264, 268, 274, 279, 282, 287, 291, 297, and 303; a heavy chain variable CDR1 sequence having at least 90% (e.g., at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100%) sequence identity to a heavy chain variable CDR1 sequence in Table 3, such as any one of SEQ ID NOs: 31-35, 43, 49,

55, 61, 95, 82, 101, 88, 177, 183, 189, 195, 217, 223, 229, 236, 242, 249, 255, 259, 265, 269, 275, 283, and 298; a heavy chain variable CDR2 sequence having at least 90% (e.g., at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or more) sequence identity to a heavy chain variable CDR2 sequence in Table 3, such as any one of SEQ ID NOs: 36-39, 12, 44, 50, 56, 62, 96, 83, 102, 89, 178, 184, 190, 196, 218, 224, 230, 237, 243, 250, 256, 260, 270, 276, 280, 284, 288, 292, 293, and 299; and a heavy chain variable CDR3 sequence having at least 90% (e.g., at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or more) sequence

identity to a heavy chain variable CDR3 sequence in Table 3, such as any one of SEQ ID NOs: 13-17, 45, 51, 57, 63, 97, 84, 103, 90, 179, 185, 191, 197, 219, 225, 233, 238, 244, 251, 382, 261, 266, 271, 277, 285, 294, and 300. In some embodiments, the ALK2 antibody or antigen binding fragment thereof includes a light chain CDR1, CDR2, and CDR3 sequence and a heavy chain CDR1, CDR2, and CDR3 sequence from the same row of Table 3 (e.g., a light chain CDR1, CDR2, and CDR3 sequence and a heavy chain CDR1, CDR2, and CDR3 sequence from any one of Antibodies 1-38).

TABLE 3

| ALK2 Antibody CDR sequences               |                                  |                                |                                |                                                   |                                              |
|-------------------------------------------|----------------------------------|--------------------------------|--------------------------------|---------------------------------------------------|----------------------------------------------|
| Anti-                                     | VL                               |                                |                                | VH                                                |                                              |
| body CDR1                                 | CDR2                             | CDR3                           | CDR1                           | CDR2                                              | CDR3                                         |
| 1 SGSSSNIGS<br>NYVS (SEQ ID NO: 1)        | VLIIYKNNHR<br>PS (SEQ ID NO: 24) | ASWDHSDR<br>FYV (SEQ ID NO: 4) | GGTFSSYG<br>VS (SEQ ID NO: 31) | WMGGIIPHF<br>GIANYAQKF<br>QG (SEQ ID NO: 36)      | EIGSLDI<br>(SEQ ID NO: 13)                   |
| 2 SGDSIPSFF<br>AS (SEQ ID NO: 18)         | LVIYRDSNR<br>PS (SEQ ID NO: 25)  | YVTAPWKSI<br>W (SEQ ID NO: 5)  | GFTFSSHA<br>MS (SEQ ID NO: 32) | WVGRIKSKA<br>DSGTTDYA<br>APVKG (SEQ ID NO: 37)    | DYGVAFAY<br>(SEQ ID NO: 14)                  |
| 3 SGDNIGTKY<br>AY (SEQ ID NO: 19)         | LVIYGDSDR<br>PS (SEQ ID NO: 26)  | YSADAQQM<br>KA (SEQ ID NO: 6)  | GFTFNSSA<br>MS (SEQ ID NO: 33) | WVGRIKSK<br>RDGYTTDY<br>AAPVKG<br>(SEQ ID NO: 38) | DYGGLKFD<br>Y (SEQ ID NO: 15)                |
| 4 SGDNLRKY<br>SAH (SEQ ID NO: 20)         | LVIYYDNKR<br>PS (SEQ ID NO: 27)  | QVYASVHR<br>M (SEQ ID NO: 7)   | GGTFSSYAI<br>H (SEQ ID NO: 34) | WMGRIIPDF<br>GTANYAQK<br>FQG (SEQ ID NO: 39)      | GPTQAIHYF<br>AY (SEQ ID NO: 16)              |
| 5 SGDSLGSK<br>SVH (SEQ ID NO: 21)         | LVIYRDSKR<br>PS (SEQ ID NO: 28)  | QTYDWSHF<br>GW (SEQ ID NO: 8)  | GFTFSSAA<br>MH (SEQ ID NO: 35) | WVSVISSDG<br>GSTYYADS<br>VKG (SEQ ID NO: 12)      | AGFILGSLG<br>VAWMDV<br>(SEQ ID NO: 17)       |
| 6 RASQGISGN<br>WLT (SEQ ID NO: 40)        | LLIYDASNL<br>QS (SEQ ID NO: 41)  | HQSYRGPM<br>(SEQ ID NO: 42)    | GFTFGRFV<br>MH (SEQ ID NO: 43) | WVSVIGYS<br>GSSTYYAD<br>SVKG (SEQ ID NO: 44)      | EPGYYYPS<br>GYYRGPGY<br>WMDV (SEQ ID NO: 45) |
| 7 SGDNIRKKY<br>VH (SEQ ID NO: 46)         | LVIYRDSNR<br>PS (SEQ ID NO: 47)  | SSAGRDNV<br>(SEQ ID NO: 48)    | GFTFSSSA<br>MH (SEQ ID NO: 49) | WVSVIHYS<br>SETYYADSV<br>KG (SEQ ID NO: 50)       | DRYFFDV<br>(SEQ ID NO: 51)                   |
| 8 SGDALRYII<br>AH (SEQ ID NO: 52)         | LVIYYNNNR<br>PS (SEQ ID NO: 53)  | QSYGPGSV<br>(SEQ ID NO: 54)    | GFTFSDYA<br>MH (SEQ ID NO: 55) | WVSSIPYSG<br>SNTYYADSV<br>KG (SEQ ID NO: 56)      | PKSYASGPF<br>AY (SEQ ID NO: 57)              |
| 9 SGSSNIGQ<br>NYVS (SEQ ID NO: 58)        | LLIYDNSKR<br>PS (SEQ ID NO: 59)  | SSWDLLSKS<br>R (SEQ ID NO: 60) | GGTFSTHAI<br>S (SEQ ID NO: 61) | WMGLIQPR<br>FGTANYAQ<br>KFQR (SEQ ID NO: 62)      | DYYGGMAY<br>(SEQ ID NO: 63)                  |
| 10 KASQNIYKY<br>LN (SEQ ID NO: 98)        | YNSNLQT<br>(SEQ ID NO: 99)       | FQYSSGPT<br>(SEQ ID NO: 100)   | GFTFSNYY<br>MY (SEQ ID NO: 95) | SINTDGGST<br>YYPDSVKG<br>(SEQ ID NO: 96)          | STPNIPLAY<br>(SEQ ID NO: 97)                 |
| 11 RANQGVSL<br>SRYNLMH<br>(SEQ ID NO: 85) | RSSNLAS<br>(SEQ ID NO: 86)       | QQSRESPFT<br>(SEQ ID NO: 87)   | GFTFSHYI<br>MA (SEQ ID NO: 82) | SITNSGGSI<br>NYRDSVKG<br>(SEQ ID NO: 83)          | EGGENYGG<br>YPPFAY<br>(SEQ ID NO: 84)        |

TABLE 3-continued

| ALK2 Antibody CDR sequences |                                        |                             |                                 |                                 |                                          |                                         |
|-----------------------------|----------------------------------------|-----------------------------|---------------------------------|---------------------------------|------------------------------------------|-----------------------------------------|
| Anti-                       | VL                                     |                             |                                 | VH                              |                                          |                                         |
| body                        | CDR1                                   | CDR2                        | CDR3                            | CDR1                            | CDR2                                     | CDR3                                    |
| 12                          | QASQDIGN<br>WLS (SEQ ID NO: 104)       | GATSLAD<br>(SEQ ID NO: 105) | LQAYSAPFT<br>(SEQ ID NO: 106)   | GFTFSYYA<br>MS (SEQ ID NO: 101) | SISRGDNT<br>YYRDTVKG<br>(SEQ ID NO: 102) | LNYYNNYFDY<br>(SEQ ID NO: 103)          |
| 13                          | LASSSVSYM<br>T (SEQ ID NO: 91)         | GTSNLAS<br>(SEQ ID NO: 92)  | LHLTSYPPY<br>T (SEQ ID NO: 93)  | GSTFSNYG<br>MK (SEQ ID NO: 88)  | SISRSSTYIY<br>YADTVKG<br>(SEQ ID NO: 89) | AISTPFYWY<br>FDF (SEQ ID NO: 90)        |
| 14                          | RANQGVSL<br>SRYNLMH<br>(SEQ ID NO: 85) | RSSNLAQ<br>(SEQ ID NO: 94)  | QQSRESPFT<br>(SEQ ID NO: 87)    | GFTFSHYIY<br>MA (SEQ ID NO: 82) | SITNSGGSI<br>NYRDSVKG<br>(SEQ ID NO: 83) | EGGENYGG<br>YPPFAY<br>(SEQ ID NO: 84)   |
| 15                          | LASQTIGTW<br>LA (SEQ ID NO: 180)       | AATSLAD<br>(SEQ ID NO: 181) | QQLYSTPW<br>T (SEQ ID NO: 182)  | GFNIKDSLM<br>H (SEQ ID NO: 177) | IDPEDGETK<br>YAPNFDQ<br>(SEQ ID NO: 178) | YTSDDYTM<br>DY (SEQ ID NO: 179)         |
| 16                          | LASQTIGTW<br>LA (SEQ ID NO: 186)       | AATSLAD<br>(SEQ ID NO: 187) | QQLYSTPW<br>T (SEQ ID NO: 188)  | GFNIKDSLM<br>H (SEQ ID NO: 183) | IDPEDGETK<br>YAPNFDQ<br>(SEQ ID NO: 184) | YTSDDYTM<br>DY (SEQ ID NO: 185)         |
| 17                          | LASQTIGTW<br>LA (SEQ ID NO: 192)       | AATSLAD<br>(SEQ ID NO: 193) | QQVYSTPW<br>T (SEQ ID NO: 194)  | GFNIKDSLM<br>H (SEQ ID NO: 189) | IDPEDGETK<br>YAPNFQS<br>(SEQ ID NO: 190) | YTSPPYTM<br>DY (SEQ ID NO: 191)         |
| 18                          | LASQTIGTW<br>LA (SEQ ID NO: 198)       | AATSLAD<br>(SEQ ID NO: 199) | QQLYWTPW<br>T (SEQ ID NO: 200)  | GFNIKDSLM<br>H (SEQ ID NO: 195) | IDPEDGETR<br>YAPNFDQ<br>(SEQ ID NO: 196) | YTSRRYTM<br>EY (SEQ ID NO: 197)         |
| 19                          | QNISSK<br>(SEQ ID NO: 220)             | GAS (SEQ ID NO: 221)        | LQYNNWWT<br>(SEQ ID NO: 222)    | GFTFSNYG<br>(SEQ ID NO: 217)    | ISYDGNK<br>(SEQ ID NO: 218)              | AKDRGLDT<br>AGDYFDY<br>(SEQ ID NO: 219) |
| 20                          | QDINIY<br>(SEQ ID NO: 226)             | DVS (SEQ ID NO: 227)        | QHYDDLPLT<br>(SEQ ID NO: 228)   | NGSFSSYY<br>(SEQ ID NO: 223)    | INHRGST<br>(SEQ ID NO: 224)              | TAHTRGFLY<br>(SEQ ID NO: 225)           |
| 21                          | QDINNY<br>(SEQ ID NO: 231)             | DVF (SEQ ID NO: 232)        | QHYDDLPLT<br>(SEQ ID NO: 228)   | NGSFSSFY<br>(SEQ ID NO: 229)    | INHRGRT<br>(SEQ ID NO: 230)              | TAHTRGFLY<br>(SEQ ID NO: 225)           |
| 22                          | QDINNY<br>(SEQ ID NO: 231)             | DAS (SEQ ID NO: 234)        | QHYDDLPLT<br>(SEQ ID NO: 235)   | NGSFSSYY<br>(SEQ ID NO: 223)    | INHRGRT<br>(SEQ ID NO: 230)              | SAHTRGFLY<br>(SEQ ID NO: 233)           |
| 23                          | QDINNY<br>(SEQ ID NO: 231)             | DAS (SEQ ID NO: 234)        | QHYDDLPLT<br>(SEQ ID NO: 235)   | NGSFSSYY<br>(SEQ ID NO: 223)    | INHRGST<br>(SEQ ID NO: 224)              | TAHTRGFLY<br>(SEQ ID NO: 225)           |
| 24                          | QSLVYSDGI<br>TY (SEQ ID NO: 239)       | QVS (SEQ ID NO: 240)        | MEGTHWPW<br>T (SEQ ID NO: 241)  | GFTFSDSA<br>(SEQ ID NO: 236)    | IRNKANTYA<br>T (SEQ ID NO: 237)          | TSDQFD<br>(SEQ ID NO: 238)              |
| 25                          | QSLVHSDG<br>DTY (SEQ ID NO: 245)       | TVS (SEQ ID NO: 246)        | MQGTQFPA<br>LT (SEQ ID NO: 247) | GFTFSDYY<br>(SEQ ID NO: 242)    | ISRTGSTK<br>(SEQ ID NO: 243)             | AREPLQH<br>(SEQ ID NO: 244)             |
| 26                          | QSLVHSDG<br>DTY (SEQ ID NO: 245)       | TIS (SEQ ID NO: 248)        | MQGTQFPA<br>LT (SEQ ID NO: 247) | GFTFSDYY<br>(SEQ ID NO: 242)    | ISRTGSTK<br>(SEQ ID NO: 243)             | AREPLQH<br>(SEQ ID NO: 244)             |

TABLE 3-continued

| ALK2 Antibody CDR sequences     |                      |                               |                              |                               |                               |
|---------------------------------|----------------------|-------------------------------|------------------------------|-------------------------------|-------------------------------|
| Anti-                           | VL                   |                               |                              | VH                            |                               |
| body CDR1                       | CDR2                 | CDR3                          | CDR1                         | CDR2                          | CDR3                          |
| 27 QSISTW<br>(SEQ ID NO: 252)   | KAS (SEQ ID NO: 253) | QQYNTYWT<br>(SEQ ID NO: 254)  | GFTLSDDS<br>(SEQ ID NO: 249) | IRSKPYSYA<br>(SEQ ID NO: 250) | TGGDDF<br>(SEQ ID NO: 251)    |
| 28 QSVSSN<br>(SEQ ID NO: 257)   | GAS (SEQ ID NO: 221) | QQYNNWPP<br>(SEQ ID NO: 258)  | GFTFSSYA<br>(SEQ ID NO: 255) | ISGSGGST<br>(SEQ ID NO: 256)  | AIPYNWNY<br>(SEQ ID NO: 382)  |
| 29 QGINNY<br>(SEQ ID NO: 262)   | AAS (SEQ ID NO: 263) | QKFNSAPLT<br>(SEQ ID NO: 264) | GYTFTGYY<br>(SEQ ID NO: 259) | INPNSVGT<br>(SEQ ID NO: 260)  | TRDGAAAG<br>(SEQ ID NO: 261)  |
| 30 QSISSY<br>(SEQ ID NO: 267)   | AAS (SEQ ID NO: 263) | QQSYSTPPI<br>(SEQ ID NO: 268) | GFTFSTYA<br>(SEQ ID NO: 265) | ISGSGGST<br>(SEQ ID NO: 256)  | ANSPSWFD<br>(SEQ ID NO: 266)  |
| 31 QSVLYSSN<br>(SEQ ID NO: 272) | WAS (SEQ ID NO: 273) | QQYYSTPLT<br>(SEQ ID NO: 274) | GFTFDDYG<br>(SEQ ID NO: 269) | ISWNSATI<br>(SEQ ID NO: 270)  | AKKNSLGW<br>(SEQ ID NO: 271)  |
| 32 QGIRND<br>(SEQ ID NO: 278)   | AAS (SEQ ID NO: 263) | LQHNSYPW<br>(SEQ ID NO: 279)  | GYTFTSYG<br>(SEQ ID NO: 275) | ISAYNGNT<br>(SEQ ID NO: 276)  | ARDINWYFD<br>(SEQ ID NO: 277) |
| 33 QGIRND<br>(SEQ ID NO: 278)   | TAS (SEQ ID NO: 281) | LOHNNYPYT<br>(SEQ ID NO: 282) | GFTFDDYG<br>(SEQ ID NO: 269) | ISWNGATI<br>(SEQ ID NO: 280)  | AKKNSLGW<br>(SEQ ID NO: 271)  |
| 34 QSVISY<br>(SEQ ID NO: 286)   | GAS (SEQ ID NO: 221) | QQNYLPPLT<br>(SEQ ID NO: 287) | GFTFSNYA<br>(SEQ ID NO: 283) | ITGNGVNT<br>(SEQ ID NO: 284)  | VKERGHSW<br>(SEQ ID NO: 285)  |
| 35 QGLVNSDG<br>(SEQ ID NO: 289) | EIS (SEQ ID NO: 290) | MQSTQFPLT<br>(SEQ ID NO: 291) | GFTFSNYA<br>(SEQ ID NO: 283) | ITGNGIKT<br>(SEQ ID NO: 288)  | VKERGHSW<br>(SEQ ID NO: 285)  |
| 36 QGIRND<br>(SEQ ID NO: 278)   | AAS (SEQ ID NO: 263) | LQHNSYPW<br>(SEQ ID NO: 279)  | GFTFSNYA<br>(SEQ ID NO: 283) | ITGNGINT<br>(SEQ ID NO: 292)  | VKERGHSW<br>(SEQ ID NO: 285)  |
| 37 QGISSH<br>(SEQ ID NO: 295)   | VAS (SEQ ID NO: 296) | QQYYSPPT<br>(SEQ ID NO: 297)  | GFTFSNYG<br>(SEQ ID NO: 217) | VSYDGSNK<br>(SEQ ID NO: 293)  | AKDRGITGT<br>(SEQ ID NO: 294) |
| 38 QSLVYSDG<br>(SEQ ID NO: 301) | KVS (SEQ ID NO: 302) | MQGTHWPP<br>(SEQ ID NO: 303)  | GFTSSRYW<br>(SEQ ID NO: 298) | INQDGSEK<br>(SEQ ID NO: 299)  | ARDEEGY<br>(SEQ ID NO: 300)   |

[0086] In some embodiments, the ALK2 antibody administered to the subject is an ALK2 antibody or an antigen-binding fragment thereof, having a variable light chain (VL) sequence and a variable heavy chain (VH) sequence of an antibody described in Table 4 (in which the CDR sequences are in bold), or an ALK2 antibody or an antigen-binding fragment thereof, having a VL and VH sequence having at least 90% sequence identity (e.g., 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or more sequence identity)

to a VH and VL sequence of an antibody listed in Table 4. In some embodiments, the ALK2 antibody or an antigen-binding fragment thereof, apart from the light chain CDR1, CDR2, and CDR3 and the heavy chain CDR1, CDR2, and CDR3, has a VL and VH sequence having at least 90% sequence identity (e.g., 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or more sequence identity) to a VH and VL sequence of an antibody listed in Table 4.



TABLE 4

| ALK2 antibody sequences |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SEQ ID NO:              | Antibody    | Sequence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 67                      | Antibody 1D | DIVLTQPPSVSGAPGQRVTISCSGSSSNIGSNYVSWYQQLPGTAPKVLIIY<br>VL: <b>KNNHRPSG</b> VPDRFSGSKSGTSASLAITGLQAEDEADYY <b>CASWDHSDRFYV</b><br>Positions 1-112 AWKADSSPVKAGVETTTPSKQSNNKYAASSYLSLTPEQWKS <hr/> SHRSYSCQVT<br>VH: HEGSTVEKTVAPTEAQVQLVQSGAEVKKPGSSVKVSCKAS <b>GGTFSSYGV</b> S<br>Positions 216-331 WVRQAPGGGLE <b>WMGGIIPHFGIANYAQKFQ</b> GRVTITADESTSTAYMELSS<br>LRS EDTAVYYCARE <b>IGSLDI</b> WGQGTTLVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSV<br>TVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSEFDYKDDDDKGAPHHHH                                                                                    |
| 68                      | Antibody 2D | IELTQPPSVSVSPGQTASITCSGDSIP <b>SFFAS</b> WYQOKPGQAPVL <b>VIYRD</b><br>VL: <b>SNRPSG</b> IPERFSGSNSGNTATLTISGTQAEDEADYY <b>CYVTA</b> PWKS <b>IW</b> VFG<br>Positions 1-110 GGTKLTVLGQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVTVAW<br>KADSSPVKAGVETTTPSKQSNNKYAASSYLSLTPEQWKS <hr/> SHRSYSCQVTHE<br>VH: GSTVEKTVAPTEAEVQLVESGGGLVQPGGSLRLSCAAS <b>GFTFSS</b> AMSWV<br>Positions 214-332 RQAPGKGLE <b>WVGRIKSKADSGTTDYAAPV</b> KGRFTISRDDSKNTLYLQMNLS<br>LKTEDTAVYYCARD <b>YGVAFAY</b> WGQGTTLVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSV<br>VTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSEFDYKDDDDKGAPHHHH     |
| 69                      | Antibody 3D | IELTQPPSVSVSPGQTASITCSGDNIG <b>TKYAY</b> WYQOKPGQAPVL <b>VIYGD</b><br>VL: <b>SDRPSG</b> IPERFSGSNSGNTATLTISGTQAEDEADYY <b>CYSA</b> DAQ <b>QMK</b> AVFG<br>Positions 1-110 GGTKLTVLGQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVTVAW<br>KADSSPVKAGVETTTPSKQSNNKYAASSYLSLTPEQWKS <hr/> SHRSYSCQVTHE<br>VH: GSTVEKTVAPTEAEVQLVESGGGLVQPGGSLRLSCAAS <b>GFTFNS</b> AMSWV<br>Positions 214-333 RQAPGKGLE <b>WVGRIKSKRDGYTTDYAAPV</b> KGRFTISRDDSKNTLYLQMNLS<br>LKTEDTAVYYCARD <b>YGGKFDY</b> WGQGTTLVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSV<br>VTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSEFDYKDDDDKGAPHHHH     |
| 70                      | Antibody 4D | IELTQPPSVSVSPGQTASITCSGDNLR <b>KYSAH</b> WYQOKPGQAPVL <b>VIYYD</b><br>VL: <b>NKRPSG</b> IPERFSGSNSGNTATLTISGTQAEDEADYY <b>CQVYA</b> SVHRMVFVG<br>Positions 1-109 GTKLTVLGQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVTVAWK<br>ADSSPVKAGVETTTPSKQSNNKYAASSYLSLTPEQWKS <hr/> SHRSYSCQVTHEG<br>VH: STVEKTVAPTEAQVQLVQSGAEVKKPGSSVKVSCKAS <b>GGTFSSYA</b> IHWVR<br>Positions 213-332 QAPGGGLE <b>WMGRIIPDFTANYAQKFQ</b> GRVTITADESTSTAYMELSSLR<br>EDTAVYYCARG <b>PTQAIHYFAY</b> WGQGTTLVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSV<br>VTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSEFDYKDDDDKGAPHHHH                |
| 71                      | Antibody 5D | IELTQPPSVSVSPGQTASITCSGDSLGS <b>SVH</b> WYQOKPGQAPVL <b>VIYRD</b><br>VL: <b>SKRPSG</b> IPERFSGSNSGNTATLTISGTQAEDEADYY <b>CQTYD</b> WSHF <b>GW</b> VFG<br>Positions 1-110 GGTKLTVLGQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVTVAW<br>KADSSPVKAGVETTTPSKQSNNKYAASSYLSLTPEQWKS <hr/> SHRSYSCQVTHE<br>VH: GSTVEKTVAPTEAEVQLLESGGGLVQPGGSLRLSCAAS <b>GFTFSSA</b> AMHWV<br>Positions 214-337 RQAPGKGLE <b>WVSVISSDGGSTYYADSV</b> KGRFTISRDNKNTLYLQMNLSL<br>AEDTAVYYCAR <b>AGFILGSLGV</b> AWMDVWGQGTTLVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSV<br>VTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSEFDYKDDDDKGAPHHHHH |
| 72                      | Antibody 6D | IQMTQSPSSLSASVGDRTITCRAS <b>QGISGN</b> WLTWYQOKPGKAP <b>KLIIY</b><br>VL: <b>DASNLQ</b> SGVPSRFSGSGSGTDFTLTISSLQPEDFATYY <b>CHQSYR</b> GPMTFG<br>Positions 1-110 QGKVEIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWK<br>VDNALQSGNSQESVTEQDSKDYSLSSLTLSKADYEKHKVYACEVTHQ<br>VH: GLSGPVTKSFNRGAEVQLLESGGGLVQPGGSLRLSCAAS <b>GFTFGR</b> FVMH<br>Positions 216-344 WVRQAPGKGLE <b>WVSVIGYSGSSSTYYADSV</b> KGRFTISRDNKNTLYLQMNLS<br>LR AEDTAVYYCARE <b>PGYYPSGYIRGPGY</b> WMDVWGQGTTLVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSV<br>VTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSEFDYKDDDDKGAPHHHHH              |
| 73                      | Antibody 7D | IELTQPPSVSVSPGQTASITCSGDNIR <b>KKYVH</b> WYQOKPGQAPVL <b>VIYRD</b><br>VL: <b>SNRPSG</b> IPERFSGSNSGNTATLTISGTQAEDEADYY <b>CSSA</b> GRDNYVFGG<br>Positions 1-108 TKLTVLGQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVTVAWKA<br>DSSPVKAGVETTTPSKQSNNKYAASSYLSLTPEQWKS <hr/> SHRSYSCQVTHEGS<br>VH: TVEKTVAPTEAEVQLLESGGGLVQPGGSLRLSCAAS <b>GFTFSSA</b> AMHWVRQ<br>Positions APGKGLE <b>WVSVIHYDSSETYYADSV</b> KGRFTISRDNKNTLYLQMNLSRAE                                                                                                                                                                                                |

TABLE 4-continued

| ALK2 antibody sequences |           |                                                                                                                                                                       |
|-------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SEQ ID NO:              | Antibody  | Sequence                                                                                                                                                              |
| 212-327                 |           | DTAVYYCARDRYFFDVWGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGTA<br>ALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPS<br>SSLTGTQTYICNVNHKPSNTKVDKKVEPKSEFDYKDDDDKGAPHHHHH           |
| 74                      | Antibody  | 8DIELTQPPSVSVSPGQTASITCSGDALRYIIAHWYQQKPGQAPVLIYYN                                                                                                                    |
|                         | VL:       | NNRPSGIPERFSGSNSGNTATLTISGTQAEDEADYYCQSYGPGSVVFGGG                                                                                                                    |
|                         | Positions | TKLTVLGQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVTVAWKA                                                                                                                    |
|                         | 1-108     | DSSPVKAGVETTTTPSKQSNKYAASSYLSLTPEQWKSHRSYSCQVTHEGS                                                                                                                    |
|                         | VH:       | TVEKTVAPTEAEVQLLESGGGLVQPGGSLRLSCAASGFTFSDYAMHWVRQ                                                                                                                    |
|                         | Positions | APGKGLEWVSSIFYSGSNTYYADSVKGRFTISRDNKNTLYLQMNSLRAE                                                                                                                     |
|                         | 212-331   | DTAVYYCARPKSYASGPFAYWGQGLTVTVSSASTKGPSVFPLAPSSKSTS<br>GGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSV<br>TVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSEFDYKDDDDKGAPHHH<br>HH    |
| 75                      | Antibody  | 9DIVLTQPPSVSGAPGQRTVITCSGSSSNIGQNYVSWYQQLPGTAPKLLIY                                                                                                                   |
|                         | VL:       | DNSKRPSGVDRFSGSKSGTSASLAITGLQAEDEADYYCSDWLLSKSRV                                                                                                                      |
|                         | Positions | FGGGLTKLTVLGQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVTV                                                                                                                   |
|                         | 1-112     | AWKADSSPVKAGVETTTTPSKQSNKYAASSYLSLTPEQWKSHRSYSCQVT                                                                                                                    |
|                         | VH:       | HEGSTVEKTVAPTEAQVQLVQSGAEVKKPGSSVKVSCKASGGTFSTHAIS                                                                                                                    |
|                         | Positions | WVRQAPGQGLEWMGLIQPRFGTANYAQKFQGRVTITADESTSTAYMELSS                                                                                                                    |
|                         | 216-332   | LRSEDTAVYYCARDYYGGMAYWGQGLTVTVSSASTKGPSVFPLAPSSKST<br>SGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSV<br>VTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSEFDYKDDDDKGAPHHH<br>HHH |

[0087] In some embodiments, the ALK2 antibody administered to the subject is an ALK2 antibody or an antigen-binding fragment thereof, having a VL and VH sequence having at least 90% sequence identity (e.g., 90%, 91%, 92%, 93%, 94%, 95%, 95%, 97%, 98%, 99% or 100% sequence identity) to a VH and VL sequence provided in Table 5. In some embodiments, the ALK2 antibody or an antigen binding fragment thereof, has a VL and VH sequence from the same row of Table 5. In some embodiments, the VH and VL have the sequence of SEQ ID NOs: 107 and 108; 109 and 110; 109 and 115; 111 and 112; 113 and 114; 116 and 118; 116 and 119; 117 and 118; 131 and 118; 117 and 119; 131 and 119; 117 and 115; 131 and 115; 117 and 120; 131 and 120; 121 and 115; 122 and 126; 132 and 126; 122 and 127; 132 and 127; 122 and 128; 132 and 128; 123 and 126; 133 and 126; 123 and 127; 133 and 127; 123 and 128; 133 and 128; 124 and 126; 124 and 127; 124 and 128; 124 and 129; 125 and 128; 125 and 129; 125 and 130; 134 and 129; 135 and 127; 139 and 129; 140 and 136; 140 and 137; 140 and 138; 201 and 202; 203 and 204; 205 and 206; 207 and 208; 304 and 305; 306 and 307; 308 and 309; 310 and 311; 306 and 312; 313 and 314; 315 and 316; 317 and 318; 319 and 320; 321 and 322; 323 and 324; 325 and 326; 327 and 328; 329 and 330; 331 and 332; 333 and 334; 335 and 336; 337

and 338; 339 and 340; or 341 and 342 (e.g., the VH has at least 90% sequence identity (e.g., 90%, 91%, 92%, 93%, 94%, 95%, 95%, 97%, 98%, 99% or 100% sequence identity) to the sequence of the first SEQ ID NO: in each pair and the VL has at least 90% sequence identity (e.g., 90%, 91%, 92%, 93%, 94%, 95%, 95%, 97%, 98%, 99% or 100% sequence identity) to the sequence of the second SEQ ID NO: in each pair). In some embodiments, the VH has the sequence of any one of SEQ ID NOs: 116, 117, 121-125, 131-135, 139, and 140 and the VL has the sequence of SEQ ID NO: 136. In some embodiments, the VH has the sequence of any one of SEQ ID NOs: 116, 117, 121-125, 131-135, 139, and 140 and the VL has the sequence of SEQ ID NO: 137. In some embodiments, the VH has the sequence of any one of SEQ ID NOs: 116, 117, 121-125, 131-135, 139, and 140 and the VL has the sequence of SEQ ID NO: 138. In some embodiments, the VH has the sequence of any one of SEQ ID NOs: 116, 117, 121-125, 131-135, 139, and 140 and the VL has the sequence of SEQ ID NO: 141. In some embodiments, the VH has the sequence of any one of SEQ ID NOs: 116, 117, 121-125, 131-135, 139, and 140 and the VL has the sequence of SEQ ID NO: 142. In some embodiments, the VH has the sequence of any one of SEQ ID NOs: 116, 117, 121-125, 131-135, 139, and 140 and the VL has the sequence of SEQ ID NO: 143.

TABLE 5

| Variable heavy and light chain sequences of exemplary ALK2 antibodies |                                      |
|-----------------------------------------------------------------------|--------------------------------------|
| Heavy chain variable region sequence                                  | Light chain variable region sequence |
| EVQLVETGGGLVQPGRSKLKSCVASGFTFSNYIM                                    | DIQMTQSPSLLSASVGDRTLSCKASQNIYKY      |
| YWIRQAPGKGLEWISSINTDGGSTYYPDSVKGRF                                    | LNWFQQLGEAPKLLIYYSNLSLTGTGIPSRFSG    |
| ATSRDNAENTVYLQMNSLRSEDTATYYCAKSTPN                                    | SGSGTDFTLTISLQPEDVAIYFCFQYSSGPT      |
| IPLAYWGQGLTVTVSS (SEQ ID NO: 107)                                     | FGPGTKLELKRA (SEQ ID NO: 108)        |

TABLE 5-continued

| Variable heavy and light chain sequences of<br>exemplary ALK2 antibodies                                                                                  |                                                                                                                                                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Heavy chain variable<br>region sequence                                                                                                                   | Light chain variable<br>region sequence                                                                                                          |
| EVQLVESGGGLVQPGRSLKLSCAASGFTFSHYIM<br>AWVRQAPTQKLEWVASITNSGGSINYRDSVKGRF<br>TISRDNASTLYLQMDSLRSEDATYYCTREGGE<br>NYGGYPPFAYWGQGLTVTVSS (SEQ ID NO:<br>109) | DIVLTQSPALAVSLGQRATISCRANQGVSLSR<br>YNLMHWYQQKPGQKPKLLIYRSSNLASGIPAR<br>FSGSGSGTDFTLTINPVQADDIATYYCQQSRE<br>SPFTFGAGTKLELKRA (SEQ ID NO:<br>110) |
| EVQLVESGGGLVQPGRSLKLSCAASGFTFSHYIM<br>AWVRQAPTQKLEWVASITNSGGSINYRDSVKGRF<br>TISRDNASTLYLQMDSLRSEDATYYCTREGGE<br>NYGGYPPFAYWGQGLTVTVSS (SEQ ID NO:<br>109) | DIVLTQSPDSLAVSLGERATINCRANQGVSLS<br>RYNLMHWYQQKPGQKPKLLIYRSSNLAQGIPA<br>RFGSGSGTDFTLTISVQADDIATYYCQQSR<br>ESPFTFGQGTKLELKRA (SEQ ID NO:<br>115)  |
| EVQLVESGGGLVQPGRSLKLSCAASGFTFSHYIM<br>SWVCQAPTQKLEWVASISRGDNTYYRDTVKGRF<br>TISRDNASTLYLQMDSLRSEDATYYCARLNYN<br>NYFDYWGQGVMTVTVSS (SEQ ID NO: 111)         | DIQMTQSPASLSASLEEIVTITCQASQDIGNW<br>LSWYQQKPGKSPQLLIYGATSLADGVPSRFGS<br>SRSGTQYSLKISRRLVEDIGIYYCLQAYSAPP<br>TFGSGTKLEIKRA (SEQ ID NO: 112)       |
| EVQLVESGGGLVQPGRSLKLSCLASGTFSTNYGM<br>KWIRQAPGKLEWVASISRSSTIYYADTVKGRF<br>TISRDNASTLYLQMDSLRSEDATYYCAAIST<br>PFYWFDFWGQGLTVTVSS (SEQ ID NO:<br>113)       | EIVLTQSPPTMAASPGKVTNLNCLASSSVSYM<br>TWYQQKSGASPKLWIYGTSNLASGVPNRFGS<br>GSGTSYSLAISMEAEVDATYYCLHLTSYPPY<br>TFGAGTKLELKRA (SEQ ID NO: 114)         |
| EVQLVESGGGLVQPGRSLKLSCAASGFTFSHYIM<br>AWVRQAPGKLEWVASITNSGGSINYRDSVKGRF<br>TISRDNASTLYLQMDSLRSEDATYYCAREGGE<br>NYGGYPPFAYWGQGLTVTVSS (SEQ ID NO:<br>116)  | DIVMTQSPDSLAVSLGERATINCRANQGVSLS<br>RYNLMHWYQQKPGQKPKLLIYRSSNLASGVDP<br>RFGSGSGTDFTLTISLQAEVDATYYCQQSR<br>ESPFTFGQGTKVEIKRA (SEQ ID NO:<br>118)  |
| EVQLVESGGGLVQPGRSLKLSCAASGFTFSHYIM<br>AWVRQAPGKLEWVASITNSGGSINYRDSVKGRF<br>TISRDNASTLYLQMDSLRSEDATYYCTREGGE<br>NYGGYPPFAYWGQGLTVTVSS (SEQ ID NO:<br>117)  | DIVLTQSPDSLAVSLGERATINCRANQGVSLS<br>RYNLMHWYQQKPGQKPKLLIYRSSNLASGIPA<br>RFGSGSGTDFTLTISVQADDIATYYCQQSR<br>ESPFTFGQGTKLELKRA (SEQ ID NO:<br>119)  |
| EVQLVESGGGLVQPGRSLKLSCAASGFTFSHYIM<br>AWVRQAPGKLEWVASITNSGGSINYRDSVKGRF<br>TISRDNASTLYLQMDSLRSEDATYYCTREGGE<br>NYGGYPPFAYWGQGLTVTVSS (SEQ ID NO:<br>131)  | DIVLTQSPDSLAVSLGERATINCRANQGVSLS<br>RYNLMHWYQQKPGQKPKLAIYRSSNLASGIPA<br>RFGSGSGTDFTLTISVQADDIATYYCQQSR<br>ESPFTFGQGTKLELKRA (SEQ ID NO:<br>120)  |
| EVQLVESGGGLVQPGRSLKLSCAASGTFSTNYGM<br>KWVRQAPGKLEWVASISRSSTIYYADTVKGRF<br>TISRDNASTLYLQMDSLRSEDATYYCARAIST<br>PFYWFDFWGQGLTVTVSS (SEQ ID NO:<br>121)      | EIVLTQSPGTLSSLSPGERATLSCLASSSVSYM<br>TWYQQKPGQAPRLIYGTSNLASGIPDRFGS<br>GSGTDFTLTISRLEPEDFATYYCLHLTSYPPY<br>TFGQGTKVEIKRA (SEQ ID NO: 126)        |
| EVQLVESGGGLVQPGRSLKLSCAASGTFSTNYGM<br>KWIRQAPGKLEWVASISRSSTIYYADTVKGRF<br>TISRDNASTLYLQMDSLRSEDATYYCAAIST<br>PFYWFDFWGQGLTVTVSS (SEQ ID NO:<br>122)       | EIVLTQSPGTLSSLSPGERATLSCLASSSVSYM<br>TWYQQKPGQSPRLWIYGTSNLASGVDPDRFGS<br>GSGTDFTLTISRLEPEDFATYYCLHLTSYPPY<br>TFGQGTKVEIKRA (SEQ ID NO: 127)      |
| EVQLVESGGGLVQPGRSLKLSCAASGTFSTNYGM<br>KWIRQAPGKLEWVASISRSSTIYYADTVKGRF<br>TISRDNASTLYLQMDSLRSEDATYYCAAIST<br>PFYWFDFWGQGLTVTVSS (SEQ ID NO:<br>132)       | EIVLTQSPGTLSSLSPGERATLSCLASSSVSYM<br>TWYQQKPGASPRLWIYGTSNLASGVDPDRFGS<br>GSGTDYTLTISRLEPEDFATYYCLHLTSYPPY<br>TFGAGTKVEIKRA (SEQ ID NO: 128)      |
| EVQLVESGGGLVQPGRSLKLSCAASGTFSTNYGM<br>KWIRQAPGKLEWVASISRSSTIYYADTVKGRF<br>TISRDNASTLYLQMDSLRSEDATYYCAAIST<br>PFYWFDFWGQGLTVTVSS (SEQ ID NO:<br>123)       | EIVLTQSPGTMSASPGERVTLNCLASSSVSYM<br>TWYQQKPGASPRLWIYGTSNLASGVDPDRFGS<br>GSGTDYTLTISRMEPEDFATYYCLHLTSYPPY<br>TFGAGTKLELKRA (SEQ ID NO: 129)       |
| EVQLVESGGGLVQPGRSLKLSCAASGTFSTNYGM<br>KWIRQAPGKLEWVASISRSSTIYYADTVKGRF<br>TISRDNASTLYLQMDSLRSEDATYYCAAIST<br>PFYWFDFWGQGLTVTVSS (SEQ ID NO:<br>133)       | EIVLTQSPGTMSASPGERVTLNCLASSSVSYM<br>TWYQQKSGASPKLWIYGTSNLASGVPNRFGS<br>GSGTSYTLTISRMEPEDFATYYCLHLTSYPPY<br>TFGAGTKLELKRA (SEQ ID NO: 130)        |

TABLE 5-continued

| Variable heavy and light chain sequences of<br>exemplary ALK2 antibodies                                                                                 |                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Heavy chain variable<br>region sequence                                                                                                                  | Light chain variable<br>region sequence                                                                                                  |
| EVQLVESGGGLVQPGGSLRLSCAASGSTFSNYGM<br>KWIRQAPGKGLEWVASISRSTYIYADTVKGRF<br>TISRDNKNTLYLQMNSLRSEDALYYCAAIST<br>PFYWYDFWGPGLTVTVSS (SEQ ID NO:<br>124)      | DIQMTQSPSSLSASVGDRTITCKASQNIYKY<br>LNWFQQKPGKAPKLLIYNSNSLQTGVPSRFSG<br>SGSGTDFTLTISLQPEDFATYYCFQYSSGPT<br>FGQGTKVEIKRT (SEQ ID NO: 136)  |
| EVQLVESGGGLVQPGGSLRLSCAASGSTFSNYGM<br>KWIRQAPGKGLEWVASISRSTYIYADTVKGRF<br>TISRDNKNTLYLQMNSLRSEDALYYCAAIST<br>PFYWYDFWGPGLTVTVSS (SEQ ID NO:<br>125)      | DIQMTQSPSSLSASVGDRTITCKASQNIYKY<br>LNWFQQKPGKAPKLLIYNSNSLQTGVPSRFSG<br>SGSGTDFTLTISLQPEDFATYYCFQYSSGPT<br>FGQGTKVEIKRT (SEQ ID NO: 137)  |
| EVQLVESGGGLVQPGGSLRLSCAASGFTFSNYGM<br>YWIRQAPGKGLEWVSSINTDGGSTYYADSVKGRF<br>TISRDNKNTLYLQMNSLRSEDALYYCAKSTPN<br>IPLAYWGQGLTVTVSS (SEQ ID NO: 134)        | DIQMTQSPSSLSASVGDRTITCKASQNIYKY<br>LNWFQQKPGKAPKLLIYNSNSLQTGVPSRFSG<br>SGSGTDFTLTISLQPEDFATYYCFQYSSGPT<br>FGPGTKVEIKRT (SEQ ID NO: 138)  |
| EVQLVESGGGLVQPGGSLRLSCAASGFTFSNYGM<br>YWIRQAPGKGLEWVSSINTDGGSTYYADSVKGRF<br>TISRDNKNTLYLQMNSLRSEDALYYCAKSTPN<br>IPLAYWGQGLTVTVSS (SEQ ID NO: 135)        | DIQMTQSPSSLSASVGDRTITCKASQDIGNW<br>LSWYQQKPGKAPKLLIYGATSLADGVPSRFSG<br>SGSGTDFTLTISLQPEDFATYYCLQAYSAPP<br>TFGQGTKVEIKRT (SEQ ID NO: 141) |
| EVQLVESGGGLVQPGGSLRLSCAASGFTFSYYAM<br>SWVRQAPGKGLEWVASISRGGDNTYYRDSVKGRF<br>TISRDNKNTLYLQMNSLRSEDALYYCARLNYN<br>NYFDYWGQGLTVTVSS (SEQ ID NO: 139)        | DIQMTQSPSSLSASVGDRTITCKASQDIGNW<br>LSWYQQKPGKAPKLLIYGATSLADGVPSRFSG<br>SRSGTDYTLTISLQPEDFATYYCLQAYSAPP<br>TFGQGTKVEIKRT (SEQ ID NO: 142) |
| EVQLVESGGGLVQPGGSLRLSCAASGFTFSYYAM<br>SWVRQAPGKGLEWVASISRGGDNTYYRDSVKGRF<br>TISRDNKNTLYLQMNSLRSEDALYYCARLNYN<br>NYFDYWGQGLTVTVSS (SEQ ID NO: 140)        | DIQMTQSPSSLSASVGDRTITCKASQDIGNW<br>LSWYQQKPGKAPKLLIYGATSLADGVPSRFSG<br>SRSGTYTLTISLQPEDFATYYCLQAYSAPP<br>TFGSGTKVEIKRT (SEQ ID NO: 143)  |
| EVQLQQSGAELVRPGASVRLSCTASGFNIKDSL<br>HWVKRPEQGLEWIGWIDPEDGETKYAPNFQDKA<br>TITAVTSSNTAYLQLSSLTSEDSAIYYCARYTSD<br>YYTMDYWGQGLTVTVSS (SEQ ID NO: 201)       | DIQMTQSPASQSASLGESVTFTCLASQTIGTW<br>LAWYQQKPGKSPQLLIYAATSLADGVPSRFSG<br>SGSGTKFSPKISLQAEDEFAFYCCQLYSTPW<br>TFGGGTKLEIK (SEQ ID NO: 202)  |
| EVQLQQSGAELVRPGASVRLSCTASGFNIKDSL<br>HWVKRPEQGLEWIGWIDPEDGETKYAPNFQDKA<br>TITAVTSSNTAYLQLSSLTSEDSAIYYCARYTSD<br>YYTMDYWGQGLTVTVSS (SEQ ID NO: 203)       | DIQMTQSPASQSASLGESVTFTCLASQTIGTW<br>LAWYQQKPGKSPQLLIYAATSLADGVPSRFSG<br>SGSGTKFSPKISLQAEDEFAFYCCQLYSTPW<br>TFGGGTKLEIK (SEQ ID NO: 204)  |
| EVQLQQSGAELVRPGASVRLSCTASGFNIKDSL<br>HWVKRPEQGLEWIGWIDPEDGETKYAPNFQSKA<br>TITAVTSSNTAYLQLSSLTSEDSAIYYCARYTSP<br>YYTMDYWGQGLTVTVSS (SEQ ID NO: 205)       | DIQMTQSPASQSASLGESVTFTCLASQTIGTW<br>LAWYQQKPGKSPQLLIYAATSLADGVPSRFSG<br>SGSGTKFSPKISLQAEDEFAFYCCQVYSTPW<br>TFGGGTKLEIK (SEQ ID NO: 206)  |
| EVQLQQSGAELVRPGASVRLSCTASGFNIKDSL<br>HWVKRPEQGLEWIGWIDPEDGETRYAPNFQDKA<br>TITAVTSSNTAYLQLSSLTSEDSAIYYCARYTSR<br>YYTMEYWGQGLTVTVSS (SEQ ID NO: 207)       | DIQMTQSPASQSASLGESVTFTCLASQTIGTW<br>LAWYQQKPGKSPQLLIYAATSLADGVPSRFSG<br>SGSGTKFSPKISLQAEDEFAFYCCQLYSTPW<br>TFGGGTKLEIK (SEQ ID NO: 208)  |
| QVHLVESGGGVVQPGSLRLSCAASGFTFSNYGI<br>HWVRQVPGKGLEWVAVISYDGNKKYYGDSVKGRF<br>TISRDNKNTLYLQMNSLRPEDTAIYYCAKDRGL<br>DTAGDYFDYWGQGLTVTVSS (SEQ ID NO:<br>304) | EIVMTQSPATLSVSTGERATLSRASQNISSK<br>LAWYQQKPGKAPKLLIYDVSNLVPGVPSRFSG<br>SGSGTEFTLTISLQSEDFAVFYCLQYNNWWT<br>FGQGTKVEIK (SEQ ID NO: 305)    |
| QVQLQQWGAGLLKPSETLSLTCVYNGSFSSYYW<br>NWIRQPPGKGLEWLGELINHRGNTNYPNLSLTSRV<br>ISVDTSKNQFSLKLTSTVTAADTAIYYCTAHTRGF<br>LYWGQGLTVTVSS (SEQ ID NO: 306)        | DIQMTQSPSSLSASVGDRTITCKASQDINIY<br>LNWYQQQPGKAPKLLIYDVSNLVPGVPSRFSG<br>TSGSTHFSFTISLQPEDFATYYCQHYDDLPL<br>TFGGGTTVAIR (SEQ ID NO: 307)   |
| QVQLQQWGAGLLKPSETLSLTCVYNGSFSSYYW<br>SWIRQPPGKGLEWLGELINHRGNTNYPNLSLTSRV<br>ISVDTSKNQFSLKLTSTVTAADTAIYYCTAHTRGF<br>LYWGQGLTVTVSS (SEQ ID NO: 308)        | DIQMTQSPSSLSASVGDRTITCKASQDINIY<br>LNWYQQKPGKAPKLLIYDVSNLVPGVPSRFSG<br>SGSGTDFSTISLQPEDFATYYCQHYDDLPL<br>TFGGGTTVAIR (SEQ ID NO: 309)    |
| QVQLQQWGAGLLKPSETLSLTCVYNGSFSSYYW<br>SWIRQPPGKGLEWLGELINHRGNTNYPNLSLTSRV<br>ISVDTSKNQFSLKLTSTVTAADTAIYYCTAHTRGF<br>LYWGQGLTVTVSS (SEQ ID NO: 309)        | DIQMTQSPSSLSASVGDRTITCKASQDINIY<br>LNWYQQKPGKAPKLLIYDVSNLVPGVPSRFSG<br>SGSGTDFSTISLQPEDFATYYCQHYDDLPL<br>TFGGGTTVAIR (SEQ ID NO: 309)    |

TABLE 5-continued

| Variable heavy and light chain sequences of<br>exemplary ALK2 antibodies                                                                                 |                                                                                                                                                   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Heavy chain variable<br>region sequence                                                                                                                  | Light chain variable<br>region sequence                                                                                                           |
| ISVDTSKNQFSLKLPSVTAADTAIYYCSAHTRGF<br>LYWGQGTLLTVTSS (SEQ ID NO: 310)                                                                                    | TGSGTDFSFSTISSLPEDIATYYCQHYIDLPL<br>TFGGGTVAIS (SEQ ID NO: 311)                                                                                   |
| QVQLQQWAGALLKPSETLSLTCAVYNGSFSSYYW<br>NWIRQAPGKGLEWLG EINHGRGSTNYNPSLTSRVT<br>ISVDTSKNQFSLKLPSVTAADTAIYYCTAHTRGF<br>LYWGQGTLLTVTSS (SEQ ID NO: 306)      | DIQMTQSPSSLSASVGDRTVITCQASQDINNY<br>LNWYQQKPGKAPKLLIYDASNLPVGPVSRFSG<br>TGSGTDFSFSTISSLPEDIATYYCQHYIDLPL<br>TFGGGTVAIR (SEQ ID NO: 312)           |
| EIQLVESGGDLVQPGGSLKLSCAASGFTFSDSAM<br>HWVRQASGKGLEWVGRIRNKANTYATSYAASVKG<br>RPTVSTDDSKTTAYLQMRYLKIEDTAIYYCTSDQ<br>FDYWGGTLLTVTSS (SEQ ID NO: 313)        | DVMTQSPSLSATLGQPASISCRSSQSLVYS<br>DGI TYLTWFPQRPQSPTRLIYQVSNRDSGVP<br>DRFSGSGAGTDFTLNI SRVEAEDVGYYCMEG<br>THWPWFQGTGKVELK (SEQ ID NO:<br>314)     |
| QVQLVESGGGLVKPGGSLRLSCAASGFTFSDYYM<br>SWIRQAPGKGLEWISFISRTGSTKYADSVKGRF<br>TISRDNKNSLNLQMSRLRAEDTAVYFCAREPLQ<br>HWGQGTLLTVTSS (SEQ ID NO: 315)           | NVLTQTPISSPVILGQPASISCRSSQSLVHS<br>DGD TYLSWLHQRPGQPPRLIYTVSNRVSGVP<br>DRFSGSGAGTDFTLKI SRVEAEDVGYYCMQG<br>TQFPALTFFGGTKVEIK (SEQ ID NO:<br>316)  |
| QVQLVESGGGLVKPGGSLRLSCAASGFTFSDYYM<br>SWIRQAPGKGLEWISYISRTGSTKYADSVKGRF<br>TISRDNKNSLNLQMSRLRAEDTAVYFCAREPLQ<br>HWGQGTLLTVTSS (SEQ ID NO: 317)           | KIVLTQTPLSSPVTLGQPASISCRSSQSLVHS<br>DGD TYLSWLHQRPGQPPRLIYTVSNRVSGVP<br>DRFSGSGAGTDFTLKI SRVEAEDVGYYCMQG<br>TQFPALTFFGGTKVEIK (SEQ ID NO:<br>318) |
| EMQLVESGGGLVQPGGSLKVS CAASGFTLSDSSI<br>HWVRQASGKGLEWIGRIRSKPYSYATAYAAVKG<br>RPTISRDDSKNTAFLQMSGLKTEDTAVYYCTGGD<br>DFWGQGTLLTVTSS (SEQ ID NO: 319)        | DIQMTQSPSTLSASVGDRTITCRASQSISTW<br>LAWYQQKPGKAPKLLIYKASSLEGGVPSRFSG<br>SGSGTDFTLTISSLPDDFATYYCQYNTYWT<br>FGQGTKEIK (SEQ ID NO: 320)               |
| EVQLVESGGGLVQPGGSLRLSCAASGFTFSSYAM<br>SWVRQAPGKGLEWVSAISGSGGSTYYADSVKGRF<br>TISRDNKNTLYLQMNSLRAEDTAVYYCAIPYNW<br>NYGGAFDIWGQGTMTVTSS (SEQ ID NO:<br>321) | EIVMTQSPATLSVSPGKRATLSRASQSVSSN<br>LAWYQQKPGQAPRLIYGASTRATGIPARFSG<br>SGSGTEFTLTISSLQSEDFAVYYCQYNNWPP<br>YTFGQGTKEIK (SEQ ID NO: 322)             |
| QVQLVQSGAEVKKPGASVKVSCASGYTFTGYM<br>HWVRQAPGQGLEWGMWINPNSVGTNFAQKFGQGRV<br>TMTSDTISITAYMELSLRSDDTAVYYCTRDGAA<br>AGLFDYWGGTLLTVTSS (SEQ ID NO: 323)       | DIQMTQSPSSLSASVGDRTVITCRASQGINNY<br>LAWYQQKPGKVPKLLIYAASLTQSGVPSRFSG<br>SGSGTDFTLTISSLPEDVATYYCQKFNSAPL<br>TFGGGTKEIK (SEQ ID NO: 324)            |
| EVQLVESGGGLVQPGGSLRLSCAASGFTFSTYAM<br>SWVRQAPGKGLEWVSAISGSGGSTYYADSVKGRF<br>TISRDNKNTLYLQMKSLRVEDTAVYYCANSPSW<br>FDPWGGTLLTVTSS (SEQ ID NO: 325)         | DIQMTQSPSSLSASVGDRTVITCRASQSISSY<br>LNWYQQKPGKAPKLLIYAASLTQSGVPSRFSG<br>SGSGTDFTLTISSLPEDFATYYCQQSSTPP<br>ITFGQGTLEIK (SEQ ID NO: 326)            |
| EVQLVESGGGLVQPGSLRLSCAASGFTFDDYGM<br>HWVRQAPGKGLEWVSVISWNSATIDYADSVKGRF<br>TISRDNKNSLFLQMNSLRPEDTALYYCAKNSL<br>GWFFDYWGQGTLLTVTSS (SEQ ID NO: 327)       | DIVMTQSPDSLAVSLGERATINCKSSQSVLYS<br>SNDNNYLAWYQQKPGQPPKLLIYWASTRESGV<br>PDRFSGSGSGTDFTLTISSLQAEADVAVYYCQ<br>YYSTPLTFGGGTKEIK (SEQ ID NO:<br>328)  |
| QVQLVQSGAEVKKPGASVKVSCASGYTFTSYGI<br>SWVRQAPGQGLEWGMWISAYNNTIYQKFGQGRV<br>TMTTDTSTAYMELSLRSDDTAVYYCARDINW<br>YFDLWGRGTLLTVTSS (SEQ ID NO: 329)           | DIQMTQSPSSLSASVGDRTVITCRASQGIRND<br>LGWFQQKPGKAPKRLIYAASLTQSGVPSRFSG<br>SGSGTEFTLTISSLPEDFATYYCQHNSTPW<br>TFGQGTKEIK (SEQ ID NO: 330)             |
| EVQLVESGGGLVQPGSLRLSCAASGFTFDDYGM<br>HWVRQAPGKGLEWVSVISWNGATIDYADSVKGRF<br>TISRDNKNSLFLQMNSLRPEDTALYYCAKNSL<br>GWFFDYWGQGTLLTVTSS (SEQ ID NO: 331)       | DIQMTQSPSSLSAFVGDRTVITCRASQGIIRND<br>LGWYQQKPGKTPKRLIYTASSLRSGVPSRFSG<br>SGSGTDFTLTISSLPEDFATYYCQHNSTPY<br>TFGQGTKEIK (SEQ ID NO: 332)            |
| EVQLVESGGNLVQPGGSLRLSCAASGFTFSNYAM<br>SWVRQAPGKGLEWVSGITGNGVNTIYVSVKGRF<br>TISRDNKNTLFLMNSLRAEDTAIYYCVKERGH<br>SWFGDWFDWGGQGTLLTVTSS (SEQ ID NO:<br>333) | DIQMTQSPSSLSASVGDRTVITCRSSQSVISY<br>LNWYQQKPGKAPKLLIFGASSLISGVPVSRFSG<br>SGSGTDFTLTIVSLQPEDFAVYFCQNNYLPPL<br>TFGGGTKEIK (SEQ ID NO: 334)          |
| EVQLVESGGNLVQPGGSLRLSCAASGFTFSNYAM<br>SWVRQAPGKGLEWVSGITGNGIKTYYSVSAKGRF                                                                                 | DIVMTQPLSSPVTLGQPASISCRSSQGLVNS<br>DGN TYLSWLHQRPGQPPRLIYIISNRFSGVP                                                                               |

TABLE 5-continued

| Variable heavy and light chain sequences of<br>exemplary ALK2 antibodies                                                                                       |                                                                                                                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Heavy chain variable<br>region sequence                                                                                                                        | Light chain variable<br>region sequence                                                                                                               |
| TISRDN SKNTLFLQMN SLRAEDTAIYYCVKERGH<br>SWFGDW FDPWQGTLVTVSS (SEQ ID NO:<br>335)                                                                               | DRFSGSGAGTDFTLKISRVEAEDVGFYYCMQS<br>TQFPLTFGGGTKVEIK (SEQ ID NO:<br>336)                                                                              |
| EVQLVESGGN LVQPGGSLRLSCAASGFTFSNYAM<br>SWVRQAPGKGLEWVSGITGNGINTYYSVVKGRF<br>TISRDN SKNTVFLQMN SLRAEDTAIYYCVKERGH<br>SWFGDW FDPWQGTLVTVSS (SEQ ID NO:<br>337)   | DIQMTQSPSSLSASVGDRTITCRASQGIRND<br>LGWYQQKPKGKAPKRLIYAASSLQSGVPSRFSG<br>SGSGTEFTLTISLQPEDFATYYCQHNSTYPW<br>TFGQGTKVEIK (SEQ ID NO: 338)               |
| QVQLVESGGGVVQPGSRSLRLSCVSGFTFSNYGM<br>HWVRQAPGKGLEWVAVVSYDGSNKYYTDSVKGRF<br>TISRDN SKNTLSLQMN SLRAEDTAVYYCAKDRGI<br>TGTSGGVFDIWGQGTMTVTVSS (SEQ ID NO:<br>339) | AIRMTQSPSFSFATTGDRVTITCRASQGISSH<br>LAWYQQKPKGKAPKFLIYVASTLQSGVPSRFSG<br>SGSGTDFTLTISLQSEDFATYYCQQYYSYPP<br>TFGQGTKVEIK (SEQ ID NO: 340)              |
| EVQLVESGGGLVQPGGSLRLSCAASGFTSSRYWM<br>SWVRQAPGKGLEWLANINQDGSEKYYLES LRGRF<br>TISRDN AKNSLYLQMN SLRAEDTAVYYCARDEEG<br>YWGQGT LVTVSS (SEQ ID NO: 341)            | DVVM TQSLLSL SVTLGQPASISCRSSQSLVYS<br>DGD TYLSW FQQRPGQSPRR LIYKVSHRDSGVP<br>DRFRGSGSGTDFTLKISRVEAEDVGIYYCMQG<br>THWPPTFGQGTKVEIK (SEQ ID NO:<br>342) |

[0088] In some embodiments, the ALK2 antibody administered to the subject is an ALK2 antibody or an antigen-binding fragment thereof, having a heavy chain and light chain sequence having at least 90% sequence identity (e.g., 90%, 91%, 92%, 93%, 94%, 95%, 95%, 97%, 98%, 99% or 100% sequence identity) to a heavy chain and light chain sequence provided in Table 6. In some embodiments, the ALK2 antibody or an antigen binding fragment thereof, has a heavy chain and light chain sequence from the same row of Table 6. In some embodiments, the heavy chain and light chain have the sequence of SEQ ID NOs: 144 and 145; 146 and 147; 148 and 149; 150 and 151; 152 and 153; 154 and 155; 156 and 157; 156 and 158; 159 and 157; 159 and 158; 159 and 149; 159 and 160; 161 and 166; 161 and 145; 161 and 167; 162 and 166; 162 and 145; 162 and 167; 163 and 166; 163 and 145; 163 and 167; 163 and 168; 164 and 167;

164 and 168; 164 and 169; 165 and 168; 170 and 168; 171 and 172; 171 and 173; 171 and 147; 146 and 172; 146 and 173; 174 and 175; 174 and 176; 174 and 151; 150 and 175; 150 and 176; 209 and 210; 211 and 212; 213 and 214; 215 and 216; 343 and 344; 345 and 346; 347 and 348; 349 and 350; 345 and 351; 352 and 353; 354 and 355; 356 and 357; 358 and 359; 360 and 361; 362 and 363; 364 and 365; 366 and 367; 368 and 369; 370 and 371; 372 and 373; 374 and 375; 376 and 377; 378 and 379; 380 and 381 (e.g., the heavy chain has at least 90% sequence identity (e.g., 90%, 91%, 92%, 93%, 94%, 95%, 95%, 97%, 98%, 99% or 100% sequence identity) to the sequence of the first SEQ ID NO: in each pair and the light chain has at least 90% sequence identity (e.g., 90%, 91%, 92%, 93%, 94%, 95%, 95%, 97%, 98%, 99% or 100% sequence identity) to the sequence of the second SEQ ID NO: in each pair).

TABLE 6

| Exemplary ALK2 antibody heavy and light chain sequences                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Heavy chain                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Light chain                                                                                                                                                                                                                                                    |
| EVQLVESGGGLVQPGGSLRLSCAASGFTFSNYG<br>MKWIRQAPGKGLEWVSSISRSSTIYYADTVKGR<br>FTISRDN SKNTLYLQMN SLRAEDTAVYYCAAIST<br>PFYWFDFWQGTLVTVSSASTKGPSVFPLAPS<br>SKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTS<br>GVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYIC<br>NVNHKPSNTKVDKRVPEPKSCDKTHTCPPCPAPE<br>AAGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVS<br>HEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTY<br>RVVSVLTVLHQDWLNGKEYCKVSNKALPAPIEK<br>TISKAKGQPREPQVYTLPPSREEMTKNQVSLTCL<br>VKGFYPSDIAVEWESNGQPENNYKTPPVLDSD<br>GSFFLYSKLTVDKSRWQQGNVFCFSVMHEALHN<br>HYTQKSLSLSPGK (SEQ ID NO: 144) | EIVLTQSPGTLSPGERATLSCLASSVSYMT<br>WYQQKPGQSPRLWIYGTSNLASGVPDFRFGS<br>GSGTDFTLTISRLEPEDFAVYCLHLTSYPPYT<br>FGQGTKVEIKRAVAAPS FIFPPSPDEQLKSGTA<br>SVVCLLN NFYPREAKVQWKVDNALQSGNSQSE<br>SVTEQDSKDSYLSSTLTLSKADYEKHKVYA<br>CEVTHQGLSPVTKSFNRGEC (SEQ ID NO:<br>145) |
| EVQLVESGGGLVQPGGSLRLSCAASGFTFSNYY<br>MYWIRQAPGKGLEWISSINTDGGSTIYPSVKGR<br>FTISRDN SKNTVYLQMN SLRAEDTAVYYCAKSTP<br>NIPLAYWQGTLVTVSSASTKGPSVFPLAPSSKS<br>TSGGTAALGCLVKDYFPEPVTVSWNSGALTSGV                                                                                                                                                                                                                                                                                                                                             | DIQMTQSPSSLSASVGDRTITCRASQNIYKYL<br>NWFQQKPGKAPKLLIYYNSLTQTGIPSRFSGS<br>GSGTDFTLTISLQPEDFAIYFCFQYSSGPTFG<br>PGTKVEIKRTVAAPS FIFPPSPDEQLKSGTASV<br>VCLLN NFYPREAKVQWKVDNALQSGNSQESV                                                                             |

TABLE 6-continued

| Exemplary ALK2 antibody heavy and light chain sequences                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Heavy chain                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Light chain                                                                                                                                                                                                                                                             |
| HTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNV<br>NHKPSNTKVDKRVEPKSCDKTHTCPPCPAPELLG<br>GPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHED<br>PEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVV<br>SVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISK<br>AKGQPREPQVYTLPPSREEMTKNQVSLTCLVKG<br>FYPYPSDIAVEWESNGQPENNYKTPPVLDSDGSFF<br>LYSKLTVDKSRWQQGNVVFSCSVMHREALNHYTQ<br>KSLSLSPGK (SEQ ID NO: 146)                                                                                                                                                                                            | TEQDSKDYSLSSSTLTLSKADYEKHKVYACE<br>VTHQGLSSPVTKSFNRGEC (SEQ ID NO: 147)                                                                                                                                                                                                 |
| EVQLVESGGGLVQPGGSLRLSCAASGFTFSHY<br>MAWVRQAPKGLEWVASITNSGGSYNYRDSVKG<br>RFTISRDNKSTLYLQMSLSRAEDTATYYCTREG<br>GENYGGYPFFAYWGQGLTVTVSSASTKGPSVFPLAP<br>LAPCSRSTSESTAALGCLVKDYFPEPVTVSWNSG<br>ALTSGVHTFPAVLQSSGLYSLSSVTVPSNFGT<br>QTYTCNVDHKPSNTKVDKTKVERKCCVECPCPA<br>PPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVS<br>HEDPEVQFNWYVDGVEVHNAKTKPREEQFNSTF<br>RVVSVLTVVLHQDWLNGKEYKCKVSNKGLPAPIEK<br>TISKTKGQPREPQVYTLPPSREEMTKNQVSLTCL<br>VKGFYPSDIAVEWESNGQPENNYKTPPMLDS<br>GSFFLYSKLTVDKSRWQQGNVVFSCSVMHREALHN<br>HYTQKSLSLSPGK (SEQ ID NO: 148) | DIVLTQSPDLSAVSLGERATINCRANQGVSLSR<br>YNLMHWYQQKPGQPKLLIYRSSNLAQGI PAR<br>FSGSGSGTDFTLTISVQADDIAVYVCQQSRE<br>SPFTFGQGTKELEKRAVAAPSVEFIPPPSDEQLK<br>SGTASVVCLLNNFYPREAKVQWKVDNALQSG<br>NSQESVTEQDSKDYSLSSSTLTLSKADYEKHK<br>KVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID<br>NO: 149)  |
| EVQLVESGGGLVQPGGSLRLSCAASGFTFSY<br>MSWVRQAPKGLEWVASITNSGGSYNYRDSVK<br>GRFTISRDNKSTLYLQMSLSRAEDTATYYCARL<br>NYNNYFDYWGQGLTVTVSSASTKGPSVFPLAPS<br>SKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTS<br>GVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYIC<br>NVNHKPSNTKVDKRVEPKSCDKTHTCPPCPAPE<br>LLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSH<br>EDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYR<br>VVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKT<br>ISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLV<br>KGFYPSDIAVEWESNGQPENNYKTPPVLDSDG<br>SFFLYSKLTVDKSRWQQGNVVFSCSVMHREALHNH<br>YTQKSLSLSPGK (SEQ ID NO: 150)     | DIQMTQSPSSLSASVGRVITTCASQDIGNW<br>LSWYQQKPGKSPKLLIYGATSLADGVPSRFSG<br>SRSGTQYTLTISLQPEDFATYYCLQAYSAPFT<br>FSGGTKEIKRTVAAPSVEFIPPPSDEQLKSGTA<br>SVVCLLNNFYPREAKVQWKVDNALQSGNSQ<br>SVTEQDSKDYSLSSSTLTLSKADYEKHKVYA<br>CEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:<br>151)           |
| EVQLVESGGGLVQPGSLRLSCAASGFTFSHY<br>MAWVRQAPKGLWVASITNSGGSYNYRDSVKG<br>RFTISRDNKSTLYLQMSLSRAEDTATYYCTREG<br>GENYGGYPFFAYWGQGLTVTVSSASTKGPSVFPLAP<br>LAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNS<br>GALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLG<br>TQTYICNVNHKPSNTKVDKRVEPKSCDKTHTCPP<br>CPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCV<br>VDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQY<br>NSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALP<br>APIEKTISKAKGQPREPQVYTLPPSREEMTKNQV<br>SLTCLVKGFYPSDIAVEWESNGQPENNYKTPPV<br>LDSDGSFFLYSKLTVDKSRWQQGNVVFSCSVMH<br>ALHNHYTQKSLSLSPGK (SEQ ID NO: 152) | DIVLTQSPALAVSLGQRATISCRANQGVSLSR<br>NLMHWYQQKPGQPKLLIYRSSNLASGIPARF<br>SGSGSGTDFTLTINPVQADDIAVYVCQQSRES<br>PFTFGAGTKLELEKRAVAAPSVEFIPPPSDEQLKS<br>GTASVVCLLNNFYPREAKVQWKVDNALQSGNS<br>NSQESVTEQDSKDYSLSSSTLTLSKADYEKHK<br>KVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID<br>NO: 153) |
| EVQLVESGGGLVQPGSLRLSCLASGFTFSNYG<br>MKWIRQAPKGLEWVASITNSGGSYNYADTVKGR<br>FTISRDNARNTLYLQMTSLRSEDALYYCAAIST<br>PFYWFDFWGPQTVTVTVSSASTKGPSVFPLAPS<br>SKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTS<br>GVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYIC<br>NVNHKPSNTKVDKRVEPKSCDKTHTCPPCPAPE<br>LLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSH<br>EDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYR<br>VVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKT<br>ISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLV<br>KGFYPSDIAVEWESNGQPENNYKTPPVLDSDG<br>SFFLYSKLTVDKSRWQQGNVVFSCSVMHREALHNH<br>YTQKSLSLSPGK (SEQ ID NO: 154)  | EIVLTQSPPTMAASPGKVTNLNCLASSSVSYM<br>TWYQQKSGASPKLWIYGTSLNLSGVPNRFSG<br>SGSGTSYSLAISMEADVATYYCLHLTSYPP<br>YTFGAGTKLELEKRAVAAPSVEFIPPPSDEQLKSG<br>TASVVCLLNNFYPREAKVQWKVDNALQSGNS<br>QESVTEQDSKDYSLSSSTLTLSKADYEKHKV<br>YACEVTHQGLSSPVTKSFNRGEC (SEQ ID<br>NO: 155)       |
| EVQLVESGGGLVQPGGSLRLSCAASGFTFSHY<br>MAWVRQAPKGLEWVASITNSGGSYNYRDSVKG<br>RFTISRDNKSTLYLQMSLSRAEDTAVYVCAREG                                                                                                                                                                                                                                                                                                                                                                                                                             | DIVMTQSPDLSAVSLGERATINCRANQGVSL<br>RYNLMHWYQQKPGQPKLLIYRSSNLASGVP<br>DRFSGSGSGTDFTLTISLQAEADVAVYVCQQS                                                                                                                                                                   |

TABLE 6-continued

| Exemplary ALK2 antibody heavy and light chain sequences                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Heavy chain                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Light chain                                                                                                                                                                                                                                                            |
| GENYGGYPPFAFWGQGT LVT VSSASTKGPSVFP<br>LAPSSKSTSGGTAALGCLVKDYFPEPVT VSWNS<br>GALTSGVHTFPAVLQSSGLYSLSVVTVPSSSLG<br>TQTYICNVNHKPSNTKVDKRVEPKSCDKTHTCPP<br>CPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVV<br>VDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQY<br>NSTYRVVSVLTVLHQDWLNGKEYCKVSNKALP<br>APIEKTISKAKGQPREPQVYTLPPSREEMTKNQV<br>SLTCLVKGFPYPSDIAVEWESNGQPENNYKTPPV<br>LDDSGSFFLYSKLTVDKSRWQQGNVFCFSVMHE<br>ALHNYHTQKSLSLSPGK (SEQ ID NO: 156)                                                                                                                | RESPFTFGQGTKVEIKRAVAAPSVFIFPPSDEQ<br>LKSGTASVVCLLNNFYPREAKVQWKVDNALQ<br>SGNSQESVTEQDSKDSITYLSSTLTLSKADYE<br>KHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ<br>ID NO: 157)                                                                                                            |
| EVQLVESGGGLVQP GGSRLRS CAASGFTFSHY<br>MAWVRQAPGKGLEWASITNSGGSIYRDSVKG<br>RFTISRDNKNTLYLQMNSLR AEDTAVYICAREG<br>GENYGGYPPFAFWGQGT LVT VSSASTKGPSVFP<br>LAPSSKSTSGGTAALGCLVKDYFPEPVT VSWNS<br>GALTSGVHTFPAVLQSSGLYSLSVVTVPSSSLG<br>TQTYICNVNHKPSNTKVDKRVEPKSCDKTHTCPP<br>CPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVV<br>VDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQY<br>NSTYRVVSVLTVLHQDWLNGKEYCKVSNKALP<br>APIEKTISKAKGQPREPQVYTLPPSREEMTKNQV<br>SLTCLVKGFPYPSDIAVEWESNGQPENNYKTPPV<br>LDDSGSFFLYSKLTVDKSRWQQGNVFCFSVMHE<br>ALHNYHTQKSLSLSPGK (SEQ ID NO: 156) | DIVLTQSPDLSAVSLGERATINCRANQGVSLSR<br>YNLMHWYQQKPGQPKLLIYRSSNLASGIPAR<br>FSGSGSGTDFTLTISVQADDIAVYICQSSRE<br>SPFTFGQGTKLELKR AVAAPSVFIFPPSDEQLK<br>SGTASVVCLLNNFYPREAKVQWKVDNALQSG<br>NSQESVTEQDSKDSITYLSSTLTLSKADYEK<br>KHYACEVTHQGLSSPVTKSFNRGEC (SEQ ID<br>NO: 158)   |
| EVQLVESGGGLVQP GGSRLRS CAASGFTFSHY<br>MAWVRQAPGKGLEWASITNSGGSIYRDSVKG<br>RFTISRDNKNTLYLQMNSLR AEDTATYYCTREG<br>GENYGGYPPFAFWGQGT LVT VSSASTKGPSVFP<br>LAPSSKSTSGGTAALGCLVKDYFPEPVT VSWNS<br>GALTSGVHTFPAVLQSSGLYSLSVVTVPSSSLG<br>TQTYICNVNHKPSNTKVDKRVEPKSCDKTHTCPP<br>CPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVV<br>VDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQY<br>NSTYRVVSVLTVLHQDWLNGKEYCKVSNKALP<br>APIEKTISKAKGQPREPQVYTLPPSREEMTKNQV<br>SLTCLVKGFPYPSDIAVEWESNGQPENNYKTPPV<br>LDDSGSFFLYSKLTVDKSRWQQGNVFCFSVMHE<br>ALHNYHTQKSLSLSPGK (SEQ ID NO: 159) | DIVMTQSPDLSAVSLGERATINCRANQGVSLSR<br>RYNLMHWYQQKPGQPKLLIYRSSNLASGVP<br>DRFGSGSGTDFTLTISVQADDIAVYICQSSRE<br>RESPFTFGQGTKVEIKRAVAAPSVFIFPPSDEQ<br>LKSGTASVVCLLNNFYPREAKVQWKVDNALQ<br>SGNSQESVTEQDSKDSITYLSSTLTLSKADYE<br>KHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ<br>ID NO: 157) |
| EVQLVESGGGLVQP GGSRLRS CAASGFTFSHY<br>MAWVRQAPGKGLEWASITNSGGSIYRDSVKG<br>RFTISRDNKNTLYLQMNSLR AEDTATYYCTREG<br>GENYGGYPPFAFWGQGT LVT VSSASTKGPSVFP<br>LAPSSKSTSGGTAALGCLVKDYFPEPVT VSWNS<br>GALTSGVHTFPAVLQSSGLYSLSVVTVPSSSLG<br>TQTYICNVNHKPSNTKVDKRVEPKSCDKTHTCPP<br>CPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVV<br>VDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQY<br>NSTYRVVSVLTVLHQDWLNGKEYCKVSNKALP<br>APIEKTISKAKGQPREPQVYTLPPSREEMTKNQV<br>SLTCLVKGFPYPSDIAVEWESNGQPENNYKTPPV<br>LDDSGSFFLYSKLTVDKSRWQQGNVFCFSVMHE<br>ALHNYHTQKSLSLSPGK (SEQ ID NO: 159) | DIVLTQSPDLSAVSLGERATINCRANQGVSLSR<br>YNLMHWYQQKPGQPKLLIYRSSNLASGIPAR<br>FSGSGSGTDFTLTISVQADDIAVYICQSSRE<br>SPFTFGQGTKLELKR AVAAPSVFIFPPSDEQLK<br>SGTASVVCLLNNFYPREAKVQWKVDNALQSG<br>NSQESVTEQDSKDSITYLSSTLTLSKADYEK<br>KHYACEVTHQGLSSPVTKSFNRGEC (SEQ ID<br>NO: 158)   |
| EVQLVESGGGLVQP GGSRLRS CAASGFTFSHY<br>MAWVRQAPGKGLEWASITNSGGSIYRDSVKG<br>RFTISRDNKNTLYLQMNSLR AEDTATYYCTREG<br>GENYGGYPPFAFWGQGT LVT VSSASTKGPSVFP<br>LAPSSKSTSGGTAALGCLVKDYFPEPVT VSWNS<br>GALTSGVHTFPAVLQSSGLYSLSVVTVPSSSLG<br>TQTYICNVNHKPSNTKVDKRVEPKSCDKTHTCPP<br>CPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVV<br>VDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQY<br>NSTYRVVSVLTVLHQDWLNGKEYCKVSNKALP<br>APIEKTISKAKGQPREPQVYTLPPSREEMTKNQV<br>SLTCLVKGFPYPSDIAVEWESNGQPENNYKTPPV<br>LDDSGSFFLYSKLTVDKSRWQQGNVFCFSVMHE<br>ALHNYHTQKSLSLSPGK (SEQ ID NO: 159) | DIVLTQSPDLSAVSLGERATINCRANQGVSLSR<br>YNLMHWYQQKPGQPKLLIYRSSNLASGIPAR<br>FSGSGSGTDFTLTISVQADDIAVYICQSSRE<br>SPFTFGQGTKLELKR AVAAPSVFIFPPSDEQLK<br>SGTASVVCLLNNFYPREAKVQWKVDNALQSG<br>NSQESVTEQDSKDSITYLSSTLTLSKADYEK<br>KHYACEVTHQGLSSPVTKSFNRGEC (SEQ ID<br>NO: 149)   |



TABLE 6-continued

| Exemplary ALK2 antibody heavy and light chain sequences                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Heavy chain                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Light chain                                                                                                                                                                                                                                                           |
| EVQLVESGGGLVQPGGSLRLSCAASGFTFSHY<br>MAWVRQAPGKGLEWVASITNSGGSIYRDSVKG<br>RFTISRDNASKNTLYLQMNLSRAEDTATYYCTREG<br>GENYGGYPPFAFWGQGTLTVTSSASTKGPSVFP<br>LAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNS<br>GALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLG<br>TQTYICNVNHKPSNTKVDKRVPEPKCDKTHTCPP<br>CPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVV<br>VDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQY<br>NSTYRVVSVLTVLHQDWLNGKEYCKVSNKALP<br>APIEKTISKAKGQPREPQVYTLPPSREEMTKNQV<br>SLTCLVKGFPYPSDIAVEWESNGQPENNYKTPPV<br>LDSGGSFFLYSKLTVDKSRWQQGNVFCSCVMHE<br>ALHNHYTQKSLSLSPGK (SEQ ID NO: 159) | DIVLTQSPDLSAVSLGERATINCRANQGVSLSR<br>YNLMHWYQQKPGQPKPLAIYRSSNLASGIPAR<br>FSGSGSGTDFTLTISVQADDIAVYYCQSSRE<br>SPFTFGQGTKELEKRAVAAPSVFIAPPDEQLK<br>SGTASVVCLLNNFYPREAKVQWKVDNALQSG<br>NSQESVTEQDSKDSSTYSLSSTLTLSKADYEKH<br>KVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID<br>NO: 160) |
| EVQLVESGGGLVQPGGSLRLSCAASGTFSTNYG<br>MKWVRQAPGKGLEWSSISRSSTIYYADTVKG<br>RFTISRDNASKNTLYLQMNLSRAEDTAVYYCARAIS<br>TPFYWYFDWGGGTLTVTSSASTKGPSVFPLAP<br>SSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALT<br>SGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYI<br>CNVNHKPSNTKVDKRVPEPKCDKTHTCPPCPAP<br>ELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVS<br>HEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTY<br>RVVSVLTVLHQDWLNGKEYCKVSNKALPAPIEK<br>TISKAKGQPREPQVYTLPPSREEMTKNQVSLTCL<br>VKGFPYPSDIAVEWESNGQPENNYKTPPVLDSD<br>GSFFLYSKLTVDKSRWQQGNVFCSCVMHEALHN<br>HYTQKSLSLSPGK (SEQ ID NO: 161)    | EIVLTQSPGTLSSLSPGERATLSCLASSSVSYMT<br>WYQQKPGQAPRLIYGTSNLASGIPDRFSGSG<br>SGTDFTLTISRLEPEDFAVYYCLHLTSYPPYTF<br>GQGTKEIKRAVAAPSVFIAPPDEQLKSGTAS<br>VVCLLNNFYPREAKVQWKVDNALQSGNSQ<br>ESVTEQDSKDSSTYSLSSTLTLSKADYEKHKVY<br>ACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:<br>166)     |
| EVQLVESGGGLVQPGGSLRLSCAASGTFSTNYG<br>MKWVRQAPGKGLEWSSISRSSTIYYADTVKG<br>RFTISRDNASKNTLYLQMNLSRAEDTAVYYCARAIS<br>TPFYWYFDWGGGTLTVTSSASTKGPSVFPLAP<br>SSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALT<br>SGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYI<br>CNVNHKPSNTKVDKRVPEPKCDKTHTCPPCPAP<br>ELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVS<br>HEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTY<br>RVVSVLTVLHQDWLNGKEYCKVSNKALPAPIEK<br>TISKAKGQPREPQVYTLPPSREEMTKNQVSLTCL<br>VKGFPYPSDIAVEWESNGQPENNYKTPPVLDSD<br>GSFFLYSKLTVDKSRWQQGNVFCSCVMHEALHN<br>HYTQKSLSLSPGK (SEQ ID NO: 161)    | EIVLTQSPGTLSSLSPGERATLSCLASSSVSYMT<br>WYQQKPGQSPRLWIYGTSNLASGVPDRFSGS<br>GSGTDFTLTISRLEPEDFAVYYCLHLTSYPPYT<br>FGQGTKEIKRAVAAPSVFIAPPDEQLKSGTA<br>SVVCLNNFYPREAKVQWKVDNALQSGNSQ<br>ESVTEQDSKDSSTYSLSSTLTLSKADYEKHKVY<br>ACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:<br>145)     |
| EVQLVESGGGLVQPGGSLRLSCAASGTFSTNYG<br>MKWVRQAPGKGLEWSSISRSSTIYYADTVKG<br>RFTISRDNASKNTLYLQMNLSRAEDTAVYYCARAIS<br>TPFYWYFDWGGGTLTVTSSASTKGPSVFPLAP<br>SSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALT<br>SGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYI<br>CNVNHKPSNTKVDKRVPEPKCDKTHTCPPCPAP<br>ELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVS<br>HEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTY<br>RVVSVLTVLHQDWLNGKEYCKVSNKALPAPIEK<br>TISKAKGQPREPQVYTLPPSREEMTKNQVSLTCL<br>VKGFPYPSDIAVEWESNGQPENNYKTPPVLDSD<br>GSFFLYSKLTVDKSRWQQGNVFCSCVMHEALHN<br>HYTQKSLSLSPGK (SEQ ID NO: 161)    | EIVLTQSPGTLSSLSPGERATLSCLASSSVSYMT<br>WYQQKPGASPRLIYGTSNLASGVPDRFSGS<br>GSGTDYTLTISRLEPEDFATYYCLHLTSYPPYT<br>FGAGTKVEIKRAVAAPSVFIAPPDEQLKSGTA<br>SVVCLNNFYPREAKVQWKVDNALQSGNSQ<br>ESVTEQDSKDSSTYSLSSTLTLSKADYEKHKVY<br>ACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:<br>167)     |
| EVQLVESGGGLVQPGGSLRLSCAASGTFSTNYG<br>MKWIRQAPGKGLEWSSISRSSTIYYADTVKGR<br>FTISRDNASKNTLYLQMNLSRAEDTAVYYCAAIST<br>PFYWYFDWGGGTLTVTSSASTKGPSVFPLAPS<br>SKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTS<br>GVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYIC<br>NVNHKPSNTKVDKRVPEPKCDKTHTCPPCPAPE<br>LLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSH<br>EDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYR<br>VVSVLTVLHQDWLNGKEYCKVSNKALPAPIEKT<br>ISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLV                                                                                                                | EIVLTQSPGTLSSLSPGERATLSCLASSSVSYMT<br>WYQQKPGQAPRLIYGTSNLASGIPDRFSGSG<br>SGTDFTLTISRLEPEDFAVYYCLHLTSYPPYTF<br>GQGTKEIKRAVAAPSVFIAPPDEQLKSGTAS<br>VVCLLNNFYPREAKVQWKVDNALQSGNSQ<br>ESVTEQDSKDSSTYSLSSTLTLSKADYEKHKVY<br>ACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:<br>166)     |

TABLE 6-continued

| Exemplary ALK2 antibody heavy and light chain sequences                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Heavy chain                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Light chain                                                                                                                                                                                                                                                            |
| KGFYPSDIAVEWESNGQPENNYKTTTPVLDSDG<br>SFFLYSKLTVDKSRWQQGNVFSVCSVMHEALHNNH<br>YTQKSLSLSPGK (SEQ ID NO: 162)                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                        |
| EVQLVESGGGLVQPGGSLRLSCAASGSTFSNYG<br>MKWIRQAPKGLEWVSSISRSTYIYYADTVKGR<br>FTISRDN SKNTLYLQMNSLRAEDTAVVYCAAIST<br>PFYWFDFWGQGTLVTVSSASTKGPSVFPLAPL<br>SKSTSGGTAALGCLVKDYFPEPTVSWNSGALTS<br>GVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYIC<br>NVNHKPSNTKVDKRVKPEKSCDKTHTCPPCPAPE<br>LLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSH<br>EDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYR<br>VVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKT<br>SKAKGQPREPQVYTLPPSREEMTKNQVSLTCLV<br>KGFYPSDIAVEWESNGQPENNYKTTTPVLDSDG<br>SFFLYSKLTVDKSRWQQGNVFSVCSVMHEALHNNH<br>YTQKSLSLSPGK (SEQ ID NO: 162) | EIVLTQSPGTLSSLSPGERATLSCLASSSVSYMT<br>WYQQKPGQSPRLWIYGTSLNLSAGVPDRFSGS<br>GSGTDFTLTISRLEPEDFAVYYCLHLTSYPPTY<br>FGQGTKVEIKRAVAAPSVEFI FPPSDEQLKSGTA<br>SVVCLLNNFYPREAKVQWKVDNALQSGNSQ<br>ESVTEQDSKSDSTYLSSTLTLSKADYEKHKVY<br>ACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:<br>145) |
| EVQLVESGGGLVQPGGSLRLSCAASGSTFSNYG<br>MKWIRQAPKGLEWVSSISRSTYIYYADTVKGR<br>FTISRDN SKNTLYLQMNSLRAEDTAVVYCAAIST<br>PFYWFDFWGQGTLVTVSSASTKGPSVFPLAPL<br>SKSTSGGTAALGCLVKDYFPEPTVSWNSGALTS<br>GVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYIC<br>NVNHKPSNTKVDKRVKPEKSCDKTHTCPPCPAPE<br>LLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSH<br>EDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYR<br>VVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKT<br>SKAKGQPREPQVYTLPPSREEMTKNQVSLTCLV<br>KGFYPSDIAVEWESNGQPENNYKTTTPVLDSDG<br>SFFLYSKLTVDKSRWQQGNVFSVCSVMHEALHNNH<br>YTQKSLSLSPGK (SEQ ID NO: 162) | EIVLTQSPGTLSSLSPGERATLSCLASSSVSYMT<br>WYQQKPGASPRLLIYGTSLNLSAGVPDRFSGS<br>GSGTDYTLTISRLEPEDFAVYYCLHLTSYPPTY<br>FGAGTKVEIKRAVAAPSVEFI FPPSDEQLKSGTA<br>SVVCLLNNFYPREAKVQWKVDNALQSGNSQ<br>ESVTEQDSKSDSTYLSSTLTLSKADYEKHKVY<br>ACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:<br>167) |
| EVQLVESGGGLVQPGGSLRLSCAASGSTFSNYG<br>MKWIRQAPKGLEWVASISRSTYIYYADTVKGR<br>FTISRDN SKNTLYLQMNSLRAEDTAVVYCAAIST<br>PFYWFDFWGQGTLVTVSSASTKGPSVFPLAPL<br>SKSTSGGTAALGCLVKDYFPEPTVSWNSGALTS<br>GVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYIC<br>NVNHKPSNTKVDKRVKPEKSCDKTHTCPPCPAPE<br>LLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSH<br>EDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYR<br>VVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKT<br>SKAKGQPREPQVYTLPPSREEMTKNQVSLTCLV<br>KGFYPSDIAVEWESNGQPENNYKTTTPVLDSDG<br>SFFLYSKLTVDKSRWQQGNVFSVCSVMHEALHNNH<br>YTQKSLSLSPGK (SEQ ID NO: 163) | EIVLTQSPGTLSSLSPGERATLSCLASSSVSYMT<br>WYQQKPGQAPRLLIYGTSLNLSAGVPDRFSGS<br>SGTDFTLTISRLEPEDFAVYYCLHLTSYPPTYF<br>GQGTKVEIKRAVAAPSVEFI FPPSDEQLKSGTAS<br>VVCLLNNFYPREAKVQWKVDNALQSGNSQ<br>ESVTEQDSKSDSTYLSSTLTLSKADYEKHKVY<br>ACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:<br>166)  |
| EVQLVESGGGLVQPGGSLRLSCAASGSTFSNYG<br>MKWIRQAPKGLEWVASISRSTYIYYADTVKGR<br>FTISRDN SKNTLYLQMNSLRAEDTAVVYCAAIST<br>PFYWFDFWGQGTLVTVSSASTKGPSVFPLAPL<br>SKSTSGGTAALGCLVKDYFPEPTVSWNSGALTS<br>GVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYIC<br>NVNHKPSNTKVDKRVKPEKSCDKTHTCPPCPAPE<br>LLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSH<br>EDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYR<br>VVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKT<br>SKAKGQPREPQVYTLPPSREEMTKNQVSLTCLV<br>KGFYPSDIAVEWESNGQPENNYKTTTPVLDSDG<br>SFFLYSKLTVDKSRWQQGNVFSVCSVMHEALHNNH<br>YTQKSLSLSPGK (SEQ ID NO: 163) | EIVLTQSPGTLSSLSPGERATLSCLASSSVSYMT<br>WYQQKPGQSPRLWIYGTSLNLSAGVPDRFSGS<br>GSGTDFTLTISRLEPEDFAVYYCLHLTSYPPTY<br>FGQGTKVEIKRAVAAPSVEFI FPPSDEQLKSGTA<br>SVVCLLNNFYPREAKVQWKVDNALQSGNSQ<br>ESVTEQDSKSDSTYLSSTLTLSKADYEKHKVY<br>ACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:<br>145) |
| EVQLVESGGGLVQPGGSLRLSCAASGSTFSNYG<br>MKWIRQAPKGLEWVASISRSTYIYYADTVKGR<br>FTISRDN SKNTLYLQMNSLRAEDTAVVYCAAIST<br>PFYWFDFWGQGTLVTVSSASTKGPSVFPLAPL<br>SKSTSGGTAALGCLVKDYFPEPTVSWNSGALTS<br>GVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYIC<br>NVNHKPSNTKVDKRVKPEKSCDKTHTCPPCPAPE<br>LLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSH<br>EDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYR<br>VVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKT<br>SKAKGQPREPQVYTLPPSREEMTKNQVSLTCLV<br>KGFYPSDIAVEWESNGQPENNYKTTTPVLDSDG<br>SFFLYSKLTVDKSRWQQGNVFSVCSVMHEALHNNH<br>YTQKSLSLSPGK (SEQ ID NO: 163) | EIVLTQSPGTLSSLSPGERATLSCLASSSVSYMT<br>WYQQKPGASPRLLIYGTSLNLSAGVPDRFSGS<br>GSGTDYTLTISRLEPEDFAVYYCLHLTSYPPTY<br>FGAGTKVEIKRAVAAPSVEFI FPPSDEQLKSGTA<br>SVVCLLNNFYPREAKVQWKVDNALQSGNSQ<br>SVTEQDSKSDSTYLSSTLTLSKADYEKHKVYA<br>CEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:<br>167)  |

TABLE 6-continued

| Exemplary ALK2 antibody heavy and light chain sequences                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Heavy chain                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Light chain                                                                                                                                                                                                                                                      |
| VVSVLTVLHQDWLNGKEYCKVSNKALPAPIEKT<br>SKAKGQPREPQVYTLPPSREEMTKNQVSLTCLV<br>KGFYPSDIAVEWESNGQPENNYKTTTPVLDSDG<br>SFFLYSKLTVDKSRWQQGNVFCSCVMHEALHNH<br>YTQKSLSLSPGK (SEQ ID NO: 163)                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                  |
| EVQLVESGGGLVQPGGSLRLS CAASGSTFSNYG<br>MKWIRQAPGKLEWVASISRSTIYYADTVKGR<br>FTISRDNSKNTLYLQMNSLRAEDTAVYYCAAIST<br>PFYWYDFWGPGLTVTVSSASTKGPSVFPLAPSS<br>SKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTS<br>GVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYIC<br>NVNHKPSNTKVDKRVPEPKCDKTHTCPPCPAPE<br>LLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSH<br>EDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYR<br>VVSVLTVLHQDWLNGKEYCKVSNKALPAPIEKT<br>SKAKGQPREPQVYTLPPSREEMTKNQVSLTCLV<br>KGFYPSDIAVEWESNGQPENNYKTTTPVLDSDG<br>SFFLYSKLTVDKSRWQQGNVFCSCVMHEALHNH<br>YTQKSLSLSPGK (SEQ ID NO: 163) | EIVLTQSPGTMSASPGERTVLSCLASSSVSYM<br>TWYQQKPGASPRLLWIYGTSNLASGVPDRFSG<br>SGSGTDYTLTISRMEPEDFATYYCLHLTSYPP<br>YTFGAGTKLELKRAVAAPSFIIPPSPDEQLKSG<br>TASVVCLLNNFYPREAKVQWKVDNALQSGNS<br>QESVTEQDSKDYSLSSLTLSKADYEKHKV<br>YACEVTHQGLSSPVTKSFNRGEC (SEQ ID<br>NO: 168) |
| EVQLVESGGGLVQPGGSLRLS CAASGSTFSNYG<br>MKWIRQAPGKLEWVASISRSTIYYADTVKGR<br>FTISRDNSKNTLYLQMNSLRAEDTAVYYCAAIST<br>PFYWYDFWGPGLTVTVSSASTKGPSVFPLAPSS<br>KSTSGGTAALGCLVKDYFPEPVTVSWNSGALTS<br>GVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYIC<br>NVNHKPSNTKVDKRVPEPKCDKTHTCPPCPAPE<br>LLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSH<br>EDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYR<br>VVSVLTVLHQDWLNGKEYCKVSNKALPAPIEKT<br>SKAKGQPREPQVYTLPPSREEMTKNQVSLTCLV<br>KGFYPSDIAVEWESNGQPENNYKTTTPVLDSDG<br>SFFLYSKLTVDKSRWQQGNVFCSCVMHEALHNH<br>YTQKSLSLSPGK (SEQ ID NO: 164)  | EIVLTQSPGTLSLSPGERATLSCLASSSVSYM<br>WYQQKPGASPRLLWIYGTSNLASGVPDRFSGS<br>SGSGTDYTLTISRLEPEDFATYYCLHLTSYPPY<br>FGAGTKVEIKRAVAAPSFIIPPSPDEQLKSGTA<br>SVVCLLNNFYPREAKVQWKVDNALQSGNSQ<br>SVTEQDSKDYSLSSLTLSKADYEKHKVYA<br>CEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:<br>167)   |
| EVQLVESGGGLVQPGGSLRLS CAASGSTFSNYG<br>MKWIRQAPGKLEWVASISRSTIYYADTVKGR<br>FTISRDNSKNTLYLQMNSLRAEDTAVYYCAAIST<br>PFYWYDFWGPGLTVTVSSASTKGPSVFPLAPSS<br>KSTSGGTAALGCLVKDYFPEPVTVSWNSGALTS<br>GVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYIC<br>NVNHKPSNTKVDKRVPEPKCDKTHTCPPCPAPE<br>LLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSH<br>EDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYR<br>VVSVLTVLHQDWLNGKEYCKVSNKALPAPIEKT<br>SKAKGQPREPQVYTLPPSREEMTKNQVSLTCLV<br>KGFYPSDIAVEWESNGQPENNYKTTTPVLDSDG<br>SFFLYSKLTVDKSRWQQGNVFCSCVMHEALHNH<br>YTQKSLSLSPGK (SEQ ID NO: 164)  | EIVLTQSPGTMSASPGERTVLSCLASSSVSYM<br>TWYQQKPGASPRLLWIYGTSNLASGVPDRFSG<br>SGSGTDYTLTISRMEPEDFATYYCLHLTSYPP<br>YTFGAGTKLELKRAVAAPSFIIPPSPDEQLKSG<br>TASVVCLLNNFYPREAKVQWKVDNALQSGNS<br>QESVTEQDSKDYSLSSLTLSKADYEKHKV<br>YACEVTHQGLSSPVTKSFNRGEC (SEQ ID<br>NO: 168) |
| EVQLVESGGGLVQPGGSLRLS CAASGSTFSNYG<br>MKWIRQAPGKLEWVASISRSTIYYADTVKGR<br>FTISRDNSKNTLYLQMNSLRAEDTAVYYCAAIST<br>PFYWYDFWGPGLTVTVSSASTKGPSVFPLAPSS<br>KSTSGGTAALGCLVKDYFPEPVTVSWNSGALTS<br>GVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYIC<br>NVNHKPSNTKVDKRVPEPKCDKTHTCPPCPAPE<br>LLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSH<br>EDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYR<br>VVSVLTVLHQDWLNGKEYCKVSNKALPAPIEKT<br>SKAKGQPREPQVYTLPPSREEMTKNQVSLTCLV<br>KGFYPSDIAVEWESNGQPENNYKTTTPVLDSDG<br>SFFLYSKLTVDKSRWQQGNVFCSCVMHEALHNH<br>YTQKSLSLSPGK (SEQ ID NO: 164)  | EIVLTQSPGTMSASPGERTVLSCLASSSVSYM<br>TWYQQKSGASPKLWIYGTSNLASGVPNRFSG<br>SGSGTSYTLTISRMEPEDFATYYCLHLTSYPPY<br>YTFGAGTKLELKRAVAAPSFIIPPSPDEQLKSGT<br>ASVVCLLNNFYPREAKVQWKVDNALQSGNSQ<br>ESVTEQDSKDYSLSSLTLSKADYEKHKV<br>ACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:<br>169)  |
| EVQLVESGGGLVQPGGSLRLS CAASGSTFSNYG<br>MKWIRQAPGKLEWVASISRSTIYYADTVKGR<br>FTISRDNAKNTLYLQMNSLRAEDTAVYYCAAIST<br>PFYWYDFWGPGLTVTVSSASTKGPSVFPLAPSS<br>KSTSGGTAALGCLVKDYFPEPVTVSWNSGALTS<br>GVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYIC<br>NVNHKPSNTKVDKRVPEPKCDKTHTCPPCPAPE<br>LLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSH<br>EDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYR<br>VVSVLTVLHQDWLNGKEYCKVSNKALPAPIEKT<br>SKAKGQPREPQVYTLPPSREEMTKNQVSLTCLV<br>KGFYPSDIAVEWESNGQPENNYKTTTPVLDSDG<br>SFFLYSKLTVDKSRWQQGNVFCSCVMHEALHNH<br>YTQKSLSLSPGK (SEQ ID NO: 164)  | EIVLTQSPGTMSASPGERTVLSCLASSSVSYM<br>TWYQQKPGASPRLLWIYGTSNLASGVPDRFSG<br>SGSGTDYTLTISRMEPEDFATYYCLHLTSYPP<br>YTFGAGTKLELKRAVAAPSFIIPPSPDEQLKSG<br>TASVVCLLNNFYPREAKVQWKVDNALQSGNS<br>QESVTEQDSKDYSLSSLTLSKADYEKHKV<br>YACEVTHQGLSSPVTKSFNRGEC (SEQ ID<br>NO: 169) |

TABLE 6-continued

| Exemplary ALK2 antibody heavy and light chain sequences                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Heavy chain                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Light chain                                                                                                                                                                                                                                                                       |
| LLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSH<br>EDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYR<br>VVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTI<br>SKAKGQPREPQVYTLPPSREEMTKNQVSLTCLV<br>KGFYPDSIAVEWESNGQPENNYKTTPPVLDSDG<br>SFFLYSKLTVDKSRWQQGNVFCSCVMHEALHNNH<br>YTQKSLSLSPGK (SEQ ID NO: 165)                                                                                                                                                                                                                                                                             | NO: 168)<br>EIVLTQSPGTMSASPGERVTLSCCLASSVSYM<br>TWYQQKPGASPRLLWIYGTSNLASGVPDRFSG<br>SGSGTDYTLTISRMEPEDFATYYCLHLTSYPP<br>YTFGAGTKLELKRAVAAPSVFIIPPSPDEQLKSG<br>TASVVCLLNNFYPREAKVQWKVDNALQSGNS<br>QESVTEQDSKSDSTYSLSSLTSLSKADYEKHKV<br>YACEVTHQGLSSPVTKSFNRGEC (SEQ ID<br>NO: 168) |
| EVQLVESGGGLVQPGGSLRLSCAASGFTFSNYG<br>MKWIRQAPGKGLIEWASISRSSTIYYADTVKGR<br>FTISRDN SKNTLYLQMNSLRAEDTAVYYCAAIST<br>PFYWFDFWGGQTLTVTVSSASTKGPSVFPPLAPS<br>SKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSG<br>GVHTFPAVLQSSGLYSLSSVTVPPSSSLGTQTYIC<br>NVNHKPSNTKVDKRVPEPKSCDKTHTCPPCPAPE<br>AAGGSPVFLFPPKPKDTLMISRTPEVTCVVVDVS<br>HEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTY<br>RVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEK<br>TISKAKGQPREPQVYTLPPSREEMTKNQVSLTCL<br>VKGFYPDSIAVEWESNGQPENNYKTTPPVLDSD<br>GSFFLYSKLTVDKSRWQQGNVFCSCVMHEALHNNH<br>HYTQKSLSLSPGK (SEQ ID NO: 170) | DIQMTQSPSSLSASVGDRTITCKASQNIYKYL<br>NWFQQKPGKAPKLLIYYNSLQTVPSRFSGS<br>GSGTDFTLTISLQPEDFATYYCFQYSSGPTF<br>GQGTKVEIKRTVAAPSVFIIPPSPDEQLKSGTAS<br>VVCLLNNFYPREAKVQWKVDNALQSGNSQ<br>SVTEQDSKSDSTYSLSSLTSLSKADYEKHKVYA<br>CEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:<br>172)                    |
| EVQLVESGGGLVQPGGSLRLSCAASGFTFSNY<br>MYWIRQAPGKGLIEWSSINTDGGSTYYADSVKG<br>RFTISRDN SKNTVYLQMNSLRAEDTAVYYCAKST<br>PNIPLAYWGQGLTVTVSSASTKGPSVFPPLAPSSK<br>STSGGTAALGCLVKDYFPEPVTVSWNSGALTSG<br>VHTFPAVLQSSGLYSLSSVTVPPSSSLGTQTYICN<br>VNHKPSNTKVDKRVPEPKSCDKTHTCPPCPAPELL<br>GGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHE<br>DPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRV<br>VSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTI<br>KAKGQPREPQVYTLPPSREEMTKNQVSLTCLVK<br>GFYPDSIAVEWESNGQPENNYKTTPPVLDSDGSF<br>FLYSLKTVDKSRWQQGNVFCSCVMHEALHNNHYT<br>QKSLSLSPGK (SEQ ID NO: 171)      | DIQMTQSPSSLSASVGDRTITCKASQNIYKYL<br>NWFQQKPGKAPKLLIYYNSLQTVPSRFSGS<br>GSGTDFTLTISLQPEDFATYYCFQYSSGPTF<br>GQGTKVEIKRTVAAPSVFIIPPSPDEQLKSGTAS<br>VVCLLNNFYPREAKVQWKVDNALQSGNSQ<br>SVTEQDSKSDSTYSLSSLTSLSKADYEKHKVYA<br>CEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:<br>173)                    |
| EVQLVESGGGLVQPGGSLRLSCAASGFTFSNY<br>MYWIRQAPGKGLIEWSSINTDGGSTYYADSVKG<br>RFTISRDN SKNTVYLQMNSLRAEDTAVYYCAKST<br>PNIPLAYWGQGLTVTVSSASTKGPSVFPPLAPSSK<br>STSGGTAALGCLVKDYFPEPVTVSWNSGALTSG<br>VHTFPAVLQSSGLYSLSSVTVPPSSSLGTQTYICN<br>VNHKPSNTKVDKRVPEPKSCDKTHTCPPCPAPELL<br>GGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHE<br>DPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRV<br>VSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTI<br>KAKGQPREPQVYTLPPSREEMTKNQVSLTCLVK<br>GFYPDSIAVEWESNGQPENNYKTTPPVLDSDGSF<br>FLYSLKTVDKSRWQQGNVFCSCVMHEALHNNHYT<br>QKSLSLSPGK (SEQ ID NO: 171)      | DIQMTQSPSSLSASVGDRTITCKASQNIYKYL<br>NWFQQKPGKAPKLLIYYNSLQTVPSRFSGS<br>GSGTDFTLTISLQPEDFAIYFCFQYSSGPTF<br>PGTKVEIKRTVAAPSVFIIPPSPDEQLKSGTASV<br>VCLLNNFYPREAKVQWKVDNALQSGNSQESV<br>TEQDSKSDSTYSLSSLTSLSKADYEKHKVYACE<br>VTHQGLSSPVTKSFNRGEC (SEQ ID NO: 147)                       |
| EVQLVESGGGLVQPGGSLRLSCAASGFTFSNY<br>MYWIRQAPGKGLIEWSSINTDGGSTYYADSVKG<br>RFTISRDN SKNTVYLQMNSLRAEDTAVYYCAKST<br>PNIPLAYWGQGLTVTVSSASTKGPSVFPPLAPSSK<br>STSGGTAALGCLVKDYFPEPVTVSWNSGALTSG<br>VHTFPAVLQSSGLYSLSSVTVPPSSSLGTQTYICN<br>VNHKPSNTKVDKRVPEPKSCDKTHTCPPCPAPELL<br>GGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHE<br>DPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRV<br>VSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTI<br>KAKGQPREPQVYTLPPSREEMTKNQVSLTCLVK<br>GFYPDSIAVEWESNGQPENNYKTTPPVLDSDGSF<br>FLYSLKTVDKSRWQQGNVFCSCVMHEALHNNHYT<br>QKSLSLSPGK (SEQ ID NO: 171)      | DIQMTQSPSSLSASVGDRTITCKASQNIYKYL<br>NWFQQKPGKAPKLLIYYNSLQTVPSRFSGS<br>GSGTDFTLTISLQPEDFAIYFCFQYSSGPTF<br>PGTKVEIKRTVAAPSVFIIPPSPDEQLKSGTASV<br>VCLLNNFYPREAKVQWKVDNALQSGNSQESV<br>TEQDSKSDSTYSLSSLTSLSKADYEKHKVYACE<br>VTHQGLSSPVTKSFNRGEC (SEQ ID NO: 147)                       |
| EVQLVESGGGLVQPGGSLRLSCAASGFTFSNY<br>MYWIRQAPGKGLIEWSSINTDGGSTYYADSVKG<br>RFTISRDN SKNTVYLQMNSLRAEDTAVYYCAKST<br>PNIPLAYWGQGLTVTVSSASTKGPSVFPPLAPSSK<br>STSGGTAALGCLVKDYFPEPVTVSWNSGALTSGV                                                                                                                                                                                                                                                                                                                                                    | DIQMTQSPSSLSASVGDRTITCKASQNIYKYL<br>NWFQQKPGKAPKLLIYYNSLQTVPSRFSGS<br>GSGTDFTLTISLQPEDFATYYCFQYSSGPTF<br>GQGTKVEIKRTVAAPSVFIIPPSPDEQLKSGTAS<br>VVCLLNNFYPREAKVQWKVDNALQSGNSQ                                                                                                      |

TABLE 6-continued

| Exemplary ALK2 antibody heavy and light chain sequences                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Heavy chain                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Light chain                                                                                                                                                                                                                                                       |
| HTFPAVLQSSGLYSLSVVTVPSSSLGTQTYICNV<br>NHKPSNTKVDKRVEPKSCDKTHTCPPCPAPELLG<br>GPSVFLFPPKPKDITLMI SRTPEVTCVVVDVSHED<br>PEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVV<br>SVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISK<br>AKGQPREPQVYTLPPSREEMTKNQVSLTCLVKG<br>FYPSPDIAVEWESNGQPENNYKTTTPVLDSDGSFF<br>LYSKLTVDKSRWQQGNVFPSCSVMHEALHNHYTQ<br>KSLSLSPGK (SEQ ID NO: 146)                                                                                                                                                                                              | SVTEQDSKDYSLSSSTLTLSKADYEKHKVYA<br>CEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:<br>172)                                                                                                                                                                                      |
| EVQLVESGGGLVQPGGSLRLSCAASGFTFSNYY<br>MYWIRQAPKGKLEWVASSIRGGSTYYPD SVKGR<br>FTISRDN SKNTLYLQMNSLRAEDTAVYYCAKSTP<br>NIPLAYWGQGLTVTVSSASTKGPSVFPLAPSSKS<br>TSGGTAALGCLVKDYFPEPVTVSWNSGALTSGV<br>HTFPAVLQSSGLYSLSVVTVPSSSLGTQTYICNV<br>NHKPSNTKVDKRVEPKSCDKTHTCPPCPAPELLG<br>GPSVFLFPPKPKDITLMI SRTPEVTCVVVDVSHED<br>PEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVV<br>SVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISK<br>AKGQPREPQVYTLPPSREEMTKNQVSLTCLVKG<br>FYPSPDIAVEWESNGQPENNYKTTTPVLDSDGSFF<br>LYSKLTVDKSRWQQGNVFPSCSVMHEALHNHYTQ<br>KSLSLSPGK (SEQ ID NO: 146) | DIQMTQSPSSLSASVGRVITITCKASQNIYKYL<br>NWFQQKPGKAPKLLIYYNSLTQGI PSRFSGS<br>SGSGTDFTLTISSLQPEDFATYYCFQYSSGPTF<br>GQGTKEIKRTVAAPSVFIFPPSDEQLKSGTAS<br>VVCLLNNFYPREAKVQWKVDNALQSGNSQ<br>SVTEQDSKDYSLSSSTLTLSKADYEKHKVYA<br>CEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:<br>173)   |
| EVQLVESGGGLVQPGGSLRLSCAASGFTFSYYA<br>MSWVRQAPKGKLEWVASSIRGGDNTYYRDSVK<br>GRFTTSTRDN SKNTLYLQMNSLRAEDTAVYYCARL<br>NYYNYFDYWGQGLTVTVSSASTKGPSVFPLAPS<br>SKSTSGGTAALGCLVKDYFPEPVTVSWNSGALT<br>GVHTTFAVLQSSGLYSLSVVTVPSSSLGTQTYIC<br>NVNHKPSNTKVDKRVEPKSCDKTHTCPPCPAPE<br>LLGGPSVFLFPPKPKDITLMI SRTPEVTCVVVDVSH<br>EDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYR<br>VVS VLTVLHQDWLNGKEYKCKVSNKALPAPIEKT<br>SKAKGQPREPQVYTLPPSREEMTKNQVSLTCLV<br>KGFYPSDIAVEWESNGQPENNYKTTTPVLDSDG<br>SFFLYSKLTVDKSRWQQGNVFPSCSVMHEALHNH<br>YTQKSLSLSPGK (SEQ ID NO: 174)  | DIQMTQSPSSLSASVGRVITITCQASQDIGNW<br>LSWYQQKPGKAPKLLIYGATSLADGVPSRFSG<br>SGSGTDFTLTISSLQPEDFATYYCLQAYSAPFT<br>FGQGTKEIKRTVAAPSVFIFPPSDEQLKSGTA<br>SVVCLLNNFYPREAKVQWKVDNALQSGNSQ<br>ESVTEQDSKDYSLSSSTLTLSKADYEKHKVY<br>ACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:<br>175)  |
| EVQLVESGGGLVQPGGSLRLSCAASGFTFSYYA<br>MSWVRQAPKGKLEWVASSIRGGDNTYYRDSVK<br>GRFTTSTRDN SKNTLYLQMNSLRAEDTAVYYCARL<br>NYYNYFDYWGQGLTVTVSSASTKGPSVFPLAPS<br>SKSTSGGTAALGCLVKDYFPEPVTVSWNSGALT<br>GVHTTFAVLQSSGLYSLSVVTVPSSSLGTQTYIC<br>NVNHKPSNTKVDKRVEPKSCDKTHTCPPCPAPE<br>LLGGPSVFLFPPKPKDITLMI SRTPEVTCVVVDVSH<br>EDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYR<br>VVS VLTVLHQDWLNGKEYKCKVSNKALPAPIEKT<br>SKAKGQPREPQVYTLPPSREEMTKNQVSLTCLV<br>KGFYPSDIAVEWESNGQPENNYKTTTPVLDSDG<br>SFFLYSKLTVDKSRWQQGNVFPSCSVMHEALHNH<br>YTQKSLSLSPGK (SEQ ID NO: 174)  | DIQMTQSPSSLSASVGRVITITCQASQDIGNW<br>LSWYQQKPGKAPKLLIYGATSLADGVPSRFSG<br>SRSGTDYTLTISSLQPEDFATYYCLQAYSAPFT<br>FGQGTKEIKRTVAAPSVFIFPPSDEQLKSGTA<br>SVVCLLNNFYPREAKVQWKVDNALQSGNSQ<br>ESVTEQDSKDYSLSSSTLTLSKADYEKHKVY<br>ACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:<br>176)  |
| EVQLVESGGGLVQPGGSLRLSCAASGFTFSYYA<br>MSWVRQAPKGKLEWVASSIRGGDNTYYRDSVK<br>GRFTTSTRDN SKNTLYLQMNSLRAEDTAVYYCARL<br>NYYNYFDYWGQGLTVTVSSASTKGPSVFPLAPS<br>SKSTSGGTAALGCLVKDYFPEPVTVSWNSGALT<br>GVHTTFAVLQSSGLYSLSVVTVPSSSLGTQTYIC<br>NVNHKPSNTKVDKRVEPKSCDKTHTCPPCPAPE<br>LLGGPSVFLFPPKPKDITLMI SRTPEVTCVVVDVSH<br>EDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYR<br>VVS VLTVLHQDWLNGKEYKCKVSNKALPAPIEKT<br>SKAKGQPREPQVYTLPPSREEMTKNQVSLTCLV<br>KGFYPSDIAVEWESNGQPENNYKTTTPVLDSDG<br>SFFLYSKLTVDKSRWQQGNVFPSCSVMHEALHNH<br>YTQKSLSLSPGK (SEQ ID NO: 174)  | DIQMTQSPSSLSASVGRVITITCQASQDIGNW<br>LSWYQQKPGKAPKLLIYGATSLADGVPSRFSG<br>SRSGTQYTLTISSLQPEDFATYYCLQAYSAPFT<br>FGSGTKVEIKRTVAAPSVFIFPPSDEQLKSGTA<br>SVVCLLNNFYPREAKVQWKVDNALQSGNSQ<br>ESVTEQDSKDYSLSSSTLTLSKADYEKHKVY<br>ACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:<br>151) |
| EVQLVESGGGLVQPGGSLRLSCAASGFTFSYYA<br>MSWVRQAPKGKLEWVASSIRGGDNTYYRDSVK<br>GRFTTSTRDN SKNTLYLQMNSLRAEDTAVYYCARL<br>NYYNYFDYWGQGLTVTVSSASTKGPSVFPLAPS<br>SKSTSGGTAALGCLVKDYFPEPVTVSWNSGALT<br>GVHTTFAVLQSSGLYSLSVVTVPSSSLGTQTYIC<br>NVNHKPSNTKVDKRVEPKSCDKTHTCPPCPAPE<br>LLGGPSVFLFPPKPKDITLMI SRTPEVTCVVVDVSH<br>EDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYR<br>VVS VLTVLHQDWLNGKEYKCKVSNKALPAPIEKT<br>SKAKGQPREPQVYTLPPSREEMTKNQVSLTCLV<br>KGFYPSDIAVEWESNGQPENNYKTTTPVLDSDG<br>SFFLYSKLTVDKSRWQQGNVFPSCSVMHEALHNH<br>YTQKSLSLSPGK (SEQ ID NO: 174)  | DIQMTQSPSSLSASVGRVITITCQASQDIGNW<br>LSWYQQKPGKAPKLLIYGATSLADGVPSRFSG<br>SGSGTDFTLTISSLQPEDFATYYCLQAYSAPFT                                                                                                                                                         |

TABLE 6-continued

| Exemplary ALK2 antibody heavy and light chain sequences                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Heavy chain                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Light chain                                                                                                                                                                                                                                                             |
| NYNNYFDYWGQGVLTIVSSASTKGPSVFPLAPS<br>SKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTS<br>GVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYIC<br>NVNHKPSNTKVKRVEPKSCDKTHTCPPCPAPE<br>LLGGPSVFLFPPKPKDTLMISRTPEVTCVVDVSH<br>EDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYR<br>VVSVLTVLHQDWLNGKEYCKVSNKALPAPIEKT<br>SKAKGQPREPQVYTLPPSREEMTKNQVSLTCLV<br>KGFYPSDIAVEWESNGQPENNYKTTTPVLDSDG<br>SFFLYSKLTVDKSRWQQGNVFCSCVMHEALHNH<br>YTQKSLSLSPGK (SEQ ID NO: 150)                                                                                                               | FGQGTKVEIKRTVAAPSVFIFPPSDEQLKSGTA<br>SVVCLLNNFYPREAKVQWKVDNALQSGNSQ<br>ESVTEQDSKSDSTYLSSTLTLSKADYEKHKVY<br>ACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:<br>175)                                                                                                                   |
| EVQLVESGGGLVQPGGSLRLSCAASGFTFSYYA<br>MSWVRQAPGKGLEWVASISRGDNTYYRDSVK<br>GRFTTSRDNSKNTLYLQMNSLRAEDTATYYCARL<br>NYNNYFDYWGQGVLTIVSSASTKGPSVFPLAPS<br>SKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTS<br>GVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYIC<br>NVNHKPSNTKVKRVEPKSCDKTHTCPPCPAPE<br>LLGGPSVFLFPPKPKDTLMISRTPEVTCVVDVSH<br>EDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYR<br>VVSVLTVLHQDWLNGKEYCKVSNKALPAPIEKT<br>SKAKGQPREPQVYTLPPSREEMTKNQVSLTCLV<br>KGFYPSDIAVEWESNGQPENNYKTTTPVLDSDG<br>SFFLYSKLTVDKSRWQQGNVFCSCVMHEALHNH<br>YTQKSLSLSPGK (SEQ ID NO: 150) | DIQMTQSPSSLSASVGRVITITCQASQDIGNW<br>LSWYQQKPGKAPKLLIYGATSLADGVPSTRFSG<br>SRSGTDYTLTISSLQPEDFATYYCLQAYSAPFT<br>FGQGTKVEIKRTVAAPSVFIFPPSDEQLKSGTA<br>SVVCLLNNFYPREAKVQWKVDNALQSGNSQ<br>ESVTEQDSKSDSTYLSSTLTLSKADYEKHKVY<br>ACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:<br>176)     |
| EVQLQQSGAELVRPGASVRLSCASGFTFSYYA<br>HWVKQRPEQGLEWIGWIDPEDGETKYAPNFQDK<br>ATITAVTSSNTAYLQLSSLTSDSAIYYCARYTSDY<br>YTMDFYWGQGSTVTVSSAKTTPPSVYPLAPGFAA<br>QTNMVTLGCLVKGYFPEPVTVWNSGSLSSGV<br>HTFPAVLQSDLYTLSSSVTVPSSTWPSQTVTCNV<br>AHPASSTKVDKKIVPRDCGCKPCICTVPEVSSVFI<br>FPPKPKDVLITITLTPKVTCTVVDISKDDPEVQFSW<br>FVDDDEVHTAQTKPREEQINSTRFSVSELPIMHQ<br>DWLNGKEFKCRVNSAAPPAPIEKTISKTGRPKA<br>PQVYTIPTPPKEQMAKDVSILTCMITNFFPEDITVE<br>WQWNGQPAENYKNTQPIIMDTDGSYFVYSKLV<br>QKSNWEAGNTFTCSVLHEGLHNHTEKSLSHSP<br>GK (SEQ ID NO: 209)    | DIQMTQSPASQASLGEVSTFTCLASQTIGTW<br>LAWYQQKPGKSPQLLIYAATSLADGVPSTRFSG<br>SGSGTKFSPFKISSLQAEDEFASYCQQLYSTPW<br>TFGGGTKLEIKRADAAPTIVSIFPPSSSEQLTSGG<br>ASVVCFLNNFYPKDINVKWKIDGSRQNGVLN<br>SWTDQDSKSDSTYSMSSTLTLTKEDEYERHNSY<br>TCEATHKTSTSPIVKSFNRENEC (SEQ ID NO:<br>210) |
| EVQLQQSGAELVRPGASVRLSCASGFTFSYYA<br>HWVKQRPEQGLEWIGWIDPEDGETKYAPNFQDK<br>ATITAVTSSNTAYLQLSSLTSDSAIYYCARYTSDY<br>YTMDFYWGQGSTVTVSSAKTTPPSVYPLAPGSA<br>QTNMVTLGCLVKGYFPEPVTVWNSGSLSSGV<br>HTFPAVLQSDLYTLSSSVTVPSSTWPSQTVTCNV<br>AHPASSTKVDKKIVPRDCGCKPCICTVPEVSSVFI<br>FPPKPKDVLITITLTPKVTCTVVDISKDDPEVQFSW<br>FVDDDEVHTAQTKPREEQINSTRFSVSELPIMHQ<br>DWLNGKEFKCRVNSAAPPAPIEKTISKTGRPKA<br>PQVYTIPTPPKEQMAKDVSILTCMITNFFPEDITVE<br>WQWNGQPAENYKNTQPIIMDTDGSYFVYSKLV<br>QKSNWEAGNTFTCSVLHEGLHNHTEKSLSHSP<br>GK (SEQ ID NO: 211)     | DIQMTQSPASQASLGEVSTFTCLASQTIGTW<br>LAWYQQKPGKSPQLLIYAATSLADGVPSTRFSG<br>SGSGTKFSPFKISSLQAEDEFASYCQQLYSTPW<br>TFGGGTKLEIKRADAAPTIVSIFPPSSSEQLTSGG<br>ASVVCFLNNFYPKDINVKWKIDGSRQNGVLN<br>SWTDQDSKSDSTYSMSSTLTLTKEDEYERHNSY<br>TCEATHKTSTSPIVKSFNRENEC (SEQ ID NO:<br>212) |
| EVQLQQSGAELVRPGASVRLSCASGFTFSYYA<br>HWVKQRPEQGLEWIGWIDPEDGETKYAPNFQSK<br>ATITAVTSSNTAYLQLSSLTSDSAIYYCARYTSPY<br>YTMDFYWGQGSTVTVSSAKTTPPSVYPLAPGSA<br>QTNMVTLGCLVKGYFPEPVTVWNSGSLSSGV<br>HTFPAVLQSDLYTLSSSVTVPSSTWPSQTVTCNV<br>AHPASSTKVDKKIVPRDCGCKPCICTVPEVSSVFI<br>FPPKPKDVLITITLTPKVTCTVVDISKDDPEVQFSW<br>FVDDDEVHTAQTKPREEQINSTRFSVSELPIMHQ<br>DWLNGKEFKCRVNSAAPPAPIEKTISKTGRPKA<br>PQVYTIPTPPKEQMAKDVSILTCMITNFFPEDITVE<br>WQWNGQPAENYKNTQPIIMDTDGSYFVYSKLV<br>QKSNWEAGNTFTCSVLHEGLHNHTEKSLSHSP<br>GK (SEQ ID NO: 213)     | DIQMTQSPASQASLGEVSTFTCLASQTIGTW<br>LAWYQQKPGKSPQLLIYAATSLADGVPSTRFSG<br>SGSGTKFSPFKISSLQAEDEFASYCQQVYSTPW<br>TFGGGTKLEIKRADAAPTIVSIFPPSSSEQLTSGG<br>ASVVCFLNNFYPKDINVKWKIDGSRQNGVLN<br>SWTDQDSKSDSTYSMSSTLTLTKEDEYERHNSY<br>TCEATHKTSTSPIVKSFNRENEC (SEQ ID NO:<br>214) |

TABLE 6-continued

| Exemplary ALK2 antibody heavy and light chain sequences                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Heavy chain                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Light chain                                                                                                                                                                                                                                                         |
| EVQLQQSGAELVRPGASVRLSCTASGFNIKDSLM<br>HWVKQRPEQGLEWIGWIDPEDGETRYAPNFQDK<br>ATITAVTSSNTAYLQLSSTLSEDSAIYYCARYTSRY<br>YTMEYWGQGTSTVTVSSAKTTPSPVYPLAPGSAA<br>QTNSMVTLGCLVKGYFPEPVTVWNSGSLSSGV<br>HTFPAVLQSDLYTLSSSVTVPSSTWPSQTVTCNV<br>AHPASSTKVDKKIVPRDCGCKPCICTVPEVSSVFI<br>FPPKPKDVLITITLTPKVTCTVVDISKDDPEVQFSW<br>FVDDVEVHTAQTKPREEQINSTFRSVSELPIMHQ<br>DWLNGKEFKCRVNSAAPPAPIEKTISKTKGRPKA<br>PQVYTITPPKPEQMAKDKVSLTCTMITNFFPEDITVE<br>WQWNGQPAENYKNTQPIMDTDGSYFVYSKLN<br>QKSNWEAGNTFTCSVLHEGLHNHTEKSLSHSP<br>GK (SEQ ID NO: 215)  | DIQMTQSPASQASLGESEVTFTCLASQTIGTW<br>LAWYQQKPGKSPQLLIYAATSLADGVPSRFSG<br>SGSGTKFSFKISSQLQAEDFASYCQQLYWTP<br>WTFGGGKLEIKRADAAPTVISIFPPSSEQLTS<br>GGASVVCFLNNFYPKDINVKKIDGSEKQNG<br>VLNSWTDQDSKDYSTYSMSSTLTITKDEYERH<br>NSYTCEATHKTSSTPIVKSFNRNEC (SEQ ID<br>NO: 216)  |
| QVHLVESGGGVQPGSRSLRSCAASGFTFSNYGI<br>HWVRQVPKGLEWVAVISYDGNKYYGDSVKG<br>RFTISRDNSTNTLYLQMNSLRPEDTAIYYCAKDRG<br>LDTAGDYFDYWGQGLVTVSSASTKGPSVFPLAP<br>CSRSTSESTAALGCLVKDYFPEPVTVSWNSGALT<br>SGVHTFPAVLQSSGLYSLSSVTVPSSSLGKTKY<br>TCNVDHKPSNTKVDKRVEKYGPPCPPCPAPEF<br>LGGPSVFLFPPKPKDITLMISRTPEVTCVVVDVSDQ<br>EDPEVQFNWYVDGVEVHNAKTKPREEQFNSTYR<br>VVSVELTVLHQDWLNGKEYCKVSNKGLPSSIEKTI<br>SKAKGQPREPQVYTLPPSQEEMTKNQVSLTCLV<br>KGFYPSDIAVEWESNGQPENNYKTTTPVLDSDG<br>SFFLYSRLTVDKSRWQEGNVFSCSVMHEALHNH<br>YTQKSLSLSLGK (SEQ ID NO: 343) | EIVMTQSPATLSVSTGERATLSRASQNISSKL<br>AWYQQKPGQAPRLLIYGASTRASGIPARFWG<br>SGSGTEFTLTISLQSEDFAVFYCLQYNNWMT<br>FGGGTKVEIKRTVAAPSVFIFPPSDEQLKSGTA<br>SVVCLLNNFYPREAKVQWKVDNALQSGNSQE<br>SVTEQDSKDYSTYSLSSTLTLSKADYEKHKVYA<br>CEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:<br>344)    |
| QVQLQQWAGALLKPSSETLSLTCVAVNGSFSSYY<br>WNWIRQPPGKLEWLGIEINHRGSTNYNPSLTSR<br>VTISVDTSKNQFSLKLTSTVTAADTAIYYCTAHTRG<br>FLYWGGQGLVTVSSASTKGPSVFPLAPCSRSTSE<br>STAALGCLVKDYFPEPVTVSWNSGALTSGVHTFP<br>AVLQSSGLYSLSSVTVPSSSLGKTKYTCNVDHK<br>PSNTKVDKRVEKYGPPCPPCPAPEFLGGPSVFL<br>FPPKPKDITLMISRTPEVTCVVVDVSDQEDPEVQFN<br>WYVDGVEVHNAKTKPREEQFNSTYRVSVELTVL<br>HQDWLNGKEYCKVSNKGLPSSIEKTIKAKGQP<br>REPQVYTLPPSQEEMTKNQVSLTCLVKGFYPSDI<br>AVEWESNGQPENNYKTTTPVLDSDGSFFLYSRLT<br>VDKSRWQEGNVFSCSVMHEALHNHYTEKSLSLSL<br>LGK (SEQ ID NO: 345)   | DIQMTQSPSSLSASVGDRTVITCQASQDINIYL<br>NWYQQKPGKAPKLLIYDVSNLVPGVPSRFSG<br>TSGGTHFSFTISLQPEDIAITYYCQHYDDLPLT<br>FGGTTVAIRRTVAAPSVFIFPPSDEQLKSGTA<br>SVVCLLNNFYPREAKVQWKVDNALQSGNSQE<br>SVTEQDSKDYSTYSLSSTLTLSKADYEKHKVYA<br>CEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:<br>346)  |
| QVQLQQWAGALLKPSSETLSLTCVAVNGSFSSFY<br>WSWIRQPPGKLEWLGIEINHRGRTNNNPSLASR<br>VTISVDTSKNSQLSLRLTSTVTAADTAIYYCTAHTRG<br>FLYWGGQGLVTVSSASTKGPSVFPLAPCSRSTSE<br>STAALGCLVKDYFPEPVTVSWNSGALTSGVHTFP<br>AVLQSSGLYSLSSVTVPSSSLGKTKYTCNVDHK<br>PSNTKVDKRVEKYGPPCPPCPAPEFLGGPSVFL<br>FPPKPKDITLMISRTPEVTCVVVDVSDQEDPEVQFN<br>WYVDGVEVHNAKTKPREEQFNSTYRVSVELTVL<br>HQDWLNGKEYCKVSNKGLPSSIEKTIKAKGQP<br>REPQVYTLPPSQEEMTKNQVSLTCLVKGFYPSDI<br>AVEWESNGQPENNYKTTTPVLDSDGSFFLYSRLT<br>VDKSRWQEGNVFSCSVMHEALHNHYTEKSLSLSL<br>LGK (SEQ ID NO: 347)  | DIQMTQSPSSLSASVGDRTVITCQASQDINNYL<br>NWYQQKPGKAPNLLIYDVFNLLPGVPSRFSG<br>SGSGTDFSTISLQPEDIAITYYCQHYDDLPLT<br>FGGTTVAIRRTVAAPSVFIFPPSDEQLKSGTA<br>SVVCLLNNFYPREAKVQWKVDNALQSGNSQE<br>SVTEQDSKDYSTYSLSSTLTLSKADYEKHKVYA<br>CEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:<br>348)   |
| QVQLQQWAGALLKPSSETLSLTCVAVNGSFSSYY<br>WSWIRQPPGKLEWLGIEINHRGRTNNNPSLTSR<br>VTISVDTSKNQFSLKLTSTVTAADTAIYYCSAHTRG<br>FLYWGGQGLVTVSSASTKGPSVFPLAPCSRSTSE<br>STAALGCLVKDYFPEPVTVSWNSGALTSGVHTFP<br>AVLQSSGLYSLSSVTVPSSSLGKTKYTCNVDHK<br>PSNTKVDKRVEKYGPPCPPCPAPEFLGGPSVFL<br>FPPKPKDITLMISRTPEVTCVVVDVSDQEDPEVQFN<br>WYVDGVEVHNAKTKPREEQFNSTYRVSVELTVL<br>HQDWLNGKEYCKVSNKGLPSSIEKTIKAKGQP<br>REPQVYTLPPSQEEMTKNQVSLTCLVKGFYPSDI<br>AVEWESNGQPENNYKTTTPVLDSDGSFFLYSRLT<br>VDKSRWQEGNVFSCSVMHEALHNHYTEKSLSLSL<br>LGK (SEQ ID NO: 347)   | DIQMTQSPSSLSASVGDRTVITCQASQDINNYL<br>NWYQQKPGKAPKLLIYDASNLVPGVPSRFSGT<br>SGSGTDFSTISLQPEDIAITYYCQHYDDLPLTF<br>GGGTTVAISRTVAAPSVFIFPPSDEQLKSGTAS<br>VVCLLNNFYPREAKVQWKVDNALQSGNSQES<br>VTEQDSKDYSTYSLSSTLTLSKADYEKHKVYAC<br>EVTHQGLSSPVTKSFNRGEC (SEQ ID NO:<br>350) |

TABLE 6-continued

| Exemplary ALK2 antibody heavy and light chain sequences                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Heavy chain                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Light chain                                                                                                                                                                                                                                                                           |
| <p>AVEWESNGQPENNYKTTTPVLDSGDSFFLYSRLT<br/>VDKSRWQEGNVFSCSVMHEALHNHYTQKSLSL<br/>LGK (SEQ ID NO: 349)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                       |
| <p>QVQLQQWAGALLKPSSETLSLTCAVYNGSFSSYY<br/>WNWIRQPPGKGLEWLGELNHRGSTNYNPSLTSR<br/>VTISVDTSKNQFSLKLTSTVTAADTAIYYCTAHTRG<br/>FLYWGGQTLVTVSSASTKGPSVFPLAPCSRSTSE<br/>STAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPP<br/>AVLQSSGLYSLSSVTVPSSSLGKTTYTCNVDPHK<br/>PSNTKVDKRVESKYGPPCPPCPAPEFLGGPSVFL<br/>FPPKPKDTLMISRTPEVTCVVVDVQEDPEVQFN<br/>WYVDGVEVHNAKTKPREEQFNSTYRVVSVLTVL<br/>HQDWLNGKEYKCKVSNKGLPSSIEKTIKAKGQP<br/>REPQVYTLPPSQEEMTKNQVSLTCLVKGFYPSDI<br/>AVEWESNGQPENNYKTTTPVLDSGDSFFLYSRLT<br/>VDKSRWQEGNVFSCSVMHEALHNHYTQKSLSL<br/>LGK (SEQ ID NO: 345)</p>   | <p>DIQMTQSPSSLSASVGDRTITTCASQDINNYL<br/>NWYQQKPGKAPKLLIYDASNLVPGVPSRFSGT<br/>GSGTDFSTISSLPQEDIATYYCQHYVDLPLTF<br/>GGGTTVAIRRTVAAPSVEIFPPSDEQLKSGTAS<br/>VVCLLNNFYPREAKVQWKVDNALQSGNSQES<br/>VTEQDSKDSYSLSTLTLSKADYEKHKVYAC<br/>EVTHQGLSSPVTKSFNRGEC (SEQ ID NO:<br/>351)</p>          |
| <p>EIQLVESGGDLVPGGSLKLSCAASGFTFSDSAM<br/>HWVRQASGKLEWVGRIKNTATYATSYAASVK<br/>GRFTVSTDDSKTAYLQMRYLKIEDTAIYYCTSDQ<br/>FDYWGGQTLVTVSSASTKGPSVFPLAPCSRSTSE<br/>STAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPP<br/>AVLQSSGLYSLSSVTVPSSSLGKTTYTCNVDPHK<br/>PSNTKVDKRVESKYGPPCPPCPAPEFLGGPSVFL<br/>FPPKPKDTLMISRTPEVTCVVVDVQEDPEVQFN<br/>WYVDGVEVHNAKTKPREEQFNSTYRVVSVLTVL<br/>HQDWLNGKEYKCKVSNKGLPSSIEKTIKAKGQP<br/>REPQVYTLPPSQEEMTKNQVSLTCLVKGFYPSDI<br/>AVEWESNGQPENNYKTTTPVLDSGDSFFLYSRLT<br/>VDKSRWQEGNVFSCSVMHEALHNHYTQKSLSL<br/>LGK (SEQ ID NO: 352)</p>        | <p>DVVMTQSPSLSATLGQPASISCRSSQSLVYS<br/>DGI TYLTFWQQRPGQSPTRLIYQVSNRDSGVP<br/>DRFNGSGSGTDFTLNISRVEAEDVGFIYCYME<br/>GTHWPWTFGQGTKEVLRKRTVAAPSVEIFPPS<br/>DEQLKSGTASVVCLLNNFYPREAKVQWKVDN<br/>ALQSGNSQESVTEQDSKDSYSLSTLTLSKA<br/>DYEKHKVYACEVTHQGLSSPVTKSFNRGEC<br/>(SEQ ID NO: 353)</p> |
| <p>QVQLVESGGGLVKPGGSLRLSCAASGFTFSDYY<br/>MSWIRQAPGKGLEWYSFISRTGSKYYADSVKGR<br/>FTISRDNAKNSLNLQMSLRAEDTAVYFCAREPL<br/>QHWGGQTLVTVSSASTKGPSVFPLAPCSRSTSE<br/>STAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPP<br/>AVLQSSGLYSLSSVTVPSSSLGKTTYTCNVDPHK<br/>PSNTKVDKRVESKYGPPCPPCPAPEFLGGPSVFL<br/>FPPKPKDTLMISRTPEVTCVVVDVQEDPEVQFN<br/>WYVDGVEVHNAKTKPREEQFNSTYRVVSVLTVL<br/>HQDWLNGKEYKCKVSNKGLPSSIEKTIKAKGQP<br/>REPQVYTLPPSQEEMTKNQVSLTCLVKGFYPSDI<br/>AVEWESNGQPENNYKTTTPVLDSGDSFFLYSRLT<br/>VDKSRWQEGNVFSCSVMHEALHNHYTQKSLSL<br/>LGK (SEQ ID NO: 354)</p>        | <p>NVLTQTPISSPVILGQPASISCRSSQSLVHSD<br/>GDTYLSWLHQRPGQPPRLLIYTVSNRVSGVP<br/>DRFSGSGAGTDFTLKISRVEAEDVGFIYCYMQ<br/>GTQFPALTFGGGTKVEIKRTVAAPSVEIFPPSD<br/>EQLKSGTASVVCLLNNFYPREAKVQWKVDNA<br/>LQSGNSQESVTEQDSKDSYSLSTLTLSKAD<br/>YEKHKVYACEVTHQGLSSPVTKSFNRGEC<br/>(SEQ ID NO: 355)</p>  |
| <p>QVQLVESGGGLVKPGGSLRLSCAASGFTFSDYY<br/>MSWIRQAPGKGLEWYSFISRTGSKYYADSVKGR<br/>RFTISRDNAKNSLNLQMSLRAEDTAVYFCAREP<br/>LQHWGGQTLVTVSSASTKGPSVFPLAPCSRSTSE<br/>STAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPP<br/>PAVLQSSGLYSLSSVTVPSSSLGKTTYTCNVDPHK<br/>KPSNTKVDKRVESKYGPPCPPCPAPEFLGGPSV<br/>FLFPPKPKDTLMISRTPEVTCVVVDVQEDPEVQ<br/>FNWYVDGVEVHNAKTKPREEQFNSTYRVVSVLTVL<br/>VLHQDWLNGKEYKCKVSNKGLPSSIEKTIKAKG<br/>QPREPQVYTLPPSQEEMTKNQVSLTCLVKGFYPS<br/>SDIAVEWESNGQPENNYKTTTPVLDSGDSFFLYS<br/>RLTVDKSRWQEGNVFSCSVMHEALHNHYTQKSL<br/>SLSLGK (SEQ ID NO: 356)</p> | <p>KIVLTQTPLSPVTLGQPASISCRSSQSLVHSD<br/>GDTYLSWLHQRPGQPPRLLIYTVSNRSGVPD<br/>RFSGSGAGTDFTLKISRVEAEDVGFIYCYMQG<br/>TQFPALTFGGGTKVEIKRTVAAPSVEIFPPSDE<br/>QLKSGTASVVCLLNNFYPREAKVQWKVDNAL<br/>QSGNSQESVTEQDSKDSYSLSTLTLSKADY<br/>EKHKVYACEVTHQGLSSPVTKSFNRGEC<br/>(SEQ ID NO: 357)</p>   |
| <p>EMQLVESGGGLVPGGSLKVS CAASGFTLSDSSI<br/>HWVRQASGKLEWIGRIKSKPYSYATAYAAVK<br/>GRFTISRDDSKNTAFLQMSGLTETDVAIYYCTGG<br/>DDFWGGQTLVTVSSASTKGPSVFPLAPCSRSTSE<br/>STAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPP<br/>PAVLQSSGLYSLSSVTVPSSSLGKTTYTCNVDPHK<br/>KPSNTKVDKRVESKYGPPCPPCPAPEFLGGPSV<br/>FLFPPKPKDTLMISRTPEVTCVVVDVQEDPEVQ<br/>FNWYVDGVEVHNAKTKPREEQFNSTYRVVSVLTVL</p>                                                                                                                                                                                       | <p>DIQMTQSPSTLSASVGDRTITTCRASQSI STLW<br/>AWYQQKPGKAPKLLIYKASLEGVPSRFSGS<br/>GSGTDFTLTISSLQPDDEFATYYCQYNTYWTF<br/>GGGTTVAIRRTVAAPSVEIFPPSDEQLKSGTAS<br/>VVCLLNNFYPREAKVQWKVDNALQSGNSQES<br/>VTEQDSKDSYSLSTLTLSKADYEKHKVYAC<br/>EVTHQGLSSPVTKSFNRGEC (SEQ ID NO:<br/>359)</p>          |



TABLE 6-continued

| Exemplary ALK2 antibody heavy and light chain sequences                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Heavy chain                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Light chain                                                                                                                                                                                                                                                          |
| VLHQDWLNGKEYKCKVSNKGLPSSIEKTIKAKG<br>QPREPQVYTLPPSQEEMTKNQVSLTCLVKGFY<br>SDIAVEWESNGQPENNYKTPPVLDSDGSFFLYS<br>RLTVDKSRWQEGNVFSCSVMEALHNHYTQKSL<br>SLSLGK (SEQ ID NO: 358)                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                      |
| EVQLVESGGGLVQPGGSLRLSCAASGFTFSSYA<br>MSWVRQAPGKLEWWSAISGSGGTYADSVK<br>GRFTISRDNKNTLYLQMNSLRAEDTAVYYCAIPY<br>NWNYGGAFDIWGGTMTVTVSSASTKGPSVFPLA<br>PCSRSTSESTAALGCLVKDYFPEPVTVSWNSGAL<br>TSGVHTFPFPAVLQSSGLYSLSVTVTPSSSLGKT<br>YTCNVDHKPSNTKVDKRVESKYGPPCPPCPAPE<br>FLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVS<br>QEDPEVQFNWYVDGVEVHNAKTKPREEQFNSTY<br>RVSVLTVLHQDWLNGKEYKCKVSNKGLPSSIEK<br>TISKAKGQPREPQVYTLPPSQEEMTKNQVSLTCL<br>VKGFYPSDIAVEWESNGQPENNYKTPPVLDSD<br>GSFFLYSRLTVDKSRWQEGNVFSCSVMEALHN<br>HYTQKSLSLSLGK (SEQ ID NO: 360) | EIVMTQSPATLSVSPGKRATLSCRASQSVSSN<br>LAWYQQKPGQAPRLLIYGASTRATGIPARFSG<br>SGSGTEFTLTISLQSEDFAVYYCQQYNWPP<br>YTFGGGTGLEIKRTVAAPSVFIFPPSDEQLKSG<br>TASVVCLLNNFYPREAKVQWKVDNALQSGNS<br>QESVTEQDSKDYSLSSLTLSKADYEKHKV<br>YACEVTHQGLSSPVTKSFNRGEC (SEQ ID<br>NO: 361)       |
| QVQLVQSGAEVKKPGASVKVSKASGYTFTGYY<br>MHWVRQAPGQGLEWMGWINPNVSGTNAQKFKQ<br>GRVTMTDRDTSISTAYMELSRRLSDDTAVYYCTR<br>GAAAGLFDYWGQGLTVTVSSASTKGPSVFPLAP<br>CSRSTSESTAALGCLVKDYFPEPVTVSWNSGALT<br>SGVHTFPFPAVLQSSGLYSLSVTVTPSSSLGKT<br>TCNVDHKPSNTKVDKRVESKYGPPCPPCPAPEF<br>LGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSQ<br>EDPEVQFNWYVDGVEVHNAKTKPREEQFNSTYR<br>VSVLTVLHQDWLNGKEYKCKVSNKGLPSSIEKTI<br>SKAKGQPREPQVYTLPPSQEEMTKNQVSLTCLV<br>KGFYPSDIAVEWESNGQPENNYKTPPVLDSDG<br>SFFLYSRLTVDKSRWQEGNVFSCSVMEALHNH<br>YTQKSLSLSLGK (SEQ ID NO: 362) | DIQMTQSPSSLSASVGRVITITCRASQGINNYL<br>AWYQQKPGKVPKLLIYAASLTQSGVPSRFGS<br>SGGTDFTLTISLQPEDVATYYCQKFNAPLTF<br>GGGKVEIKRTVAAPSVFIFPPSDEQLKSGTAS<br>VVCLLNNFYPREAKVQWKVDNALQSGNSQES<br>VTEQDSKDYSLSSLTLSKADYEKHKVYAC<br>EVTHQGLSSPVTKSFNRGEC (SEQ ID NO:<br>363)          |
| EVQLVESGGGLVQPGGSLRLSCAASGFTFSTYA<br>MSWVRQAPGKLEWWSAISGSGGTYADSVK<br>GRFTISRDNKNTLYLQMNSLRVEDTAVYYCANS<br>PSWFDPPWGGTGLTVTVSSASTKGPSVFPLAPCSR<br>STSESTAALGCLVKDYFPEPVTVSWNSGALTSGV<br>HTFPFPAVLQSSGLYSLSVTVTPSSSLGKT<br>VDHKPSNTKVDKRVESKYGPPCPPCPAPEFLGG<br>PSVFLFPPKPKDTLMISRTPEVTCVVVDVSQEDP<br>EVQFNWYVDGVEVHNAKTKPREEQFNSTYRVV<br>VSVLTVLHQDWLNGKEYKCKVSNKGLPSSIEKTI<br>KAGQPREPQVYTLPPSQEEMTKNQVSLTCLVKGF<br>YPSDIAVEWESNGQPENNYKTPPVLDSDGSFFL<br>YSRLTVDKSRWQEGNVFSCSVMEALHNHYTQK<br>SLSLSLGK (SEQ ID NO: 364)        | DIQMTQSPSSLSASVGRVITITCRASQSISSYL<br>NWYQQKPGKAPKLLIYAASSLQSGVPSRFGS<br>SGSGTDFTLTISLQPEDFATYYCQQSYSTPPI<br>TFGGGTGLEIKRTVAAPSVFIFPPSDEQLKSGT<br>ASVVCLLNNFYPREAKVQWKVDNALQSGNSQ<br>ESVTEQDSKDYSLSSLTLSKADYEKHKVY<br>ACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:<br>365)      |
| EVQLVESGGGLVQPGSLRLSCAASGFTFDDYG<br>MHWVRQAPGKLEWWSVISWNSATIDYADSVK<br>RFTISRDNKNTLYLQMNSLRPEDTALYYCAKKN<br>SLGWFFDYWGQGLTVTVSSASTKGPSVFPLAPC<br>SRSTSESTAALGCLVKDYFPEPVTVSWNSGALT<br>GVTTPFPAVLQSSGLYSLSVTVTPSSSLGKT<br>CNVDHKPSNTKVDKRVESKYGPPCPPCPAPEFL<br>GGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSQ<br>DPEVQFNWYVDGVEVHNAKTKPREEQFNSTYR<br>VSVLTVLHQDWLNGKEYKCKVSNKGLPSSIEKTI<br>KAKGQPREPQVYTLPPSQEEMTKNQVSLTCLVK<br>GFYPSDIAVEWESNGQPENNYKTPPVLDSDGSF<br>FLYSRLTVDKSRWQEGNVFSCSVMEALHNHYT<br>QKSLSLSLGK (SEQ ID NO: 366)         | DIVMTQSPDLSAVSLGERATINCKSSQSVLYSS<br>NDNNYLAWYQQKPGQPPKLLIYWASTRESGV<br>PDRFSGSGSGTDFTLTISLQAEADVAYYCQQ<br>YYSTPLTFGGGTGLEIKRTVAAPSVFIFPPSDE<br>QLKSGTASVVCLLNNFYPREAKVQWKVDNAL<br>QSGNSQESVTEQDSKDYSLSSLTLSKADY<br>EKHKVYACEVTHQGLSSPVTKSFNRGEC<br>(SEQ ID NO: 367) |
| QVQLVQSGAEVKKPGASVKVSKASGYTFTSYGI<br>SWVRQAPGQGLEWMGWI SAYNGNTIYAQKFKQ<br>RFTMTTDTSTSTAYMELSRRLSDDTAVYYCARDI<br>NWFYDLWGRGTLTVTVSSASTKGPSVFPLAPCSR<br>STSESTAALGCLVKDYFPEPVTVSWNSGALTSGV<br>HTFPFPAVLQSSGLYSLSVTVTPSSSLGKT<br>VDHKPSNTKVDKRVESKYGPPCPPCPAPEFLGG                                                                                                                                                                                                                                                                | DIQMTQSPSSLSASVGRVITITCRASQGIKND<br>LGWFQQKPGKAPKRLIYAASSLQSGVPSRFS<br>SGSGTEFTLTISLQPEDFATYYCQLHNSYP<br>WTFGGGTGLEIKRTVAAPSVFIFPPSDEQLKS<br>GTASVVCLLNNFYPREAKVQWKVDNALQSGN<br>SQESVTEQDSKDYSLSSLTLSKADYEKHK<br>VYACEVTHQGLSSPVTKSFNRGEC (SEQ ID                    |

TABLE 6-continued

| Exemplary ALK2 antibody heavy and light chain sequences                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Heavy chain                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Light chain                                                                                                                                                                                                                                                                     |
| PSVFLFPPKPKDTLMISRTPEVTCVVVDVSQEDP<br>EVQFNWYVDGVEVHNAKTKPREEQFNSTYRVVS<br>VLTVLHQDWLNGKEYKCKVSNKGLPSSIEKTIKSKA<br>KGQPREPQVYITLPPSQEEMTKNQVSLTCLVKGF<br>YPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFL<br>YSRLTVDKSRWQEGNVFSCVMHEALHNHYTQK<br>SLSSLGLGK (SEQ ID NO: 368)                                                                                                                                                                                                                                                                           | NO: 369)<br><br>DIQMTQSPSSLSAFVGDRTITCRASQGIRNDL<br>GWYQQKPGKTPKRLIYTASSLRSGVPSRFSG<br>SSGTDFTLTITSLQPEDFATYYCLQHNNYPY<br>TFGQGTKEIKRTVAAPSVFIFPPSDEQLKSGT<br>ASVVCLLNNFYPREAKVQWKVDNALQSGNSQ<br>ESVTEQDSKSDSTYLSSTLTLSKADYEKHKVY<br>ACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:<br>371) |
| EVQLVESGGGLVQPGRSLRLSCAASGFTFDDYG<br>MHWVRQAPGKGLIEWSVISWNGATIDYADSVKG<br>RFTISRDNKNSLFLQMNSLRPEDTALYYCAKKN<br>SLGWFPDYWGQGTLLVTVSSASTKGPSVFPLAPC<br>SRSTSESTAALGCLVKDYFPEPVTVSWNSGALTS<br>GVHTFPAVLQSSGLYSLSSVTVPPSSSLGKT<br>CNVDHKPSNTKVDKRVESKYGPPCPPCPAPEFL<br>GGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSQ<br>DPEVQFNWYVDGVEVHNAKTKPREEQFNSTYRV<br>VSVLTVLHQDWLNGKEYKCKVSNKGLPSSIEKTI<br>KAKGQPREPQVYITLPPSQEEMTKNQVSLTCLVK<br>GFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSF<br>FLYSRLTVDKSRWQEGNVFSCVMHEALHNHYT<br>QKSLSSLGLGK (SEQ ID NO: 370)         | DIQMTQSPSSLSASVGDRTITCRSSQSVISYL<br>NWYQQKPGKAPKLLIFGASSLRSGVPSRFSGS<br>GSGTDFTLTIVSLQPEDFAVYFCQQNYLPPLTF<br>GGGTKVEIKRTVAAPSVFIFPPSDEQLKSGTAS<br>VVCLLNNFYPREAKVQWKVDNALQSGNSQES<br>VTEQDSKSDSTYLSSTLTLSKADYEKHKVYAC<br>EVTHQGLSSPVTKSFNRGEC (SEQ ID NO:<br>373)               |
| EVQLVESGGNLVQPGGSLRLSCAASGFTFSNYA<br>MSWVRQAPGKGLIEWSGITGNVNTYYSVSVK<br>GRFTISRDNKNTLFLQMNSLRADTAIYYCVKER<br>GHSWFGDWPDPWGQGTLLVTVSSASTKGPSVFPP<br>LAPCSRSTSESTAALGCLVKDYFPEPVTVSWNSG<br>ALTSGVHTFPAVLQSSGLYSLSSVTVPPSSSLGT<br>KTYTCNVVDHKPSNTKVDKRVESKYGPPCPPCPA<br>PEFLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDV<br>SQEDPEVQFNWYVDGVEVHNAKTKPREEQFNST<br>YRVVSVLTVLHQDWLNGKEYKCKVSNKGLPSSIE<br>KTISKAKGQPREPQVYITLPPSQEEMTKNQVSLTC<br>LVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSD<br>GSFFLYSRLTVDKSRWQEGNVFSCVMHEALHN<br>HYTQKSLSSLGLGK (SEQ ID NO: 372) | DIVMTQTPPLSSPVTLGQPASISCRSSQGLVNS<br>DGNTYLSWLHQRPGQPPRLLIYEISNRFSGVP<br>DRFSGSGAGTDFTLKISRVEAEDVGFYYCMQ<br>STQFPLTFGGGTKEIKRTVAAPSVFIFPPSDE<br>QLKSGTASVVCLLNNFYPREAKVQWKVDNAL<br>QSGNSQESVTEQDSKSDSTYLSSTLTLSKADY<br>EKHKVVYACEVTHQGLSSPVTKSFNRGEC<br>(SEQ ID NO: 375)        |
| EVQLVESGGNLVQPGGSLRLSCAASGFTFSNYA<br>MSWVRQAPGKGLIEWSGITGNINTYYSVSVKG<br>RFTISRDNKNTVFLQMNSLRADTAIYYCVKER<br>HSWFGDWPDPWGQGTLLVTVSSASTKGPSVFPLA<br>PCSRSTSESTAALGCLVKDYFPEPVTVSWNSGAL<br>TSGVHTFPAVLQSSGLYSLSSVTVPPSSSLGKT<br>YTCNVVDHKPSNTKVDKRVESKYGPPCPPCPAPE<br>FLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVS<br>QEDPEVQFNWYVDGVEVHNAKTKPREEQFNSTY<br>RVVSVLTVLHQDWLNGKEYKCKVSNKGLPSSIEK<br>TISKAKGQPREPQVYITLPPSQEEMTKNQVSLTCL<br>VKGFYPSDIAVEWESNGQPENNYKTTTPVLDSD<br>GSFFLYSRLTVDKSRWQEGNVFSCVMHEALHN<br>HYTQKSLSSLGLGK (SEQ ID NO: 374)    | DIQMTQSPSSLSASVGDRTITCRASQGIRND<br>LGWYQQKPGKAPKRLIYAASSLQSGVPSRFS<br>GSGSGTEFTLTITSLQPEDFATYYCLQHNSYP<br>WTFGQGTKEIKRTVAAPSVFIFPPSDEQLKS<br>GTASVVCLLNNFYPREAKVQWKVDNALQSGN<br>SQESVTEQDSKSDSTYLSSTLTLSKADYEKHK<br>VYACEVTHQGLSSPVTKSFNRGEC (SEQ ID<br>NO: 377)                |
| EVQLVESGGVVPQGRSLRLSCAASGFTFSNYG<br>MHWVRQAPGKGLIEWAVVSYDGSNKYYTDSVK<br>GRFTISRDNKNTLSLQMNSLRADTAIYYCAKD<br>RGITGTSGGVFDIWGGQMTVTVSSASTKGPSVFPP<br>LAPCSRSTSESTAALGCLVKDYFPEPVTVSWNSG                                                                                                                                                                                                                                                                                                                                                   | AIRMTQSPSSFSATITGDRVTITCRASQGISHL<br>AWYQQKPGKAPKRLIYVASTLQSGVPSRFSGS<br>GSGTDFTLTITSLQSEDFATYYCQYYSPPT<br>FGQGTKEIKRTVAAPSVFIFPPSDEQLKSGTA<br>SVVCLLNNFYPREAKVQWKVDNALQSGNSQ                                                                                                   |

TABLE 6-continued

| Exemplary ALK2 antibody heavy and light chain sequences                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Heavy chain                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Light chain                                                                                                                                                                                                                                                           |
| ALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGT<br>KTYTCNVDHKPSNTKVDKRVESKYGPPCPPCPA<br>PEFLGGPSVFLPPPKDITLMISRTPEVTCVVVDV<br>SQEDPEVQFNWYVDGVEVHNAKTKPREEQFNST<br>YRVVSVLTVHLQDNLNGKEYKCKVSNKGLPSSIE<br>KTISKAKGQPREPQVYTLPPSQEEMTKNQVSLTCL<br>LVKGFPYSDIAVEWESNGQPENNYKTTPPVLDSD<br>GSFFLYSRLTVDKSRWQEGNVFSCSVMHEALHN<br>HYTQKSLSLSLGK (SEQ ID NO: 378)                                                                                                                                                                                  | SVTEQDSKDYSLSSSTLTLSKADYEKHKVYA<br>CEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:<br>379)                                                                                                                                                                                          |
| EVQLVESGGGLVQPGGSLRLSCAASGFTSSRYW<br>MSWVRQAPGKGLLEWLANINQDGESEKYYLESRLG<br>RPTISRDNAKNSLYLQMNNLRADTAVYYCARDE<br>EGYWGQGTLLVTVSSASTKGPSVFPLAPCSRSTS<br>ESTAALGCLVKDYFPEPVTVSWNSGALTSGVHTF<br>PAVLQSSGLYSLSSVVTVPSSSLGTITCTCNVDH<br>KPSNTKVDKRVESKYGPPCPPCPAPEFLGGPSV<br>FLFPPKPKDTLMISRTPEVTCVVVDVSDPEVQ<br>FNWYVDGVEVHNAKTKPREEQFNSTYRVVSVLT<br>VLHQQDNLNGKEYKCKVSNKGLPSSIEKTIKAKG<br>QPREPQVYTLPPSQEEMTKNQVSLTCLVKGFY<br>SDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYS<br>RLTVDKSRWQEGNVFSCSVMHEALHNHYTQKSL<br>SLSLGK (SEQ ID NO: 380) | DVVMTQSLLSLSTLTGQPASISCRSSQSLVYS<br>DGDYLSWVQQRPGQSPRRLIYKVSHRDSGV<br>PDRFRGSGSGTDFTLKISRVEAEDVGIYYCMQ<br>GTHWPPPTFGQGTKEIKRTVAAPSVFIFPPSD<br>EQLKSGTASVVCCLNNFYPREAKVQWKVDNA<br>LQSGNSQESVTEQDSKDYSLSSSTLTLSKAD<br>YEKHKVYACEVTHQGLSSPVTKSFNRGEC<br>(SEQ ID NO: 381) |

**[0089]** In some embodiments, the subject may be administered an isolated ALK3 antibody, or antigen-binding fragment thereof. The ALK3 antibody or antigen binding fragment thereof can contain an antigen binding fragment (Fab) described in Harth et al., PLoS ONE 5: e13049, 2010, such as AbD1556 or AbD1564, both of which were found to have high nanomolar affinities for BMPR1 and to neutralize BMP2 activity.

**[0090]** In some embodiments, the ALK3 antibody or an antigen binding fragment thereof specifically binds to an extracellular domain of human ALK3 (BMPR1A) and contains: (a) a heavy chain CDR1 including TGYMK (SEQ ID NO: 79); (b) a heavy chain CDR2 including RINPDNG-GRYTNQIFKDK (SEQ ID NO: 80); and (c) a heavy chain CDR3 including RERGQYGNYGGSFSD (SEQ ID NO: 81).

**[0091]** In some embodiments, the ALK3 antibody or an antigen binding fragment thereof contains a heavy chain variable region having at least 90% (e.g., at least 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or more) sequence identity to SEQ ID NO: 77 or SEQ ID NO: 78, shown below:

(SEQ ID NO: 77)  
 MEWSWIFLFLSGTAGVLSEVQLQQSGPELVKPGTSVKISKASGYSFTG  
 YYMHWKQSQVKSLEWIGRINPDNGGRYTNQIFKDKASLTVHKSSSTAYM  
 ELHSLTSDDSAVYYCTRRERGQYGNYGGSFSDWGQGTTLVT  
 (SEQ ID NO: 78)  
 EVQLQQSGPELVKPGTSVKISKASGYSFTGYMHWKQSQVKSLEWIGR  
 INPDNGGRYTNQIFKDKASLTVHKSSSTAYMELHSLTSDDSAVYYCTRR  
 GQYGNYGGSFSDWGQGTTLVT

In some embodiments, the ALK3 antibody or an antigen binding fragment thereof contains a heavy chain variable region having at least 95% (e.g., at 95%, 96%, 97%, 98%, 99%, or more), at least 97% (e.g., at least 97%, 98%, 99%, or more), or at least 99% sequence identity to SEQ ID NO:

77 or SEQ ID NO: 78. In some embodiments, the antibody contains a heavy chain variable region having the sequence of SEQ ID NO: 77 or SEQ ID NO: 78. Such antibodies are described in U.S. Patent Application Publication No. US20130089560A1, which is incorporated herein by reference.

#### Vectors, Host Cells, and Protein Production

**[0092]** The ALK2 and ALK3 antibodies and antigen binding fragments thereof (e.g., ALK2 or ALK3 binding fragments) described herein can be produced from a host cell. A host cell refers to a vehicle that includes the necessary cellular components, e.g., organelles, needed to express the antibodies and antigen binding fragments thereof, described herein from their corresponding nucleic acids. The nucleic acids may be included in nucleic acid vectors that can be introduced into the host cell by conventional techniques known in the art (e.g., transformation, transfection, electroporation, calcium phosphate precipitation, direct microinjection, infection, or the like). The choice of nucleic acid vectors depends in part on the host cells to be used. Generally, preferred host cells are of either eukaryotic (e.g., mammalian) or prokaryotic (e.g., bacterial) origin.

#### Nucleic Acid Vector Construction and Host Cells

**[0093]** A nucleic acid sequence encoding the amino acid sequence of an ALK2 or ALK3 antibody or antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment), described herein may be prepared by a variety of methods known in the art. These methods include, but are not limited to, oligonucleotide-mediated (or site-directed) mutagenesis and PCR mutagenesis. A nucleic acid molecule encoding an ALK2 or ALK3 antibody, or antigen binding fragment thereof, described herein may be obtained using standard techniques, e.g., gene synthesis. Alternatively, a nucleic acid molecule encoding an ALK2 or ALK3 antibody or antigen binding fragment thereof, described herein may

be mutated to include specific amino acid substitutions using standard techniques in the art, e.g., QuikChange™ mutagenesis. Nucleic acid molecules can be synthesized using a nucleotide synthesizer or PCR techniques.

**[0094]** A nucleic acid sequence encoding an ALK2 or ALK3 antibody or antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment), described herein may be inserted into a vector capable of replicating and expressing the nucleic acid molecule in prokaryotic or eukaryotic host cells. Many vectors are available in the art and can be used for the purpose of the invention. Each vector may include various components that may be adjusted and optimized for compatibility with the particular host cell. For example, the vector components may include, but are not limited to, an origin of replication, a selection marker gene, a promoter, a ribosome binding site, a signal sequence, the nucleic acid sequence encoding the protein of interest, and a transcription termination sequence.

**[0095]** In some embodiments, mammalian cells may be used as host cells for the invention. Examples of mammalian cell types include, but are not limited to, human embryonic kidney (HEK) (e.g., HEK293, HEK 293F), Chinese hamster ovary (CHO), HeLa, COS, PC3, Vero, MC3T3, NS0, Sp2/0, VERY, BHK, MDCK, W138, BT483, Hs578T, HTB2, BT20, T47D, NS0 (a murine myeloma cell line that does not endogenously produce any immunoglobulin chains), CRL7030, and HsS78Bst cells. In some embodiments, *E. coli* cells may also be used as host cells for the invention. Examples of *E. coli* strains include, but are not limited to, *E. coli* 294 (ATCC®31,446), *E. coli* A 1776 (ATCC®31,537), *E. coli* BL21 (DE3) (ATCC® BAA-1025), and *E. coli* RV308 (ATCC®31,608). Different host cells have characteristic and specific mechanisms for the posttranslational processing and modification of protein products (e.g., glycosylation). Appropriate cell lines or host systems may be chosen to ensure the correct modification and processing of the polypeptide expressed. The above-described expression vectors may be introduced into appropriate host cells using conventional techniques in the art, e.g., transformation, transfection, electroporation, calcium phosphate precipitation, and direct microinjection. Once the vectors are introduced into host cells for protein production, host cells are cultured in conventional nutrient media modified as appropriate for inducing promoters, selecting transformants, or amplifying the genes encoding the desired sequences. Methods for expression of therapeutic proteins are known in the art, see, for example, Paulina Balbas, Argelia Lorence (eds.) *Recombinant Gene Expression: Reviews and Protocols* (Methods in Molecular Biology), Humana Press; 2nd ed. 2004 and Vladimir Voynov and Justin A. Caravella (eds.) *Therapeutic Proteins: Methods and Protocols* (Methods in Molecular Biology) Humana Press; 2nd ed. 2012.

#### Protein Production, Recovery, and Purification

**[0096]** Host cells used to produce the ALK2 or ALK3 antibodies or antigen binding fragments (e.g., ALK2 or ALK3 binding fragments) thereof, described herein may be grown in media known in the art and suitable for culturing of the selected host cells. Examples of suitable media for mammalian host cells include Minimal Essential Medium (MEM), Dulbecco's Modified Eagle's Medium (DMEM), Expi293™ Expression Medium, DMEM with supplemented fetal bovine serum (FBS), and RPMI-1640. Examples of suitable media for bacterial host cells include Luria broth

(LB) plus necessary supplements, such as a selection agent, e.g., ampicillin. Host cells are cultured at suitable temperatures, such as from about 20° C. to about 39° C., e.g., from 25° C. to about 37° C., preferably 37° C., and CO<sub>2</sub> levels, such as 5 to 10%. The pH of the medium is generally from about 6.8 to 7.4, e.g., 7.0, depending mainly on the host organism. If an inducible promoter is used in the expression vector of the invention, protein expression is induced under conditions suitable for the activation of the promoter.

**[0097]** In some embodiments, depending on the expression vector and the host cells used, the expressed protein may be secreted from the host cells (e.g., mammalian host cells) into the cell culture media. Protein recovery may involve filtering the cell culture media to remove cell debris. The proteins may be further purified. An ALK2 or ALK3 antibody or antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment) described herein may be purified by any method known in the art of protein purification, for example, by chromatography (e.g., ion exchange, affinity, and size-exclusion column chromatography), centrifugation, differential solubility, or by any other standard technique for the purification of proteins. For example, the protein can be isolated and purified by appropriately selecting and combining affinity columns such as Protein A column (e.g., POROS Protein A chromatography) with chromatography columns (e.g., POROS HS-50 cation exchange chromatography), filtration, ultrafiltration, salting-out and dialysis procedures.

**[0098]** In other embodiments, host cells may be disrupted, e.g., by osmotic shock, sonication, or lysis, to recover the expressed protein. Once the cells are disrupted, cell debris may be removed by centrifugation or filtration. In some instances, a polypeptide can be conjugated to marker sequences, such as a peptide to facilitate purification. An example of a marker amino acid sequence is a hexa-histidine peptide (His-tag), which binds to nickel-functionalized agarose affinity column with micromolar affinity. Other peptide tags useful for purification include, but are not limited to, the hemagglutinin "HA" tag, which corresponds to an epitope derived from influenza hemagglutinin protein (Wilson et al., *Cell* 37:767, 1984).

**[0099]** Alternatively, the ALK2 or ALK3 antibodies or antigen binding fragments thereof (e.g., ALK2 or ALK3 binding fragments), described herein can be produced by the cells of a subject (e.g., a human), e.g., in the context of gene therapy, by administering a vector (such as a viral vector (e.g., a retroviral vector, adenoviral vector, poxviral vector (e.g., vaccinia viral vector, such as Modified Vaccinia Ankara (MVA)), adeno-associated viral vector, and alphaviral vector)) containing a nucleic acid molecule encoding an antibody or antigen binding fragment thereof, described herein. The vector, once inside a cell of the subject (e.g., by transformation, transfection, electroporation, calcium phosphate precipitation, direct microinjection, infection, etc.) will promote expression of the antibody or antigen binding fragment thereof, which is then secreted from the cell. If treatment of a disease or disorder is the desired outcome, no further action may be required. If collection of the protein is desired, blood may be collected from the subject and the protein purified from the blood by methods known in the art.

#### Pharmaceutical Compositions and Preparations

**[0100]** The ALK2 or ALK3 antibodies or antigen binding fragments thereof (e.g., ALK2 or ALK3 binding fragments),

described herein can be incorporated into a vehicle for administration into a patient, such as a human patient suffering from bone disease, low red blood cell levels (e.g., low hemoglobin levels or low red blood cell count, e.g., anemia), heterotopic ossification (e.g., heterotopic ossification resulting from FOP), Sjogren's syndrome (e.g., dry eye associated with Sjogren's syndrome), MO, DIPG, PCO, or cardiac hypertrophy and/or cardiac fibrosis. In some embodiments, a pharmaceutical composition including an ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment), described herein may be used in combination with other agents (e.g., therapeutic biologics and/or small molecules) or compositions in a therapy. Pharmaceutical compositions containing ALK2 or ALK3 antibodies or antigen binding fragments thereof, can be prepared using methods known in the art. For example, such compositions can be prepared using, e.g., physiologically acceptable carriers, excipients or stabilizers (Remington: The Science and Practice of Pharmacology 22nd edition, Allen, L. Ed. (2013); incorporated herein by reference), and in a desired form, e.g., in the form of lyophilized formulations or aqueous solutions. In some embodiments, a pharmaceutical composition of the invention includes a nucleic acid molecule (DNA or RNA, e.g., mRNA) encoding an antibody or ALK2 or ALK3 binding fragment thereof, described herein, or a vector containing such a nucleic acid molecule.

**[0101]** Acceptable carriers and excipients in the pharmaceutical compositions are nontoxic to recipients at the dosages and concentrations employed. Acceptable carriers and excipients may include buffers such as phosphate, citrate, HEPES, and TAE, antioxidants such as ascorbic acid and methionine, preservatives such as hexamethonium chloride, octadecyldimethylbenzyl ammonium chloride, resorcinol, and benzalkonium chloride, proteins such as human serum albumin, gelatin, dextran, and immunoglobulins, hydrophilic polymers such as polyvinylpyrrolidone, amino acids such as glycine, glutamine, histidine, and lysine, and carbohydrates such as glucose, mannose, sucrose, and sorbitol. Pharmaceutical compositions of the invention can be administered parenterally in the form of an injectable formulation. Pharmaceutical compositions for injection can be formulated using a sterile solution or any pharmaceutically acceptable liquid as a vehicle. Pharmaceutically acceptable vehicles include, but are not limited to, sterile water, physiological saline, and cell culture media (e.g., Dulbecco's Modified Eagle Medium (DMEM),  $\alpha$ -Modified Eagles Medium ( $\alpha$ -MEMI

**[0102]** Mixtures of ALK2 or ALK3 antibodies or antigen binding fragments thereof (e.g., ALK2 or ALK3 binding fragments), may be prepared in water suitably mixed with one or more excipients, carriers, or diluents. Dispersions may also be prepared in glycerol, liquid polyethylene glycols, and mixtures thereof, and in oils. Under ordinary conditions of storage and use, these preparations may contain a preservative to prevent the growth of microorganisms. The pharmaceutical forms suitable for injectable use include sterile aqueous solutions or dispersions and sterile powders for the extemporaneous preparation of sterile injectable solutions or dispersions (described in U.S. Pat. No. 5,466, 468, the disclosure of which is incorporated herein by reference). In any case the formulation may be sterile and may be fluid to the extent that easy syringability exists. Formulations may be stable under the conditions of manu-

facture and storage and may be preserved against the contaminating action of microorganisms, such as bacteria and fungi. The carrier can be a solvent or dispersion medium containing, for example, water, ethanol, polyol (e.g., glycerol, propylene glycol, and liquid polyethylene glycol, and the like), suitable mixtures thereof, and/or vegetable oils. Proper fluidity may be maintained, for example, by the use of a coating, such as lecithin, by the maintenance of the required particle size in the case of dispersion and by the use of surfactants. The prevention of the action of microorganisms can be brought about by various antibacterial and antifungal agents, for example, parabens, chlorobutanol, phenol, sorbic acid, thimerosal, and the like. In many cases, it will be preferable to include isotonic agents, for example, sugars or sodium chloride. Prolonged absorption of the injectable compositions can be brought about by the use in the compositions of agents delaying absorption, for example, aluminum monostearate and gelatin.

**[0103]** For example, a solution containing a pharmaceutical composition described herein may be suitably buffered, if necessary, and the liquid diluent first rendered isotonic with sufficient saline or glucose. These particular aqueous solutions are especially suitable for intravenous, intramuscular, subcutaneous, and intraperitoneal administration. In this connection, sterile aqueous media that can be employed will be known to those of skill in the art in light of the present disclosure. For example, one dosage may be dissolved in 1 ml of isotonic NaCl solution and either added to 1000 ml of hypodermoclysis fluid or injected at the proposed site of infusion. Some variation in dosage will necessarily occur depending on the condition of the subject being treated.

**[0104]** The pharmaceutical compositions of the invention may be prepared in microcapsules, such as hydroxymethylcellulose or gelatin-microcapsule and poly-(methylmethacrylate) microcapsule. The pharmaceutical compositions of the invention may also be prepared in other drug delivery systems such as liposomes, albumin microspheres, microemulsions, nanoparticles, and nanocapsules. Such techniques are described in Remington: The Science and Practice of Pharmacology 22nd edition, Allen, L. Ed. (2013). The pharmaceutical compositions to be used for in vivo administration must be sterile. This is readily accomplished by filtration through sterile filtration membranes.

**[0105]** The pharmaceutical compositions of the invention may also be prepared as a sustained-release formulation. Suitable examples of sustained-release preparations include semipermeable matrices of solid hydrophobic polymers containing the polypeptides of the invention. Examples of sustained release matrices include polyesters, hydrogels, polylactides, copolymers of L-glutamic acid and  $\gamma$  ethyl-L-glutamate, non-degradable ethylene-vinyl acetate, degradable lactic acid-glycolic acid copolymers such as LUPRON DEPOT<sup>TM</sup>, and poly-D-(-)-3-hydroxybutyric acid. Some sustained-release formulations enable release of molecules over a few months, e.g., one to six months, while other formulations release pharmaceutical compositions of the invention for shorter time periods, e.g., days to weeks.

**[0106]** The pharmaceutical composition may be formed in a unit dose form as needed. The amount of active component, e.g., an ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment), described herein, included in the pharmaceutical preparations is such that a suitable dose within the desig-

nated range is provided (e.g., a dose within the range of 0.002 mg/kg to 3 mg/kg of body weight).

**[0107]** The pharmaceutical composition for gene therapy can be in an acceptable diluent or can include a slow-release matrix in which the gene delivery vehicle is imbedded. If hydrodynamic injection is used as the delivery method, the pharmaceutical composition containing a nucleic acid molecule encoding an antibody or ALK2 or ALK3 binding fragment thereof, described herein or a vector (e.g., a viral vector) containing the nucleic acid molecule is delivered rapidly in a large fluid volume intravenously. Vectors that may be used as in vivo gene delivery vehicle include, but are not limited to, retroviral vectors, adenoviral vectors, poxviral vectors (e.g., vaccinia viral vectors, such as Modified Vaccinia Ankara), adeno-associated viral vectors, and alphaviral vectors.

#### Routes, Dosage, and Administration

**[0108]** Pharmaceutical compositions that include the ALK2 or ALK3 antibodies or antigen binding fragments thereof (e.g., ALK2 or ALK3 binding fragments), of the invention as the therapeutic agents may be by a variety of routes, such as intravenous, parenteral, intradermal, transdermal, intramuscular, intranasal, subcutaneous, percutaneous, topical, intratracheal, intraperitoneal, intraarterial, intravascular, intrathecal, intracerebroventricular, inhalation, perfusion, lavage, and oral administration. The pharmaceutical composition may also be formulated for, or administered via, oral, ocular, nasal, spray, aerosol, rectal, or vaginal administration. In specific embodiments, the pharmaceutical composition is administered subcutaneously. For injectable formulations, various effective pharmaceutical carriers are known in the art. See, e.g., ASHP Handbook on Injectable Drugs, Toissel, 18th ed. (2014). For ocular administration, the formulation may be delivered systemically, by injection (e.g., intraocular injection), or topically (e.g., as a solution, suspension, or ointment, such as by instillation (e.g., an eye drop)).

**[0109]** In some embodiments, a pharmaceutical composition that includes a nucleic acid molecule encoding an ALK2 or ALK3 antibody or an ALK2 or ALK3 binding fragment thereof, described herein or a vector containing such nucleic acid molecule may be administered by way of gene delivery. Methods of gene delivery are well-known to one of skill in the art. Vectors that may be used for in vivo gene delivery and expression include, but are not limited to, retroviral vectors, adenoviral vectors, poxviral vectors (e.g., vaccinia viral vectors, such as Modified Vaccinia Ankara (MVA)), adeno-associated viral vectors, and alphaviral vectors. In some embodiments, mRNA molecules encoding polypeptides of the invention may be administered directly to a subject.

**[0110]** In some embodiments of the present invention, nucleic acid molecules encoding a polypeptide described herein or vectors containing such nucleic acid molecules may be administered using a hydrodynamic injection platform. In the hydrodynamic injection method, a nucleic acid molecule encoding an antibody or ALK2 or ALK3 binding fragment thereof, described herein is put under the control of a strong promoter in an engineered plasmid (e.g., a viral plasmid). The plasmid is often delivered rapidly in a large fluid volume intravenously. Hydrodynamic injection uses controlled hydrodynamic pressure in veins to enhance cell permeability such that the elevated pressure from the rapid

injection of the large fluid volume results in fluid and plasmid extravasation from the vein. The expression of the nucleic acid molecule is driven primarily by the liver. In mice, hydrodynamic injection is often performed by injection of the plasmid into the tail vein. In certain embodiments, mRNA molecules encoding an antibody or ALK2 or ALK3 binding fragment thereof, described herein may be administered using hydrodynamic injection. The most suitable route and dosage for administration in any given case will depend on the particular composition administered, the patient, pharmaceutical formulation methods, administration methods (e.g., administration time and administration route), the patient's age, body weight, sex, severity of the disease being treated, the patient's diet, and the patient's excretion rate. A pharmaceutical composition of the invention may include a dosage of an ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment), of the invention ranging from 0.002 mg/kg to 3 mg/kg (e.g., 0.005 mg/kg to 3 mg/kg, 0.01 mg/kg to 3 mg/kg, 0.05 mg/kg to 3 mg/kg, 0.1 mg/kg to 3 mg/kg, 0.5 mg/kg to 3 mg/kg, 1 mg/kg to 3 mg/kg, 1.5 mg/kg to 3 mg/kg, 2 mg/kg to 3 mg/kg, 2.5 mg/kg to 3 mg/kg, 0.002 mg/kg to 2.5 mg/kg, 0.002 mg/kg to 2 mg/kg, 0.002 mg/kg to 1.5 mg/kg, 0.002 mg/kg to 1 mg/kg, 0.002 mg/kg to 0.5 mg/kg, 0.002 mg/kg to 0.25 mg/kg, 0.002 mg/kg to 0.1 mg/kg, 0.002 mg/kg to 0.05 mg/kg, 0.002 mg/kg to 0.01 mg/kg, 0.002 mg/kg to 0.005 mg/kg, 0.1 mg/kg to 0.3 mg/kg, or 0.1 mg/kg to 1 mg/kg), and, in a more specific embodiment, 0.01 mg/kg to 2 mg/kg (e.g., 0.05 mg/kg to 2 mg/kg, 0.1 mg/kg to 2 mg/kg, 0.5 mg/kg to 2 mg/kg, 1 mg/kg to 2 mg/kg, 1.5 mg/kg to 2 mg/kg, 0.01 mg/kg to 1.5 mg/kg, 0.01 mg/kg to 1 mg/kg, 0.01 mg/kg to 0.5 mg/kg, 0.01 mg/kg to 0.25 mg/kg, 0.01 mg/kg to 0.1 mg/kg, or 0.01 mg/kg to 0.05 mg/kg), and, in a more specific embodiment, 0.01 mg/kg to 1 mg/kg (e.g., 0.05 mg/kg to 1 mg/kg, 0.1 mg/kg to 1 mg/kg, 0.2 mg/kg to 1 mg/kg, 0.3 mg/kg to 1 mg/kg, 0.4 mg/kg to 1 mg/kg, 0.5 mg/kg to 1 mg/kg, 0.6 mg/kg to 1 mg/kg, 0.7 mg/kg to 1 mg/kg, 0.8 mg/kg to 1 mg/kg, 0.9 mg/kg to 1 mg/kg, 0.01 mg/kg to 0.9 mg/kg, 0.01 mg/kg to 0.8 mg/kg, 0.01 mg/kg to 0.7 mg/kg, 0.01 mg/kg to 0.6 mg/kg, 0.01 mg/kg to 0.5 mg/kg, 0.01 mg/kg to 0.4 mg/kg, 0.01 mg/kg to 0.3 mg/kg, 0.01 mg/kg to 0.2 mg/kg, 0.01 mg/kg to 0.1 mg/kg, or 0.01 mg/kg to 0.05 mg/kg), and, in a more specific embodiment, 0.05 mg/kg to 1 mg/kg (e.g., 0.1 mg/kg to 1 mg/kg, 0.2 mg/kg to 1 mg/kg, 0.3 mg/kg to 1 mg/kg, 0.4 mg/kg to 1 mg/kg, 0.5 mg/kg to 1 mg/kg, 0.6 mg/kg to 1 mg/kg, 0.7 mg/kg to 1 mg/kg, 0.8 mg/kg to 1 mg/kg, 0.9 mg/kg to 1 mg/kg, 0.05 mg/kg to 0.9 mg/kg, 0.05 mg/kg to 0.8 mg/kg, 0.05 mg/kg to 0.7 mg/kg, 0.05 mg/kg to 0.6 mg/kg, 0.05 mg/kg to 0.5 mg/kg, 0.05 mg/kg to 0.4 mg/kg, 0.05 mg/kg to 0.3 mg/kg, 0.05 mg/kg to 0.2 mg/kg, or 0.05 mg/kg to 0.1 mg/kg) or 0.01 mg/kg to 0.3 mg/kg (e.g., 0.01 mg/kg to 3 mg/kg, 0.025 mg/kg to 0.3 mg/kg, 0.05 mg/kg to 0.3 mg/kg, 0.075 mg/kg to 0.3 mg/kg, 0.1 mg/kg to 0.3 mg/kg, 0.125 mg/kg to 0.3 mg/kg, 0.15 mg/kg to 0.3 mg/kg, 0.175 mg/kg to 0.3 mg/kg, 0.2 mg/kg to 0.3 mg/kg, 0.225 mg/kg to 0.3 mg/kg, 0.25 mg/kg to 0.3 mg/kg, 0.275 mg/kg to 0.3 mg/kg, 0.01 mg/kg to 0.275 mg/kg, 0.01 mg/kg to 0.25 mg/kg, 0.01 mg/kg to 0.225 mg/kg, 0.01 mg/kg to 0.2 mg/kg, 0.01 mg/kg to 0.175 mg/kg, 0.01 mg/kg to 0.15 mg/kg, 0.01 mg/kg to 0.125 mg/kg, 0.01 mg/kg to 0.1 mg/kg, 0.01 mg/kg to 0.075 mg/kg, 0.01 mg/kg to 0.05 mg/kg, or 0.01 mg/kg to 0.025

mg/kg), and, in a more specific embodiment, 0.1 mg/kg to 0.3 mg/kg (e.g., 0.1 mg/kg to 0.25 mg/kg, 0.1 mg/kg to 0.2 mg/kg, 0.1 mg/kg to 0.15 mg/kg, 0.15 mg/kg to 0.3 mg/kg, 0.2 mg/kg to 0.3 mg/kg, and 0.25 mg/kg to 0.3 mg/kg). The dosage may be adapted by the physician in accordance with conventional factors such as the extent of the disease and different parameters of the subject.

**[0111]** The pharmaceutical compositions are administered in a manner compatible with the dosage formulation and in such amount as is therapeutically effective to result in an improvement or remediation of the symptoms. The pharmaceutical compositions are administered in a variety of dosage forms, e.g., intravenous dosage forms, subcutaneous dosage forms, and oral dosage forms (e.g., ingestible solutions, drug release capsules). In some embodiments, the pharmaceutical composition is administered subcutaneously. In some embodiments, the pharmaceutical composition is administered intravenously. Pharmaceutical compositions that include an ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment), of the invention may be administered to a subject in need thereof, at a frequency that is weekly to semi-annually. For example, the antibody or antigen binding fragment thereof may be administered once a week, once every 14 days, once a month, once every two months, once every three months, once every four months, once every five months, or once every six months. In some embodiments, the antibody may be administered to the subject at a frequency of once every 6 months. In some embodiments, the antibody may be administered to the subject at a frequency of once every 4 months. In some embodiments, the antibody may be administered to the subject at a frequency of once every 2 months. In some embodiments, the antibody may be administered to the subject at a frequency of once every month. In some embodiments, the antibody may be administered to the subject at a frequency of once biweekly. In some embodiments, the frequency is once every 10 to 14 days (e.g., once every 10 days, once every 11 days, once every 12 days, once every 13 days, or once every 14 days). In some embodiments, the frequency is once a week.

**[0112]** In some embodiments, the ALK2 or ALK3 antibody or an antigen binding fragment thereof is administered using a microneedle injector. In some embodiments, the ALK2 or ALK3 antibody or an antigen binding fragment thereof is administered using needle-free injection technology. Such technology is described in Ravi et al., *Int J Pharm Investig.* 5:192-199, 2015. For example, an ALK2 or ALK3 antibody or an antigen binding fragment thereof can be administered subcutaneously using any one of the Biojector® 2000, Vitajet™ 3, Tev-Tropin®, Sumavel®DosePro™, Bioject® ZetaJet™, PharmaJet Stratis®, or Jupiter Jet™, devices or other similar needle-free injection technology that is designed for subcutaneous administration of a small volume of a pharmaceutical composition.

**[0113]** In some embodiments, pharmaceutical compositions that include an ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment), of the invention may be administered to a subject in need thereof weekly, biweekly, monthly, bimonthly, or quarterly. Dosages may be provided in either a single or multiple dosage regimens. The timing between administrations may increase as the medical condition improves or decrease as the health of the patient declines.

## Methods of Treatment

**[0114]** The compositions and methods described herein can be used to treat and/or prevent (e.g., prevent the development of or treat a subject diagnosed with) medical conditions, e.g., bone disease, anemia or low red blood cell levels (e.g., low hemoglobin levels or low red blood cell count), heterotopic ossification (e.g., heterotopic ossification resulting from FOP), Sjogren's syndrome (e.g., dry eye associated with Sjogren's syndrome), DIPG, MO, PCO, or cardiac hypertrophy and/or cardiac fibrosis. In some embodiments, the ALK2 or ALK3 antibodies (e.g., neutralizing antibodies) or antigen binding fragments thereof (e.g., ALK2 or ALK3 binding fragments), described herein are administered to increase bone mineral density, increase bone formation, increase bone strength, reduce bone resorption (e.g., bone loss), or reduce the risk of bone fracture in a subject in need thereof. The ALK2 or ALK3 antibodies or antigen binding fragments thereof (e.g., ALK2 or ALK3 binding fragments), described herein may increase bone mineral density, increase bone formation, increase bone strength, reduce bone resorption (e.g., bone loss), or reduce the risk of bone fracture compared to measurements obtained prior to treatment or compared to bone mineral density, bone strength, bone formation, bone resorption, or risk of bone fracture typically observed in untreated subjects. In some embodiments, the subject may have or be at risk of developing a disease that results in bone damage (e.g., osteoporosis or osteopenia). In some embodiments of the methods described herein, the subject has or is at risk of developing a disease or condition involving bone damage (e.g., primary osteoporosis, secondary osteoporosis, osteopenia, osteopetrosis, osteogenesis imperfecta, bone fracture, bone cancer or cancer metastasis-related bone loss, Paget's disease, renal osteodystrophy, treatment-related bone loss, neuromuscular disease-related bone loss, burn-induced bone loss, anorexia-related bone loss, diet-related bone loss, bone loss associated with the treatment of obesity, low gravity-related bone loss, and immobility-related bone loss). The invention also includes methods of treating a subject having or at risk of developing primary osteoporosis, secondary osteoporosis, osteopenia, osteopetrosis, osteogenesis imperfecta, bone fracture, bone cancer or cancer metastasis-related bone loss, Paget's disease, renal osteodystrophy, treatment-related bone loss, neuromuscular disease-related bone loss, burn-induced bone loss, anorexia-related bone loss, diet-related bone loss, bone loss associated with the treatment of obesity, low gravity-related bone loss, and immobility-related bone loss, by administering to the subject an effective amount of an ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment), described herein.

**[0115]** In any of the methods described herein, a subject having or at risk of developing bone disease (e.g., bone damage) has or is at risk of developing a disease or condition including osteoporosis, osteopenia, osteopetrosis, osteogenesis imperfecta, bone fracture, bone cancer or cancer metastasis-related bone loss, Paget's disease, renal osteodystrophy, treatment-related bone loss, neuromuscular disease-related bone loss, burn-induced bone loss, anorexia-related bone loss, diet-related bone loss, bone loss associated with the treatment of obesity, low gravity-related bone loss, or immobility-related bone loss. The osteoporosis may be primary osteoporosis or secondary osteoporosis. In some embodiments, the primary osteoporosis is age-related or

hormone-related osteoporosis (e.g., related to a decline in estrogen). In some embodiments, the secondary osteoporosis is immobilization-induced or glucocorticoid-induced osteoporosis. In some embodiments, the bone cancer is multiple myeloma, or the cancer metastasis-related bone loss is caused by multiple myeloma. In some embodiments, the treatment-related bone loss occurs due to treatment with FGF-21 or GLP-1, due to treatment with an FGF-21 or GLP-1 containing therapeutic, due to treatment of Type-2 diabetes and/or obesity, or due to cancer therapy (e.g., chemotherapy or radiation). In some embodiments, the diet-related bone loss is rickets (e.g., vitamin D deficiency). In some embodiments, the low-gravity related bone loss is lack of load-related bone loss. In some embodiments, the methods described herein increase bone mineral density (e.g., increase bone mass). In some embodiments, the methods described herein reduce bone resorption (e.g., reduce bone catabolic activity or reduce bone loss), e.g., reduce bone resorption compared to measurements obtained prior to treatment or compared to bone resorption typically observed in untreated subjects. In some embodiments, the methods described herein increase bone formation (e.g., increase bone anabolic activity or increase osteogenesis), e.g., increase bone formation compared to measurements obtained prior to treatment or compared to bone formation typically observed in untreated subjects. In some embodiments, the methods described herein increase osteoblast activity or osteoblastogenesis, e.g., increase osteoblast activity or osteoblastogenesis compared to measurements obtained prior to treatment or compared to osteoblast activity or osteoblastogenesis typically observed in untreated subjects. In some embodiments, the methods described herein decrease osteoclast activity or osteoclastogenesis, e.g., decrease osteoclast activity or osteoclastogenesis compared to measurements obtained prior to treatment or compared to osteoclast activity or osteoclastogenesis typically observed in untreated subjects. In some embodiments, the bone is cortical or trabecular bone. In some embodiments, a subject having or at risk of developing a bone disease may be administered an ALK2 or ALK3 antibody, or an antigen-binding fragment thereof, once a month, every two months, every three months, every four months, every five months, or every six months. In an embodiment, the ALK2 or ALK3 antibody, or an antigen-binding fragment thereof, may be administered to the subject every 4 months. In an embodiment, the ALK2 or ALK3 antibody, or an antigen-binding fragment thereof, may be administered to the subject every 6 months.

**[0116]** In some embodiments, the ALK2 or ALK3 antibodies or antigen binding fragments thereof (e.g., ALK2 or ALK3 binding fragments), described herein are administered to increase red blood cell levels (e.g., increase hemoglobin levels, increase red blood cell count, or increase red blood cell formation or production) in a subject in need thereof. The ALK2 or ALK3 antibodies or antigen binding fragments thereof (e.g., ALK2 or ALK3 binding fragments), described herein may increase red blood cell levels (e.g., increase hemoglobin levels, red blood cell count, or red blood cell formation) compared to measurements obtained prior to treatment or compared to red blood cell levels typically observed in untreated subjects. In some embodiments, the subject may have a disease or condition associated with low red blood cell levels (e.g., anemia or blood loss). In some embodiments of the methods described

herein, the subject has or is at risk of developing a disease or condition involving low red blood cell levels (e.g., anemia or blood loss, such as aplastic anemia, iron deficiency anemia, iron-refractory iron deficiency anemia (IRIDA), or anemia of inflammation (also called anemia of chronic disease, e.g., anemia caused by inflammatory diseases or conditions, such as infection (e.g., chronic infection, such as HIV/AIDS or tuberculosis), autoimmune disease (e.g., rheumatoid arthritis or lupus), cancer (e.g., cancer or cancer treatment), inflammatory bowel disease (e.g., Crohn's disease or ulcerative colitis), and chronic kidney disease). The invention also includes methods of treating a subject having or at risk of developing low red blood cell levels (e.g., low hemoglobin levels or low red blood cell count, e.g., anemia) by administering to the subject an effective amount of an ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment), described herein. The methods described herein may include a step of screening a subject for one or more mutations in genes known to be associated with anemia (e.g., mutations in *TMPRSS6*, which are associated with IRIDA) prior to treatment with or administration of the compositions described herein. A subject can be screened for a genetic mutation using standard methods known to those of skill in the art (e.g., genetic testing).

**[0117]** The invention also includes methods of treating a subject having or at risk of developing anemia or blood loss by administering to the subject an effective amount of an ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment), described herein. In any of the methods described herein, a subject having or at risk of developing low red blood cell levels (e.g., low hemoglobin levels, low hematocrit, or low red blood cell counts) has or is at risk of developing anemia or blood loss. In some embodiments, the anemia is associated with a nutritional deficit (e.g., a vitamin deficiency, such as vitamin B-12 deficiency, folate deficiency), a bone marrow defect (e.g., paroxysmal nocturnal hemoglobinuria), adverse reaction to medication (e.g., anti-retroviral HIV drugs), a myelodysplastic syndrome, bone marrow transplantation, myelofibrosis, ineffective hematopoiesis, cancer (e.g., a solid tumor, such as breast cancer, lung cancer, colon cancer; a tumor of the lymphatic system, such as chronic lymphocytic leukemia, non-Hodgkin's lymphoma, Hodgkin's lymphoma; or a tumor of the hematopoietic system, such as leukemia or multiple myeloma), cancer treatment (e.g., radiation or chemotherapy, e.g., chemotherapy with platinum-containing agents), myelofibrosis treatment (e.g., treatment with a JAK inhibitor, such as ruxolitinib or fedratinib), an inflammatory or autoimmune disease (e.g., rheumatoid arthritis, other inflammatory arthritides, systemic lupus erythematosus (SLE), an acute or chronic skin disease (e.g. psoriasis), or inflammatory bowel disease (e.g., Crohn's disease or ulcerative colitis), cystitis, gastritis), acute or chronic renal disease or failure (e.g., chronic kidney disease) including idiopathic or congenital conditions, diabetes, acute or chronic liver disease, acute or chronic bleeding, infection (e.g., malaria, osteomyelitis), splenomegaly, porphyria, vasculitis, hemolysis, urinary tract infection, hemoglobinopathy (e.g., sickle cell disease), thalassemia (e.g.,  $\alpha$ - or  $\beta$ -thalassemia), Churg-Strauss syndrome, Felty syndrome, Pearson syndrome, dyskeratosis congenita, graft versus host disease, hematopoietic stem cell transplantation, osteomyelofibrosis, pancytopenia, pure red-cell aplasia, pur-



pura Schoenlein-Henoch, Shwachman syndrome (e.g., Shwachman-Diamond syndrome), drug use or abuse (e.g., alcohol abuse), or contraindication to transfusion (e.g., patients of advanced age, patients with allo- or auto-antibodies, pediatric patients, patients with cardiopulmonary disease, patients who object to transfusion for religious reasons (e.g., some Jehovah's Witnesses)). The myelodysplastic syndrome may be myelodysplastic syndrome with unilineage dysplasia (MDS-SLD), myelodysplastic syndrome with multilineage dysplasia (MDS-MLD), myelodysplastic syndrome with ring sideroblasts (MDS-RS, which includes single lineage dysplasia (MDS-RS-SLD) and multilineage dysplasia (MDS-RS-MLD)), myelodysplastic syndrome associated with isolated del chromosome abnormality (MDS with isolated del(5q)), myelodysplastic syndrome with excess blasts (MDS-EB; which includes myelodysplastic syndrome with excess blasts—type 1 (MDS-EB-1) and myelodysplastic syndrome with excess blasts—type 2 (MDS-EB-2)), myelodysplastic syndrome, unclassifiable (MDS-U), or myelodysplastic syndrome/myeloproliferative neoplasm with ring sideroblasts and thrombocytosis (MDS/MPN-RS-T). The myelodysplastic syndrome may be a very low, low, or intermediate risk MDS as determined by the Revised International Prognostic Scoring System (IPSS-R). The myelodysplastic syndrome may be an RS-positive myelodysplastic syndrome (e.g., the subject with a myelodysplastic syndrome may have ring sideroblasts) or a non-RS myelodysplastic syndrome (e.g., the subject with a myelodysplastic syndrome may lack ring sideroblasts). In some embodiments, the RS-positive myelodysplastic syndrome is associated with a splicing factor mutation, such as a mutation in SF3B1. In some embodiments, the MDS is associated with a defect in terminal maturation (often observed in RS-positive MDS and in subjects having splicing factor mutations, such a subject may have increased erythroid progenitor cells in the bone marrow relative to a healthy subject). In some embodiments, the MDS is associated with a defect in early-stage hematopoiesis (e.g., early-stage erythroid cell development, such as commitment or early differentiation, such a subject may have fewer erythroid progenitor cells in the bone marrow compared to a healthy subject or to a subject with a defect in terminal maturation). In some embodiments, the MDS is associated with elevated endogenous erythropoietin levels. In some embodiments, the myelodysplastic syndrome is associated with hypocellular bone marrow (e.g., a subject with MDS has hypocellular bone marrow). The subject may have a low transfusion burden or a high transfusion burden. In some embodiments, the subject has a low transfusion burden and received 1-3 RBC units in the eight weeks prior to treatment with an antibody described herein. In some embodiments, the subject has a low transfusion burden and did not receive a transfusion (received 0 RBC units) in the eight weeks prior to treatment with an antibody described herein. In some embodiments, the anemia is aplastic anemia, iron deficiency anemia, iron-refractory iron deficiency anemia (IRIDA), vitamin deficiency anemia, anemia of inflammation (also called anemia of chronic disease, e.g., anemia caused by an inflammatory disease or condition, such as an infection (e.g., a chronic infection, such as HIV/AIDS or tuberculosis), an autoimmune disease (e.g., rheumatoid arthritis or lupus), cancer, inflammatory bowel disease (e.g., Crohn's disease or ulcerative colitis), and chronic kidney disease), anemia associated with bone marrow disease, hemolytic anemia,

sickle cell anemia, microcytic anemia, hypochromic anemia, sideroblastic anemia, congenital dyserythropoietic anemia, Diamond Blackfan anemia, Fanconi anemia, or refractory anemia with excess of blasts. The sideroblastic anemia can be acquired sideroblastic anemia or congenital sideroblastic anemia. In some embodiments, the congenital sideroblastic anemia is associated with a mutation in ALAS2, SLC25A38, FECH, GLRX5, HSPA9, HSCB, SLC25A38, or ABCB7. In some embodiments, the congenital sideroblastic anemia is associated with a mutation in PUS1, YARS2, LARS2, TRNT1, MT-ATP6, NDUFB11, or SLC19A2, or with an mtDNA mutation. In some embodiments, the anemia is associated with elevated hepcidin levels (e.g., elevated compared to hepcidin levels in a subject that does not have anemia). The compositions and methods described herein can also be used to treat subjects that do not respond well to erythropoietin (EPO) or that are susceptible to adverse effects of EPO (e.g., hypertension, headaches, vascular thrombosis, influenza-like syndrome, obstruction of shunts, and myocardial infarction) or to treat subjects that do not respond to an erythroid maturation agent. In some embodiments, the subject has previously been treated with an ESA. In some embodiments, the subject has not previously been treated with an ESA. In some embodiments, the blood loss is due to surgery, trauma, a wound, an ulcer, urinary tract bleeding, digestive tract bleeding, frequent blood donation, or heavy menstrual bleeding (e.g., menorrhagia). In some embodiments, the methods described herein increase red blood cell levels (e.g., hemoglobin levels or red blood cell counts) compared to measurements obtained prior to treatment or compared to red blood cell levels typically observed in untreated subjects. In some embodiments, the methods described herein increase or induce red blood cell formation compared to measurements obtained prior to treatment or compared to red blood cell formation typically observed in untreated subjects. In some embodiments, the compositions and methods described herein reduce the need of a subject for a blood transfusion (e.g., the subject no longer needs blood transfusions, or the subject needs less frequent blood transfusion than before treatment with the compositions and methods described herein). Subjects with normal red blood cell levels can also be treated using the methods and compositions described herein to increase red blood cell levels so that blood can be drawn and stored for later use in transfusions. In some embodiments, a subject having or at risk of developing anemia may be administered an ALK2 or ALK3 antibody, or an antigen-binding fragment thereof, once a month, every two months, every three months, every four months, every five months, or every six months. In an embodiment, the ALK2 or ALK3 antibody, or an antigen-binding fragment thereof, may be administered to the subject every 4 months. In an embodiment, the ALK2 or ALK3 antibody, or an antigen-binding fragment thereof, may be administered to the subject every 6 months.

**[0118]** In some embodiments, the ALK2 or ALK3 antibodies or antigen binding fragments thereof (e.g., ALK2 or ALK3 binding fragments), described herein may be administered to a subject to treat FOP, to slow or stop the progression of FOP, to delay the onset of FOP (e.g., delay the development of heterotopic ossification), or to prevent or reduce heterotopic ossification (e.g., the formation of bone in muscle, tendons, ligaments, or other connective tissues, e.g., prevent or reduce heterotopic ossification in a subject with FOP who has already experienced heterotopic ossifi-

cation, or prevent the development of heterotopic ossification in a subject who has not yet exhibited heterotopic ossification), such as heterotopic ossification in a subject with FOP. The FOP can be inherited FOP (e.g., FOP related to germline transmission of a mutation, such as an autosomal dominant mutation) or sporadic FOP (e.g., FOP related to a spontaneous, non-inherited mutation). The ALK2 or ALK3 antibodies or antigen binding fragments thereof (e.g., ALK2 or ALK3 binding fragments), described herein may reduce the amount of heterotopic bone, reduce the formation of heterotopic bone, or reduce the recurrence of heterotopic bone (e.g., recurrence after surgical resection) compared to the amount of heterotopic bone, formation of heterotopic bone, or recurrence of heterotopic bone observed in untreated or control treated subjects. The ALK2 or ALK3 antibodies or antigen binding fragments thereof (e.g., ALK2 or ALK3 binding fragments), described herein may prevent the formation of heterotopic bone (e.g., in a subject at risk of developing heterotopic ossification, such as a subject with FOP, or prevent the recurrence of heterotopic bone, such as recurrence after surgical resection). In some embodiments, the subject is at risk of developing heterotopic ossification (e.g., the subject has a genetic mutation associated with FOP but has not yet exhibited symptoms of FOP, e.g., has not yet exhibited heterotopic ossification). The methods described herein may include a step of screening a subject for one or more mutations in genes known to be associated with FOP (e.g., mutations in ALK2, also known as Activin A type I receptor (ACVR1)) prior to treatment with or administration of the compositions described herein. A subject can be screened for a genetic mutation using standard methods known to those of skill in the art (e.g., genetic testing). In some embodiments, a subject having FOP may be administered an ALK2 or ALK3 antibody, or an antigen-binding fragment thereof, once a week, every two weeks, every three weeks, or once a month. In an embodiment, the ALK2 or ALK3 antibody, or an antigen-binding fragment thereof, may be administered to the subject once a week. In an embodiment, the ALK2 or ALK3 antibody, or an antigen-binding fragment thereof, may be administered to the subject once every 10 to 14 days. In an embodiment, the ALK2 or ALK3 antibody, or an antigen-binding fragment thereof, may be administered to the subject once a month.

**[0119]** In some embodiments, the ALK2 or ALK3 antibodies or antigen binding fragments thereof (e.g., ALK2 or ALK3 binding fragments), described herein may be administered to a subject to treat Sjogren's syndrome (e.g., to treat dry eye associated with Sjogren's syndrome) or to reduce one or more symptom of Sjogren's syndrome (e.g., to reduce dry eye or dry mouth associated with Sjogren's syndrome). The ALK2 or ALK3 antibodies or antigen binding fragments thereof (e.g., ALK2 or ALK3 binding fragments), described herein may reduce eye irritation, reduce fibrosis of the lacrimal gland, reduce inflammation of the lacrimal gland, improve vision, increase the production of saliva, increase the production of tears (e.g., increase tear volume production), reduce systemic inflammation, reduce joint pain, or reduce fatigue in a subject with Sjogren's syndrome, or reduce the severity or occurrence of one or more symptom of Sjogren's syndrome (e.g., reduce dry eye, dry mouth, or joint pain associated with Sjogren's syndrome) compared to eye irritation, vision, inflammation or fibrosis of the lacrimal gland, the production of saliva, the production of tears, systemic inflammation, joint pain, fatigue, or the severity or

occurrence of symptoms of Sjogren's syndrome in the subject prior to treatment or in an untreated subject with Sjogren's syndrome. In some embodiments, a subject having Sjogren's syndrome may be administered an ALK2 or ALK3 antibody, or an antigen-binding fragment thereof, once a week, every two weeks, every three weeks, or once a month. In an embodiment, the ALK2 or ALK3 antibody, or an antigen-binding fragment thereof, may be administered to the subject once a week. In an embodiment, the ALK2 or ALK3 antibody, or an antigen-binding fragment thereof, may be administered to the subject once every 10 to 14 days. In an embodiment, the ALK2 or ALK3 antibody, or an antigen-binding fragment thereof, may be administered to the subject once a month.

**[0120]** In some embodiments, the ALK2 or ALK3 antibodies or antigen binding fragments thereof (e.g., ALK2 or ALK3 binding fragments), described herein may be administered to a subject to treat DIPG, increase the survival (e.g., survival time, e.g., lifespan) of a subject having DIPG, increase progression free survival of a subject having DIPG, reduce DIPG tumor growth, reduce DIPG tumor size or volume, or prevent or reduce DIPG tumor metastasis. The ALK2 or ALK3 antibodies or antigen binding fragments thereof (e.g., ALK2 or ALK3 binding fragments), described herein may increase the survival (e.g., survival time, e.g., lifespan) of a subject having DIPG, increase progression free survival of a subject having DIPG, reduce DIPG tumor growth, reduce DIPG tumor size or volume, or prevent or reduce DIPG tumor metastasis compared survival time, progression free survival, tumor growth, tumor size or volume, or tumor metastasis compared to untreated or control treated subjects or compared to tumor growth, tumor size or volume, or tumor metastasis in the subject prior to treatment. In some embodiments, the subject has an activating mutation in ALK2 (ACVR1). The methods described herein may include a step of screening a subject for a mutation in ALK2 (ACVR1) prior to treatment with or administration of the compositions described herein. A subject can be screened for a genetic mutation using standard methods known to those of skill in the art (e.g., genetic testing). In some embodiments, a subject having DIPG may be administered an ALK2 or ALK3 antibody, or an antigen-binding fragment thereof, once a week, every two weeks, every three weeks, or once a month. In an embodiment, the ALK2 or ALK3 antibody, or an antigen-binding fragment thereof, may be administered to the subject once a week. In an embodiment, the ALK2 or ALK3 antibody, or an antigen-binding fragment thereof, may be administered to the subject once every 10 to 14 days. In an embodiment, the ALK2 or ALK3 antibody, or an antigen-binding fragment thereof, may be administered to the subject once a month.

**[0121]** In some embodiments, the ALK2 or ALK3 antibodies or antigen binding fragments thereof (e.g., ALK2 or ALK3 binding fragments), described herein may be administered to a subject to treat MO, reduce or prevent the formation of an osteochondroma in a subject with MO, reduce the number of osteochondromas in a subject with MO, reduce the size of an osteochondroma in a subject with MO, or slow the growth of an osteochondroma in a subject with MO. The osteochondroma can be formed on the growing end (metaphysis) of a bone, on a long bone (e.g., a long bone of the leg, arm, or digit), and/or on a flat bone (e.g., a hip bone (pelvic bone), rib bone, or shoulder blade). The ALK2 or ALK3 antibodies or antigen binding fragments

thereof (e.g., ALK2 or ALK3 binding fragments), described herein may reduce or prevent the formation of an osteochondroma (e.g., prevent the formation of an osteochondroma in a subject with MO who has not yet developed osteochondroma or reduce or prevent the formation of osteochondroma in a subject with MO who has already developed one or more osteochondroma), reduce the number of osteochondromas, reduce the size of an osteochondroma, or slow the growth of an osteochondroma compared to the formation of an osteochondroma, the number of osteochondromas, the size of an osteochondroma, or the growth of an osteochondroma in the subject prior to treatment or in an untreated subject with MO. The methods described herein may include a step of screening a subject for a mutation in EXT1 or EXT2 prior to treatment with or administration of the compositions described herein. A subject can be screened for a genetic mutation using standard methods known to those of skill in the art (e.g., genetic testing). In some embodiments, a subject having MO may be administered an ALK2 or ALK3 antibody, or an antigen-binding fragment thereof, once a week, every two weeks, every three weeks, or once a month. In an embodiment, the ALK2 or ALK3 antibody, or an antigen-binding fragment thereof, may be administered to the subject once a week. In an embodiment, the ALK2 or ALK3 antibody, or an antigen-binding fragment thereof, may be administered to the subject once every 10 to 14 days. In an embodiment, the ALK2 or ALK3 antibody, or an antigen-binding fragment thereof, may be administered to the subject once a month.

**[0122]** In some embodiments, the ALK2 or ALK3 antibodies or antigen binding fragments thereof (e.g., ALK2 or ALK3 binding fragments), described herein may be administered to a subject to treat PCO, reduce or prevent the development of PCO (e.g., to prevent the development of PCO in a subject who has undergone cataract surgery but has not yet developed PCO or in a subject who is soon to undergo cataract surgery e.g., a subject who will undergo cataract surgery in 6 months, 5 months, 4 months, 3 months, 2 months, 1 month, 2 weeks, 1 week, or less, e.g., a subject at risk of developing PCO), improve visual acuity (e.g., reduce blurry or cloudy vision) in a subject with PCO, reduce light sensitivity or glare in a subject with PCO, reduce or prevent fibrosis (e.g., fibrosis near the implanted lens, e.g., fibrosis on or near the posterior capsule), reduce or inhibit lens epithelial cell proliferation, reduce or prevent lens fiber differentiation, or reduce or prevent inflammation in the eye in a subject having PCO or at risk of developing PCO (e.g., in a subject who has undergone or is soon to undergo cataract surgery e.g., a subject who will undergo cataract surgery in 6 months, 5 months, 4 months, 3 months, 2 months, 1 month, 2 weeks, 1 week, or less). The ALK2 or ALK3 antibodies or antigen binding fragments thereof (e.g., ALK2 or ALK3 binding fragments), described herein may reduce or prevent the development of PCO, reduce or prevent fibrosis (e.g., fibrosis near the implanted lens, e.g., fibrosis on or near the posterior capsule), reduce or prevent lens fiber differentiation, reduce or inhibit lens epithelial cell proliferation, or reduce or prevent eye inflammation in a subject at risk of developing PCO compared the development of PCO, fibrosis (e.g., fibrosis near the implanted lens, e.g., fibrosis on or near the posterior capsule), lens fiber differentiation, lens epithelial cell proliferation, or eye inflammation in an untreated subject, or may improve visual acuity (e.g., reduce blurry or cloudy vision) or reduce light

sensitivity or glare, reduce fibrosis, reduce eye inflammation, reduce lens epithelial cell proliferation, or reduce lens fiber differentiation compared to visual acuity, light sensitivity or glare, fibrosis, eye inflammation, lens epithelial cell proliferation, or lens fiber differentiation in the subject prior to treatment.

**[0123]** In some embodiments, the ALK2 or ALK3 antibodies or antigen binding fragments thereof (e.g., ALK2 or ALK3 binding fragments), described herein may be administered to a subject to treat cardiac hypertrophy, treat cardiac fibrosis, slow or prevent the development of cardiac hypertrophy (e.g., in a subject at risk of developing cardiac hypertrophy, such as a subject with hypertension or valvular disease), slow or prevent the development of cardiac fibrosis (e.g., in a subject at risk of developing cardiac fibrosis, such as a subject with hypertension, a subject who has had a myocardial infarction, a subject with diabetic hypertrophic cardiomyopathy, or a subject with idiopathic dilated cardiomyopathy), reverse cardiac fibrosis, reduce or inhibit cardiac scar formation, increase or induce cardiac regeneration, or improve one or more symptoms of cardiac hypertrophy or cardiac fibrosis (e.g., increase exercise capacity, increase blood ejection volume, reduce left ventricular end diastolic pressure, reduce pulmonary capillary wedge pressure, increase cardiac output, increase cardiac index, reduce pulmonary artery pressures, reduce left ventricular end systolic and diastolic dimensions, reduce left and right ventricular wall stress, reduce wall tension and/or wall thickness, increase myocardial contractility, reduce cardiomyocyte area, reduce extracellular matrix deposition in the cardiac muscle, improve quality of life, and/or reduce disease-related morbidity and mortality). The ALK2 or ALK3 antibodies or antigen binding fragments thereof (e.g., ALK2 or ALK3 binding fragments), described herein slow or prevent the development of cardiac hypertrophy, slow or prevent the development of cardiac fibrosis, reverse cardiac fibrosis, reduce or inhibit cardiac scar formation, increase or induce cardiac regeneration, or improve one or more symptoms of cardiac hypertrophy or cardiac fibrosis in a subject having or risk of developing cardiac hypertrophy or cardiac fibrosis compared to the development of cardiac hypertrophy, the development of cardiac fibrosis, the extent of cardiac fibrosis, cardiac scar formation, cardiac regeneration, or one or more symptoms of cardiac hypertrophy or cardiac fibrosis in the subject prior to treatment or in an untreated or control treated subject. In some embodiments, an ALK2 or ALK3 antibody, or an antigen-binding fragment thereof, may be administered to a subject to treat hypertrophy, treat cardiac fibrosis, slow or prevent the development of cardiac hypertrophy once a week, every two weeks, every three weeks, or once a month. In an embodiment, the ALK2 or ALK3 antibody, or an antigen-binding fragment thereof, may be administered to the subject once a week. In an embodiment, the ALK2 or ALK3 antibody, or an antigen-binding fragment thereof, may be administered to the subject once every 10 to 14 days. In an embodiment, the ALK2 or ALK3 antibody, or an antigen-binding fragment thereof, may be administered to the subject once a month.

**[0124]** The compositions described herein are administered in an amount sufficient to increase bone mineral density, increase bone strength, density, reduce bone resorption (e.g., bone loss), reduce the rate of bone resorption (e.g., bone loss), increase bone formation, increase the rate of bone formation, reduce osteoclast activity, increase osteo-

blast activity, reduce the risk of bone fracture, increase red blood cell levels, increase hemoglobin levels, reduce the need for a transfusion, increase red blood cell formation, or increase red blood cell count, treat anemia, increase iron levels, reduce iron deficiency, decrease elevated hepcidin levels, treat FOP, prevent heterotopic ossification (e.g., prevent the initial formation of ectopic bone in a subject or prevent the recurrence of ectopic bone after surgical resection), slow or stop the progression of FOP, delay the onset of FOP (e.g., delay the onset of heterotopic ossification), reduce heterotopic ossification (e.g., reduce the amount of ectopic bone in a subject or reduce the formation of ectopic bone, such as the initial formation of ectopic bone or the recurrence of ectopic bone after surgical resection), treat Sjogren's syndrome, reduce dry eye associated with Sjogren's syndrome, reduce dry mouth associated with Sjogren's syndrome, reduce eye inflammation, increase the production of saliva, increase the production of tears, reduce systemic inflammation, reduce joint pain, reduce fatigue, reduce the severity or occurrence of symptoms of Sjogren's syndrome (e.g., reduce dry eye, dry mouth, or joint pain associated with Sjogren's syndrome), treat DIPG, increase the survival (e.g., survival time, e.g., lifespan) of a subject having DIPG, increase progression free survival of a subject having DIPG, reduce DIPG tumor growth, reduce DIPG tumor size or volume, prevent or reduce DIPG tumor metastasis, treat MO, reduce or prevent the formation of an osteochondroma in a subject with MO, reduce the number of osteochondromas in a subject with MO, reduce the size of an osteochondroma in a subject with MO, slow the growth of an osteochondroma in a subject with MO, reduce or prevent the development of PCO, improve visual acuity (e.g., reduce blurry or cloudy vision) in a subject with PCO, reduce light sensitivity or glare in a subject with PCO, reduce or prevent fibrosis (e.g., fibrosis near the implanted lens, e.g., fibrosis on or near the posterior capsule), reduce or inhibit lens epithelial cell proliferation, reduce or prevent lens fiber differentiation, or reduce or prevent inflammation in the eye of a subject with PCO, treat cardiac hypertrophy, treat cardiac fibrosis, slow or prevent the development of cardiac hypertrophy, slow or prevent the development of cardiac fibrosis, reverse cardiac fibrosis, reduce or inhibit cardiac scar formation, increase or induce cardiac regeneration, or improve one or more symptoms of cardiac hypertrophy or cardiac fibrosis (e.g., increase exercise capacity, increase blood ejection volume, reduce left ventricular end diastolic pressure, reduce pulmonary capillary wedge pressure, increase cardiac output, increase cardiac index, reduce pulmonary artery pressures, reduce left ventricular end systolic and diastolic dimensions, reduce left and right ventricular wall stress, reduce wall tension and/or wall thickness, increase myocardial contractility, reduce extracellular matrix deposition in the cardiac muscle, improve quality of life, and/or reduce disease-related morbidity and mortality). Bone mineral density can be evaluated using well-established clinical techniques known to one of skill in the art (e.g., by dual-energy x-ray absorptiometry). Red blood cell levels can be assessed using a standard blood test, which measures red blood cell counts and hemoglobin levels. FOP symptoms (e.g., heterotopic ossification) and MO (e.g., osteochondromas) can be evaluated using standard imaging methods, such as radiographs (e.g., X-rays), CT (computed tomography), and/or MRI (magnetic resonance imaging). DIPG tumor size, growth, volume, or metastasis, may be

assessed using imaging methods, such as MRI. Symptoms of cardiac hypertrophy and/or cardiac fibrosis may be evaluated using clinical approaches to evaluate heart function and morphology (e.g., a stress test, echocardiogram, CT, MRI, cardiac magnetic resonance imaging, single photon emission computed tomography, positron emission tomography, a balloon catheter-based approach, or histological analysis of a biopsy). Sjogren's syndrome and PCO symptoms may be assessed by evaluating visual function and imaging the eye to assess inflammation and/or fibrosis (e.g., using slit lamp imaging to evaluate fibrosis forming on the posterior capsule). The methods described herein may also include a step of assessing bone mineral density, red blood cell levels, or symptoms of FOP, DIPG, MO, PCO, Sjogren's syndrome, or cardiac hypertrophy and/or cardiac fibrosis in a subject prior to treatment with or administration of the compositions described herein or after administration of or treatment with the compositions described herein. The subject may be evaluated 1 month, 2 months, 3 months, 4 months, 5 months, 6 months or more following administration of the ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment), or pharmaceutical composition depending on the dose and route of administration used for treatment. Depending on the outcome of the evaluation, the subject may receive additional treatments.

**[0125]** Nucleic acids encoding the ALK2 or ALK3 antibodies or antigen binding fragments thereof (e.g., ALK2 or ALK3 binding fragments), described herein, or expression vectors containing said nucleic acids can also be administered according to any of the methods described herein. In any of the methods described herein, the polypeptide, nucleic acid, or vector can be administered as part of a pharmaceutical composition. Compositions that can be administered to a subject according to the methods described herein are provided in Table 2.

#### Kits

**[0126]** The compositions described herein can be provided in a kit for use in treating bone disease, anemia, FOP, Sjogren's syndrome (e.g., dry eye associated with Sjogren's syndrome), MO, DIPG, PCO, or cardiac hypertrophy and/or cardiac fibrosis. Compositions may include an ALK2 antibody or antigen binding fragment thereof (e.g., an ALK2 binding fragment), described herein, and may be provided in unit dosage form, optionally in a pharmaceutically acceptable excipient (e.g., saline), in an amount sufficient to treat bone disease, anemia, FOP, Sjogren's syndrome (e.g., dry eye associated with Sjogren's syndrome), MO, DIPG, PCO, or cardiac hypertrophy and/or cardiac fibrosis. The kit can further include a package insert that instructs a user of the kit, such as a physician, to perform the methods described herein. The kit may optionally include a syringe or other device for administering the composition.

#### EXAMPLES

**[0127]** The following examples are provided to further illustrate some embodiments of the present invention, but are not intended to limit the scope of the invention; it will be understood by their exemplary nature that other procedures, methodologies, or techniques known to those skilled in the art may alternatively be used.

Example 1—Effect of ALK2 Antibodies in a  
Non-Human Primate

[0128] To determine the pharmacokinetic and pharmacodynamic properties of Antibody 2 and Antibody 4 in cynomolgus monkeys, a preclinical model highly representative of human biology, ad libitum-fed, female cynomolgus monkeys ranging in age from 2-4 years old were randomly assigned to receive Antibody 2 or Antibody 4 administered as a single subcutaneous dose of 3 mg/kg. Serum was sampled intermittently over a 56-day period and assessed for drug exposure (FIGS. 1A and 1B). Analysis of the exposure data using Excel PK Solver resulted in the parameters summarized in Table 7. Serum sampled intermittently over a 56-day period was also assessed for serum hepcidin levels (FIGS. 2A and 2B) measured using Intrinsic LifeSciences Primate-Hepcidin Compete ELISA kit as per manufacturer's instructions, serum iron levels (FIGS. 3A and 3B) which were measured using BioAssay Systems Iron Assay Kit, and transferrin saturation (FIGS. 4A and 4B) measured at IDEXX. At a subset of timepoints, blood was sampled and assessed for reticulocyte (FIG. 5A) and red blood cell hemoglobin content (FIG. 5B) (RET-Hgb and RBC-Hgb, respectively) and mean corpuscular hemoglobin concentration (FIG. 5C) with the use of a IDEXX Procyte analyzer. The hematological data was combined across the therapeutic groups that were administered Antibody 2 or Antibody 4.

RBC-Hgb content were observed initially 35 days post dose and remained 4.2% higher at study termination at day 56 (FIG. 5B).

[0130] These data demonstrate that both Antibody 2 and Antibody 4 are efficacious at reducing serum hepcidin and increasing circulating iron in non-human primates, a pre-clinical model highly representative of human biology. Furthermore, the mobilized iron resulted in increased RET-Hgb and RBC-Hgb. These results provide evidence that treatment with Antibody 2 and Antibody 4 may be a viable approach to treating anemias such as IRIDA and inflammation of anemia that arise from elevated hepcidin.

Example 2—Effect of ALK2 Antibodies in  
Wild-Type Mice

[0131] To determine the pharmacokinetic and pharmacodynamic properties of Antibody 2 and Antibody 4 in mice, ad libitum-fed, 8-week-old, male CD1 mice were randomly assigned to receive IgG1 isotype control, Antibody 2, or Antibody 4 administered as a single subcutaneous dose of 0.5 mg/kg, 1.0 mg/kg, or 3 mg/kg. Serum was collected from whole blood 24 hours post-dose. Serum was assessed for drug exposure, hepcidin levels (Intrinsic LifeSciences Hepcidin Murine-Compete ELISA kit as per manufacturer's instructions), and iron levels (Pointe Scientific Iron/TIBC

TABLE 7

| Summary of pharmacokinetic parameters of drug exposure |                      |                     |                     |
|--------------------------------------------------------|----------------------|---------------------|---------------------|
| Parameter                                              | Unit                 | Antibody 2<br>Value | Antibody 4<br>Value |
| Lambda <sub>z</sub>                                    | 1/h                  | 0.014113837         | 0.020451            |
| t <sub>1/2</sub>                                       | h                    | 49.11117886         | 33.893021           |
| Tmax                                                   | h                    | 48                  | 24                  |
| Cmax                                                   | µg/ml                | 11.24               | 17.466667           |
| Tlag                                                   | h                    | 0                   | 0                   |
| Clast <sub>obs</sub> /Cmax                             |                      | 0.045077106         | 0.0129771           |
| AUC 0-t                                                | µg/ml*h              | 1852.255            | 2551.89             |
| AUC 0-inf <sub>obs</sub>                               | µg/ml*h              | 1888.153577         | 2562.9734           |
| AUC 0-t/0-inf <sub>obs</sub>                           |                      | 0.98098747          | 0.9956756           |
| AUMC 0-inf <sub>obs</sub>                              | µg/ml*h <sup>2</sup> | 216906.6341         | 245569.71           |
| MRT 0-inf <sub>obs</sub>                               | h                    | 114.8776438         | 95.81438            |
| Vz/F <sub>obs</sub>                                    | (mg/kg)/(µg/ml)      | 0.112574192         | 0.057235            |
| Cl/F <sub>obs</sub>                                    | (mg/kg)/(µg/ml)/h    | 0.001588854         | 0.0011705           |

[0129] Antibody 2 and Antibody 4 were rapidly absorbed and reached C<sub>max</sub> within 48 hrs and had half-lives of 33.9 hours and 49.1 hours, respectively (FIGS. 1A and 1B). Robust and profound effects on hepcidin and serum iron were observed with both antibodies. Within 6 hours of administration of Antibody 2 or Antibody 4, serum hepcidin was reduced by 55.6% and 50.3%, respectively (FIGS. 2A and 2B). The peak effect was observed beginning after 48 hours with reductions of 77.2% in the Antibody 2 treated group, and 77.8% in the Antibody 4 group and continued through day 10 before returning to baseline by day 14. The reduction in hepcidin corresponded to increased circulating iron with maximal changes occurring within 24 hours of antibody administration. A 63.3% and 54.2% increase in iron was achieved with Antibody 2 and Antibody 4, respectively (FIGS. 3A and 3B). Similar to hepcidin, the response was sustained through 10 days, returning to baseline by day 14. RET-Hgb increased by 4.9% at 3 days post dose and remained elevated for 10 days (FIG. 5A). Increases in

Reagent Set). Results are shown in FIGS. 6A-6C. ns—not significant; \*p<0.05; \*\*p<0.01; \*\*\*p<0.001.

Example 3—Treatment of Anemia by  
Administration of an ALK2 or ALK3 Antibody

[0132] According to the methods disclosed herein, a physician of skill in the art can treat a subject, such as a human patient, having anemia (e.g., IRIDA or anemia of inflammation) so as to increase hemoglobin levels, increase red blood cell counts, or reduce iron deficiency. The method of treatment can include diagnosing or identifying a subject as a candidate for treatment based on a blood test measuring hematological parameters. To treat the subject, a physician of skill in the art can administer to the subject a composition containing an ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment). The composition containing the ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an

ALK2 or ALK3 binding fragment), may be administered to the subject, for example, subcutaneously to treat anemia. The ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment), is administered in a therapeutically effective amount, such as from 0.002 mg/kg to 3 mg/kg (e.g., 0.005 mg/kg to 3 mg/kg, 0.01 mg/kg to 3 mg/kg, 0.05 mg/kg to 3 mg/kg, 0.1 mg/kg to 3 mg/kg, 0.5 mg/kg to 3 mg/kg, 1 mg/kg to 3 mg/kg, 1.5 mg/kg to 3 mg/kg, 2 mg/kg to 3 mg/kg, 2.5 mg/kg to 3 mg/kg, 0.002 mg/kg to 2.5 mg/kg, 0.002 mg/kg to 2 mg/kg, 0.002 mg/kg to 1.5 mg/kg, 0.002 mg/kg to 1 mg/kg, 0.002 mg/kg to 0.5 mg/kg, 0.002 mg/kg to 0.25 mg/kg, 0.002 mg/kg to 0.1 mg/kg, 0.002 mg/kg to 0.05 mg/kg, 0.002 mg/kg to 0.01 mg/kg, 0.002 mg/kg to 0.005 mg/kg, 0.1 mg/kg to 0.3 mg/kg, or 0.1 mg/kg to 1 mg/kg). In some embodiments, the ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment), is administered once a month, once every two months, once every three months, once every four months, once every five months, or once every six months. The ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment), is administered in an amount sufficient to increase hemoglobin levels, increase red blood cell counts, or reduce iron deficiency.

**[0133]** Following administration of the composition to a patient, a practitioner of skill in the art can monitor the patient's improvement in response to the therapy by a variety of methods. For example, a physician can monitor the patient's hemoglobin levels, red blood cell counts, or iron deficiency by performing a blood test. A finding that the patient exhibits improved hemoglobin levels, red blood cell counts, or iron deficiency following administration of the composition compared to test results prior to administration of the composition indicates that the patient is responding favorably to the treatment. Subsequent doses can be determined and administered as needed.

#### Example 4—Treatment of Bone Disease by Administration of an ALK2 or ALK3 Antibody

**[0134]** According to the methods disclosed herein, a physician of skill in the art can treat a subject, such as a human patient, having bone disease (e.g., osteoporosis or osteopenia) so as to increase bone mineral density, increase bone formation, reduce bone resorption (e.g., bone loss), or reduce the risk of bone fracture. The method of treatment can include diagnosing or identifying a subject as a candidate for treatment based on standard clinical tests for bone mineral density (e.g., dual X-ray absorptiometry). To treat the subject, a physician of skill in the art can administer to the subject a composition containing an ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment). The composition containing the ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 binding fragment), may be administered to the subject, for example, by parenteral injection (e.g., intravenous injection) to treat bone disease. The ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment), is administered in a therapeutically effective amount, such as from 0.002 mg/kg to 3 mg/kg (e.g., 0.005 mg/kg to 3 mg/kg, 0.01 mg/kg to 3 mg/kg, 0.05 mg/kg to 3 mg/kg, 0.1 mg/kg to 3 mg/kg, 0.5 mg/kg to 3 mg/kg, 1 mg/kg to 3 mg/kg, 1.5 mg/kg to 3 mg/kg, 2 mg/kg to 3 mg/kg, 2.5 mg/kg to 3

mg/kg, 0.002 mg/kg to 2.5 mg/kg, 0.002 mg/kg to 2 mg/kg, 0.002 mg/kg to 1.5 mg/kg, 0.002 mg/kg to 1 mg/kg, 0.002 mg/kg to 0.5 mg/kg, 0.002 mg/kg to 0.25 mg/kg, 0.002 mg/kg to 0.1 mg/kg, 0.002 mg/kg to 0.05 mg/kg, 0.002 mg/kg to 0.01 mg/kg, 0.002 mg/kg to 0.005 mg/kg, 0.1 mg/kg to 0.3 mg/kg, or 0.1 mg/kg to 1 mg/kg). In some embodiments, the ALK2 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment), is administered once a month, once every two months, once every three months, once every four months, once every five months, or once every six months. The ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment), is administered in an amount sufficient to increase bone mineral density, increase bone formation, reduce bone resorption (e.g., bone loss), or reduce the risk of bone fracture.

**[0135]** Following administration of the composition to a patient, a practitioner of skill in the art can monitor the patient's improvement in response to the therapy by a variety of methods. For example, a physician can monitor the patient's bone mineral density by performing dual X-ray absorptiometry. A finding that the patient exhibits increased bone mineral density, increased bone formation, reduced bone resorption (e.g., bone loss), or a reduced risk of bone fracture following administration of the composition compared to test results prior to administration of the composition indicates that the patient is responding favorably to the treatment. Subsequent doses can be determined and administered as needed.

#### Example 5—Treatment of FOP by Administration of an ALK2 Antibody

**[0136]** According to the methods disclosed herein, a physician of skill in the art can treat a subject, such as a human patient, having FOP (e.g., a patient with a gain of function mutation in ALK2) so as to prevent or reduce heterotopic ossification, slow the progression of FOP, or delay the onset of FOP. The method of treatment can include diagnosing or identifying a subject as a candidate for treatment based on genetic testing for a mutation in ALK2 or performing radiographic imaging, a CT, or an MRI. To treat the subject, a physician of skill in the art can administer to the subject a composition containing an ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment). The composition containing the ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment), may be administered to the subject, for example, by parenteral injection (e.g., intravenous injection) to treat FOP. The ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment), is administered in a therapeutically effective amount, such as from 0.002 to 3 mg/kg (e.g., 0.005 mg/kg to 3 mg/kg, 0.01 mg/kg to 3 mg/kg, 0.05 mg/kg to 3 mg/kg, 0.1 mg/kg to 3 mg/kg, 0.5 mg/kg to 3 mg/kg, 1 mg/kg to 3 mg/kg, 1.5 mg/kg to 3 mg/kg, 2 mg/kg to 3 mg/kg, 2.5 mg/kg to 3 mg/kg, 0.002 mg/kg to 2.5 mg/kg, 0.002 mg/kg to 2 mg/kg, 0.002 mg/kg to 1.5 mg/kg, 0.002 mg/kg to 1 mg/kg, 0.002 mg/kg to 0.5 mg/kg, 0.002 mg/kg to 0.25 mg/kg, 0.002 mg/kg to 0.1 mg/kg, 0.002 mg/kg to 0.05 mg/kg, 0.002 mg/kg to 0.01 mg/kg, 0.002 mg/kg to 0.005 mg/kg, 0.1 mg/kg to 0.3 mg/kg, or 0.1 mg/kg to 1 mg/kg). In some embodiments, the ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment), is

administered once a week, once every two weeks, once every three weeks, or once a month. The ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment), is administered in an amount sufficient to reduce heterotopic ossification, slow the progression of FOP, or delay the onset of FOP.

**[0137]** Following administration of the composition to a patient, a practitioner of skill in the art can monitor the patient's improvement in response to the therapy by a variety of methods. For example, a physician can monitor the patient's heterotopic ossification or disease progression using radiography, CT, and/or MRI. A finding that the patient exhibits reduced heterotopic ossification (e.g., a reduction in the amount of heterotopic ossification or the formation of additional heterotopic bone), a failure to develop heterotopic ossification, or delayed onset or slowed progression of FOP following administration of the composition compared to test results prior to administration of the composition or compared to an untreated subject indicates that the patient is responding favorably to the treatment. Subsequent doses can be determined and administered as needed.

#### Example 6—Treatment of Sjogren's Syndrome by Administration of an ALK2 or ALK3 Antibody

**[0138]** According to the methods disclosed herein, a physician of skill in the art can treat a subject, such as a human patient, having Sjogren's syndrome so as to ameliorate or reduce one or more symptom of Sjogren's syndrome (e.g., dry eye, dry mouth, eye irritation, blurred vision, lacrimal gland fibrosis, lacrimal gland inflammation, joint pain, or fatigue). To treat the subject, a physician of skill in the art can administer to the subject a composition containing an ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment). The composition containing the ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment), may be administered to the subject, for example, by parenteral injection (e.g., intravenous injection and subcutaneous injection) or by administration to the eye (e.g., intraocular injection or topical administration) to treat Sjogren's syndrome. The ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment), is administered in a therapeutically effective amount, such as from 0.002 to 3 mg/kg (e.g., 0.005 mg/kg to 3 mg/kg, 0.01 mg/kg to 3 mg/kg, 0.05 mg/kg to 3 mg/kg, 0.1 mg/kg to 3 mg/kg, 0.5 mg/kg to 3 mg/kg, 1 mg/kg to 3 mg/kg, 1.5 mg/kg to 3 mg/kg, 2 mg/kg to 3 mg/kg, 2.5 mg/kg to 3 mg/kg, 0.002 mg/kg to 2.5 mg/kg, 0.002 mg/kg to 2 mg/kg, 0.002 mg/kg to 1.5 mg/kg, 0.002 mg/kg to 1 mg/kg, 0.002 mg/kg to 0.5 mg/kg, 0.002 mg/kg to 0.25 mg/kg, 0.002 mg/kg to 0.1 mg/kg, 0.002 mg/kg to 0.05 mg/kg, 0.002 mg/kg to 0.01 mg/kg, 0.002 mg/kg to 0.005 mg/kg, 0.1 mg/kg to 0.3 mg/kg, or 0.1 mg/kg to 1 mg/kg). In some embodiments, the ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment), is administered once a week, once every two weeks, once every three weeks, or once a month. The ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment), is administered in an amount sufficient to ameliorate or reduce one or more symptom of Sjogren's syndrome (e.g., dry eye, dry mouth, eye irritation, blurred vision, lacrimal gland

fibrosis, lacrimal gland inflammation, joint pain, or fatigue, e.g., by increasing the production of tears or saliva).

**[0139]** Following administration of the composition to a patient, a practitioner of skill in the art can monitor the patient's improvement in response to the therapy by a variety of methods. For example, a physician can monitor the patient's symptoms of Sjogren's syndrome by evaluating the production of tears (e.g., tear volume production), the production of saliva, vision, eye irritation, inflammation or fibrosis of the lacrimal gland, or the patient's reports of joint pain or fatigue. A finding that the patient exhibits a reduction in one or more symptom of Sjogren's syndrome (e.g., dry eye, dry mouth, eye irritation, blurred vision, lacrimal gland fibrosis, lacrimal gland inflammation, joint pain, or fatigue) following administration of the composition compared to the symptom prior to administration of the composition or compared to an untreated subject indicates that the patient is responding favorably to the treatment. Subsequent doses can be determined and administered as needed.

#### Example 7—Treatment of MO by Administration of an ALK2 or ALK3 Antibody

**[0140]** According to the methods disclosed herein, a physician of skill in the art can treat a subject, such as a human patient, having MO so as to prevent or reduce the formation of an osteochondroma, reduce the size of an osteochondroma, reduce the number of osteochondromas, or slow the growth of an osteochondroma. The method of treatment can include diagnosing or identifying a subject as a candidate for treatment based on genetic testing for a mutation in EXT1 or EXT2 or performing radiographic imaging, a CT, or an MRI. To treat the subject, a physician of skill in the art can administer to the subject a composition containing an ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment). The composition containing the ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment), may be administered to the subject, for example, by parenteral injection (e.g., intravenous injection and subcutaneous injection) to treat MO. The ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment), is administered in a therapeutically effective amount, such as from 0.002 to 3 mg/kg (e.g., 0.005 mg/kg to 3 mg/kg, 0.01 mg/kg to 3 mg/kg, 0.05 mg/kg to 3 mg/kg, 0.1 mg/kg to 3 mg/kg, 0.5 mg/kg to 3 mg/kg, 1 mg/kg to 3 mg/kg, 1.5 mg/kg to 3 mg/kg, 2 mg/kg to 3 mg/kg, 2.5 mg/kg to 3 mg/kg, 0.002 mg/kg to 2.5 mg/kg, 0.002 mg/kg to 2 mg/kg, 0.002 mg/kg to 1.5 mg/kg, 0.002 mg/kg to 1 mg/kg, 0.002 mg/kg to 0.5 mg/kg, 0.002 mg/kg to 0.25 mg/kg, 0.002 mg/kg to 0.1 mg/kg, 0.002 mg/kg to 0.05 mg/kg, 0.002 mg/kg to 0.01 mg/kg, 0.002 mg/kg to 0.005 mg/kg, 0.1 mg/kg to 0.3 mg/kg, or 0.1 mg/kg to 1 mg/kg). In some embodiments, the ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment), is administered once a week, once every two weeks, once every three weeks, or once a month. The ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment), is administered in an amount sufficient to reduce the formation of an osteochondroma, reduce the size of an osteochondroma, reduce the number of osteochondromas, or slow the growth of an osteochondroma.

**[0141]** Following administration of the composition to a patient, a practitioner of skill in the art can monitor the patient's improvement in response to the therapy by a variety of methods. For example, a physician can monitor the growth, number, or size of osteochondromas in the patient using radiography, CT, and/or MRI. A finding that the patient exhibits reduced osteochondroma formation (e.g., does not form new osteochondromas or forms fewer osteochondromas compared to an untreated subject or osteochondroma formation prior to treatment), reduced osteochondroma size, reduced osteochondroma number, or slower osteochondroma growth following administration of the composition compared to test results prior to administration of the composition or compared to an untreated subject indicates that the patient is responding favorably to the treatment. Subsequent doses can be determined and administered as needed.

Example 8—Treatment of DIPG by Administration of an ALK2 or ALK3 Antibody

**[0142]** According to the methods disclosed herein, a physician of skill in the art can treat a subject, such as a human patient, having DIPG so as to improve survival time, reduce tumor growth, reduce tumor size or volume, or prevent or reduce tumor metastasis. The method of treatment can include diagnosing or identifying a subject as a candidate for treatment based on genetic testing for an activating mutation in ALK2. To treat the subject, a physician of skill in the art can administer to the subject a composition containing an ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment). The composition containing the ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment), may be administered to the subject, for example, by parenteral injection (e.g., intravenous or subcutaneous injection) or by local administration to the central nervous system (e.g., intracerebroventricular injection, intrathecal injection, or intra-cisternal injection). The ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment), is administered in a therapeutically effective amount, such as from 0.002 to 3 mg/kg (e.g., 0.005 mg/kg to 3 mg/kg, 0.01 mg/kg to 3 mg/kg, 0.05 mg/kg to 3 mg/kg, 0.1 mg/kg to 3 mg/kg, 0.5 mg/kg to 3 mg/kg, 1 mg/kg to 3 mg/kg, 1.5 mg/kg to 3 mg/kg, 2 mg/kg to 3 mg/kg, 2.5 mg/kg to 3 mg/kg, 0.002 mg/kg to 2.5 mg/kg, 0.002 mg/kg to 2 mg/kg, 0.002 mg/kg to 1.5 mg/kg, 0.002 mg/kg to 1 mg/kg, 0.002 mg/kg to 0.5 mg/kg, 0.002 mg/kg to 0.25 mg/kg, 0.002 mg/kg to 0.1 mg/kg, 0.002 mg/kg to 0.05 mg/kg, 0.002 mg/kg to 0.01 mg/kg, 0.002 mg/kg to 0.005 mg/kg, 0.1 mg/kg to 0.3 mg/kg, or 0.1 mg/kg to 1 mg/kg). In some embodiments, the ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment), is administered once a week, once every two weeks, once every three weeks, or once a month. The ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment), is administered in an amount sufficient to improve survival time, reduce tumor growth, reduce tumor size or volume, or prevent or reduce tumor metastasis.

**[0143]** Following administration of the composition to a patient, a practitioner of skill in the art can monitor the patient's improvement in response to the therapy by a variety of methods. For example, a physician can monitor

the patient's survival time or tumor size, growth, or metastasis using imaging (e.g., MRI). A finding that the patient exhibits reduced tumor growth, reduced tumor size or volume, reduced tumor metastasis, or increased survival time compared to an untreated subject or compared to tumor growth, size, or metastasis in the subject prior to administration indicates that the patient is responding favorably to the treatment. Subsequent doses can be determined and administered as needed.

Example 9—Treatment or Prevention of PCO by Administration of an ALK2 or ALK3 Antibody

**[0144]** According to the methods disclosed herein, a physician of skill in the art can treat a subject, such as a human patient, having or at risk of developing PCO (e.g., a patient having PCO or a patient that has undergone cataract surgery or is soon to undergo cataract surgery, e.g., a patient that will undergo cataract surgery in 6 months, 5 months, 4 months, 3 months, 2 months, 1 month, 2 weeks, 1 week, or less) so as to prevent the development of PCO, reduce PCO, improve visual acuity (e.g., reduce blurry or cloudy vision), reduce light sensitivity or glare, reduce or prevent fibrosis (e.g., fibrosis on or near the posterior capsule, e.g., near the implanted lens), reduce or prevent lens fiber differentiation, reduce or inhibit lens epithelial cell proliferation, or reduce or prevent inflammation in the eye. To treat the subject, a physician of skill in the art can administer to the subject a composition containing an ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment). The composition containing the ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment), may be administered to the subject, for example, by parenteral injection (e.g., intravenous or subcutaneous injection) or by administration to the eye (e.g., intraocular injection or topical administration) to treat PCO. The ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment) is administered in a therapeutically effective amount, such as from 0.002 to 3 mg/kg (e.g., 0.005 mg/kg to 3 mg/kg, 0.01 mg/kg to 3 mg/kg, 0.05 mg/kg to 3 mg/kg, 0.1 mg/kg to 3 mg/kg, 0.5 mg/kg to 3 mg/kg, 1 mg/kg to 3 mg/kg, 1.5 mg/kg to 3 mg/kg, 2 mg/kg to 3 mg/kg, 2.5 mg/kg to 3 mg/kg, 0.002 mg/kg to 2.5 mg/kg, 0.002 mg/kg to 2 mg/kg, 0.002 mg/kg to 1.5 mg/kg, 0.002 mg/kg to 1 mg/kg, 0.002 mg/kg to 0.5 mg/kg, 0.002 mg/kg to 0.25 mg/kg, 0.002 mg/kg to 0.1 mg/kg, 0.002 mg/kg to 0.05 mg/kg, 0.002 mg/kg to 0.01 mg/kg, 0.002 mg/kg to 0.005 mg/kg, 0.1 mg/kg to 0.3 mg/kg, or 0.1 mg/kg to 1 mg/kg). In some embodiments, the ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment), is administered once a week, once every two weeks, once every three weeks, or once a month. The ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment), is administered in an amount sufficient to prevent the development of PCO, reduce PCO, improve visual acuity (e.g., reduce blurry or cloudy vision), reduce light sensitivity or glare, reduce or prevent fibrosis (e.g., fibrosis on or near the posterior capsule, e.g., near the implanted lens), reduce or prevent lens fiber differentiation, reduce or inhibit lens epithelial cell proliferation, or reduce or prevent inflammation in the eye.

**[0145]** Following administration of the composition to a patient, a practitioner of skill in the art can monitor the



patient's improvement in response to the therapy by a variety of methods. For example, a physician can monitor the patient's visual acuity, sensitivity to light or glare, opacity of the posterior capsule, or eye inflammation. A finding that the patient does not develop PCO or exhibits improved visual acuity, reduced sensitivity to light or glare, reduced fibrosis, or reduced inflammation in the eye following administration of the composition compared to an untreated subject or compared to measurements from the patient prior to treatment indicates that the patient is responding favorably to the treatment. Subsequent doses can be determined and administered as needed.

Example 10—Treatment of Cardiac Hypertrophy and/or Cardiac Fibrosis by Administration of an ALK2 or ALK3 Antibody

**[0146]** According to the methods disclosed herein, a physician of skill in the art can treat a subject, such as a human patient, having cardiac hypertrophy and/or cardiac fibrosis so as to slow or prevent the development of cardiac hypertrophy or cardiac fibrosis, reverse cardiac fibrosis, reduce or inhibit cardiac scar formation, increase or induce cardiac regeneration, or improve one or more symptoms of cardiac hypertrophy or cardiac fibrosis (e.g., increase exercise capacity, increase blood ejection volume, reduce left ventricular end diastolic pressure, reduce pulmonary capillary wedge pressure, increase cardiac output, increase cardiac index, reduce pulmonary artery pressures, reduce left ventricular end systolic and diastolic dimensions, reduce left and right ventricular wall stress, reduce wall tension and/or wall thickness, increase myocardial contractility, reduce extracellular matrix deposition in the cardiac muscle, or reduce fibrosis). To treat the subject, a physician of skill in the art can administer to the subject a composition containing an ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment). The composition containing the ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment), may be administered to the subject, for example, by parenteral injection (e.g., intravenous or subcutaneous injection) to treat cardiac hypertrophy and/or cardiac fibrosis. The ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment), is administered in a therapeutically effective amount, such as from 0.002 to 3 mg/kg (e.g., 0.005 mg/kg to 3 mg/kg, 0.01 mg/kg to 3 mg/kg, 0.05 mg/kg to 3 mg/kg, 0.1 mg/kg to 3 mg/kg, 0.5 mg/kg to 3 mg/kg, 1 mg/kg to 3 mg/kg, 1.5 mg/kg to 3 mg/kg, 2 mg/kg to 3 mg/kg, 2.5 mg/kg to 3 mg/kg, 0.002 mg/kg to 2.5 mg/kg, 0.002 mg/kg to 2 mg/kg, 0.002 mg/kg to 1.5 mg/kg, 0.002 mg/kg to 1 mg/kg, 0.002 mg/kg to 0.5 mg/kg, 0.002 mg/kg to 0.25 mg/kg, 0.002 mg/kg to 0.1 mg/kg, 0.002 mg/kg to 0.05 mg/kg, 0.002 mg/kg to 0.01 mg/kg, 0.002 mg/kg to 0.005 mg/kg, 0.1 mg/kg to 0.3 mg/kg, or 0.1 mg/kg to 1 mg/kg). In some embodiments, the ALK2 or ALK3 antibody or an antigen binding fragment thereof (e.g., an ALK2 or ALK3 binding fragment), is administered once a week, once every two weeks, once every three weeks, or once a month. The ALK2 or ALK3 antibody or an antigen binding frag-

ment thereof (e.g., an ALK2 or ALK3 binding fragment), is administered in an amount sufficient to slow or prevent the development of cardiac hypertrophy or cardiac fibrosis, reverse cardiac fibrosis, reduce or inhibit cardiac scar formation, increase or induce cardiac regeneration, or improve one or more symptoms of cardiac hypertrophy or cardiac fibrosis (e.g., increase exercise capacity, increase blood ejection volume, reduce left ventricular end diastolic pressure, reduce pulmonary capillary wedge pressure, increase cardiac output, increase cardiac index, reduce pulmonary artery pressures, reduce left ventricular end systolic and diastolic dimensions, reduce left and right ventricular wall stress, reduce wall tension and/or wall thickness, increase myocardial contractility, reduce extracellular matrix deposition in the cardiac muscle, or reduce fibrosis).

**[0147]** Following administration of the composition to a patient, a practitioner of skill in the art can monitor the patient's improvement in response to the therapy by a variety of methods. For example, a physician can monitor the patient's symptoms of cardiac hypertrophy or cardiac fibrosis (e.g., exercise capacity, blood ejection volume, left ventricular end diastolic pressure, pulmonary capillary wedge pressure, cardiac output, cardiac index, pulmonary artery pressures, left ventricular end systolic and diastolic dimensions, left and right ventricular wall stress, wall tension and/or wall thickness, or myocardial contractility) using a stress test, echocardiogram, MRI, or other approaches. A finding that the patient exhibits reduced cardiac fibrosis, increased cardiac regeneration, or an improvement of one or more symptoms of cardiac hypertrophy or cardiac fibrosis (e.g., increase exercise capacity, increase blood ejection volume, reduce left ventricular end diastolic pressure, reduce pulmonary capillary wedge pressure, increase cardiac output, increase cardiac index, reduce pulmonary artery pressures, reduce left ventricular end systolic and diastolic dimensions, reduce left and right ventricular wall stress, reduce wall tension and/or wall thickness, increase myocardial contractility, reduce extracellular matrix deposition in the cardiac muscle, or reduce fibrosis) following administration of the composition compared to test results prior to administration of the composition or compared to an untreated subject indicates that the patient is responding favorably to the treatment. Subsequent doses can be determined and administered as needed.

Other Embodiments

**[0148]** While the invention has been described in connection with specific embodiments thereof, it will be understood that it is capable of further modifications and this application is intended to cover any variations, uses, or adaptations of the invention following, in general, the principles of the invention and including such departures from the present disclosure come within known or customary practice within the art to which the invention pertains and may be applied to the essential features hereinbefore set forth.

**[0149]** All publications, patents, and patent applications are herein incorporated by reference in their entirety to the same extent as if each individual publication, patent, or patent application was specifically and individually indicated to be incorporated by reference in its entirety.

**[0150]** Other embodiments are within the following claims.

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Lys Phe Gln Gly  
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|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Trp | Met | Gly | Gly | Ile | Ile | Pro | His | Phe | Gly | Ile | Ala | Asn | Tyr | Ala | Gln |
| 1   |     |     |     | 5   |     |     |     |     | 10  |     |     |     |     | 15  |     |

|     |     |     |     |
|-----|-----|-----|-----|
| Lys | Phe | Gln | Gly |
|     |     |     | 20  |

<210> SEQ ID NO 37  
<211> LENGTH: 22  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 37

|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Trp | Val | Gly | Arg | Ile | Lys | Ser | Lys | Ala | Asp | Ser | Gly | Thr | Thr | Asp | Tyr |
| 1   |     |     |     | 5   |     |     |     |     | 10  |     |     |     |     | 15  |     |

|     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|
| Ala | Ala | Pro | Val | Lys | Gly |
|     |     |     |     |     | 20  |

<210> SEQ ID NO 38  
<211> LENGTH: 22  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 38

|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Trp | Val | Gly | Arg | Ile | Lys | Ser | Lys | Arg | Asp | Gly | Tyr | Thr | Thr | Asp | Tyr |
| 1   |     |     |     | 5   |     |     |     |     | 10  |     |     |     |     | 15  |     |

|     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|
| Ala | Ala | Pro | Val | Lys | Gly |
|     |     |     |     |     | 20  |

<210> SEQ ID NO 39  
<211> LENGTH: 20  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 39

|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Trp | Met | Gly | Arg | Ile | Ile | Pro | Asp | Phe | Gly | Thr | Ala | Asn | Tyr | Ala | Gln |
| 1   |     |     |     | 5   |     |     |     |     | 10  |     |     |     |     | 15  |     |

|     |     |     |     |
|-----|-----|-----|-----|
| Lys | Phe | Gln | Gly |
|     |     |     | 20  |

<210> SEQ ID NO 40  
<211> LENGTH: 12  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 40

|     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Arg | Ala | Ser | Gln | Gly | Ile | Ser | Gly | Asn | Trp | Leu | Thr |
| 1   |     |     |     | 5   |     |     |     |     | 10  |     |     |

<210> SEQ ID NO 41  
<211> LENGTH: 11  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 41

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Leu Leu Ile Tyr Asp Ala Ser Asn Leu Gln Ser  
1 5 10

<210> SEQ ID NO 42  
<211> LENGTH: 8  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 42

His Gln Ser Tyr Arg Gly Pro Met  
1 5

<210> SEQ ID NO 43  
<211> LENGTH: 10  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 43

Gly Phe Thr Phe Gly Arg Phe Val Met His  
1 5 10

<210> SEQ ID NO 44  
<211> LENGTH: 20  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 44

Trp Val Ser Val Ile Gly Tyr Ser Gly Ser Ser Thr Tyr Tyr Ala Asp  
1 5 10 15

Ser Val Lys Gly  
20

<210> SEQ ID NO 45  
<211> LENGTH: 20  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 45

Glu Pro Gly Tyr Tyr Tyr Pro Ser Gly Tyr Tyr Arg Gly Pro Gly Tyr  
1 5 10 15

Trp Met Asp Val  
20

<210> SEQ ID NO 46  
<211> LENGTH: 11  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 46

Ser Gly Asp Asn Ile Arg Lys Lys Tyr Val His  
1 5 10

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<210> SEQ ID NO 47  
<211> LENGTH: 11  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 47

Leu Val Ile Tyr Arg Asp Ser Asn Arg Pro Ser  
1 5 10

<210> SEQ ID NO 48  
<211> LENGTH: 8  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 48

Ser Ser Ala Gly Arg Asp Asn Tyr  
1 5

<210> SEQ ID NO 49  
<211> LENGTH: 10  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 49

Gly Phe Thr Phe Ser Ser Ser Ala Met His  
1 5 10

<210> SEQ ID NO 50  
<211> LENGTH: 20  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 50

Trp Val Ser Val Ile His Tyr Asp Ser Ser Glu Thr Tyr Tyr Ala Asp  
1 5 10 15

Ser Val Lys Gly  
20

<210> SEQ ID NO 51  
<211> LENGTH: 7  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 51

Asp Arg Tyr Phe Phe Asp Val  
1 5

<210> SEQ ID NO 52  
<211> LENGTH: 11  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 52

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Ser Gly Asp Ala Leu Arg Tyr Tyr Ile Ala His  
1 5 10

<210> SEQ ID NO 53  
<211> LENGTH: 11  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 53

Leu Val Ile Tyr Tyr Asn Asn Asn Arg Pro Ser  
1 5 10

<210> SEQ ID NO 54  
<211> LENGTH: 8  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 54

Gln Ser Tyr Gly Pro Gly Ser Val  
1 5

<210> SEQ ID NO 55  
<211> LENGTH: 10  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 55

Gly Phe Thr Phe Ser Asp Tyr Ala Met His  
1 5 10

<210> SEQ ID NO 56  
<211> LENGTH: 20  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 56

Trp Val Ser Ser Ile Phe Tyr Ser Gly Ser Asn Thr Tyr Tyr Ala Asp  
1 5 10 15

Ser Val Lys Gly  
20

<210> SEQ ID NO 57  
<211> LENGTH: 11  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 57

Pro Lys Ser Tyr Ala Ser Gly Pro Phe Ala Tyr  
1 5 10

<210> SEQ ID NO 58  
<211> LENGTH: 13  
<212> TYPE: PRT

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<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 58

Ser Gly Ser Ser Ser Asn Ile Gly Gln Asn Tyr Val Ser  
1 5 10

<210> SEQ ID NO 59  
<211> LENGTH: 11  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 59

Leu Leu Ile Tyr Asp Asn Ser Lys Arg Pro Ser  
1 5 10

<210> SEQ ID NO 60  
<211> LENGTH: 10  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 60

Ser Ser Trp Asp Leu Leu Ser Lys Ser Arg  
1 5 10

<210> SEQ ID NO 61  
<211> LENGTH: 10  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 61

Gly Gly Thr Phe Ser Thr His Ala Ile Ser  
1 5 10

<210> SEQ ID NO 62  
<211> LENGTH: 20  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 62

Trp Met Gly Leu Ile Gln Pro Arg Phe Gly Thr Ala Asn Tyr Ala Gln  
1 5 10 15

Lys Phe Gln Arg  
20

<210> SEQ ID NO 63  
<211> LENGTH: 8  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 63

Asp Tyr Tyr Gly Gly Met Ala Tyr  
1 5

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<210> SEQ ID NO 64  
<211> LENGTH: 11  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct  
<220> FEATURE:  
<221> NAME/KEY: MISC\_FEATURE  
<222> LOCATION: (4)..(4)  
<223> OTHER INFORMATION: Xaa is Asn or Ala  
<220> FEATURE:  
<221> NAME/KEY: MISC\_FEATURE  
<222> LOCATION: (5)..(5)  
<223> OTHER INFORMATION: Xaa is Ile or Leu  
<220> FEATURE:  
<221> NAME/KEY: MISC\_FEATURE  
<222> LOCATION: (7)..(8)  
<223> OTHER INFORMATION: Xaa is Lys or Tyr  
<220> FEATURE:  
<221> NAME/KEY: MISC\_FEATURE  
<222> LOCATION: (9)..(9)  
<223> OTHER INFORMATION: Xaa is Tyr or Ile  
<220> FEATURE:  
<221> NAME/KEY: MISC\_FEATURE  
<222> LOCATION: (10)..(10)  
<223> OTHER INFORMATION: Xaa is Val or Ala  
  
<400> SEQUENCE: 64

Ser Gly Asp Xaa Xaa Arg Xaa Xaa Xaa Xaa His  
1 5 10

<210> SEQ ID NO 65  
<211> LENGTH: 11  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct  
<220> FEATURE:  
<221> NAME/KEY: MISC\_FEATURE  
<222> LOCATION: (2)..(2)  
<223> OTHER INFORMATION: Xaa is Val or Leu  
<220> FEATURE:  
<221> NAME/KEY: MISC\_FEATURE  
<222> LOCATION: (5)..(5)  
<223> OTHER INFORMATION: Xaa is Asp, Arg, or Tyr  
<220> FEATURE:  
<221> NAME/KEY: MISC\_FEATURE  
<222> LOCATION: (6)..(6)  
<223> OTHER INFORMATION: Xaa is Ala, Asp, or Asn  
<220> FEATURE:  
<221> NAME/KEY: MISC\_FEATURE  
<222> LOCATION: (7)..(7)  
<223> OTHER INFORMATION: Xaa is Ser or Asn  
<220> FEATURE:  
<221> NAME/KEY: MISC\_FEATURE  
<222> LOCATION: (8)..(8)  
<223> OTHER INFORMATION: Xaa is Lys or Asn  
<220> FEATURE:  
<221> NAME/KEY: MISC\_FEATURE  
<222> LOCATION: (9)..(9)  
<223> OTHER INFORMATION: Xaa is Leu or Arg  
<220> FEATURE:  
<221> NAME/KEY: MISC\_FEATURE  
<222> LOCATION: (10)..(10)  
<223> OTHER INFORMATION: Xaa is Gln or Pro  
  
<400> SEQUENCE: 65

Leu Xaa Ile Tyr Xaa Xaa Xaa Xaa Xaa Ser  
1 5 10

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<210> SEQ ID NO 66
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (2)..(2)
<223> OTHER INFORMATION: Xaa is Phe or Gly
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (5)..(5)
<223> OTHER INFORMATION: Xaa is Gly or Ser
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (6)..(6)
<223> OTHER INFORMATION: Xaa is Arg, Ser, Asp, or Thr
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (7)..(7)
<223> OTHER INFORMATION: Xaa is Phe, Ser, Tyr, or His
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (8)..(8)
<223> OTHER INFORMATION: Xaa is Val or Ala
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (9)..(9)
<223> OTHER INFORMATION: Xaa is Met or Ile
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (10)..(10)
<223> OTHER INFORMATION: Xaa is His or Ser

<400> SEQUENCE: 66

Gly Xaa Thr Phe Xaa Xaa Xaa Xaa Xaa Xaa
1           5           10

<210> SEQ ID NO 67
<211> LENGTH: 452
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 67

Asp Ile Val Leu Thr Gln Pro Pro Ser Val Ser Gly Ala Pro Gly Gln
1           5           10           15

Arg Val Thr Ile Ser Cys Ser Gly Ser Ser Ser Asn Ile Gly Ser Asn
20          25          30

Tyr Val Ser Trp Tyr Gln Gln Leu Pro Gly Thr Ala Pro Lys Val Leu
35          40          45

Ile Tyr Lys Asn Asn His Arg Pro Ser Gly Val Pro Asp Arg Phe Ser
50          55          60

Gly Ser Lys Ser Gly Thr Ser Ala Ser Leu Ala Ile Thr Gly Leu Gln
65          70          75          80

Ala Glu Asp Glu Ala Asp Tyr Tyr Cys Ala Ser Trp Asp His Ser Asp
85          90          95

Arg Phe Tyr Val Phe Gly Gly Gly Thr Lys Leu Thr Val Leu Gly Gln
100         105         110

Pro Lys Ala Ala Pro Ser Val Thr Leu Phe Pro Pro Ser Ser Glu Glu
115         120         125

Leu Gln Ala Asn Lys Ala Thr Leu Val Cys Leu Ile Ser Asp Phe Tyr
130         135         140

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Pro Gly Ala Val Thr Val Ala Trp Lys Ala Asp Ser Ser Pro Val Lys  
 145 150 155 160  
 Ala Gly Val Glu Thr Thr Thr Pro Ser Lys Gln Ser Asn Asn Lys Tyr  
 165 170 175  
 Ala Ala Ser Ser Tyr Leu Ser Leu Thr Pro Glu Gln Trp Lys Ser His  
 180 185 190  
 Arg Ser Tyr Ser Cys Gln Val Thr His Glu Gly Ser Thr Val Glu Lys  
 195 200 205  
 Thr Val Ala Pro Thr Glu Ala Gln Val Gln Leu Val Gln Ser Gly Ala  
 210 215 220  
 Glu Val Lys Lys Pro Gly Ser Ser Val Lys Val Ser Cys Lys Ala Ser  
 225 230 235 240  
 Gly Gly Thr Phe Ser Ser Tyr Gly Val Ser Trp Val Arg Gln Ala Pro  
 245 250 255  
 Gly Gln Gly Leu Glu Trp Met Gly Gly Ile Ile Pro His Phe Gly Ile  
 260 265 270  
 Ala Asn Tyr Ala Gln Lys Phe Gln Gly Arg Val Thr Ile Thr Ala Asp  
 275 280 285  
 Glu Ser Thr Ser Thr Ala Tyr Met Glu Leu Ser Ser Leu Arg Ser Glu  
 290 295 300  
 Asp Thr Ala Val Tyr Tyr Cys Ala Arg Glu Ile Gly Ser Leu Asp Ile  
 305 310 315 320  
 Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly  
 325 330 335  
 Pro Ser Val Phe Pro Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly  
 340 345 350  
 Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val  
 355 360 365  
 Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe  
 370 375 380  
 Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val  
 385 390 395 400  
 Thr Val Pro Ser Ser Ser Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val  
 405 410 415  
 Asn His Lys Pro Ser Asn Thr Lys Val Asp Lys Lys Val Glu Pro Lys  
 420 425 430  
 Ser Glu Phe Asp Tyr Lys Asp Asp Asp Asp Lys Gly Ala Pro His His  
 435 440 445  
 His His His His  
 450

&lt;210&gt; SEQ ID NO 68

&lt;211&gt; LENGTH: 453

&lt;212&gt; TYPE: PRT

&lt;213&gt; ORGANISM: Artificial Sequence

&lt;220&gt; FEATURE:

&lt;223&gt; OTHER INFORMATION: Synthetic Construct

&lt;400&gt; SEQUENCE: 68

Asp Ile Glu Leu Thr Gln Pro Pro Ser Val Ser Val Ser Pro Gly Gln  
 1 5 10 15  
 Thr Ala Ser Ile Thr Cys Ser Gly Asp Ser Ile Pro Ser Phe Phe Ala  
 20 25 30

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|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Ser | Trp | Tyr | Gln | Gln | Lys | Pro | Gly | Gln | Ala | Pro | Val | Leu | Val | Ile | Tyr |
|     | 35  |     |     |     |     |     | 40  |     |     |     |     | 45  |     |     |     |
| Arg | Asp | Ser | Asn | Arg | Pro | Ser | Gly | Ile | Pro | Glu | Arg | Phe | Ser | Gly | Ser |
|     | 50  |     |     |     |     | 55  |     |     |     |     | 60  |     |     |     |     |
| Asn | Ser | Gly | Asn | Thr | Ala | Thr | Leu | Thr | Ile | Ser | Gly | Thr | Gln | Ala | Glu |
| 65  |     |     |     |     | 70  |     |     |     |     | 75  |     |     |     |     | 80  |
| Asp | Glu | Ala | Asp | Tyr | Tyr | Cys | Tyr | Val | Thr | Ala | Pro | Trp | Lys | Ser | Ile |
|     |     |     |     | 85  |     |     |     |     | 90  |     |     |     |     | 95  |     |
| Trp | Val | Phe | Gly | Gly | Gly | Thr | Lys | Leu | Thr | Val | Leu | Gly | Gln | Pro | Lys |
|     |     |     | 100 |     |     |     |     | 105 |     |     |     |     | 110 |     |     |
| Ala | Ala | Pro | Ser | Val | Thr | Leu | Phe | Pro | Pro | Ser | Ser | Glu | Glu | Leu | Gln |
|     |     | 115 |     |     |     |     | 120 |     |     |     |     | 125 |     |     |     |
| Ala | Asn | Lys | Ala | Thr | Leu | Val | Cys | Leu | Ile | Ser | Asp | Phe | Tyr | Pro | Gly |
|     | 130 |     |     |     |     |     | 135 |     |     |     | 140 |     |     |     |     |
| Ala | Val | Thr | Val | Ala | Trp | Lys | Ala | Asp | Ser | Ser | Pro | Val | Lys | Ala | Gly |
| 145 |     |     |     |     | 150 |     |     |     | 155 |     |     |     |     |     | 160 |
| Val | Glu | Thr | Thr | Thr | Pro | Ser | Lys | Gln | Ser | Asn | Asn | Lys | Tyr | Ala | Ala |
|     |     |     |     | 165 |     |     |     |     | 170 |     |     |     |     | 175 |     |
| Ser | Ser | Tyr | Leu | Ser | Leu | Thr | Pro | Glu | Gln | Trp | Lys | Ser | His | Arg | Ser |
|     |     |     | 180 |     |     |     |     | 185 |     |     |     |     | 190 |     |     |
| Tyr | Ser | Cys | Gln | Val | Thr | His | Glu | Gly | Ser | Thr | Val | Glu | Lys | Thr | Val |
|     | 195 |     |     |     |     |     | 200 |     |     |     |     | 205 |     |     |     |
| Ala | Pro | Thr | Glu | Ala | Glu | Val | Gln | Leu | Val | Glu | Ser | Gly | Gly | Gly | Leu |
|     | 210 |     |     |     |     | 215 |     |     |     | 220 |     |     |     |     |     |
| Val | Lys | Pro | Gly | Gly | Ser | Leu | Arg | Leu | Ser | Cys | Ala | Ala | Ser | Gly | Phe |
| 225 |     |     |     |     | 230 |     |     |     | 235 |     |     |     |     |     | 240 |
| Thr | Phe | Ser | Ser | His | Ala | Met | Ser | Trp | Val | Arg | Gln | Ala | Pro | Gly | Lys |
|     |     |     |     | 245 |     |     |     | 250 |     |     |     |     |     | 255 |     |
| Gly | Leu | Glu | Trp | Val | Gly | Arg | Ile | Lys | Ser | Lys | Ala | Asp | Ser | Gly | Thr |
|     | 260 |     |     |     |     |     |     | 265 |     |     |     |     | 270 |     |     |
| Thr | Asp | Tyr | Ala | Ala | Pro | Val | Lys | Gly | Arg | Phe | Thr | Ile | Ser | Arg | Asp |
|     | 275 |     |     |     |     |     | 280 |     |     |     |     | 285 |     |     |     |
| Asp | Ser | Lys | Asn | Thr | Leu | Tyr | Leu | Gln | Met | Asn | Ser | Leu | Lys | Thr | Glu |
|     | 290 |     |     |     |     | 295 |     |     |     |     | 300 |     |     |     |     |
| Asp | Thr | Ala | Val | Tyr | Tyr | Cys | Ala | Arg | Asp | Tyr | Gly | Val | Ala | Phe | Ala |
| 305 |     |     |     |     | 310 |     |     |     |     | 315 |     |     |     |     | 320 |
| Tyr | Trp | Gly | Gln | Gly | Thr | Leu | Val | Thr | Val | Ser | Ser | Ala | Ser | Thr | Lys |
|     |     |     |     | 325 |     |     |     |     | 330 |     |     |     |     | 335 |     |
| Gly | Pro | Ser | Val | Phe | Pro | Leu | Ala | Pro | Ser | Ser | Lys | Ser | Thr | Ser | Gly |
|     |     |     | 340 |     |     |     |     | 345 |     |     |     |     | 350 |     |     |
| Gly | Thr | Ala | Ala | Leu | Gly | Cys | Leu | Val | Lys | Asp | Tyr | Phe | Pro | Glu | Pro |
|     | 355 |     |     |     |     | 360 |     |     |     |     |     | 365 |     |     |     |
| Val | Thr | Val | Ser | Trp | Asn | Ser | Gly | Ala | Leu | Thr | Ser | Gly | Val | His | Thr |
|     | 370 |     |     |     |     | 375 |     |     |     |     | 380 |     |     |     |     |
| Phe | Pro | Ala | Val | Leu | Gln | Ser | Ser | Gly | Leu | Tyr | Ser | Leu | Ser | Ser | Val |
| 385 |     |     |     |     | 390 |     |     |     |     | 395 |     |     |     |     | 400 |
| Val | Thr | Val | Pro | Ser | Ser | Ser | Leu | Gly | Thr | Gln | Thr | Tyr | Ile | Cys | Asn |
|     |     |     |     | 405 |     |     |     |     | 410 |     |     |     |     | 415 |     |
| Val | Asn | His | Lys | Pro | Ser | Asn | Thr | Lys | Val | Asp | Lys | Lys | Val | Glu | Pro |
|     |     |     | 420 |     |     |     |     | 425 |     |     |     |     | 430 |     |     |

|                                              |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|----------------------------------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Lys                                          | Ser | Glu | Phe | Asp | Tyr | Lys | Asp | Asp | Asp | Lys | Gly | Ala | Pro | His |
| 435                                          |     |     |     |     |     | 440 |     |     |     | 445 |     |     |     |     |
| His                                          | His | His | His | His |     |     |     |     |     |     |     |     |     |     |
| 450                                          |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| <210> SEQ ID NO 69                           |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| <211> LENGTH: 454                            |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| <212> TYPE: PRT                              |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| <213> ORGANISM: Artificial Sequence          |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| <220> FEATURE:                               |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| <223> OTHER INFORMATION: Synthetic Construct |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| <400> SEQUENCE: 69                           |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| Asp                                          | Ile | Glu | Leu | Thr | Gln | Pro | Pro | Ser | Val | Ser | Val | Ser | Pro | Gln |
| 1                                            |     |     |     | 5   |     |     |     | 10  |     |     |     | 15  |     |     |
| Thr                                          | Ala | Ser | Ile | Thr | Cys | Ser | Gly | Asp | Asn | Ile | Gly | Thr | Lys | Ala |
|                                              |     |     | 20  |     |     |     | 25  |     |     |     | 30  |     |     |     |
| Tyr                                          | Trp | Tyr | Gln | Gln | Lys | Pro | Gly | Gln | Ala | Pro | Val | Leu | Val | Tyr |
|                                              |     | 35  |     |     |     | 40  |     |     |     | 45  |     |     |     |     |
| Gly                                          | Asp | Ser | Asp | Arg | Pro | Ser | Gly | Ile | Pro | Glu | Arg | Phe | Ser | Ser |
|                                              |     | 50  |     |     |     | 55  |     |     |     | 60  |     |     |     |     |
| Asn                                          | Ser | Gly | Asn | Thr | Ala | Thr | Leu | Thr | Ile | Ser | Gly | Thr | Gln | Glu |
| 65                                           |     |     |     | 70  |     |     |     | 75  |     |     |     | 80  |     |     |
| Asp                                          | Glu | Ala | Asp | Tyr | Tyr | Cys | Tyr | Ser | Ala | Asp | Ala | Gln | Gln | Lys |
|                                              |     |     | 85  |     |     |     | 90  |     |     |     | 95  |     |     |     |
| Ala                                          | Val | Phe | Gly | Gly | Gly | Thr | Lys | Leu | Thr | Val | Leu | Gly | Gln | Lys |
|                                              |     |     | 100 |     |     |     | 105 |     |     |     | 110 |     |     |     |
| Ala                                          | Ala | Pro | Ser | Val | Thr | Leu | Phe | Pro | Pro | Ser | Ser | Glu | Glu | Gln |
|                                              |     | 115 |     |     |     | 120 |     |     |     | 125 |     |     |     |     |
| Ala                                          | Asn | Lys | Ala | Thr | Leu | Val | Cys | Leu | Ile | Ser | Asp | Phe | Tyr | Gly |
|                                              |     | 130 |     |     |     | 135 |     |     |     | 140 |     |     |     |     |
| Ala                                          | Val | Thr | Val | Ala | Trp | Lys | Ala | Asp | Ser | Ser | Pro | Val | Lys | Gly |
| 145                                          |     |     |     | 150 |     |     |     | 155 |     |     |     | 160 |     |     |
| Val                                          | Glu | Thr | Thr | Thr | Pro | Ser | Lys | Gln | Ser | Asn | Asn | Lys | Tyr | Ala |
|                                              |     |     | 165 |     |     |     | 170 |     |     |     | 175 |     |     |     |
| Ser                                          | Ser | Tyr | Leu | Ser | Leu | Thr | Pro | Glu | Gln | Trp | Lys | Ser | His | Ser |
|                                              |     |     | 180 |     |     |     | 185 |     |     |     | 190 |     |     |     |
| Tyr                                          | Ser | Cys | Gln | Val | Thr | His | Glu | Gly | Ser | Thr | Val | Glu | Lys | Val |
|                                              |     | 195 |     |     |     | 200 |     |     |     | 205 |     |     |     |     |
| Ala                                          | Pro | Thr | Glu | Ala | Glu | Val | Gln | Leu | Val | Glu | Ser | Gly | Gly | Leu |
|                                              |     | 210 |     |     |     | 215 |     |     |     | 220 |     |     |     |     |
| Val                                          | Lys | Pro | Gly | Gly | Ser | Leu | Arg | Leu | Ser | Cys | Ala | Ala | Ser | Phe |
| 225                                          |     |     |     | 230 |     |     |     | 235 |     |     |     | 240 |     |     |
| Thr                                          | Phe | Asn | Ser | Ser | Ala | Met | Ser | Trp | Val | Arg | Gln | Ala | Pro | Lys |
|                                              |     |     | 245 |     |     |     | 250 |     |     |     | 255 |     |     |     |
| Gly                                          | Leu | Glu | Trp | Val | Gly | Arg | Ile | Lys | Ser | Lys | Arg | Asp | Gly | Thr |
|                                              |     |     | 260 |     |     |     | 265 |     |     |     | 270 |     |     |     |
| Thr                                          | Asp | Tyr | Ala | Ala | Pro | Val | Lys | Gly | Arg | Phe | Thr | Ile | Ser | Asp |
|                                              |     | 275 |     |     |     | 280 |     |     |     | 285 |     |     |     |     |
| Asp                                          | Ser | Lys | Asn | Thr | Leu | Tyr | Leu | Gln | Met | Asn | Ser | Leu | Lys | Glu |
|                                              |     | 290 |     |     |     | 295 |     |     |     | 300 |     |     |     |     |
| Asp                                          | Thr | Ala | Val | Tyr | Tyr | Cys | Ala | Arg | Asp | Tyr | Gly | Gly | Leu | Phe |
| 305                                          |     |     |     | 310 |     |     |     | 315 |     |     |     | 320 |     |     |

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|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Asp | Tyr | Trp | Gly | Gln | Gly | Thr | Leu | Val | Thr | Val | Ser | Ser | Ala | Ser | Thr |
|     |     |     | 325 |     |     |     |     |     | 330 |     |     |     |     | 335 |     |
| Lys | Gly | Pro | Ser | Val | Phe | Pro | Leu | Ala | Pro | Ser | Ser | Lys | Ser | Thr | Ser |
|     |     | 340 |     |     |     |     | 345 |     |     |     |     |     | 350 |     |     |
| Gly | Gly | Thr | Ala | Ala | Leu | Gly | Cys | Leu | Val | Lys | Asp | Tyr | Phe | Pro | Glu |
|     |     | 355 |     |     |     | 360 |     |     |     |     |     | 365 |     |     |     |
| Pro | Val | Thr | Val | Ser | Trp | Asn | Ser | Gly | Ala | Leu | Thr | Ser | Gly | Val | His |
|     | 370 |     |     |     |     | 375 |     |     |     |     | 380 |     |     |     |     |
| Thr | Phe | Pro | Ala | Val | Leu | Gln | Ser | Ser | Gly | Leu | Tyr | Ser | Leu | Ser | Ser |
| 385 |     |     |     |     | 390 |     |     |     |     | 395 |     |     |     |     | 400 |
| Val | Val | Thr | Val | Pro | Ser | Ser | Ser | Leu | Gly | Thr | Gln | Thr | Tyr | Ile | Cys |
|     |     |     | 405 |     |     |     |     |     | 410 |     |     |     |     | 415 |     |
| Asn | Val | Asn | His | Lys | Pro | Ser | Asn | Thr | Lys | Val | Asp | Lys | Lys | Val | Glu |
|     |     |     | 420 |     |     |     |     | 425 |     |     |     |     | 430 |     |     |
| Pro | Lys | Ser | Glu | Phe | Asp | Tyr | Lys | Asp | Asp | Asp | Asp | Lys | Gly | Ala | Pro |
|     |     | 435 |     |     |     |     | 440 |     |     |     |     | 445 |     |     |     |
| His | His | His | His | His | His |     |     |     |     |     |     |     |     |     |     |
|     |     |     | 450 |     |     |     |     |     |     |     |     |     |     |     |     |

<210> SEQ ID NO 70  
 <211> LENGTH: 453  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 70

|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Asp | Ile | Glu | Leu | Thr | Gln | Pro | Pro | Ser | Val | Ser | Val | Ser | Pro | Gly | Gln |
| 1   |     |     |     | 5   |     |     |     |     | 10  |     |     |     |     | 15  |     |
| Thr | Ala | Ser | Ile | Thr | Cys | Ser | Gly | Asp | Asn | Leu | Arg | Lys | Tyr | Ser | Ala |
|     |     | 20  |     |     |     |     | 25  |     |     |     |     |     | 30  |     |     |
| His | Trp | Tyr | Gln | Gln | Lys | Pro | Gly | Gln | Ala | Pro | Val | Leu | Val | Ile | Tyr |
|     |     | 35  |     |     |     | 40  |     |     |     |     |     | 45  |     |     |     |
| Tyr | Asp | Asn | Lys | Arg | Pro | Ser | Gly | Ile | Pro | Glu | Arg | Phe | Ser | Gly | Ser |
|     | 50  |     |     |     |     | 55  |     |     |     | 60  |     |     |     |     |     |
| Asn | Ser | Gly | Asn | Thr | Ala | Thr | Leu | Thr | Ile | Ser | Gly | Thr | Gln | Ala | Glu |
| 65  |     |     |     |     | 70  |     |     |     |     | 75  |     |     |     |     | 80  |
| Asp | Glu | Ala | Asp | Tyr | Tyr | Cys | Gln | Val | Tyr | Ala | Ser | Val | His | Arg | Met |
|     |     |     | 85  |     |     |     |     |     | 90  |     |     |     |     | 95  |     |
| Val | Phe | Gly | Gly | Gly | Thr | Lys | Leu | Thr | Val | Leu | Gly | Gln | Pro | Lys | Ala |
|     |     | 100 |     |     |     |     |     | 105 |     |     |     |     | 110 |     |     |
| Ala | Pro | Ser | Val | Thr | Leu | Phe | Pro | Pro | Ser | Ser | Glu | Glu | Leu | Gln | Ala |
|     |     | 115 |     |     |     | 120 |     |     |     |     |     | 125 |     |     |     |
| Asn | Lys | Ala | Thr | Leu | Val | Cys | Leu | Ile | Ser | Asp | Phe | Tyr | Pro | Gly | Ala |
|     | 130 |     |     |     |     | 135 |     |     |     |     | 140 |     |     |     |     |
| Val | Thr | Val | Ala | Trp | Lys | Ala | Asp | Ser | Ser | Pro | Val | Lys | Ala | Gly | Val |
| 145 |     |     |     |     | 150 |     |     |     |     | 155 |     |     |     |     | 160 |
| Glu | Thr | Thr | Thr | Pro | Ser | Lys | Gln | Ser | Asn | Asn | Lys | Tyr | Ala | Ala | Ser |
|     |     |     | 165 |     |     |     |     |     | 170 |     |     |     |     | 175 |     |
| Ser | Tyr | Leu | Ser | Leu | Thr | Pro | Glu | Gln | Trp | Lys | Ser | His | Arg | Ser | Tyr |
|     |     | 180 |     |     |     |     |     | 185 |     |     |     |     | 190 |     |     |
| Ser | Cys | Gln | Val | Thr | His | Glu | Gly | Ser | Thr | Val | Glu | Lys | Thr | Val | Ala |
|     |     | 195 |     |     |     |     | 200 |     |     |     |     | 205 |     |     |     |

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|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Pro | Thr | Glu | Ala | Gln | Val | Gln | Leu | Val | Gln | Ser | Gly | Ala | Glu | Val | Lys |
| 210 |     |     |     |     |     | 215 |     |     |     |     | 220 |     |     |     |     |
| Lys | Pro | Gly | Ser | Ser | Val | Lys | Val | Ser | Cys | Lys | Ala | Ser | Gly | Gly | Thr |
| 225 |     |     |     |     | 230 |     |     |     |     | 235 |     |     |     |     | 240 |
| Phe | Ser | Ser | Tyr | Ala | Ile | His | Trp | Val | Arg | Gln | Ala | Pro | Gly | Gln | Gly |
|     |     |     |     | 245 |     |     |     |     | 250 |     |     |     |     | 255 |     |
| Leu | Glu | Trp | Met | Gly | Arg | Ile | Ile | Pro | Asp | Phe | Gly | Thr | Ala | Asn | Tyr |
|     |     |     | 260 |     |     |     |     | 265 |     |     |     |     | 270 |     |     |
| Ala | Gln | Lys | Phe | Gln | Gly | Arg | Val | Thr | Ile | Thr | Ala | Asp | Glu | Ser | Thr |
|     |     | 275 |     |     |     |     | 280 |     |     |     |     | 285 |     |     |     |
| Ser | Thr | Ala | Tyr | Met | Glu | Leu | Ser | Ser | Leu | Arg | Ser | Glu | Asp | Thr | Ala |
|     | 290 |     |     |     |     | 295 |     |     |     |     | 300 |     |     |     |     |
| Val | Tyr | Tyr | Cys | Ala | Arg | Gly | Pro | Thr | Gln | Ala | Ile | His | Tyr | Phe | Ala |
| 305 |     |     |     |     | 310 |     |     |     |     | 315 |     |     |     |     | 320 |
| Tyr | Trp | Gly | Gln | Gly | Thr | Leu | Val | Thr | Val | Ser | Ser | Ala | Ser | Thr | Lys |
|     |     |     | 325 |     |     |     |     |     | 330 |     |     |     |     | 335 |     |
| Gly | Pro | Ser | Val | Phe | Pro | Leu | Ala | Pro | Ser | Ser | Lys | Ser | Thr | Ser | Gly |
|     |     |     | 340 |     |     |     |     | 345 |     |     |     |     | 350 |     |     |
| Gly | Thr | Ala | Ala | Leu | Gly | Cys | Leu | Val | Lys | Asp | Tyr | Phe | Pro | Glu | Pro |
|     |     | 355 |     |     |     |     | 360 |     |     |     |     | 365 |     |     |     |
| Val | Thr | Val | Ser | Trp | Asn | Ser | Gly | Ala | Leu | Thr | Ser | Gly | Val | His | Thr |
|     | 370 |     |     |     |     | 375 |     |     |     |     | 380 |     |     |     |     |
| Phe | Pro | Ala | Val | Leu | Gln | Ser | Ser | Gly | Leu | Tyr | Ser | Leu | Ser | Ser | Val |
| 385 |     |     |     |     | 390 |     |     |     |     | 395 |     |     |     |     | 400 |
| Val | Thr | Val | Pro | Ser | Ser | Ser | Leu | Gly | Thr | Gln | Thr | Tyr | Ile | Cys | Asn |
|     |     |     | 405 |     |     |     |     |     | 410 |     |     |     |     | 415 |     |
| Val | Asn | His | Lys | Pro | Ser | Asn | Thr | Lys | Val | Asp | Lys | Lys | Val | Glu | Pro |
|     |     |     | 420 |     |     |     |     | 425 |     |     |     |     | 430 |     |     |
| Lys | Ser | Glu | Phe | Asp | Tyr | Lys | Asp | Asp | Asp | Asp | Lys | Gly | Ala | Pro | His |
|     |     | 435 |     |     |     |     | 440 |     |     |     |     | 445 |     |     |     |
| His | His | His | His | His |     |     |     |     |     |     |     |     |     |     |     |
|     |     |     |     |     | 450 |     |     |     |     |     |     |     |     |     |     |

&lt;210&gt; SEQ ID NO 71

&lt;211&gt; LENGTH: 458

&lt;212&gt; TYPE: PRT

&lt;213&gt; ORGANISM: Artificial Sequence

&lt;220&gt; FEATURE:

&lt;223&gt; OTHER INFORMATION: Synthetic Construct

&lt;400&gt; SEQUENCE: 71

|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Asp | Ile | Glu | Leu | Thr | Gln | Pro | Pro | Ser | Val | Ser | Val | Ser | Pro | Gly | Gln |
| 1   |     |     |     | 5   |     |     |     | 10  |     |     |     |     | 15  |     |     |
| Thr | Ala | Ser | Ile | Thr | Cys | Ser | Gly | Asp | Ser | Leu | Gly | Ser | Lys | Ser | Val |
|     |     |     | 20  |     |     |     | 25  |     |     |     |     | 30  |     |     |     |
| His | Trp | Tyr | Gln | Gln | Lys | Pro | Gly | Gln | Ala | Pro | Val | Leu | Val | Ile | Tyr |
|     |     | 35  |     |     |     | 40  |     |     |     |     | 45  |     |     |     |     |
| Arg | Asp | Ser | Lys | Arg | Pro | Ser | Gly | Ile | Pro | Glu | Arg | Phe | Ser | Gly | Ser |
|     | 50  |     |     |     |     | 55  |     |     |     | 60  |     |     |     |     |     |
| Asn | Ser | Gly | Asn | Thr | Ala | Thr | Leu | Thr | Ile | Ser | Gly | Thr | Gln | Ala | Glu |
| 65  |     |     |     | 70  |     |     |     |     | 75  |     |     |     | 80  |     |     |
| Asp | Glu | Ala | Asp | Tyr | Tyr | Cys | Gln | Thr | Tyr | Asp | Trp | Ser | His | Phe | Gly |
|     |     |     | 85  |     |     |     |     | 90  |     |     |     |     | 95  |     |     |

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|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Trp | Val | Phe | Gly | Gly | Gly | Thr | Lys | Leu | Thr | Val | Leu | Gly | Gln | Pro | Lys |
|     |     |     | 100 |     |     |     |     | 105 |     |     |     |     | 110 |     |     |
| Ala | Ala | Pro | Ser | Val | Thr | Leu | Phe | Pro | Pro | Ser | Ser | Glu | Glu | Leu | Gln |
|     |     | 115 |     |     |     |     | 120 |     |     |     |     | 125 |     |     |     |
| Ala | Asn | Lys | Ala | Thr | Leu | Val | Cys | Leu | Ile | Ser | Asp | Phe | Tyr | Pro | Gly |
|     | 130 |     |     |     |     | 135 |     |     |     |     | 140 |     |     |     |     |
| Ala | Val | Thr | Val | Ala | Trp | Lys | Ala | Asp | Ser | Ser | Pro | Val | Lys | Ala | Gly |
| 145 |     |     |     |     | 150 |     |     |     |     | 155 |     |     |     |     | 160 |
| Val | Glu | Thr | Thr | Thr | Pro | Ser | Lys | Gln | Ser | Asn | Asn | Lys | Tyr | Ala | Ala |
|     |     |     | 165 |     |     |     |     |     | 170 |     |     |     |     | 175 |     |
| Ser | Ser | Tyr | Leu | Ser | Leu | Thr | Pro | Glu | Gln | Trp | Lys | Ser | His | Arg | Ser |
|     |     | 180 |     |     |     |     |     | 185 |     |     |     |     | 190 |     |     |
| Tyr | Ser | Cys | Gln | Val | Thr | His | Glu | Gly | Ser | Thr | Val | Glu | Lys | Thr | Val |
|     |     | 195 |     |     |     |     | 200 |     |     |     |     | 205 |     |     |     |
| Ala | Pro | Thr | Glu | Ala | Glu | Val | Gln | Leu | Leu | Glu | Ser | Gly | Gly | Gly | Leu |
|     | 210 |     |     |     |     | 215 |     |     |     |     | 220 |     |     |     |     |
| Val | Gln | Pro | Gly | Gly | Ser | Leu | Arg | Leu | Ser | Cys | Ala | Ala | Ser | Gly | Phe |
| 225 |     |     |     |     | 230 |     |     |     |     | 235 |     |     |     |     | 240 |
| Thr | Phe | Ser | Ser | Ala | Ala | Met | His | Trp | Val | Arg | Gln | Ala | Pro | Gly | Lys |
|     |     |     | 245 |     |     |     |     |     | 250 |     |     |     |     | 255 |     |
| Gly | Leu | Glu | Trp | Val | Ser | Val | Ile | Ser | Ser | Asp | Gly | Gly | Ser | Thr | Tyr |
|     |     | 260 |     |     |     |     |     | 265 |     |     |     |     | 270 |     |     |
| Tyr | Ala | Asp | Ser | Val | Lys | Gly | Arg | Phe | Thr | Ile | Ser | Arg | Asp | Asn | Ser |
|     |     | 275 |     |     |     |     | 280 |     |     |     |     | 285 |     |     |     |
| Lys | Asn | Thr | Leu | Tyr | Leu | Gln | Met | Asn | Ser | Leu | Arg | Ala | Glu | Asp | Thr |
|     | 290 |     |     |     |     | 295 |     |     |     |     | 300 |     |     |     |     |
| Ala | Val | Tyr | Tyr | Cys | Ala | Arg | Ala | Gly | Phe | Ile | Leu | Gly | Ser | Leu | Gly |
| 305 |     |     |     |     | 310 |     |     |     |     | 315 |     |     |     |     | 320 |
| Val | Ala | Trp | Met | Asp | Val | Trp | Gly | Gln | Gly | Thr | Leu | Val | Thr | Val | Ser |
|     |     |     | 325 |     |     |     |     |     | 330 |     |     |     |     | 335 |     |
| Ser | Ala | Ser | Thr | Lys | Gly | Pro | Ser | Val | Phe | Pro | Leu | Ala | Pro | Ser | Ser |
|     |     |     | 340 |     |     |     |     | 345 |     |     |     |     | 350 |     |     |
| Lys | Ser | Thr | Ser | Gly | Gly | Thr | Ala | Ala | Leu | Gly | Cys | Leu | Val | Lys | Asp |
|     |     | 355 |     |     |     |     | 360 |     |     |     |     | 365 |     |     |     |
| Tyr | Phe | Pro | Glu | Pro | Val | Thr | Val | Ser | Trp | Asn | Ser | Gly | Ala | Leu | Thr |
|     | 370 |     |     |     |     | 375 |     |     |     |     | 380 |     |     |     |     |
| Ser | Gly | Val | His | Thr | Phe | Pro | Ala | Val | Leu | Gln | Ser | Ser | Gly | Leu | Tyr |
| 385 |     |     |     |     | 390 |     |     |     |     | 395 |     |     |     |     | 400 |
| Ser | Leu | Ser | Ser | Val | Val | Thr | Val | Pro | Ser | Ser | Ser | Leu | Gly | Thr | Gln |
|     |     |     | 405 |     |     |     |     |     | 410 |     |     |     |     | 415 |     |
| Thr | Tyr | Ile | Cys | Asn | Val | Asn | His | Lys | Pro | Ser | Asn | Thr | Lys | Val | Asp |
|     |     |     | 420 |     |     |     |     | 425 |     |     |     |     | 430 |     |     |
| Lys | Lys | Val | Glu | Pro | Lys | Ser | Glu | Phe | Asp | Tyr | Lys | Asp | Asp | Asp | Asp |
|     | 435 |     |     |     |     |     | 440 |     |     |     |     | 445 |     |     |     |
| Lys | Gly | Ala | Pro | His | His | His | His | His | His | His | His | His | His | His | His |
|     | 450 |     |     |     |     | 455 |     |     |     |     |     |     |     |     |     |

&lt;210&gt; SEQ ID NO 72

&lt;211&gt; LENGTH: 465

&lt;212&gt; TYPE: PRT

&lt;213&gt; ORGANISM: Artificial Sequence

&lt;220&gt; FEATURE:

&lt;223&gt; OTHER INFORMATION: Synthetic Construct

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&lt;400&gt; SEQUENCE: 72

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Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1           5           10           15
Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Gly Ile Ser Gly Asn
20           25           30
Trp Leu Thr Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu
35           40           45
Ile Tyr Asp Ala Ser Asn Leu Gln Ser Gly Val Pro Ser Arg Phe Ser
50           55           60
Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln
65           70           75           80
Pro Glu Asp Phe Ala Thr Tyr Tyr Cys His Gln Ser Tyr Arg Gly Pro
85           90           95
Met Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Arg Thr Val Ala
100          105          110
Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser
115          120          125
Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu
130          135          140
Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly Asn Ser
145          150          155          160
Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu
165          170          175
Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys His Lys Val
180          185          190
Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro Val Thr Lys
195          200          205
Ser Phe Asn Arg Gly Glu Ala Glu Val Gln Leu Leu Glu Ser Gly Gly
210          215          220
Gly Leu Val Gln Pro Gly Gly Ser Leu Arg Leu Ser Cys Ala Ala Ser
225          230          235          240
Gly Phe Thr Phe Gly Arg Phe Val Met His Trp Val Arg Gln Ala Pro
245          250          255
Gly Lys Gly Leu Glu Trp Val Ser Val Ile Gly Tyr Ser Gly Ser Ser
260          265          270
Thr Tyr Tyr Ala Asp Ser Val Lys Gly Arg Phe Thr Ile Ser Arg Asp
275          280          285
Asn Ser Lys Asn Thr Leu Tyr Leu Gln Met Asn Ser Leu Arg Ala Glu
290          295          300
Asp Thr Ala Val Tyr Tyr Cys Ala Arg Glu Pro Gly Tyr Tyr Tyr Pro
305          310          315          320
Ser Gly Tyr Tyr Arg Gly Pro Gly Tyr Trp Met Asp Val Trp Gly Gln
325          330          335
Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro Ser Val
340          345          350
Phe Pro Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly Thr Ala Ala
355          360          365
Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val Ser
370          375          380
Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala Val

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|                                                                 |     |     |     |
|-----------------------------------------------------------------|-----|-----|-----|
| 385                                                             | 390 | 395 | 400 |
| Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val Pro | 405 | 410 | 415 |
| Ser Ser Ser Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val Asn His Lys | 420 | 425 | 430 |
| Pro Ser Asn Thr Lys Val Asp Lys Lys Val Glu Pro Lys Ser Glu Phe | 435 | 440 | 445 |
| Asp Tyr Lys Asp Asp Asp Asp Lys Gly Ala Pro His His His His His | 450 | 455 | 460 |

His  
465

<210> SEQ ID NO 73  
 <211> LENGTH: 448  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 73

|                                                                 |     |     |     |     |
|-----------------------------------------------------------------|-----|-----|-----|-----|
| Asp Ile Glu Leu Thr Gln Pro Pro Ser Val Ser Val Ser Pro Gly Gln | 1   | 5   | 10  | 15  |
| Thr Ala Ser Ile Thr Cys Ser Gly Asp Asn Ile Arg Lys Lys Tyr Val | 20  | 25  | 30  |     |
| His Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Val Leu Val Ile Tyr | 35  | 40  | 45  |     |
| Arg Asp Ser Asn Arg Pro Ser Gly Ile Pro Glu Arg Phe Ser Gly Ser | 50  | 55  | 60  |     |
| Asn Ser Gly Asn Thr Ala Thr Leu Thr Ile Ser Gly Thr Gln Ala Glu | 65  | 70  | 75  | 80  |
| Asp Glu Ala Asp Tyr Tyr Cys Ser Ser Ala Gly Arg Asp Asn Tyr Val | 85  | 90  | 95  |     |
| Phe Gly Gly Gly Thr Lys Leu Thr Val Leu Gly Gln Pro Lys Ala Ala | 100 | 105 | 110 |     |
| Pro Ser Val Thr Leu Phe Pro Pro Ser Ser Glu Glu Leu Gln Ala Asn | 115 | 120 | 125 |     |
| Lys Ala Thr Leu Val Cys Leu Ile Ser Asp Phe Tyr Pro Gly Ala Val | 130 | 135 | 140 |     |
| Thr Val Ala Trp Lys Ala Asp Ser Ser Pro Val Lys Ala Gly Val Glu | 145 | 150 | 155 | 160 |
| Thr Thr Thr Pro Ser Lys Gln Ser Asn Asn Lys Tyr Ala Ala Ser Ser | 165 | 170 | 175 |     |
| Tyr Leu Ser Leu Thr Pro Glu Gln Trp Lys Ser His Arg Ser Tyr Ser | 180 | 185 | 190 |     |
| Cys Gln Val Thr His Glu Gly Ser Thr Val Glu Lys Thr Val Ala Pro | 195 | 200 | 205 |     |
| Thr Glu Ala Glu Val Gln Leu Leu Glu Ser Gly Gly Gly Leu Val Gln | 210 | 215 | 220 |     |
| Pro Gly Gly Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe | 225 | 230 | 235 | 240 |
| Ser Ser Ser Ala Met His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu | 245 | 250 | 255 |     |

Glu Trp Val Ser Val Ile His Tyr Asp Ser Ser Glu Thr Tyr Tyr Ala



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| 260 |     |     |     |     | 265 |     |     |     |     | 270 |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Asp | Ser | Val | Lys | Gly | Arg | Phe | Thr | Ile | Ser | Arg | Asp | Asn | Ser | Lys | Asn |
|     | 275 |     |     |     |     |     | 280 |     |     |     |     | 285 |     |     |     |
| Thr | Leu | Tyr | Leu | Gln | Met | Asn | Ser | Leu | Arg | Ala | Glu | Asp | Thr | Ala | Val |
|     | 290 |     |     |     |     | 295 |     |     |     |     | 300 |     |     |     |     |
| Tyr | Tyr | Cys | Ala | Arg | Asp | Arg | Tyr | Phe | Phe | Asp | Val | Trp | Gly | Gln | Gly |
|     | 305 |     |     |     |     | 310 |     |     |     |     | 315 |     |     |     | 320 |
| Thr | Leu | Val | Thr | Val | Ser | Ser | Ala | Ser | Thr | Lys | Gly | Pro | Ser | Val | Phe |
|     |     |     | 325 |     |     |     |     |     | 330 |     |     |     |     | 335 |     |
| Pro | Leu | Ala | Pro | Ser | Ser | Lys | Ser | Thr | Ser | Gly | Gly | Thr | Ala | Ala | Leu |
|     |     | 340 |     |     |     |     | 345 |     |     |     |     | 350 |     |     |     |
| Gly | Cys | Leu | Val | Lys | Asp | Tyr | Phe | Pro | Glu | Pro | Val | Thr | Val | Ser | Trp |
|     | 355 |     |     |     |     |     | 360 |     |     |     |     | 365 |     |     |     |
| Asn | Ser | Gly | Ala | Leu | Thr | Ser | Gly | Val | His | Thr | Phe | Pro | Ala | Val | Leu |
|     | 370 |     |     |     |     | 375 |     |     |     |     | 380 |     |     |     |     |
| Gln | Ser | Ser | Gly | Leu | Tyr | Ser | Leu | Ser | Ser | Val | Val | Thr | Val | Pro | Ser |
|     | 385 |     |     |     |     | 390 |     |     |     |     | 395 |     |     |     | 400 |
| Ser | Ser | Leu | Gly | Thr | Gln | Thr | Tyr | Ile | Cys | Asn | Val | Asn | His | Lys | Pro |
|     |     | 405 |     |     |     |     |     | 410 |     |     |     |     |     | 415 |     |
| Ser | Asn | Thr | Lys | Val | Asp | Lys | Lys | Val | Glu | Pro | Lys | Ser | Glu | Phe | Asp |
|     |     | 420 |     |     |     |     | 425 |     |     |     |     | 430 |     |     |     |
| Tyr | Lys | Asp | Asp | Asp | Asp | Lys | Gly | Ala | Pro | His | His | His | His | His | His |
|     | 435 |     |     |     |     | 440 |     |     |     |     | 445 |     |     |     |     |

&lt;210&gt; SEQ ID NO 74

&lt;211&gt; LENGTH: 452

&lt;212&gt; TYPE: PRT

&lt;213&gt; ORGANISM: Artificial Sequence

&lt;220&gt; FEATURE:

&lt;223&gt; OTHER INFORMATION: Synthetic Construct

&lt;400&gt; SEQUENCE: 74

|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Asp | Ile | Glu | Leu | Thr | Gln | Pro | Pro | Ser | Val | Ser | Val | Ser | Pro | Gly | Gln |
| 1   |     |     |     | 5   |     |     |     |     | 10  |     |     |     |     | 15  |     |
| Thr | Ala | Ser | Ile | Thr | Cys | Ser | Gly | Asp | Ala | Leu | Arg | Tyr | Tyr | Ile | Ala |
|     | 20  |     |     |     |     |     | 25  |     |     |     |     |     | 30  |     |     |
| His | Trp | Tyr | Gln | Gln | Lys | Pro | Gly | Gln | Ala | Pro | Val | Leu | Val | Ile | Tyr |
|     | 35  |     |     |     |     | 40  |     |     |     |     | 45  |     |     |     |     |
| Tyr | Asn | Asn | Asn | Arg | Pro | Ser | Gly | Ile | Pro | Glu | Arg | Phe | Ser | Gly | Ser |
|     | 50  |     |     |     | 55  |     |     |     |     | 60  |     |     |     |     |     |
| Asn | Ser | Gly | Asn | Thr | Ala | Thr | Leu | Thr | Ile | Ser | Gly | Thr | Gln | Ala | Glu |
|     | 65  |     |     | 70  |     |     |     |     | 75  |     |     |     |     | 80  |     |
| Asp | Glu | Ala | Asp | Tyr | Tyr | Cys | Gln | Ser | Tyr | Gly | Pro | Gly | Ser | Val | Val |
|     |     | 85  |     |     |     |     |     | 90  |     |     |     |     | 95  |     |     |
| Phe | Gly | Gly | Gly | Thr | Lys | Leu | Thr | Val | Leu | Gly | Gln | Pro | Lys | Ala | Ala |
|     | 100 |     |     |     |     | 105 |     |     |     |     |     |     | 110 |     |     |
| Pro | Ser | Val | Thr | Leu | Phe | Pro | Pro | Ser | Ser | Glu | Glu | Leu | Gln | Ala | Asn |
|     | 115 |     |     |     |     | 120 |     |     |     |     |     | 125 |     |     |     |
| Lys | Ala | Thr | Leu | Val | Cys | Leu | Ile | Ser | Asp | Phe | Tyr | Pro | Gly | Ala | Val |
|     | 130 |     |     |     | 135 |     |     |     |     | 140 |     |     |     |     |     |
| Thr | Val | Ala | Trp | Lys | Ala | Asp | Ser | Ser | Pro | Val | Lys | Ala | Gly | Val | Glu |
|     | 145 |     |     | 150 |     |     |     |     | 155 |     |     |     |     | 160 |     |
| Thr | Thr | Thr | Pro | Ser | Lys | Gln | Ser | Asn | Asn | Lys | Tyr | Ala | Ala | Ser | Ser |

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| 165 |     |     |     |     | 170 |     |     |     |     | 175 |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Tyr | Leu | Ser | Leu | Thr | Pro | Glu | Gln | Trp | Lys | Ser | His | Arg | Ser | Tyr | Ser |
|     |     |     | 180 |     |     |     |     | 185 |     |     |     |     | 190 |     |     |
| Cys | Gln | Val | Thr | His | Glu | Gly | Ser | Thr | Val | Glu | Lys | Thr | Val | Ala | Pro |
|     |     | 195 |     |     |     |     | 200 |     |     |     |     |     | 205 |     |     |
| Thr | Glu | Ala | Glu | Val | Gln | Leu | Leu | Glu | Ser | Gly | Gly | Gly | Leu | Val | Gln |
|     | 210 |     |     |     |     | 215 |     |     |     |     | 220 |     |     |     |     |
| Pro | Gly | Gly | Ser | Leu | Arg | Leu | Ser | Cys | Ala | Ala | Ser | Gly | Phe | Thr | Phe |
| 225 |     |     |     |     |     | 230 |     |     |     |     | 235 |     |     |     | 240 |
| Ser | Asp | Tyr | Ala | Met | His | Trp | Val | Arg | Gln | Ala | Pro | Gly | Lys | Gly | Leu |
|     |     |     |     | 245 |     |     |     |     | 250 |     |     |     |     | 255 |     |
| Glu | Trp | Val | Ser | Ser | Ile | Phe | Tyr | Ser | Gly | Ser | Asn | Thr | Tyr | Tyr | Ala |
|     |     |     | 260 |     |     |     |     |     | 265 |     |     |     |     | 270 |     |
| Asp | Ser | Val | Lys | Gly | Arg | Phe | Thr | Ile | Ser | Arg | Asp | Asn | Ser | Lys | Asn |
|     |     | 275 |     |     |     |     |     |     | 280 |     |     |     |     | 285 |     |
| Thr | Leu | Tyr | Leu | Gln | Met | Asn | Ser | Leu | Arg | Ala | Glu | Asp | Thr | Ala | Val |
|     | 290 |     |     |     |     | 295 |     |     |     |     | 300 |     |     |     |     |
| Tyr | Tyr | Cys | Ala | Arg | Pro | Lys | Ser | Tyr | Ala | Ser | Gly | Pro | Phe | Ala | Tyr |
| 305 |     |     |     |     |     | 310 |     |     |     |     | 315 |     |     |     | 320 |
| Trp | Gly | Gln | Gly | Thr | Leu | Val | Thr | Val | Ser | Ser | Ala | Ser | Thr | Lys | Gly |
|     |     |     |     | 325 |     |     |     |     | 330 |     |     |     |     | 335 |     |
| Pro | Ser | Val | Phe | Pro | Leu | Ala | Pro | Ser | Ser | Lys | Ser | Thr | Ser | Gly | Gly |
|     |     |     | 340 |     |     |     |     |     | 345 |     |     |     |     | 350 |     |
| Thr | Ala | Ala | Leu | Gly | Cys | Leu | Val | Lys | Asp | Tyr | Phe | Pro | Glu | Pro | Val |
|     |     | 355 |     |     |     |     |     |     | 360 |     |     |     |     | 365 |     |
| Thr | Val | Ser | Trp | Asn | Ser | Gly | Ala | Leu | Thr | Ser | Gly | Val | His | Thr | Phe |
|     | 370 |     |     |     |     | 375 |     |     |     |     | 380 |     |     |     |     |
| Pro | Ala | Val | Leu | Gln | Ser | Ser | Gly | Leu | Tyr | Ser | Leu | Ser | Ser | Val | Val |
| 385 |     |     |     |     |     | 390 |     |     |     |     | 395 |     |     |     | 400 |
| Thr | Val | Pro | Ser | Ser | Ser | Leu | Gly | Thr | Gln | Thr | Tyr | Ile | Cys | Asn | Val |
|     |     |     |     | 405 |     |     |     |     | 410 |     |     |     |     | 415 |     |
| Asn | His | Lys | Pro | Ser | Asn | Thr | Lys | Val | Asp | Lys | Lys | Val | Glu | Pro | Lys |
|     |     |     | 420 |     |     |     |     |     | 425 |     |     |     |     | 430 |     |
| Ser | Glu | Phe | Asp | Tyr | Lys | Asp | Asp | Asp | Asp | Lys | Gly | Ala | Pro | His | His |
|     |     | 435 |     |     |     |     | 440 |     |     |     |     | 445 |     |     |     |
| His | His | His |     |     |     |     |     |     |     |     |     |     |     |     |     |
|     |     |     | 450 |     |     |     |     |     |     |     |     |     |     |     |     |

&lt;210&gt; SEQ ID NO 75

&lt;211&gt; LENGTH: 453

&lt;212&gt; TYPE: PRT

&lt;213&gt; ORGANISM: Artificial Sequence

&lt;220&gt; FEATURE:

&lt;223&gt; OTHER INFORMATION: Synthetic Construct

&lt;400&gt; SEQUENCE: 75

|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Asp | Ile | Val | Leu | Thr | Gln | Pro | Pro | Ser | Val | Ser | Gly | Ala | Pro | Gly | Gln |
| 1   |     |     |     | 5   |     |     |     |     | 10  |     |     |     |     | 15  |     |
| Arg | Val | Thr | Ile | Ser | Cys | Ser | Gly | Ser | Ser | Ser | Asn | Ile | Gly | Gln | Asn |
|     |     | 20  |     |     |     |     |     | 25  |     |     |     |     | 30  |     |     |
| Tyr | Val | Ser | Trp | Tyr | Gln | Gln | Leu | Pro | Gly | Thr | Ala | Pro | Lys | Leu | Leu |
|     | 35  |     |     |     |     |     | 40  |     |     |     |     | 45  |     |     |     |
| Ile | Tyr | Asp | Asn | Ser | Lys | Arg | Pro | Ser | Gly | Val | Pro | Asp | Arg | Phe | Ser |

|           |     |     |     |     |     |     |     |     |     |     |     |     |     |     |           |  |
|-----------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----------|--|
| 50        |     |     |     | 55  |     |     |     | 60  |     |     |     |     |     |     |           |  |
| Gly<br>65 | Ser | Lys | Ser | Gly | Thr | Ser | Ala | Ser | Leu | Ala | Ile | Thr | Gly | Leu | Gln<br>80 |  |
|           |     |     |     | 85  |     |     |     |     | 90  |     |     |     |     | 95  |           |  |
| Ala       | Glu | Asp | Glu | Ala | Asp | Tyr | Tyr | Cys | Ser | Ser | Trp | Asp | Leu | Leu | Ser       |  |
|           |     |     |     | 100 |     |     |     |     | 105 |     |     |     |     | 110 |           |  |
| Lys       | Ser | Arg | Val | Phe | Gly | Gly | Gly | Thr | Lys | Leu | Thr | Val | Leu | Gly | Gln       |  |
|           |     |     |     | 115 |     |     |     |     | 120 |     |     |     |     | 125 |           |  |
| Pro       | Lys | Ala | Ala | Pro | Ser | Val | Thr | Leu | Phe | Pro | Pro | Ser | Ser | Glu | Glu       |  |
|           |     |     |     | 130 |     |     |     |     | 135 |     |     |     |     | 140 |           |  |
| Leu       | Gln | Ala | Asn | Lys | Ala | Thr | Leu | Val | Cys | Leu | Ile | Ser | Asp | Phe | Tyr       |  |
|           |     |     |     | 145 |     |     |     |     | 150 |     |     |     |     | 155 |           |  |
| Pro       | Gly | Ala | Val | Thr | Val | Ala | Trp | Lys | Ala | Asp | Ser | Ser | Pro | Val | Lys       |  |
|           |     |     |     | 165 |     |     |     |     | 170 |     |     |     |     | 175 |           |  |
| Ala       | Gly | Val | Glu | Thr | Thr | Thr | Pro | Ser | Lys | Gln | Ser | Asn | Asn | Lys | Tyr       |  |
|           |     |     |     | 180 |     |     |     |     | 185 |     |     |     |     | 190 |           |  |
| Ala       | Ala | Ser | Ser | Tyr | Leu | Ser | Leu | Thr | Pro | Glu | Gln | Trp | Lys | Ser | His       |  |
|           |     |     |     | 195 |     |     |     |     | 200 |     |     |     |     | 205 |           |  |
| Arg       | Ser | Tyr | Ser | Cys | Gln | Val | Thr | His | Glu | Gly | Ser | Thr | Val | Glu | Lys       |  |
|           |     |     |     | 210 |     |     |     |     | 215 |     |     |     |     | 220 |           |  |
| Thr       | Val | Ala | Pro | Thr | Glu | Ala | Gln | Val | Gln | Leu | Val | Gln | Ser | Gly | Ala       |  |
|           |     |     |     | 225 |     |     |     |     | 230 |     |     |     |     | 235 |           |  |
| Glu       | Val | Lys | Lys | Pro | Gly | Ser | Ser | Val | Lys | Val | Ser | Cys | Lys | Ala | Ser       |  |
|           |     |     |     | 245 |     |     |     |     | 250 |     |     |     |     | 255 |           |  |
| Gly       | Gly | Thr | Phe | Ser | Thr | His | Ala | Ile | Ser | Trp | Val | Arg | Gln | Ala | Pro       |  |
|           |     |     |     | 260 |     |     |     |     | 265 |     |     |     |     | 270 |           |  |
| Gly       | Gln | Gly | Leu | Glu | Trp | Met | Gly | Leu | Ile | Gln | Pro | Arg | Phe | Gly | Thr       |  |
|           |     |     |     | 275 |     |     |     |     | 280 |     |     |     |     | 285 |           |  |
| Ala       | Asn | Tyr | Ala | Gln | Lys | Phe | Gln | Gly | Arg | Val | Thr | Ile | Thr | Ala | Asp       |  |
|           |     |     |     | 290 |     |     |     |     | 295 |     |     |     |     | 300 |           |  |
| Glu       | Ser | Thr | Ser | Thr | Ala | Tyr | Met | Glu | Leu | Ser | Ser | Leu | Arg | Ser | Glu       |  |
|           |     |     |     | 305 |     |     |     |     | 310 |     |     |     |     | 315 |           |  |
| Asp       | Thr | Ala | Val | Tyr | Tyr | Cys | Ala | Arg | Asp | Tyr | Tyr | Gly | Gly | Met | Ala       |  |
|           |     |     |     | 325 |     |     |     |     | 330 |     |     |     |     | 335 |           |  |
| Tyr       | Trp | Gly | Gln | Gly | Thr | Leu | Val | Thr | Val | Ser | Ser | Ala | Ser | Thr | Lys       |  |
|           |     |     |     | 340 |     |     |     |     | 345 |     |     |     |     | 350 |           |  |
| Gly       | Pro | Ser | Val | Phe | Pro | Leu | Ala | Pro | Ser | Ser | Lys | Ser | Thr | Ser | Gly       |  |
|           |     |     |     | 355 |     |     |     |     | 360 |     |     |     |     | 365 |           |  |
| Gly       | Thr | Ala | Ala | Leu | Gly | Cys | Leu | Val | Lys | Asp | Tyr | Phe | Pro | Glu | Pro       |  |
|           |     |     |     | 370 |     |     |     |     | 375 |     |     |     |     | 380 |           |  |
| Val       | Thr | Val | Ser | Trp | Asn | Ser | Gly | Ala | Leu | Thr | Ser | Gly | Val | His | Thr       |  |
|           |     |     |     | 385 |     |     |     |     | 390 |     |     |     |     | 395 |           |  |
| Phe       | Pro | Ala | Val | Leu | Gln | Ser | Ser | Gly | Leu | Tyr | Ser | Leu | Ser | Ser | Val       |  |
|           |     |     |     | 405 |     |     |     |     | 410 |     |     |     |     | 415 |           |  |
| Val       | Thr | Val | Pro | Ser | Ser | Ser | Leu | Gly | Thr | Gln | Thr | Tyr | Ile | Cys | Asn       |  |
|           |     |     |     | 420 |     |     |     |     | 425 |     |     |     |     | 430 |           |  |
| Val       | Asn | His | Lys | Pro | Ser | Asn | Thr | Lys | Val | Asp | Lys | Lys | Val | Glu | Pro       |  |
|           |     |     |     | 435 |     |     |     |     | 440 |     |     |     |     | 445 |           |  |
| Lys       | Ser | Glu | Phe | Asp | Tyr | Lys | Asp | Asp | Asp | Asp | Lys | Gly | Ala | Pro | His       |  |
|           |     |     |     | 450 |     |     |     |     |     |     |     |     |     |     |           |  |
| His       | His | His | His | His |     |     |     |     |     |     |     |     |     |     |           |  |

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<210> SEQ ID NO 76  
<211> LENGTH: 20  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct  
<220> FEATURE:  
<221> NAME/KEY: MISC\_FEATURE  
<222> LOCATION: (4)..(4)  
<223> OTHER INFORMATION: Xaa is Val or Ser  
<220> FEATURE:  
<221> NAME/KEY: MISC\_FEATURE  
<222> LOCATION: (6)..(6)  
<223> OTHER INFORMATION: Xaa is Gly, His, or Phe  
<220> FEATURE:  
<221> NAME/KEY: MISC\_FEATURE  
<222> LOCATION: (8)..(8)  
<223> OTHER INFORMATION: Xaa is Ser or Asp  
<220> FEATURE:  
<221> NAME/KEY: MISC\_FEATURE  
<222> LOCATION: (9)..(9)  
<223> OTHER INFORMATION: Xaa is Gly or Ser  
<220> FEATURE:  
<221> NAME/KEY: MISC\_FEATURE  
<222> LOCATION: (11)..(11)  
<223> OTHER INFORMATION: Xaa is Ser, Glu, or Asn

<400> SEQUENCE: 76

Trp Val Ser Xaa Ile Xaa Tyr Xaa Xaa Ser Xaa Thr Tyr Tyr Ala Asp  
1 5 10 15  
Ser Val Lys Gly  
20

<210> SEQ ID NO 77  
<211> LENGTH: 138  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 77

Met Glu Trp Ser Trp Ile Phe Leu Phe Leu Leu Ser Gly Thr Ala Gly  
1 5 10 15  
Val Leu Ser Glu Val Gln Leu Gln Gln Ser Gly Pro Glu Leu Val Lys  
20 25 30  
Pro Gly Thr Ser Val Lys Ile Ser Cys Lys Ala Ser Gly Tyr Ser Phe  
35 40 45  
Thr Gly Tyr Tyr Met His Trp Val Lys Gln Ser Gln Val Lys Ser Leu  
50 55 60  
Glu Trp Ile Gly Arg Ile Asn Pro Asp Asn Gly Gly Arg Thr Tyr Asn  
65 70 75 80  
Gln Ile Phe Lys Asp Lys Ala Ser Leu Thr Val His Lys Ser Ser Ser  
85 90 95  
Thr Ala Tyr Met Glu Leu His Ser Leu Thr Ser Asp Asp Ser Ala Val  
100 105 110  
Tyr Tyr Cys Thr Arg Glu Arg Gly Gln Tyr Gly Asn Tyr Gly Gly Phe  
115 120 125  
Ser Asp Trp Gly Gln Gly Thr Leu Val Thr  
130 135

<210> SEQ ID NO 78

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<211> LENGTH: 119  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct  
  
<400> SEQUENCE: 78  
  
Glu Val Gln Leu Gln Gln Ser Gly Pro Glu Leu Val Lys Pro Gly Thr  
1                   5                   10                   15  
  
Ser Val Lys Ile Ser Cys Lys Ala Ser Gly Tyr Ser Phe Thr Gly Tyr  
          20                   25                   30  
  
Tyr Met His Trp Val Lys Gln Ser Gln Val Lys Ser Leu Glu Trp Ile  
          35                   40                   45  
  
Gly Arg Ile Asn Pro Asp Asn Gly Gly Arg Thr Tyr Asn Gln Ile Phe  
50                   55                   60  
  
Lys Asp Lys Ala Ser Leu Thr Val His Lys Ser Ser Ser Thr Ala Tyr  
65                   70                   75                   80  
  
Met Glu Leu His Ser Leu Thr Ser Asp Asp Ser Ala Val Tyr Tyr Cys  
          85                   90                   95  
  
Thr Arg Glu Arg Gly Gln Tyr Gly Asn Tyr Gly Gly Phe Ser Asp Trp  
100                   105                   110  
  
Gly Gln Gly Thr Leu Val Thr  
115

<210> SEQ ID NO 79  
<211> LENGTH: 6  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 79

Thr Gly Tyr Tyr Met Lys  
1                   5

<210> SEQ ID NO 80  
<211> LENGTH: 18  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 80

Arg Ile Asn Pro Asp Asn Gly Gly Arg Thr Tyr Asn Gln Ile Phe Lys  
1                   5                   10                   15

Asp Lys

<210> SEQ ID NO 81  
<211> LENGTH: 14  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 81

Arg Glu Arg Gly Gln Tyr Gly Asn Tyr Gly Gly Phe Ser Asp  
1                   5                   10

<210> SEQ ID NO 82  
<211> LENGTH: 10

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<212> TYPE: PRT  
<213> ORGANISM: Rattus norvegicus

<400> SEQUENCE: 82

Gly Phe Thr Phe Ser His Tyr Tyr Met Ala  
1 5 10

<210> SEQ ID NO 83  
<211> LENGTH: 17  
<212> TYPE: PRT  
<213> ORGANISM: Rattus norvegicus

<400> SEQUENCE: 83

Ser Ile Thr Asn Ser Gly Gly Ser Ile Asn Tyr Arg Asp Ser Val Lys  
1 5 10 15

Gly

<210> SEQ ID NO 84  
<211> LENGTH: 14  
<212> TYPE: PRT  
<213> ORGANISM: Rattus norvegicus

<400> SEQUENCE: 84

Glu Gly Gly Glu Asn Tyr Gly Gly Tyr Pro Pro Phe Ala Tyr  
1 5 10

<210> SEQ ID NO 85  
<211> LENGTH: 15  
<212> TYPE: PRT  
<213> ORGANISM: Rattus norvegicus

<400> SEQUENCE: 85

Arg Ala Asn Gln Gly Val Ser Leu Ser Arg Tyr Asn Leu Met His  
1 5 10 15

<210> SEQ ID NO 86  
<211> LENGTH: 7  
<212> TYPE: PRT  
<213> ORGANISM: Rattus norvegicus

<400> SEQUENCE: 86

Arg Ser Ser Asn Leu Ala Ser  
1 5

<210> SEQ ID NO 87  
<211> LENGTH: 9  
<212> TYPE: PRT  
<213> ORGANISM: Rattus norvegicus

<400> SEQUENCE: 87

Gln Gln Ser Arg Glu Ser Pro Phe Thr  
1 5

<210> SEQ ID NO 88  
<211> LENGTH: 10  
<212> TYPE: PRT  
<213> ORGANISM: Rattus norvegicus

<400> SEQUENCE: 88

Gly Ser Thr Phe Ser Asn Tyr Gly Met Lys  
1 5 10

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<210> SEQ ID NO 89  
<211> LENGTH: 17  
<212> TYPE: PRT  
<213> ORGANISM: Rattus norvegicus

<400> SEQUENCE: 89

|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Ser | Ile | Ser | Arg | Ser | Ser | Thr | Tyr | Ile | Tyr | Tyr | Ala | Asp | Thr | Val | Lys |
| 1   |     |     | 5   |     |     |     |     | 10  |     |     |     |     |     | 15  |     |

Gly

<210> SEQ ID NO 90  
<211> LENGTH: 12  
<212> TYPE: PRT  
<213> ORGANISM: Rattus norvegicus

<400> SEQUENCE: 90

|     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Ala | Ile | Ser | Thr | Pro | Phe | Tyr | Trp | Tyr | Phe | Asp | Phe |
| 1   |     |     | 5   |     |     |     |     | 10  |     |     |     |

<210> SEQ ID NO 91  
<211> LENGTH: 10  
<212> TYPE: PRT  
<213> ORGANISM: Rattus norvegicus

<400> SEQUENCE: 91

|     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Leu | Ala | Ser | Ser | Ser | Val | Ser | Tyr | Met | Thr |
| 1   |     |     | 5   |     |     |     |     | 10  |     |

<210> SEQ ID NO 92  
<211> LENGTH: 7  
<212> TYPE: PRT  
<213> ORGANISM: Rattus norvegicus

<400> SEQUENCE: 92

|     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|
| Gly | Thr | Ser | Asn | Leu | Ala | Ser |
| 1   |     |     | 5   |     |     |     |

<210> SEQ ID NO 93  
<211> LENGTH: 10  
<212> TYPE: PRT  
<213> ORGANISM: Rattus norvegicus

<400> SEQUENCE: 93

|     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Leu | His | Leu | Thr | Ser | Tyr | Pro | Pro | Tyr | Thr |
| 1   |     |     | 5   |     |     |     |     | 10  |     |

<210> SEQ ID NO 94  
<211> LENGTH: 7  
<212> TYPE: PRT  
<213> ORGANISM: Rattus norvegicus

<400> SEQUENCE: 94

|     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|
| Arg | Ser | Ser | Asn | Leu | Ala | Gln |
| 1   |     |     | 5   |     |     |     |

<210> SEQ ID NO 95  
<211> LENGTH: 10  
<212> TYPE: PRT  
<213> ORGANISM: Rattus norvegicus

<400> SEQUENCE: 95

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Gly Phe Thr Phe Ser Asn Tyr Tyr Met Tyr  
1 5 10

<210> SEQ ID NO 96  
<211> LENGTH: 17  
<212> TYPE: PRT  
<213> ORGANISM: Rattus norvegicus

<400> SEQUENCE: 96

Ser Ile Asn Thr Asp Gly Gly Ser Thr Tyr Tyr Pro Asp Ser Val Lys  
1 5 10 15

Gly

<210> SEQ ID NO 97  
<211> LENGTH: 9  
<212> TYPE: PRT  
<213> ORGANISM: Rattus norvegicus

<400> SEQUENCE: 97

Ser Thr Pro Asn Ile Pro Leu Ala Tyr  
1 5

<210> SEQ ID NO 98  
<211> LENGTH: 11  
<212> TYPE: PRT  
<213> ORGANISM: Rattus norvegicus

<400> SEQUENCE: 98

Lys Ala Ser Gln Asn Ile Tyr Lys Tyr Leu Asn  
1 5 10

<210> SEQ ID NO 99  
<211> LENGTH: 7  
<212> TYPE: PRT  
<213> ORGANISM: Rattus norvegicus

<400> SEQUENCE: 99

Tyr Ser Asn Ser Leu Gln Thr  
1 5

<210> SEQ ID NO 100  
<211> LENGTH: 8  
<212> TYPE: PRT  
<213> ORGANISM: Rattus norvegicus

<400> SEQUENCE: 100

Phe Gln Tyr Ser Ser Gly Pro Thr  
1 5

<210> SEQ ID NO 101  
<211> LENGTH: 10  
<212> TYPE: PRT  
<213> ORGANISM: Rattus norvegicus

<400> SEQUENCE: 101

Gly Phe Thr Phe Ser Tyr Tyr Ala Met Ser  
1 5 10

<210> SEQ ID NO 102  
<211> LENGTH: 17  
<212> TYPE: PRT  
<213> ORGANISM: Rattus norvegicus



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<400> SEQUENCE: 102

Ser Ile Ser Arg Gly Gly Asp Asn Thr Tyr Tyr Arg Asp Thr Val Lys  
1 5 10 15

Gly

<210> SEQ ID NO 103

<211> LENGTH: 9

<212> TYPE: PRT

<213> ORGANISM: Rattus norvegicus

<400> SEQUENCE: 103

Leu Asn Tyr Asn Asn Tyr Phe Asp Tyr  
1 5

<210> SEQ ID NO 104

<211> LENGTH: 11

<212> TYPE: PRT

<213> ORGANISM: Rattus norvegicus

<400> SEQUENCE: 104

Gln Ala Ser Gln Asp Ile Gly Asn Trp Leu Ser  
1 5 10

<210> SEQ ID NO 105

<211> LENGTH: 7

<212> TYPE: PRT

<213> ORGANISM: Rattus norvegicus

<400> SEQUENCE: 105

Gly Ala Thr Ser Leu Ala Asp  
1 5

<210> SEQ ID NO 106

<211> LENGTH: 9

<212> TYPE: PRT

<213> ORGANISM: Rattus norvegicus

<400> SEQUENCE: 106

Leu Gln Ala Tyr Ser Ala Pro Phe Thr  
1 5

<210> SEQ ID NO 107

<211> LENGTH: 118

<212> TYPE: PRT

<213> ORGANISM: Rattus norvegicus

<400> SEQUENCE: 107

Glu Val Gln Leu Val Glu Thr Gly Gly Gly Leu Val Gln Pro Gly Arg  
1 5 10 15

Ser Leu Lys Leu Ser Cys Val Ala Ser Gly Phe Thr Phe Ser Asn Tyr  
20 25 30

Tyr Met Tyr Trp Ile Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile  
35 40 45

Ser Ser Ile Asn Thr Asp Gly Gly Ser Thr Tyr Tyr Pro Asp Ser Val  
50 55 60

Lys Gly Arg Phe Ala Ile Ser Arg Asp Asn Ala Glu Asn Thr Val Tyr  
65 70 75 80

Leu Gln Met Asn Ser Leu Arg Ser Glu Asp Thr Ala Thr Tyr Tyr Cys

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|                                                                 |     |  |     |  |     |
|-----------------------------------------------------------------|-----|--|-----|--|-----|
|                                                                 | 85  |  | 90  |  | 95  |
| Ala Lys Ser Thr Pro Asn Ile Pro Leu Ala Tyr Trp Gly Gln Gly Thr |     |  |     |  |     |
|                                                                 | 100 |  | 105 |  | 110 |
| Leu Val Thr Val Ser Ser                                         |     |  |     |  |     |
|                                                                 | 115 |  |     |  |     |

<210> SEQ ID NO 108  
 <211> LENGTH: 108  
 <212> TYPE: PRT  
 <213> ORGANISM: Rattus norvegicus

<400> SEQUENCE: 108

|                                                                 |     |  |     |  |    |
|-----------------------------------------------------------------|-----|--|-----|--|----|
| Asp Ile Gln Met Thr Gln Ser Pro Ser Leu Leu Ser Ala Ser Val Gly |     |  |     |  |    |
| 1                                                               | 5   |  | 10  |  | 15 |
| Asp Arg Val Thr Leu Ser Cys Lys Ala Ser Gln Asn Ile Tyr Lys Tyr |     |  |     |  |    |
|                                                                 | 20  |  | 25  |  | 30 |
| Leu Asn Trp Phe Gln Gln Lys Leu Gly Glu Ala Pro Lys Leu Leu Ile |     |  |     |  |    |
|                                                                 | 35  |  | 40  |  | 45 |
| Tyr Tyr Ser Asn Ser Leu Gln Thr Gly Ile Pro Ser Arg Phe Ser Gly |     |  |     |  |    |
|                                                                 | 50  |  | 55  |  | 60 |
| Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro |     |  |     |  |    |
| 65                                                              | 70  |  | 75  |  | 80 |
| Glu Asp Val Ala Ile Tyr Phe Cys Phe Gln Tyr Ser Ser Gly Pro Thr |     |  |     |  |    |
|                                                                 | 85  |  | 90  |  | 95 |
| Phe Gly Pro Gly Thr Lys Leu Glu Leu Lys Arg Ala                 |     |  |     |  |    |
|                                                                 | 100 |  | 105 |  |    |

<210> SEQ ID NO 109  
 <211> LENGTH: 123  
 <212> TYPE: PRT  
 <213> ORGANISM: Rattus norvegicus

<400> SEQUENCE: 109

|                                                                 |     |  |     |  |     |
|-----------------------------------------------------------------|-----|--|-----|--|-----|
| Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Arg |     |  |     |  |     |
| 1                                                               | 5   |  | 10  |  | 15  |
| Ser Leu Lys Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser His Tyr |     |  |     |  |     |
|                                                                 | 20  |  | 25  |  | 30  |
| Tyr Met Ala Trp Val Arg Gln Ala Pro Thr Lys Gly Leu Glu Trp Val |     |  |     |  |     |
|                                                                 | 35  |  | 40  |  | 45  |
| Ala Ser Ile Thr Asn Ser Gly Gly Ser Ile Asn Tyr Arg Asp Ser Val |     |  |     |  |     |
|                                                                 | 50  |  | 55  |  | 60  |
| Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Ser Thr Leu Tyr |     |  |     |  |     |
| 65                                                              | 70  |  | 75  |  | 80  |
| Leu Gln Met Asp Ser Leu Arg Ser Glu Asp Thr Ala Thr Tyr Tyr Cys |     |  |     |  |     |
|                                                                 | 85  |  | 90  |  | 95  |
| Thr Arg Glu Gly Gly Glu Asn Tyr Gly Gly Tyr Pro Pro Phe Ala Tyr |     |  |     |  |     |
|                                                                 | 100 |  | 105 |  | 110 |
| Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser                     |     |  |     |  |     |
|                                                                 | 115 |  | 120 |  |     |

<210> SEQ ID NO 110  
 <211> LENGTH: 112  
 <212> TYPE: PRT  
 <213> ORGANISM: Rattus norvegicus

<400> SEQUENCE: 110

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Asp Ile Val Leu Thr Gln Ser Pro Ala Leu Ala Val Ser Leu Gly Gln
1           5           10           15
Arg Ala Thr Ile Ser Cys Arg Ala Asn Gln Gly Val Ser Leu Ser Arg
           20           25           30
Tyr Asn Leu Met His Trp Tyr Gln Gln Lys Pro Gly Gln Lys Pro Lys
           35           40           45
Leu Leu Ile Tyr Arg Ser Ser Asn Leu Ala Ser Gly Ile Pro Ala Arg
           50           55           60
Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Asn Pro
65           70           75           80
Val Gln Ala Asp Asp Ile Ala Thr Tyr Tyr Cys Gln Gln Ser Arg Glu
           85           90           95
Ser Pro Phe Thr Phe Gly Ala Gly Thr Lys Leu Glu Leu Lys Arg Ala
           100          105          110

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<210> SEQ ID NO 111
<211> LENGTH: 118
<212> TYPE: PRT
<213> ORGANISM: Rattus norvegicus

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<400> SEQUENCE: 111

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Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1           5           10           15
Ser Leu Lys Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Tyr Tyr
           20           25           30
Ala Met Ser Trp Val Cys Gln Ala Pro Thr Lys Gly Leu Glu Trp Val
           35           40           45
Ala Ser Ile Ser Arg Gly Gly Asp Asn Thr Tyr Tyr Arg Asp Thr Val
           50           55           60
Lys Gly Arg Phe Thr Thr Ser Arg Asp Asn Ala Lys Asn Thr Leu Tyr
65           70           75           80
Leu Gln Met Asp Ser Leu Arg Ser Glu Asp Thr Ala Thr Tyr Tyr Cys
           85           90           95
Ala Arg Leu Asn Tyr Asn Asn Tyr Phe Asp Tyr Trp Gly Gln Gly Val
           100          105          110
Met Val Thr Val Ser Ser
           115

```

```

<210> SEQ ID NO 112
<211> LENGTH: 109
<212> TYPE: PRT
<213> ORGANISM: Rattus norvegicus

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```

<400> SEQUENCE: 112

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```

Asp Ile Gln Met Thr Gln Ser Pro Ala Ser Leu Ser Ala Ser Leu Glu
1           5           10           15
Glu Ile Val Thr Ile Thr Cys Gln Ala Ser Gln Asp Ile Gly Asn Trp
           20           25           30
Leu Ser Trp Tyr Gln Gln Lys Pro Gly Lys Ser Pro Gln Leu Leu Ile
           35           40           45
Tyr Gly Ala Thr Ser Leu Ala Asp Gly Val Pro Ser Arg Phe Ser Gly
           50           55           60
Ser Arg Ser Gly Thr Gln Tyr Ser Leu Lys Ile Ser Arg Leu Arg Val
65           70           75           80

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Glu Asp Ile Gly Ile Tyr Tyr Cys Leu Gln Ala Tyr Ser Ala Pro Phe  
85 90 95

Thr Phe Gly Ser Gly Thr Lys Leu Glu Ile Lys Arg Ala  
100 105

<210> SEQ ID NO 113  
<211> LENGTH: 121  
<212> TYPE: PRT  
<213> ORGANISM: Rattus norvegicus

<400> SEQUENCE: 113

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Arg  
1 5 10 15

Ser Leu Lys Leu Ser Cys Leu Ala Ser Gly Ser Thr Phe Ser Asn Tyr  
20 25 30

Gly Met Lys Trp Ile Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val  
35 40 45

Ala Ser Ile Ser Arg Ser Ser Thr Tyr Ile Tyr Tyr Ala Asp Thr Val  
50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Arg Asn Thr Leu Tyr  
65 70 75 80

Leu Gln Met Thr Ser Leu Arg Ser Glu Asp Thr Ala Leu Tyr Tyr Cys  
85 90 95

Ala Ala Ala Ile Ser Thr Pro Phe Tyr Trp Tyr Phe Asp Phe Trp Gly  
100 105 110

Pro Gly Thr Val Val Thr Val Ser Ser  
115 120

<210> SEQ ID NO 114  
<211> LENGTH: 109  
<212> TYPE: PRT  
<213> ORGANISM: Rattus norvegicus

<400> SEQUENCE: 114

Glu Ile Val Leu Thr Gln Ser Pro Thr Thr Met Ala Ala Ser Pro Gly  
1 5 10 15

Glu Lys Val Thr Leu Asn Cys Leu Ala Ser Ser Ser Val Ser Tyr Met  
20 25 30

Thr Trp Tyr Gln Gln Lys Ser Gly Ala Ser Pro Lys Leu Trp Ile Tyr  
35 40 45

Gly Thr Ser Asn Leu Ala Ser Gly Val Pro Asn Arg Phe Ser Gly Ser  
50 55 60

Gly Ser Gly Thr Ser Tyr Ser Leu Ala Ile Ser Ser Met Glu Ala Glu  
65 70 75 80

Asp Val Ala Thr Tyr Tyr Cys Leu His Leu Thr Ser Tyr Pro Pro Tyr  
85 90 95

Thr Phe Gly Ala Gly Thr Lys Leu Glu Leu Lys Arg Ala  
100 105

<210> SEQ ID NO 115  
<211> LENGTH: 113  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

-continued

&lt;400&gt; SEQUENCE: 115

```

Asp Ile Val Leu Thr Gln Ser Pro Asp Ser Leu Ala Val Ser Leu Gly
1           5           10           15
Glu Arg Ala Thr Ile Asn Cys Arg Ala Asn Gln Gly Val Ser Leu Ser
20          25          30
Arg Tyr Asn Leu Met His Trp Tyr Gln Gln Lys Pro Gly Gln Lys Pro
35          40          45
Lys Leu Leu Ile Tyr Arg Ser Ser Asn Leu Ala Gln Gly Ile Pro Ala
50          55          60
Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser
65          70          75          80
Ser Val Gln Ala Asp Asp Ile Ala Val Tyr Tyr Cys Gln Gln Ser Arg
85          90          95
Glu Ser Pro Phe Thr Phe Gly Gln Gly Thr Lys Leu Glu Leu Lys Arg
100         105         110

```

Ala

&lt;210&gt; SEQ ID NO 116

&lt;211&gt; LENGTH: 123

&lt;212&gt; TYPE: PRT

&lt;213&gt; ORGANISM: Artificial Sequence

&lt;220&gt; FEATURE:

&lt;223&gt; OTHER INFORMATION: Synthetic Construct

&lt;400&gt; SEQUENCE: 116

```

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1           5           10           15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser His Tyr
20          25          30
Tyr Met Ala Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35          40          45
Ala Ser Ile Thr Asn Ser Gly Gly Ser Ile Asn Tyr Arg Asp Ser Val
50          55          60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65          70          75          80
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85          90          95
Ala Arg Glu Gly Gly Glu Asn Tyr Gly Gly Tyr Pro Pro Phe Ala Tyr
100         105         110
Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser
115         120

```

&lt;210&gt; SEQ ID NO 117

&lt;211&gt; LENGTH: 123

&lt;212&gt; TYPE: PRT

&lt;213&gt; ORGANISM: Artificial Sequence

&lt;220&gt; FEATURE:

&lt;223&gt; OTHER INFORMATION: Synthetic Construct

&lt;400&gt; SEQUENCE: 117

```

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1           5           10           15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser His Tyr
20          25          30
Tyr Met Ala Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val

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|                                                                 |     |     |
|-----------------------------------------------------------------|-----|-----|
| 35                                                              | 40  | 45  |
| Ala Ser Ile Thr Asn Ser Gly Gly Ser Ile Asn Tyr Arg Asp Ser Val |     |     |
| 50                                                              | 55  | 60  |
| Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Ser Thr Leu Tyr |     |     |
| 65                                                              | 70  | 75  |
| Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Thr Tyr Tyr Cys |     |     |
|                                                                 | 85  | 90  |
| Thr Arg Glu Gly Gly Glu Asn Tyr Gly Gly Tyr Pro Pro Phe Ala Tyr |     |     |
|                                                                 | 100 | 105 |
| Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser                     |     |     |
|                                                                 | 115 | 120 |

<210> SEQ ID NO 118  
 <211> LENGTH: 113  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 118

|                                                                 |
|-----------------------------------------------------------------|
| Asp Ile Val Met Thr Gln Ser Pro Asp Ser Leu Ala Val Ser Leu Gly |
| 1 5 10 15                                                       |
| Glu Arg Ala Thr Ile Asn Cys Arg Ala Asn Gln Gly Val Ser Leu Ser |
| 20 25 30                                                        |
| Arg Tyr Asn Leu Met His Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro |
| 35 40 45                                                        |
| Lys Leu Leu Ile Tyr Arg Ser Ser Asn Leu Ala Ser Gly Val Pro Asp |
| 50 55 60                                                        |
| Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser |
| 65 70 75 80                                                     |
| Ser Leu Gln Ala Glu Asp Val Ala Val Tyr Tyr Cys Gln Gln Ser Arg |
| 85 90 95                                                        |
| Glu Ser Pro Phe Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Arg |
| 100 105 110                                                     |

Ala

<210> SEQ ID NO 119  
 <211> LENGTH: 113  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 119

|                                                                 |
|-----------------------------------------------------------------|
| Asp Ile Val Leu Thr Gln Ser Pro Asp Ser Leu Ala Val Ser Leu Gly |
| 1 5 10 15                                                       |
| Glu Arg Ala Thr Ile Asn Cys Arg Ala Asn Gln Gly Val Ser Leu Ser |
| 20 25 30                                                        |
| Arg Tyr Asn Leu Met His Trp Tyr Gln Gln Lys Pro Gly Gln Lys Pro |
| 35 40 45                                                        |
| Lys Leu Leu Ile Tyr Arg Ser Ser Asn Leu Ala Ser Gly Ile Pro Ala |
| 50 55 60                                                        |
| Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser |
| 65 70 75 80                                                     |
| Ser Val Gln Ala Asp Asp Ile Ala Val Tyr Tyr Cys Gln Gln Ser Arg |

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Ala

<400> SEQUENCE: 120

Ala

<400> SEQUENCE: 121

<210> SEQ ID NO 122

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<211> LENGTH: 121  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct  
  
<400> SEQUENCE: 122  
  
Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly  
1                   5                   10                   15  
  
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Ser Thr Phe Ser Asn Tyr  
          20                   25                   30  
  
Gly Met Lys Trp Ile Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val  
          35                   40                   45  
  
Ser Ser Ile Ser Arg Ser Ser Thr Tyr Ile Tyr Tyr Ala Asp Thr Val  
50                   55                   60  
  
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr  
65                   70                   75                   80  
  
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys  
          85                   90                   95  
  
Ala Ala Ala Ile Ser Thr Pro Phe Tyr Trp Tyr Phe Asp Phe Trp Gly  
100                   105                   110  
  
Gln Gly Thr Leu Val Thr Val Ser Ser  
115                   120

<210> SEQ ID NO 123  
<211> LENGTH: 121  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct  
  
<400> SEQUENCE: 123  
  
Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly  
1                   5                   10                   15  
  
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Ser Thr Phe Ser Asn Tyr  
          20                   25                   30  
  
Gly Met Lys Trp Ile Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val  
          35                   40                   45  
  
Ala Ser Ile Ser Arg Ser Ser Thr Tyr Ile Tyr Tyr Ala Asp Thr Val  
50                   55                   60  
  
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr  
65                   70                   75                   80  
  
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys  
          85                   90                   95  
  
Ala Ala Ala Ile Ser Thr Pro Phe Tyr Trp Tyr Phe Asp Phe Trp Gly  
100                   105                   110  
  
Gln Gly Thr Leu Val Thr Val Ser Ser  
115                   120

<210> SEQ ID NO 124  
<211> LENGTH: 121  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct  
  
<400> SEQUENCE: 124



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Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1          5          10          15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Ser Thr Phe Ser Asn Tyr
          20          25          30
Gly Met Lys Trp Ile Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
          35          40          45
Ala Ser Ile Ser Arg Ser Ser Thr Tyr Ile Tyr Tyr Ala Asp Thr Val
          50          55          60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65          70          75          80
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Leu Tyr Tyr Cys
          85          90          95
Ala Ala Ala Ile Ser Thr Pro Phe Tyr Trp Tyr Phe Asp Phe Trp Gly
          100          105          110
Pro Gly Thr Leu Val Thr Val Ser Ser
          115          120

```

```

<210> SEQ ID NO 125
<211> LENGTH: 121
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

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<400> SEQUENCE: 125

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```

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1          5          10          15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Ser Thr Phe Ser Asn Tyr
          20          25          30
Gly Met Lys Trp Ile Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
          35          40          45
Ala Ser Ile Ser Arg Ser Ser Thr Tyr Ile Tyr Tyr Ala Asp Thr Val
          50          55          60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Asn Thr Leu Tyr
65          70          75          80
Leu Gln Met Asn Ser Leu Arg Ser Glu Asp Thr Ala Leu Tyr Tyr Cys
          85          90          95
Ala Ala Ala Ile Ser Thr Pro Phe Tyr Trp Tyr Phe Asp Phe Trp Gly
          100          105          110
Pro Gly Thr Leu Val Thr Val Ser Ser
          115          120

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<210> SEQ ID NO 126
<211> LENGTH: 109
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

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<400> SEQUENCE: 126

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Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Leu Ser Pro Gly
1          5          10          15
Glu Arg Ala Thr Leu Ser Cys Leu Ala Ser Ser Ser Val Ser Tyr Met
          20          25          30
Thr Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile Tyr
          35          40          45

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Gly Thr Ser Asn Leu Ala Ser Gly Ile Pro Asp Arg Phe Ser Gly Ser  
 50 55 60

Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Arg Leu Glu Pro Glu  
 65 70 75 80

Asp Phe Ala Val Tyr Tyr Cys Leu His Leu Thr Ser Tyr Pro Pro Tyr  
 85 90 95

Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Arg Ala  
 100 105

<210> SEQ ID NO 127  
 <211> LENGTH: 109  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 127

Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Leu Ser Pro Gly  
 1 5 10 15

Glu Arg Ala Thr Leu Ser Cys Leu Ala Ser Ser Ser Val Ser Tyr Met  
 20 25 30

Thr Trp Tyr Gln Gln Lys Pro Gly Gln Ser Pro Arg Leu Trp Ile Tyr  
 35 40 45

Gly Thr Ser Asn Leu Ala Ser Gly Val Pro Asp Arg Phe Ser Gly Ser  
 50 55 60

Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Arg Leu Glu Pro Glu  
 65 70 75 80

Asp Phe Ala Val Tyr Tyr Cys Leu His Leu Thr Ser Tyr Pro Pro Tyr  
 85 90 95

Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Arg Ala  
 100 105

<210> SEQ ID NO 128  
 <211> LENGTH: 109  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 128

Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Leu Ser Pro Gly  
 1 5 10 15

Glu Arg Ala Thr Leu Ser Cys Leu Ala Ser Ser Ser Val Ser Tyr Met  
 20 25 30

Thr Trp Tyr Gln Gln Lys Pro Gly Ala Ser Pro Arg Leu Trp Ile Tyr  
 35 40 45

Gly Thr Ser Asn Leu Ala Ser Gly Val Pro Asp Arg Phe Ser Gly Ser  
 50 55 60

Gly Ser Gly Thr Asp Tyr Thr Leu Thr Ile Ser Arg Leu Glu Pro Glu  
 65 70 75 80

Asp Phe Ala Thr Tyr Tyr Cys Leu His Leu Thr Ser Tyr Pro Pro Tyr  
 85 90 95

Thr Phe Gly Ala Gly Thr Lys Val Glu Ile Lys Arg Ala  
 100 105

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<210> SEQ ID NO 129  
<211> LENGTH: 109  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 129

Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Met Ser Ala Ser Pro Gly  
1 5 10 15  
Glu Arg Val Thr Leu Ser Cys Leu Ala Ser Ser Ser Val Ser Tyr Met  
20 25 30  
Thr Trp Tyr Gln Gln Lys Pro Gly Ala Ser Pro Arg Leu Trp Ile Tyr  
35 40 45  
Gly Thr Ser Asn Leu Ala Ser Gly Val Pro Asp Arg Phe Ser Gly Ser  
50 55 60  
Gly Ser Gly Thr Asp Tyr Thr Leu Thr Ile Ser Arg Met Glu Pro Glu  
65 70 75 80  
Asp Phe Ala Thr Tyr Tyr Cys Leu His Leu Thr Ser Tyr Pro Pro Tyr  
85 90 95  
Thr Phe Gly Ala Gly Thr Lys Leu Glu Leu Lys Arg Ala  
100 105

<210> SEQ ID NO 130  
<211> LENGTH: 109  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 130

Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Met Ser Ala Ser Pro Gly  
1 5 10 15  
Glu Arg Val Thr Leu Asn Cys Leu Ala Ser Ser Ser Val Ser Tyr Met  
20 25 30  
Thr Trp Tyr Gln Gln Lys Ser Gly Ala Ser Pro Lys Leu Trp Ile Tyr  
35 40 45  
Gly Thr Ser Asn Leu Ala Ser Gly Val Pro Asn Arg Phe Ser Gly Ser  
50 55 60  
Gly Ser Gly Thr Ser Tyr Thr Leu Thr Ile Ser Arg Met Glu Pro Glu  
65 70 75 80  
Asp Phe Ala Thr Tyr Tyr Cys Leu His Leu Thr Ser Tyr Pro Pro Tyr  
85 90 95  
Thr Phe Gly Ala Gly Thr Lys Leu Glu Leu Lys Arg Ala  
100 105

<210> SEQ ID NO 131  
<211> LENGTH: 123  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 131

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly  
1 5 10 15  
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser His Tyr  
20 25 30

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Tyr Met Ala Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val  
           35                          40                          45  
 Ala Ser Ile Thr Asn Ser Gly Gly Ser Ile Asn Tyr Arg Asp Ser Val  
           50                          55                          60  
 Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Ser Thr Leu Tyr  
           65                          70                          75                          80  
 Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Thr Tyr Tyr Cys  
                           85                          90                          95  
 Thr Arg Glu Gly Gly Glu Asn Tyr Gly Gly Tyr Pro Pro Phe Ala Tyr  
                           100                          105                          110  
 Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser  
           115                          120

<210> SEQ ID NO 132  
 <211> LENGTH: 121  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 132

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly  
 1                          5                          10                          15  
 Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Ser Thr Phe Ser Asn Tyr  
           20                          25                          30  
 Gly Met Lys Trp Ile Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val  
           35                          40                          45  
 Ser Ser Ile Ser Arg Ser Ser Thr Tyr Ile Tyr Tyr Ala Asp Thr Val  
           50                          55                          60  
 Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr  
           65                          70                          75                          80  
 Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys  
                           85                          90                          95  
 Ala Ala Ala Ile Ser Thr Pro Phe Tyr Trp Tyr Phe Asp Phe Trp Gly  
           100                          105                          110  
 Gln Gly Thr Leu Val Thr Val Ser Ser  
           115                          120

<210> SEQ ID NO 133  
 <211> LENGTH: 121  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 133

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly  
 1                          5                          10                          15  
 Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Ser Thr Phe Ser Asn Tyr  
           20                          25                          30  
 Gly Met Lys Trp Ile Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val  
           35                          40                          45  
 Ala Ser Ile Ser Arg Ser Ser Thr Tyr Ile Tyr Tyr Ala Asp Thr Val  
           50                          55                          60  
 Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr

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|                                                                 |     |     |     |
|-----------------------------------------------------------------|-----|-----|-----|
| 65                                                              | 70  | 75  | 80  |
| Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys | 85  | 90  | 95  |
| Ala Ala Ala Ile Ser Thr Pro Phe Tyr Trp Tyr Phe Asp Phe Trp Gly | 100 | 105 | 110 |
| Gln Gly Thr Leu Val Thr Val Ser Ser                             | 115 | 120 |     |

<210> SEQ ID NO 134  
 <211> LENGTH: 118  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 134

|                                                                 |     |     |     |    |
|-----------------------------------------------------------------|-----|-----|-----|----|
| Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly | 1   | 5   | 10  | 15 |
| Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Asn Tyr | 20  | 25  | 30  |    |
| Tyr Met Tyr Trp Ile Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val | 35  | 40  | 45  |    |
| Ser Ser Ile Asn Thr Asp Gly Gly Ser Thr Tyr Tyr Ala Asp Ser Val | 50  | 55  | 60  |    |
| Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Val Tyr | 65  | 70  | 75  | 80 |
| Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys | 85  | 90  | 95  |    |
| Ala Lys Ser Thr Pro Asn Ile Pro Leu Ala Tyr Trp Gly Gln Gly Thr | 100 | 105 | 110 |    |
| Leu Val Thr Val Ser Ser                                         | 115 |     |     |    |

<210> SEQ ID NO 135  
 <211> LENGTH: 118  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 135

|                                                                 |     |     |     |    |
|-----------------------------------------------------------------|-----|-----|-----|----|
| Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly | 1   | 5   | 10  | 15 |
| Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Asn Tyr | 20  | 25  | 30  |    |
| Tyr Met Tyr Trp Ile Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile | 35  | 40  | 45  |    |
| Ser Ser Ile Asn Thr Asp Gly Gly Ser Thr Tyr Tyr Pro Asp Ser Val | 50  | 55  | 60  |    |
| Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Val Tyr | 65  | 70  | 75  | 80 |
| Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys | 85  | 90  | 95  |    |
| Ala Lys Ser Thr Pro Asn Ile Pro Leu Ala Tyr Trp Gly Gln Gly Thr | 100 | 105 | 110 |    |

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Leu Val Thr Val Ser Ser  
115

<210> SEQ ID NO 136  
<211> LENGTH: 108  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 136

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly  
1 5 10 15  
Asp Arg Val Thr Ile Thr Cys Lys Ala Ser Gln Asn Ile Tyr Lys Tyr  
20 25 30  
Leu Asn Trp Phe Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile  
35 40 45  
Tyr Tyr Ser Asn Ser Leu Gln Thr Gly Val Pro Ser Arg Phe Ser Gly  
50 55 60  
Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro  
65 70 75 80  
Glu Asp Phe Ala Thr Tyr Tyr Cys Phe Gln Tyr Ser Ser Gly Pro Thr  
85 90 95  
Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Arg Thr  
100 105

<210> SEQ ID NO 137  
<211> LENGTH: 108  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 137

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly  
1 5 10 15  
Asp Arg Val Thr Ile Thr Cys Lys Ala Ser Gln Asn Ile Tyr Lys Tyr  
20 25 30  
Leu Asn Trp Phe Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile  
35 40 45  
Tyr Tyr Ser Asn Ser Leu Gln Thr Gly Ile Pro Ser Arg Phe Ser Gly  
50 55 60  
Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro  
65 70 75 80  
Glu Asp Phe Ala Thr Tyr Tyr Cys Phe Gln Tyr Ser Ser Gly Pro Thr  
85 90 95  
Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Arg Thr  
100 105

<210> SEQ ID NO 138  
<211> LENGTH: 108  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 138

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly

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|                                                                 |     |     |    |
|-----------------------------------------------------------------|-----|-----|----|
| 1                                                               | 5   | 10  | 15 |
| Asp Arg Val Thr Ile Ser Cys Lys Ala Ser Gln Asn Ile Tyr Lys Tyr | 20  | 25  | 30 |
| Leu Asn Trp Phe Gln Gln Lys Pro Gly Glu Ala Pro Lys Leu Leu Ile | 35  | 40  | 45 |
| Tyr Tyr Ser Asn Ser Leu Gln Thr Gly Ile Pro Ser Arg Phe Ser Gly | 50  | 55  | 60 |
| Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro | 65  | 70  | 75 |
| Glu Asp Phe Ala Ile Tyr Phe Cys Phe Gln Tyr Ser Ser Gly Pro Thr | 85  | 90  | 95 |
| Phe Gly Pro Gly Thr Lys Val Glu Ile Lys Arg Thr                 | 100 | 105 |    |

<210> SEQ ID NO 139  
 <211> LENGTH: 118  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 139

|                                                                 |     |     |     |    |
|-----------------------------------------------------------------|-----|-----|-----|----|
| Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly | 1   | 5   | 10  | 15 |
| Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Tyr Tyr | 20  | 25  | 30  |    |
| Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val | 35  | 40  | 45  |    |
| Ala Ser Ile Ser Arg Gly Gly Asp Asn Thr Tyr Tyr Arg Asp Ser Val | 50  | 55  | 60  |    |
| Lys Gly Arg Phe Thr Thr Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr | 65  | 70  | 75  | 80 |
| Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys | 85  | 90  | 95  |    |
| Ala Arg Leu Asn Tyr Asn Asn Tyr Phe Asp Tyr Trp Gly Gln Gly Thr | 100 | 105 | 110 |    |
| Leu Val Thr Val Ser Ser                                         | 115 |     |     |    |

<210> SEQ ID NO 140  
 <211> LENGTH: 118  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 140

|                                                                 |    |    |    |    |
|-----------------------------------------------------------------|----|----|----|----|
| Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly | 1  | 5  | 10 | 15 |
| Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Tyr Tyr | 20 | 25 | 30 |    |
| Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val | 35 | 40 | 45 |    |
| Ala Ser Ile Ser Arg Gly Gly Asp Asn Thr Tyr Tyr Arg Asp Ser Val | 50 | 55 | 60 |    |

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Lys Gly Arg Phe Thr Thr Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr  
65 70 75 80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Thr Tyr Tyr Cys  
85 90 95

Ala Arg Leu Asn Tyr Asn Asn Tyr Phe Asp Tyr Trp Gly Gln Gly Val  
100 105 110

Leu Val Thr Val Ser Ser  
115

<210> SEQ ID NO 141  
 <211> LENGTH: 109  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 141

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly  
1 5 10 15

Asp Arg Val Thr Ile Thr Cys Gln Ala Ser Gln Asp Ile Gly Asn Trp  
20 25 30

Leu Ser Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile  
35 40 45

Tyr Gly Ala Thr Ser Leu Ala Asp Gly Val Pro Ser Arg Phe Ser Gly  
50 55 60

Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro  
65 70 75 80

Glu Asp Phe Ala Thr Tyr Tyr Cys Leu Gln Ala Tyr Ser Ala Pro Phe  
85 90 95

Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Arg Thr  
100 105

<210> SEQ ID NO 142  
 <211> LENGTH: 109  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 142

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly  
1 5 10 15

Asp Arg Val Thr Ile Thr Cys Gln Ala Ser Gln Asp Ile Gly Asn Trp  
20 25 30

Leu Ser Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile  
35 40 45

Tyr Gly Ala Thr Ser Leu Ala Asp Gly Val Pro Ser Arg Phe Ser Gly  
50 55 60

Ser Arg Ser Gly Thr Asp Tyr Thr Leu Thr Ile Ser Ser Leu Gln Pro  
65 70 75 80

Glu Asp Phe Ala Thr Tyr Tyr Cys Leu Gln Ala Tyr Ser Ala Pro Phe  
85 90 95

Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Arg Thr  
100 105

<210> SEQ ID NO 143



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<211> LENGTH: 109
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 143
Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1      5      10      15
Asp Arg Val Thr Ile Thr Cys Gln Ala Ser Gln Asp Ile Gly Asn Trp
20     25     30
Leu Ser Trp Tyr Gln Gln Lys Pro Gly Lys Ser Pro Lys Leu Leu Ile
35     40     45
Tyr Gly Ala Thr Ser Leu Ala Asp Gly Val Pro Ser Arg Phe Ser Gly
50     55     60
Ser Arg Ser Gly Thr Gln Tyr Thr Leu Thr Ile Ser Ser Leu Gln Pro
65     70     75     80
Glu Asp Phe Ala Thr Tyr Tyr Cys Leu Gln Ala Tyr Ser Ala Pro Phe
85     90     95
Thr Phe Gly Ser Gly Thr Lys Val Glu Ile Lys Arg Thr
100    105

<210> SEQ ID NO 144
<211> LENGTH: 451
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 144
Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1      5      10      15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Ser Thr Phe Ser Asn Tyr
20     25     30
Gly Met Lys Trp Ile Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35     40     45
Ser Ser Ile Ser Arg Ser Ser Thr Tyr Ile Tyr Tyr Ala Asp Thr Val
50     55     60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65     70     75     80
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85     90     95
Ala Ala Ala Ile Ser Thr Pro Phe Tyr Trp Tyr Phe Asp Phe Trp Gly
100    105    110
Gln Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro Ser
115    120    125
Val Phe Pro Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly Thr Ala
130    135    140
Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val
145    150    155    160
Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala
165    170    175
Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val
180    185    190
Pro Ser Ser Ser Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val Asn His

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| 195 |     |     |     |     | 200 |     |     |     |     | 205 |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Lys | Pro | Ser | Asn | Thr | Lys | Val | Asp | Lys | Arg | Val | Glu | Pro | Lys | Ser | Cys |
| 210 |     |     |     |     | 215 |     |     |     |     |     | 220 |     |     |     |     |
| Asp | Lys | Thr | His | Thr | Cys | Pro | Pro | Cys | Pro | Ala | Pro | Glu | Ala | Ala | Gly |
| 225 |     |     |     |     | 230 |     |     |     |     | 235 |     |     |     |     | 240 |
| Gly | Pro | Ser | Val | Phe | Leu | Phe | Pro | Pro | Lys | Pro | Lys | Asp | Thr | Leu | Met |
|     |     |     | 245 |     |     |     |     |     | 250 |     |     |     |     | 255 |     |
| Ile | Ser | Arg | Thr | Pro | Glu | Val | Thr | Cys | Val | Val | Val | Asp | Val | Ser | His |
|     |     |     | 260 |     |     |     |     | 265 |     |     |     |     | 270 |     |     |
| Glu | Asp | Pro | Glu | Val | Lys | Phe | Asn | Trp | Tyr | Val | Asp | Gly | Val | Glu | Val |
|     |     | 275 |     |     |     |     | 280 |     |     |     |     | 285 |     |     |     |
| His | Asn | Ala | Lys | Thr | Lys | Pro | Arg | Glu | Glu | Gln | Tyr | Asn | Ser | Thr | Tyr |
|     | 290 |     |     |     |     | 295 |     |     |     |     | 300 |     |     |     |     |
| Arg | Val | Val | Ser | Val | Leu | Thr | Val | Leu | His | Gln | Asp | Trp | Leu | Asn | Gly |
| 305 |     |     |     |     | 310 |     |     |     |     | 315 |     |     |     |     | 320 |
| Lys | Glu | Tyr | Lys | Cys | Lys | Val | Ser | Asn | Lys | Ala | Leu | Pro | Ala | Pro | Ile |
|     |     |     | 325 |     |     |     |     |     | 330 |     |     |     |     | 335 |     |
| Glu | Lys | Thr | Ile | Ser | Lys | Ala | Lys | Gly | Gln | Pro | Arg | Glu | Pro | Gln | Val |
|     |     |     | 340 |     |     |     |     | 345 |     |     |     |     | 350 |     |     |
| Tyr | Thr | Leu | Pro | Pro | Ser | Arg | Glu | Glu | Met | Thr | Lys | Asn | Gln | Val | Ser |
|     |     | 355 |     |     |     |     | 360 |     |     |     |     | 365 |     |     |     |
| Leu | Thr | Cys | Leu | Val | Lys | Gly | Phe | Tyr | Pro | Ser | Asp | Ile | Ala | Val | Glu |
|     | 370 |     |     |     |     | 375 |     |     |     |     | 380 |     |     |     |     |
| Trp | Glu | Ser | Asn | Gly | Gln | Pro | Glu | Asn | Asn | Tyr | Lys | Thr | Thr | Pro | Pro |
| 385 |     |     |     |     | 390 |     |     |     |     | 395 |     |     |     |     | 400 |
| Val | Leu | Asp | Ser | Asp | Gly | Ser | Phe | Phe | Leu | Tyr | Ser | Lys | Leu | Thr | Val |
|     |     |     | 405 |     |     |     |     |     | 410 |     |     |     |     | 415 |     |
| Asp | Lys | Ser | Arg | Trp | Gln | Gln | Gly | Asn | Val | Phe | Ser | Cys | Ser | Val | Met |
|     |     |     | 420 |     |     |     |     | 425 |     |     |     |     | 430 |     |     |
| His | Glu | Ala | Leu | His | Asn | His | Tyr | Thr | Gln | Lys | Ser | Leu | Ser | Leu | Ser |
|     | 435 |     |     |     |     | 440 |     |     |     |     |     | 445 |     |     |     |
| Pro | Gly | Lys |     |     |     |     |     |     |     |     |     |     |     |     |     |
|     | 450 |     |     |     |     |     |     |     |     |     |     |     |     |     |     |

&lt;210&gt; SEQ ID NO 145

&lt;211&gt; LENGTH: 214

&lt;212&gt; TYPE: PRT

&lt;213&gt; ORGANISM: Artificial Sequence

&lt;220&gt; FEATURE:

&lt;223&gt; OTHER INFORMATION: Synthetic Construct

&lt;400&gt; SEQUENCE: 145

|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Glu | Ile | Val | Leu | Thr | Gln | Ser | Pro | Gly | Thr | Leu | Ser | Leu | Ser | Pro | Gly |
| 1   |     |     |     | 5   |     |     |     |     | 10  |     |     |     |     | 15  |     |
| Glu | Arg | Ala | Thr | Leu | Ser | Cys | Leu | Ala | Ser | Ser | Ser | Val | Ser | Tyr | Met |
|     |     |     | 20  |     |     |     |     | 25  |     |     |     |     | 30  |     |     |
| Thr | Trp | Tyr | Gln | Gln | Lys | Pro | Gly | Gln | Ser | Pro | Arg | Leu | Trp | Ile | Tyr |
|     |     | 35  |     |     |     |     | 40  |     |     |     |     | 45  |     |     |     |
| Gly | Thr | Ser | Asn | Leu | Ala | Ser | Gly | Val | Pro | Asp | Arg | Phe | Ser | Gly | Ser |
|     | 50  |     |     |     |     | 55  |     |     |     |     | 60  |     |     |     |     |
| Gly | Ser | Gly | Thr | Asp | Phe | Thr | Leu | Thr | Ile | Ser | Arg | Leu | Glu | Pro | Glu |
| 65  |     |     |     | 70  |     |     |     |     | 75  |     |     |     |     | 80  |     |
| Asp | Phe | Ala | Val | Tyr | Tyr | Cys | Leu | His | Leu | Thr | Ser | Tyr | Pro | Pro | Tyr |

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|                                                                 |     |     |
|-----------------------------------------------------------------|-----|-----|
| 85                                                              | 90  | 95  |
| Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Arg Ala Val Ala Ala |     |     |
| 100                                                             | 105 | 110 |
| Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser Gly |     |     |
| 115                                                             | 120 | 125 |
| Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu Ala |     |     |
| 130                                                             | 135 | 140 |
| Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly Asn Ser Gln |     |     |
| 145                                                             | 150 | 155 |
| Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu Ser |     |     |
| 165                                                             | 170 | 175 |
| Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys His Lys Val Tyr |     |     |
| 180                                                             | 185 | 190 |
| Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro Val Thr Lys Ser |     |     |
| 195                                                             | 200 | 205 |
| Phe Asn Arg Gly Glu Cys                                         |     |     |
| 210                                                             |     |     |

&lt;210&gt; SEQ ID NO 146

&lt;211&gt; LENGTH: 448

&lt;212&gt; TYPE: PRT

&lt;213&gt; ORGANISM: Artificial Sequence

&lt;220&gt; FEATURE:

&lt;223&gt; OTHER INFORMATION: Synthetic Construct

&lt;400&gt; SEQUENCE: 146

|                                                                 |     |     |
|-----------------------------------------------------------------|-----|-----|
| Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly |     |     |
| 1                                                               | 5   | 10  |
| Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Asn Tyr |     |     |
| 20                                                              | 25  | 30  |
| Tyr Met Tyr Trp Ile Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile |     |     |
| 35                                                              | 40  | 45  |
| Ser Ser Ile Asn Thr Asp Gly Gly Ser Thr Tyr Tyr Pro Asp Ser Val |     |     |
| 50                                                              | 55  | 60  |
| Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Val Tyr |     |     |
| 65                                                              | 70  | 75  |
| Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys |     |     |
| 85                                                              | 90  | 95  |
| Ala Lys Ser Thr Pro Asn Ile Pro Leu Ala Tyr Trp Gly Gln Gly Thr |     |     |
| 100                                                             | 105 | 110 |
| Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro Ser Val Phe Pro |     |     |
| 115                                                             | 120 | 125 |
| Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly Thr Ala Ala Leu Gly |     |     |
| 130                                                             | 135 | 140 |
| Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val Ser Trp Asn |     |     |
| 145                                                             | 150 | 155 |
| Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala Val Leu Gln |     |     |
| 165                                                             | 170 | 175 |
| Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val Pro Ser Ser |     |     |
| 180                                                             | 185 | 190 |
| Ser Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val Asn His Lys Pro Ser |     |     |
| 195                                                             | 200 | 205 |
| Asn Thr Lys Val Asp Lys Arg Val Glu Pro Lys Ser Cys Asp Lys Thr |     |     |

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| 210                                          |     |     |     |     | 215 |     |     |     |     | 220 |     |     |     |     |     |
|----------------------------------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| His                                          | Thr | Cys | Pro | Pro | Cys | Pro | Ala | Pro | Glu | Leu | Leu | Gly | Gly | Pro | Ser |
| 225                                          |     |     |     |     | 230 |     |     |     |     | 235 |     |     |     | 240 |     |
| Val                                          | Phe | Leu | Phe | Pro | Pro | Lys | Pro | Lys | Asp | Thr | Leu | Met | Ile | Ser | Arg |
|                                              |     |     |     | 245 |     |     |     |     | 250 |     |     |     |     | 255 |     |
| Thr                                          | Pro | Glu | Val | Thr | Cys | Val | Val | Val | Asp | Val | Ser | His | Glu | Asp | Pro |
|                                              |     |     | 260 |     |     |     |     |     | 265 |     |     |     |     | 270 |     |
| Glu                                          | Val | Lys | Phe | Asn | Trp | Tyr | Val | Asp | Gly | Val | Glu | Val | His | Asn | Ala |
|                                              |     |     | 275 |     |     |     |     |     | 280 |     |     |     |     | 285 |     |
| Lys                                          | Thr | Lys | Pro | Arg | Glu | Glu | Gln | Tyr | Asn | Ser | Thr | Tyr | Arg | Val | Val |
|                                              |     |     | 290 |     |     |     |     |     | 295 |     |     |     |     | 300 |     |
| Ser                                          | Val | Leu | Thr | Val | Leu | His | Gln | Asp | Trp | Leu | Asn | Gly | Lys | Glu | Tyr |
| 305                                          |     |     |     |     | 310 |     |     |     |     | 315 |     |     |     | 320 |     |
| Lys                                          | Cys | Lys | Val | Ser | Asn | Lys | Ala | Leu | Pro | Ala | Pro | Ile | Glu | Lys | Thr |
|                                              |     |     | 325 |     |     |     |     |     | 330 |     |     |     |     | 335 |     |
| Ile                                          | Ser | Lys | Ala | Lys | Gly | Gln | Pro | Arg | Glu | Pro | Gln | Val | Tyr | Thr | Leu |
|                                              |     |     | 340 |     |     |     |     |     | 345 |     |     |     |     | 350 |     |
| Pro                                          | Pro | Ser | Arg | Glu | Glu | Met | Thr | Lys | Asn | Gln | Val | Ser | Leu | Thr | Cys |
|                                              |     |     | 355 |     |     |     |     |     | 360 |     |     |     |     | 365 |     |
| Leu                                          | Val | Lys | Gly | Phe | Tyr | Pro | Ser | Asp | Ile | Ala | Val | Glu | Trp | Glu | Ser |
|                                              |     |     | 370 |     |     |     |     |     | 375 |     |     |     |     | 380 |     |
| Asn                                          | Gly | Gln | Pro | Glu | Asn | Asn | Tyr | Lys | Thr | Thr | Pro | Pro | Val | Leu | Asp |
| 385                                          |     |     |     |     | 390 |     |     |     |     | 395 |     |     |     | 400 |     |
| Ser                                          | Asp | Gly | Ser | Phe | Phe | Leu | Tyr | Ser | Lys | Leu | Thr | Val | Asp | Lys | Ser |
|                                              |     |     | 405 |     |     |     |     |     | 410 |     |     |     |     | 415 |     |
| Arg                                          | Trp | Gln | Gln | Gly | Asn | Val | Phe | Ser | Cys | Ser | Val | Met | His | Glu | Ala |
|                                              |     |     | 420 |     |     |     |     |     | 425 |     |     |     |     | 430 |     |
| Leu                                          | His | Asn | His | Tyr | Thr | Gln | Lys | Ser | Leu | Ser | Leu | Ser | Pro | Gly | Lys |
|                                              |     |     | 435 |     |     |     |     |     | 440 |     |     |     |     | 445 |     |
| <210> SEQ ID NO 147                          |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| <211> LENGTH: 213                            |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| <212> TYPE: PRT                              |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| <213> ORGANISM: Artificial Sequence          |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| <220> FEATURE:                               |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| <223> OTHER INFORMATION: Synthetic Construct |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| <400> SEQUENCE: 147                          |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| Asp                                          | Ile | Gln | Met | Thr | Gln | Ser | Pro | Ser | Ser | Leu | Ser | Ala | Ser | Val | Gly |
| 1                                            |     |     |     | 5   |     |     |     |     | 10  |     |     |     |     | 15  |     |
| Asp                                          | Arg | Val | Thr | Ile | Ser | Cys | Lys | Ala | Ser | Gln | Asn | Ile | Tyr | Lys | Tyr |
|                                              |     |     | 20  |     |     |     |     | 25  |     |     |     |     | 30  |     |     |
| Leu                                          | Asn | Trp | Phe | Gln | Gln | Lys | Pro | Gly | Glu | Ala | Pro | Lys | Leu | Leu | Ile |
|                                              |     |     | 35  |     |     |     | 40  |     |     |     |     | 45  |     |     |     |
| Tyr                                          | Tyr | Ser | Asn | Ser | Leu | Gln | Thr | Gly | Ile | Pro | Ser | Arg | Phe | Ser | Gly |
|                                              |     |     | 50  |     |     |     | 55  |     |     |     |     | 60  |     |     |     |
| Ser                                          | Gly | Ser | Gly | Thr | Asp | Phe | Thr | Leu | Thr | Ile | Ser | Ser | Leu | Gln | Pro |
| 65                                           |     |     |     | 70  |     |     |     |     | 75  |     |     |     |     | 80  |     |
| Glu                                          | Asp | Phe | Ala | Ile | Tyr | Phe | Cys | Phe | Gln | Tyr | Ser | Ser | Gly | Pro | Thr |
|                                              |     |     | 85  |     |     |     |     | 90  |     |     |     |     |     | 95  |     |
| Phe                                          | Gly | Pro | Gly | Thr | Lys | Val | Glu | Ile | Lys | Arg | Thr | Val | Ala | Ala | Pro |
|                                              |     |     | 100 |     |     |     |     | 105 |     |     |     |     | 110 |     |     |
| Ser                                          | Val | Phe | Ile | Phe | Pro | Pro | Ser | Asp | Glu | Gln | Leu | Lys | Ser | Gly | Thr |

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|                                                                 |     |     |
|-----------------------------------------------------------------|-----|-----|
| 115                                                             | 120 | 125 |
| Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu Ala Lys |     |     |
| 130                                                             | 135 | 140 |
| Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly Asn Ser Gln Glu |     |     |
| 145                                                             | 150 | 155 |
| Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu Ser Ser |     |     |
| 165                                                             | 170 | 175 |
| Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys His Lys Val Tyr Ala |     |     |
| 180                                                             | 185 | 190 |
| Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro Val Thr Lys Ser Phe |     |     |
| 195                                                             | 200 | 205 |
| Asn Arg Gly Glu Cys                                             |     |     |
| 210                                                             |     |     |

<210> SEQ ID NO 148  
 <211> LENGTH: 449  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 148

|                                                                 |     |     |
|-----------------------------------------------------------------|-----|-----|
| Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly |     |     |
| 1                                                               | 5   | 10  |
| Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser His Tyr |     |     |
| 20                                                              | 25  | 30  |
| Tyr Met Ala Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val |     |     |
| 35                                                              | 40  | 45  |
| Ala Ser Ile Thr Asn Ser Gly Gly Ser Ile Asn Tyr Arg Asp Ser Val |     |     |
| 50                                                              | 55  | 60  |
| Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Ser Thr Leu Tyr |     |     |
| 65                                                              | 70  | 75  |
| Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Thr Tyr Tyr Cys |     |     |
| 85                                                              | 90  | 95  |
| Thr Arg Glu Gly Gly Glu Asn Tyr Gly Gly Tyr Pro Pro Phe Ala Tyr |     |     |
| 100                                                             | 105 | 110 |
| Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly |     |     |
| 115                                                             | 120 | 125 |
| Pro Ser Val Phe Pro Leu Ala Pro Cys Ser Arg Ser Thr Ser Glu Ser |     |     |
| 130                                                             | 135 | 140 |
| Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val |     |     |
| 145                                                             | 150 | 155 |
| Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe |     |     |
| 165                                                             | 170 | 175 |
| Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val |     |     |
| 180                                                             | 185 | 190 |
| Thr Val Pro Ser Ser Asn Phe Gly Thr Gln Thr Tyr Thr Cys Asn Val |     |     |
| 195                                                             | 200 | 205 |
| Asp His Lys Pro Ser Asn Thr Lys Val Asp Lys Thr Val Glu Arg Lys |     |     |
| 210                                                             | 215 | 220 |
| Cys Cys Val Glu Cys Pro Pro Cys Pro Ala Pro Pro Val Ala Gly Pro |     |     |
| 225                                                             | 230 | 235 |
| Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser |     |     |

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| 245 |     |     |     | 250 |     |     |     | 255 |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Arg | Thr | Pro | Glu | Val | Thr | Cys | Val | Val | Val | Asp | Val | Ser | His | Glu | Asp |
|     |     |     | 260 |     |     |     |     |     |     | 265 |     |     |     | 270 |     |
| Pro | Glu | Val | Gln | Phe | Asn | Trp | Tyr | Val | Asp | Gly | Val | Glu | Val | His | Asn |
|     |     |     | 275 |     |     |     | 280 |     |     |     |     |     |     | 285 |     |
| Ala | Lys | Thr | Lys | Pro | Arg | Glu | Glu | Gln | Phe | Asn | Ser | Thr | Phe | Arg | Val |
|     |     |     | 290 |     |     |     | 295 |     |     |     | 300 |     |     |     |     |
| Val | Ser | Val | Leu | Thr | Val | Val | His | Gln | Asp | Trp | Leu | Asn | Gly | Lys | Glu |
|     |     |     | 305 |     |     |     | 310 |     |     |     | 315 |     |     |     | 320 |
| Tyr | Lys | Cys | Lys | Val | Ser | Asn | Lys | Gly | Leu | Pro | Ala | Pro | Ile | Glu | Lys |
|     |     |     | 325 |     |     |     |     |     |     | 330 |     |     |     | 335 |     |
| Thr | Ile | Ser | Lys | Thr | Lys | Gly | Gln | Pro | Arg | Glu | Pro | Gln | Val | Tyr | Thr |
|     |     |     | 340 |     |     |     |     |     |     | 345 |     |     |     | 350 |     |
| Leu | Pro | Pro | Ser | Arg | Glu | Glu | Met | Thr | Lys | Asn | Gln | Val | Ser | Leu | Thr |
|     |     |     | 355 |     |     |     | 360 |     |     |     |     |     |     | 365 |     |
| Cys | Leu | Val | Lys | Gly | Phe | Tyr | Pro | Ser | Asp | Ile | Ala | Val | Glu | Trp | Glu |
|     |     |     | 370 |     |     |     | 375 |     |     |     | 380 |     |     |     |     |
| Ser | Asn | Gly | Gln | Pro | Glu | Asn | Asn | Tyr | Lys | Thr | Thr | Pro | Pro | Met | Leu |
|     |     |     | 385 |     |     |     | 390 |     |     |     | 395 |     |     |     | 400 |
| Asp | Ser | Asp | Gly | Ser | Phe | Phe | Leu | Tyr | Ser | Lys | Leu | Thr | Val | Asp | Lys |
|     |     |     | 405 |     |     |     |     |     |     | 410 |     |     |     | 415 |     |
| Ser | Arg | Trp | Gln | Gln | Gly | Asn | Val | Phe | Ser | Cys | Ser | Val | Met | His | Glu |
|     |     |     | 420 |     |     |     |     |     |     | 425 |     |     |     | 430 |     |
| Ala | Leu | His | Asn | His | Tyr | Thr | Gln | Lys | Ser | Leu | Ser | Leu | Ser | Pro | Gly |
|     |     |     | 435 |     |     |     | 440 |     |     |     |     |     |     | 445 |     |

Lys

&lt;210&gt; SEQ ID NO 149

&lt;211&gt; LENGTH: 218

&lt;212&gt; TYPE: PRT

&lt;213&gt; ORGANISM: Artificial Sequence

&lt;220&gt; FEATURE:

&lt;223&gt; OTHER INFORMATION: Synthetic Construct

&lt;400&gt; SEQUENCE: 149

|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Asp | Ile | Val | Leu | Thr | Gln | Ser | Pro | Asp | Ser | Leu | Ala | Val | Ser | Leu | Gly |
| 1   |     |     | 5   |     |     |     |     |     | 10  |     |     |     |     | 15  |     |
| Glu | Arg | Ala | Thr | Ile | Asn | Cys | Arg | Ala | Asn | Gln | Gly | Val | Ser | Leu | Ser |
|     |     |     | 20  |     |     |     |     |     | 25  |     |     |     |     | 30  |     |
| Arg | Tyr | Asn | Leu | Met | His | Trp | Tyr | Gln | Gln | Lys | Pro | Gly | Gln | Lys | Pro |
|     |     | 35  |     |     |     |     | 40  |     |     |     |     |     | 45  |     |     |
| Lys | Leu | Leu | Ile | Tyr | Arg | Ser | Ser | Asn | Leu | Ala | Gln | Gly | Ile | Pro | Ala |
|     |     | 50  |     |     |     | 55  |     |     |     |     | 60  |     |     |     |     |
| Arg | Phe | Ser | Gly | Ser | Gly | Ser | Gly | Thr | Asp | Phe | Thr | Leu | Thr | Ile | Ser |
|     |     | 65  |     |     | 70  |     |     |     | 75  |     |     |     |     | 80  |     |
| Ser | Val | Gln | Ala | Asp | Asp | Ile | Ala | Val | Tyr | Tyr | Cys | Gln | Gln | Ser | Arg |
|     |     |     | 85  |     |     |     |     |     | 90  |     |     |     |     | 95  |     |
| Glu | Ser | Pro | Phe | Thr | Phe | Gly | Gln | Gly | Thr | Lys | Leu | Glu | Leu | Lys | Arg |
|     |     |     | 100 |     |     |     |     |     | 105 |     |     |     |     | 110 |     |
| Ala | Val | Ala | Ala | Pro | Ser | Val | Phe | Ile | Phe | Pro | Pro | Ser | Asp | Glu | Gln |
|     |     |     | 115 |     |     |     |     | 120 |     |     |     |     | 125 |     |     |
| Leu | Lys | Ser | Gly | Thr | Ala | Ser | Val | Val | Cys | Leu | Leu | Asn | Asn | Phe | Tyr |
|     |     |     | 130 |     |     |     | 135 |     |     |     |     | 140 |     |     |     |

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Pro Arg Glu Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser  
145 150 155 160

Gly Asn Ser Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr  
165 170 175

Tyr Ser Leu Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys  
180 185 190

His Lys Val Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro  
195 200 205

Val Thr Lys Ser Phe Asn Arg Gly Glu Cys  
210 215

<210> SEQ ID NO 150

<211> LENGTH: 448

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 150

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly  
1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Tyr Tyr  
20 25 30

Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val  
35 40 45

Ala Ser Ile Ser Arg Gly Gly Asp Asn Thr Tyr Tyr Arg Asp Ser Val  
50 55 60

Lys Gly Arg Phe Thr Thr Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr  
65 70 75 80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Thr Tyr Tyr Cys  
85 90 95

Ala Arg Leu Asn Tyr Asn Asn Tyr Phe Asp Tyr Trp Gly Gln Gly Val  
100 105 110

Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro Ser Val Phe Pro  
115 120 125

Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly Thr Ala Ala Leu Gly  
130 135 140

Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val Ser Trp Asn  
145 150 155 160

Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala Val Leu Gln  
165 170 175

Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val Pro Ser Ser  
180 185 190

Ser Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val Asn His Lys Pro Ser  
195 200 205

Asn Thr Lys Val Asp Lys Arg Val Glu Pro Lys Ser Cys Asp Lys Thr  
210 215 220

His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly Gly Pro Ser  
225 230 235 240

Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser Arg  
245 250 255

Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His Glu Asp Pro  
260 265 270

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Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val His Asn Ala  
 275 280 285  
 Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr Arg Val Val  
 290 295 300  
 Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys Glu Tyr  
 305 310 315 320  
 Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro Ile Glu Lys Thr  
 325 330 335  
 Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu  
 340 345 350  
 Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val Ser Leu Thr Cys  
 355 360 365  
 Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser  
 370 375 380  
 Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp  
 385 390 395 400  
 Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val Asp Lys Ser  
 405 410 415  
 Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His Glu Ala  
 420 425 430  
 Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly Lys  
 435 440 445

<210> SEQ ID NO 151  
 <211> LENGTH: 214  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 151

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly  
 1 5 10 15  
 Asp Arg Val Thr Ile Thr Cys Gln Ala Ser Gln Asp Ile Gly Asn Trp  
 20 25 30  
 Leu Ser Trp Tyr Gln Gln Lys Pro Gly Lys Ser Pro Lys Leu Leu Ile  
 35 40 45  
 Tyr Gly Ala Thr Ser Leu Ala Asp Gly Val Pro Ser Arg Phe Ser Gly  
 50 55 60  
 Ser Arg Ser Gly Thr Gln Tyr Thr Leu Thr Ile Ser Ser Leu Gln Pro  
 65 70 75 80  
 Glu Asp Phe Ala Thr Tyr Tyr Cys Leu Gln Ala Tyr Ser Ala Pro Phe  
 85 90 95  
 Thr Phe Gly Ser Gly Thr Lys Val Glu Ile Lys Arg Thr Val Ala Ala  
 100 105 110  
 Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser Gly  
 115 120 125  
 Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu Ala  
 130 135 140  
 Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly Asn Ser Gln  
 145 150 155 160  
 Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu Ser  
 165 170 175



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Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys His Lys Val Tyr  
 180 185 190

Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro Val Thr Lys Ser  
 195 200 205

Phe Asn Arg Gly Glu Cys  
 210

<210> SEQ ID NO 152  
 <211> LENGTH: 453  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 152

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Arg  
 1 5 10 15

Ser Leu Lys Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser His Tyr  
 20 25 30

Tyr Met Ala Trp Val Arg Gln Ala Pro Thr Lys Gly Leu Glu Trp Val  
 35 40 45

Ala Ser Ile Thr Asn Ser Gly Gly Ser Ile Asn Tyr Arg Asp Ser Val  
 50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Ser Thr Leu Tyr  
 65 70 75 80

Leu Gln Met Asp Ser Leu Arg Ser Glu Asp Thr Ala Thr Tyr Tyr Cys  
 85 90 95

Thr Arg Glu Gly Gly Glu Asn Tyr Gly Gly Tyr Pro Pro Phe Ala Tyr  
 100 105 110

Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly  
 115 120 125

Pro Ser Val Phe Pro Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly  
 130 135 140

Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val  
 145 150 155 160

Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe  
 165 170 175

Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val  
 180 185 190

Thr Val Pro Ser Ser Ser Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val  
 195 200 205

Asn His Lys Pro Ser Asn Thr Lys Val Asp Lys Arg Val Glu Pro Lys  
 210 215 220

Ser Cys Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu  
 225 230 235 240

Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr  
 245 250 255

Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val  
 260 265 270

Ser His Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val  
 275 280 285

Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser  
 290 295 300

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Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu  
 305 310 315 320  
 Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala  
 325 330 335  
 Pro Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro  
 340 345 350  
 Gln Val Tyr Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln  
 355 360 365  
 Val Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala  
 370 375 380  
 Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr  
 385 390 395 400  
 Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu  
 405 410 415  
 Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser  
 420 425 430  
 Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser  
 435 440 445  
 Leu Ser Pro Gly Lys  
 450

<210> SEQ ID NO 153  
 <211> LENGTH: 217  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 153

Asp Ile Val Leu Thr Gln Ser Pro Ala Leu Ala Val Ser Leu Gly Gln  
 1 5 10 15  
 Arg Ala Thr Ile Ser Cys Arg Ala Asn Gln Gly Val Ser Leu Ser Arg  
 20 25 30  
 Tyr Asn Leu Met His Trp Tyr Gln Gln Lys Pro Gly Gln Lys Pro Lys  
 35 40 45  
 Leu Leu Ile Tyr Arg Ser Ser Asn Leu Ala Ser Gly Ile Pro Ala Arg  
 50 55 60  
 Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Asn Pro  
 65 70 75 80  
 Val Gln Ala Asp Asp Ile Ala Thr Tyr Tyr Cys Gln Gln Ser Arg Glu  
 85 90 95  
 Ser Pro Phe Thr Phe Gly Ala Gly Thr Lys Leu Glu Leu Lys Arg Ala  
 100 105 110  
 Val Ala Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln Leu  
 115 120 125  
 Lys Ser Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr Pro  
 130 135 140  
 Arg Glu Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly  
 145 150 155 160  
 Asn Ser Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr Tyr  
 165 170 175  
 Ser Leu Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys His  
 180 185 190

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Lys Val Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro Val  
195 200 205

Thr Lys Ser Phe Asn Arg Gly Glu Cys  
210 215

<210> SEQ ID NO 154  
<211> LENGTH: 451  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 154

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Arg  
1 5 10 15

Ser Leu Lys Leu Ser Cys Leu Ala Ser Gly Ser Thr Phe Ser Asn Tyr  
20 25 30

Gly Met Lys Trp Ile Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val  
35 40 45

Ala Ser Ile Ser Arg Ser Ser Thr Tyr Ile Tyr Tyr Ala Asp Thr Val  
50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Arg Asn Thr Leu Tyr  
65 70 75 80

Leu Gln Met Thr Ser Leu Arg Ser Glu Asp Thr Ala Leu Tyr Tyr Cys  
85 90 95

Ala Ala Ala Ile Ser Thr Pro Phe Tyr Trp Tyr Phe Asp Phe Trp Gly  
100 105 110

Pro Gly Thr Val Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro Ser  
115 120 125

Val Phe Pro Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly Thr Ala  
130 135 140

Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val  
145 150 155 160

Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala  
165 170 175

Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val  
180 185 190

Pro Ser Ser Ser Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val Asn His  
195 200 205

Lys Pro Ser Asn Thr Lys Val Asp Lys Arg Val Glu Pro Lys Ser Cys  
210 215 220

Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly  
225 230 235 240

Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met  
245 250 255

Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His  
260 265 270

Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val  
275 280 285

His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr  
290 295 300

Arg Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly  
305 310 315 320

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[illegible]

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<210> SEQ ID NO 155
<211> LENGTH: 214
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct
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<400> SEQUENCE: 155

|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Glu | Ile | Val | Leu | Thr | Gln | Ser | Pro | Thr | Thr | Met | Ala | Ala | Ser | Pro | Gly |
| 1   |     |     | 5   |     |     |     |     | 10  |     |     |     |     |     | 15  |     |
| Glu | Lys | Val | Thr | Leu | Asn | Cys | Leu | Ala | Ser | Ser | Ser | Val | Ser | Tyr | Met |
|     |     |     | 20  |     |     |     | 25  |     |     |     |     | 30  |     |     |     |
| Thr | Trp | Tyr | Gln | Gln | Lys | Ser | Gly | Ala | Ser | Pro | Lys | Leu | Trp | Ile | Tyr |
|     |     | 35  |     |     |     |     | 40  |     |     |     |     | 45  |     |     |     |
| Gly | Thr | Ser | Asn | Leu | Ala | Ser | Gly | Val | Pro | Asn | Arg | Phe | Ser | Gly | Ser |
|     | 50  |     |     |     |     | 55  |     |     |     |     | 60  |     |     |     |     |
| Gly | Ser | Gly | Thr | Ser | Tyr | Ser | Leu | Ala | Ile | Ser | Ser | Met | Glu | Ala | Glu |
| 65  |     |     |     |     | 70  |     |     |     |     | 75  |     |     |     |     | 80  |
| Asp | Val | Ala | Thr | Tyr | Tyr | Cys | Leu | His | Leu | Thr | Ser | Tyr | Pro | Pro | Tyr |
|     |     |     |     | 85  |     |     |     |     | 90  |     |     |     |     | 95  |     |
| Thr | Phe | Gly | Ala | Gly | Thr | Lys | Leu | Glu | Leu | Lys | Arg | Ala | Val | Ala | Ala |
|     |     |     | 100 |     |     |     |     | 105 |     |     |     |     | 110 |     |     |
| Pro | Ser | Val | Phe | Ile | Phe | Pro | Pro | Ser | Asp | Glu | Gln | Leu | Lys | Ser | Gly |
|     |     | 115 |     |     |     |     | 120 |     |     |     |     | 125 |     |     |     |
| Thr | Ala | Ser | Val | Val | Cys | Leu | Leu | Asn | Asn | Phe | Tyr | Pro | Arg | Glu | Ala |
|     | 130 |     |     |     |     | 135 |     |     |     |     | 140 |     |     |     |     |
| Lys | Val | Gln | Trp | Lys | Val | Asp | Asn | Ala | Leu | Gln | Ser | Gly | Asn | Ser | Gln |
| 145 |     |     |     | 150 |     |     |     |     |     | 155 |     |     |     |     | 160 |
| Glu | Ser | Val | Thr | Glu | Gln | Asp | Ser | Lys | Asp | Ser | Thr | Tyr | Ser | Leu | Ser |
|     |     |     |     | 165 |     |     |     |     | 170 |     |     |     |     | 175 |     |
| Ser | Thr | Leu | Thr | Leu | Ser | Lys | Ala | Asp | Tyr | Glu | Lys | His | Lys | Val | Tyr |
|     |     |     | 180 |     |     |     |     | 185 |     |     |     |     | 190 |     |     |
| Ala | Cys | Glu | Val | Thr | His | Gln | Gly | Leu | Ser | Ser | Pro | Val | Thr | Lys | Ser |
|     |     | 195 |     |     |     |     | 200 |     |     |     |     | 205 |     |     |     |

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Phe Asn Arg Gly Glu Cys  
210

<210> SEQ ID NO 156

<211> LENGTH: 453

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 156

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly  
1 5 10 15  
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser His Tyr  
20 25 30  
Tyr Met Ala Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val  
35 40 45  
Ala Ser Ile Thr Asn Ser Gly Gly Ser Ile Asn Tyr Arg Asp Ser Val  
50 55 60  
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr  
65 70 75 80  
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys  
85 90 95  
Ala Arg Glu Gly Gly Glu Asn Tyr Gly Gly Tyr Pro Pro Phe Ala Tyr  
100 105 110  
Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly  
115 120 125  
Pro Ser Val Phe Pro Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly  
130 135 140  
Thr Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val  
145 150 155 160  
Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe  
165 170 175  
Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val  
180 185 190  
Thr Val Pro Ser Ser Ser Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val  
195 200 205  
Asn His Lys Pro Ser Asn Thr Lys Val Asp Lys Arg Val Glu Pro Lys  
210 215 220  
Ser Cys Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu  
225 230 235 240  
Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr  
245 250 255  
Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val  
260 265 270  
Ser His Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val  
275 280 285  
Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser  
290 295 300  
Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu  
305 310 315 320  
Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala  
325 330 335

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Pro Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro  
                   340                  345                  350

Gln Val Tyr Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln  
                   355                  360                  365

Val Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala  
                   370                  375                  380

Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr  
                   385                  390                  395                  400

Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu  
                   405                  410                  415

Thr Val Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser  
                   420                  425                  430

Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser  
                   435                  440                  445

Leu Ser Pro Gly Lys  
                   450

&lt;210&gt; SEQ ID NO 157

&lt;211&gt; LENGTH: 218

&lt;212&gt; TYPE: PRT

&lt;213&gt; ORGANISM: Artificial Sequence

&lt;220&gt; FEATURE:

&lt;223&gt; OTHER INFORMATION: Synthetic Construct

&lt;400&gt; SEQUENCE: 157

Asp Ile Val Met Thr Gln Ser Pro Asp Ser Leu Ala Val Ser Leu Gly  
   1                  5                  10                  15

Glu Arg Ala Thr Ile Asn Cys Arg Ala Asn Gln Gly Val Ser Leu Ser  
                   20                  25                  30

Arg Tyr Asn Leu Met His Trp Tyr Gln Gln Lys Pro Gly Gln Pro Pro  
                   35                  40                  45

Lys Leu Leu Ile Tyr Arg Ser Ser Asn Leu Ala Ser Gly Val Pro Asp  
                   50                  55                  60

Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser  
                   65                  70                  75                  80

Ser Leu Gln Ala Glu Asp Val Ala Val Tyr Tyr Cys Gln Gln Ser Arg  
                   85                  90                  95

Glu Ser Pro Phe Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Arg  
                   100                  105                  110

Ala Val Ala Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln  
                   115                  120                  125

Leu Lys Ser Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr  
                   130                  135                  140

Pro Arg Glu Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser  
                   145                  150                  155                  160

Gly Asn Ser Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr  
                   165                  170                  175

Tyr Ser Leu Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys  
                   180                  185                  190

His Lys Val Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro  
                   195                  200                  205

Val Thr Lys Ser Phe Asn Arg Gly Glu Cys  
                   210                  215

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<210> SEQ ID NO 158  
<211> LENGTH: 218  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 158

Asp Ile Val Leu Thr Gln Ser Pro Asp Ser Leu Ala Val Ser Leu Gly  
1 5 10 15  
Glu Arg Ala Thr Ile Asn Cys Arg Ala Asn Gln Gly Val Ser Leu Ser  
20 25 30  
Arg Tyr Asn Leu Met His Trp Tyr Gln Gln Lys Pro Gly Gln Lys Pro  
35 40 45  
Lys Leu Leu Ile Tyr Arg Ser Ser Asn Leu Ala Ser Gly Ile Pro Ala  
50 55 60  
Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser  
65 70 75 80  
Ser Val Gln Ala Asp Asp Ile Ala Val Tyr Tyr Cys Gln Gln Ser Arg  
85 90 95  
Glu Ser Pro Phe Thr Phe Gly Gln Gly Thr Lys Leu Glu Leu Lys Arg  
100 105 110  
Ala Val Ala Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln  
115 120 125  
Leu Lys Ser Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr  
130 135 140  
Pro Arg Glu Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser  
145 150 155 160  
Gly Asn Ser Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr  
165 170 175  
Tyr Ser Leu Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys  
180 185 190  
His Lys Val Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro  
195 200 205  
Val Thr Lys Ser Phe Asn Arg Gly Glu Cys  
210 215

<210> SEQ ID NO 159  
<211> LENGTH: 453  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 159

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly  
1 5 10 15  
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser His Tyr  
20 25 30  
Tyr Met Ala Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val  
35 40 45  
Ala Ser Ile Thr Asn Ser Gly Gly Ser Ile Asn Tyr Arg Asp Ser Val  
50 55 60  
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Ser Thr Leu Tyr

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|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| 65  |     | 70  |     | 75  |     | 80  |     |     |     |     |     |     |     |     |     |
| Leu | Gln | Met | Asn | Ser | Leu | Arg | Ala | Glu | Asp | Thr | Ala | Thr | Tyr | Tyr | Cys |
|     |     |     | 85  |     |     |     |     |     | 90  |     |     |     |     | 95  |     |
| Thr | Arg | Glu | Gly | Gly | Glu | Asn | Tyr | Gly | Gly | Tyr | Pro | Pro | Phe | Ala | Tyr |
|     |     |     | 100 |     |     |     |     | 105 |     |     |     |     | 110 |     |     |
| Trp | Gly | Gln | Gly | Thr | Leu | Val | Thr | Val | Ser | Ser | Ala | Ser | Thr | Lys | Gly |
|     |     | 115 |     |     |     |     | 120 |     |     |     |     | 125 |     |     |     |
| Pro | Ser | Val | Phe | Pro | Leu | Ala | Pro | Ser | Ser | Lys | Ser | Thr | Ser | Gly | Gly |
|     |     | 130 |     |     |     | 135 |     |     |     |     | 140 |     |     |     |     |
| Thr | Ala | Ala | Leu | Gly | Cys | Leu | Val | Lys | Asp | Tyr | Phe | Pro | Glu | Pro | Val |
| 145 |     |     |     |     | 150 |     |     |     |     | 155 |     |     |     |     | 160 |
| Thr | Val | Ser | Trp | Asn | Ser | Gly | Ala | Leu | Thr | Ser | Gly | Val | His | Thr | Phe |
|     |     |     |     | 165 |     |     |     |     | 170 |     |     |     |     | 175 |     |
| Pro | Ala | Val | Leu | Gln | Ser | Ser | Gly | Leu | Tyr | Ser | Leu | Ser | Ser | Val | Val |
|     |     |     | 180 |     |     |     |     | 185 |     |     |     |     |     | 190 |     |
| Thr | Val | Pro | Ser | Ser | Ser | Leu | Gly | Thr | Gln | Thr | Tyr | Ile | Cys | Asn | Val |
|     |     | 195 |     |     |     |     | 200 |     |     |     |     | 205 |     |     |     |
| Asn | His | Lys | Pro | Ser | Asn | Thr | Lys | Val | Asp | Lys | Arg | Val | Glu | Pro | Lys |
|     |     | 210 |     |     |     | 215 |     |     |     |     | 220 |     |     |     |     |
| Ser | Cys | Asp | Lys | Thr | His | Thr | Cys | Pro | Pro | Cys | Pro | Ala | Pro | Glu | Leu |
| 225 |     |     |     |     | 230 |     |     |     |     | 235 |     |     |     |     | 240 |
| Leu | Gly | Gly | Pro | Ser | Val | Phe | Leu | Phe | Pro | Pro | Lys | Pro | Lys | Asp | Thr |
|     |     |     |     | 245 |     |     |     |     | 250 |     |     |     |     | 255 |     |
| Leu | Met | Ile | Ser | Arg | Thr | Pro | Glu | Val | Thr | Cys | Val | Val | Val | Asp | Val |
|     |     |     | 260 |     |     |     |     | 265 |     |     |     |     |     | 270 |     |
| Ser | His | Glu | Asp | Pro | Glu | Val | Lys | Phe | Asn | Trp | Tyr | Val | Asp | Gly | Val |
|     |     | 275 |     |     |     |     | 280 |     |     |     |     | 285 |     |     |     |
| Glu | Val | His | Asn | Ala | Lys | Thr | Lys | Pro | Arg | Glu | Glu | Gln | Tyr | Asn | Ser |
|     |     | 290 |     |     |     | 295 |     |     |     |     | 300 |     |     |     |     |
| Thr | Tyr | Arg | Val | Val | Ser | Val | Leu | Thr | Val | Leu | His | Gln | Asp | Trp | Leu |
| 305 |     |     |     |     | 310 |     |     |     |     | 315 |     |     |     |     | 320 |
| Asn | Gly | Lys | Glu | Tyr | Lys | Cys | Lys | Val | Ser | Asn | Lys | Ala | Leu | Pro | Ala |
|     |     |     |     | 325 |     |     |     |     | 330 |     |     |     |     | 335 |     |
| Pro | Ile | Glu | Lys | Thr | Ile | Ser | Lys | Ala | Lys | Gly | Gln | Pro | Arg | Glu | Pro |
|     |     | 340 |     |     |     |     |     | 345 |     |     |     |     | 350 |     |     |
| Gln | Val | Tyr | Thr | Leu | Pro | Pro | Ser | Arg | Glu | Glu | Met | Thr | Lys | Asn | Gln |
|     |     | 355 |     |     |     |     | 360 |     |     |     |     | 365 |     |     |     |
| Val | Ser | Leu | Thr | Cys | Leu | Val | Lys | Gly | Phe | Tyr | Pro | Ser | Asp | Ile | Ala |
|     |     | 370 |     |     |     | 375 |     |     |     |     | 380 |     |     |     |     |
| Val | Glu | Trp | Glu | Ser | Asn | Gly | Gln | Pro | Glu | Asn | Asn | Tyr | Lys | Thr | Thr |
| 385 |     |     |     |     | 390 |     |     |     |     | 395 |     |     |     |     | 400 |
| Pro | Pro | Val | Leu | Asp | Ser | Asp | Gly | Ser | Phe | Phe | Leu | Tyr | Ser | Lys | Leu |
|     |     |     |     | 405 |     |     |     |     | 410 |     |     |     |     | 415 |     |
| Thr | Val | Asp | Lys | Ser | Arg | Trp | Gln | Gln | Gly | Asn | Val | Phe | Ser | Cys | Ser |
|     |     |     | 420 |     |     |     |     | 425 |     |     |     |     | 430 |     |     |
| Val | Met | His | Glu | Ala | Leu | His | Asn | His | Tyr | Thr | Gln | Lys | Ser | Leu | Ser |
|     |     | 435 |     |     |     |     | 440 |     |     |     |     | 445 |     |     |     |
| Leu | Ser | Pro | Gly | Lys |     |     |     |     |     |     |     |     |     |     |     |
|     |     |     | 450 |     |     |     |     |     |     |     |     |     |     |     |     |



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<211> LENGTH: 218
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 160
Asp Ile Val Leu Thr Gln Ser Pro Asp Ser Leu Ala Val Ser Leu Gly
1      5      10      15
Glu Arg Ala Thr Ile Asn Cys Arg Ala Asn Gln Gly Val Ser Leu Ser
20     25     30
Arg Tyr Asn Leu Met His Trp Tyr Gln Gln Lys Pro Gly Gln Lys Pro
35     40     45
Lys Leu Ala Ile Tyr Arg Ser Ser Asn Leu Ala Ser Gly Ile Pro Ala
50     55     60
Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser
65     70     75     80
Ser Val Gln Ala Asp Asp Ile Ala Val Tyr Tyr Cys Gln Gln Ser Arg
85     90     95
Glu Ser Pro Phe Thr Phe Gly Gln Gly Thr Lys Leu Glu Leu Lys Arg
100    105    110
Ala Val Ala Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln
115    120    125
Leu Lys Ser Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr
130    135    140
Pro Arg Glu Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser
145    150    155    160
Gly Asn Ser Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr
165    170    175
Tyr Ser Leu Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys
180    185    190
His Lys Val Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro
195    200    205
Val Thr Lys Ser Phe Asn Arg Gly Glu Cys
210    215

<210> SEQ ID NO 161
<211> LENGTH: 451
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 161
Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1      5      10      15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Ser Thr Phe Ser Asn Tyr
20     25     30
Gly Met Lys Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35     40     45
Ser Ser Ile Ser Arg Ser Ser Thr Tyr Ile Tyr Tyr Ala Asp Thr Val
50     55     60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65     70     75     80
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys

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| 85  |     |     |     |     | 90  |     |     |     |     | 95  |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Ala | Arg | Ala | Ile | Ser | Thr | Pro | Phe | Tyr | Trp | Tyr | Phe | Asp | Phe | Trp | Gly |
|     |     |     | 100 |     |     |     |     | 105 |     |     |     |     | 110 |     |     |
| Gln | Gly | Thr | Leu | Val | Thr | Val | Ser | Ser | Ala | Ser | Thr | Lys | Gly | Pro | Ser |
|     |     | 115 |     |     |     |     | 120 |     |     |     |     | 125 |     |     |     |
| Val | Phe | Pro | Leu | Ala | Pro | Ser | Ser | Lys | Ser | Thr | Ser | Gly | Gly | Thr | Ala |
|     | 130 |     |     |     |     | 135 |     |     |     |     | 140 |     |     |     |     |
| Ala | Leu | Gly | Cys | Leu | Val | Lys | Asp | Tyr | Phe | Pro | Glu | Pro | Val | Thr | Val |
| 145 |     |     |     |     | 150 |     |     |     |     | 155 |     |     |     |     | 160 |
| Ser | Trp | Asn | Ser | Gly | Ala | Leu | Thr | Ser | Gly | Val | His | Thr | Phe | Pro | Ala |
|     |     |     | 165 |     |     |     |     |     | 170 |     |     |     |     | 175 |     |
| Val | Leu | Gln | Ser | Ser | Gly | Leu | Tyr | Ser | Leu | Ser | Ser | Val | Val | Thr | Val |
|     |     | 180 |     |     |     |     |     | 185 |     |     |     |     | 190 |     |     |
| Pro | Ser | Ser | Ser | Leu | Gly | Thr | Gln | Thr | Tyr | Ile | Cys | Asn | Val | Asn | His |
|     | 195 |     |     |     |     |     | 200 |     |     |     |     | 205 |     |     |     |
| Lys | Pro | Ser | Asn | Thr | Lys | Val | Asp | Lys | Arg | Val | Glu | Pro | Lys | Ser | Cys |
|     | 210 |     |     |     | 215 |     |     |     |     |     | 220 |     |     |     |     |
| Asp | Lys | Thr | His | Thr | Cys | Pro | Pro | Cys | Pro | Ala | Pro | Glu | Leu | Leu | Gly |
| 225 |     |     |     |     | 230 |     |     |     |     | 235 |     |     |     |     | 240 |
| Gly | Pro | Ser | Val | Phe | Leu | Phe | Pro | Pro | Lys | Pro | Lys | Asp | Thr | Leu | Met |
|     |     |     | 245 |     |     |     |     |     | 250 |     |     |     |     | 255 |     |
| Ile | Ser | Arg | Thr | Pro | Glu | Val | Thr | Cys | Val | Val | Val | Asp | Val | Ser | His |
|     |     | 260 |     |     |     |     |     | 265 |     |     |     |     | 270 |     |     |
| Glu | Asp | Pro | Glu | Val | Lys | Phe | Asn | Trp | Tyr | Val | Asp | Gly | Val | Glu | Val |
|     | 275 |     |     |     |     |     | 280 |     |     |     |     | 285 |     |     |     |
| His | Asn | Ala | Lys | Thr | Lys | Pro | Arg | Glu | Glu | Gln | Tyr | Asn | Ser | Thr | Tyr |
|     | 290 |     |     |     |     | 295 |     |     |     |     | 300 |     |     |     |     |
| Arg | Val | Val | Ser | Val | Leu | Thr | Val | Leu | His | Gln | Asp | Trp | Leu | Asn | Gly |
| 305 |     |     |     |     | 310 |     |     |     |     | 315 |     |     |     |     | 320 |
| Lys | Glu | Tyr | Lys | Cys | Lys | Val | Ser | Asn | Lys | Ala | Leu | Pro | Ala | Pro | Ile |
|     |     |     | 325 |     |     |     |     |     | 330 |     |     |     |     | 335 |     |
| Glu | Lys | Thr | Ile | Ser | Lys | Ala | Lys | Gly | Gln | Pro | Arg | Glu | Pro | Gln | Val |
|     |     | 340 |     |     |     |     |     | 345 |     |     |     |     | 350 |     |     |
| Tyr | Thr | Leu | Pro | Pro | Ser | Arg | Glu | Glu | Met | Thr | Lys | Asn | Gln | Val | Ser |
|     | 355 |     |     |     |     |     | 360 |     |     |     |     | 365 |     |     |     |
| Leu | Thr | Cys | Leu | Val | Lys | Gly | Phe | Tyr | Pro | Ser | Asp | Ile | Ala | Val | Glu |
|     | 370 |     |     |     |     | 375 |     |     |     |     | 380 |     |     |     |     |
| Trp | Glu | Ser | Asn | Gly | Gln | Pro | Glu | Asn | Asn | Tyr | Lys | Thr | Thr | Pro | Pro |
| 385 |     |     |     |     | 390 |     |     |     |     | 395 |     |     |     |     | 400 |
| Val | Leu | Asp | Ser | Asp | Gly | Ser | Phe | Phe | Leu | Tyr | Ser | Lys | Leu | Thr | Val |
|     |     |     | 405 |     |     |     |     |     | 410 |     |     |     |     | 415 |     |
| Asp | Lys | Ser | Arg | Trp | Gln | Gln | Gly | Asn | Val | Phe | Ser | Cys | Ser | Val | Met |
|     |     |     | 420 |     |     |     |     | 425 |     |     |     | 430 |     |     |     |
| His | Glu | Ala | Leu | His | Asn | His | Tyr | Thr | Gln | Lys | Ser | Leu | Ser | Leu | Ser |
|     |     | 435 |     |     |     |     | 440 |     |     |     |     | 445 |     |     |     |
| Pro | Gly | Lys |     |     |     |     |     |     |     |     |     |     |     |     |     |
|     |     | 450 |     |     |     |     |     |     |     |     |     |     |     |     |     |

&lt;210&gt; SEQ ID NO 162

&lt;211&gt; LENGTH: 451

&lt;212&gt; TYPE: PRT

&lt;213&gt; ORGANISM: Artificial Sequence

-continued

&lt;220&gt; FEATURE:

&lt;223&gt; OTHER INFORMATION: Synthetic Construct

&lt;400&gt; SEQUENCE: 162

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Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1          5          10          15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Ser Thr Phe Ser Asn Tyr
20          25          30
Gly Met Lys Trp Ile Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35          40          45
Ser Ser Ile Ser Arg Ser Ser Thr Tyr Ile Tyr Tyr Ala Asp Thr Val
50          55          60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65          70          75          80
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85          90          95
Ala Ala Ala Ile Ser Thr Pro Phe Tyr Trp Tyr Phe Asp Phe Trp Gly
100         105         110
Gln Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro Ser
115         120         125
Val Phe Pro Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly Thr Ala
130         135         140
Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val
145         150         155         160
Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala
165         170         175
Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val
180         185         190
Pro Ser Ser Ser Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val Asn His
195         200         205
Lys Pro Ser Asn Thr Lys Val Asp Lys Arg Val Glu Pro Lys Ser Cys
210         215         220
Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly
225         230         235         240
Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met
245         250         255
Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His
260         265         270
Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val
275         280         285
His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr
290         295         300
Arg Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly
305         310         315         320
Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro Ile
325         330         335
Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val
340         345         350
Tyr Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val Ser
355         360         365
Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu
370         375         380

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Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro  
 385 390 395 400

Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val  
 405 410 415

Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met  
 420 425 430

His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser  
 435 440 445

Pro Gly Lys  
 450

<210> SEQ ID NO 163  
 <211> LENGTH: 451  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 163

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly  
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Ser Thr Phe Ser Asn Tyr  
 20 25 30

Gly Met Lys Trp Ile Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val  
 35 40 45

Ala Ser Ile Ser Arg Ser Ser Thr Tyr Ile Tyr Tyr Ala Asp Thr Val  
 50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr  
 65 70 75 80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys  
 85 90 95

Ala Ala Ala Ile Ser Thr Pro Phe Tyr Trp Tyr Phe Asp Phe Trp Gly  
 100 105 110

Gln Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro Ser  
 115 120 125

Val Phe Pro Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly Thr Ala  
 130 135 140

Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val  
 145 150 155 160

Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala  
 165 170 175

Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val  
 180 185 190

Pro Ser Ser Ser Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val Asn His  
 195 200 205

Lys Pro Ser Asn Thr Lys Val Asp Lys Arg Val Glu Pro Lys Ser Cys  
 210 215 220

Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly  
 225 230 235 240

Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met  
 245 250 255

Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His  
 260 265 270

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<210> SEQ ID NO 164
<211> LENGTH: 451
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 164
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|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Glu | Val | Gln | Leu | Val | Glu | Ser | Gly | Gly | Gly | Leu | Val | Gln | Pro | Gly | Gly |
| 1   |     |     | 5   |     |     |     |     | 10  |     |     |     |     |     | 15  |     |
| Ser | Leu | Arg | Leu | Ser | Cys | Ala | Ala | Ser | Gly | Ser | Thr | Phe | Ser | Asn | Tyr |
|     |     |     | 20  |     |     |     |     | 25  |     |     |     |     | 30  |     |     |
| Gly | Met | Lys | Trp | Ile | Arg | Gln | Ala | Pro | Gly | Lys | Gly | Leu | Glu | Trp | Val |
|     |     | 35  |     |     |     |     | 40  |     |     |     |     | 45  |     |     |     |
| Ala | Ser | Ile | Ser | Arg | Ser | Ser | Thr | Tyr | Ile | Tyr | Tyr | Ala | Asp | Thr | Val |
|     | 50  |     |     |     |     | 55  |     |     |     |     | 60  |     |     |     |     |
| Lys | Gly | Arg | Phe | Thr | Ile | Ser | Arg | Asp | Asn | Ser | Lys | Asn | Thr | Leu | Tyr |
| 65  |     |     |     |     | 70  |     |     |     |     | 75  |     |     |     |     | 80  |
| Leu | Gln | Met | Asn | Ser | Leu | Arg | Ala | Glu | Asp | Thr | Ala | Leu | Tyr | Tyr | Cys |
|     |     |     | 85  |     |     |     |     |     | 90  |     |     |     |     | 95  |     |
| Ala | Ala | Ala | Ile | Ser | Thr | Pro | Phe | Tyr | Trp | Tyr | Phe | Asp | Phe | Trp | Gly |
|     |     |     | 100 |     |     |     |     | 105 |     |     |     |     | 110 |     |     |
| Pro | Gly | Thr | Leu | Val | Thr | Val | Ser | Ser | Ala | Ser | Thr | Lys | Gly | Pro | Ser |
|     |     | 115 |     |     |     |     | 120 |     |     |     |     | 125 |     |     |     |
| Val | Phe | Pro | Leu | Ala | Pro | Ser | Ser | Lys | Ser | Thr | Ser | Gly | Gly | Thr | Ala |
|     | 130 |     |     |     |     | 135 |     |     |     |     | 140 |     |     |     |     |
| Ala | Leu | Gly | Cys | Leu | Val | Lys | Asp | Tyr | Phe | Pro | Glu | Pro | Val | Thr | Val |
| 145 |     |     |     |     | 150 |     |     |     |     | 155 |     |     |     |     | 160 |

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Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala  
 165 170 175  
 Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val  
 180 185 190  
 Pro Ser Ser Ser Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val Asn His  
 195 200 205  
 Lys Pro Ser Asn Thr Lys Val Asp Lys Arg Val Glu Pro Lys Ser Cys  
 210 215 220  
 Asp Lys Thr His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly  
 225 230 235 240  
 Gly Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met  
 245 250 255  
 Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His  
 260 265 270  
 Glu Asp Pro Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val  
 275 280 285  
 His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr  
 290 295 300  
 Arg Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly  
 305 310 315 320  
 Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro Ile  
 325 330 335  
 Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val  
 340 345 350  
 Tyr Thr Leu Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val Ser  
 355 360 365  
 Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu  
 370 375 380  
 Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro  
 385 390 395 400  
 Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val  
 405 410 415  
 Asp Lys Ser Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met  
 420 425 430  
 His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser  
 435 440 445  
 Pro Gly Lys  
 450

&lt;210&gt; SEQ ID NO 165

&lt;211&gt; LENGTH: 451

&lt;212&gt; TYPE: PRT

&lt;213&gt; ORGANISM: Artificial Sequence

&lt;220&gt; FEATURE:

&lt;223&gt; OTHER INFORMATION: Synthetic Construct

&lt;400&gt; SEQUENCE: 165

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly  
 1 5 10 15  
 Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Ser Thr Phe Ser Asn Tyr  
 20 25 30  
 Gly Met Lys Trp Ile Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val  
 35 40 45

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|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Ala | Ser | Ile | Ser | Arg | Ser | Ser | Thr | Tyr | Ile | Tyr | Tyr | Ala | Asp | Thr | Val |
| 50  |     |     |     |     |     | 55  |     |     |     |     | 60  |     |     |     |     |
| Lys | Gly | Arg | Phe | Thr | Ile | Ser | Arg | Asp | Asn | Ala | Lys | Asn | Thr | Leu | Tyr |
| 65  |     |     |     |     | 70  |     |     |     |     | 75  |     |     |     |     | 80  |
| Leu | Gln | Met | Asn | Ser | Leu | Arg | Ser | Glu | Asp | Thr | Ala | Leu | Tyr | Tyr | Cys |
|     |     |     | 85  |     |     |     |     |     | 90  |     |     |     |     | 95  |     |
| Ala | Ala | Ala | Ile | Ser | Thr | Pro | Phe | Tyr | Trp | Tyr | Phe | Asp | Phe | Trp | Gly |
|     |     |     | 100 |     |     |     |     | 105 |     |     |     |     | 110 |     |     |
| Pro | Gly | Thr | Leu | Val | Thr | Val | Ser | Ser | Ala | Ser | Thr | Lys | Gly | Pro | Ser |
|     |     |     | 115 |     |     |     | 120 |     |     |     |     | 125 |     |     |     |
| Val | Phe | Pro | Leu | Ala | Pro | Ser | Ser | Lys | Ser | Thr | Ser | Gly | Gly | Thr | Ala |
|     |     |     | 130 |     |     | 135 |     |     |     |     | 140 |     |     |     |     |
| Ala | Leu | Gly | Cys | Leu | Val | Lys | Asp | Tyr | Phe | Pro | Glu | Pro | Val | Thr | Val |
| 145 |     |     |     |     | 150 |     |     |     |     | 155 |     |     |     |     | 160 |
| Ser | Trp | Asn | Ser | Gly | Ala | Leu | Thr | Ser | Gly | Val | His | Thr | Phe | Pro | Ala |
|     |     |     |     | 165 |     |     |     |     | 170 |     |     |     |     | 175 |     |
| Val | Leu | Gln | Ser | Ser | Gly | Leu | Tyr | Ser | Leu | Ser | Ser | Val | Val | Thr | Val |
|     |     |     | 180 |     |     |     |     | 185 |     |     |     |     | 190 |     |     |
| Pro | Ser | Ser | Ser | Leu | Gly | Thr | Gln | Thr | Tyr | Ile | Cys | Asn | Val | Asn | His |
|     |     |     | 195 |     |     |     | 200 |     |     |     |     | 205 |     |     |     |
| Lys | Pro | Ser | Asn | Thr | Lys | Val | Asp | Lys | Arg | Val | Glu | Pro | Lys | Ser | Cys |
|     |     |     | 210 |     |     | 215 |     |     |     |     | 220 |     |     |     |     |
| Asp | Lys | Thr | His | Thr | Cys | Pro | Pro | Cys | Pro | Ala | Pro | Glu | Leu | Leu | Gly |
| 225 |     |     |     |     | 230 |     |     |     |     | 235 |     |     |     |     | 240 |
| Gly | Pro | Ser | Val | Phe | Leu | Phe | Pro | Pro | Lys | Pro | Lys | Asp | Thr | Leu | Met |
|     |     |     |     | 245 |     |     |     |     | 250 |     |     |     |     | 255 |     |
| Ile | Ser | Arg | Thr | Pro | Glu | Val | Thr | Cys | Val | Val | Val | Asp | Val | Ser | His |
|     |     |     | 260 |     |     |     |     | 265 |     |     |     |     | 270 |     |     |
| Glu | Asp | Pro | Glu | Val | Lys | Phe | Asn | Trp | Tyr | Val | Asp | Gly | Val | Glu | Val |
|     |     |     | 275 |     |     |     | 280 |     |     |     |     | 285 |     |     |     |
| His | Asn | Ala | Lys | Thr | Lys | Pro | Arg | Glu | Glu | Gln | Tyr | Asn | Ser | Thr | Tyr |
|     |     |     | 290 |     |     | 295 |     |     |     |     | 300 |     |     |     |     |
| Arg | Val | Val | Ser | Val | Leu | Thr | Val | Leu | His | Gln | Asp | Trp | Leu | Asn | Gly |
| 305 |     |     |     |     | 310 |     |     |     |     | 315 |     |     |     |     | 320 |
| Lys | Glu | Tyr | Lys | Cys | Lys | Val | Ser | Asn | Lys | Ala | Leu | Pro | Ala | Pro | Ile |
|     |     |     |     | 325 |     |     |     |     | 330 |     |     |     |     | 335 |     |
| Glu | Lys | Thr | Ile | Ser | Lys | Ala | Lys | Gly | Gln | Pro | Arg | Glu | Pro | Gln | Val |
|     |     |     | 340 |     |     |     |     | 345 |     |     |     |     | 350 |     |     |
| Tyr | Thr | Leu | Pro | Pro | Ser | Arg | Glu | Glu | Met | Thr | Lys | Asn | Gln | Val | Ser |
|     |     |     | 355 |     |     |     | 360 |     |     |     |     | 365 |     |     |     |
| Leu | Thr | Cys | Leu | Val | Lys | Gly | Phe | Tyr | Pro | Ser | Asp | Ile | Ala | Val | Glu |
|     |     |     | 370 |     |     | 375 |     |     |     |     | 380 |     |     |     |     |
| Trp | Glu | Ser | Asn | Gly | Gln | Pro | Glu | Asn | Asn | Tyr | Lys | Thr | Thr | Pro | Pro |
| 385 |     |     |     |     | 390 |     |     |     |     | 395 |     |     |     |     | 400 |
| Val | Leu | Asp | Ser | Asp | Gly | Ser | Phe | Phe | Leu | Tyr | Ser | Lys | Leu | Thr | Val |
|     |     |     |     | 405 |     |     |     |     | 410 |     |     |     |     | 415 |     |
| Asp | Lys | Ser | Arg | Trp | Gln | Gln | Gly | Asn | Val | Phe | Ser | Cys | Ser | Val | Met |
|     |     |     |     | 420 |     |     |     | 425 |     |     |     | 430 |     |     |     |
| His | Glu | Ala | Leu | His | Asn | His | Tyr | Thr | Gln | Lys | Ser | Leu | Ser | Leu | Ser |
|     |     |     | 435 |     |     |     | 440 |     |     |     |     | 445 |     |     |     |

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Pro Gly Lys  
450

<210> SEQ ID NO 166  
 <211> LENGTH: 214  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 166

Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Leu Ser Pro Gly  
 1 5 10 15  
 Glu Arg Ala Thr Leu Ser Cys Leu Ala Ser Ser Ser Val Ser Tyr Met  
 20 25 30  
 Thr Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile Tyr  
 35 40 45  
 Gly Thr Ser Asn Leu Ala Ser Gly Ile Pro Asp Arg Phe Ser Gly Ser  
 50 55 60  
 Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Arg Leu Glu Pro Glu  
 65 70 75 80  
 Asp Phe Ala Val Tyr Tyr Cys Leu His Leu Thr Ser Tyr Pro Pro Tyr  
 85 90 95  
 Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Arg Ala Val Ala Ala  
 100 105 110  
 Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser Gly  
 115 120 125  
 Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu Ala  
 130 135 140  
 Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly Asn Ser Gln  
 145 150 155 160  
 Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu Ser  
 165 170 175  
 Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys His Lys Val Tyr  
 180 185 190  
 Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro Val Thr Lys Ser  
 195 200 205  
 Phe Asn Arg Gly Glu Cys  
 210

<210> SEQ ID NO 167  
 <211> LENGTH: 214  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 167

Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Leu Ser Leu Ser Pro Gly  
 1 5 10 15  
 Glu Arg Ala Thr Leu Ser Cys Leu Ala Ser Ser Ser Val Ser Tyr Met  
 20 25 30  
 Thr Trp Tyr Gln Gln Lys Pro Gly Ala Ser Pro Arg Leu Trp Ile Tyr  
 35 40 45  
 Gly Thr Ser Asn Leu Ala Ser Gly Val Pro Asp Arg Phe Ser Gly Ser  
 50 55 60



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Gly Ser Gly Thr Asp Tyr Thr Leu Thr Ile Ser Arg Leu Glu Pro Glu  
 65 70 75 80  
 Asp Phe Ala Thr Tyr Tyr Cys Leu His Leu Thr Ser Tyr Pro Pro Tyr  
 85 90 95  
 Thr Phe Gly Ala Gly Thr Lys Val Glu Ile Lys Arg Ala Val Ala Ala  
 100 105 110  
 Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser Gly  
 115 120 125  
 Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu Ala  
 130 135 140  
 Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly Asn Ser Gln  
 145 150 155 160  
 Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu Ser  
 165 170 175  
 Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys His Lys Val Tyr  
 180 185 190  
 Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro Val Thr Lys Ser  
 195 200 205  
 Phe Asn Arg Gly Glu Cys  
 210

<210> SEQ ID NO 168  
 <211> LENGTH: 214  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 168

Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Met Ser Ala Ser Pro Gly  
 1 5 10 15  
 Glu Arg Val Thr Leu Ser Cys Leu Ala Ser Ser Ser Val Ser Tyr Met  
 20 25 30  
 Thr Trp Tyr Gln Gln Lys Pro Gly Ala Ser Pro Arg Leu Trp Ile Tyr  
 35 40 45  
 Gly Thr Ser Asn Leu Ala Ser Gly Val Pro Asp Arg Phe Ser Gly Ser  
 50 55 60  
 Gly Ser Gly Thr Asp Tyr Thr Leu Thr Ile Ser Arg Met Glu Pro Glu  
 65 70 75 80  
 Asp Phe Ala Thr Tyr Tyr Cys Leu His Leu Thr Ser Tyr Pro Pro Tyr  
 85 90 95  
 Thr Phe Gly Ala Gly Thr Lys Leu Glu Leu Lys Arg Ala Val Ala Ala  
 100 105 110  
 Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser Gly  
 115 120 125  
 Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu Ala  
 130 135 140  
 Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly Asn Ser Gln  
 145 150 155 160  
 Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu Ser  
 165 170 175  
 Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys His Lys Val Tyr  
 180 185 190

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Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro Val Thr Lys Ser  
195 200 205

Phe Asn Arg Gly Glu Cys  
210

<210> SEQ ID NO 169  
<211> LENGTH: 214  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 169

Glu Ile Val Leu Thr Gln Ser Pro Gly Thr Met Ser Ala Ser Pro Gly  
1 5 10 15

Glu Arg Val Thr Leu Asn Cys Leu Ala Ser Ser Ser Val Ser Tyr Met  
20 25 30

Thr Trp Tyr Gln Gln Lys Ser Gly Ala Ser Pro Lys Leu Trp Ile Tyr  
35 40 45

Gly Thr Ser Asn Leu Ala Ser Gly Val Pro Asn Arg Phe Ser Gly Ser  
50 55 60

Gly Ser Gly Thr Ser Tyr Thr Leu Thr Ile Ser Arg Met Glu Pro Glu  
65 70 75 80

Asp Phe Ala Thr Tyr Tyr Cys Leu His Leu Thr Ser Tyr Pro Pro Tyr  
85 90 95

Thr Phe Gly Ala Gly Thr Lys Leu Glu Leu Lys Arg Ala Val Ala Ala  
100 105 110

Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser Gly  
115 120 125

Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu Ala  
130 135 140

Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly Asn Ser Gln  
145 150 155 160

Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu Ser  
165 170 175

Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys His Lys Val Tyr  
180 185 190

Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro Val Thr Lys Ser  
195 200 205

Phe Asn Arg Gly Glu Cys  
210

<210> SEQ ID NO 170  
<211> LENGTH: 451  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 170

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly  
1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Ser Thr Phe Ser Asn Tyr  
20 25 30

Gly Met Lys Trp Ile Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val

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| 35      |         |         |         |         | 40      |         |         |         |         | 45      |         |         |         |         |         |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| Ala 50  | Ser     | Ile     | Ser     | Arg     | Ser 55  | Ser     | Thr     | Tyr     | Ile     | Tyr 60  | Tyr     | Ala     | Asp     | Thr     | Val     |
| Lys 65  | Gly     | Arg     | Phe     | Thr 70  | Ile     | Ser     | Arg     | Asp     | Asn 75  | Ser     | Lys     | Asn     | Thr     | Leu     | Tyr 80  |
| Leu     | Gln     | Met     | Asn 85  | Ser     | Leu     | Arg     | Ala     | Glu     | Asp 90  | Thr     | Ala     | Val     | Tyr     | Tyr 95  | Cys     |
| Ala     | Ala     | Ala 100 | Ile     | Ser     | Thr     | Pro     | Phe     | Tyr 105 | Trp     | Tyr     | Phe     | Asp     | Phe 110 | Trp     | Gly     |
| Gln     | Gly     | Thr 115 | Leu     | Val     | Thr     | Val     | Ser 120 | Ser     | Ala     | Ser     | Thr     | Lys 125 | Gly     | Pro     | Ser     |
| Val     | Phe 130 | Pro     | Leu     | Ala     | Pro 135 | Ser     | Ser     | Lys     | Ser     | Thr     | Ser 140 | Gly     | Gly     | Thr     | Ala     |
| Ala 145 | Leu     | Gly     | Cys     | Leu 150 | Val     | Lys     | Asp     | Tyr     | Phe     | Pro 155 | Glu     | Pro     | Val     | Thr     | Val 160 |
| Ser     | Trp     | Asn     | Ser     | Gly 165 | Ala     | Leu     | Thr     | Ser     | Gly 170 | Val     | His     | Thr     | Phe     | Pro 175 | Ala     |
| Val     | Leu     | Gln     | Ser 180 | Ser     | Gly     | Leu     | Tyr     | Ser 185 | Leu     | Ser     | Ser     | Val     | Val 190 | Thr     | Val     |
| Pro     | Ser 195 | Ser     | Ser     | Leu     | Gly     | Thr     | Gln 200 | Thr     | Tyr     | Ile     | Cys     | Asn 205 | Val     | Asn     | His     |
| Lys     | Pro 210 | Ser     | Asn     | Thr     | Lys 215 | Val     | Asp     | Lys     | Arg     | Val     | Glu 220 | Pro     | Lys     | Ser     | Cys     |
| Asp 225 | Lys     | Thr     | His     | Thr     | Cys 230 | Pro     | Pro     | Cys     | Pro     | Ala 235 | Pro     | Glu     | Ala     | Ala     | Gly 240 |
| Gly     | Pro     | Ser     | Val     | Phe 245 | Leu     | Phe     | Pro     | Pro     | Lys 250 | Pro     | Lys     | Asp     | Thr     | Leu     | Met 255 |
| Ile     | Ser     | Arg     | Thr 260 | Pro     | Glu     | Val     | Thr     | Cys 265 | Val     | Val     | Val     | Asp     | Val 270 | Ser     | His     |
| Glu     | Asp     | Pro 275 | Glu     | Val     | Lys     | Phe     | Asn 280 | Trp     | Tyr     | Val     | Asp     | Gly 285 | Val     | Glu     | Val     |
| His     | Asn 290 | Ala     | Lys     | Thr     | Lys     | Pro 295 | Arg     | Glu     | Glu     | Gln     | Tyr 300 | Asn     | Ser     | Thr     | Tyr     |
| Arg 305 | Val     | Val     | Ser     | Val     | Leu 310 | Thr     | Val     | Leu     | His     | Gln 315 | Asp     | Trp     | Leu     | Asn     | Gly 320 |
| Lys     | Glu     | Tyr     | Lys     | Cys 325 | Lys     | Val     | Ser     | Asn     | Lys 330 | Ala     | Leu     | Pro     | Ala     | Pro 335 | Ile     |
| Glu     | Lys     | Thr     | Ile 340 | Ser     | Lys     | Ala     | Lys     | Gly 345 | Gln     | Pro     | Arg     | Glu     | Pro 350 | Gln     | Val     |
| Tyr     | Thr     | Leu 355 | Pro     | Pro     | Ser     | Arg     | Glu 360 | Glu     | Met     | Thr     | Lys     | Asn 365 | Gln     | Val     | Ser     |
| Leu     | Thr 370 | Cys     | Leu     | Val     | Lys     | Gly 375 | Phe     | Tyr     | Pro     | Ser     | Asp     | Ile     | Ala     | Val     | Glu     |
| Trp 385 | Glu     | Ser     | Asn     | Gly     | Gln 390 | Pro     | Glu     | Asn     | Asn     | Tyr 395 | Lys     | Thr     | Thr     | Pro     | Pro     |
| Val     | Leu     | Asp     | Ser     | Asp 405 | Gly     | Ser     | Phe     | Phe     | Leu     | Tyr 410 | Ser     | Lys     | Leu     | Thr     | Val     |
| Asp     | Lys     | Ser     | Arg     | Trp 420 | Gln     | Gln     | Gly     | Asn 425 | Val     | Phe     | Ser     | Cys     | Ser     | Val     | Met     |
| His     | Glu     | Ala 435 | Leu     | His     | Asn     | His     | Tyr 440 | Thr     | Gln     | Lys     | Ser     | Leu     | Ser     | Leu     | Ser     |

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Pro Gly Lys  
450

<210> SEQ ID NO 171  
<211> LENGTH: 448  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 171

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly  
1 5 10 15  
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Asn Tyr  
20 25 30  
Tyr Met Tyr Trp Ile Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val  
35 40 45  
Ser Ser Ile Asn Thr Asp Gly Gly Ser Thr Tyr Tyr Ala Asp Ser Val  
50 55 60  
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Val Tyr  
65 70 75 80  
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys  
85 90 95  
Ala Lys Ser Thr Pro Asn Ile Pro Leu Ala Tyr Trp Gly Gln Gly Thr  
100 105 110  
Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro Ser Val Phe Pro  
115 120 125  
Leu Ala Pro Ser Ser Lys Ser Thr Ser Gly Gly Thr Ala Ala Leu Gly  
130 135 140  
Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val Ser Trp Asn  
145 150 155 160  
Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala Val Leu Gln  
165 170 175  
Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val Pro Ser Ser  
180 185 190  
Ser Leu Gly Thr Gln Thr Tyr Ile Cys Asn Val Asn His Lys Pro Ser  
195 200 205  
Asn Thr Lys Val Asp Lys Arg Val Glu Pro Lys Ser Cys Asp Lys Thr  
210 215 220  
His Thr Cys Pro Pro Cys Pro Ala Pro Glu Leu Leu Gly Gly Pro Ser  
225 230 235 240  
Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser Arg  
245 250 255  
Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser His Glu Asp Pro  
260 265 270  
Glu Val Lys Phe Asn Trp Tyr Val Asp Gly Val Glu Val His Asn Ala  
275 280 285  
Lys Thr Lys Pro Arg Glu Glu Gln Tyr Asn Ser Thr Tyr Arg Val Val  
290 295 300  
Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys Glu Tyr  
305 310 315 320  
Lys Cys Lys Val Ser Asn Lys Ala Leu Pro Ala Pro Ile Glu Lys Thr  
325 330 335

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Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu
      340                      345                      350

Pro Pro Ser Arg Glu Glu Met Thr Lys Asn Gln Val Ser Leu Thr Cys
      355                      360                      365

Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser
      370                      375                      380

Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp
385                      390                      395                      400

Ser Asp Gly Ser Phe Phe Leu Tyr Ser Lys Leu Thr Val Asp Lys Ser
      405                      410                      415

Arg Trp Gln Gln Gly Asn Val Phe Ser Cys Ser Val Met His Glu Ala
      420                      425                      430

Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly Lys
      435                      440                      445

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<210> SEQ ID NO 172
<211> LENGTH: 213
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

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<400> SEQUENCE: 172

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Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1      5      10      15

Asp Arg Val Thr Ile Thr Cys Lys Ala Ser Gln Asn Ile Tyr Lys Tyr
20     25     30

Leu Asn Trp Phe Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile
35     40     45

Tyr Tyr Ser Asn Ser Leu Gln Thr Gly Val Pro Ser Arg Phe Ser Gly
50     55     60

Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
65     70     75     80

Glu Asp Phe Ala Thr Tyr Tyr Cys Phe Gln Tyr Ser Ser Gly Pro Thr
85     90     95

Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Arg Thr Val Ala Ala Pro
100    105    110

Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser Gly Thr
115    120    125

Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu Ala Lys
130    135    140

Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly Asn Ser Gln Glu
145    150    155    160

Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu Ser Ser
165    170    175

Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys His Lys Val Tyr Ala
180    185    190

Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro Val Thr Lys Ser Phe
195    200    205

Asn Arg Gly Glu Cys
210

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<210> SEQ ID NO 173

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<211> LENGTH: 213
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 173
Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1      5      10      15
Asp Arg Val Thr Ile Thr Cys Lys Ala Ser Gln Asn Ile Tyr Lys Tyr
20     25     30
Leu Asn Trp Phe Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile
35     40     45
Tyr Tyr Ser Asn Ser Leu Gln Thr Gly Ile Pro Ser Arg Phe Ser Gly
50     55     60
Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
65     70     75     80
Glu Asp Phe Ala Thr Tyr Tyr Cys Phe Gln Tyr Ser Ser Gly Pro Thr
85     90     95
Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Arg Thr Val Ala Ala Pro
100    105    110
Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser Gly Thr
115    120    125
Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu Ala Lys
130    135    140
Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly Asn Ser Gln Glu
145    150    155    160
Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu Ser Ser
165    170    175
Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys His Lys Val Tyr Ala
180    185    190
Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro Val Thr Lys Ser Phe
195    200    205
Asn Arg Gly Glu Cys
210

<210> SEQ ID NO 174
<211> LENGTH: 448
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 174
Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1      5      10      15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Tyr Tyr
20     25     30
Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35     40     45
Ala Ser Ile Ser Arg Gly Gly Asp Asn Thr Tyr Tyr Arg Asp Ser Val
50     55     60
Lys Gly Arg Phe Thr Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65     70     75     80
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys

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| 85  |     |     |     | 90  |     |     |     | 95  |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Ala | Arg | Leu | Asn | Tyr | Asn | Asn | Tyr | Phe | Asp | Tyr | Trp | Gly | Gln | Gly | Thr |
|     |     |     | 100 |     |     |     |     |     |     |     | 105 |     |     |     | 110 |
| Leu | Val | Thr | Val | Ser | Ser | Ala | Ser | Thr | Lys | Gly | Pro | Ser | Val | Phe | Pro |
|     |     |     | 115 |     |     |     |     |     |     |     | 120 |     |     |     | 125 |
| Leu | Ala | Pro | Ser | Ser | Lys | Ser | Thr | Ser | Gly | Gly | Thr | Ala | Ala | Leu | Gly |
|     |     |     | 130 |     |     |     |     |     |     |     | 135 |     |     |     | 140 |
| Cys | Leu | Val | Lys | Asp | Tyr | Phe | Pro | Glu | Pro | Val | Thr | Val | Ser | Trp | Asn |
|     |     |     | 145 |     |     |     |     |     |     |     | 150 |     |     |     | 155 |
| Ser | Gly | Ala | Leu | Thr | Ser | Gly | Val | His | Thr | Phe | Pro | Ala | Val | Leu | Gln |
|     |     |     | 165 |     |     |     |     |     |     |     | 170 |     |     |     | 175 |
| Ser | Ser | Gly | Leu | Tyr | Ser | Leu | Ser | Ser | Val | Val | Thr | Val | Pro | Ser | Ser |
|     |     |     | 180 |     |     |     |     |     |     |     | 185 |     |     |     | 190 |
| Ser | Leu | Gly | Thr | Gln | Thr | Tyr | Ile | Cys | Asn | Val | Asn | His | Lys | Pro | Ser |
|     |     |     | 195 |     |     |     |     |     |     |     | 200 |     |     |     | 205 |
| Asn | Thr | Lys | Val | Asp | Lys | Arg | Val | Glu | Pro | Lys | Ser | Cys | Asp | Lys | Thr |
|     |     |     | 210 |     |     |     |     |     |     |     | 215 |     |     |     | 220 |
| His | Thr | Cys | Pro | Pro | Cys | Pro | Ala | Pro | Glu | Leu | Leu | Gly | Gly | Pro | Ser |
|     |     |     | 225 |     |     |     |     |     |     |     | 230 |     |     |     | 235 |
| Val | Phe | Leu | Phe | Pro | Pro | Lys | Pro | Lys | Asp | Thr | Leu | Met | Ile | Ser | Arg |
|     |     |     | 245 |     |     |     |     |     |     |     | 250 |     |     |     | 255 |
| Thr | Pro | Glu | Val | Thr | Cys | Val | Val | Val | Asp | Val | Ser | His | Glu | Asp | Pro |
|     |     |     | 260 |     |     |     |     |     |     |     | 265 |     |     |     | 270 |
| Glu | Val | Lys | Phe | Asn | Trp | Tyr | Val | Asp | Gly | Val | Glu | Val | His | Asn | Ala |
|     |     |     | 275 |     |     |     |     |     |     |     | 280 |     |     |     | 285 |
| Lys | Thr | Lys | Pro | Arg | Glu | Glu | Gln | Tyr | Asn | Ser | Thr | Tyr | Arg | Val | Val |
|     |     |     | 290 |     |     |     |     |     |     |     | 295 |     |     |     | 300 |
| Ser | Val | Leu | Thr | Val | Leu | His | Gln | Asp | Trp | Leu | Asn | Gly | Lys | Glu | Tyr |
|     |     |     | 305 |     |     |     |     |     |     |     | 310 |     |     |     | 315 |
| Lys | Cys | Lys | Val | Ser | Asn | Lys | Ala | Leu | Pro | Ala | Pro | Ile | Glu | Lys | Thr |
|     |     |     | 325 |     |     |     |     |     |     |     | 330 |     |     |     | 335 |
| Ile | Ser | Lys | Ala | Lys | Gly | Gln | Pro | Arg | Glu | Pro | Gln | Val | Tyr | Thr | Leu |
|     |     |     | 340 |     |     |     |     |     |     |     | 345 |     |     |     | 350 |
| Pro | Pro | Ser | Arg | Glu | Glu | Met | Thr | Lys | Asn | Gln | Val | Ser | Leu | Thr | Cys |
|     |     |     | 355 |     |     |     |     |     |     |     | 360 |     |     |     | 365 |
| Leu | Val | Lys | Gly | Phe | Tyr | Pro | Ser | Asp | Ile | Ala | Val | Glu | Trp | Glu | Ser |
|     |     |     | 370 |     |     |     |     |     |     |     | 375 |     |     |     | 380 |
| Asn | Gly | Gln | Pro | Glu | Asn | Asn | Tyr | Lys | Thr | Thr | Pro | Pro | Val | Leu | Asp |
|     |     |     | 385 |     |     |     |     |     |     |     | 390 |     |     |     | 395 |
| Ser | Asp | Gly | Ser | Phe | Phe | Leu | Tyr | Ser | Lys | Leu | Thr | Val | Asp | Lys | Ser |
|     |     |     | 405 |     |     |     |     |     |     |     | 410 |     |     |     | 415 |
| Arg | Trp | Gln | Gln | Gly | Asn | Val | Phe | Ser | Cys | Ser | Val | Met | His | Glu | Ala |
|     |     |     | 420 |     |     |     |     |     |     |     | 425 |     |     |     | 430 |
| Leu | His | Asn | His | Tyr | Thr | Gln | Lys | Ser | Leu | Ser | Leu | Ser | Pro | Gly | Lys |
|     |     |     | 435 |     |     |     |     |     |     |     | 440 |     |     |     | 445 |

&lt;210&gt; SEQ ID NO 175

&lt;211&gt; LENGTH: 214

&lt;212&gt; TYPE: PRT

&lt;213&gt; ORGANISM: Artificial Sequence

&lt;220&gt; FEATURE:

&lt;223&gt; OTHER INFORMATION: Synthetic Construct

-continued

&lt;400&gt; SEQUENCE: 175

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Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1           5           10           15
Asp Arg Val Thr Ile Thr Cys Gln Ala Ser Gln Asp Ile Gly Asn Trp
20           25           30
Leu Ser Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile
35           40           45
Tyr Gly Ala Thr Ser Leu Ala Asp Gly Val Pro Ser Arg Phe Ser Gly
50           55           60
Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
65           70           75           80
Glu Asp Phe Ala Thr Tyr Tyr Cys Leu Gln Ala Tyr Ser Ala Pro Phe
85           90           95
Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Arg Thr Val Ala Ala
100          105          110
Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser Gly
115          120          125
Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu Ala
130          135          140
Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly Asn Ser Gln
145          150          155          160
Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu Ser
165          170          175
Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys His Lys Val Tyr
180          185          190
Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro Val Thr Lys Ser
195          200          205
Phe Asn Arg Gly Glu Cys
210

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&lt;210&gt; SEQ ID NO 176

&lt;211&gt; LENGTH: 214

&lt;212&gt; TYPE: PRT

&lt;213&gt; ORGANISM: Artificial Sequence

&lt;220&gt; FEATURE:

&lt;223&gt; OTHER INFORMATION: Synthetic Construct

&lt;400&gt; SEQUENCE: 176

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Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1           5           10           15
Asp Arg Val Thr Ile Thr Cys Gln Ala Ser Gln Asp Ile Gly Asn Trp
20           25           30
Leu Ser Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile
35           40           45
Tyr Gly Ala Thr Ser Leu Ala Asp Gly Val Pro Ser Arg Phe Ser Gly
50           55           60
Ser Arg Ser Gly Thr Asp Tyr Thr Leu Thr Ile Ser Ser Leu Gln Pro
65           70           75           80
Glu Asp Phe Ala Thr Tyr Tyr Cys Leu Gln Ala Tyr Ser Ala Pro Phe
85           90           95
Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Arg Thr Val Ala Ala
100          105          110
Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser Gly

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<210> SEQ ID NO 181
<211> LENGTH: 7
<212> TYPE: PRT
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<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 181

Ala Ala Thr Ser Leu Ala Asp  
1 5

<210> SEQ ID NO 182  
<211> LENGTH: 9  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 182

Gln Gln Leu Tyr Ser Thr Pro Trp Thr  
1 5

<210> SEQ ID NO 183  
<211> LENGTH: 10  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 183

Gly Phe Asn Ile Lys Asp Ser Leu Met His  
1 5 10

<210> SEQ ID NO 184  
<211> LENGTH: 16  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 184

Ile Asp Pro Glu Asp Gly Glu Thr Lys Tyr Ala Pro Asn Phe Gln Asp  
1 5 10 15

<210> SEQ ID NO 185  
<211> LENGTH: 10  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 185

Tyr Thr Ser Asp Tyr Tyr Thr Met Asp Tyr  
1 5 10

<210> SEQ ID NO 186  
<211> LENGTH: 11  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 186

Leu Ala Ser Gln Thr Ile Gly Thr Trp Leu Ala  
1 5 10

<210> SEQ ID NO 187

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-continued

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<211> LENGTH: 7  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 187

Ala Ala Thr Ser Leu Ala Asp  
1 5

<210> SEQ ID NO 188  
<211> LENGTH: 9  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 188

Gln Gln Leu Tyr Ser Thr Pro Trp Thr  
1 5

<210> SEQ ID NO 189  
<211> LENGTH: 10  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 189

Gly Phe Asn Ile Lys Asp Ser Leu Met His  
1 5 10

<210> SEQ ID NO 190  
<211> LENGTH: 16  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 190

Ile Asp Pro Glu Asp Gly Glu Thr Lys Tyr Ala Pro Asn Phe Gln Ser  
1 5 10 15

<210> SEQ ID NO 191  
<211> LENGTH: 10  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 191

Tyr Thr Ser Pro Tyr Tyr Thr Met Asp Tyr  
1 5 10

<210> SEQ ID NO 192  
<211> LENGTH: 11  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 192

Leu Ala Ser Gln Thr Ile Gly Thr Trp Leu Ala  
1 5 10

-continued

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<210> SEQ ID NO 193  
<211> LENGTH: 7  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 193

Ala Ala Thr Ser Leu Ala Asp  
1 5

<210> SEQ ID NO 194  
<211> LENGTH: 9  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 194

Gln Gln Val Tyr Ser Thr Pro Trp Thr  
1 5

<210> SEQ ID NO 195  
<211> LENGTH: 10  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 195

Gly Phe Asn Ile Lys Asp Ser Leu Met His  
1 5 10

<210> SEQ ID NO 196  
<211> LENGTH: 16  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 196

Ile Asp Pro Glu Asp Gly Glu Thr Arg Tyr Ala Pro Asn Phe Gln Asp  
1 5 10 15

<210> SEQ ID NO 197  
<211> LENGTH: 10  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 197

Tyr Thr Ser Arg Tyr Tyr Thr Met Glu Tyr  
1 5 10

<210> SEQ ID NO 198  
<211> LENGTH: 11  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 198

Leu Ala Ser Gln Thr Ile Gly Thr Trp Leu Ala

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|   |   |    |
|---|---|----|
| 1 | 5 | 10 |
|---|---|----|

<210> SEQ ID NO 199  
 <211> LENGTH: 7  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct  
  
 <400> SEQUENCE: 199  
  
 Ala Ala Thr Ser Leu Ala Asp  
 1 5

<210> SEQ ID NO 200  
 <211> LENGTH: 9  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct  
  
 <400> SEQUENCE: 200  
  
 Gln Gln Leu Tyr Trp Thr Pro Trp Thr  
 1 5

<210> SEQ ID NO 201  
 <211> LENGTH: 119  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct  
  
 <400> SEQUENCE: 201  
  
 Glu Val Gln Leu Gln Gln Ser Gly Ala Glu Leu Val Arg Pro Gly Ala  
 1 5 10 15  
  
 Ser Val Arg Leu Ser Cys Thr Ala Ser Gly Phe Asn Ile Lys Asp Ser  
 20 25 30  
  
 Leu Met His Trp Val Lys Gln Arg Pro Glu Gln Gly Leu Glu Trp Ile  
 35 40 45  
  
 Gly Trp Ile Asp Pro Glu Asp Gly Glu Thr Lys Tyr Ala Pro Asn Phe  
 50 55 60  
  
 Gln Asp Lys Ala Thr Ile Thr Ala Val Thr Ser Ser Asn Thr Ala Tyr  
 65 70 75 80  
  
 Leu Gln Leu Ser Ser Leu Thr Ser Glu Asp Ser Ala Ile Tyr Tyr Cys  
 85 90 95  
  
 Ala Arg Tyr Thr Ser Asp Tyr Tyr Thr Met Asp Tyr Trp Gly Gln Gly  
 100 105 110  
  
 Thr Ser Val Thr Val Ser Ser  
 115

<210> SEQ ID NO 202  
 <211> LENGTH: 107  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct  
  
 <400> SEQUENCE: 202  
  
 Asp Ile Gln Met Thr Gln Ser Pro Ala Ser Gln Ser Ala Ser Leu Gly  
 1 5 10 15  
  
 Glu Ser Val Thr Phe Thr Cys Leu Ala Ser Gln Thr Ile Gly Thr Trp

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|                             |                     |                 |
|-----------------------------|---------------------|-----------------|
| 20                          | 25                  | 30              |
| Leu Ala Trp Tyr Gln Gln Lys | Pro Gly Lys Ser Pro | Gln Leu Leu Ile |
| 35                          | 40                  | 45              |
| Tyr Ala Ala Thr Ser Leu Ala | Asp Gly Val Pro Ser | Arg Phe Ser Gly |
| 50                          | 55                  | 60              |
| Ser Gly Ser Gly Thr Lys Phe | Ser Phe Lys Ile Ser | Ser Leu Gln Ala |
| 65                          | 70                  | 75              |
| Glu Asp Phe Ala Ser Tyr Tyr | Cys Gln Gln Leu Tyr | Ser Thr Pro Trp |
| 85                          | 90                  | 95              |
| Thr Phe Gly Gly Gly Thr Lys | Leu Glu Ile Lys     |                 |
| 100                         | 105                 |                 |

<210> SEQ ID NO 203  
 <211> LENGTH: 119  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 203

|                             |                         |                 |
|-----------------------------|-------------------------|-----------------|
| Glu Val Gln Leu Gln Gln Ser | Gly Ala Glu Leu Val Arg | Pro Gly Ala     |
| 1                           | 5                       | 10              |
| Ser Val Arg Leu Ser Cys Thr | Ala Ser Gly Phe Asn Ile | Lys Asp Ser     |
| 20                          | 25                      | 30              |
| Leu Met His Trp Val Lys Gln | Arg Pro Glu Gln Gly     | Leu Glu Trp Ile |
| 35                          | 40                      | 45              |
| Gly Trp Ile Asp Pro Glu Asp | Gly Glu Thr Lys Tyr     | Ala Pro Asn Phe |
| 50                          | 55                      | 60              |
| Gln Asp Lys Ala Thr Ile Thr | Ala Val Thr Ser Ser     | Asn Thr Ala Tyr |
| 65                          | 70                      | 75              |
| Leu Gln Leu Ser Ser Leu Thr | Ser Glu Asp Ser Ala Ile | Tyr Tyr Cys     |
| 85                          | 90                      | 95              |
| Ala Arg Tyr Thr Ser Asp Tyr | Tyr Thr Met Asp Tyr Trp | Gly Gln Gly     |
| 100                         | 105                     | 110             |
| Thr Ser Val Thr Val Ser Ser |                         |                 |
| 115                         |                         |                 |

<210> SEQ ID NO 204  
 <211> LENGTH: 107  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 204

|                             |                     |                 |
|-----------------------------|---------------------|-----------------|
| Asp Ile Gln Met Thr Gln Ser | Pro Ala Ser Gln Ser | Ala Ser Leu Gly |
| 1                           | 5                   | 10              |
| Glu Ser Val Thr Phe Thr Cys | Leu Ala Ser Gln Thr | Ile Gly Thr Trp |
| 20                          | 25                  | 30              |
| Leu Ala Trp Tyr Gln Gln Lys | Pro Gly Lys Ser Pro | Gln Leu Leu Ile |
| 35                          | 40                  | 45              |
| Tyr Ala Ala Thr Ser Leu Ala | Asp Gly Val Pro Ser | Arg Phe Ser Gly |
| 50                          | 55                  | 60              |
| Ser Gly Ser Gly Thr Lys Phe | Ser Phe Lys Ile Ser | Ser Leu Gln Ala |
| 65                          | 70                  | 75              |
|                             |                     | 80              |

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Glu Asp Phe Ala Ser Tyr Tyr Cys Gln Gln Leu Tyr Ser Thr Pro Trp  
85 90 95

Thr Phe Gly Gly Gly Thr Lys Leu Glu Ile Lys  
100 105

<210> SEQ ID NO 205  
<211> LENGTH: 119  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 205

Glu Val Gln Leu Gln Gln Ser Gly Ala Glu Leu Val Arg Pro Gly Ala  
1 5 10 15

Ser Val Arg Leu Ser Cys Thr Ala Ser Gly Phe Asn Ile Lys Asp Ser  
20 25 30

Leu Met His Trp Val Lys Gln Arg Pro Glu Gln Gly Leu Glu Trp Ile  
35 40 45

Gly Trp Ile Asp Pro Glu Asp Gly Glu Thr Lys Tyr Ala Pro Asn Phe  
50 55 60

Gln Ser Lys Ala Thr Ile Thr Ala Val Thr Ser Ser Asn Thr Ala Tyr  
65 70 75 80

Leu Gln Leu Ser Ser Leu Thr Ser Glu Asp Ser Ala Ile Tyr Tyr Cys  
85 90 95

Ala Arg Tyr Thr Ser Pro Tyr Tyr Thr Met Asp Tyr Trp Gly Gln Gly  
100 105 110

Thr Ser Val Thr Val Ser Ser  
115

<210> SEQ ID NO 206  
<211> LENGTH: 107  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 206

Asp Ile Gln Met Thr Gln Ser Pro Ala Ser Gln Ser Ala Ser Leu Gly  
1 5 10 15

Glu Ser Val Thr Phe Thr Cys Leu Ala Ser Gln Thr Ile Gly Thr Trp  
20 25 30

Leu Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ser Pro Gln Leu Leu Ile  
35 40 45

Tyr Ala Ala Thr Ser Leu Ala Asp Gly Val Pro Ser Arg Phe Ser Gly  
50 55 60

Ser Gly Ser Gly Thr Lys Phe Ser Phe Lys Ile Ser Ser Leu Gln Ala  
65 70 75 80

Glu Asp Phe Ala Ser Tyr Tyr Cys Gln Gln Val Tyr Ser Thr Pro Trp  
85 90 95

Thr Phe Gly Gly Gly Thr Lys Leu Glu Ile Lys  
100 105

<210> SEQ ID NO 207  
<211> LENGTH: 119  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence

-continued

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 207

Glu Val Gln Leu Gln Gln Ser Gly Ala Glu Leu Val Arg Pro Gly Ala  
1 5 10 15  
Ser Val Arg Leu Ser Cys Thr Ala Ser Gly Phe Asn Ile Lys Asp Ser  
20 25 30  
Leu Met His Trp Val Lys Gln Arg Pro Glu Gln Gly Leu Glu Trp Ile  
35 40 45  
Gly Trp Ile Asp Pro Glu Asp Gly Glu Thr Arg Tyr Ala Pro Asn Phe  
50 55 60  
Gln Asp Lys Ala Thr Ile Thr Ala Val Thr Ser Ser Asn Thr Ala Tyr  
65 70 75 80  
Leu Gln Leu Ser Ser Leu Thr Ser Glu Asp Ser Ala Ile Tyr Tyr Cys  
85 90 95  
Ala Arg Tyr Thr Ser Arg Tyr Tyr Thr Met Glu Tyr Trp Gly Gln Gly  
100 105 110  
Thr Ser Val Thr Val Ser Ser  
115

<210> SEQ ID NO 208

<211> LENGTH: 107

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 208

Asp Ile Gln Met Thr Gln Ser Pro Ala Ser Gln Ser Ala Ser Leu Gly  
1 5 10 15  
Glu Ser Val Thr Phe Thr Cys Leu Ala Ser Gln Thr Ile Gly Thr Trp  
20 25 30  
Leu Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ser Pro Gln Leu Leu Ile  
35 40 45  
Tyr Ala Ala Thr Ser Leu Ala Asp Gly Val Pro Ser Arg Phe Ser Gly  
50 55 60  
Ser Gly Ser Gly Thr Lys Phe Ser Phe Lys Ile Ser Ser Leu Gln Ala  
65 70 75 80  
Glu Asp Phe Ala Ser Tyr Tyr Cys Gln Gln Leu Tyr Trp Thr Pro Trp  
85 90 95  
Thr Phe Gly Gly Gly Thr Lys Leu Glu Ile Lys  
100 105

<210> SEQ ID NO 209

<211> LENGTH: 443

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 209

Glu Val Gln Leu Gln Gln Ser Gly Ala Glu Leu Val Arg Pro Gly Ala  
1 5 10 15  
Ser Val Arg Leu Ser Cys Thr Ala Ser Gly Phe Asn Ile Lys Asp Ser  
20 25 30



|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |  |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|--|
| Leu | Met | His | Trp | Val | Lys | Gln | Arg | Pro | Glu | Gln | Gly | Leu | Glu | Trp | Ile |  |
| 35  |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |  |
| Gly | Trp | Ile | Asp | Pro | Glu | Asp | Gly | Glu | Thr | Lys | Tyr | Ala | Pro | Asn | Phe |  |
| 50  |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |  |
| Gln | Asp | Lys | Ala | Thr | Ile | Thr | Ala | Val | Thr | Ser | Ser | Asn | Thr | Ala | Tyr |  |
| 65  |     | 70  |     |     |     |     | 75  |     |     |     |     | 80  |     |     |     |  |
| Leu | Gln | Leu | Ser | Ser | Leu | Thr | Ser | Glu | Asp | Ser | Ala | Ile | Tyr | Tyr | Cys |  |
| 85  |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |  |
| Ala | Arg | Tyr | Thr | Ser | Asp | Tyr | Tyr | Thr | Met | Asp | Tyr | Trp | Gly | Gln | Gly |  |
| 100 |     |     |     | 105 |     |     |     |     | 110 |     |     |     |     |     |     |  |
| Thr | Ser | Val | Thr | Val | Ser | Ser | Ala | Lys | Thr | Thr | Pro | Pro | Ser | Val | Tyr |  |
| 115 |     |     |     | 120 |     |     |     |     | 125 |     |     |     |     |     |     |  |
| Pro | Leu | Ala | Pro | Gly | Phe | Ala | Ala | Gln | Thr | Asn | Ser | Met | Val | Thr | Leu |  |
| 130 |     | 135 |     |     |     |     | 140 |     |     |     |     |     |     |     |     |  |
| Gly | Cys | Leu | Val | Lys | Gly | Tyr | Phe | Pro | Glu | Pro | Val | Thr | Val | Thr | Trp |  |
| 145 |     | 150 |     |     |     |     | 155 |     |     |     |     | 160 |     |     |     |  |
| Asn | Ser | Gly | Ser | Leu | Ser | Ser | Gly | Val | His | Thr | Phe | Pro | Ala | Val | Leu |  |
| 165 |     |     |     | 170 |     |     |     |     | 175 |     |     |     |     |     |     |  |
| Gln | Ser | Asp | Leu | Tyr | Thr | Leu | Ser | Ser | Ser | Val | Thr | Val | Pro | Ser | Ser |  |
| 180 |     |     |     | 185 |     |     |     |     | 190 |     |     |     |     |     |     |  |
| Thr | Trp | Pro | Ser | Gln | Thr | Val | Thr | Cys | Asn | Val | Ala | His | Pro | Ala | Ser |  |
| 195 |     | 200 |     |     |     |     | 205 |     |     |     |     |     |     |     |     |  |
| Ser | Thr | Lys | Val | Asp | Lys | Lys | Ile | Val | Pro | Arg | Asp | Cys | Gly | Cys | Lys |  |
| 210 |     | 215 |     |     |     |     | 220 |     |     |     |     |     |     |     |     |  |
| Pro | Cys | Ile | Cys | Thr | Val | Pro | Glu | Val | Ser | Ser | Val | Phe | Ile | Phe | Pro |  |
| 225 |     | 230 |     |     |     |     | 235 |     |     |     |     | 240 |     |     |     |  |
| Pro | Lys | Pro | Lys | Asp | Val | Leu | Thr | Ile | Thr | Leu | Thr | Pro | Lys | Val | Thr |  |
| 245 |     |     |     | 250 |     |     |     |     | 255 |     |     |     |     |     |     |  |
| Cys | Val | Val | Val | Asp | Ile | Ser | Lys | Asp | Asp | Pro | Glu | Val | Gln | Phe | Ser |  |
| 260 |     |     |     | 265 |     |     |     |     | 270 |     |     |     |     |     |     |  |
| Trp | Phe | Val | Asp | Asp | Val | Glu | Val | His | Thr | Ala | Gln | Thr | Lys | Pro | Arg |  |
| 275 |     | 280 |     |     |     |     | 285 |     |     |     |     |     |     |     |     |  |
| Glu | Glu | Gln | Ile | Asn | Ser | Thr | Phe | Arg | Ser | Val | Ser | Glu | Leu | Pro | Ile |  |
| 290 |     | 295 |     |     |     |     | 300 |     |     |     |     |     |     |     |     |  |
| Met | His | Gln | Asp | Trp | Leu | Asn | Gly | Lys | Glu | Phe | Lys | Cys | Arg | Val | Asn |  |
| 305 |     | 310 |     |     |     |     | 315 |     |     |     |     | 320 |     |     |     |  |
| Ser | Ala | Ala | Phe | Pro | Ala | Pro | Ile | Glu | Lys | Thr | Ile | Ser | Lys | Thr | Lys |  |
| 325 |     |     |     | 330 |     |     |     |     | 335 |     |     |     |     |     |     |  |
| Gly | Arg | Pro | Lys | Ala | Pro | Gln | Val | Tyr | Thr | Ile | Pro | Pro | Pro | Lys | Glu |  |
| 340 |     |     |     | 345 |     |     |     |     | 350 |     |     |     |     |     |     |  |
| Gln | Met | Ala | Lys | Asp | Lys | Val | Ser | Leu | Thr | Cys | Met | Ile | Thr | Asn | Phe |  |
| 355 |     | 360 |     |     |     |     | 365 |     |     |     |     |     |     |     |     |  |
| Phe | Pro | Glu | Asp | Ile | Thr | Val | Glu | Trp | Gln | Trp | Asn | Gly | Gln | Pro | Ala |  |
| 370 |     | 375 |     |     |     |     | 380 |     |     |     |     |     |     |     |     |  |
| Glu | Asn | Tyr | Lys | Asn | Thr | Gln | Pro | Ile | Met | Asp | Thr | Asp | Gly | Ser | Tyr |  |
| 385 |     | 390 |     |     |     |     | 395 |     |     |     |     | 400 |     |     |     |  |
| Phe | Val | Tyr | Ser | Lys | Leu | Asn | Val | Gln | Lys | Ser | Asn | Trp | Glu | Ala | Gly |  |
| 405 |     |     |     | 410 |     |     |     |     | 415 |     |     |     |     |     |     |  |
| Asn | Thr | Phe | Thr | Cys | Ser | Val | Leu | His | Glu | Gly | Leu | His | Asn | His | His |  |
| 420 |     |     |     | 425 |     |     |     |     | 430 |     |     |     |     |     |     |  |
| Thr | Glu | Lys | Ser | Leu | Ser | His | Ser | Pro | Gly | Lys |     |     |     |     |     |  |

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| 435                                                             | 440 |
|-----------------------------------------------------------------|-----|
| <210> SEQ ID NO 210                                             |     |
| <211> LENGTH: 214                                               |     |
| <212> TYPE: PRT                                                 |     |
| <213> ORGANISM: Artificial Sequence                             |     |
| <220> FEATURE:                                                  |     |
| <223> OTHER INFORMATION: Synthetic Construct                    |     |
| <400> SEQUENCE: 210                                             |     |
| Asp Ile Gln Met Thr Gln Ser Pro Ala Ser Gln Ser Ala Ser Leu Gly |     |
| 1 5 10 15                                                       |     |
| Glu Ser Val Thr Phe Thr Cys Leu Ala Ser Gln Thr Ile Gly Thr Trp |     |
| 20 25 30                                                        |     |
| Leu Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ser Pro Gln Leu Leu Ile |     |
| 35 40 45                                                        |     |
| Tyr Ala Ala Thr Ser Leu Ala Asp Gly Val Pro Ser Arg Phe Ser Gly |     |
| 50 55 60                                                        |     |
| Ser Gly Ser Gly Thr Lys Phe Ser Phe Lys Ile Ser Ser Leu Gln Ala |     |
| 65 70 75 80                                                     |     |
| Glu Asp Phe Ala Ser Tyr Tyr Cys Gln Gln Leu Tyr Ser Thr Pro Trp |     |
| 85 90 95                                                        |     |
| Thr Phe Gly Gly Gly Thr Lys Leu Glu Ile Lys Arg Ala Asp Ala Ala |     |
| 100 105 110                                                     |     |
| Pro Thr Val Ser Ile Phe Pro Pro Ser Ser Glu Gln Leu Thr Ser Gly |     |
| 115 120 125                                                     |     |
| Gly Ala Ser Val Val Cys Phe Leu Asn Asn Phe Tyr Pro Lys Asp Ile |     |
| 130 135 140                                                     |     |
| Asn Val Lys Trp Lys Ile Asp Gly Ser Glu Arg Gln Asn Gly Val Leu |     |
| 145 150 155 160                                                 |     |
| Asn Ser Trp Thr Asp Gln Asp Ser Lys Asp Ser Thr Tyr Ser Met Ser |     |
| 165 170 175                                                     |     |
| Ser Thr Leu Thr Leu Thr Lys Asp Glu Tyr Glu Arg His Asn Ser Tyr |     |
| 180 185 190                                                     |     |
| Thr Cys Glu Ala Thr His Lys Thr Ser Thr Ser Pro Ile Val Lys Ser |     |
| 195 200 205                                                     |     |
| Phe Asn Arg Asn Glu Cys                                         |     |
| 210                                                             |     |
| <210> SEQ ID NO 211                                             |     |
| <211> LENGTH: 443                                               |     |
| <212> TYPE: PRT                                                 |     |
| <213> ORGANISM: Artificial Sequence                             |     |
| <220> FEATURE:                                                  |     |
| <223> OTHER INFORMATION: Synthetic Construct                    |     |
| <400> SEQUENCE: 211                                             |     |
| Glu Val Gln Leu Gln Gln Ser Gly Ala Glu Leu Val Arg Pro Gly Ala |     |
| 1 5 10 15                                                       |     |
| Ser Val Arg Leu Ser Cys Thr Ala Ser Gly Phe Asn Ile Lys Asp Ser |     |
| 20 25 30                                                        |     |
| Leu Met His Trp Val Lys Gln Arg Pro Glu Gln Gly Leu Glu Trp Ile |     |
| 35 40 45                                                        |     |
| Gly Trp Ile Asp Pro Glu Asp Gly Glu Thr Lys Tyr Ala Pro Asn Phe |     |
| 50 55 60                                                        |     |

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<210> SEQ ID NO 212
<211> LENGTH: 214
<212> TYPE: PRT
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<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 212
Asp Ile Gln Met Thr Gln Ser Pro Ala Ser Gln Ser Ala Ser Leu Gly
1           5           10           15
Glu Ser Val Thr Phe Thr Cys Leu Ala Ser Gln Thr Ile Gly Thr Trp
20           25           30
Leu Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ser Pro Gln Leu Leu Ile
35           40           45
Tyr Ala Ala Thr Ser Leu Ala Asp Gly Val Pro Ser Arg Phe Ser Gly
50           55           60
Ser Gly Ser Gly Thr Lys Phe Ser Phe Lys Ile Ser Ser Leu Gln Ala
65           70           75           80
Glu Asp Phe Ala Ser Tyr Tyr Cys Gln Gln Leu Tyr Ser Thr Pro Trp
85           90           95
Thr Phe Gly Gly Gly Thr Lys Leu Glu Ile Lys Arg Ala Asp Ala Ala
100          105          110
Pro Thr Val Ser Ile Phe Pro Pro Ser Ser Glu Gln Leu Thr Ser Gly
115          120          125
Gly Ala Ser Val Val Cys Phe Leu Asn Asn Phe Tyr Pro Lys Asp Ile
130          135          140
Asn Val Lys Trp Lys Ile Asp Gly Ser Glu Arg Gln Asn Gly Val Leu
145          150          155          160
Asn Ser Trp Thr Asp Gln Asp Ser Lys Asp Ser Thr Tyr Ser Met Ser
165          170          175
Ser Thr Leu Thr Leu Thr Lys Asp Glu Tyr Glu Arg His Asn Ser Tyr
180          185          190
Thr Cys Glu Ala Thr His Lys Thr Ser Thr Ser Pro Ile Val Lys Ser
195          200          205
Phe Asn Arg Asn Glu Cys
210

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<210> SEQ ID NO 213
<211> LENGTH: 443
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 213
Glu Val Gln Leu Gln Gln Ser Gly Ala Glu Leu Val Arg Pro Gly Ala
1           5           10           15
Ser Val Arg Leu Ser Cys Thr Ala Ser Gly Phe Asn Ile Lys Asp Ser
20           25           30
Leu Met His Trp Val Lys Gln Arg Pro Glu Gln Gly Leu Glu Trp Ile
35           40           45
Gly Trp Ile Asp Pro Glu Asp Gly Glu Thr Lys Tyr Ala Pro Asn Phe
50           55           60
Gln Ser Lys Ala Thr Ile Thr Ala Val Thr Ser Ser Asn Thr Ala Tyr
65           70           75           80
Leu Gln Leu Ser Ser Leu Thr Ser Glu Asp Ser Ala Ile Tyr Tyr Cys
85           90           95

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|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Ala | Arg | Tyr | Thr | Ser | Pro | Tyr | Tyr | Thr | Met | Asp | Tyr | Trp | Gly | Gln | Gly | 100 | 105 | 110 |
| Thr | Ser | Val | Thr | Val | Ser | Ser | Ala | Lys | Thr | Thr | Pro | Pro | Ser | Val | Tyr | 115 | 120 | 125 |
| Pro | Leu | Ala | Pro | Gly | Ser | Ala | Ala | Gln | Thr | Asn | Ser | Met | Val | Thr | Leu | 130 | 135 | 140 |
| Gly | Cys | Leu | Val | Lys | Gly | Tyr | Phe | Pro | Glu | Pro | Val | Thr | Val | Thr | Trp | 145 | 150 | 155 |
| Asn | Ser | Gly | Ser | Leu | Ser | Ser | Gly | Val | His | Thr | Phe | Pro | Ala | Val | Leu | 165 | 170 | 175 |
| Gln | Ser | Asp | Leu | Tyr | Thr | Leu | Ser | Ser | Ser | Val | Thr | Val | Pro | Ser | Ser | 180 | 185 | 190 |
| Thr | Trp | Pro | Ser | Gln | Thr | Val | Thr | Cys | Asn | Val | Ala | His | Pro | Ala | Ser | 195 | 200 | 205 |
| Ser | Thr | Lys | Val | Asp | Lys | Lys | Ile | Val | Pro | Arg | Asp | Cys | Gly | Cys | Lys | 210 | 215 | 220 |
| Pro | Cys | Ile | Cys | Thr | Val | Pro | Glu | Val | Ser | Ser | Val | Phe | Ile | Phe | Pro | 225 | 230 | 235 |
| Pro | Lys | Pro | Lys | Asp | Val | Leu | Thr | Ile | Thr | Leu | Thr | Pro | Lys | Val | Thr | 245 | 250 | 255 |
| Cys | Val | Val | Val | Asp | Ile | Ser | Lys | Asp | Asp | Pro | Glu | Val | Gln | Phe | Ser | 260 | 265 | 270 |
| Trp | Phe | Val | Asp | Asp | Val | Glu | Val | His | Thr | Ala | Gln | Thr | Lys | Pro | Arg | 275 | 280 | 285 |
| Glu | Glu | Gln | Ile | Asn | Ser | Thr | Phe | Arg | Ser | Val | Ser | Glu | Leu | Pro | Ile | 290 | 295 | 300 |
| Met | His | Gln | Asp | Trp | Leu | Asn | Gly | Lys | Glu | Phe | Lys | Cys | Arg | Val | Asn | 305 | 310 | 315 |
| Ser | Ala | Ala | Phe | Pro | Ala | Pro | Ile | Glu | Lys | Thr | Ile | Ser | Lys | Thr | Lys | 325 | 330 | 335 |
| Gly | Arg | Pro | Lys | Ala | Pro | Gln | Val | Tyr | Thr | Ile | Pro | Pro | Pro | Lys | Glu | 340 | 345 | 350 |
| Gln | Met | Ala | Lys | Asp | Lys | Val | Ser | Leu | Thr | Cys | Met | Ile | Thr | Asn | Phe | 355 | 360 | 365 |
| Phe | Pro | Glu | Asp | Ile | Thr | Val | Glu | Trp | Gln | Trp | Asn | Gly | Gln | Pro | Ala | 370 | 375 | 380 |
| Glu | Asn | Tyr | Lys | Asn | Thr | Gln | Pro | Ile | Met | Asp | Thr | Asp | Gly | Ser | Tyr | 385 | 390 | 395 |
| Phe | Val | Tyr | Ser | Lys | Leu | Asn | Val | Gln | Lys | Ser | Asn | Trp | Glu | Ala | Gly | 405 | 410 | 415 |
| Asn | Thr | Phe | Thr | Cys | Ser | Val | Leu | His | Glu | Gly | Leu | His | Asn | His | His | 420 | 425 | 430 |
| Thr | Glu | Lys | Ser | Leu | Ser | His | Ser | Pro | Gly | Lys |     |     |     |     |     | 435 | 440 |     |

&lt;210&gt; SEQ ID NO 214

&lt;211&gt; LENGTH: 214

&lt;212&gt; TYPE: PRT

&lt;213&gt; ORGANISM: Artificial Sequence

&lt;220&gt; FEATURE:

&lt;223&gt; OTHER INFORMATION: Synthetic Construct

&lt;400&gt; SEQUENCE: 214

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|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Asp | Ile | Gln | Met | Thr | Gln | Ser | Pro | Ala | Ser | Gln | Ser | Ala | Ser | Leu | Gly |
| 1   |     |     |     | 5   |     |     |     |     | 10  |     |     |     |     | 15  |     |
| Glu | Ser | Val | Thr | Phe | Thr | Cys | Leu | Ala | Ser | Gln | Thr | Ile | Gly | Thr | Trp |
|     |     | 20  |     |     |     |     | 25  |     |     |     |     |     | 30  |     |     |
| Leu | Ala | Trp | Tyr | Gln | Gln | Lys | Pro | Gly | Lys | Ser | Pro | Gln | Leu | Leu | Ile |
|     | 35  |     |     |     |     | 40  |     |     |     |     | 45  |     |     |     |     |
| Tyr | Ala | Ala | Thr | Ser | Leu | Ala | Asp | Gly | Val | Pro | Ser | Arg | Phe | Ser | Gly |
|     | 50  |     |     |     | 55  |     |     |     |     | 60  |     |     |     |     |     |
| Ser | Gly | Ser | Gly | Thr | Lys | Phe | Ser | Phe | Lys | Ile | Ser | Ser | Leu | Gln | Ala |
| 65  |     |     |     | 70  |     |     |     |     | 75  |     |     |     |     | 80  |     |
| Glu | Asp | Phe | Ala | Ser | Tyr | Tyr | Cys | Gln | Gln | Val | Tyr | Ser | Thr | Pro | Trp |
|     |     | 85  |     |     |     |     | 90  |     |     |     |     |     | 95  |     |     |
| Thr | Phe | Gly | Gly | Gly | Thr | Lys | Leu | Glu | Ile | Lys | Arg | Ala | Asp | Ala | Ala |
|     | 100 |     |     |     |     |     | 105 |     |     |     |     |     | 110 |     |     |
| Pro | Thr | Val | Ser | Ile | Phe | Pro | Pro | Ser | Ser | Glu | Gln | Leu | Thr | Ser | Gly |
|     | 115 |     |     |     |     | 120 |     |     |     |     |     | 125 |     |     |     |
| Gly | Ala | Ser | Val | Val | Cys | Phe | Leu | Asn | Asn | Phe | Tyr | Pro | Lys | Asp | Ile |
|     | 130 |     |     |     | 135 |     |     |     |     |     | 140 |     |     |     |     |
| Asn | Val | Lys | Trp | Lys | Ile | Asp | Gly | Ser | Glu | Arg | Gln | Asn | Gly | Val | Leu |
| 145 |     |     |     | 150 |     |     |     |     | 155 |     |     |     |     | 160 |     |
| Asn | Ser | Trp | Thr | Asp | Gln | Asp | Ser | Lys | Asp | Ser | Thr | Tyr | Ser | Met | Ser |
|     |     | 165 |     |     |     |     | 170 |     |     |     |     |     |     | 175 |     |
| Ser | Thr | Leu | Thr | Leu | Thr | Lys | Asp | Glu | Tyr | Glu | Arg | His | Asn | Ser | Tyr |
|     | 180 |     |     |     |     |     | 185 |     |     |     |     | 190 |     |     |     |
| Thr | Cys | Glu | Ala | Thr | His | Lys | Thr | Ser | Thr | Ser | Pro | Ile | Val | Lys | Ser |
|     | 195 |     |     |     |     | 200 |     |     |     |     | 205 |     |     |     |     |
| Phe | Asn | Arg | Asn | Glu | Cys |     |     |     |     |     |     |     |     |     |     |
|     | 210 |     |     |     |     |     |     |     |     |     |     |     |     |     |     |

&lt;210&gt; SEQ ID NO 215

&lt;211&gt; LENGTH: 443

&lt;212&gt; TYPE: PRT

&lt;213&gt; ORGANISM: Artificial Sequence

&lt;220&gt; FEATURE:

&lt;223&gt; OTHER INFORMATION: Synthetic Construct

&lt;400&gt; SEQUENCE: 215

|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Glu | Val | Gln | Leu | Gln | Gln | Ser | Gly | Ala | Glu | Leu | Val | Arg | Pro | Gly | Ala |
| 1   |     |     | 5   |     |     |     |     | 10  |     |     |     |     | 15  |     |     |
| Ser | Val | Arg | Leu | Ser | Cys | Thr | Ala | Ser | Gly | Phe | Asn | Ile | Lys | Asp | Ser |
|     | 20  |     |     |     |     |     | 25  |     |     |     |     | 30  |     |     |     |
| Leu | Met | His | Trp | Val | Lys | Gln | Arg | Pro | Glu | Gln | Gly | Leu | Glu | Trp | Ile |
|     | 35  |     |     |     |     | 40  |     |     |     |     | 45  |     |     |     |     |
| Gly | Trp | Ile | Asp | Pro | Glu | Asp | Gly | Glu | Thr | Arg | Tyr | Ala | Pro | Asn | Phe |
|     | 50  |     |     |     | 55  |     |     |     | 60  |     |     |     |     |     |     |
| Gln | Asp | Lys | Ala | Thr | Ile | Thr | Ala | Val | Thr | Ser | Ser | Asn | Thr | Ala | Tyr |
| 65  |     |     |     | 70  |     |     |     |     | 75  |     |     |     |     | 80  |     |
| Leu | Gln | Leu | Ser | Ser | Leu | Thr | Ser | Glu | Asp | Ser | Ala | Ile | Tyr | Tyr | Cys |
|     |     | 85  |     |     |     |     | 90  |     |     |     |     |     | 95  |     |     |
| Ala | Arg | Tyr | Thr | Ser | Arg | Tyr | Tyr | Thr | Met | Glu | Tyr | Trp | Gly | Gln | Gly |
|     | 100 |     |     |     |     |     | 105 |     |     |     |     |     | 110 |     |     |
| Thr | Ser | Val | Thr | Val | Ser | Ser | Ala | Lys | Thr | Thr | Pro | Pro | Ser | Val | Tyr |
|     | 115 |     |     |     |     | 120 |     |     |     |     | 125 |     |     |     |     |

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<210> SEQ ID NO 216
<211> LENGTH: 214
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 216

Asp Ile Gln Met Thr Gln Ser Pro Ala Ser Gln Ser Ala Ser Leu Gly
1             5             10             15
Glu Ser Val Thr Phe Thr Cys Leu Ala Ser Gln Thr Ile Gly Thr Trp
          20             25             30

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|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Leu | Ala | Trp | Tyr | Gln | Gln | Lys | Pro | Gly | Lys | Ser | Pro | Gln | Leu | Leu | Ile |
| 35  |     |     |     |     |     | 40  |     |     |     |     |     | 45  |     |     |     |
| Tyr | Ala | Ala | Thr | Ser | Leu | Ala | Asp | Gly | Val | Pro | Ser | Arg | Phe | Ser | Gly |
| 50  |     |     |     |     |     | 55  |     |     |     | 60  |     |     |     |     |     |
| Ser | Gly | Ser | Gly | Thr | Lys | Phe | Ser | Phe | Lys | Ile | Ser | Ser | Leu | Gln | Ala |
| 65  |     |     |     | 70  |     |     |     |     |     | 75  |     |     |     | 80  |     |
| Glu | Asp | Phe | Ala | Ser | Tyr | Tyr | Cys | Gln | Gln | Leu | Tyr | Trp | Thr | Pro | Trp |
|     |     |     |     | 85  |     |     |     | 90  |     |     |     |     |     | 95  |     |
| Thr | Phe | Gly | Gly | Gly | Thr | Lys | Leu | Glu | Ile | Lys | Arg | Ala | Asp | Ala | Ala |
|     |     | 100 |     |     |     |     |     | 105 |     |     |     | 110 |     |     |     |
| Pro | Thr | Val | Ser | Ile | Phe | Pro | Pro | Ser | Ser | Glu | Gln | Leu | Thr | Ser | Gly |
|     |     | 115 |     |     |     | 120 |     |     |     |     |     | 125 |     |     |     |
| Gly | Ala | Ser | Val | Val | Cys | Phe | Leu | Asn | Asn | Phe | Tyr | Pro | Lys | Asp | Ile |
| 130 |     |     |     |     |     | 135 |     |     |     |     |     | 140 |     |     |     |
| Asn | Val | Lys | Trp | Lys | Ile | Asp | Gly | Ser | Glu | Arg | Gln | Asn | Gly | Val | Leu |
| 145 |     |     |     | 150 |     |     |     |     |     | 155 |     |     |     | 160 |     |
| Asn | Ser | Trp | Thr | Asp | Gln | Asp | Ser | Lys | Asp | Ser | Thr | Tyr | Ser | Met | Ser |
|     |     |     |     | 165 |     |     |     | 170 |     |     |     |     |     | 175 |     |
| Ser | Thr | Leu | Thr | Leu | Thr | Lys | Asp | Glu | Tyr | Glu | Arg | His | Asn | Ser | Tyr |
|     |     | 180 |     |     |     |     |     | 185 |     |     |     | 190 |     |     |     |
| Thr | Cys | Glu | Ala | Thr | His | Lys | Thr | Ser | Thr | Ser | Pro | Ile | Val | Lys | Ser |
|     |     | 195 |     |     |     | 200 |     |     |     |     |     | 205 |     |     |     |
| Phe | Asn | Arg | Asn | Glu | Cys |     |     |     |     |     |     |     |     |     |     |
| 210 |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |

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<210> SEQ ID NO 217
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct
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<400> SEQUENCE: 217

Gly Phe Thr Phe Ser Asn Tyr Gly  
1 5

```
<210> SEQ ID NO 218
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct
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<400> SEQUENCE: 218

Ile Ser Tyr Asp Gly Asn Asn Lys  
1 5

```
<210> SEQ ID NO 219
<211> LENGTH: 15
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct
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<400> SEQUENCE: 219

Ala Lys Asp Arg Gly Leu Asp Thr Ala Gly Asp Tyr Phe Asp Tyr  
1                   5                   10                   15



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<210> SEQ ID NO 220  
<211> LENGTH: 6  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 220

Gln Asn Ile Ser Ser Lys  
1 5

<210> SEQ ID NO 221  
<211> LENGTH: 3  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 221

Gly Ala Ser  
1

<210> SEQ ID NO 222  
<211> LENGTH: 8  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 222

Leu Gln Tyr Asn Asn Trp Trp Thr  
1 5

<210> SEQ ID NO 223  
<211> LENGTH: 8  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 223

Asn Gly Ser Phe Ser Ser Tyr Tyr  
1 5

<210> SEQ ID NO 224  
<211> LENGTH: 7  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 224

Ile Asn His Arg Gly Ser Thr  
1 5

<210> SEQ ID NO 225  
<211> LENGTH: 9  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 225

Thr Ala His Thr Arg Gly Phe Leu Tyr  
1 5

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<210> SEQ ID NO 226  
<211> LENGTH: 6  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 226

Gln Asp Ile Asn Ile Tyr  
1 5

<210> SEQ ID NO 227  
<211> LENGTH: 3  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 227

Asp Val Ser  
1

<210> SEQ ID NO 228  
<211> LENGTH: 9  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 228

Gln His Tyr Asp Asp Leu Pro Leu Thr  
1 5

<210> SEQ ID NO 229  
<211> LENGTH: 8  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 229

Asn Gly Ser Phe Ser Ser Phe Tyr  
1 5

<210> SEQ ID NO 230  
<211> LENGTH: 7  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 230

Ile Asn His Arg Gly Arg Thr  
1 5

<210> SEQ ID NO 231  
<211> LENGTH: 6  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 231

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Gln Asp Ile Asn Asn Tyr  
1 5

<210> SEQ ID NO 232  
<211> LENGTH: 3  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 232

Asp Val Phe  
1

<210> SEQ ID NO 233  
<211> LENGTH: 9  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 233

Ser Ala His Thr Arg Gly Phe Leu Tyr  
1 5

<210> SEQ ID NO 234  
<211> LENGTH: 3  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 234

Asp Ala Ser  
1

<210> SEQ ID NO 235  
<211> LENGTH: 9  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 235

Gln His Tyr Tyr Asp Leu Pro Leu Thr  
1 5

<210> SEQ ID NO 236  
<211> LENGTH: 8  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 236

Gly Phe Thr Phe Ser Asp Ser Ala  
1 5

<210> SEQ ID NO 237  
<211> LENGTH: 10  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

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&lt;400&gt; SEQUENCE: 237

|     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Ile | Arg | Asn | Lys | Ala | Asn | Thr | Tyr | Ala | Thr |
| 1   |     |     |     | 5   |     |     |     |     | 10  |

&lt;210&gt; SEQ ID NO 238

&lt;211&gt; LENGTH: 6

&lt;212&gt; TYPE: PRT

&lt;213&gt; ORGANISM: Artificial Sequence

&lt;220&gt; FEATURE:

&lt;223&gt; OTHER INFORMATION: Synthetic Construct

&lt;400&gt; SEQUENCE: 238

|     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|
| Thr | Ser | Asp | Gln | Phe | Asp |
| 1   |     |     | 5   |     |     |

&lt;210&gt; SEQ ID NO 239

&lt;211&gt; LENGTH: 11

&lt;212&gt; TYPE: PRT

&lt;213&gt; ORGANISM: Artificial Sequence

&lt;220&gt; FEATURE:

&lt;223&gt; OTHER INFORMATION: Synthetic Construct

&lt;400&gt; SEQUENCE: 239

|     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Gln | Ser | Leu | Val | Tyr | Ser | Asp | Gly | Ile | Thr | Tyr |
| 1   |     |     | 5   |     |     |     |     |     | 10  |     |

&lt;210&gt; SEQ ID NO 240

&lt;211&gt; LENGTH: 3

&lt;212&gt; TYPE: PRT

&lt;213&gt; ORGANISM: Artificial Sequence

&lt;220&gt; FEATURE:

&lt;223&gt; OTHER INFORMATION: Synthetic Construct

&lt;400&gt; SEQUENCE: 240

|     |     |     |
|-----|-----|-----|
| Gln | Val | Ser |
| 1   |     |     |

&lt;210&gt; SEQ ID NO 241

&lt;211&gt; LENGTH: 9

&lt;212&gt; TYPE: PRT

&lt;213&gt; ORGANISM: Artificial Sequence

&lt;220&gt; FEATURE:

&lt;223&gt; OTHER INFORMATION: Synthetic Construct

&lt;400&gt; SEQUENCE: 241

|     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Met | Glu | Gly | Thr | His | Trp | Pro | Trp | Thr |
| 1   |     |     |     | 5   |     |     |     |     |

&lt;210&gt; SEQ ID NO 242

&lt;211&gt; LENGTH: 8

&lt;212&gt; TYPE: PRT

&lt;213&gt; ORGANISM: Artificial Sequence

&lt;220&gt; FEATURE:

&lt;223&gt; OTHER INFORMATION: Synthetic Construct

&lt;400&gt; SEQUENCE: 242

|     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|
| Gly | Phe | Thr | Phe | Ser | Asp | Tyr | Tyr |
| 1   |     |     |     | 5   |     |     |     |

&lt;210&gt; SEQ ID NO 243

&lt;211&gt; LENGTH: 8

&lt;212&gt; TYPE: PRT

&lt;213&gt; ORGANISM: Artificial Sequence

&lt;220&gt; FEATURE:

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&lt;223&gt; OTHER INFORMATION: Synthetic Construct

&lt;400&gt; SEQUENCE: 243

Ile Ser Arg Thr Gly Ser Thr Lys  
1 5

&lt;210&gt; SEQ ID NO 244

&lt;211&gt; LENGTH: 7

&lt;212&gt; TYPE: PRT

&lt;213&gt; ORGANISM: Artificial Sequence

&lt;220&gt; FEATURE:

&lt;223&gt; OTHER INFORMATION: Synthetic Construct

&lt;400&gt; SEQUENCE: 244

Ala Arg Glu Pro Leu Gln His  
1 5

&lt;210&gt; SEQ ID NO 245

&lt;211&gt; LENGTH: 11

&lt;212&gt; TYPE: PRT

&lt;213&gt; ORGANISM: Artificial Sequence

&lt;220&gt; FEATURE:

&lt;223&gt; OTHER INFORMATION: Synthetic Construct

&lt;400&gt; SEQUENCE: 245

Gln Ser Leu Val His Ser Asp Gly Asp Thr Tyr  
1 5 10

&lt;210&gt; SEQ ID NO 246

&lt;211&gt; LENGTH: 3

&lt;212&gt; TYPE: PRT

&lt;213&gt; ORGANISM: Artificial Sequence

&lt;220&gt; FEATURE:

&lt;223&gt; OTHER INFORMATION: Synthetic Construct

&lt;400&gt; SEQUENCE: 246

Thr Val Ser  
1

&lt;210&gt; SEQ ID NO 247

&lt;211&gt; LENGTH: 10

&lt;212&gt; TYPE: PRT

&lt;213&gt; ORGANISM: Artificial Sequence

&lt;220&gt; FEATURE:

&lt;223&gt; OTHER INFORMATION: Synthetic Construct

&lt;400&gt; SEQUENCE: 247

Met Gln Gly Thr Gln Phe Pro Ala Leu Thr  
1 5 10

&lt;210&gt; SEQ ID NO 248

&lt;211&gt; LENGTH: 3

&lt;212&gt; TYPE: PRT

&lt;213&gt; ORGANISM: Artificial Sequence

&lt;220&gt; FEATURE:

&lt;223&gt; OTHER INFORMATION: Synthetic Construct

&lt;400&gt; SEQUENCE: 248

Thr Ile Ser  
1

&lt;210&gt; SEQ ID NO 249

&lt;211&gt; LENGTH: 8

&lt;212&gt; TYPE: PRT

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<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 249

Gly Phe Thr Leu Ser Asp Ser Ser  
1 5

<210> SEQ ID NO 250  
<211> LENGTH: 10  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 250

Ile Arg Ser Lys Pro Tyr Ser Tyr Ala Thr  
1 5 10

<210> SEQ ID NO 251  
<211> LENGTH: 6  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 251

Thr Gly Gly Asp Asp Phe  
1 5

<210> SEQ ID NO 252  
<211> LENGTH: 6  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 252

Gln Ser Ile Ser Thr Trp  
1 5

<210> SEQ ID NO 253  
<211> LENGTH: 3  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 253

Lys Ala Ser  
1

<210> SEQ ID NO 254  
<211> LENGTH: 8  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 254

Gln Gln Tyr Asn Thr Tyr Trp Thr  
1 5

<210> SEQ ID NO 255

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<211> LENGTH: 8  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 255

Gly Phe Thr Phe Ser Ser Tyr Ala  
1 5

<210> SEQ ID NO 256  
<211> LENGTH: 8  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 256

Ile Ser Gly Ser Gly Gly Ser Thr  
1 5

<210> SEQ ID NO 257  
<211> LENGTH: 6  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 257

Gln Ser Val Ser Ser Asn  
1 5

<210> SEQ ID NO 258  
<211> LENGTH: 10  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 258

Gln Gln Tyr Asn Asn Trp Pro Pro Tyr Thr  
1 5 10

<210> SEQ ID NO 259  
<211> LENGTH: 8  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 259

Gly Tyr Thr Phe Thr Gly Tyr Tyr  
1 5

<210> SEQ ID NO 260  
<211> LENGTH: 8  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 260

Ile Asn Pro Asn Ser Val Gly Thr  
1 5

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<210> SEQ ID NO 261  
<211> LENGTH: 12  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct  
  
<400> SEQUENCE: 261  
  
Thr Arg Asp Gly Ala Ala Ala Gly Leu Phe Asp Tyr  
1                  5                  10

<210> SEQ ID NO 262  
<211> LENGTH: 6  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct  
  
<400> SEQUENCE: 262

Gln Gly Ile Asn Asn Tyr  
1                  5

<210> SEQ ID NO 263  
<211> LENGTH: 3  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct  
  
<400> SEQUENCE: 263

Ala Ala Ser  
1

<210> SEQ ID NO 264  
<211> LENGTH: 9  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct  
  
<400> SEQUENCE: 264

Gln Lys Phe Asn Ser Ala Pro Leu Thr  
1                  5

<210> SEQ ID NO 265  
<211> LENGTH: 8  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct  
  
<400> SEQUENCE: 265

Gly Phe Thr Phe Ser Thr Tyr Ala  
1                  5

<210> SEQ ID NO 266  
<211> LENGTH: 9  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct  
  
<400> SEQUENCE: 266

Ala Asn Ser Pro Ser Trp Phe Asp Pro



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1                      5

<210> SEQ ID NO 267  
<211> LENGTH: 6  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 267

Gln Ser Ile Ser Ser Tyr  
1                      5

<210> SEQ ID NO 268  
<211> LENGTH: 10  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 268

Gln Gln Ser Tyr Ser Thr Pro Pro Ile Thr  
1                      5                      10

<210> SEQ ID NO 269  
<211> LENGTH: 8  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 269

Gly Phe Thr Phe Asp Asp Tyr Gly  
1                      5

<210> SEQ ID NO 270  
<211> LENGTH: 8  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 270

Ile Ser Trp Asn Ser Ala Thr Ile  
1                      5

<210> SEQ ID NO 271  
<211> LENGTH: 12  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 271

Ala Lys Lys Asn Ser Leu Gly Trp Phe Phe Asp Tyr  
1                      5                      10

<210> SEQ ID NO 272  
<211> LENGTH: 12  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 272

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Gln Ser Val Leu Tyr Ser Ser Asn Asp Asn Asn Tyr  
1 5 10

<210> SEQ ID NO 273  
<211> LENGTH: 3  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 273

Trp Ala Ser  
1

<210> SEQ ID NO 274  
<211> LENGTH: 9  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 274

Gln Gln Tyr Tyr Ser Thr Pro Leu Thr  
1 5

<210> SEQ ID NO 275  
<211> LENGTH: 8  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 275

Gly Tyr Thr Phe Thr Ser Tyr Gly  
1 5

<210> SEQ ID NO 276  
<211> LENGTH: 8  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 276

Ile Ser Ala Tyr Asn Gly Asn Thr  
1 5

<210> SEQ ID NO 277  
<211> LENGTH: 10  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 277

Ala Arg Asp Ile Asn Trp Tyr Phe Asp Leu  
1 5 10

<210> SEQ ID NO 278  
<211> LENGTH: 6  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

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<400> SEQUENCE: 278

Gln Gly Ile Arg Asn Asp  
1 5

<210> SEQ ID NO 279

<211> LENGTH: 9

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 279

Leu Gln His Asn Ser Tyr Pro Trp Thr  
1 5

<210> SEQ ID NO 280

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 280

Ile Ser Trp Asn Gly Ala Thr Ile  
1 5

<210> SEQ ID NO 281

<211> LENGTH: 3

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 281

Thr Ala Ser  
1

<210> SEQ ID NO 282

<211> LENGTH: 9

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 282

Leu Gln His Asn Asn Tyr Pro Tyr Thr  
1 5

<210> SEQ ID NO 283

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 283

Gly Phe Thr Phe Ser Asn Tyr Ala  
1 5

<210> SEQ ID NO 284

<211> LENGTH: 8

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

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<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 284

Ile Thr Gly Asn Gly Val Asn Thr  
1 5

<210> SEQ ID NO 285  
<211> LENGTH: 15  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 285

Val Lys Glu Arg Gly His Ser Trp Phe Gly Asp Trp Phe Asp Pro  
1 5 10 15

<210> SEQ ID NO 286  
<211> LENGTH: 6  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 286

Gln Ser Val Ile Ser Tyr  
1 5

<210> SEQ ID NO 287  
<211> LENGTH: 9  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 287

Gln Gln Asn Tyr Leu Pro Pro Leu Thr  
1 5

<210> SEQ ID NO 288  
<211> LENGTH: 8  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 288

Ile Thr Gly Asn Gly Ile Lys Thr  
1 5

<210> SEQ ID NO 289  
<211> LENGTH: 11  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 289

Gln Gly Leu Val Asn Ser Asp Gly Asn Thr Tyr  
1 5 10

<210> SEQ ID NO 290  
<211> LENGTH: 3

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<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 290

Glu Ile Ser  
1

<210> SEQ ID NO 291  
<211> LENGTH: 9  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 291

Met Gln Ser Thr Gln Phe Pro Leu Thr  
1 5

<210> SEQ ID NO 292  
<211> LENGTH: 8  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 292

Ile Thr Gly Asn Gly Ile Asn Thr  
1 5

<210> SEQ ID NO 293  
<211> LENGTH: 8  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 293

Val Ser Tyr Asp Gly Ser Asn Lys  
1 5

<210> SEQ ID NO 294  
<211> LENGTH: 15  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 294

Ala Lys Asp Arg Gly Ile Thr Gly Thr Ser Gly Gly Val Phe Asp  
1 5 10 15

<210> SEQ ID NO 295  
<211> LENGTH: 6  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 295

Gln Gly Ile Ser Ser His  
1 5

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<210> SEQ ID NO 296  
<211> LENGTH: 3  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 296

Val Ala Ser  
1

<210> SEQ ID NO 297  
<211> LENGTH: 9  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 297

Gln Gln Tyr Tyr Ser Tyr Pro Pro Thr  
1 5

<210> SEQ ID NO 298  
<211> LENGTH: 8  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 298

Gly Phe Thr Ser Ser Arg Tyr Trp  
1 5

<210> SEQ ID NO 299  
<211> LENGTH: 8  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 299

Ile Asn Gln Asp Gly Ser Glu Lys  
1 5

<210> SEQ ID NO 300  
<211> LENGTH: 7  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 300

Ala Arg Asp Glu Glu Gly Tyr  
1 5

<210> SEQ ID NO 301  
<211> LENGTH: 11  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 301

Gln Ser Leu Val Tyr Ser Asp Gly Asp Thr Tyr  
1 5 10

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<210> SEQ ID NO 302  
<211> LENGTH: 3  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 302

Lys Val Ser  
1

<210> SEQ ID NO 303  
<211> LENGTH: 9  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 303

Met Gln Gly Thr His Trp Pro Pro Thr  
1 5

<210> SEQ ID NO 304  
<211> LENGTH: 122  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 304

Gln Val His Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Arg  
1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Asn Tyr  
20 25 30

Gly Ile His Trp Val Arg Gln Val Pro Gly Lys Gly Leu Glu Trp Val  
35 40 45

Ala Val Ile Ser Tyr Asp Gly Asn Asn Lys Tyr Tyr Gly Asp Ser Val  
50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr  
65 70 75 80

Leu Gln Met Asn Ser Leu Arg Pro Glu Asp Thr Ala Ile Tyr Tyr Cys  
85 90 95

Ala Lys Asp Arg Gly Leu Asp Thr Ala Gly Asp Tyr Phe Asp Tyr Trp  
100 105 110

Gly Gln Gly Thr Leu Val Thr Val Ser Ser  
115 120

<210> SEQ ID NO 305  
<211> LENGTH: 106  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 305

Glu Ile Val Met Thr Gln Ser Pro Ala Thr Leu Ser Val Ser Thr Gly  
1 5 10 15

Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Asn Ile Ser Ser Lys  
20 25 30

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Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile  
 35 40 45  
 Tyr Gly Ala Ser Thr Arg Ala Ser Gly Ile Pro Ala Arg Phe Trp Gly  
 50 55 60  
 Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Ser Ser Leu Gln Ser  
 65 70 75 80  
 Glu Asp Phe Ala Val Phe Tyr Cys Leu Gln Tyr Asn Asn Trp Trp Thr  
 85 90 95  
 Phe Gly Gln Gly Thr Lys Val Glu Ile Lys  
 100 105

<210> SEQ ID NO 306  
 <211> LENGTH: 115  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 306

Gln Val Gln Leu Gln Gln Trp Gly Ala Gly Leu Leu Lys Pro Ser Glu  
 1 5 10 15  
 Thr Leu Ser Leu Thr Cys Ala Val Tyr Asn Gly Ser Phe Ser Ser Tyr  
 20 25 30  
 Tyr Trp Asn Trp Ile Arg Gln Pro Pro Gly Lys Gly Leu Glu Trp Leu  
 35 40 45  
 Gly Glu Ile Asn His Arg Gly Ser Thr Asn Tyr Asn Pro Ser Leu Thr  
 50 55 60  
 Ser Arg Val Thr Ile Ser Val Asp Thr Ser Lys Asn Gln Phe Ser Leu  
 65 70 75 80  
 Lys Leu Thr Ser Val Thr Ala Ala Asp Thr Ala Ile Tyr Tyr Cys Thr  
 85 90 95  
 Ala His Thr Arg Gly Phe Leu Tyr Trp Gly Gln Gly Thr Leu Val Thr  
 100 105 110  
 Val Ser Ser  
 115

<210> SEQ ID NO 307  
 <211> LENGTH: 107  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 307

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly  
 1 5 10 15  
 Asp Arg Val Thr Ile Thr Cys Gln Ala Ser Gln Asp Ile Asn Ile Tyr  
 20 25 30  
 Leu Asn Trp Tyr Gln Gln Gln Pro Gly Lys Ala Pro Lys Leu Leu Ile  
 35 40 45  
 Tyr Asp Val Ser Asn Leu Val Pro Gly Val Pro Ser Arg Phe Ser Gly  
 50 55 60  
 Thr Gly Ser Gly Thr His Phe Ser Phe Thr Ile Ser Ser Leu Gln Pro  
 65 70 75 80  
 Glu Asp Ile Ala Thr Tyr Tyr Cys Gln His Tyr Asp Asp Leu Pro Leu



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|                                             |     |     |    |
|---------------------------------------------|-----|-----|----|
|                                             | 85  | 90  | 95 |
| Thr Phe Gly Gly Gly Thr Thr Val Ala Ile Arg |     |     |    |
|                                             | 100 | 105 |    |

<210> SEQ ID NO 308  
 <211> LENGTH: 115  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct  
  
 <400> SEQUENCE: 308

|                                                                 |     |     |     |
|-----------------------------------------------------------------|-----|-----|-----|
| Gln Val Gln Leu Gln Gln Trp Gly Ala Gly Leu Leu Lys Pro Ser Glu |     |     |     |
| 1                                                               | 5   | 10  | 15  |
| Thr Leu Ser Leu Thr Cys Ala Val Tyr Asn Gly Ser Phe Ser Ser Phe |     |     |     |
|                                                                 | 20  | 25  | 30  |
| Tyr Trp Ser Trp Ile Arg Gln Pro Pro Gly Lys Gly Leu Glu Trp Leu |     |     |     |
|                                                                 | 35  | 40  | 45  |
| Gly Glu Ile Asn His Arg Gly Arg Thr Asn Asn Asn Pro Ser Leu Ala |     |     |     |
|                                                                 | 50  | 55  | 60  |
| Ser Arg Val Thr Ile Ser Val Asp Thr Ser Lys Ser Gln Phe Ser Leu |     |     |     |
| 65                                                              | 70  | 75  | 80  |
| Arg Leu Thr Ser Val Thr Ala Ala Asp Thr Ala Thr Tyr Tyr Cys Thr |     |     |     |
|                                                                 | 85  | 90  | 95  |
| Ala His Thr Arg Gly Phe Leu Tyr Trp Gly Gln Gly Thr Arg Val Thr |     |     |     |
|                                                                 | 100 | 105 | 110 |
| Val Ser Ser                                                     |     |     |     |
|                                                                 | 115 |     |     |

<210> SEQ ID NO 309  
 <211> LENGTH: 107  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct  
  
 <400> SEQUENCE: 309

|                                                                 |     |     |    |
|-----------------------------------------------------------------|-----|-----|----|
| Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly |     |     |    |
| 1                                                               | 5   | 10  | 15 |
| Asp Arg Val Thr Ile Thr Cys Gln Ala Ser Gln Asp Ile Asn Asn Tyr |     |     |    |
|                                                                 | 20  | 25  | 30 |
| Leu Asn Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Asn Leu Leu Ile |     |     |    |
|                                                                 | 35  | 40  | 45 |
| Tyr Asp Val Phe Asn Leu Leu Pro Gly Val Pro Ser Arg Phe Ser Gly |     |     |    |
|                                                                 | 50  | 55  | 60 |
| Ser Gly Ser Gly Thr Asp Phe Ser Phe Thr Ile Ser Ser Leu Gln Pro |     |     |    |
| 65                                                              | 70  | 75  | 80 |
| Glu Asp Ile Ala Thr Tyr Tyr Cys Gln His Tyr Asp Asp Leu Pro Leu |     |     |    |
|                                                                 | 85  | 90  | 95 |
| Thr Phe Gly Gly Gly Thr Thr Val Ala Ile Arg                     |     |     |    |
|                                                                 | 100 | 105 |    |

<210> SEQ ID NO 310  
 <211> LENGTH: 115  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:

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&lt;223&gt; OTHER INFORMATION: Synthetic Construct

&lt;400&gt; SEQUENCE: 310

Gln Val Gln Leu Gln Gln Trp Gly Ala Gly Leu Leu Lys Pro Ser Glu  
 1 5 10 15

Thr Leu Ser Leu Thr Cys Ala Val Tyr Asn Gly Ser Phe Ser Ser Tyr  
 20 25 30

Tyr Trp Ser Trp Ile Arg Gln Pro Pro Gly Lys Gly Leu Glu Trp Leu  
 35 40 45

Gly Glu Ile Asn His Arg Gly Arg Thr Asn Tyr Asn Pro Ser Leu Thr  
 50 55 60

Ser Arg Val Thr Ile Ser Val Asp Thr Ser Lys Asn Gln Phe Ser Leu  
 65 70 75 80

Lys Leu Pro Ser Val Thr Ala Ala Asp Thr Ala Ile Tyr Tyr Cys Ser  
 85 90 95

Ala His Thr Arg Gly Phe Leu Tyr Trp Gly Gln Gly Thr Leu Val Thr  
 100 105 110

Val Ser Ser  
 115

&lt;210&gt; SEQ ID NO 311

&lt;211&gt; LENGTH: 107

&lt;212&gt; TYPE: PRT

&lt;213&gt; ORGANISM: Artificial Sequence

&lt;220&gt; FEATURE:

&lt;223&gt; OTHER INFORMATION: Synthetic Construct

&lt;400&gt; SEQUENCE: 311

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly  
 1 5 10 15

Asp Arg Val Thr Ile Thr Cys Gln Ala Ser Gln Asp Ile Asn Asn Tyr  
 20 25 30

Leu Asn Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile  
 35 40 45

Tyr Asp Ala Ser Asn Leu Val Pro Gly Val Pro Ser Arg Phe Ser Gly  
 50 55 60

Thr Gly Ser Gly Thr Asp Phe Ser Phe Thr Ile Ser Ser Leu Gln Pro  
 65 70 75 80

Glu Asp Ile Ala Thr Tyr Tyr Cys Gln His Tyr Tyr Asp Leu Pro Leu  
 85 90 95

Thr Phe Gly Gly Gly Thr Thr Val Ala Ile Ser  
 100 105

&lt;210&gt; SEQ ID NO 312

&lt;211&gt; LENGTH: 107

&lt;212&gt; TYPE: PRT

&lt;213&gt; ORGANISM: Artificial Sequence

&lt;220&gt; FEATURE:

&lt;223&gt; OTHER INFORMATION: Synthetic Construct

&lt;400&gt; SEQUENCE: 312

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly  
 1 5 10 15

Asp Arg Val Thr Ile Thr Cys Gln Ala Ser Gln Asp Ile Asn Asn Tyr  
 20 25 30

Leu Asn Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile

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|                                                                 |     |       |
|-----------------------------------------------------------------|-----|-------|
| 35                                                              | 40  | 45    |
| Tyr Asp Ala Ser Asn Leu Val Pro Gly Val Pro Ser Arg Phe Ser Gly |     |       |
| 50                                                              | 55  | 60    |
| Thr Gly Ser Gly Thr Asp Phe Ser Phe Thr Ile Ser Ser Leu Gln Pro |     |       |
| 65                                                              | 70  | 75 80 |
| Glu Asp Ile Ala Thr Tyr Tyr Cys Gln His Tyr Tyr Asp Leu Pro Leu |     |       |
|                                                                 | 85  | 90 95 |
| Thr Phe Gly Gly Gly Thr Thr Val Ala Ile Arg                     |     |       |
| 100                                                             | 105 |       |

<210> SEQ ID NO 313  
 <211> LENGTH: 116  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 313

|                                                                 |     |         |
|-----------------------------------------------------------------|-----|---------|
| Glu Ile Gln Leu Val Glu Ser Gly Gly Asp Leu Val Gln Pro Gly Gly |     |         |
| 1                                                               | 5   | 10 15   |
| Ser Leu Lys Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Asp Ser |     |         |
|                                                                 | 20  | 25 30   |
| Ala Met His Trp Val Arg Gln Ala Ser Gly Lys Gly Leu Glu Trp Val |     |         |
|                                                                 | 35  | 40 45   |
| Gly Arg Ile Arg Asn Lys Ala Asn Thr Tyr Ala Thr Ser Tyr Ala Ala |     |         |
|                                                                 | 50  | 55 60   |
| Ser Val Lys Gly Arg Phe Thr Val Ser Thr Asp Asp Ser Lys Thr Thr |     |         |
| 65                                                              | 70  | 75 80   |
| Ala Tyr Leu Gln Met Arg Tyr Leu Lys Ile Glu Asp Thr Ala Ile Tyr |     |         |
|                                                                 | 85  | 90 95   |
| Tyr Cys Thr Ser Asp Gln Phe Asp Tyr Trp Gly Gln Gly Thr Leu Val |     |         |
|                                                                 | 100 | 105 110 |
| Thr Val Ser Ser                                                 |     |         |
| 115                                                             |     |         |

<210> SEQ ID NO 314  
 <211> LENGTH: 112  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 314

|                                                                 |    |       |
|-----------------------------------------------------------------|----|-------|
| Asp Val Val Met Thr Gln Ser Pro Leu Ser Leu Ser Ala Thr Leu Gly |    |       |
| 1                                                               | 5  | 10 15 |
| Gln Pro Ala Ser Ile Ser Cys Arg Ser Ser Gln Ser Leu Val Tyr Ser |    |       |
|                                                                 | 20 | 25 30 |
| Asp Gly Ile Thr Tyr Leu Thr Trp Phe Gln Gln Arg Pro Gly Gln Ser |    |       |
|                                                                 | 35 | 40 45 |
| Pro Thr Arg Leu Ile Tyr Gln Val Ser Asn Arg Asp Ser Gly Val Pro |    |       |
|                                                                 | 50 | 55 60 |
| Asp Arg Phe Asn Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Asn Ile |    |       |
| 65                                                              | 70 | 75 80 |
| Ser Arg Val Glu Ala Glu Asp Val Gly Phe Tyr Tyr Cys Met Glu Gly |    |       |
|                                                                 | 85 | 90 95 |

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|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Thr | His | Trp | Pro | Trp | Thr | Phe | Gly | Gln | Gly | Thr | Lys | Val | Glu | Leu | Lys |
|     |     |     | 100 |     |     |     |     | 105 |     |     |     |     | 110 |     |     |

<210> SEQ ID NO 315  
 <211> LENGTH: 114  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 315

|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Gln | Val | Gln | Leu | Val | Glu | Ser | Gly | Gly | Gly | Leu | Val | Lys | Pro | Gly | Gly |
| 1   |     |     | 5   |     |     |     |     | 10  |     |     |     |     | 15  |     |     |
| Ser | Leu | Arg | Leu | Ser | Cys | Ala | Ala | Ser | Gly | Phe | Thr | Phe | Ser | Asp | Tyr |
|     |     | 20  |     |     |     |     |     | 25  |     |     |     |     | 30  |     |     |
| Tyr | Met | Ser | Trp | Ile | Arg | Gln | Ala | Pro | Gly | Lys | Gly | Leu | Glu | Trp | Ile |
|     |     | 35  |     |     |     | 40  |     |     |     |     |     | 45  |     |     |     |
| Ser | Phe | Ile | Ser | Arg | Thr | Gly | Ser | Thr | Lys | Tyr | Tyr | Ala | Asp | Ser | Val |
|     | 50  |     |     |     |     | 55  |     |     |     | 60  |     |     |     |     |     |
| Lys | Gly | Arg | Phe | Thr | Ile | Ser | Arg | Asp | Asn | Ala | Lys | Asn | Ser | Leu | Asn |
| 65  |     |     |     | 70  |     |     |     |     | 75  |     |     |     |     | 80  |     |
| Leu | Gln | Met | Ser | Ser | Leu | Arg | Ala | Glu | Asp | Thr | Ala | Val | Tyr | Phe | Cys |
|     |     |     | 85  |     |     |     |     | 90  |     |     |     |     | 95  |     |     |
| Ala | Arg | Glu | Pro | Leu | Gln | His | Trp | Gly | Gln | Gly | Thr | Leu | Val | Thr | Val |
|     |     |     | 100 |     |     |     |     | 105 |     |     |     |     | 110 |     |     |

Ser Ser

<210> SEQ ID NO 316  
 <211> LENGTH: 113  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 316

|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Asn | Val | Val | Leu | Thr | Gln | Thr | Pro | Ile | Ser | Ser | Pro | Val | Ile | Leu | Gly |
| 1   |     |     | 5   |     |     |     |     | 10  |     |     |     |     | 15  |     |     |
| Gln | Pro | Ala | Ser | Ile | Ser | Cys | Arg | Ser | Ser | Gln | Ser | Leu | Val | His | Ser |
|     |     | 20  |     |     |     |     |     | 25  |     |     |     | 30  |     |     |     |
| Asp | Gly | Asp | Thr | Tyr | Leu | Ser | Trp | Leu | His | Gln | Arg | Pro | Gly | Gln | Pro |
|     |     | 35  |     |     |     |     | 40  |     |     |     |     | 45  |     |     |     |
| Pro | Arg | Leu | Leu | Ile | Tyr | Thr | Val | Ser | Asn | Arg | Val | Ser | Gly | Val | Pro |
|     | 50  |     |     |     |     | 55  |     |     |     | 60  |     |     |     |     |     |
| Asp | Arg | Phe | Ser | Gly | Ser | Gly | Ala | Gly | Thr | Asp | Phe | Thr | Leu | Lys | Ile |
| 65  |     |     |     | 70  |     |     |     |     | 75  |     |     |     | 80  |     |     |
| Ser | Arg | Val | Glu | Ala | Glu | Asp | Val | Gly | Val | Tyr | Tyr | Cys | Met | Gln | Gly |
|     |     |     | 85  |     |     |     |     | 90  |     |     |     |     | 95  |     |     |
| Thr | Gln | Phe | Pro | Ala | Leu | Thr | Phe | Gly | Gly | Gly | Thr | Lys | Val | Glu | Ile |
|     |     |     | 100 |     |     |     |     | 105 |     |     |     |     | 110 |     |     |

Lys

<210> SEQ ID NO 317  
 <211> LENGTH: 114  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct

-continued

&lt;400&gt; SEQUENCE: 317

Gln Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Lys Pro Gly Gly  
 1 5 10 15  
 Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Asp Tyr  
 20 25 30  
 Tyr Met Ser Trp Ile Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val  
 35 40 45  
 Ser Tyr Ile Ser Arg Thr Gly Ser Thr Lys Tyr Tyr Ala Asp Ser Val  
 50 55 60  
 Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Asn Ser Leu Asn  
 65 70 75 80  
 Leu Gln Met Ser Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Phe Cys  
 85 90 95  
 Ala Arg Glu Pro Leu Gln His Trp Gly Gln Gly Thr Leu Val Thr Val  
 100 105 110  
 Ser Ser

&lt;210&gt; SEQ ID NO 318

&lt;211&gt; LENGTH: 113

&lt;212&gt; TYPE: PRT

&lt;213&gt; ORGANISM: Artificial Sequence

&lt;220&gt; FEATURE:

&lt;223&gt; OTHER INFORMATION: Synthetic Construct

&lt;400&gt; SEQUENCE: 318

Lys Ile Val Leu Thr Gln Thr Pro Leu Ser Ser Pro Val Thr Leu Gly  
 1 5 10 15  
 Gln Pro Ala Ser Ile Ser Cys Arg Ser Ser Gln Ser Leu Val His Ser  
 20 25 30  
 Asp Gly Asp Thr Tyr Leu Ser Trp Leu His Gln Arg Pro Gly Gln Pro  
 35 40 45  
 Pro Arg Leu Leu Ile Tyr Thr Ile Ser Asn Arg Leu Ser Gly Val Pro  
 50 55 60  
 Asp Arg Phe Ser Gly Ser Gly Ala Gly Thr Asp Phe Thr Leu Lys Ile  
 65 70 75 80  
 Ser Arg Val Glu Ala Glu Asp Val Gly Val Tyr Tyr Cys Met Gln Gly  
 85 90 95  
 Thr Gln Phe Pro Ala Leu Thr Phe Gly Gly Gly Thr Lys Val Glu Ile  
 100 105 110  
 Lys

&lt;210&gt; SEQ ID NO 319

&lt;211&gt; LENGTH: 115

&lt;212&gt; TYPE: PRT

&lt;213&gt; ORGANISM: Artificial Sequence

&lt;220&gt; FEATURE:

&lt;223&gt; OTHER INFORMATION: Synthetic Construct

&lt;400&gt; SEQUENCE: 319

Glu Met Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly  
 1 5 10 15  
 Ser Leu Lys Val Ser Cys Ala Ala Ser Gly Phe Thr Leu Ser Asp Ser  
 20 25 30  
 Ser Ile His Trp Val Arg Gln Ala Ser Gly Lys Gly Leu Glu Trp Ile

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|                                                                 |     |         |
|-----------------------------------------------------------------|-----|---------|
| 35                                                              | 40  | 45      |
| Gly Arg Ile Arg Ser Lys Pro Tyr Ser Tyr Ala Thr Ala Tyr Ala Ala |     |         |
| 50                                                              | 55  | 60      |
| Ser Val Lys Gly Arg Phe Thr Ile Ser Arg Asp Asp Ser Lys Asn Thr |     |         |
| 65                                                              | 70  | 75 80   |
| Ala Phe Leu Gln Met Ser Gly Leu Lys Thr Glu Asp Thr Ala Val Tyr |     |         |
|                                                                 | 85  | 90 95   |
| Tyr Cys Thr Gly Gly Asp Asp Phe Trp Gly Gln Gly Thr Leu Val Thr |     |         |
|                                                                 | 100 | 105 110 |
| Val Ser Ser                                                     |     |         |
| 115                                                             |     |         |

<210> SEQ ID NO 320  
 <211> LENGTH: 106  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 320

|                                                                 |     |       |
|-----------------------------------------------------------------|-----|-------|
| Asp Ile Gln Met Thr Gln Ser Pro Ser Thr Leu Ser Ala Ser Val Gly |     |       |
| 1                                                               | 5   | 10 15 |
| Asp Arg Ile Thr Ile Thr Cys Arg Ala Ser Gln Ser Ile Ser Thr Trp |     |       |
|                                                                 | 20  | 25 30 |
| Leu Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile |     |       |
|                                                                 | 35  | 40 45 |
| Tyr Lys Ala Ser Ser Leu Glu Gly Gly Val Pro Ser Arg Phe Ser Gly |     |       |
|                                                                 | 50  | 55 60 |
| Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro |     |       |
| 65                                                              | 70  | 75 80 |
| Asp Asp Phe Ala Thr Tyr Tyr Cys Gln Gln Tyr Asn Thr Tyr Trp Thr |     |       |
|                                                                 | 85  | 90 95 |
| Phe Gly Gln Gly Thr Lys Val Glu Ile Lys                         |     |       |
|                                                                 | 100 | 105   |

<210> SEQ ID NO 321  
 <211> LENGTH: 121  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 321

|                                                                 |    |       |
|-----------------------------------------------------------------|----|-------|
| Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly |    |       |
| 1                                                               | 5  | 10 15 |
| Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr |    |       |
|                                                                 | 20 | 25 30 |
| Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val |    |       |
|                                                                 | 35 | 40 45 |
| Ser Ala Ile Ser Gly Ser Gly Gly Ser Thr Tyr Tyr Ala Asp Ser Val |    |       |
|                                                                 | 50 | 55 60 |
| Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr |    |       |
| 65                                                              | 70 | 75 80 |
| Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys |    |       |
|                                                                 | 85 | 90 95 |

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Ala Ile Pro Tyr Asn Trp Asn Tyr Gly Gly Ala Phe Asp Ile Trp Gly  
                   100                  105                  110

Gln Gly Thr Met Val Thr Val Ser Ser  
           115                  120

<210> SEQ ID NO 322  
 <211> LENGTH: 108  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 322

Glu Ile Val Met Thr Gln Ser Pro Ala Thr Leu Ser Val Ser Pro Gly  
 1                  5                  10                  15

Lys Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser Val Ser Ser Asn  
           20                  25                  30

Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile  
           35                  40                  45

Tyr Gly Ala Ser Thr Arg Ala Thr Gly Ile Pro Ala Arg Phe Ser Gly  
           50                  55                  60

Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Ser Ser Leu Gln Ser  
 65                  70                  75                  80

Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Asn Asn Trp Pro Pro  
           85                  90                  95

Tyr Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile Lys  
           100                  105

<210> SEQ ID NO 323  
 <211> LENGTH: 119  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 323

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ala  
 1                  5                  10                  15

Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Thr Phe Thr Gly Tyr  
           20                  25                  30

Tyr Met His Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Glu Trp Met  
           35                  40                  45

Gly Trp Ile Asn Pro Asn Ser Val Gly Thr Asn Phe Ala Gln Lys Phe  
           50                  55                  60

Gln Gly Arg Val Thr Met Thr Arg Asp Thr Ser Ile Ser Thr Ala Tyr  
 65                  70                  75                  80

Met Glu Leu Ser Arg Leu Arg Ser Asp Asp Thr Ala Val Tyr Tyr Cys  
           85                  90                  95

Thr Arg Asp Gly Ala Ala Ala Gly Leu Phe Asp Tyr Trp Gly Gln Gly  
           100                  105                  110

Thr Leu Val Thr Val Ser Ser  
           115

<210> SEQ ID NO 324  
 <211> LENGTH: 107  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence

-continued

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 324

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly  
1 5 10 15  
Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Gly Ile Asn Asn Tyr  
20 25 30  
Leu Ala Trp Tyr Gln Gln Lys Pro Gly Lys Val Pro Lys Leu Leu Ile  
35 40 45  
Tyr Ala Ala Ser Thr Leu Gln Ser Gly Val Pro Ser Arg Phe Ser Gly  
50 55 60  
Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro  
65 70 75 80  
Glu Asp Val Ala Thr Tyr Tyr Cys Gln Lys Phe Asn Ser Ala Pro Leu  
85 90 95  
Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys  
100 105

<210> SEQ ID NO 325

<211> LENGTH: 116

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 325

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly  
1 5 10 15  
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Thr Tyr  
20 25 30  
Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val  
35 40 45  
Ser Ala Ile Ser Gly Ser Gly Gly Ser Thr Tyr Tyr Ala Asp Ser Val  
50 55 60  
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr  
65 70 75 80  
Leu Gln Met Lys Ser Leu Arg Val Glu Asp Thr Ala Val Tyr Tyr Cys  
85 90 95  
Ala Asn Ser Pro Ser Trp Phe Asp Pro Trp Gly Gln Gly Thr Leu Val  
100 105 110  
Thr Val Ser Ser  
115

<210> SEQ ID NO 326

<211> LENGTH: 108

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 326

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly  
1 5 10 15  
Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Ser Ile Ser Ser Tyr  
20 25 30



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Leu Asn Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile
   35                               40           45

Tyr Ala Ala Ser Ser Leu Gln Ser Gly Val Pro Ser Arg Phe Ser Gly
   50                               55           60

Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
  65                               70           75           80

Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln Ser Tyr Ser Thr Pro Pro
   85                               90           95

Ile Thr Phe Gly Gln Gly Thr Arg Leu Glu Ile Lys
   100                               105

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<210> SEQ ID NO 327
<211> LENGTH: 119
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

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<400> SEQUENCE: 327

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Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Arg
 1                               10           15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Asp Asp Tyr
 20                               25           30

Gly Met His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
 35                               40           45

Ser Val Ile Ser Trp Asn Ser Ala Thr Ile Asp Tyr Ala Asp Ser Val
 50                               55           60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Asn Ser Leu Phe
 65                               70           75           80

Leu Gln Met Asn Ser Leu Arg Pro Glu Asp Thr Ala Leu Tyr Tyr Cys
 85                               90           95

Ala Lys Lys Asn Ser Leu Gly Trp Phe Phe Asp Tyr Trp Gly Gln Gly
 100                               105           110

Thr Leu Val Thr Val Ser Ser
 115

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<210> SEQ ID NO 328
<211> LENGTH: 113
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

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<400> SEQUENCE: 328

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Asp Ile Val Met Thr Gln Ser Pro Asp Ser Leu Ala Val Ser Leu Gly
 1                               10           15

Glu Arg Ala Thr Ile Asn Cys Lys Ser Ser Gln Ser Val Leu Tyr Ser
 20                               25           30

Ser Asn Asp Asn Asn Tyr Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln
 35                               40           45

Pro Pro Lys Leu Leu Ile Tyr Trp Ala Ser Thr Arg Glu Ser Gly Val
 50                               55           60

Pro Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr
 65                               70           75           80

Ile Ser Ser Leu Gln Ala Glu Asp Val Ala Val Tyr Tyr Cys Gln Gln
 85                               90           95

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Tyr Tyr Ser Thr Pro Leu Thr Phe Gly Gly Gly Thr Lys Val Glu Ile  
 100 105 110

Lys

<210> SEQ ID NO 329  
 <211> LENGTH: 117  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 329

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ala  
 1 5 10 15  
 Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Thr Phe Thr Ser Tyr  
 20 25 30  
 Gly Ile Ser Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Glu Trp Met  
 35 40 45  
 Gly Trp Ile Ser Ala Tyr Asn Gly Asn Thr Ile Tyr Ala Gln Lys Phe  
 50 55 60  
 Gln Gly Arg Val Thr Met Thr Thr Asp Thr Ser Thr Ser Thr Ala Tyr  
 65 70 75 80  
 Met Glu Leu Arg Ser Leu Arg Ser Asp Asp Thr Ala Val Tyr Tyr Cys  
 85 90 95  
 Ala Arg Asp Ile Asn Trp Tyr Phe Asp Leu Trp Gly Arg Gly Thr Leu  
 100 105 110  
 Val Thr Val Ser Ser  
 115

<210> SEQ ID NO 330  
 <211> LENGTH: 107  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 330

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly  
 1 5 10 15  
 Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Gly Ile Arg Asn Asp  
 20 25 30  
 Leu Gly Trp Phe Gln Gln Lys Pro Gly Lys Ala Pro Lys Arg Leu Ile  
 35 40 45  
 Tyr Ala Ala Ser Ser Leu Gln Ser Gly Val Pro Ser Arg Phe Ser Gly  
 50 55 60  
 Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro  
 65 70 75 80  
 Glu Asp Phe Ala Thr Tyr Tyr Cys Leu Gln His Asn Ser Tyr Pro Trp  
 85 90 95  
 Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys  
 100 105

<210> SEQ ID NO 331  
 <211> LENGTH: 119  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence

-continued

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 331

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Arg  
1 5 10 15  
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Asp Asp Tyr  
20 25 30  
Gly Met His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val  
35 40 45  
Ser Val Ile Ser Trp Asn Gly Ala Thr Ile Asp Tyr Ala Asp Ser Val  
50 55 60  
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Asn Ser Leu Phe  
65 70 75 80  
Leu Gln Met Asn Ser Leu Arg Pro Glu Asp Thr Ala Leu Tyr Tyr Cys  
85 90 95  
Ala Lys Lys Asn Ser Leu Gly Trp Phe Phe Asp Tyr Trp Gly Gln Gly  
100 105 110  
Thr Leu Val Thr Val Ser Ser  
115

<210> SEQ ID NO 332

<211> LENGTH: 107

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 332

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Phe Val Gly  
1 5 10 15  
Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Gly Ile Arg Asn Asp  
20 25 30  
Leu Gly Trp Tyr Gln Gln Lys Pro Gly Lys Thr Pro Lys Arg Leu Ile  
35 40 45  
Tyr Thr Ala Ser Ser Leu Arg Ser Gly Val Pro Ser Arg Phe Ser Gly  
50 55 60  
Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro  
65 70 75 80  
Glu Asp Phe Ala Thr Tyr Tyr Cys Leu Gln His Asn Asn Tyr Pro Tyr  
85 90 95  
Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile Lys  
100 105

<210> SEQ ID NO 333

<211> LENGTH: 122

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 333

Glu Val Gln Leu Val Glu Ser Gly Gly Asn Leu Val Gln Pro Gly Gly  
1 5 10 15  
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Asn Tyr  
20 25 30

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|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Ala | Met | Ser | Trp | Val | Arg | Gln | Ala | Pro | Gly | Lys | Gly | Leu | Glu | Trp | Val |
|     | 35  |     |     |     |     |     | 40  |     |     |     |     | 45  |     |     |     |
| Ser | Gly | Ile | Thr | Gly | Asn | Gly | Val | Asn | Thr | Tyr | Tyr | Ser | Val | Ser | Val |
|     | 50  |     |     |     |     | 55  |     |     |     |     | 60  |     |     |     |     |
| Lys | Gly | Arg | Phe | Thr | Ile | Ser | Arg | Asp | Asn | Ser | Lys | Asn | Thr | Leu | Phe |
|     | 65  |     |     |     | 70  |     |     |     |     | 75  |     |     |     |     | 80  |
| Leu | Glu | Met | Asn | Ser | Leu | Arg | Ala | Glu | Asp | Thr | Ala | Ile | Tyr | Tyr | Cys |
|     |     |     | 85  |     |     |     |     |     | 90  |     |     |     |     | 95  |     |
| Val | Lys | Glu | Arg | Gly | His | Ser | Trp | Phe | Gly | Asp | Trp | Phe | Asp | Pro | Trp |
|     |     |     | 100 |     |     |     |     | 105 |     |     |     |     | 110 |     |     |
| Gly | Gln | Gly | Thr | Leu | Val | Thr | Val | Ser | Ser |     |     |     |     |     |     |
|     | 115 |     |     |     |     |     | 120 |     |     |     |     |     |     |     |     |

<210> SEQ ID NO 334  
 <211> LENGTH: 107  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 334

|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Asp | Ile | Gln | Met | Thr | Gln | Ser | Pro | Ser | Ser | Leu | Ser | Ala | Ser | Val | Gly |
| 1   |     |     |     | 5   |     |     |     |     |     | 10  |     |     |     | 15  |     |
| Asp | Arg | Val | Thr | Ile | Thr | Cys | Arg | Ser | Ser | Gln | Ser | Val | Ile | Ser | Tyr |
|     |     | 20  |     |     |     |     |     | 25  |     |     |     |     | 30  |     |     |
| Leu | Asn | Trp | Tyr | Gln | Gln | Lys | Pro | Gly | Lys | Ala | Pro | Lys | Leu | Leu | Ile |
|     |     | 35  |     |     |     |     | 40  |     |     |     |     | 45  |     |     |     |
| Phe | Gly | Ala | Ser | Ser | Leu | Ile | Ser | Gly | Val | Pro | Ser | Arg | Phe | Ser | Gly |
|     | 50  |     |     |     |     | 55  |     |     |     |     | 60  |     |     |     |     |
| Ser | Gly | Ser | Gly | Thr | Asp | Phe | Thr | Leu | Thr | Ile | Val | Ser | Leu | Gln | Pro |
|     | 65  |     |     |     | 70  |     |     |     |     | 75  |     |     |     |     | 80  |
| Glu | Asp | Phe | Ala | Val | Tyr | Phe | Cys | Gln | Gln | Asn | Tyr | Leu | Pro | Pro | Leu |
|     |     |     | 85  |     |     |     |     |     | 90  |     |     |     |     | 95  |     |
| Thr | Phe | Gly | Gly | Gly | Thr | Lys | Val | Glu | Ile | Lys |     |     |     |     |     |
|     |     | 100 |     |     |     |     |     | 105 |     |     |     |     |     |     |     |

<210> SEQ ID NO 335  
 <211> LENGTH: 122  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 335

|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Glu | Val | Gln | Leu | Val | Glu | Ser | Gly | Gly | Asn | Leu | Val | Gln | Pro | Gly | Gly |
| 1   |     |     |     | 5   |     |     |     |     | 10  |     |     |     |     | 15  |     |
| Ser | Leu | Arg | Leu | Ser | Cys | Ala | Ala | Ser | Gly | Phe | Thr | Phe | Ser | Asn | Tyr |
|     |     | 20  |     |     |     |     |     | 25  |     |     |     |     | 30  |     |     |
| Ala | Met | Ser | Trp | Val | Arg | Gln | Ala | Pro | Gly | Lys | Gly | Leu | Glu | Trp | Val |
|     |     | 35  |     |     |     |     | 40  |     |     |     |     | 45  |     |     |     |
| Ser | Gly | Ile | Thr | Gly | Asn | Gly | Ile | Lys | Thr | Tyr | Tyr | Ser | Val | Ser | Ala |
|     | 50  |     |     |     |     | 55  |     |     |     |     | 60  |     |     |     |     |
| Lys | Gly | Arg | Phe | Thr | Ile | Ser | Arg | Asp | Asn | Ser | Lys | Asn | Thr | Leu | Phe |
|     | 65  |     |     |     | 70  |     |     |     |     | 75  |     |     |     |     | 80  |
| Leu | Gln | Met | Asn | Ser | Leu | Arg | Ala | Glu | Asp | Thr | Ala | Ile | Tyr | Tyr | Cys |
|     |     |     | 85  |     |     |     |     |     | 90  |     |     |     |     | 95  |     |

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Val Lys Glu Arg Gly His Ser Trp Phe Gly Asp Trp Phe Asp Pro Trp  
                   100                                  105                                  110

Gly Gln Gly Thr Leu Val Thr Val Ser Ser  
                   115                                  120

<210> SEQ ID NO 336  
 <211> LENGTH: 112  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 336

Asp Ile Val Met Thr Gln Thr Pro Leu Ser Ser Pro Val Thr Leu Gly  
 1                  5                                  10                                  15

Gln Pro Ala Ser Ile Ser Cys Arg Ser Ser Gln Gly Leu Val Asn Ser  
                   20                                  25                                  30

Asp Gly Asn Thr Tyr Leu Ser Trp Leu His Gln Arg Pro Gly Gln Pro  
                   35                                  40                                  45

Pro Arg Leu Leu Ile Tyr Glu Ile Ser Asn Arg Phe Ser Gly Val Pro  
                   50                                  55                                  60

Asp Arg Phe Ser Gly Ser Gly Ala Gly Thr Asp Phe Thr Leu Lys Ile  
 65                  70                                  75                                  80

Ser Arg Val Glu Ala Glu Asp Val Gly Phe Tyr Tyr Cys Met Gln Ser  
                   85                                  90                                  95

Thr Gln Phe Pro Leu Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys  
                   100                                  105                                  110

<210> SEQ ID NO 337  
 <211> LENGTH: 122  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 337

Glu Val Gln Leu Val Glu Ser Gly Gly Asn Leu Val Gln Pro Gly Gly  
 1                  5                                  10                                  15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Asn Tyr  
                   20                                  25                                  30

Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val  
                   35                                  40                                  45

Ser Gly Ile Thr Gly Asn Gly Ile Asn Thr Tyr Tyr Ser Val Ser Val  
                   50                                  55                                  60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Val Phe  
 65                  70                                  75                                  80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Ile Tyr Tyr Cys  
                   85                                  90                                  95

Val Lys Glu Arg Gly His Ser Trp Phe Gly Asp Trp Phe Asp Pro Trp  
                   100                                  105                                  110

Gly Gln Gly Thr Leu Val Thr Val Ser Ser  
                   115                                  120

<210> SEQ ID NO 338  
 <211> LENGTH: 107  
 <212> TYPE: PRT

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<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct  
  
<400> SEQUENCE: 338  
Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly  
1 5 10 15  
Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Gly Ile Arg Asn Asp  
20 25 30  
Leu Gly Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Arg Leu Ile  
35 40 45  
Tyr Ala Ala Ser Ser Leu Gln Ser Gly Val Pro Ser Arg Phe Ser Gly  
50 55 60  
Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro  
65 70 75 80  
Glu Asp Phe Ala Thr Tyr Tyr Cys Leu Gln His Asn Ser Tyr Pro Trp  
85 90 95  
Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys  
100 105

<210> SEQ ID NO 339  
<211> LENGTH: 123  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct  
  
<400> SEQUENCE: 339  
Gln Val Gln Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Arg  
1 5 10 15  
Ser Leu Arg Leu Ser Cys Glu Val Ser Gly Phe Thr Phe Ser Asn Tyr  
20 25 30  
Gly Met His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val  
35 40 45  
Ala Val Val Ser Tyr Asp Gly Ser Asn Lys Tyr Tyr Thr Asp Ser Val  
50 55 60  
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Ser  
65 70 75 80  
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys  
85 90 95  
Ala Lys Asp Arg Gly Ile Thr Gly Thr Ser Gly Gly Val Phe Asp Ile  
100 105 110  
Trp Gly Gln Gly Thr Met Val Thr Val Ser Ser  
115 120

<210> SEQ ID NO 340  
<211> LENGTH: 107  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct  
  
<400> SEQUENCE: 340

Ala Ile Arg Met Thr Gln Ser Pro Ser Ser Phe Ser Ala Thr Thr Gly  
1 5 10 15  
Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Gly Ile Ser Ser His  
20 25 30

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<210> SEQ ID NO 341
<211> LENGTH: 114
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 341
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<210> SEQ ID NO 342
<211> LENGTH: 112
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 342
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|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Asp | Val | Val | Met | Thr | Gln | Ser | Leu | Leu | Ser | Leu | Ser | Val | Thr | Leu | Gly |
| 1   |     |     |     | 5   |     |     |     |     | 10  |     |     |     |     | 15  |     |
| Gln | Pro | Ala | Ser | Ile | Ser | Cys | Arg | Ser | Ser | Gln | Ser | Leu | Val | Tyr | Ser |
|     |     |     | 20  |     |     |     |     | 25  |     |     |     |     | 30  |     |     |
| Asp | Gly | Asp | Thr | Tyr | Leu | Ser | Trp | Phe | Gln | Gln | Arg | Pro | Gly | Gln | Ser |
|     |     | 35  |     |     |     |     | 40  |     |     |     |     | 45  |     |     |     |
| Pro | Arg | Arg | Leu | Ile | Tyr | Lys | Val | Ser | His | Arg | Asp | Ser | Gly | Val | Pro |
|     | 50  |     |     |     |     | 55  |     |     |     |     | 60  |     |     |     |     |
| Asp | Arg | Phe | Arg | Gly | Ser | Gly | Ser | Gly | Thr | Asp | Phe | Thr | Leu | Lys | Ile |
| 65  |     |     |     |     | 70  |     |     |     |     | 75  |     |     |     | 80  |     |
| Ser | Arg | Val | Glu | Ala | Glu | Asp | Val | Gly | Ile | Tyr | Tyr | Cys | Met | Gln | Gly |
|     |     |     | 85  |     |     |     |     | 90  |     |     |     |     |     | 95  |     |

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Thr His Trp Pro Pro Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys  
 100 105 110

<210> SEQ ID NO 343  
 <211> LENGTH: 449  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 343

Gln Val His Leu Val Glu Ser Gly Gly Gly Val Val Gln Pro Gly Arg  
 1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Asn Tyr  
 20 25 30

Gly Ile His Trp Val Arg Gln Val Pro Gly Lys Gly Leu Glu Trp Val  
 35 40 45

Ala Val Ile Ser Tyr Asp Gly Asn Asn Lys Tyr Tyr Gly Asp Ser Val  
 50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr  
 65 70 75 80

Leu Gln Met Asn Ser Leu Arg Pro Glu Asp Thr Ala Ile Tyr Tyr Cys  
 85 90 95

Ala Lys Asp Arg Gly Leu Asp Thr Ala Gly Asp Tyr Phe Asp Tyr Trp  
 100 105 110

Gly Gln Gly Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro  
 115 120 125

Ser Val Phe Pro Leu Ala Pro Cys Ser Arg Ser Thr Ser Glu Ser Thr  
 130 135 140

Ala Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr  
 145 150 155 160

Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro  
 165 170 175

Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr  
 180 185 190

Val Pro Ser Ser Ser Leu Gly Thr Lys Thr Tyr Thr Cys Asn Val Asp  
 195 200 205

His Lys Pro Ser Asn Thr Lys Val Asp Lys Arg Val Glu Ser Lys Tyr  
 210 215 220

Gly Pro Pro Cys Pro Pro Cys Pro Ala Pro Glu Phe Leu Gly Gly Pro  
 225 230 235 240

Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser  
 245 250 255

Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser Gln Glu Asp  
 260 265 270

Pro Glu Val Gln Phe Asn Trp Tyr Val Asp Gly Val Glu Val His Asn  
 275 280 285

Ala Lys Thr Lys Pro Arg Glu Glu Gln Phe Asn Ser Thr Tyr Arg Val  
 290 295 300

Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys Glu  
 305 310 315 320

Tyr Lys Cys Lys Val Ser Asn Lys Gly Leu Pro Ser Ser Ile Glu Lys  
 325 330 335



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Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr  
 340 345 350  
 Leu Pro Pro Ser Gln Glu Glu Met Thr Lys Asn Gln Val Ser Leu Thr  
 355 360 365  
 Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu  
 370 375 380  
 Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu  
 385 390 395 400  
 Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Arg Leu Thr Val Asp Lys  
 405 410 415  
 Ser Arg Trp Gln Glu Gly Asn Val Phe Ser Cys Ser Val Met His Glu  
 420 425 430  
 Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Leu Gly  
 435 440 445

Lys

<210> SEQ ID NO 344  
 <211> LENGTH: 213  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct

&lt;400&gt; SEQUENCE: 344

Glu Ile Val Met Thr Gln Ser Pro Ala Thr Leu Ser Val Ser Thr Gly  
 1 5 10 15  
 Glu Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Asn Ile Ser Ser Lys  
 20 25 30  
 Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile  
 35 40 45  
 Tyr Gly Ala Ser Thr Arg Ala Ser Gly Ile Pro Ala Arg Phe Trp Gly  
 50 55 60  
 Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Ser Ser Leu Gln Ser  
 65 70 75 80  
 Glu Asp Phe Ala Val Phe Tyr Cys Leu Gln Tyr Asn Asn Trp Trp Thr  
 85 90 95  
 Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Arg Thr Val Ala Ala Pro  
 100 105 110  
 Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser Gly Thr  
 115 120 125  
 Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu Ala Lys  
 130 135 140  
 Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly Asn Ser Gln Glu  
 145 150 155 160  
 Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu Ser Ser  
 165 170 175  
 Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys His Lys Val Tyr Ala  
 180 185 190  
 Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro Val Thr Lys Ser Phe  
 195 200 205  
 Asn Arg Gly Glu Cys  
 210

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<210> SEQ ID NO 345  
<211> LENGTH: 442  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 345

Gln Val Gln Leu Gln Gln Trp Gly Ala Gly Leu Leu Lys Pro Ser Glu  
1 5 10 15  
Thr Leu Ser Leu Thr Cys Ala Val Tyr Asn Gly Ser Phe Ser Ser Tyr  
20 25 30  
Tyr Trp Asn Trp Ile Arg Gln Pro Pro Gly Lys Gly Leu Glu Trp Leu  
35 40 45  
Gly Glu Ile Asn His Arg Gly Ser Thr Asn Tyr Asn Pro Ser Leu Thr  
50 55 60  
Ser Arg Val Thr Ile Ser Val Asp Thr Ser Lys Asn Gln Phe Ser Leu  
65 70 75 80  
Lys Leu Thr Ser Val Thr Ala Ala Asp Thr Ala Ile Tyr Tyr Cys Thr  
85 90 95  
Ala His Thr Arg Gly Phe Leu Tyr Trp Gly Gln Gly Thr Leu Val Thr  
100 105 110  
Val Ser Ser Ala Ser Thr Lys Gly Pro Ser Val Phe Pro Leu Ala Pro  
115 120 125  
Cys Ser Arg Ser Thr Ser Glu Ser Thr Ala Ala Leu Gly Cys Leu Val  
130 135 140  
Lys Asp Tyr Phe Pro Glu Pro Val Thr Val Ser Trp Asn Ser Gly Ala  
145 150 155 160  
Leu Thr Ser Gly Val His Thr Phe Pro Ala Val Leu Gln Ser Ser Gly  
165 170 175  
Leu Tyr Ser Leu Ser Ser Val Val Thr Val Pro Ser Ser Ser Leu Gly  
180 185 190  
Thr Lys Thr Tyr Thr Cys Asn Val Asp His Lys Pro Ser Asn Thr Lys  
195 200 205  
Val Asp Lys Arg Val Glu Ser Lys Tyr Gly Pro Pro Cys Pro Pro Cys  
210 215 220  
Pro Ala Pro Glu Phe Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro  
225 230 235 240  
Lys Pro Lys Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys  
245 250 255  
Val Val Val Asp Val Ser Gln Glu Asp Pro Glu Val Gln Phe Asn Trp  
260 265 270  
Tyr Val Asp Gly Val Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu  
275 280 285  
Glu Gln Phe Asn Ser Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu  
290 295 300  
His Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn  
305 310 315 320  
Lys Gly Leu Pro Ser Ser Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly  
325 330 335  
Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Gln Glu Glu  
340 345 350

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|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Met | Thr | Lys | Asn | Gln | Val | Ser | Leu | Thr | Cys | Leu | Val | Lys | Gly | Phe | Tyr |
|     |     | 355 |     |     |     |     | 360 |     |     |     |     | 365 |     |     |     |
|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| Pro | Ser | Asp | Ile | Ala | Val | Glu | Trp | Glu | Ser | Asn | Gly | Gln | Pro | Glu | Asn |
|     | 370 |     |     |     |     | 375 |     |     |     |     | 380 |     |     |     |     |
|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| Asn | Tyr | Lys | Thr | Thr | Pro | Pro | Val | Leu | Asp | Ser | Asp | Gly | Ser | Phe | Phe |
| 385 |     |     |     |     | 390 |     |     |     |     | 395 |     |     |     |     | 400 |
|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| Leu | Tyr | Ser | Arg | Leu | Thr | Val | Asp | Lys | Ser | Arg | Trp | Gln | Glu | Gly | Asn |
|     |     |     | 405 |     |     |     |     | 410 |     |     |     |     |     | 415 |     |
|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| Val | Phe | Ser | Cys | Ser | Val | Met | His | Glu | Ala | Leu | His | Asn | His | Tyr | Thr |
|     |     |     | 420 |     |     |     |     | 425 |     |     |     |     | 430 |     |     |
|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| Gln | Lys | Ser | Leu | Ser | Leu | Ser | Leu | Gly | Lys |     |     |     |     |     |     |
|     | 435 |     |     |     |     |     | 440 |     |     |     |     |     |     |     |     |

<210> SEQ ID NO 346  
 <211> LENGTH: 214  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 346

|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Asp | Ile | Gln | Met | Thr | Gln | Ser | Pro | Ser | Ser | Leu | Ser | Ala | Ser | Val | Gly |
| 1   |     |     |     | 5   |     |     |     | 10  |     |     |     |     | 15  |     |     |
|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| Asp | Arg | Val | Thr | Ile | Thr | Cys | Gln | Ala | Ser | Gln | Asp | Ile | Asn | Ile | Tyr |
|     |     | 20  |     |     |     |     | 25  |     |     |     |     |     | 30  |     |     |
|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| Leu | Asn | Trp | Tyr | Gln | Gln | Gln | Pro | Gly | Lys | Ala | Pro | Lys | Leu | Leu | Ile |
|     | 35  |     |     |     |     |     | 40  |     |     |     |     | 45  |     |     |     |
|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| Tyr | Asp | Val | Ser | Asn | Leu | Val | Pro | Gly | Val | Pro | Ser | Arg | Phe | Ser | Gly |
|     | 50  |     |     |     | 55  |     |     |     |     | 60  |     |     |     |     |     |
|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| Thr | Gly | Ser | Gly | Thr | His | Phe | Ser | Phe | Thr | Ile | Ser | Ser | Leu | Gln | Pro |
| 65  |     |     |     |     | 70  |     |     |     |     | 75  |     |     |     |     | 80  |
|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| Glu | Asp | Ile | Ala | Thr | Tyr | Tyr | Cys | Gln | His | Tyr | Asp | Asp | Leu | Pro | Leu |
|     |     |     | 85  |     |     |     |     | 90  |     |     |     |     | 95  |     |     |
|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| Thr | Phe | Gly | Gly | Gly | Thr | Thr | Val | Ala | Ile | Arg | Arg | Thr | Val | Ala | Ala |
|     |     | 100 |     |     |     |     |     | 105 |     |     |     |     | 110 |     |     |
|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| Pro | Ser | Val | Phe | Ile | Phe | Pro | Pro | Ser | Asp | Glu | Gln | Leu | Lys | Ser | Gly |
|     |     | 115 |     |     |     | 120 |     |     |     |     |     | 125 |     |     |     |
|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| Thr | Ala | Ser | Val | Val | Cys | Leu | Leu | Asn | Asn | Phe | Tyr | Pro | Arg | Glu | Ala |
|     | 130 |     |     |     | 135 |     |     |     |     |     | 140 |     |     |     |     |
|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| Lys | Val | Gln | Trp | Lys | Val | Asp | Asn | Ala | Leu | Gln | Ser | Gly | Asn | Ser | Gln |
| 145 |     |     |     |     | 150 |     |     |     |     | 155 |     |     |     |     | 160 |
|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| Glu | Ser | Val | Thr | Glu | Gln | Asp | Ser | Lys | Asp | Ser | Thr | Tyr | Ser | Leu | Ser |
|     |     |     | 165 |     |     |     |     | 170 |     |     |     |     |     | 175 |     |
|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| Ser | Thr | Leu | Thr | Leu | Ser | Lys | Ala | Asp | Tyr | Glu | Lys | His | Lys | Val | Tyr |
|     |     | 180 |     |     |     |     | 185 |     |     |     |     |     | 190 |     |     |
|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| Ala | Cys | Glu | Val | Thr | His | Gln | Gly | Leu | Ser | Ser | Pro | Val | Thr | Lys | Ser |
|     |     | 195 |     |     |     |     | 200 |     |     |     |     | 205 |     |     |     |
|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
| Phe | Asn | Arg | Gly | Glu | Cys |     |     |     |     |     |     |     |     |     |     |
|     | 210 |     |     |     |     |     |     |     |     |     |     |     |     |     |     |

<210> SEQ ID NO 347  
 <211> LENGTH: 442  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:

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&lt;223&gt; OTHER INFORMATION: Synthetic Construct

&lt;400&gt; SEQUENCE: 347

Gln Val Gln Leu Gln Gln Trp Gly Ala Gly Leu Leu Lys Pro Ser Glu  
 1 5 10 15  
 Thr Leu Ser Leu Thr Cys Ala Val Tyr Asn Gly Ser Phe Ser Ser Phe  
 20 25 30  
 Tyr Trp Ser Trp Ile Arg Gln Pro Pro Gly Lys Gly Leu Glu Trp Leu  
 35 40 45  
 Gly Glu Ile Asn His Arg Gly Arg Thr Asn Asn Asn Pro Ser Leu Ala  
 50 55 60  
 Ser Arg Val Thr Ile Ser Val Asp Thr Ser Lys Ser Gln Phe Ser Leu  
 65 70 75 80  
 Arg Leu Thr Ser Val Thr Ala Ala Asp Thr Ala Thr Tyr Tyr Cys Thr  
 85 90 95  
 Ala His Thr Arg Gly Phe Leu Tyr Trp Gly Gln Gly Thr Arg Val Thr  
 100 105 110  
 Val Ser Ser Ala Ser Thr Lys Gly Pro Ser Val Phe Pro Leu Ala Pro  
 115 120 125  
 Cys Ser Arg Ser Thr Ser Glu Ser Thr Ala Ala Leu Gly Cys Leu Val  
 130 135 140  
 Lys Asp Tyr Phe Pro Glu Pro Val Thr Val Ser Trp Asn Ser Gly Ala  
 145 150 155 160  
 Leu Thr Ser Gly Val His Thr Phe Pro Ala Val Leu Gln Ser Ser Gly  
 165 170 175  
 Leu Tyr Ser Leu Ser Ser Val Val Thr Val Pro Ser Ser Ser Leu Gly  
 180 185 190  
 Thr Lys Thr Tyr Thr Cys Asn Val Asp His Lys Pro Ser Asn Thr Lys  
 195 200 205  
 Val Asp Lys Arg Val Glu Ser Lys Tyr Gly Pro Pro Cys Pro Pro Cys  
 210 215 220  
 Pro Ala Pro Glu Phe Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro  
 225 230 235 240  
 Lys Pro Lys Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys  
 245 250 255  
 Val Val Val Asp Val Ser Gln Glu Asp Pro Glu Val Gln Phe Asn Trp  
 260 265 270  
 Tyr Val Asp Gly Val Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu  
 275 280 285  
 Glu Gln Phe Asn Ser Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu  
 290 295 300  
 His Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn  
 305 310 315 320  
 Lys Gly Leu Pro Ser Ser Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly  
 325 330 335  
 Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Gln Glu Glu  
 340 345 350  
 Met Thr Lys Asn Gln Val Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr  
 355 360 365  
 Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn  
 370 375 380

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Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe
385                      390                      395                      400

Leu Tyr Ser Arg Leu Thr Val Asp Lys Ser Arg Trp Gln Glu Gly Asn
                      405                      410                      415

Val Phe Ser Cys Ser Val Met His Glu Ala Leu His Asn His Tyr Thr
                      420                      425                      430

Gln Lys Ser Leu Ser Leu Ser Leu Gly Lys
                      435                      440

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<210> SEQ ID NO 348
<211> LENGTH: 214
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

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<400> SEQUENCE: 348

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Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1                      5                      10                      15

Asp Arg Val Thr Ile Thr Cys Gln Ala Ser Gln Asp Ile Asn Asn Tyr
                      20                      25                      30

Leu Asn Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Asn Leu Leu Ile
                      35                      40                      45

Tyr Asp Val Phe Asn Leu Leu Pro Gly Val Pro Ser Arg Phe Ser Gly
50                      55                      60

Ser Gly Ser Gly Thr Asp Phe Ser Phe Thr Ile Ser Ser Leu Gln Pro
65                      70                      75                      80

Glu Asp Ile Ala Thr Tyr Tyr Cys Gln His Tyr Asp Asp Leu Pro Leu
                      85                      90                      95

Thr Phe Gly Gly Gly Thr Thr Val Ala Ile Arg Arg Thr Val Ala Ala
100                     105                     110

Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser Gly
115                     120                     125

Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu Ala
130                     135                     140

Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly Asn Ser Gln
145                     150                     155                     160

Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu Ser
165                     170                     175

Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys His Lys Val Tyr
180                     185                     190

Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro Val Thr Lys Ser
195                     200                     205

Phe Asn Arg Gly Glu Cys
210

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<210> SEQ ID NO 349
<211> LENGTH: 442
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

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<400> SEQUENCE: 349

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```

Gln Val Gln Leu Gln Gln Trp Gly Ala Gly Leu Leu Lys Pro Ser Glu
1                      5                      10                      15

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|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |  |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|--|
| Thr | Leu | Ser | Leu | Thr | Cys | Ala | Val | Tyr | Asn | Gly | Ser | Phe | Ser | Ser | Tyr |  |
|     |     |     | 20  |     |     |     | 25  |     |     |     |     |     | 30  |     |     |  |
| Tyr | Trp | Ser | Trp | Ile | Arg | Gln | Pro | Pro | Gly | Lys | Gly | Leu | Glu | Trp | Leu |  |
|     |     |     | 35  |     |     |     | 40  |     |     |     | 45  |     |     |     |     |  |
| Gly | Glu | Ile | Asn | His | Arg | Gly | Arg | Thr | Asn | Tyr | Asn | Pro | Ser | Leu | Thr |  |
|     |     |     | 50  |     |     |     | 55  |     |     |     | 60  |     |     |     |     |  |
| Ser | Arg | Val | Thr | Ile | Ser | Val | Asp | Thr | Ser | Lys | Asn | Gln | Phe | Ser | Leu |  |
|     |     |     | 65  |     |     |     | 70  |     |     |     | 75  |     |     | 80  |     |  |
| Lys | Leu | Pro | Ser | Val | Thr | Ala | Ala | Asp | Thr | Ala | Ile | Tyr | Tyr | Cys | Ser |  |
|     |     |     | 85  |     |     |     |     |     | 90  |     |     | 95  |     |     |     |  |
| Ala | His | Thr | Arg | Gly | Phe | Leu | Tyr | Trp | Gly | Gln | Gly | Thr | Leu | Val | Thr |  |
|     |     |     | 100 |     |     |     |     |     | 105 |     |     | 110 |     |     |     |  |
| Val | Ser | Ser | Ala | Ser | Thr | Lys | Gly | Pro | Ser | Val | Phe | Pro | Leu | Ala | Pro |  |
|     |     |     | 115 |     |     |     |     |     | 120 |     |     | 125 |     |     |     |  |
| Cys | Ser | Arg | Ser | Thr | Ser | Glu | Ser | Thr | Ala | Ala | Leu | Gly | Cys | Leu | Val |  |
|     |     |     | 130 |     |     | 135 |     |     |     |     |     | 140 |     |     |     |  |
| Lys | Asp | Tyr | Phe | Pro | Glu | Pro | Val | Thr | Val | Ser | Trp | Asn | Ser | Gly | Ala |  |
|     |     |     | 145 |     |     | 150 |     |     | 155 |     |     | 160 |     |     |     |  |
| Leu | Thr | Ser | Gly | Val | His | Thr | Phe | Pro | Ala | Val | Leu | Gln | Ser | Ser | Gly |  |
|     |     |     | 165 |     |     |     |     |     | 170 |     |     | 175 |     |     |     |  |
| Leu | Tyr | Ser | Leu | Ser | Ser | Val | Val | Thr | Val | Pro | Ser | Ser | Ser | Leu | Gly |  |
|     |     |     | 180 |     |     |     |     |     | 185 |     |     | 190 |     |     |     |  |
| Thr | Lys | Thr | Tyr | Thr | Cys | Asn | Val | Asp | His | Lys | Pro | Ser | Asn | Thr | Lys |  |
|     |     |     | 195 |     |     | 200 |     |     |     |     |     | 205 |     |     |     |  |
| Val | Asp | Lys | Arg | Val | Glu | Ser | Lys | Tyr | Gly | Pro | Pro | Cys | Pro | Pro | Cys |  |
|     |     |     | 210 |     |     | 215 |     |     |     |     |     | 220 |     |     |     |  |
| Pro | Ala | Pro | Glu | Phe | Leu | Gly | Gly | Pro | Ser | Val | Phe | Leu | Phe | Pro | Pro |  |
|     |     |     | 225 |     |     | 230 |     |     | 235 |     |     | 240 |     |     |     |  |
| Lys | Pro | Lys | Asp | Thr | Leu | Met | Ile | Ser | Arg | Thr | Pro | Glu | Val | Thr | Cys |  |
|     |     |     | 245 |     |     |     |     |     | 250 |     |     | 255 |     |     |     |  |
| Val | Val | Val | Asp | Val | Ser | Gln | Glu | Asp | Pro | Glu | Val | Gln | Phe | Asn | Trp |  |
|     |     |     | 260 |     |     | 265 |     |     |     |     |     | 270 |     |     |     |  |
| Tyr | Val | Asp | Gly | Val | Glu | Val | His | Asn | Ala | Lys | Thr | Lys | Pro | Arg | Glu |  |
|     |     |     | 275 |     |     | 280 |     |     |     |     |     | 285 |     |     |     |  |
| Glu | Gln | Phe | Asn | Ser | Thr | Tyr | Arg | Val | Val | Ser | Val | Leu | Thr | Val | Leu |  |
|     |     |     | 290 |     |     | 295 |     |     |     |     |     | 300 |     |     |     |  |
| His | Gln | Asp | Trp | Leu | Asn | Gly | Lys | Glu | Tyr | Lys | Cys | Lys | Val | Ser | Asn |  |
|     |     |     | 305 |     |     | 310 |     |     | 315 |     |     | 320 |     |     |     |  |
| Lys | Gly | Leu | Pro | Ser | Ser | Ile | Glu | Lys | Thr | Ile | Ser | Lys | Ala | Lys | Gly |  |
|     |     |     | 325 |     |     |     |     |     | 330 |     |     | 335 |     |     |     |  |
| Gln | Pro | Arg | Glu | Pro | Gln | Val | Tyr | Thr | Leu | Pro | Pro | Ser | Gln | Glu | Glu |  |
|     |     |     | 340 |     |     | 345 |     |     |     |     |     | 350 |     |     |     |  |
| Met | Thr | Lys | Asn | Gln | Val | Ser | Leu | Thr | Cys | Leu | Val | Lys | Gly | Phe | Tyr |  |
|     |     |     | 355 |     |     | 360 |     |     |     |     |     | 365 |     |     |     |  |
| Pro | Ser | Asp | Ile | Ala | Val | Glu | Trp | Glu | Ser | Asn | Gly | Gln | Pro | Glu | Asn |  |
|     |     |     | 370 |     |     | 375 |     |     |     |     |     | 380 |     |     |     |  |
| Asn | Tyr | Lys | Thr | Thr | Pro | Pro | Val | Leu | Asp | Ser | Asp | Gly | Ser | Phe | Phe |  |
|     |     |     | 385 |     |     | 390 |     |     | 395 |     |     | 400 |     |     |     |  |
| Leu | Tyr | Ser | Arg | Leu | Thr | Val | Asp | Lys | Ser | Arg | Trp | Gln | Glu | Gly | Asn |  |
|     |     |     | 405 |     |     | 410 |     |     |     |     |     | 415 |     |     |     |  |

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|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Val | Phe | Ser | Cys | Ser | Val | Met | His | Glu | Ala | Leu | His | Asn | His | Tyr | Thr |
|     |     |     | 420 |     |     |     |     | 425 |     |     |     |     | 430 |     |     |

|     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Gln | Lys | Ser | Leu | Ser | Leu | Ser | Leu | Gly | Lys |
|     | 435 |     |     |     |     | 440 |     |     |     |

<210> SEQ ID NO 350  
 <211> LENGTH: 214  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 350

|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Asp | Ile | Gln | Met | Thr | Gln | Ser | Pro | Ser | Ser | Leu | Ser | Ala | Ser | Val | Gly |
| 1   |     |     |     | 5   |     |     |     |     |     | 10  |     |     |     | 15  |     |

|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Asp | Arg | Val | Thr | Ile | Thr | Cys | Gln | Ala | Ser | Gln | Asp | Ile | Asn | Asn | Tyr |
|     |     | 20  |     |     |     |     |     | 25  |     |     |     |     | 30  |     |     |

|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Leu | Asn | Trp | Tyr | Gln | Gln | Lys | Pro | Gly | Lys | Ala | Pro | Lys | Leu | Leu | Ile |
|     |     | 35  |     |     |     |     | 40  |     |     |     |     | 45  |     |     |     |

|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Tyr | Asp | Ala | Ser | Asn | Leu | Val | Pro | Gly | Val | Pro | Ser | Arg | Phe | Ser | Gly |
|     | 50  |     |     |     |     | 55  |     |     |     |     | 60  |     |     |     |     |

|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Thr | Gly | Ser | Gly | Thr | Asp | Phe | Ser | Phe | Thr | Ile | Ser | Ser | Leu | Gln | Pro |
| 65  |     |     |     |     | 70  |     |     |     |     | 75  |     |     |     | 80  |     |

|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Glu | Asp | Ile | Ala | Thr | Tyr | Tyr | Cys | Gln | His | Tyr | Tyr | Asp | Leu | Pro | Leu |
|     |     |     |     | 85  |     |     |     |     | 90  |     |     |     |     | 95  |     |

|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Thr | Phe | Gly | Gly | Gly | Thr | Thr | Val | Ala | Ile | Ser | Arg | Thr | Val | Ala | Ala |
|     |     | 100 |     |     |     |     |     | 105 |     |     |     |     | 110 |     |     |

|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Pro | Ser | Val | Phe | Ile | Phe | Pro | Pro | Ser | Asp | Glu | Gln | Leu | Lys | Ser | Gly |
|     |     | 115 |     |     |     |     | 120 |     |     |     |     | 125 |     |     |     |

|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Thr | Ala | Ser | Val | Val | Cys | Leu | Leu | Asn | Asn | Phe | Tyr | Pro | Arg | Glu | Ala |
|     | 130 |     |     |     |     | 135 |     |     |     |     | 140 |     |     |     |     |

|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Lys | Val | Gln | Trp | Lys | Val | Asp | Asn | Ala | Leu | Gln | Ser | Gly | Asn | Ser | Gln |
| 145 |     |     |     |     | 150 |     |     |     |     | 155 |     |     |     | 160 |     |

|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Glu | Ser | Val | Thr | Glu | Gln | Asp | Ser | Lys | Asp | Ser | Thr | Tyr | Ser | Leu | Ser |
|     |     |     | 165 |     |     |     |     |     | 170 |     |     |     |     | 175 |     |

|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Ser | Thr | Leu | Thr | Leu | Ser | Lys | Ala | Asp | Tyr | Glu | Lys | His | Lys | Val | Tyr |
|     |     | 180 |     |     |     |     |     | 185 |     |     |     |     | 190 |     |     |

|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Ala | Cys | Glu | Val | Thr | His | Gln | Gly | Leu | Ser | Ser | Pro | Val | Thr | Lys | Ser |
|     |     | 195 |     |     |     |     | 200 |     |     |     |     | 205 |     |     |     |

|     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|
| Phe | Asn | Arg | Gly | Glu | Cys |
|     |     |     |     |     | 210 |

<210> SEQ ID NO 351  
 <211> LENGTH: 214  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 351

|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Asp | Ile | Gln | Met | Thr | Gln | Ser | Pro | Ser | Ser | Leu | Ser | Ala | Ser | Val | Gly |
| 1   |     |     |     | 5   |     |     |     |     |     | 10  |     |     |     | 15  |     |

|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Asp | Arg | Val | Thr | Ile | Thr | Cys | Gln | Ala | Ser | Gln | Asp | Ile | Asn | Asn | Tyr |
|     |     | 20  |     |     |     |     |     | 25  |     |     |     |     | 30  |     |     |

|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Leu | Asn | Trp | Tyr | Gln | Gln | Lys | Pro | Gly | Lys | Ala | Pro | Lys | Leu | Leu | Ile |
|     |     | 35  |     |     |     |     | 40  |     |     |     |     | 45  |     |     |     |

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Tyr Asp Ala Ser Asn Leu Val Pro Gly Val Pro Ser Arg Phe Ser Gly  
 50 55 60  
 Thr Gly Ser Gly Thr Asp Phe Ser Phe Thr Ile Ser Ser Leu Gln Pro  
 65 70 75 80  
 Glu Asp Ile Ala Thr Tyr Tyr Cys Gln His Tyr Tyr Asp Leu Pro Leu  
 85 90 95  
 Thr Phe Gly Gly Gly Thr Thr Val Ala Ile Arg Arg Thr Val Ala Ala  
 100 105 110  
 Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser Gly  
 115 120 125  
 Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu Ala  
 130 135 140  
 Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly Asn Ser Gln  
 145 150 155 160  
 Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu Ser  
 165 170 175  
 Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys His Lys Val Tyr  
 180 185 190  
 Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro Val Thr Lys Ser  
 195 200 205  
 Phe Asn Arg Gly Glu Cys  
 210

&lt;210&gt; SEQ ID NO 352

&lt;211&gt; LENGTH: 443

&lt;212&gt; TYPE: PRT

&lt;213&gt; ORGANISM: Artificial Sequence

&lt;220&gt; FEATURE:

&lt;223&gt; OTHER INFORMATION: Synthetic Construct

&lt;400&gt; SEQUENCE: 352

Glu Ile Gln Leu Val Glu Ser Gly Gly Asp Leu Val Gln Pro Gly Gly  
 1 5 10 15  
 Ser Leu Lys Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Asp Ser  
 20 25 30  
 Ala Met His Trp Val Arg Gln Ala Ser Gly Lys Gly Leu Glu Trp Val  
 35 40 45  
 Gly Arg Ile Arg Asn Lys Ala Asn Thr Tyr Ala Thr Ser Tyr Ala Ala  
 50 55 60  
 Ser Val Lys Gly Arg Phe Thr Val Ser Thr Asp Ser Lys Thr Thr  
 65 70 75 80  
 Ala Tyr Leu Gln Met Arg Tyr Leu Lys Ile Glu Asp Thr Ala Ile Tyr  
 85 90 95  
 Tyr Cys Thr Ser Asp Gln Phe Asp Tyr Trp Gly Gln Gly Thr Leu Val  
 100 105 110  
 Thr Val Ser Ser Ala Ser Thr Lys Gly Pro Ser Val Phe Pro Leu Ala  
 115 120 125  
 Pro Cys Ser Arg Ser Thr Ser Glu Ser Thr Ala Ala Leu Gly Cys Leu  
 130 135 140  
 Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val Ser Trp Asn Ser Gly  
 145 150 155 160  
 Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala Val Leu Gln Ser Ser  
 165 170 175



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Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val Pro Ser Ser Ser Leu  
 180 185 190  
 Gly Thr Lys Thr Tyr Thr Cys Asn Val Asp His Lys Pro Ser Asn Thr  
 195 200 205  
 Lys Val Asp Lys Arg Val Glu Ser Lys Tyr Gly Pro Pro Cys Pro Pro  
 210 215 220  
 Cys Pro Ala Pro Glu Phe Leu Gly Gly Pro Ser Val Phe Leu Phe Pro  
 225 230 235 240  
 Pro Lys Pro Lys Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr  
 245 250 255  
 Cys Val Val Val Asp Val Ser Gln Glu Asp Pro Glu Val Gln Phe Asn  
 260 265 270  
 Trp Tyr Val Asp Gly Val Glu Val His Asn Ala Lys Thr Lys Pro Arg  
 275 280 285  
 Glu Glu Gln Phe Asn Ser Thr Tyr Arg Val Val Ser Val Leu Thr Val  
 290 295 300  
 Leu His Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser  
 305 310 315 320  
 Asn Lys Gly Leu Pro Ser Ser Ile Glu Lys Thr Ile Ser Lys Ala Lys  
 325 330 335  
 Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Gln Glu  
 340 345 350  
 Glu Met Thr Lys Asn Gln Val Ser Leu Thr Cys Leu Val Lys Gly Phe  
 355 360 365  
 Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu  
 370 375 380  
 Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe  
 385 390 395 400  
 Phe Leu Tyr Ser Arg Leu Thr Val Asp Lys Ser Arg Trp Gln Glu Gly  
 405 410 415  
 Asn Val Phe Ser Cys Ser Val Met His Glu Ala Leu His Asn His Tyr  
 420 425 430  
 Thr Gln Lys Ser Leu Ser Leu Ser Leu Gly Lys  
 435 440

&lt;210&gt; SEQ ID NO 353

&lt;211&gt; LENGTH: 219

&lt;212&gt; TYPE: PRT

&lt;213&gt; ORGANISM: Artificial Sequence

&lt;220&gt; FEATURE:

&lt;223&gt; OTHER INFORMATION: Synthetic Construct

&lt;400&gt; SEQUENCE: 353

Asp Val Val Met Thr Gln Ser Pro Leu Ser Leu Ser Ala Thr Leu Gly  
 1 5 10 15  
 Gln Pro Ala Ser Ile Ser Cys Arg Ser Ser Gln Ser Leu Val Tyr Ser  
 20 25 30  
 Asp Gly Ile Thr Tyr Leu Thr Trp Phe Gln Gln Arg Pro Gly Gln Ser  
 35 40 45  
 Pro Thr Arg Leu Ile Tyr Gln Val Ser Asn Arg Asp Ser Gly Val Pro  
 50 55 60  
 Asp Arg Phe Asn Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Asn Ile  
 65 70 75 80

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<210> SEQ ID NO 354
<211> LENGTH: 441
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 354

Gln Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Lys Pro Gly Gly
1          5          10          15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Asp Tyr
20          25          30
Tyr Met Ser Trp Ile Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Ile
35          40          45
Ser Phe Ile Ser Arg Thr Gly Ser Thr Lys Tyr Tyr Ala Asp Ser Val
50          55          60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Asn Ser Leu Asn
65          70          75          80
Leu Gln Met Ser Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Phe Cys
85          90          95
Ala Arg Glu Pro Leu Gln His Trp Gly Gln Gly Thr Leu Val Thr Val
100         105         110
Ser Ser Ala Ser Thr Lys Gly Pro Ser Val Phe Pro Leu Ala Pro Cys
115         120         125
Ser Arg Ser Thr Ser Glu Ser Thr Ala Ala Leu Gly Cys Leu Val Lys
130         135         140
Asp Tyr Phe Pro Glu Pro Val Thr Val Ser Trp Asn Ser Gly Ala Leu
145         150         155         160
Thr Ser Gly Val His Thr Phe Pro Ala Val Leu Gln Ser Ser Gly Leu
165         170         175
Tyr Ser Leu Ser Ser Val Val Thr Val Pro Ser Ser Ser Leu Gly Thr
180         185         190
Lys Thr Tyr Thr Cys Asn Val Asp His Lys Pro Ser Asn Thr Lys Val
195         200         205

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Asp Lys Arg Val Glu Ser Lys Tyr Gly Pro Pro Cys Pro Pro Cys Pro
 210                215                220

Ala Pro Glu Phe Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro Lys
 225                230                235                240

Pro Lys Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val
                245                250                255

Val Val Asp Val Ser Gln Glu Asp Pro Glu Val Gln Phe Asn Trp Tyr
 260                265                270

Val Asp Gly Val Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu
 275                280                285

Gln Phe Asn Ser Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu His
 290                295                300

Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys
 305                310                315                320

Gly Leu Pro Ser Ser Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln
                325                330                335

Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Gln Glu Glu Met
 340                345                350

Thr Lys Asn Gln Val Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro
 355                360                365

Ser Asp Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn
 370                375                380

Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu
 385                390                395                400

Tyr Ser Arg Leu Thr Val Asp Lys Ser Arg Trp Gln Glu Gly Asn Val
 405                410                415

Phe Ser Cys Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln
 420                425                430

Lys Ser Leu Ser Leu Ser Leu Gly Lys
 435                440

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<210> SEQ ID NO 355
<211> LENGTH: 220
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

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<400> SEQUENCE: 355

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Asn Val Val Leu Thr Gln Thr Pro Ile Ser Ser Pro Val Ile Leu Gly
 1                5                10                15

Gln Pro Ala Ser Ile Ser Cys Arg Ser Ser Gln Ser Leu Val His Ser
 20                25                30

Asp Gly Asp Thr Tyr Leu Ser Trp Leu His Gln Arg Pro Gly Gln Pro
 35                40                45

Pro Arg Leu Leu Ile Tyr Thr Val Ser Asn Arg Val Ser Gly Val Pro
 50                55                60

Asp Arg Phe Ser Gly Ser Gly Ala Gly Thr Asp Phe Thr Leu Lys Ile
 65                70                75                80

Ser Arg Val Glu Ala Glu Asp Val Gly Val Tyr Tyr Cys Met Gln Gly
 85                90                95

Thr Gln Phe Pro Ala Leu Thr Phe Gly Gly Gly Thr Lys Val Glu Ile
 100               105               110

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Lys Arg Thr Val Ala Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp  
 115 120 125  
 Glu Gln Leu Lys Ser Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn  
 130 135 140  
 Phe Tyr Pro Arg Glu Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu  
 145 150 155 160  
 Gln Ser Gly Asn Ser Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp  
 165 170 175  
 Ser Thr Tyr Ser Leu Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr  
 180 185 190  
 Glu Lys His Lys Val Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser  
 195 200 205  
 Ser Pro Val Thr Lys Ser Phe Asn Arg Gly Glu Cys  
 210 215 220

<210> SEQ ID NO 356  
 <211> LENGTH: 441  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 356

Gln Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Lys Pro Gly Gly  
 1 5 10 15  
 Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Asp Tyr  
 20 25 30  
 Tyr Met Ser Trp Ile Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val  
 35 40 45  
 Ser Tyr Ile Ser Arg Thr Gly Ser Thr Lys Tyr Tyr Ala Asp Ser Val  
 50 55 60  
 Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Lys Asn Ser Leu Asn  
 65 70 75 80  
 Leu Gln Met Ser Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Phe Cys  
 85 90 95  
 Ala Arg Glu Pro Leu Gln His Trp Gly Gln Gly Thr Leu Val Thr Val  
 100 105 110  
 Ser Ser Ala Ser Thr Lys Gly Pro Ser Val Phe Pro Leu Ala Pro Cys  
 115 120 125  
 Ser Arg Ser Thr Ser Glu Ser Thr Ala Ala Leu Gly Cys Leu Val Lys  
 130 135 140  
 Asp Tyr Phe Pro Glu Pro Val Thr Val Ser Trp Asn Ser Gly Ala Leu  
 145 150 155 160  
 Thr Ser Gly Val His Thr Phe Pro Ala Val Leu Gln Ser Ser Gly Leu  
 165 170 175  
 Tyr Ser Leu Ser Ser Val Val Thr Val Pro Ser Ser Ser Leu Gly Thr  
 180 185 190  
 Lys Thr Tyr Thr Cys Asn Val Asp His Lys Pro Ser Asn Thr Lys Val  
 195 200 205  
 Asp Lys Arg Val Glu Ser Lys Tyr Gly Pro Pro Cys Pro Pro Cys Pro  
 210 215 220  
 Ala Pro Glu Phe Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro Lys  
 225 230 235 240

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<210> SEQ ID NO 357
<211> LENGTH: 220
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 357

Lys Ile Val Leu Thr Gln Thr Pro Leu Ser Ser Pro Val Thr Leu Gly
1             5             10             15
Gln Pro Ala Ser Ile Ser Cys Arg Ser Ser Gln Ser Leu Val His Ser
                20             25             30
Asp Gly Asp Thr Tyr Leu Ser Trp Leu His Gln Arg Pro Gly Gln Pro
        35             40             45
Pro Arg Leu Leu Ile Tyr Thr Ile Ser Asn Arg Leu Ser Gly Val Pro
        50             55             60
Asp Arg Phe Ser Gly Ser Gly Ala Gly Thr Asp Phe Thr Leu Lys Ile
65             70             75             80
Ser Arg Val Glu Ala Glu Asp Val Gly Val Tyr Tyr Cys Met Gln Gly
        85             90             95
Thr Gln Phe Pro Ala Leu Thr Phe Gly Gly Gly Thr Lys Val Glu Ile
        100            105            110
Lys Arg Thr Val Ala Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp
        115            120            125
Glu Gln Leu Lys Ser Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn
        130            135            140

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Phe Tyr Pro Arg Glu Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu  
 145 150 155 160  
 Gln Ser Gly Asn Ser Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp  
 165 170 175  
 Ser Thr Tyr Ser Leu Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr  
 180 185 190  
 Glu Lys His Lys Val Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser  
 195 200 205  
 Ser Pro Val Thr Lys Ser Phe Asn Arg Gly Glu Cys  
 210 215 220

<210> SEQ ID NO 358  
 <211> LENGTH: 442  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 358

Glu Met Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly  
 1 5 10 15  
 Ser Leu Lys Val Ser Cys Ala Ala Ser Gly Phe Thr Leu Ser Asp Ser  
 20 25 30  
 Ser Ile His Trp Val Arg Gln Ala Ser Gly Lys Gly Leu Glu Trp Ile  
 35 40 45  
 Gly Arg Ile Arg Ser Lys Pro Tyr Ser Tyr Ala Thr Ala Tyr Ala Ala  
 50 55 60  
 Ser Val Lys Gly Arg Phe Thr Ile Ser Arg Asp Asp Ser Lys Asn Thr  
 65 70 75 80  
 Ala Phe Leu Gln Met Ser Gly Leu Lys Thr Glu Asp Thr Ala Val Tyr  
 85 90 95  
 Tyr Cys Thr Gly Gly Asp Asp Phe Trp Gly Gln Gly Thr Leu Val Thr  
 100 105 110  
 Val Ser Ser Ala Ser Thr Lys Gly Pro Ser Val Phe Pro Leu Ala Pro  
 115 120 125  
 Cys Ser Arg Ser Thr Ser Glu Ser Thr Ala Ala Leu Gly Cys Leu Val  
 130 135 140  
 Lys Asp Tyr Phe Pro Glu Pro Val Thr Val Ser Trp Asn Ser Gly Ala  
 145 150 155 160  
 Leu Thr Ser Gly Val His Thr Phe Pro Ala Val Leu Gln Ser Ser Gly  
 165 170 175  
 Leu Tyr Ser Leu Ser Ser Val Val Thr Val Pro Ser Ser Ser Leu Gly  
 180 185 190  
 Thr Lys Thr Tyr Thr Cys Asn Val Asp His Lys Pro Ser Asn Thr Lys  
 195 200 205  
 Val Asp Lys Arg Val Glu Ser Lys Tyr Gly Pro Pro Cys Pro Pro Cys  
 210 215 220  
 Pro Ala Pro Glu Phe Leu Gly Gly Pro Ser Val Phe Leu Phe Pro Pro  
 225 230 235 240  
 Lys Pro Lys Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys  
 245 250 255  
 Val Val Val Asp Val Ser Gln Glu Asp Pro Glu Val Gln Phe Asn Trp  
 260 265 270

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Tyr Val Asp Gly Val Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu  
 275 280 285  
 Glu Gln Phe Asn Ser Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu  
 290 295 300  
 His Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn  
 305 310 315 320  
 Lys Gly Leu Pro Ser Ser Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly  
 325 330 335  
 Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Gln Glu Glu  
 340 345 350  
 Met Thr Lys Asn Gln Val Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr  
 355 360 365  
 Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn  
 370 375 380  
 Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe  
 385 390 395 400  
 Leu Tyr Ser Arg Leu Thr Val Asp Lys Ser Arg Trp Gln Glu Gly Asn  
 405 410 415  
 Val Phe Ser Cys Ser Val Met His Glu Ala Leu His Asn His Tyr Thr  
 420 425 430  
 Gln Lys Ser Leu Ser Leu Ser Leu Gly Lys  
 435 440

<210> SEQ ID NO 359  
 <211> LENGTH: 213  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 359

Asp Ile Gln Met Thr Gln Ser Pro Ser Thr Leu Ser Ala Ser Val Gly  
 1 5 10 15  
 Asp Arg Ile Thr Ile Thr Cys Arg Ala Ser Gln Ser Ile Ser Thr Trp  
 20 25 30  
 Leu Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile  
 35 40 45  
 Tyr Lys Ala Ser Ser Leu Glu Gly Gly Val Pro Ser Arg Phe Ser Gly  
 50 55 60  
 Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro  
 65 70 75 80  
 Asp Asp Phe Ala Thr Tyr Tyr Cys Gln Gln Tyr Asn Thr Tyr Trp Thr  
 85 90 95  
 Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Arg Thr Val Ala Ala Pro  
 100 105 110  
 Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser Gly Thr  
 115 120 125  
 Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu Ala Lys  
 130 135 140  
 Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly Asn Ser Gln Glu  
 145 150 155 160  
 Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu Ser Ser  
 165 170 175

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Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys His Lys Val Tyr Ala  
                   180                  185                  190

Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro Val Thr Lys Ser Phe  
                   195                  200                  205

Asn Arg Gly Glu Cys  
                   210

<210> SEQ ID NO 360  
 <211> LENGTH: 448  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 360

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly  
 1                  5                  10                  15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Ser Tyr  
                   20                  25                  30

Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val  
                   35                  40                  45

Ser Ala Ile Ser Gly Ser Gly Gly Ser Thr Tyr Tyr Ala Asp Ser Val  
                   50                  55                  60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr  
                   65                  70                  75                  80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys  
                   85                  90                  95

Ala Ile Pro Tyr Asn Trp Asn Tyr Gly Gly Ala Phe Asp Ile Trp Gly  
                   100                  105                  110

Gln Gly Thr Met Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro Ser  
                   115                  120                  125

Val Phe Pro Leu Ala Pro Cys Ser Arg Ser Thr Ser Glu Ser Thr Ala  
                   130                  135                  140

Ala Leu Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val  
                   145                  150                  155                  160

Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala  
                   165                  170                  175

Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val  
                   180                  185                  190

Pro Ser Ser Ser Leu Gly Thr Lys Thr Tyr Thr Cys Asn Val Asp His  
                   195                  200                  205

Lys Pro Ser Asn Thr Lys Val Asp Lys Arg Val Glu Ser Lys Tyr Gly  
                   210                  215                  220

Pro Pro Cys Pro Pro Cys Pro Ala Pro Glu Phe Leu Gly Gly Pro Ser  
                   225                  230                  235                  240

Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser Arg  
                   245                  250                  255

Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser Gln Glu Asp Pro  
                   260                  265                  270

Glu Val Gln Phe Asn Trp Tyr Val Asp Gly Val Glu Val His Asn Ala  
                   275                  280                  285

Lys Thr Lys Pro Arg Glu Glu Gln Phe Asn Ser Thr Tyr Arg Val Val  
                   290                  295                  300



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Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys Glu Tyr  
 305 310 315 320

Lys Cys Lys Val Ser Asn Lys Gly Leu Pro Ser Ser Ile Glu Lys Thr  
 325 330 335

Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu  
 340 345 350

Pro Pro Ser Gln Glu Glu Met Thr Lys Asn Gln Val Ser Leu Thr Cys  
 355 360 365

Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser  
 370 375 380

Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp  
 385 390 395 400

Ser Asp Gly Ser Phe Phe Leu Tyr Ser Arg Leu Thr Val Asp Lys Ser  
 405 410 415

Arg Trp Gln Glu Gly Asn Val Phe Ser Cys Ser Val Met His Glu Ala  
 420 425 430

Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Leu Gly Lys  
 435 440 445

<210> SEQ ID NO 361  
 <211> LENGTH: 215  
 <212> TYPE: PRT  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 361

Glu Ile Val Met Thr Gln Ser Pro Ala Thr Leu Ser Val Ser Pro Gly  
 1 5 10 15

Lys Arg Ala Thr Leu Ser Cys Arg Ala Ser Gln Ser Val Ser Ser Asn  
 20 25 30

Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln Ala Pro Arg Leu Leu Ile  
 35 40 45

Tyr Gly Ala Ser Thr Arg Ala Thr Gly Ile Pro Ala Arg Phe Ser Gly  
 50 55 60

Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Ser Ser Leu Gln Ser  
 65 70 75 80

Glu Asp Phe Ala Val Tyr Tyr Cys Gln Gln Tyr Asn Asn Trp Pro Pro  
 85 90 95

Tyr Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile Lys Arg Thr Val Ala  
 100 105 110

Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser  
 115 120 125

Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu  
 130 135 140

Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly Asn Ser  
 145 150 155 160

Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu  
 165 170 175

Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys His Lys Val  
 180 185 190

Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro Val Thr Lys  
 195 200 205

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Ser Phe Asn Arg Gly Glu Cys  
210 215

<210> SEQ ID NO 362  
<211> LENGTH: 446  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 362

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ala  
1 5 10 15  
Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Thr Phe Thr Gly Tyr  
20 25 30  
Tyr Met His Trp Val Arg Gln Ala Pro Gly Gln Gly Leu Glu Trp Met  
35 40 45  
Gly Trp Ile Asn Pro Asn Ser Val Gly Thr Asn Phe Ala Gln Lys Phe  
50 55 60  
Gln Gly Arg Val Thr Met Thr Arg Asp Thr Ser Ile Ser Thr Ala Tyr  
65 70 75 80  
Met Glu Leu Ser Arg Leu Arg Ser Asp Asp Thr Ala Val Tyr Tyr Cys  
85 90 95  
Thr Arg Asp Gly Ala Ala Ala Gly Leu Phe Asp Tyr Trp Gly Gln Gly  
100 105 110  
Thr Leu Val Thr Val Ser Ser Ala Ser Thr Lys Gly Pro Ser Val Phe  
115 120 125  
Pro Leu Ala Pro Cys Ser Arg Ser Thr Ser Glu Ser Thr Ala Ala Leu  
130 135 140  
Gly Cys Leu Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val Ser Trp  
145 150 155 160  
Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala Val Leu  
165 170 175  
Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val Pro Ser  
180 185 190  
Ser Ser Leu Gly Thr Lys Thr Tyr Thr Cys Asn Val Asp His Lys Pro  
195 200 205  
Ser Asn Thr Lys Val Asp Lys Arg Val Glu Ser Lys Tyr Gly Pro Pro  
210 215 220  
Cys Pro Pro Cys Pro Ala Pro Glu Phe Leu Gly Gly Pro Ser Val Phe  
225 230 235 240  
Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser Arg Thr Pro  
245 250 255  
Glu Val Thr Cys Val Val Val Asp Val Ser Gln Glu Asp Pro Glu Val  
260 265 270  
Gln Phe Asn Trp Tyr Val Asp Gly Val Glu Val His Asn Ala Lys Thr  
275 280 285  
Lys Pro Arg Glu Glu Gln Phe Asn Ser Thr Tyr Arg Val Val Ser Val  
290 295 300  
Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys  
305 310 315 320  
Lys Val Ser Asn Lys Gly Leu Pro Ser Ser Ile Glu Lys Thr Ile Ser  
325 330 335

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Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro  
                   340                                  345                                  350  
 Ser Gln Glu Glu Met Thr Lys Asn Gln Val Ser Leu Thr Cys Leu Val  
                   355                                  360                                  365  
 Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn Gly  
                   370                                  375                                  380  
 Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser Asp  
                   385                                  390                                  395                                  400  
 Gly Ser Phe Phe Leu Tyr Ser Arg Leu Thr Val Asp Lys Ser Arg Trp  
                                   405                                  410                                  415  
 Gln Glu Gly Asn Val Phe Ser Cys Ser Val Met His Glu Ala Leu His  
                                   420                                  425                                  430  
 Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Leu Gly Lys  
                   435                                  440                                  445

&lt;210&gt; SEQ ID NO 363

&lt;211&gt; LENGTH: 214

&lt;212&gt; TYPE: PRT

&lt;213&gt; ORGANISM: Artificial Sequence

&lt;220&gt; FEATURE:

&lt;223&gt; OTHER INFORMATION: Synthetic Construct

&lt;400&gt; SEQUENCE: 363

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly  
 1                  5                                  10                                  15  
 Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Gly Ile Asn Asn Tyr  
                   20                                  25                                  30  
 Leu Ala Trp Tyr Gln Gln Lys Pro Gly Lys Val Pro Lys Leu Leu Ile  
                   35                                  40                                  45  
 Tyr Ala Ala Ser Thr Leu Gln Ser Gly Val Pro Ser Arg Phe Ser Gly  
                   50                                  55                                  60  
 Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro  
                   65                                  70                                  75                                  80  
 Glu Asp Val Ala Thr Tyr Tyr Cys Gln Lys Phe Asn Ser Ala Pro Leu  
                   85                                  90                                  95  
 Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys Arg Thr Val Ala Ala  
                   100                                  105                                  110  
 Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser Gly  
                   115                                  120                                  125  
 Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu Ala  
                   130                                  135                                  140  
 Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly Asn Ser Gln  
                   145                                  150                                  155                                  160  
 Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu Ser  
                   165                                  170                                  175  
 Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys His Lys Val Tyr  
                   180                                  185                                  190  
 Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro Val Thr Lys Ser  
                   195                                  200                                  205  
 Phe Asn Arg Gly Glu Cys  
                   210

&lt;210&gt; SEQ ID NO 364

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<211> LENGTH: 443
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 364

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1          5          10          15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Thr Tyr
20          25          30
Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35          40          45
Ser Ala Ile Ser Gly Ser Gly Gly Ser Thr Tyr Tyr Ala Asp Ser Val
50          55          60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65          70          75          80
Leu Gln Met Lys Ser Leu Arg Val Glu Asp Thr Ala Val Tyr Tyr Cys
85          90          95
Ala Asn Ser Pro Ser Trp Phe Asp Pro Trp Gly Gln Gly Thr Leu Val
100         105         110
Thr Val Ser Ser Ala Ser Thr Lys Gly Pro Ser Val Phe Pro Leu Ala
115         120         125
Pro Cys Ser Arg Ser Thr Ser Glu Ser Thr Ala Ala Leu Gly Cys Leu
130         135         140
Val Lys Asp Tyr Phe Pro Glu Pro Val Thr Val Ser Trp Asn Ser Gly
145         150         155         160
Ala Leu Thr Ser Gly Val His Thr Phe Pro Ala Val Leu Gln Ser Ser
165         170         175
Gly Leu Tyr Ser Leu Ser Ser Val Val Thr Val Pro Ser Ser Ser Leu
180         185         190
Gly Thr Lys Thr Tyr Thr Cys Asn Val Asp His Lys Pro Ser Asn Thr
195         200         205
Lys Val Asp Lys Arg Val Glu Ser Lys Tyr Gly Pro Pro Cys Pro Pro
210         215         220
Cys Pro Ala Pro Glu Phe Leu Gly Gly Pro Ser Val Phe Leu Phe Pro
225         230         235         240
Pro Lys Pro Lys Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr
245         250         255
Cys Val Val Val Asp Val Ser Gln Glu Asp Pro Glu Val Gln Phe Asn
260         265         270
Trp Tyr Val Asp Gly Val Glu Val His Asn Ala Lys Thr Lys Pro Arg
275         280         285
Glu Glu Gln Phe Asn Ser Thr Tyr Arg Val Val Ser Val Leu Thr Val
290         295         300
Leu His Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser
305         310         315         320
Asn Lys Gly Leu Pro Ser Ser Ile Glu Lys Thr Ile Ser Lys Ala Lys
325         330         335
Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Gln Glu
340         345         350
Glu Met Thr Lys Asn Gln Val Ser Leu Thr Cys Leu Val Lys Gly Phe
355         360         365

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Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu  
370 375 380  
Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe  
385 390 395 400  
Phe Leu Tyr Ser Arg Leu Thr Val Asp Lys Ser Arg Trp Gln Glu Gly  
405 410 415  
Asn Val Phe Ser Cys Ser Val Met His Glu Ala Leu His Asn His Tyr  
420 425 430  
Thr Gln Lys Ser Leu Ser Leu Ser Leu Gly Lys  
435 440

<210> SEQ ID NO 365  
<211> LENGTH: 215  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 365

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly  
1 5 10 15  
Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Ser Ile Ser Ser Tyr  
20 25 30  
Leu Asn Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile  
35 40 45  
Tyr Ala Ala Ser Ser Leu Gln Ser Gly Val Pro Ser Arg Phe Ser Gly  
50 55 60  
Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro  
65 70 75 80  
Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln Ser Tyr Ser Thr Pro Pro  
85 90 95  
Ile Thr Phe Gly Gln Gly Thr Arg Leu Glu Ile Lys Arg Thr Val Ala  
100 105 110  
Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser  
115 120 125  
Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu  
130 135 140  
Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly Asn Ser  
145 150 155 160  
Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu  
165 170 175  
Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys His Lys Val  
180 185 190  
Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro Val Thr Lys  
195 200 205  
Ser Phe Asn Arg Gly Glu Cys  
210 215

<210> SEQ ID NO 366  
<211> LENGTH: 446  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

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&lt;400&gt; SEQUENCE: 366

|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Glu | Val | Gln | Leu | Val | Glu | Ser | Gly | Gly | Gly | Leu | Val | Gln | Pro | Gly | Arg | 1   | 5   | 10  | 15  |
| Ser | Leu | Arg | Leu | Ser | Cys | Ala | Ala | Ser | Gly | Phe | Thr | Phe | Asp | Asp | Tyr | 20  | 25  | 30  |     |
| Gly | Met | His | Trp | Val | Arg | Gln | Ala | Pro | Gly | Lys | Gly | Leu | Glu | Trp | Val | 35  | 40  | 45  |     |
| Ser | Val | Ile | Ser | Trp | Asn | Ser | Ala | Thr | Ile | Asp | Tyr | Ala | Asp | Ser | Val | 50  | 55  | 60  |     |
| Lys | Gly | Arg | Phe | Thr | Ile | Ser | Arg | Asp | Asn | Ala | Lys | Asn | Ser | Leu | Phe | 65  | 70  | 75  | 80  |
| Leu | Gln | Met | Asn | Ser | Leu | Arg | Pro | Glu | Asp | Thr | Ala | Leu | Tyr | Tyr | Cys | 85  | 90  | 95  |     |
| Ala | Lys | Lys | Asn | Ser | Leu | Gly | Trp | Phe | Phe | Asp | Tyr | Trp | Gly | Gln | Gly | 100 | 105 | 110 |     |
| Thr | Leu | Val | Thr | Val | Ser | Ser | Ala | Ser | Thr | Lys | Gly | Pro | Ser | Val | Phe | 115 | 120 | 125 |     |
| Pro | Leu | Ala | Pro | Cys | Ser | Arg | Ser | Thr | Ser | Glu | Ser | Thr | Ala | Ala | Leu | 130 | 135 | 140 |     |
| Gly | Cys | Leu | Val | Lys | Asp | Tyr | Phe | Pro | Glu | Pro | Val | Thr | Val | Ser | Trp | 145 | 150 | 155 | 160 |
| Asn | Ser | Gly | Ala | Leu | Thr | Ser | Gly | Val | His | Thr | Phe | Pro | Ala | Val | Leu | 165 | 170 | 175 |     |
| Gln | Ser | Ser | Gly | Leu | Tyr | Ser | Leu | Ser | Ser | Val | Val | Thr | Val | Pro | Ser | 180 | 185 | 190 |     |
| Ser | Ser | Leu | Gly | Thr | Lys | Thr | Tyr | Thr | Cys | Asn | Val | Asp | His | Lys | Pro | 195 | 200 | 205 |     |
| Ser | Asn | Thr | Lys | Val | Asp | Lys | Arg | Val | Glu | Ser | Lys | Tyr | Gly | Pro | Pro | 210 | 215 | 220 |     |
| Cys | Pro | Pro | Cys | Pro | Ala | Pro | Glu | Phe | Leu | Gly | Gly | Pro | Ser | Val | Phe | 225 | 230 | 235 | 240 |
| Leu | Phe | Pro | Pro | Lys | Pro | Lys | Asp | Thr | Leu | Met | Ile | Ser | Arg | Thr | Pro | 245 | 250 | 255 |     |
| Glu | Val | Thr | Cys | Val | Val | Val | Asp | Val | Ser | Gln | Glu | Asp | Pro | Glu | Val | 260 | 265 | 270 |     |
| Gln | Phe | Asn | Trp | Tyr | Val | Asp | Gly | Val | Glu | Val | His | Asn | Ala | Lys | Thr | 275 | 280 | 285 |     |
| Lys | Pro | Arg | Glu | Glu | Gln | Phe | Asn | Ser | Thr | Tyr | Arg | Val | Val | Ser | Val | 290 | 295 | 300 |     |
| Leu | Thr | Val | Leu | His | Gln | Asp | Trp | Leu | Asn | Gly | Lys | Glu | Tyr | Lys | Cys | 305 | 310 | 315 | 320 |
| Lys | Val | Ser | Asn | Lys | Gly | Leu | Pro | Ser | Ser | Ile | Glu | Lys | Thr | Ile | Ser | 325 | 330 | 335 |     |
| Lys | Ala | Lys | Gly | Gln | Pro | Arg | Glu | Pro | Gln | Val | Tyr | Thr | Leu | Pro | Pro | 340 | 345 | 350 |     |
| Ser | Gln | Glu | Glu | Met | Thr | Lys | Asn | Gln | Val | Ser | Leu | Thr | Cys | Leu | Val | 355 | 360 | 365 |     |
| Lys | Gly | Phe | Tyr | Pro | Ser | Asp | Ile | Ala | Val | Glu | Trp | Glu | Ser | Asn | Gly | 370 | 375 | 380 |     |
| Gln | Pro | Glu | Asn | Asn | Tyr | Lys | Thr | Thr | Pro | Pro | Val | Leu | Asp | Ser | Asp | 385 | 390 | 395 | 400 |

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Gly Ser Phe Phe Leu Tyr Ser Arg Leu Thr Val Asp Lys Ser Arg Trp  
405 410 415

Gln Glu Gly Asn Val Phe Ser Cys Ser Val Met His Glu Ala Leu His  
420 425 430

Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Leu Gly Lys  
435 440 445

<210> SEQ ID NO 367

<211> LENGTH: 220

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 367

Asp Ile Val Met Thr Gln Ser Pro Asp Ser Leu Ala Val Ser Leu Gly  
1 5 10 15

Glu Arg Ala Thr Ile Asn Cys Lys Ser Ser Gln Ser Val Leu Tyr Ser  
20 25 30

Ser Asn Asp Asn Asn Tyr Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln  
35 40 45

Pro Pro Lys Leu Leu Ile Tyr Trp Ala Ser Thr Arg Glu Ser Gly Val  
50 55 60

Pro Asp Arg Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr  
65 70 75 80

Ile Ser Ser Leu Gln Ala Glu Asp Val Ala Val Tyr Tyr Cys Gln Gln  
85 90 95

Tyr Tyr Ser Thr Pro Leu Thr Phe Gly Gly Gly Thr Lys Val Glu Ile  
100 105 110

Lys Arg Thr Val Ala Ala Pro Ser Val Phe Ile Phe Pro Pro Ser Asp  
115 120 125

Glu Gln Leu Lys Ser Gly Thr Ala Ser Val Val Cys Leu Leu Asn Asn  
130 135 140

Phe Tyr Pro Arg Glu Ala Lys Val Gln Trp Lys Val Asp Asn Ala Leu  
145 150 155 160

Gln Ser Gly Asn Ser Gln Glu Ser Val Thr Glu Gln Asp Ser Lys Asp  
165 170 175

Ser Thr Tyr Ser Leu Ser Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr  
180 185 190

Glu Lys His Lys Val Tyr Ala Cys Glu Val Thr His Gln Gly Leu Ser  
195 200 205

Ser Pro Val Thr Lys Ser Phe Asn Arg Gly Glu Cys  
210 215 220

<210> SEQ ID NO 368

<211> LENGTH: 444

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 368

Gln Val Gln Leu Val Gln Ser Gly Ala Glu Val Lys Lys Pro Gly Ala  
1 5 10 15

Ser Val Lys Val Ser Cys Lys Ala Ser Gly Tyr Thr Phe Thr Ser Tyr

-continued

| 20  |     |     |     |     |     | 25  |     |     |     |     | 30  |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Gly | Ile | Ser | Trp | Val | Arg | Gln | Ala | Pro | Gly | Gln | Gly | Leu | Glu | Trp | Met |
|     |     | 35  |     |     |     |     | 40  |     |     |     |     | 45  |     |     |     |
| Gly | Trp | Ile | Ser | Ala | Tyr | Asn | Gly | Asn | Thr | Ile | Tyr | Ala | Gln | Lys | Phe |
|     | 50  |     |     |     |     | 55  |     |     |     |     | 60  |     |     |     |     |
| Gln | Gly | Arg | Val | Thr | Met | Thr | Thr | Asp | Thr | Ser | Thr | Ser | Thr | Ala | Tyr |
| 65  |     |     |     |     | 70  |     |     |     |     | 75  |     |     |     |     | 80  |
| Met | Glu | Leu | Arg | Ser | Leu | Arg | Ser | Asp | Asp | Thr | Ala | Val | Tyr | Tyr | Cys |
|     |     |     |     | 85  |     |     |     |     | 90  |     |     |     |     | 95  |     |
| Ala | Arg | Asp | Ile | Asn | Trp | Tyr | Phe | Asp | Leu | Trp | Gly | Arg | Gly | Thr | Leu |
|     |     |     | 100 |     |     |     |     | 105 |     |     |     |     | 110 |     |     |
| Val | Thr | Val | Ser | Ser | Ala | Ser | Thr | Lys | Gly | Pro | Ser | Val | Phe | Pro | Leu |
|     |     | 115 |     |     |     |     | 120 |     |     |     |     | 125 |     |     |     |
| Ala | Pro | Cys | Ser | Arg | Ser | Thr | Ser | Glu | Ser | Thr | Ala | Ala | Leu | Gly | Cys |
|     | 130 |     |     |     |     | 135 |     |     |     |     | 140 |     |     |     |     |
| Leu | Val | Lys | Asp | Tyr | Phe | Pro | Glu | Pro | Val | Thr | Val | Ser | Trp | Asn | Ser |
| 145 |     |     |     |     | 150 |     |     |     |     | 155 |     |     |     |     | 160 |
| Gly | Ala | Leu | Thr | Ser | Gly | Val | His | Thr | Phe | Pro | Ala | Val | Leu | Gln | Ser |
|     |     |     | 165 |     |     |     |     |     | 170 |     |     |     |     | 175 |     |
| Ser | Gly | Leu | Tyr | Ser | Leu | Ser | Ser | Val | Val | Thr | Val | Pro | Ser | Ser | Ser |
|     |     | 180 |     |     |     |     |     | 185 |     |     |     |     | 190 |     |     |
| Leu | Gly | Thr | Lys | Thr | Tyr | Thr | Cys | Asn | Val | Asp | His | Lys | Pro | Ser | Asn |
|     | 195 |     |     |     |     |     | 200 |     |     |     |     | 205 |     |     |     |
| Thr | Lys | Val | Asp | Lys | Arg | Val | Glu | Ser | Lys | Tyr | Gly | Pro | Pro | Cys | Pro |
|     | 210 |     |     |     |     | 215 |     |     |     |     | 220 |     |     |     |     |
| Pro | Cys | Pro | Ala | Pro | Glu | Phe | Leu | Gly | Gly | Pro | Ser | Val | Phe | Leu | Phe |
|     | 225 |     |     |     | 230 |     |     |     |     | 235 |     |     |     |     | 240 |
| Pro | Pro | Lys | Pro | Lys | Asp | Thr | Leu | Met | Ile | Ser | Arg | Thr | Pro | Glu | Val |
|     |     |     | 245 |     |     |     |     |     | 250 |     |     |     |     | 255 |     |
| Thr | Cys | Val | Val | Val | Asp | Val | Ser | Gln | Glu | Asp | Pro | Glu | Val | Gln | Phe |
|     |     | 260 |     |     |     |     |     | 265 |     |     |     |     | 270 |     |     |
| Asn | Trp | Tyr | Val | Asp | Gly | Val | Glu | Val | His | Asn | Ala | Lys | Thr | Lys | Pro |
|     | 275 |     |     |     |     |     | 280 |     |     |     |     | 285 |     |     |     |
| Arg | Glu | Glu | Gln | Phe | Asn | Ser | Thr | Tyr | Arg | Val | Val | Ser | Val | Leu | Thr |
|     | 290 |     |     |     |     | 295 |     |     |     |     | 300 |     |     |     |     |
| Val | Leu | His | Gln | Asp | Trp | Leu | Asn | Gly | Lys | Glu | Tyr | Lys | Cys | Lys | Val |
|     | 305 |     |     |     | 310 |     |     |     |     | 315 |     |     |     |     | 320 |
| Ser | Asn | Lys | Gly | Leu | Pro | Ser | Ser | Ile | Glu | Lys | Thr | Ile | Ser | Lys | Ala |
|     |     |     | 325 |     |     |     |     |     | 330 |     |     |     |     | 335 |     |
| Lys | Gly | Gln | Pro | Arg | Glu | Pro | Gln | Val | Tyr | Thr | Leu | Pro | Pro | Ser | Gln |
|     |     | 340 |     |     |     |     |     | 345 |     |     |     |     | 350 |     |     |
| Glu | Glu | Met | Thr | Lys | Asn | Gln | Val | Ser | Leu | Thr | Cys | Leu | Val | Lys | Gly |
|     | 355 |     |     |     |     |     | 360 |     |     |     |     | 365 |     |     |     |
| Phe | Tyr | Pro | Ser | Asp | Ile | Ala | Val | Glu | Trp | Glu | Ser | Asn | Gly | Gln | Pro |
|     | 370 |     |     |     |     | 375 |     |     |     |     | 380 |     |     |     |     |
| Glu | Asn | Asn | Tyr | Lys | Thr | Thr | Pro | Pro | Val | Leu | Asp | Ser | Asp | Gly | Ser |
|     | 385 |     |     |     | 390 |     |     |     |     | 395 |     |     |     | 400 |     |
| Phe | Phe | Leu | Tyr | Ser | Arg | Leu | Thr | Val | Asp | Lys | Ser | Arg | Trp | Gln | Glu |
|     |     |     | 405 |     |     |     |     |     | 410 |     |     |     |     | 415 |     |
| Gly | Asn | Val | Phe | Ser | Cys | Ser | Val | Met | His | Glu | Ala | Leu | His | Asn | His |
|     |     | 420 |     |     |     |     |     | 425 |     |     |     |     | 430 |     |     |



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Tyr Thr Gln Lys Ser Leu Ser Leu Ser Leu Gly Lys  
435 440

<210> SEQ ID NO 369  
<211> LENGTH: 214  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 369

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly  
1 5 10 15  
Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Gly Ile Arg Asn Asp  
20 25 30  
Leu Gly Trp Phe Gln Gln Lys Pro Gly Lys Ala Pro Lys Arg Leu Ile  
35 40 45  
Tyr Ala Ala Ser Ser Leu Gln Ser Gly Val Pro Ser Arg Phe Ser Gly  
50 55 60  
Ser Gly Ser Gly Thr Glu Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro  
65 70 75 80  
Glu Asp Phe Ala Thr Tyr Tyr Cys Leu Gln His Asn Ser Tyr Pro Trp  
85 90 95  
Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Arg Thr Val Ala Ala  
100 105 110  
Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser Gly  
115 120 125  
Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu Ala  
130 135 140  
Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly Asn Ser Gln  
145 150 155 160  
Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu Ser  
165 170 175  
Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys His Lys Val Tyr  
180 185 190  
Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro Val Thr Lys Ser  
195 200 205  
Phe Asn Arg Gly Glu Cys  
210

<210> SEQ ID NO 370  
<211> LENGTH: 446  
<212> TYPE: PRT  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 370

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Arg  
1 5 10 15  
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Asp Asp Tyr  
20 25 30  
Gly Met His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val  
35 40 45  
Ser Val Ile Ser Trp Asn Gly Ala Thr Ile Asp Tyr Ala Asp Ser Val

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| 50  |     |     |     |     | 55  |     |     |     |     | 60  |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Lys | Gly | Arg | Phe | Thr | Ile | Ser | Arg | Asp | Asn | Ala | Lys | Asn | Ser | Leu | Phe |
| 65  |     |     |     | 70  |     |     |     |     | 75  |     |     |     |     | 80  |     |
| Leu | Gln | Met | Asn | Ser | Leu | Arg | Pro | Glu | Asp | Thr | Ala | Leu | Tyr | Tyr | Cys |
|     |     |     | 85  |     |     |     |     |     | 90  |     |     |     |     | 95  |     |
| Ala | Lys | Lys | Asn | Ser | Leu | Gly | Trp | Phe | Phe | Asp | Tyr | Trp | Gly | Gln | Gly |
|     |     |     | 100 |     |     |     |     | 105 |     |     |     |     | 110 |     |     |
| Thr | Leu | Val | Thr | Val | Ser | Ser | Ala | Ser | Thr | Lys | Gly | Pro | Ser | Val | Phe |
|     | 115 |     |     |     |     |     | 120 |     |     |     |     | 125 |     |     |     |
| Pro | Leu | Ala | Pro | Cys | Ser | Arg | Ser | Thr | Ser | Glu | Ser | Thr | Ala | Ala | Leu |
|     | 130 |     |     |     |     | 135 |     |     |     |     | 140 |     |     |     |     |
| Gly | Cys | Leu | Val | Lys | Asp | Tyr | Phe | Pro | Glu | Pro | Val | Thr | Val | Ser | Trp |
| 145 |     |     |     | 150 |     |     |     |     | 155 |     |     |     |     |     | 160 |
| Asn | Ser | Gly | Ala | Leu | Thr | Ser | Gly | Val | His | Thr | Phe | Pro | Ala | Val | Leu |
|     |     |     | 165 |     |     |     |     | 170 |     |     |     |     |     | 175 |     |
| Gln | Ser | Ser | Gly | Leu | Tyr | Ser | Leu | Ser | Ser | Val | Val | Thr | Val | Pro | Ser |
|     | 180 |     |     |     |     |     | 185 |     |     |     |     |     | 190 |     |     |
| Ser | Ser | Leu | Gly | Thr | Lys | Thr | Tyr | Thr | Cys | Asn | Val | Asp | His | Lys | Pro |
|     | 195 |     |     |     |     | 200 |     |     |     |     |     | 205 |     |     |     |
| Ser | Asn | Thr | Lys | Val | Asp | Lys | Arg | Val | Glu | Ser | Lys | Tyr | Gly | Pro | Pro |
|     | 210 |     |     |     |     | 215 |     |     |     |     | 220 |     |     |     |     |
| Cys | Pro | Pro | Cys | Pro | Ala | Pro | Glu | Phe | Leu | Gly | Gly | Pro | Ser | Val | Phe |
| 225 |     |     |     | 230 |     |     |     |     | 235 |     |     |     |     |     | 240 |
| Leu | Phe | Pro | Pro | Lys | Pro | Lys | Asp | Thr | Leu | Met | Ile | Ser | Arg | Thr | Pro |
|     |     |     | 245 |     |     |     |     | 250 |     |     |     |     |     | 255 |     |
| Glu | Val | Thr | Cys | Val | Val | Val | Asp | Val | Ser | Gln | Glu | Asp | Pro | Glu | Val |
|     | 260 |     |     |     |     |     | 265 |     |     |     |     |     | 270 |     |     |
| Gln | Phe | Asn | Trp | Tyr | Val | Asp | Gly | Val | Glu | Val | His | Asn | Ala | Lys | Thr |
|     | 275 |     |     |     |     |     | 280 |     |     |     |     | 285 |     |     |     |
| Lys | Pro | Arg | Glu | Gln | Gln | Phe | Asn | Ser | Thr | Tyr | Arg | Val | Val | Ser | Val |
|     | 290 |     |     |     |     | 295 |     |     |     |     | 300 |     |     |     |     |
| Leu | Thr | Val | Leu | His | Gln | Asp | Trp | Leu | Asn | Gly | Lys | Glu | Tyr | Lys | Cys |
| 305 |     |     |     | 310 |     |     |     |     | 315 |     |     |     |     |     | 320 |
| Lys | Val | Ser | Asn | Lys | Gly | Leu | Pro | Ser | Ser | Ile | Glu | Lys | Thr | Ile | Ser |
|     |     |     | 325 |     |     |     |     | 330 |     |     |     |     |     | 335 |     |
| Lys | Ala | Lys | Gly | Gln | Pro | Arg | Glu | Pro | Gln | Val | Tyr | Thr | Leu | Pro | Pro |
|     | 340 |     |     |     |     |     | 345 |     |     |     |     |     | 350 |     |     |
| Ser | Gln | Glu | Glu | Met | Thr | Lys | Asn | Gln | Val | Ser | Leu | Thr | Cys | Leu | Val |
|     | 355 |     |     |     |     |     | 360 |     |     |     |     | 365 |     |     |     |
| Lys | Gly | Phe | Tyr | Pro | Ser | Asp | Ile | Ala | Val | Glu | Trp | Glu | Ser | Asn | Gly |
|     | 370 |     |     |     |     | 375 |     |     |     |     | 380 |     |     |     |     |
| Gln | Pro | Glu | Asn | Asn | Tyr | Lys | Thr | Thr | Pro | Pro | Val | Leu | Asp | Ser | Asp |
| 385 |     |     |     | 390 |     |     |     |     | 395 |     |     |     |     |     | 400 |
| Gly | Ser | Phe | Phe | Leu | Tyr | Ser | Arg | Leu | Thr | Val | Asp | Lys | Ser | Arg | Trp |
|     |     |     | 405 |     |     |     |     | 410 |     |     |     |     |     | 415 |     |
| Gln | Glu | Gly | Asn | Val | Phe | Ser | Cys | Ser | Val | Met | His | Glu | Ala | Leu | His |
|     |     |     | 420 |     |     |     | 425 |     |     |     |     |     | 430 |     |     |
| Asn | His | Tyr | Thr | Gln | Lys | Ser | Leu | Ser | Leu | Ser | Leu | Gly | Lys |     |     |
|     | 435 |     |     |     |     |     | 440 |     |     |     |     | 445 |     |     |     |

&lt;210&gt; SEQ ID NO 371

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<211> LENGTH: 214
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 371
Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Phe Val Gly
1      5      10      15
Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Gly Ile Arg Asn Asp
20     25     30
Leu Gly Trp Tyr Gln Gln Lys Pro Gly Lys Thr Pro Lys Arg Leu Ile
35     40     45
Tyr Thr Ala Ser Ser Leu Arg Ser Gly Val Pro Ser Arg Phe Ser Gly
50     55     60
Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
65     70     75     80
Glu Asp Phe Ala Thr Tyr Tyr Cys Leu Gln His Asn Asn Tyr Pro Tyr
85     90     95
Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile Lys Arg Thr Val Ala Ala
100    105    110
Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser Gly
115    120    125
Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu Ala
130    135    140
Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly Asn Ser Gln
145    150    155    160
Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu Ser
165    170    175
Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys His Lys Val Tyr
180    185    190
Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro Val Thr Lys Ser
195    200    205
Phe Asn Arg Gly Glu Cys
210

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<210> SEQ ID NO 372
<211> LENGTH: 449
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 372
Glu Val Gln Leu Val Glu Ser Gly Gly Asn Leu Val Gln Pro Gly Gly
1      5      10      15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Asn Tyr
20     25     30
Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35     40     45
Ser Gly Ile Thr Gly Asn Gly Val Asn Thr Tyr Tyr Ser Val Ser Val
50     55     60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Phe
65     70     75     80
Leu Glu Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Ile Tyr Tyr Cys

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| 85  |     |     |     |     | 90  |     |     |     |     | 95  |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Val | Lys | Glu | Arg | Gly | His | Ser | Trp | Phe | Gly | Asp | Trp | Phe | Asp | Pro | Trp |
|     |     |     | 100 |     |     |     |     | 105 |     |     |     |     | 110 |     |     |
| Gly | Gln | Gly | Thr | Leu | Val | Thr | Val | Ser | Ser | Ala | Ser | Thr | Lys | Gly | Pro |
|     |     | 115 |     |     |     |     | 120 |     |     |     |     | 125 |     |     |     |
| Ser | Val | Phe | Pro | Leu | Ala | Pro | Cys | Ser | Arg | Ser | Thr | Ser | Glu | Ser | Thr |
|     | 130 |     |     |     |     | 135 |     |     |     |     | 140 |     |     |     |     |
| Ala | Ala | Leu | Gly | Cys | Leu | Val | Lys | Asp | Tyr | Phe | Pro | Glu | Pro | Val | Thr |
| 145 |     |     |     |     |     | 150 |     |     |     |     | 155 |     |     |     | 160 |
| Val | Ser | Trp | Asn | Ser | Gly | Ala | Leu | Thr | Ser | Gly | Val | His | Thr | Phe | Pro |
|     |     |     | 165 |     |     |     |     |     | 170 |     |     |     |     | 175 |     |
| Ala | Val | Leu | Gln | Ser | Ser | Gly | Leu | Tyr | Ser | Leu | Ser | Ser | Val | Val | Thr |
|     |     | 180 |     |     |     |     | 185 |     |     |     |     |     | 190 |     |     |
| Val | Pro | Ser | Ser | Ser | Leu | Gly | Thr | Lys | Thr | Tyr | Thr | Cys | Asn | Val | Asp |
|     |     | 195 |     |     |     |     | 200 |     |     |     |     | 205 |     |     |     |
| His | Lys | Pro | Ser | Asn | Thr | Lys | Val | Asp | Lys | Arg | Val | Glu | Ser | Lys | Tyr |
|     | 210 |     |     |     |     | 215 |     |     |     |     | 220 |     |     |     |     |
| Gly | Pro | Pro | Cys | Pro | Pro | Cys | Pro | Ala | Pro | Glu | Phe | Leu | Gly | Gly | Pro |
| 225 |     |     |     |     |     | 230 |     |     |     |     | 235 |     |     |     | 240 |
| Ser | Val | Phe | Leu | Phe | Pro | Pro | Lys | Pro | Lys | Asp | Thr | Leu | Met | Ile | Ser |
|     |     |     | 245 |     |     |     |     |     | 250 |     |     |     |     | 255 |     |
| Arg | Thr | Pro | Glu | Val | Thr | Cys | Val | Val | Val | Asp | Val | Ser | Gln | Glu | Asp |
|     |     | 260 |     |     |     |     | 265 |     |     |     |     |     | 270 |     |     |
| Pro | Glu | Val | Gln | Phe | Asn | Trp | Tyr | Val | Asp | Gly | Val | Glu | Val | His | Asn |
|     |     | 275 |     |     |     |     | 280 |     |     |     |     | 285 |     |     |     |
| Ala | Lys | Thr | Lys | Pro | Arg | Glu | Glu | Gln | Phe | Asn | Ser | Thr | Tyr | Arg | Val |
|     | 290 |     |     |     |     | 295 |     |     |     |     | 300 |     |     |     |     |
| Val | Ser | Val | Leu | Thr | Val | Leu | His | Gln | Asp | Trp | Leu | Asn | Gly | Lys | Glu |
| 305 |     |     |     |     |     | 310 |     |     |     |     | 315 |     |     |     | 320 |
| Tyr | Lys | Cys | Lys | Val | Ser | Asn | Lys | Gly | Leu | Pro | Ser | Ser | Ile | Glu | Lys |
|     |     |     | 325 |     |     |     |     |     | 330 |     |     |     |     | 335 |     |
| Thr | Ile | Ser | Lys | Ala | Lys | Gly | Gln | Pro | Arg | Glu | Pro | Gln | Val | Tyr | Thr |
|     |     | 340 |     |     |     |     |     | 345 |     |     |     |     | 350 |     |     |
| Leu | Pro | Pro | Ser | Gln | Glu | Glu | Met | Thr | Lys | Asn | Gln | Val | Ser | Leu | Thr |
|     |     | 355 |     |     |     |     | 360 |     |     |     |     | 365 |     |     |     |
| Cys | Leu | Val | Lys | Gly | Phe | Tyr | Pro | Ser | Asp | Ile | Ala | Val | Glu | Trp | Glu |
|     | 370 |     |     |     |     | 375 |     |     |     |     | 380 |     |     |     |     |
| Ser | Asn | Gly | Gln | Pro | Glu | Asn | Asn | Tyr | Lys | Thr | Thr | Pro | Pro | Val | Leu |
| 385 |     |     |     |     |     | 390 |     |     |     |     | 395 |     |     |     | 400 |
| Asp | Ser | Asp | Gly | Ser | Phe | Phe | Leu | Tyr | Ser | Arg | Leu | Thr | Val | Asp | Lys |
|     |     |     | 405 |     |     |     |     |     | 410 |     |     |     |     | 415 |     |
| Ser | Arg | Trp | Gln | Glu | Gly | Asn | Val | Phe | Ser | Cys | Ser | Val | Met | His | Glu |
|     |     | 420 |     |     |     |     | 425 |     |     |     |     | 430 |     |     |     |
| Ala | Leu | His | Asn | His | Tyr | Thr | Gln | Lys | Ser | Leu | Ser | Leu | Ser | Leu | Gly |
|     |     | 435 |     |     |     |     | 440 |     |     |     |     | 445 |     |     |     |

Lys

&lt;210&gt; SEQ ID NO 373

&lt;211&gt; LENGTH: 214

&lt;212&gt; TYPE: PRT

&lt;213&gt; ORGANISM: Artificial Sequence

&lt;220&gt; FEATURE:

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<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 373

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly  
1 5 10 15  
Asp Arg Val Thr Ile Thr Cys Arg Ser Ser Gln Ser Val Ile Ser Tyr  
20 25 30  
Leu Asn Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile  
35 40 45  
Phe Gly Ala Ser Ser Leu Ile Ser Gly Val Pro Ser Arg Phe Ser Gly  
50 55 60  
Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Val Ser Leu Gln Pro  
65 70 75 80  
Glu Asp Phe Ala Val Tyr Phe Cys Gln Gln Asn Tyr Leu Pro Pro Leu  
85 90 95  
Thr Phe Gly Gly Gly Thr Lys Val Glu Ile Lys Arg Thr Val Ala Ala  
100 105 110  
Pro Ser Val Phe Ile Phe Pro Pro Ser Asp Glu Gln Leu Lys Ser Gly  
115 120 125  
Thr Ala Ser Val Val Cys Leu Leu Asn Asn Phe Tyr Pro Arg Glu Ala  
130 135 140  
Lys Val Gln Trp Lys Val Asp Asn Ala Leu Gln Ser Gly Asn Ser Gln  
145 150 155 160  
Glu Ser Val Thr Glu Gln Asp Ser Lys Asp Ser Thr Tyr Ser Leu Ser  
165 170 175  
Ser Thr Leu Thr Leu Ser Lys Ala Asp Tyr Glu Lys His Lys Val Tyr  
180 185 190  
Ala Cys Glu Val Thr His Gln Gly Leu Ser Ser Pro Val Thr Lys Ser  
195 200 205  
Phe Asn Arg Gly Glu Cys  
210

<210> SEQ ID NO 374

<211> LENGTH: 449

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 374

Glu Val Gln Leu Val Glu Ser Gly Gly Asn Leu Val Gln Pro Gly Gly  
1 5 10 15  
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Phe Ser Asn Tyr  
20 25 30  
Ala Met Ser Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val  
35 40 45  
Ser Gly Ile Thr Gly Asn Gly Ile Lys Thr Tyr Tyr Ser Val Ser Ala  
50 55 60  
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Phe  
65 70 75 80  
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Ile Tyr Tyr Cys  
85 90 95  
Val Lys Glu Arg Gly His Ser Trp Phe Gly Asp Trp Phe Asp Pro Trp  
100 105 110

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|         |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|---------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Gly 115 | Gln | Thr | Leu | Val | Thr | Val | Ser | Ser | Ala | Ser | Thr | Lys | Gly | Pro |     |
| Ser 130 | Val | Phe | Pro | Leu | Ala | Pro | Cys | Ser | Arg | Ser | Thr | Ser | Glu | Ser | Thr |
| Ala 145 | Ala | Leu | Gly | Cys | Leu | Val | Lys | Asp | Tyr | Phe | Pro | Glu | Pro | Val | Thr |
| Val     | Ser | Trp | Asn | Ser | Gly | Ala | Leu | Thr | Ser | Gly | Val | His | Thr | Phe | Pro |
| Ala     | Val | Leu | Gln | Ser | Ser | Gly | Leu | Tyr | Ser | Leu | Ser | Ser | Val | Val | Thr |
| Val     | Pro | Ser | Ser | Ser | Leu | Gly | Thr | Lys | Thr | Tyr | Thr | Cys | Asn | Val | Asp |
| His     | Lys | Pro | Ser | Asn | Thr | Lys | Val | Asp | Lys | Arg | Val | Glu | Ser | Lys | Tyr |
| Gly 225 | Pro | Pro | Cys | Pro | Pro | Cys | Pro | Ala | Pro | Glu | Phe | Leu | Gly | Gly | Pro |
| Ser     | Val | Phe | Leu | Phe | Pro | Pro | Lys | Pro | Lys | Asp | Thr | Leu | Met | Ile | Ser |
| Arg     | Thr | Pro | Glu | Val | Thr | Cys | Val | Val | Val | Asp | Val | Ser | Gln | Glu | Asp |
| Pro     | Glu | Val | Gln | Phe | Asn | Trp | Tyr | Val | Asp | Gly | Val | Glu | Val | His | Asn |
| Ala     | Lys | Thr | Lys | Pro | Arg | Glu | Glu | Gln | Phe | Asn | Ser | Thr | Tyr | Arg | Val |
| Val 305 | Ser | Val | Leu | Thr | Val | Leu | His | Gln | Asp | Trp | Leu | Asn | Gly | Lys | Glu |
| Tyr     | Lys | Cys | Lys | Val | Ser | Asn | Lys | Gly | Leu | Pro | Ser | Ser | Ile | Glu | Lys |
| Thr     | Ile | Ser | Lys | Ala | Lys | Gly | Gln | Pro | Arg | Glu | Pro | Gln | Val | Tyr | Thr |
| Leu     | Pro | Pro | Ser | Gln | Glu | Glu | Met | Thr | Lys | Asn | Gln | Val | Ser | Leu | Thr |
| Cys     | Leu | Val | Lys | Gly | Phe | Tyr | Pro | Ser | Asp | Ile | Ala | Val | Glu | Trp | Glu |
| Ser 385 | Asn | Gly | Gln | Pro | Glu | Asn | Asn | Tyr | Lys | Thr | Thr | Pro | Pro | Val | Leu |
| Asp     | Ser | Asp | Gly | Ser | Phe | Phe | Leu | Tyr | Ser | Arg | Leu | Thr | Val | Asp | Lys |
| Ser     | Arg | Trp | Gln | Glu | Gly | Asn | Val | Phe | Ser | Cys | Ser | Val | Met | His | Glu |
| Ala     | Leu | His | Asn | His | Tyr | Thr | Gln | Lys | Ser | Leu | Ser | Leu | Ser | Leu | Gly |

Lys

```
<210> SEQ ID NO 375
<211> LENGTH: 219
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 375
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|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Asp | Ile | Val | Met | Thr | Gln | Thr | Pro | Leu | Ser | Ser | Pro | Val | Thr | Leu | Gly | 1   | 5   | 10  | 15  |
| Gln | Pro | Ala | Ser | Ile | Ser | Cys | Arg | Ser | Ser | Gln | Gly | Leu | Val | Asn | Ser | 20  | 25  | 30  |     |
| Asp | Gly | Asn | Thr | Tyr | Leu | Ser | Trp | Leu | His | Gln | Arg | Pro | Gly | Gln | Pro | 35  | 40  | 45  |     |
| Pro | Arg | Leu | Leu | Ile | Tyr | Glu | Ile | Ser | Asn | Arg | Phe | Ser | Gly | Val | Pro | 50  | 55  | 60  |     |
| Asp | Arg | Phe | Ser | Gly | Ser | Gly | Ala | Gly | Thr | Asp | Phe | Thr | Leu | Lys | Ile | 65  | 70  | 75  | 80  |
| Ser | Arg | Val | Glu | Ala | Glu | Asp | Val | Gly | Phe | Tyr | Tyr | Cys | Met | Gln | Ser | 85  | 90  | 95  |     |
| Thr | Gln | Phe | Pro | Leu | Thr | Phe | Gly | Gly | Gly | Thr | Lys | Val | Glu | Ile | Lys | 100 | 105 | 110 |     |
| Arg | Thr | Val | Ala | Ala | Pro | Ser | Val | Phe | Ile | Phe | Pro | Pro | Ser | Asp | Glu | 115 | 120 | 125 |     |
| Gln | Leu | Lys | Ser | Gly | Thr | Ala | Ser | Val | Val | Cys | Leu | Leu | Asn | Asn | Phe | 130 | 135 | 140 |     |
| Tyr | Pro | Arg | Glu | Ala | Lys | Val | Gln | Trp | Lys | Val | Asp | Asn | Ala | Leu | Gln | 145 | 150 | 155 | 160 |
| Ser | Gly | Asn | Ser | Gln | Glu | Ser | Val | Thr | Glu | Gln | Asp | Ser | Lys | Asp | Ser | 165 | 170 | 175 |     |
| Thr | Tyr | Ser | Leu | Ser | Ser | Thr | Leu | Thr | Leu | Ser | Lys | Ala | Asp | Tyr | Glu | 180 | 185 | 190 |     |
| Lys | His | Lys | Val | Tyr | Ala | Cys | Glu | Val | Thr | His | Gln | Gly | Leu | Ser | Ser | 195 | 200 | 205 |     |
| Pro | Val | Thr | Lys | Ser | Phe | Asn | Arg | Gly | Glu | Cys | 210 | 215 |     |     |     |     |     |     |     |

&lt;210&gt; SEQ ID NO 376

&lt;211&gt; LENGTH: 449

&lt;212&gt; TYPE: PRT

&lt;213&gt; ORGANISM: Artificial Sequence

&lt;220&gt; FEATURE:

&lt;223&gt; OTHER INFORMATION: Synthetic Construct

&lt;400&gt; SEQUENCE: 376

|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |    |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|----|
| Glu | Val | Gln | Leu | Val | Glu | Ser | Gly | Gly | Asn | Leu | Val | Gln | Pro | Gly | Gly | 1   | 5   | 10  | 15 |
| Ser | Leu | Arg | Leu | Ser | Cys | Ala | Ala | Ser | Gly | Phe | Thr | Phe | Ser | Asn | Tyr | 20  | 25  | 30  |    |
| Ala | Met | Ser | Trp | Val | Arg | Gln | Ala | Pro | Gly | Lys | Gly | Leu | Glu | Trp | Val | 35  | 40  | 45  |    |
| Ser | Gly | Ile | Thr | Gly | Asn | Gly | Ile | Asn | Thr | Tyr | Tyr | Ser | Val | Ser | Val | 50  | 55  | 60  |    |
| Lys | Gly | Arg | Phe | Thr | Ile | Ser | Arg | Asp | Asn | Ser | Lys | Asn | Thr | Val | Phe | 65  | 70  | 75  | 80 |
| Leu | Gln | Met | Asn | Ser | Leu | Arg | Ala | Glu | Asp | Thr | Ala | Ile | Tyr | Tyr | Cys | 85  | 90  | 95  |    |
| Val | Lys | Glu | Arg | Gly | His | Ser | Trp | Phe | Gly | Asp | Trp | Phe | Asp | Pro | Trp | 100 | 105 | 110 |    |
| Gly | Gln | Gly | Thr | Leu | Val | Thr | Val | Ser | Ser | Ala | Ser | Thr | Lys | Gly | Pro | 115 | 120 | 125 |    |

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|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Ser | Val | Phe | Pro | Leu | Ala | Pro | Cys | Ser | Arg | Ser | Thr | Ser | Glu | Ser | Thr | 130 | 135 | 140 |
| Ala | Ala | Leu | Gly | Cys | Leu | Val | Lys | Asp | Tyr | Phe | Pro | Glu | Pro | Val | Thr | 145 | 150 | 155 |
| Val | Ser | Trp | Asn | Ser | Gly | Ala | Leu | Thr | Ser | Gly | Val | His | Thr | Phe | Pro | 165 | 170 | 175 |
| Ala | Val | Leu | Gln | Ser | Ser | Gly | Leu | Tyr | Ser | Leu | Ser | Ser | Val | Val | Thr | 180 | 185 | 190 |
| Val | Pro | Ser | Ser | Ser | Leu | Gly | Thr | Lys | Thr | Tyr | Thr | Cys | Asn | Val | Asp | 195 | 200 | 205 |
| His | Lys | Pro | Ser | Asn | Thr | Lys | Val | Asp | Lys | Arg | Val | Glu | Ser | Lys | Tyr | 210 | 215 | 220 |
| Gly | Pro | Pro | Cys | Pro | Pro | Cys | Pro | Ala | Pro | Glu | Phe | Leu | Gly | Gly | Pro | 225 | 230 | 235 |
| Ser | Val | Phe | Leu | Phe | Pro | Pro | Lys | Pro | Lys | Asp | Thr | Leu | Met | Ile | Ser | 245 | 250 | 255 |
| Arg | Thr | Pro | Glu | Val | Thr | Cys | Val | Val | Val | Asp | Val | Ser | Gln | Glu | Asp | 260 | 265 | 270 |
| Pro | Glu | Val | Gln | Phe | Asn | Trp | Tyr | Val | Asp | Gly | Val | Glu | Val | His | Asn | 275 | 280 | 285 |
| Ala | Lys | Thr | Lys | Pro | Arg | Glu | Glu | Gln | Phe | Asn | Ser | Thr | Tyr | Arg | Val | 290 | 295 | 300 |
| Val | Ser | Val | Leu | Thr | Val | Leu | His | Gln | Asp | Trp | Leu | Asn | Gly | Lys | Glu | 305 | 310 | 315 |
| Tyr | Lys | Cys | Lys | Val | Ser | Asn | Lys | Gly | Leu | Pro | Ser | Ser | Ile | Glu | Lys | 325 | 330 | 335 |
| Thr | Ile | Ser | Lys | Ala | Lys | Gly | Gln | Pro | Arg | Glu | Pro | Gln | Val | Tyr | Thr | 340 | 345 | 350 |
| Leu | Pro | Pro | Ser | Gln | Glu | Glu | Met | Thr | Lys | Asn | Gln | Val | Ser | Leu | Thr | 355 | 360 | 365 |
| Cys | Leu | Val | Lys | Gly | Phe | Tyr | Pro | Ser | Asp | Ile | Ala | Val | Glu | Trp | Glu | 370 | 375 | 380 |
| Ser | Asn | Gly | Gln | Pro | Glu | Asn | Asn | Tyr | Lys | Thr | Thr | Pro | Pro | Val | Leu | 385 | 390 | 395 |
| Asp | Ser | Asp | Gly | Ser | Phe | Phe | Leu | Tyr | Ser | Arg | Leu | Thr | Val | Asp | Lys | 405 | 410 | 415 |
| Ser | Arg | Trp | Gln | Glu | Gly | Asn | Val | Phe | Ser | Cys | Ser | Val | Met | His | Glu | 420 | 425 | 430 |
| Ala | Leu | His | Asn | His | Tyr | Thr | Gln | Lys | Ser | Leu | Ser | Leu | Ser | Leu | Gly | 435 | 440 | 445 |

Lys

&lt;210&gt; SEQ ID NO 377

&lt;211&gt; LENGTH: 214

&lt;212&gt; TYPE: PRT

&lt;213&gt; ORGANISM: Artificial Sequence

&lt;220&gt; FEATURE:

&lt;223&gt; OTHER INFORMATION: Synthetic Construct

&lt;400&gt; SEQUENCE: 377

|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Asp | Ile | Gln | Met | Thr | Gln | Ser | Pro | Ser | Ser | Leu | Ser | Ala | Ser | Val | Gly |
| 1   |     |     |     | 5   |     |     |     |     |     | 10  |     |     |     | 15  |     |

Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Gly Ile Arg Asn Asp



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| 20  |     |     |     |     | 25  |     |     |     |     | 30  |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Leu | Gly | Trp | Tyr | Gln | Gln | Lys | Pro | Gly | Lys | Ala | Pro | Lys | Arg | Leu | Ile |
|     | 35  |     |     |     |     | 40  |     |     |     |     | 45  |     |     |     |     |
| Tyr | Ala | Ala | Ser | Ser | Leu | Gln | Ser | Gly | Val | Pro | Ser | Arg | Phe | Ser | Gly |
|     | 50  |     |     |     |     | 55  |     |     |     |     | 60  |     |     |     |     |
| Ser | Gly | Ser | Gly | Thr | Glu | Phe | Thr | Leu | Thr | Ile | Ser | Ser | Leu | Gln | Pro |
|     | 65  |     |     |     |     | 70  |     |     |     |     | 75  |     |     |     | 80  |
| Glu | Asp | Phe | Ala | Thr | Tyr | Tyr | Cys | Leu | Gln | His | Asn | Ser | Tyr | Pro | Trp |
|     |     |     | 85  |     |     |     |     |     | 90  |     |     |     |     | 95  |     |
| Thr | Phe | Gly | Gln | Gly | Thr | Lys | Val | Glu | Ile | Lys | Arg | Thr | Val | Ala | Ala |
|     |     |     | 100 |     |     |     |     | 105 |     |     |     |     | 110 |     |     |
| Pro | Ser | Val | Phe | Ile | Phe | Pro | Pro | Ser | Asp | Glu | Gln | Leu | Lys | Ser | Gly |
|     |     | 115 |     |     |     |     | 120 |     |     |     |     | 125 |     |     |     |
| Thr | Ala | Ser | Val | Val | Cys | Leu | Leu | Asn | Asn | Phe | Tyr | Pro | Arg | Glu | Ala |
|     |     | 130 |     |     |     | 135 |     |     |     |     | 140 |     |     |     |     |
| Lys | Val | Gln | Trp | Lys | Val | Asp | Asn | Ala | Leu | Gln | Ser | Gly | Asn | Ser | Gln |
|     | 145 |     |     |     |     | 150 |     |     |     |     | 155 |     |     |     | 160 |
| Glu | Ser | Val | Thr | Glu | Gln | Asp | Ser | Lys | Asp | Ser | Thr | Tyr | Ser | Leu | Ser |
|     |     |     | 165 |     |     |     |     |     | 170 |     |     |     |     | 175 |     |
| Ser | Thr | Leu | Thr | Leu | Ser | Lys | Ala | Asp | Tyr | Glu | Lys | His | Lys | Val | Tyr |
|     |     | 180 |     |     |     |     |     | 185 |     |     |     |     | 190 |     |     |
| Ala | Cys | Glu | Val | Thr | His | Gln | Gly | Leu | Ser | Ser | Pro | Val | Thr | Lys | Ser |
|     |     | 195 |     |     |     |     | 200 |     |     |     |     | 205 |     |     |     |
| Phe | Asn | Arg | Gly | Glu | Cys |     |     |     |     |     |     |     |     |     |     |
|     | 210 |     |     |     |     |     |     |     |     |     |     |     |     |     |     |

&lt;210&gt; SEQ ID NO 378

&lt;211&gt; LENGTH: 450

&lt;212&gt; TYPE: PRT

&lt;213&gt; ORGANISM: Artificial Sequence

&lt;220&gt; FEATURE:

&lt;223&gt; OTHER INFORMATION: Synthetic Construct

&lt;400&gt; SEQUENCE: 378

|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Gln | Val | Gln | Leu | Val | Glu | Ser | Gly | Gly | Gly | Val | Val | Gln | Pro | Gly | Arg |
| 1   |     |     | 5   |     |     |     |     | 10  |     |     |     |     | 15  |     |     |
| Ser | Leu | Arg | Leu | Ser | Cys | Glu | Val | Ser | Gly | Phe | Thr | Phe | Ser | Asn | Tyr |
|     |     | 20  |     |     |     |     |     | 25  |     |     |     |     | 30  |     |     |
| Gly | Met | His | Trp | Val | Arg | Gln | Ala | Pro | Gly | Lys | Gly | Leu | Glu | Trp | Val |
|     |     | 35  |     |     |     |     | 40  |     |     |     |     | 45  |     |     |     |
| Ala | Val | Val | Ser | Tyr | Asp | Gly | Ser | Asn | Lys | Tyr | Tyr | Thr | Asp | Ser | Val |
|     |     | 50  |     |     |     | 55  |     |     |     |     | 60  |     |     |     |     |
| Lys | Gly | Arg | Phe | Thr | Ile | Ser | Arg | Asp | Asn | Ser | Lys | Asn | Thr | Leu | Ser |
|     | 65  |     |     |     | 70  |     |     |     | 75  |     |     |     |     | 80  |     |
| Leu | Gln | Met | Asn | Ser | Leu | Arg | Ala | Glu | Asp | Thr | Ala | Val | Tyr | Tyr | Cys |
|     |     |     | 85  |     |     |     |     | 90  |     |     |     |     | 95  |     |     |
| Ala | Lys | Asp | Arg | Gly | Ile | Thr | Gly | Thr | Ser | Gly | Gly | Val | Phe | Asp | Ile |
|     |     | 100 |     |     |     |     | 105 |     |     |     |     |     | 110 |     |     |
| Trp | Gly | Gln | Gly | Thr | Met | Val | Thr | Val | Ser | Ser | Ala | Ser | Thr | Lys | Gly |
|     |     | 115 |     |     |     | 120 |     |     |     |     |     | 125 |     |     |     |
| Pro | Ser | Val | Phe | Pro | Leu | Ala | Pro | Cys | Ser | Arg | Ser | Thr | Ser | Glu | Ser |
|     |     | 130 |     |     |     | 135 |     |     |     |     |     | 140 |     |     |     |
| Thr | Ala | Ala | Leu | Gly | Cys | Leu | Val | Lys | Asp | Tyr | Phe | Pro | Glu | Pro | Val |

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|                                                                 |     |     |     |
|-----------------------------------------------------------------|-----|-----|-----|
| 145                                                             | 150 | 155 | 160 |
| Thr Val Ser Trp Asn Ser Gly Ala Leu Thr Ser Gly Val His Thr Phe |     |     |     |
|                                                                 | 165 | 170 | 175 |
| Pro Ala Val Leu Gln Ser Ser Gly Leu Tyr Ser Leu Ser Ser Val Val |     |     |     |
|                                                                 | 180 | 185 | 190 |
| Thr Val Pro Ser Ser Ser Leu Gly Thr Lys Thr Tyr Thr Cys Asn Val |     |     |     |
|                                                                 | 195 | 200 | 205 |
| Asp His Lys Pro Ser Asn Thr Lys Val Asp Lys Arg Val Glu Ser Lys |     |     |     |
|                                                                 | 210 | 215 | 220 |
| Tyr Gly Pro Pro Cys Pro Pro Cys Pro Ala Pro Glu Phe Leu Gly Gly |     |     |     |
|                                                                 | 225 | 230 | 235 |
| Pro Ser Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile |     |     |     |
|                                                                 | 245 | 250 | 255 |
| Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser Gln Glu |     |     |     |
|                                                                 | 260 | 265 | 270 |
| Asp Pro Glu Val Gln Phe Asn Trp Tyr Val Asp Gly Val Glu Val His |     |     |     |
|                                                                 | 275 | 280 | 285 |
| Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Phe Asn Ser Thr Tyr Arg |     |     |     |
|                                                                 | 290 | 295 | 300 |
| Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys |     |     |     |
|                                                                 | 305 | 310 | 315 |
| Glu Tyr Lys Cys Lys Val Ser Asn Lys Gly Leu Pro Ser Ser Ile Glu |     |     |     |
|                                                                 | 325 | 330 | 335 |
| Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr |     |     |     |
|                                                                 | 340 | 345 | 350 |
| Thr Leu Pro Pro Ser Gln Glu Glu Met Thr Lys Asn Gln Val Ser Leu |     |     |     |
|                                                                 | 355 | 360 | 365 |
| Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp |     |     |     |
|                                                                 | 370 | 375 | 380 |
| Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val |     |     |     |
|                                                                 | 385 | 390 | 395 |
| Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Arg Leu Thr Val Asp |     |     |     |
|                                                                 | 405 | 410 | 415 |
| Lys Ser Arg Trp Gln Glu Gly Asn Val Phe Ser Cys Ser Val Met His |     |     |     |
|                                                                 | 420 | 425 | 430 |
| Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Leu |     |     |     |
|                                                                 | 435 | 440 | 445 |
| Gly Lys                                                         |     |     |     |
| 450                                                             |     |     |     |

&lt;210&gt; SEQ ID NO 379

&lt;211&gt; LENGTH: 214

&lt;212&gt; TYPE: PRT

&lt;213&gt; ORGANISM: Artificial Sequence

&lt;220&gt; FEATURE:

&lt;223&gt; OTHER INFORMATION: Synthetic Construct

&lt;400&gt; SEQUENCE: 379

|                                                                 |    |    |    |
|-----------------------------------------------------------------|----|----|----|
| Ala Ile Arg Met Thr Gln Ser Pro Ser Ser Phe Ser Ala Thr Thr Gly |    |    |    |
| 1                                                               | 5  | 10 | 15 |
| Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Gly Ile Ser Ser His |    |    |    |
|                                                                 | 20 | 25 | 30 |
| Leu Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Phe Leu Ile |    |    |    |

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| 35  |     |     |     |     | 40  |     |     |     |     | 45  |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Tyr | Val | Ala | Ser | Thr | Leu | Gln | Ser | Gly | Val | Pro | Ser | Arg | Phe | Ser | Gly |
| 50  |     |     |     |     | 55  |     |     |     |     | 60  |     |     |     |     |     |
| Ser | Gly | Ser | Gly | Thr | Asp | Phe | Thr | Leu | Thr | Ile | Ser | Ser | Leu | Gln | Ser |
| 65  |     |     |     |     | 70  |     |     |     |     | 75  |     |     |     |     | 80  |
| Glu | Asp | Phe | Ala | Thr | Tyr | Tyr | Cys | Gln | Gln | Tyr | Tyr | Ser | Tyr | Pro | Pro |
|     |     |     |     | 85  |     |     |     |     | 90  |     |     |     |     | 95  |     |
| Thr | Phe | Gly | Gln | Gly | Thr | Lys | Val | Glu | Ile | Lys | Arg | Thr | Val | Ala | Ala |
|     |     |     |     | 100 |     |     |     |     | 105 |     |     |     |     | 110 |     |
| Pro | Ser | Val | Phe | Ile | Phe | Pro | Pro | Ser | Asp | Glu | Gln | Leu | Lys | Ser | Gly |
|     |     |     |     | 115 |     |     |     |     | 120 |     |     |     |     | 125 |     |
| Thr | Ala | Ser | Val | Val | Cys | Leu | Leu | Asn | Asn | Phe | Tyr | Pro | Arg | Glu | Ala |
|     |     |     |     | 130 |     |     |     |     | 135 |     |     |     |     | 140 |     |
| Lys | Val | Gln | Trp | Lys | Val | Asp | Asn | Ala | Leu | Gln | Ser | Gly | Asn | Ser | Gln |
| 145 |     |     |     |     | 150 |     |     |     |     | 155 |     |     |     |     | 160 |
| Glu | Ser | Val | Thr | Glu | Gln | Asp | Ser | Lys | Asp | Ser | Thr | Tyr | Ser | Leu | Ser |
|     |     |     |     | 165 |     |     |     |     | 170 |     |     |     |     | 175 |     |
| Ser | Thr | Leu | Thr | Leu | Ser | Lys | Ala | Asp | Tyr | Glu | Lys | His | Lys | Val | Tyr |
|     |     |     |     | 180 |     |     |     |     | 185 |     |     |     |     | 190 |     |
| Ala | Cys | Glu | Val | Thr | His | Gln | Gly | Leu | Ser | Ser | Pro | Val | Thr | Lys | Ser |
|     |     |     |     | 195 |     |     |     |     | 200 |     |     |     |     | 205 |     |
| Phe | Asn | Arg | Gly | Glu | Cys |     |     |     |     |     |     |     |     |     |     |
|     |     |     |     | 210 |     |     |     |     |     |     |     |     |     |     |     |

&lt;210&gt; SEQ ID NO 380

&lt;211&gt; LENGTH: 441

&lt;212&gt; TYPE: PRT

&lt;213&gt; ORGANISM: Artificial Sequence

&lt;220&gt; FEATURE:

&lt;223&gt; OTHER INFORMATION: Synthetic Construct

&lt;400&gt; SEQUENCE: 380

|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Glu | Val | Gln | Leu | Val | Glu | Ser | Gly | Gly | Gly | Leu | Val | Gln | Pro | Gly | Gly |
| 1   |     |     |     | 5   |     |     |     |     | 10  |     |     |     |     | 15  |     |
| Ser | Leu | Arg | Leu | Ser | Cys | Ala | Ala | Ser | Gly | Phe | Thr | Ser | Ser | Arg | Tyr |
|     |     |     | 20  |     |     |     |     | 25  |     |     |     |     | 30  |     |     |
| Trp | Met | Ser | Trp | Val | Arg | Gln | Ala | Pro | Gly | Lys | Gly | Leu | Glu | Trp | Leu |
|     |     | 35  |     |     |     | 40  |     |     |     |     |     | 45  |     |     |     |
| Ala | Asn | Ile | Asn | Gln | Asp | Gly | Ser | Glu | Lys | Tyr | Tyr | Leu | Glu | Ser | Leu |
|     |     | 50  |     |     |     | 55  |     |     |     |     |     | 60  |     |     |     |
| Arg | Gly | Arg | Phe | Thr | Ile | Ser | Arg | Asp | Asn | Ala | Lys | Asn | Ser | Leu | Tyr |
| 65  |     |     |     |     | 70  |     |     |     |     | 75  |     |     |     |     | 80  |
| Leu | Gln | Met | Asn | Asn | Leu | Arg | Ala | Glu | Asp | Thr | Ala | Val | Tyr | Tyr | Cys |
|     |     |     |     | 85  |     |     |     |     | 90  |     |     |     |     | 95  |     |
| Ala | Arg | Asp | Glu | Glu | Gly | Tyr | Trp | Gly | Gln | Gly | Thr | Leu | Val | Thr | Val |
|     |     |     | 100 |     |     |     |     | 105 |     |     |     |     | 110 |     |     |
| Ser | Ser | Ala | Ser | Thr | Lys | Gly | Pro | Ser | Val | Phe | Pro | Leu | Ala | Pro | Cys |
|     |     |     | 115 |     |     |     | 120 |     |     |     |     | 125 |     |     |     |
| Ser | Arg | Ser | Thr | Ser | Glu | Ser | Thr | Ala | Ala | Leu | Gly | Cys | Leu | Val | Lys |
|     |     |     | 130 |     |     |     | 135 |     |     |     |     | 140 |     |     |     |
| Asp | Tyr | Phe | Pro | Glu | Pro | Val | Thr | Val | Ser | Trp | Asn | Ser | Gly | Ala | Leu |
| 145 |     |     |     |     | 150 |     |     |     |     | 155 |     |     |     |     | 160 |
| Thr | Ser | Gly | Val | His | Thr | Phe | Pro | Ala | Val | Leu | Gln | Ser | Ser | Gly | Leu |

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<210> SEQ ID NO 381
<211> LENGTH: 219
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic Construct

<400> SEQUENCE: 381

Asp Val Val Met Thr Gln Ser Leu Leu Ser Leu Ser Val Thr Leu Gly
1             5             10             15

Gln Pro Ala Ser Ile Ser Cys Arg Ser Ser Gln Ser Leu Val Tyr Ser
20             25             30

Asp Gly Asp Thr Tyr Leu Ser Trp Phe Gln Gln Arg Pro Gly Gln Ser
35             40             45

Pro Arg Arg Leu Ile Tyr Lys Val Ser His Arg Asp Ser Gly Val Pro
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Asp Arg Phe Arg Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Lys Ile

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1             5             10
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10-11. (canceled)

12. The method of claim 8, wherein the frequency is once every month.

13. The method of claim 8, wherein the frequency is once biweekly, once every 10 to 14 days, or once a week.

14-15. (canceled)

16. The method of claim 1, wherein the antibody, or antigen-binding fragment thereof, is administered subcutaneously.

17. The method of claim 1, wherein the disease or condition that can be treated with an ALK2 or ALK3 inhibitor is anemia.

18. The method of claim 17, wherein the anemia is associated with cancer, cancer treatment, myelofibrosis treatment, chronic kidney disease, acute renal disease or failure, chronic renal disease or failure, a myelodysplastic syndrome, thalassemia, a nutritional deficit, adverse reaction to medication, ineffective hematopoiesis, an inflammatory or autoimmune disease, splenomegaly, porphyria, vasculitis, hemolysis, a bone marrow defect, bone marrow transplantation, myelofibrosis, diabetes, acute liver disease, chronic liver disease, acute bleeding, chronic bleeding, an infection, hemoglobinopathy, drug use, alcohol abuse, Churg-Strauss syndrome, Felty syndrome, graft versus host disease, hematopoietic stem cell transplantation, osteomyelofibrosis, pancytopenia, pure red-cell aplasia, purpura Schoenlein-Henoch, Schwachman Diamond syndrome, advanced age.

contraindication to transfusion, surgery, trauma, a wound, an ulcer, urinary tract bleeding, digestive tract bleeding, frequent blood donation, or heavy menstrual bleeding.

**19.** The method of claim **17**, wherein the anemia is aplastic anemia, iron deficiency anemia, vitamin deficiency anemia, anemia of inflammation, IRIDA, anemia associated with a bone marrow disease, hemolytic anemia, sickle cell anemia, microcytic anemia, hypochromic anemia, sideroblastic anemia, Diamond Blackfan anemia, Fanconi anemia, or refractory anemia with excess of blasts.

**20-21.** (canceled)

**22.** The method of claim **17**, wherein the anemia is associated with elevated hepcidin levels.

**23.** The method of claim **1**, wherein the disease or condition that can be treated with an ALK2 or ALK3 inhibitor is fibrodysplasia ossificans progressiva (FOP).

**24.** (canceled)

**25.** The method of claim **1**, wherein the disease or condition that can be treated with an ALK2 or ALK3 inhibitor is a bone disease.

**26.** The method of claim **25**, wherein the bone disease is osteoporosis, osteopenia, osteopetrosis, osteogenesis imperfecta, bone fracture, bone cancer or cancer metastasis-related bone loss, Paget's disease, renal osteodystrophy, treatment-related bone loss, neuromuscular disease-related bone loss, burn-induced bone loss, anorexia-related bone loss, diet-related bone loss, bone loss associated with the treatment of obesity, low gravity-related bone loss, or immobility-related bone loss.

**27.** The method of claim **1**, wherein the disease or condition that can be treated with an ALK2 or ALK3 inhibitor is Sjogren's syndrome, multiple osteochondroma, diffuse intrinsic pontine glioma, posterior capsule opacification, cardiac hypertrophy, or cardiac fibrosis.

**28.** (canceled)

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